

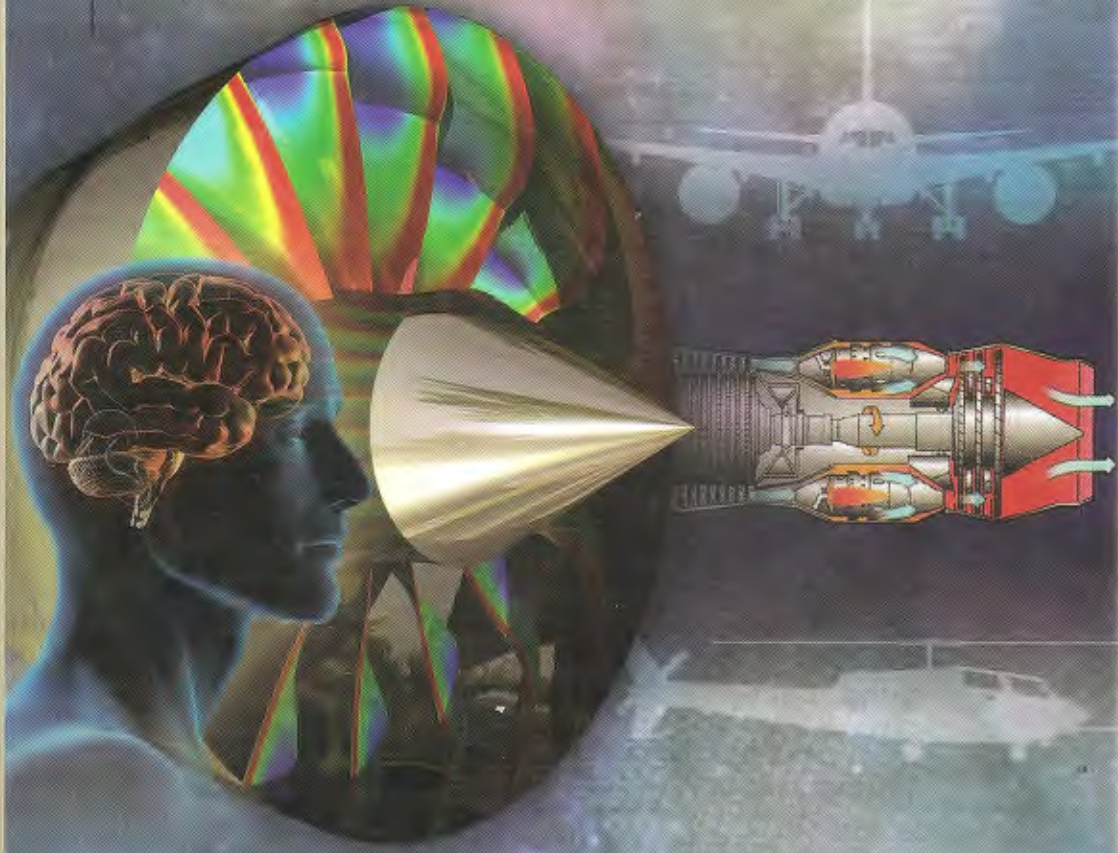
AIRCRAFT ENGINE CONTROLS

DESIGN, SYSTEM ANALYSIS, AND HEALTH MONITORING

Link C. Jaw

with

Jack D. Mattingly



American Institute of
Aeronautics and Astronautics

AIAA EDUCATION SERIES

JOSEPH A. SCHETZ
EDITOR-IN-CHIEF

Aircraft Engine Controls

Design, System Analysis, and Health Monitoring

Link C. Jaw

**Scientific Monitoring, Inc.
Scottsdale, Arizona**

With

Jack D. Mattingly

**Mattingly Consulting
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Foreword

Starting in the late 1970s, the U.S. Air Force began an effort to provide up-to-date educational material on aircraft gas turbine engines. This book is the most recent result of that effort. Early on, three subject areas were identified: aerothermodynamics, controls, and structures. Sponsored by the Aero Propulsion Directorate of the Wright Labs at Wright-Patterson Air Force Base, Ohio, the first efforts resulted in three volumes on aerothermodynamics by Gordon C. Oates and a volume on aircraft engine design by Mattingly, Heiser, and Pratt. All are part of this AIAA Education Series.

This up-to-date textbook on gas turbine engine controls was sponsored by the Propulsion Directorate at Tinker Air Force Base, Oklahoma, under the leadership of Timothy Dues. In 2003, Dues asked me to initiate this effort by finding an author for this book. After an extensive search, I found Link Jaw who was highly recommended to me by Sanjay Garg of NASA Glenn Research Center and Timothy Lewis of U.S. Air Force Research Laboratory. Dr. Jaw is extremely well-qualified to write this textbook because of his expertise in both gas turbine controls and health monitoring systems. A majority of the book is based on a one-week course on gas turbine engine controls developed and taught by Jaw during 2005–2007 to Air Force, Navy, and Army engineers. In addition to six chapters on engine controls, chapters are included on integration with aircraft control, advanced engine control concepts, and engine monitoring and health management.

Jack D. Mattingly

Table of Contents

Preface	xv
Acknowledgments	xvii
Nomenclature	xix
1. Overview of Engine Control Systems	1
1.1 Terminology for Control Systems.....	5
1.2 Introduction to Gas Turbine Engine Control Systems.....	7
1.3 Historical Development of Engine Control Systems.....	9
References	32
2. Engine Modeling and Simulation	37
2.1 Steady-State Engine Models	37
2.2 Dynamic Engine Models.....	39
2.3 Modeling of Complete Engine Dynamics.....	47
2.4 Modeling of Actuator and Sensor Dynamics	50
2.5 High-Fidelity Engine Simulations.....	57
2.6 Derivation of Linear Engine Models.....	61
References	66
3. Model Reduction and Dynamic Analysis	69
3.1 Example of Engine Dynamics.....	70
3.2 Frequency Spectrums of Interest.....	71
3.3 Dominant Dynamics in the Primary Control Bandwidth	72
3.4 Integrated Engine and Vehicle Dynamics	79
3.5 Variations in Engine Dynamics.....	90
References	92

4. Design of Set-Point Controllers	95
4.1 Controller Design for One-Spool Engines	96
4.2 Controller Design for Two-Spool Engines	104
4.3 Control Design for Turboshaft Engines	109
4.4 Some Practical Considerations for Set-Point Controls	114
References	118
5. Design of Transient and Limit Controllers	119
5.1 Schedule-Based Transient Controller Design	120
5.2 Nonlinear Considerations in Transient Controls	124
5.3 Acceleration-Based Transient Controller Design	128
5.4 Controller Design for Limit Protections	136
5.5 Rotorcraft Engine Considerations	139
References	141
6. Control System Integration	143
6.1 Quick Review	143
6.2 Power Setting	144
6.3 Transient Schedules	150
6.4 Control Modes	152
6.5 Engine Accessories	154
6.6 Controller Synthesis Examples	158
References	170
7. Advanced Control Concepts	171
7.1 Multivariable Control	171
7.2 Active Clearance Control	181
7.3 Active Stall and Surge Control	187
7.4 Active Combustion Control	196
References	204
8. Engine Monitoring and Health Management	207
8.1 Basic Concepts	207
8.2 Monitoring System Design	215
8.3 Monitoring Algorithm Design	220
8.4 Trend Monitoring from Periodically Recorded Data	227
References	235
9. Integrated Control and Health Monitoring	239
9.1 Integration Architecture, Capabilities, and Requirements	240
9.2 Life-Extending Control	245
9.3 Safety Assurance Control	249
9.4 Fast-Response Control for Helicopter Engines	253
References	263

Appendix A. Fundamentals of Automatic Control Systems.....	265
A.1 Open-Loop and Closed-Loop Controls	265
A.2 Representations of Linear Dynamic Systems	267
A.3 Basic Control Laws	276
A.4 Analysis of Linear Dynamic Systems	279
A.5 Some Linear Control Design Methods	291
A.6 Nonlinear Dynamic Systems.....	296
References	301
Appendix B. Gas Turbine Engine Performance and Operability	303
B.1 Introduction	303
B.2 Operational Envelopes and Standard Atmosphere	304
B.3 Basic Thermodynamics of Perfect Gas	306
B.4 Total Properties and Mass Flow Parameter.....	310
B.5 Dimensionless/Corrected Parameters and Component Maps	315
B.6 Basic Engine Performance Model.....	319
B.7 Operability Analysis	336
References	345
Index	347
Supporting Materials	363

Preface

Since its invention in the late 1930s and early 1940s, the gas turbine engine has become the primary source of propulsive power for aircraft by the 1950s. Today, the gas turbine engine and its related technologies represent one of the most efficient forms of power generation; the gas turbine engine is also the predominant choice for propulsion in aviation, not only for its high efficiency, but also for its ability to demonstrate the reliability of operation necessary to meet stringent flight safety requirements.

Design of a gas turbine engine requires the knowledge of multiple academic disciplines including aerodynamics, fluid mechanics, solid mechanics, thermodynamics, chemistry, and material sciences. Controlling such complex machinery requires a thorough understanding of the performance of the engine “system” as a whole. For some aviation applications, a gas turbine engine must provide a wide range of predictable and repeatable thrust performance over the entire operating envelope of the engine, which can cover the altitude from sub sea level to tens of thousand feet. These altitude changes along with variations in flight speed from takeoff to supersonic velocities result in large, simultaneous variations in engine inlet temperature, inlet pressure, and exhaust pressure. These large variations in engine operating conditions and the demand for precise thrust control, coupled with the demand for highly reliable operation, despite the complexity of the engine itself, create a significant challenge for the design of engine control systems.

Historically, when unexpected problems are encountered in the field, modifications to the control system becomes the desired “path to a fix” because it is easier to change the control logic and a few lines of code rather than to redesign, remanufacture, and reinstall new engine components.

Given the significance of the control system on a gas turbine engine, there have been very few organized, comprehensive publications about the design of engine control systems since the gas turbine engine was invented some 70 years ago. The last known book about this subject was written by Sobey and Suggs and published in 1963. Although the book provided a very good foundation for the control of aircraft powerplants, its contents could not have covered some of the most interesting and significant advancements in engine control and monitoring systems that have occurred in the last 50 years. Hence, this book is written to provide a continuum of knowledge in gas turbine engine control and monitoring systems.

In the early 2000s, NASA John H. Glenn Research Center was interested in publishing a technical report on the historical development of aircraft engine

controls in the United States to commemorate the 100th anniversary of the first Wright brothers' flight in 1903. I was fortunate to be chosen as the primary author for this work. Shortly after the engine controls report was completed, I was again given an opportunity by NASA to conduct a survey on the development of engine health management technologies. These two survey reports prompted the U. S. Air Force to ask me to prepare and conduct two short courses on the design of aircraft engine control and monitoring systems in 2005 and 2007, respectively. This book is based on the materials covered in these short courses. I took upon the challenge to write this book as a way to thank all of those who have taught and inspired me, although I believe there are other qualified people to undertake the "assignment." The major challenge in writing this book was to assimilate more than 50 years of knowledge in aircraft engine controls, and yet to present this knowledge in a thought-stimulating manner for readers with a wide range of backgrounds. During the process of preparing the book manuscript, I was often torn between including and discarding a certain topic, or a certain reference, in order to fit the book in an adequate length. Hence, I recognize that the book inevitably leaves out items that some readers would consider relevant, or vice versa.

This book is *not* about the theories of control and monitoring systems; rather, it is about applying control theories and engineering knowledge to real-world problems. In many cases, it is about "knowing" the problem rather than "solving" the problem completely. The book uses aircraft engines as the "target" system of application. Because the turbine engine represents one of the most complex physical systems in the world, and because aircraft engines have pioneered many of the control and monitoring technologies used by other industries to date, I believe that the modeling techniques, design methodologies, and problem-solving approaches discussed in this book can be applied to other engineering systems and industries. Hence, the intended readers of this book are practicing engineers in controls and monitoring system design, students in their senior year of an undergraduate study, and students in their first year of a graduate study.

Link C. Jaw
August 2009

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I am grateful for the Garrett Corporation (now a part of Honeywell International) for awarding me an unprecedented doctoral fellowship, which allowed me to take an extended leave from work to study at Stanford University. Specifically, I want to thank Jack Fredlake, John Mason, and Joan Cudahy for their efforts in helping me get this award. While at Stanford, I had both the honor and pleasure to study under Arthur E. Bryson. A master of multidisciplinary optimization, Prof. Bryson has set great examples for me in productivity and humility.

I am also grateful to the U.S. Air Force for providing generous financial support for the development of two short courses on engine control and monitoring system designs, which provided a foundation for the manuscript of this book. I would like to thank the participants of the short courses for their feedback and suggestions. Their positive feedback has been a constant motivation for me to complete this book. In particular, I thank Fred Kimler and Douglas Decker for their discussions and suggestions.

This book would not have been published without Jack Mattingly. Jack has not only shown me the way of writing a technical book, but also recreated many figures in the book to ensure a professional quality. Moreover, he wrote Appendix B of this book on engine performance and operability to make the book more self-contained. I would also like to thank AIAA editor Pat DuMoulin and her team for their work in editing the manuscript.

I thank my daughter Jessica for contacting the publishers for their permission to reuse some of the figures and images in this book. I also thank my children

Timothy and Stacey for their patience and encouragement while writing this book. I am indebted to my mother for instilling in me the value of dedication, perseverance, and continuous learning. Lastly, I am grateful to my wife for being always supportive, loving, and cheerful. She is one of the great blessings in my life.

Nomenclature

<i>A</i>	=	area, in. ² or m ² ; state dynamic distribution matrix
<i>a</i>	=	speed of sound, fps or m/s
<i>A(s)</i>	=	actuator transfer function
<i>B</i>	=	input-to-state distribution matrix
<i>C</i>	=	state-to-output distribution matrix
<i>c_d</i>	=	discharge coefficient
<i>c_p</i>	=	specific heat of the air or a gas at constant pressure, BTU/lb-°R or J/Kg-K
<i>c_m</i>	=	specific heat of the turbine metal, BTU/lb-°R or J/Kg-K
<i>D</i>	=	input-to-output distribution matrix
<i>F</i>	=	force or thrust, lb or N; discrete-time state dynamic matrix
<i>f</i>	=	state function; fuel-to-air mass flow ratio
<i>G</i>	=	discrete-time input-state distribution matrix
<i>G(s)</i>	=	system or process transfer function
<i>g</i>	=	output function
<i>H</i>	=	discrete state-to-output distribution matrix (same as <i>C</i> for continuous systems)
<i>H(s)</i>	=	measurement feedback transfer function
<i>h</i>	=	enthalpy, in. or cm
<i>I</i>	=	current, inertia, or impulse function, A; moment of inertia or mass moment of inertia, lb-ft ² or Kg-m ² ; identity matrix
<i>I_{z0}</i>	=	moment of inertia of each blade about the lead-lag hinge, in. ⁴ or m ⁴
<i>IGV</i>	=	inlet guide vane angle or position, degree, radian (or rad or r), in, or m
<i>K(s)</i>	=	controller transfer function
<i>K_d</i>	=	derivative control gain
<i>K_e</i>	=	engine gain constant
<i>K_i</i>	=	integral control gain
<i>K_p</i>	=	proportional control gain
<i>k</i>	=	gain constant; stiffness (or spring constant)
<i>L</i>	=	length, in. or cm
<i>M</i>	=	Mach number
<i>$\dot{m}$</i>	=	mass flow rate, pps or g/s
<i>$\dot{m}_{ci}$</i>	=	corrected mass flow rate at station <i>i</i>
<i>N</i>	=	rotational speed, rpm or %

N_{ci}	= corrected engine speed at station i
N_{cmd}	= speed command, rpm or %
N_g	= rotation speed of gas generator
N_H	= rotational speed of high-pressure spool, rpm or %
N_L	= rotational speed of low-pressure spool, rpm or %
N_p	= rotation speed of power turbine
N_1	= rotational speed of low-pressure spool, rpm or %
N_2	= rotational speed of high-pressure spool, rpm or %
P	= pressure, psi or N/m ² ; power HP or W; cumulative probability
p	= probability density function
P_i	= pressure at station i
P_{ti}	= total (stagnation) pressure at station i
Q	= torque, ft-lb, N-m, or J; volumetric flow rate, m ³ /s, ft ³ /s, or gpm; weighting matrix
R	= gas constant, ft-lb/lb-mol-°R or J/K-mol; weighting matrix
s	= frequency variable, 1/rad; entropy, BTU/°R or J/K
sp	= specific gravity
T	= temperature, °F, °R, °C, or K; torque, ft-lb or N-m
t	= time, s
T_{ti}	= total (or stagnation)
u	= input (control) variable
V	= velocity, fps or m/s; volume, in. ³ or m ³ ; voltage, V
v	= measurement noise
W	= weighted flow rate, lb/s or Kg/s
W_f	= fuel flow rate, pph or Kg/h
x	= state variable
$\dot{x}$	= rate of change of state variable
$\bar{x}$	= mean (or expected) value of x
y	= output variable
z	= measured output variable
Γ_k	= multistage (or discrete) input matrix for stage (or interval) k
γ	= ratio of specific heat
Δ	= finite change
δ	= small (or infinitesimal) change
δ_i	= dimensionless total pressure, P_i/P_{std}
∂	= derivative symbol
Λ	= diagonal matrix
η_i	= adiabatic efficiency of component i
θ	= dimensionless temperature, T/T_{std}
θ_i	= dimensionless total temperature, T_i/T_{std}
ρ	= density
%	= standard deviation
σ	= dimensionless density, ρ/ρ_{std}
τ	= time constant; characteristic time (second, s)
Φ_k	= multi-stage (or discrete) dynamic matrix for stage (or interval) k
ω	= frequency (radian, rad)
$\wedge$	= estimated value

Overview of Engine Control Systems

The gas turbine engine and its related technologies represent one of the most efficient forms of propulsion and power generation, with applications ranging from land-based power generation, ground-based vehicle propulsion, on-board power and propulsion sources for marine ships, to aircraft propulsion and auxiliary power systems. Arguably gas turbine technology has also found its way into turbopumps and turbochargers for automobiles, trucks, off-road vehicles, and even the turbomachinery used in space applications such as the space shuttle main engine. Continuous improvements in demonstrated gas turbine reliability now enable safe, two-engine aircraft flights over ocean water for extended periods of time. This has been accomplished even though the gas turbine engine [1, 2] is one of the most complex types of machines ever created by humans.

A major challenge facing the design of turbine engine control systems is that engine performance varies from engine to engine and even on the same engine over time. Although engine operation and reliability have continuously been improving, no two engines are exactly alike. Engines vary because of the reality of manufacturing tolerances and material properties, whereas time-elapsd performance variations come from deteriorations and the mix-and-match nature of engine components when an engine is rebuilt after repair or overhaul. Because of the harsh operating environment of the gas turbine engine, hot-section components like turbines and nozzles deteriorate faster than fan and compressor components, so that they are more frequently refurbished or replaced. Because a gas turbine engine control system can last for several engine overhaul cycles, the control system must deal with not only slow degradations in engine/component performance but also with sudden shifts in performance after repair. This means that the control system must be easily trimmed, or even capable of *autotrim* (or self-calibration).

Design of engine control systems faces yet another challenge. Historically, when unexpected problems are encountered in the field, modifications to the control system become the desired “path to a fix” because it is easier to change the control logic and a few lines of code rather than to redesign, remanufacture, and reinstall new engine components. Thus, control implementations need to have the flexibility of adapting to these in-field modifications.

Finally, the engine control system must handle some engine monitoring functions. Traditionally, engine monitoring functions have been a part of the control system functionality. Because each engine has its own performance signature in

time, there is a practical need for continuous monitoring of engine health state so as to use feedback control to accommodate performance degradations. Continuous monitoring is also needed to manage engine fleet so that they are available to support vehicle-level dispatch reliability (or mission success). To monitor engine health state and maintain high engine availability, an engine monitoring system must be able to detect incipient failures and predict how much longer the engine can operate with the "known" degradation before the failure becomes so severe that the engine performance becomes unacceptable. Because most of the failures in gas turbine engines are either in control components like actuators, sensors, and controllers, or in engine modules that could be controlled to avoid or accommodate the failure through some actions of the control system, the monitoring system must work closely with the control system to sample operational signals and assess engine and component health. Furthermore, the sensors, actuators, and controllers on modern gas turbine engines are dual or triple redundant, and they require substantial cross-channel communications and fault-tolerant logics; hence, coordinating between the control system and the monitoring system becomes essential for failure detection and accommodation.

A survey of the historical development of aircraft engine control systems in the United States [3] has revealed remarkable accomplishments by control system designers and engineers in meeting these challenges. Another survey [4] of engine monitoring capabilities indicates that aviation gas turbine engines have led the way in the development and maturation of modern monitoring and condition-based maintenance capabilities for all industrial equipment and systems.

Given the significance of the control system on a gas turbine engine, there have been very few organized, comprehensive publications about the design of engine control systems since the gas turbine engine was invented some 70 years ago. The last known book about this subject was written by Sobey and Suggs and published in 1963 [5]. Although the book provided a very good foundation for the control of aircraft powerplants, its contents could not have covered some of the most interesting and significant advancements in engine control and monitoring systems that have occurred in the last 50 years. Although various papers and book chapters have provided good updates and summaries for engine controls (e.g., [6] and Appendix O of [2]), they are limited in explaining the design methodologies of control laws and monitoring algorithms, and they primarily cover high-level design principles for turbofan engines. Hence, this book is written to provide a continuum of knowledge in the design of engine control and monitoring systems, with a dual interest in both turbofan and turboshaft engines. In doing so, the contents are focused mostly on four areas of interest: 1) modeling of engine dynamics, 2) application of specific control design methods to gas turbine engines, 3) advanced control concepts, and 4) engine condition monitoring. The material in this book is applicable to all air, land, and sea-based gas turbines, although the majority of the treatment in the book is concerned with aviation gas turbines. The emphasis is on aviation gas turbine controls because it is believed that most of the issues in the control design of other gas turbine control systems will be included in the substantial challenges inherent in aviation gas turbine control designs. The book is intended to provide a one-stop resource for understanding the design of modern turbine engine control and monitoring systems. Because gas turbine engine control and monitoring systems have successfully employed

applications of *nonlinear control* and *system health management*, this book can also be used as a reference for the engineers and researchers who are designing control and monitoring systems for other industrial equipment and systems.

A control system generally consists of four types of components: a plant (or physical system) to be controlled, a number of sensors, a number of actuators, and a controller. This book primarily deals with the controller and associated control and monitoring algorithms. Conventional and high-response actuators are covered in Chapters 2 and 7, respectively. Sensors are the least covered subject in this book because they can be approximated by simple dynamic systems in control law design and because sensor technologies vary widely.

The controller, like a human brain, collects and manipulates sensory inputs, determines what control actions are needed in terms of control commands, and sends the control commands to appropriate actuators. The rules that a controller goes by in making control decisions are called control laws. Similarly, the rules that a monitoring system goes by in making health management decisions are called monitoring algorithms. In this book, design of monitoring system follows the design of control system because 80% of the effort in designing a monitoring system is spent on estimating the expected or predicted values of the physical system to be monitored, just as 80% of the effort in designing a control system is spent on modeling and understanding the dynamics of the physical system. Experience tell us that a control system is only designed to provide satisfactory performance in the frequency spectrum of interest, and the performance and stability characteristics will always be changed by the nonlinearity in the system.

This book contains nine chapters and two appendices. Appendix A reviews the fundamental principles of automatic control systems, and Appendix B reviews the fundamental principles of engine performance and operability. These two appendices and the first three chapters of the book are intended to provide the reader with sufficient knowledge to handle the design issues facing engine control and monitoring systems. In the first chapter, a brief history of the development of engine control systems (mostly in the United States) is introduced. This historical perspective of engine control design is divided into four periods: the startup period (from 1940 to 1949), the growth period (from 1950 to 1969), the electronics period (from 1970 to 1989), and the integration period (from 1990 to mid-2000s). After the introduction of the engine control system, we will discuss engine modeling and simulation in Chapter 2. Both the steady-state and dynamic models of generic gas turbine engines are covered, although dynamic models are more relevant to control and monitoring system designs. The models of actuators and sensors are also derived in this chapter, with an emphasis on hydraulic and pneumatic actuators. Two nonlinear simulation models of the entire engine are briefly described. This is followed by a discussion of deriving linear engine models, at various nominal operating points, from a nonlinear simulation model, or from actual engine test or flight data. In Chapter 3, we will discuss two model reduction techniques and analyze engine dynamics using reduced models. We will start the analysis with a two-spool turbofan engine. Then we will analyze a turboshaft engine (for helicopters). We will also analyze the dynamics of an integrated system of engine and airplane or helicopter. This leads to the discussion of an important challenge in helicopter engine control design, that is, the airframe-engine torsional stability problem.

The next three chapters deal with engine control designs. Chapter 4 covers the design of set-point controllers by using two classical design methods, the root locus method and the frequency response method. These methods are applied to a generic one-spool engine and a generic two-spool engine. Furthermore, set-point controller design for turboshaft engines is discussed at the end of the chapter. In Chapter 5, we will discuss transient control and limit-protection control designs. Designing these controls is more challenging than designing set-point controls because a set-point control for a turbine engine is primarily a linear control problem, whereas transient and limit controls are nonlinear design problems. There are basically two different methodologies in controlling engine transients: schedule-based control and acceleration control. These two methodologies will be explained, and their advantages will be compared. As in the preceding chapter, transient control for turboshaft engines will be discussed. After learning the set-point and transient control designs, we will be in a position to integrate these controllers to make them work over the entire engine operating envelope. In Chapter 6, the methods of controlling the engine power (or thrust) throughout its operating envelope, or the so-called power-setting logic, will be discussed. Then two simplified, operating-line engine models, that is, a one-spool engine model and a two-spool engine model, will be described. These models will be used to illustrate the design of integrated controllers over the entire operating line (from idle to TO power).

The last three chapters deal with advanced control concepts and engine monitoring related topics. In Chapter 7, we will first review the modern (or multivariable) control design concept. Examples of applying some multivariable design techniques to the turbine engine in both the time domain and the frequency domain will be described. We will then devote the rest of this chapter to active controls; specifically we will look at the three widely studied active control problems for turbine engine, namely, active tip clearance control, active stall/surge control, and active combustion control. Because actuation has been identified as the key enabling technology for active controls, we will focus our discussions on actuation and use three specific design examples to illustrate the challenges and the work involved in designing high-response actuators.

Design of engine monitoring and health management system will be covered in Chapter 8. Because monitoring and controls are closely related by sharing the same methodology in modeling and estimation, it is appropriate to cover monitoring and health management related topics after the reader has learned engine modeling and controls. Also because aircraft engine condition monitoring has pioneered a lot of the modern health management and condition-based maintenance capabilities, it is necessary to start the chapter with the general concepts of safety, reliability, fault, diagnostics, prognostics, maintenance, and support, before we dive into monitoring system architecture and algorithms. The subject of monitoring system design is broad enough to warrant a textbook of its own; however, presenting a concise treatment of the subject in this book, which is primarily about engine controls, puts forth the right framework for monitoring system design, and that is that the design must be made in the same context of control system design involving embedded sensors, system responses, state estimations, control commands, and corrective actions. Lastly, we will bring together the control and monitoring capabilities in Chapter 9 to look at some integrated control and health

management problems for turbine engines. First, we will discuss the affordability concept in evaluating engine capabilities. We will also discuss the requirements for diagnostic and prognostic systems. Then, three integrated control problems will be discussed: a life-extending control, a safety-assurance control, and a fast-response control for helicopter engines. As in the discussion of active controls in Chapter 7, these integrated control and health management problems are presented to give an introduction to the reader about the challenges, potential benefits, and the converging trend of future engine control and monitoring systems.

In the remainder of this introductory chapter, sections on terminology, history, and operation are provided. For the most part, this book will follow the terminology of [2]. Control-system-related terminology and the naming convention for symbols are described in Sec. 1.1. Turbine engine control systems are introduced briefly in Sec. 1.2. A historical perspective of the development of turbine engine control systems is covered in Sec. 1.3.

1.1 Terminology for Control Systems

A system can mean a physical or an abstract object. This object contains certain entities, and each entity represents a certain property. In this book, system refers to a physical object. It can mean a physical system like a gas turbine engine, or it can mean a control system that makes a physical system behave as desired. Whenever possible, the term *physical system* or *process* is used as the object to be controlled like a jet engine, and the term *control system* is used as the controlling object.

A control system can be manual or automatic. In this book, we are primarily interested in *automatic* control systems, that is, the control system that automatically makes the physical system follow desired behavior based on an external, high-level command. Automatic control systems fall in two categories: *open loop* (or feedforward) and *closed loop* (or feedback) [7]. Figure 1.1 shows the two types of control systems in block diagram form. Some fundamentals of automatic control systems can be found in Appendix A. For additional reading and background information, the readers are referred to the textbooks by Dorf and Bishop [7] and Brogan [8] on modern control theories and systems.

As seen in Fig. 1.1, a closed-loop control system consists of four types of components: 1) the physical system to be controlled (or the process), 2) the controller (or the computational unit), 3) the actuator, and 4) the sensor. The controller determines desired control actions by executing a series of computations and checking a set of rules, which are collectively called the *control law*. Before electronic controls became available, controllers were primarily of the electrical or the mechanical type, for instance, cams and linkages were the computational elements of the mechanical type. For gas turbine engines, the hydromechanical fuel control performed both the controller and the actuator functions. Note that the actuator includes not only the effector that changes a control variable, but also a working medium that provides the energy to effect the change. For instance, a valve is an effector for controlling positions. It is moved by a working fluid (in either gas or liquid form), and the working fluid needs to be stored and pressurized before it flows through the valve. Similarly, an electric motor can be an

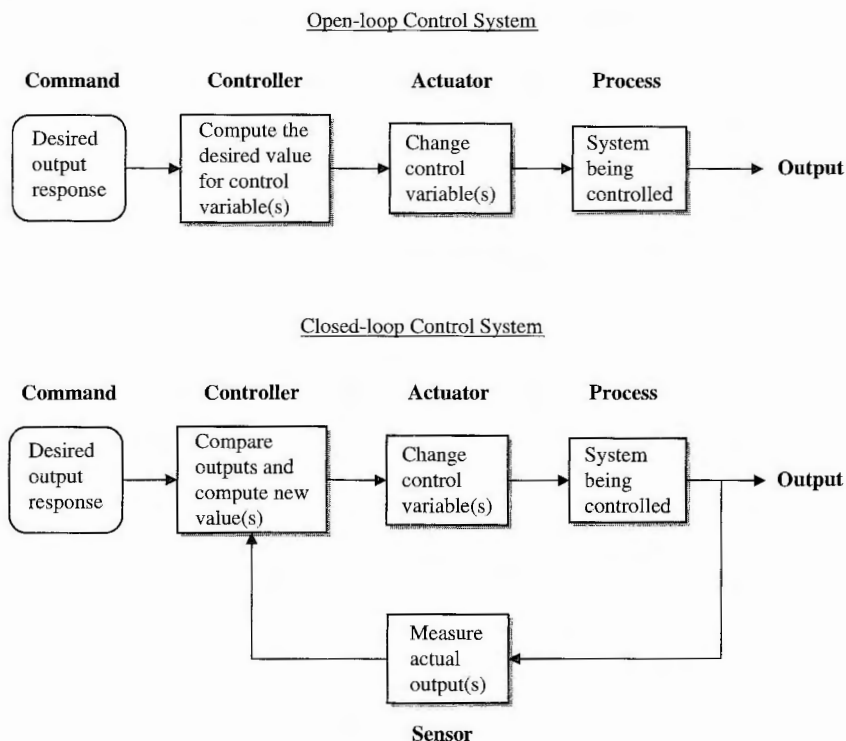


Fig. 1.1 Block diagrams of open-loop and closed-loop control systems.

effector for position control, but it needs current to drive it, and the current must come from a power source like a battery. The term accessory sometimes is used in the aviation industry to include the sensor and the actuator components.

An automatic control system provides three basic functions:

1) Set-point control (*thrust control mode* in [5]) is used to keep the physical system near desired operating conditions over time. It is also called steady-state control or a regulation.

2) Transient control is used to move the system's steady-state operating condition from one to another within a given time period and within prescribed bounds.

3) Limit protection control (*physical limiting control mode* in [5]) is used to protect the physical system from entering unsafe or undesirable operating regions.

Because each of these control functions requires a specific approach to accomplishing its goal, we can consider each function a control mode. Switching between two different control functions (or modes) is called a mode switching. Occasionally we can encounter a control mode that is defined as the output variable to be regulated or tracked; for instance, a speed control mode, a pressure control mode, or a temperature control mode. In general, *functional* control modes or *output*

control modes are used interchangeably. In Chapter 6, a more unified concept of control mode, the *system* control mode, will be introduced.

The naming convention for the symbols representing the engine and the control system will follow the commonly used notations of the engine manufacturing industry (as given in Nomenclature for this book); for instance, upper-case letter P refers to pressure, T refers to temperature, N refers to engine shaft speed, A refers to area, Q refers to torque in a rotational part, and F refers to the net thrust produced by the engine. However, some symbols are used to represent more than one type of physical quantity; for instance, V can represent the volume of an open space or a collection of mass, the velocity of air or gas flow, and the voltage of an electrical system; similarly, I can represent electrical current, inertia of a mass, or impulse response.

The naming convention for the symbols representing engine models and control laws will follow the convention defined as follows: 1) lower-case letter x refers to the state variable; 2) lower-case letter u refers to the control (or input) variable; 3) lower-case letter y refers to the output variable; 4) upper-case letters A, B, C, D , etc. refer to matrices used in the state equation and control laws; 5) lower-case letters f, g, h , etc. represent functions of time (or responses in time domain); 6) upper-case letters F, G, H , etc. represent Laplace transforms (or functions in frequency domain); and 7) K and k represent control gains.

1.2 Introduction to Gas Turbine Engine Control Systems

Modern gas turbine engine control systems are closed-loop control systems that consist of all four types of control components: controller, sensor, actuator, and accessory. The simplest engine control system is one that produces desired engine thrust (or shaft power) by changing the fuel flow. Because reliable, in-flight engine thrust measurement is not currently practical, the engine shaft rotational speed N or the engine pressure ratio (EPR) has been used effectively as an indicator of engine thrust (or power). Hence, for this simplified control system, the command variable (or the desired output variable) is shaft speed (or engine pressure ratio), the control variable is actuator position, the actuator is fuel metering valve, the output of the metering valve is the fuel flow that is injected in the combustor, the output of the engine is engine power setting variable (shaft speed or engine pressure ratio); furthermore, fuel control accessory components are the fuel tank and the fuel pump and the sensors (tachometer or pressure transducers). Figure 1.2 illustrates this simplified engine control system. The system accomplishes closed-loop control of engine thrust *through* speed, and so thrust is the control mode, speed is the output variable, and the speed feedback loop is the *primary control loop* of the engine. Note in the figure that the speed command is converted from the pilot's throttle input signal, which is commonly referred to as the power lever angle. Similarly, the closed-loop control of engine pressure ratio is called pressure ratio control mode.

If the "power request" in the block diagram is changed to "turbine temperature," then the process represents a temperature control loop, or a temperature mode. Likewise, if the "power" and "fuel flow" are changed to inlet guide vane (IGV) or variable stator vane (VSV) angle, then the process becomes an IGV control loop or IGV control.

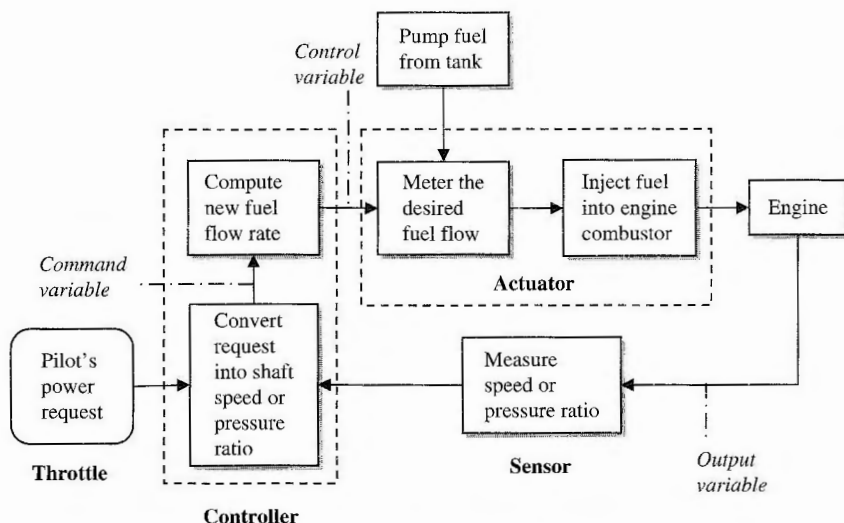


Fig. 1.2 Functional diagram of a simplified engine control system.

An aircraft engine is designed to operate in a wide operating envelope (to support aircraft mission requirements). The operating envelope is typically defined as a graph of flight altitude vs flight Mach number. Actually the operating envelope for a turbojet or turbofan engine has a third dimension, the ambient temperature, as shown in Fig. 1.3 [5].

For a modern engine, the altitude ranges from near sea level to 50,000 feet or higher, and the Mach number ranges from static condition (or Mach zero) to high subsonic speed for civil engines and to a few multiples of the speed of sound for military engines. Moreover, even at the same altitude and Mach number, the ambient air temperature can vary from a hot summer day in a desert climate to a cold winter night in a polar region. These variations in the operating condition impose severe challenges on the engine control system as it tries to provide satisfactory engine performance with adequate safety margins. From the control design perspective, these challenges translate into the *engine control envelope*, which is the allowable operating region defined on the plane of the primary control variable fuel flow vs engine shaft speed (or pressure ratio in the case of EPR control) as shown in Fig. 1.4. Note that the fuel ratio unit has been almost universally used as the primary control variable rather than the fuel flow. The fuel ratio unit is defined as the ratio of fuel flow to compressor exit pressure, that is, $RU \equiv W_f/P_3$. The reason for this practice in the engine industry will be explained in Chapter 4, and in Chapter 5.

After an engine starts and reaches its idle speed, the engine's "control envelope," in a simplistic view, is the "box" in Fig. 1.4. The box is bounded by the maximum flow limit on the top, the minimum flow limit on the bottom, the idle speed limit on the left, and the maximum speed limit on the right. The maximum flow limit prevents the engine from surging, whereas the minimum flow limit

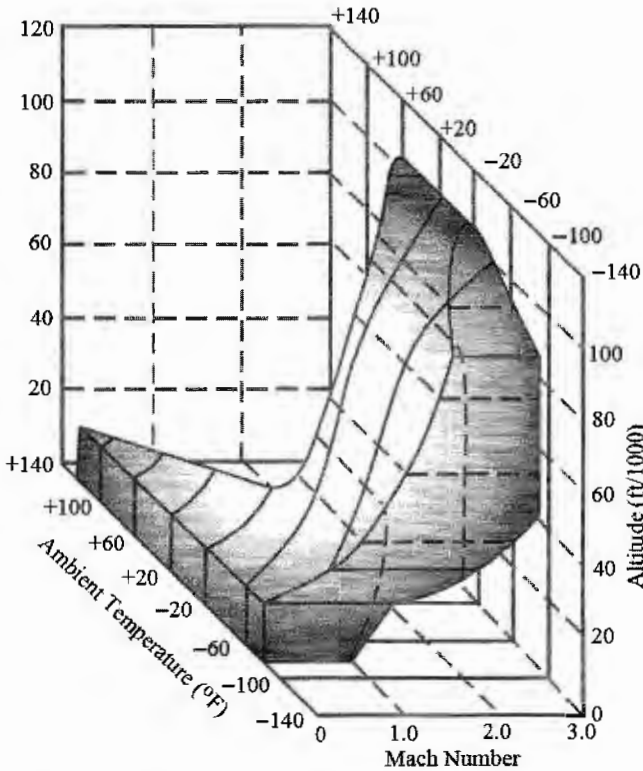


Fig. 1.3 Typical turbojet and turbofan engine operating envelope (from [5], reprinted with permission).

prevents the engine from flaming out. All of these limits vary with the operating condition of the engine. The primary goal of an engine control system is to keep the fuel flow within the box while changing the engine speed up and down in response to pilot's throttle command. Hence, the key to successful engine control design is to define these limits accurately and to keep the engine fuel flow (and other control variables) within these limits at all times.

1.3 Historical Development of Engine Control Systems

The world's first jet-powered flight was made on 27 August 1939 by the German Heinkel HE-178. This aircraft was powered by the HeS-3B turbojet, designed by Hans von Ohain. About the same time in England, Sir Frank Whittle made the first ground run of his W.1 engine in April 1937. The W.1 engine started England's jet-powered age on the Gloster E28/39 aircraft on 15 May 1941 [9].

Since the birth of jet-powered flight in the late 1930s and early 1940s, control technologies for turbine engines, particularly control design methodologies, have advanced significantly. The W.1 engine used a physical stop on the throttle to limit

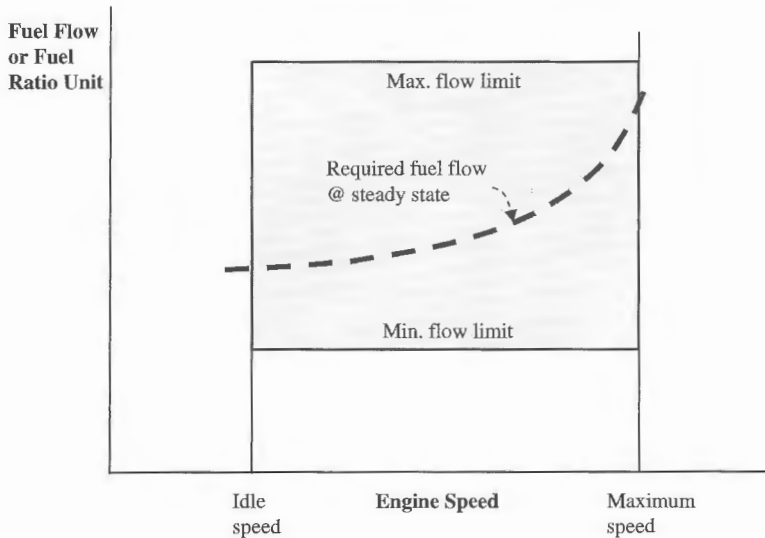


Fig. 1.4 Control envelope for an aircraft turbine engine.

engine speed to 17,000 rpm. The first U.S. jet engine was built in 1942 by General Electric (GE). This engine, GE I-A (see Fig. 1.5), was designed based on the British Whittle turbojet engine. The control for the GE I-A was a hydromechanical governor, which metered the fuel flow going into the engine to be proportional to the difference between the desired speed and the actual speed of the turbine. To prevent the engine from flame-out, a minimum-flow stop was added to the fuel metering valve. To prevent the engine from overtemperature, a maximum-flow schedule was also incorporated. This system represents the basic functionality of



Fig. 1.5 First jet engine built in the United States—GE I-A (courtesy of General Electric Company).

the control system for a single-spool engine including fuel metering, overspeed protection, flame-out protection, and overtemperature protection.

In contrast, a modern jet engine is controlled by a full-authority digital electronic control (FADEC) that automatically sets engine power with unrestricted throttle movement.

From the historical perspective, the advancement of engine control technologies can be divided into four periods [3]: 1) infancy or start-up period (from 1940 to 1949), 2) growth period (from 1950 to 1969), 3) electronics period (from 1970 to 1989), and 4) integration period (from 1990 to mid-2000s). Major advancements in control design methodology and modeling in each of these phases are summarized in the subsections that follow.

1.3.1 Infancy Period

The computing elements of engine controls during this period were primarily mechanical. These mechanical computing devices fell into three general categories [5]: gear trains, linkages, and cams. These devices were used independently or in combination to provide the best output results. Engine control laws in this period were primarily designed by the frequency response method, which uses Bode plots as a way to “shape” the frequency response of the controlled system to achieve desired gain and phase margins (see Appendix A). Time-domain design methods, like the step response method and the root locus method, were also used to improve the control system performance at high altitude.

In 1948, GE tested the world’s first afterburning turbojet engine, the J47 (see Fig. 1.6). The engine incorporated a hydromechanical fuel control for the main combustion chamber and an electronic (vacuum tube) fuel control for the afterburner. Unfortunately, the reliability of the vacuum-tube-based electronic control was poor as a result of the harsh operating environment of the jet engine.

Engine modeling in this period was focused primarily on steady-state performance analysis. The performance analysis relied on engine module (or component) performance maps, derived from module rig tests. Assumed constant specific-heat gas properties were used to calculate the thermodynamic properties of each module.

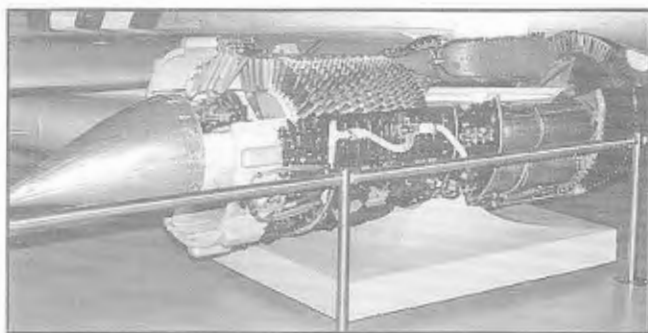


Fig. 1.6 First afterburning engine—GE J47 (courtesy of U.S. Air Force Museum, Dayton, Ohio).

Because engine operating parameters at each engine operating point could take several hours to calculate with a slide rule, techniques to use corrected parameters, that is, the engine performance parameters corrected to the sea level, standard day condition [1], were developed to reduce the number of operating points needed to estimate engine performance over the entire operating envelope.

Analysis of engine dynamics focused on the shaft dynamic models only, and it assumed that the engine was at the quasi-static condition, where the shaft power consumed by the compressor and the shaft power produced by the turbine is *unbalanced*. This kind of analysis yielded a first-order linear dynamic system for a turbojet engine [10].

1.3.2 Growth Period

The computing elements of fuel controls during this period were primarily hydromechanical because the majority of engine control functions required some form of regulation of a liquid (or fluid). Hydromechanical controls (HMCs) were self-contained with metering valves, and they normally did not depend on external power besides the hydraulic “working” fluid. They could generate substantial forces available for computing and actuation, and consequently, offered rapid response characteristics [5].

In 1951, Pratt and Whitney (PW) flight tested the world’s first two-spool turbojet engine, the J57 (see Fig. 1.7), and achieved supersonic speed on a YF-100 aircraft. The J57 control system consisted of a hydromechanical fuel control (HMFC or HMC) for the engine’s main combustion chambers, an HMFC for the afterburner, and separate anti-ice and ignition controls.

During this period, engine design technology advancements include high-compression-ratio compressors and high-bypass-flow turbofans. Corresponding control technology also matured to include variable geometry controls, for example,

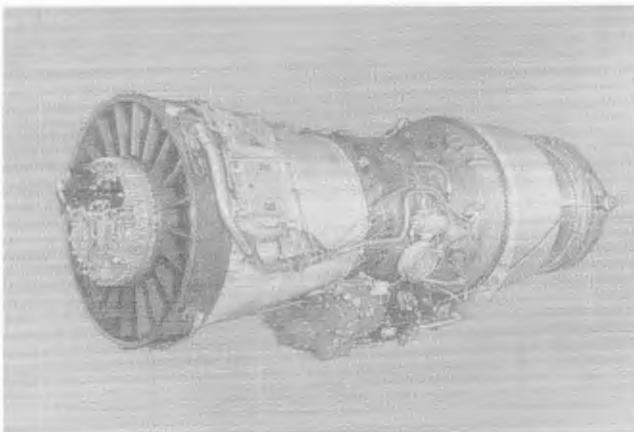


Fig. 1.7 First aerodynamically connected two-spool engine—PW J57 (courtesy of the Aeronautical Systems Center (ASC) History Office of the Wright Patterson Air Force Base).

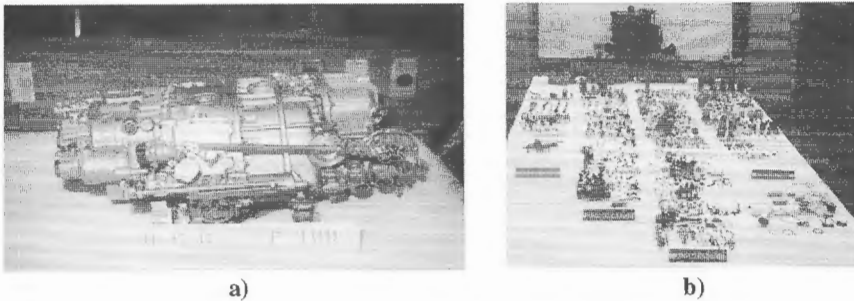


Fig. 1.8 (a) F100 hydromechanical fuel control (UFC). (b) F100 UFC parts (U.S. Air Force file photos, courtesy of B. A. Richter).

the compressor stator control, intake and nozzle controls. By 1969, a number of well-known engines, such as GE's J79 and F101 and PW's TF30 and F100, had seen long service and had demonstrated impressive reliability records. Figure 1.8 shows the hydromechanical control unit, called the UFC (unified fuel control), for the F100 engine. The UFC has close to 6000 parts (as shown in Fig. 1.8b), making it the most complicated engine control unit of its time.

During this period, engine control laws were designed primarily using *classical* control design methods, that is, single-loop, frequency-response, and time-domain methods. In fact, classical design methods were used for the control laws in all production engines during this period. The successive loop-closure technique was used by the industry to separate the coupling effects between two control loops, for instance, the shaft speed control loop and the inlet guide vane (IGV) control loop. As the name implies, the successive loop-closure design technique enables the control laws of multiple control loops to be designed one at a time by assuming other loops are kept at or near their steady-state (or quasi-steady) conditions. Modern control design methods started to develop in the early 1960s. They were first based on the state-space formulation of a system, and their theoretical foundation was based on optimal estimation and control [11, 12]. However, modern control design methods would not find their way into engine control law design until the next period of engine technology development.

On the engine modeling side, the first computerized, component-level, *steady-state* engine performance model for a turbojet was developed in early 1950s for constant specific heats. It was later extended to variable specific heats for different modules. Computer engine modeling was further extended to a turbofan engine in the late 1950s. Initial development of the simulation of *dynamic* performance of gas turbine engines was also achieved in early 1950s through the use of electronic analog computers at NACA Lewis [now the NASA John H. Glenn Research Center at Lewis Field (NASA Glenn)]. The response of a turbojet engine to a step change in fuel flow was simulated, and the results were validated by experiments [13]. An example of the results is shown in Fig. 1.9.

A more efficient numerical solution technique for calculating steady-state engine performance was developed in the mid-1960s by the Air Force Aero Propulsion Laboratory. This new technique used the Newton–Raphson method

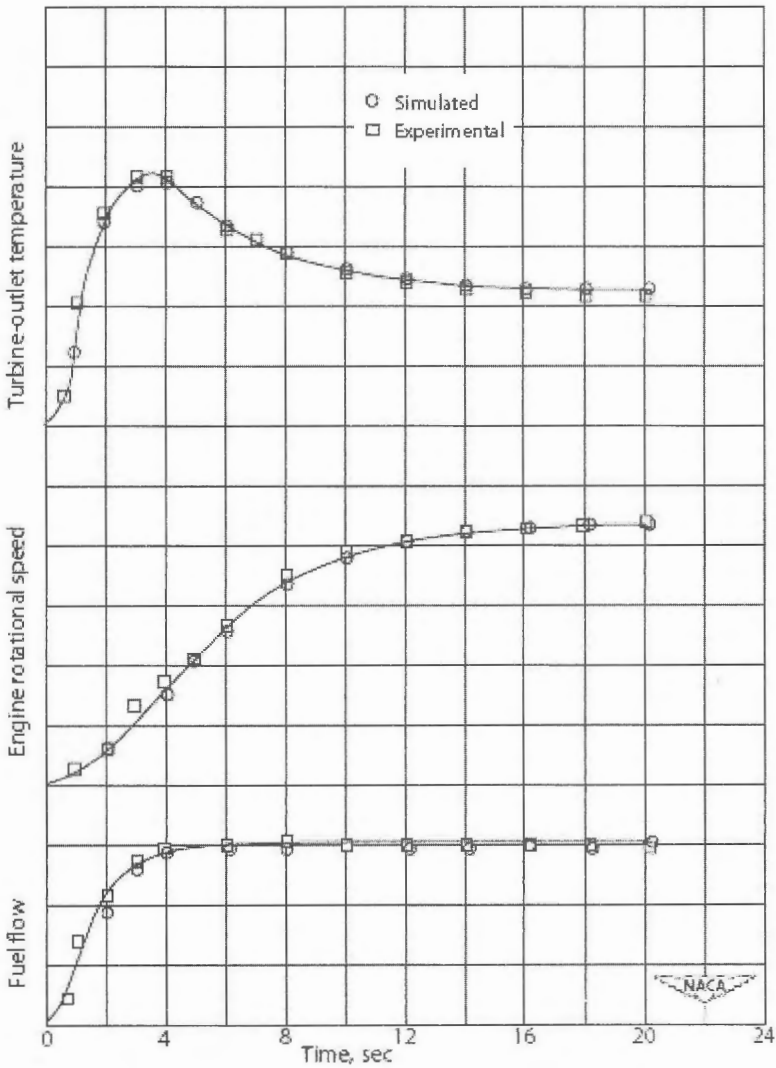


Fig. 1.9 Corrected engine time constant as a function of corrected engine speed (from [13]; courtesy of Sanjay Garg of NASA Glenn).

for solving a set of nonlinear equations to reduce significantly the time to compute a design point from weeks to days [14].

A simulation of dynamic engine performance usually requires a control system model in addition to the engine model. By mid-1960s, 90% of the analysis of engine dynamic (or transient) performance was handled by analog or digital computer simulations, while the remaining 10% was handled by manual analysis. Almost every known jet engine's transient performance was analyzed through dynamic simulations.

1.3.3 Electronics Period

As engine functions and consequently the engine control functionality expanded, the traditional hydromechanical fuel control and its related servo components were quickly reaching a practical limit. Although existing HMCs had demonstrated high reliability, the improved HMCs that would meet expanded requirements would be too large, heavy, and expensive to be practical for aircraft engine control applications.

Thus, in the early 1970s, analog and digital electronic control units (ECU) were designed to provide high-level supervisory or trim functions. *Supervisory* control units primarily calculated speed or temperature set points throughout the engine operating envelope. For some noncritical control variables, they would have sole control responsibilities. Early production supervisory control units included: the digital electronic engine control (EEC) unit for the PW F100 and the analog augmentor fan temperature (AFT) control for the GE F101. The EEC on the F100 engine, as shown in Fig. 1.10, was the first supervisory engine control.

An ECU is considered *full authority* when it controls the entire operation of an engine from startup to shutdown for the engine thrust (or power) according to the pilot's throttle command (or power request). To avoid confusion with the supervisory control, a full-authority control must manage the power setting functions, on which civil certification or military qualification standards must be based, regardless of the backup control mode. If engine certification can be achieved without an electronic control, which is subsequently added to achieve improved operational efficiency in the aircraft, then the control is supervisory [15].

The first full-authority ECU was introduced into service in 1972 on the Garrett/AirResearch (now a division of Honeywell) TFE731 engine [16]. The TFE731 ECU was a single-channel *analog* control. In case of a failure or pilot override, the electronic control switched to a backup HMC with conservative, fixed maximum and minimum fuel-ratio schedules. The electronic control provided the following key functions: minimal pilot attention and fail-safe operation; unrestricted power

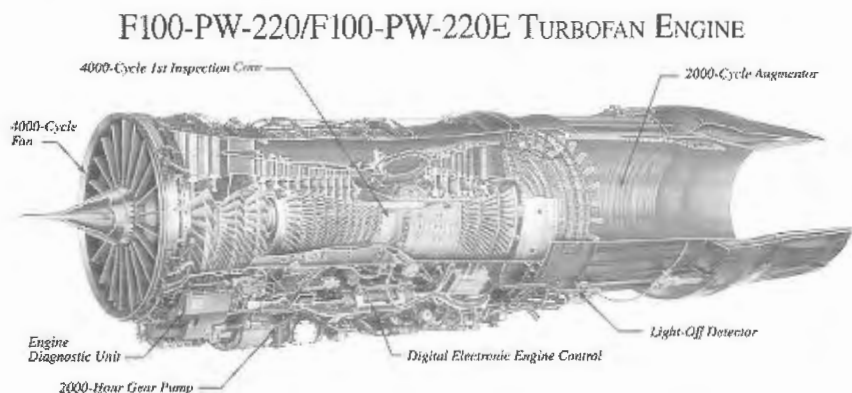


Fig. 1.10 First supervisory EEC unit on the F100-220 (courtesy of Pratt and Whitney).

lever movement, that is, the electronic control computes all control schedules, power setting, and engine operating limits; no-throttle dead-band; maximum power lever always gave the maximum allowable thrust; automatic flat rating for thrust; and rapid surge-free acceleration through a modulated on-off bleed valve.

Improving the mean time between failure, especially for a given mean time between unscheduled removal, and the ease of field maintenance were the prime motivations behind the initial transition from analog to digital controls. Other advantages were quickly discovered, such as the flexibility of the digital software for complex computation and software revision, the capability of fault isolation to a line replaceable unit (LRU) level through self-testing, and simplified and more accurate control adjustments. All of these factors insured the successful introduction of digital engine controls.

The F100 DEEC (digital electronic engine control) was the first FADEC to be flight tested. The DEEC evolved from the supervisory EEC of the F100 engine HMC and from a digital electronic breadboard for automating the design of the cam schedule used in the HMC. This electronic breadboard significantly reduced the development time from months to weeks, demonstrating the capability of electronic controls to improve the overall engine development cycle time.

During the past 30 years, ECUs have proven the following indisputable benefits: significant size and weight reduction for the same control functionality; unrestricted power lever movement, especially in military applications, thus reducing pilot workload; elimination of engine trim after any control component change; easy software update (through a laptop computer); compatibility with flight control, fire control, and other onboard and monitoring systems; and ability to record engine data and better maintainability caused by increased diagnostic capabilities from electronics (fault flags and fault isolation).

During this period, substantial research was undertaken to apply multivariable control design methods to turbine engine control law designs. Although a lot of interest was shown in advanced control design technologies, all production engine controls were still designed by the single-input and single-output (SISO) approach for each control loop. These single-loop designs were then "pieced" together using the successive loop-closure technique discussed earlier.

Multivariable control design methods had matured during this phase. Both frequency-domain and time-domain methods were studied extensively. A comprehensive summary of these research activities was published in [17], which divided the work into five categories: 1) linear quadratic regulator (LQR) methods; 2) frequency-domain methods; 3) identification, estimation, and model reduction; 4) detection, isolation, and accommodation; and 5) miscellaneous papers on engine control.

Multivariable frequency-domain methods were led by an application of the inverse nyquist array (INA) method to a gas turbine engine [18, 19]. Later, applications of the INA and the characteristic locus (CL) methods to an aircraft turbine engine were studied [20]. However, due to computational complexity, multivariable frequency-domain design methods have not been adopted as widely as their time-domain counterparts.

In the area of multivariable time-domain design methods, the United Aircraft Research Laboratories (now the United Technologies Research Center) appeared to be the first to apply a modern control design method, or more specifically, the

LQR design, to an aircraft engine [21] in early 1970s. The U.S. Air Force Wright Aeronautical Laboratory (AFWAL, now the Air Force Research Laboratory) started to apply multivariable control design methods to the J85 engine in 1973 [22]. AFWAL and NASA Glenn then jointly sponsored the Multivariable Control Synthesis program from 1975 to 1978. The program developed a series of LQRs and connected them with simple transition logic and several operating limits to achieve effective large-transient controls. The control design [23] was verified extensively at NASA's hybrid simulation facility [24] and followed by a successful testing on an F100 engine in NASA's altitude test facilities [25]. Figure 1.11 shows the block diagram of the LQR-based, multivariable control law from [23].

The multivariable control methodology lends itself to complex, integrated control problems. One of these problems is the integrated flight and propulsion control (IFPC). IFPC relieves pilot workload and improves aircraft maneuverability. Improving maneuverability can be achieved by thrust vectoring and jet diversion. In the mid-1980s, AFWAL sponsored the Design Methods for Integrated Control Systems program [26, 27] to study different design methods. The natural approach was to combine the flight control and the propulsion control into one integrated system and to design an integrated control law for the combined system [28]. This approach had some practical concerns: 1) airframe and engine manufacturers traditionally design their control systems separately before system integration, and 2) early and close coordination and cooperation between flight and propulsion system designers could lengthen the design cycle and thus increase costs. A less intuitive approach was to treat the IFPC control design problem as a hierarchical, decentralized system and integrate the functions of the flight and propulsion subsystems through cross-coupling effects and flight command translation [29].

By early 1970s, steady-state and transient performance simulations (commonly called the steady-state cycle deck and the transient cycle deck, respectively) of jet engines were routinely used in engine and control system designs. Furthermore,

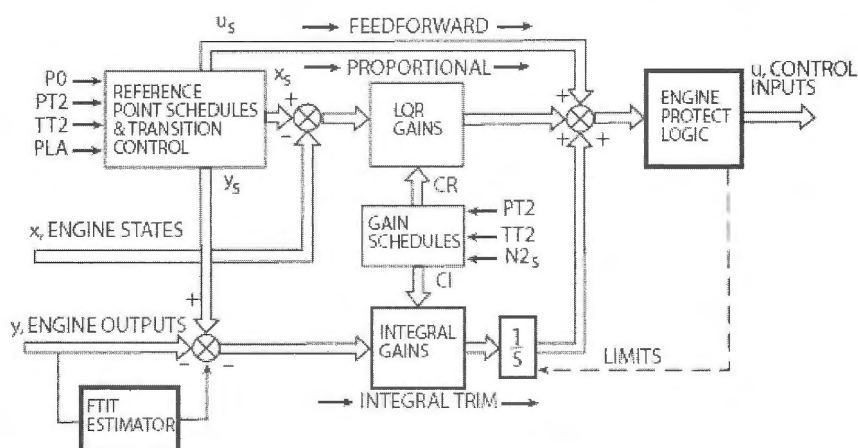


Fig. 1.11 LQR-based, multivariable control law for the F100 engine (from [22]).

the combined-cycle deck proved effective in improving engine test quality by ensuring engine and test cell compatibility, reducing test requirements, and extending the understanding of engine performance [30].

During this period, simulation capabilities were extended further to include real-time and hardware-in-the-loop simulations. In the early 1980s, NASA Glenn initiated an effort to build a real-time, digital engine simulator. In the mid-1980s, a joint NASA Glenn and NASA Ames Research Center (NASA Ames) research program, with the objective of developing and validating technologies for integrated flight-propulsion controls, was conducted. An essential element of the integrated controls research program is a real-time simulator of integrated aircraft and propulsion systems [31]. Around this time, an integrated, real-time simulation of vertical/short-takeoff-and-landing aircraft was also developed [32]. These integrated flight and propulsion control programs for control concept studies and simulation capability buildup laid the ground work for today's Joint Strike Fighter (JSF) propulsion control technologies.

1.3.4 Integration Period

During this period, dual-channel FADECs became the standard control system for jet engines. Compared to the "first-generation" FADECs in the preceding period, they have more built-in test functions; they have some form of embedded engine models to improve engine performance and diagnostics; they have some type of usage and life-tracking algorithms; and they are much lighter and smaller per control variable because of spectacular improvements in microprocessor speeds and memory technologies. Electric actuation systems gained wider acceptance. In addition to stepper motors, electromechanical actuators have been installed on engines. New sensors, especially diagnostic-specific sensors, are being tested for enhanced engine health management capabilities.

The fundamental issues facing the design of IFPC systems were studied in a NASA-sponsored short-takeoff-and-vertical-landing (STOVL) control technology program. In this program, a modified GE F110 engine with an ejector-augmentor was used as the propulsion system. The engine not only generated a forward thrust during cruise, but also provided an upward lift as well as lateral reaction control forces during hover and landing [33]. To optimize the maneuverability of the IFPC aircraft, multivariable control design techniques were used to determine the control laws for various actuators on the engine and the aircraft.

Increased propulsion system functionality and capability sparked an interest in model-based controls [34]. The increased capability also facilitated a migration of the multivariable control design methodology from laboratories to production engines. Indeed, nearly 30 years after the initial research on multivariable control applications for aircraft engines, the control law of the PW F135 engine was designed by using multivariable control techniques. The design was then implemented in the engine's FADEC and flight tested in 2000.

Also during this phase, subsystem integration for improved aircraft maneuverability was a key motivator for engine control designs. Integration of primary engine control functions with engine inlet and exhaust controls on the PW F119 engine delivered unprecedented air superiority for the F-22 fighter. The F119 engine, as shown in Fig. 1.12, is equipped with a unique two-dimensional pitch vectoring

exhaust nozzle. This convergent/divergent nozzle changes the angle of the thrust as much as 20 deg up or down. Furthermore, the nozzle position control loop is integrated with the F-22 flight control system and is automatically regulated by the engine's FADEC. The FADEC also features advanced diagnostics and on-condition management capability for maintenance awareness, autonomic logistics support, and automatic field and test data processing.

The F135 engine, which powers the JSF (or the F-135) aircraft, is the most advanced jet engine produced to date. The F135 engine is a low-bypass-ratio, augmented turbofan engine. The engine itself was designed using an integrated product development approach to ensure proper balance among performance, safety, reliability, maintainability, and life-cycle cost. The engine is capable of cruising at supersonic speeds without using the afterburner. The F135 engine-based propulsion system consists of two subsystems: a conventional thrust generation system and a lift fan system for STOVL [35]. Figure 1.12 shows the F135 propulsion system in the STOVL configuration.

During this period, block-diagram-based modeling and simulation tools with easy-to-use graphical user interfaces reached a state of maturity. Some of these tools are even supplemented with control system design toolboxes that can calculate the parameters of a control law, based on the desired format defined by the user, in the order of a second. These graphical tools were used routinely by control engineers to design and analyze control systems. Two most widely used such tools are the MATRIXx[®]/SystemBuild and the MATLAB[®]/Simulink[®].

Automatic software- (or code-) generation tools became a new standard for real-time simulation and embedded applications. Aside from the autocode-generation tool offered by SystemBuild and Simulink[®], proprietary tools such as PW's Pictures-to-Code and GE's BEACON were both proven to reduce control and diagnostic software development time significantly. However, the overwhelming cost of maintaining these in-house tools has limited the use of proprietary code-generation tools.

Control-system rapid prototyping capabilities were used widely to reduce development risk and costs by adopting a vertically integrated design process. The process includes the following steps: 1) building a block-diagram model of the engine and its control system, 2) performing analysis and non-real-time simulation

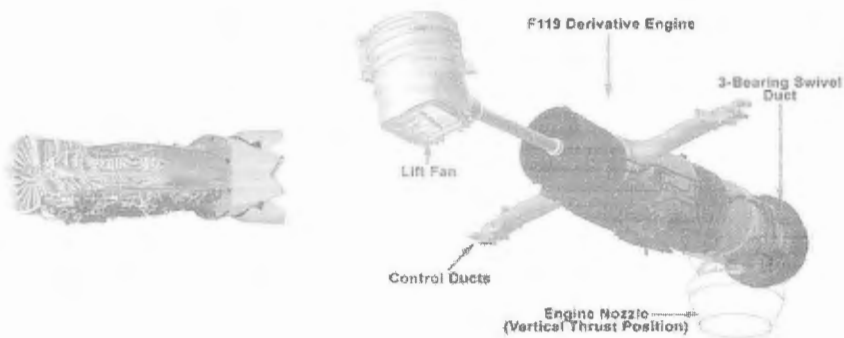


Fig. 1.12 PW F119 engine (left) and PW F135 engine (right) (from [35]).

of the model, 3) generating the source code for the control law from the block-diagram model automatically, 4) compiling the source code and downloading the compiled code to a control hardware (typically a board-level computer), and 5) executing the control law and letting it interact with the engine model, or the actual hardware including the engine, actuators, and sensors, in real time.

1.3.5 Flight Research and Advanced Controls

Propulsion flight research with the purposes of investigating engine control-related problems and testing new control concepts began in the late 1960s as a joint effort between NASA and the U.S. Air Force. Since that time, propulsion flight research in the United States has been conducted mainly at Edwards Air Force Base and the NASA Dryden Flight Research Center (NASA Dryden) in Edwards, California, with some level of support from NASA Glenn. The U.S. Air Force asked NASA in 1973 to assist in developing and flight testing a digital integrated propulsion control system (IPCS). The IPCS, installed on an F-111E airplane, was the first flight test of an integrated engine control system [36]. The IPCS consists of the afterburner ECU from the PW TF-30 engine and the variable-geometry external compression inlet control. In the late 1970s, NASA Dryden continued its pioneering research in integrated control with the flight test of a cooperative control system, where all functions of inlet control, autopilot, auto-throttle, and navigation systems were combined in a single digital computer on a YF-12 airplane.

In 1981, flight testing of the F100 DEEC began at NASA Dryden on the NASA F-15 aircraft. The success of the flight-test program allowed the Air Force to put the DEEC into full-scale development and production. The success also stimulated a series of flight research projects in the mid- and late 1980s to study advanced control modes. These projects, centered on the Highly Integrated Digital Electronic Control (HIDEC) program, included integrated flight and propulsion control, adaptive engine control, extended engine life control, and inlet integration.

In the advanced control research area, the study of the characteristics of compressor surge and rotating stall during this phase enabled the blossoming work in active compressor stability controls of the 1990s. In the late 1970s, theoretical and experimental works on surge and rotating stall were summarized for compressors [37]. These works prompted NASA to conduct research in surge and rotating stall. A hybrid computer simulation was developed at NASA Glenn to extract the in-stall pumping characteristics of a multistage axial-flow compressor from engine surge data [38].

The advances in multivariable control design and FADEC had stimulated the concept of model-based controls. Because the engine model typically represents the nominal performance of the engine fleet, on-line estimation of model parameters allows the model to be adjusted for each specific engine (i.e., off-nominal performance), even with engine deterioration. This on-line tuning (or adaptive) model is then able to improve both the controlled engine performance and the diagnostic accuracy. In the late 1980s, the Kalman-filtering approach to adaptive modeling was first applied to jet engines. A piecewise linear, state-variable model of a turbofan engine was adjusted by five estimated parameters, representing five implicit engine component variations (e.g., compressor and turbine

efficiency and flow variations), using a Kalman filter with in-flight engine measurements [39]. This work paved the way for the development and flight test of the performance-seeking control (PSC) [40].

Flight test of the PSC on the NASA F-15 aircraft started in the late 1980s and finished in the mid-1990s. The PSC took the HIDEDEC to the next step—onboard real-time adaptive optimization of engine and airplane parameters. A Kalman filter was used to estimate key engine parameters in near steady-state conditions. This program provided valuable information for the self-tuning, onboard engine model that was implemented in the ECU of the F119 engine on the F-22 airplane.

A distortion-tolerant control system was developed and flight tested in 1997 on the High Stability Engine Control (HISTEC) program [41]. The control system included a distortion estimation system, which assimilated information from engine face pressure sensors and flight-control-estimated angle-of-attack and side-slip angle. The estimated distortion level was then used to trim engine actuator commands for stability management. Figure 1.13 illustrates the concept of the HISTEC program.

Propulsion-controlled aircraft (PCA) research represented another contribution in the area of flight test research of engine controls. The PCA used computer-controlled engine thrust to provide a safe landing capability in case of major flight-control-system failure. Later tests verified PCA operation over the full flight envelope, even in upset conditions with all hydraulic systems turned off, and automatic landing with the ILS (instrument landing system).

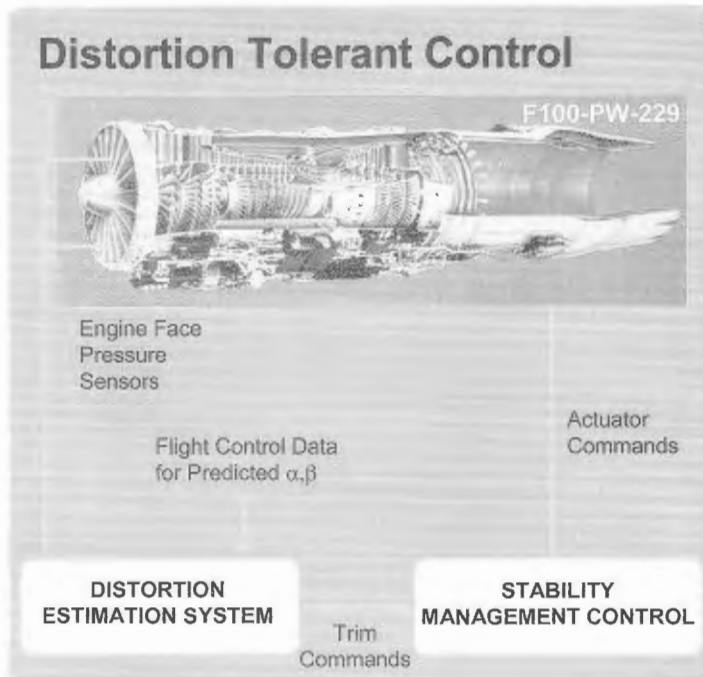


Fig. 1.13 Engine control research in extending inlet distortion tolerance (from [41]).

Flight testing of engine control technologies was extended to development and validation of health management technologies in the early 2000s. NASA Dryden, in cooperation with NASA Glenn and NASA Ames, conducted a vehicle health management program to test a number of engine and vehicle health management technologies on an Air Force C-17 airplane with its four PW F117 engines.

Life-extending controls (LEC) were investigated for both large and small engines. NASA pioneered the concept of LEC and demonstrated, in simulation, the objectives of achieving high performance in transient responses and damage mitigation with increased structural durability for rocket engines [42]. The LEC concept was applied to aircraft turbine engines in the mid-1990s. GE [43], Honeywell, and PW were investigating different methods to reduce damage and increase engine life. One method showed, in a production-quality FADEC-in-the-loop simulation, that thermal-mechanical fatigue and creep/rupture life can be extended significantly by small increases in acceleration transient time and cruise flight time, respectively. The small increase in acceleration time did not violate the FAA's (Federal Aviation Administration) engine acceleration time requirement because the time increase, resulting from reduced fuel flow and turbine temperature, occurred in the final 10% of the acceleration [44].

Active controls represented new opportunities to improve engine performance, operability, and reliability. Research programs in high-response controls (e.g., hundreds of cycles per second) of surge and stall, tip clearance, and combustion instabilities were sponsored by the U.S. Air Force, Army, Navy, and NASA. Although the technology readiness levels of active control technologies have not reached the implementation stage, they could significantly increase the number of control variables and complexity because of their distributed nature. Figure 1.14 illustrates the potential benefit of extending the operating regime, that is, the engine pressure ratio, of a compressor by active stall/surge controls. Figure 1.15 shows a high-response (600+ Hz) fuel valve under interface testing on a hardware-in-the-loop simulation bench at the U.S. Air Force Research Laboratory [45].

1.3.6 Control System Complexity

Control system complexity can be measured by the number of control variables or by the number of measured variables in the system. Typically, the number of control variables corresponds directly to the number of actuators and the number of measured variables to the number of sensors. However, selected actuators and sensors can be repeated, hence increasing complexity, to achieve increased control system reliability through redundancy.

1.3.6.1 Control complexity in the number of control variables. One way to assess the complexity of an engine control system is by the number of control variables [46]. Typical engine control variables include main burner and augmentor fuel flows, stator/vane positions, bleed valve positions, and nozzle area. In addition to these multivalued control variables, there are also discrete, or on/off, control variables, such as engine ignition and anti-ice. Table 1.1 lists the control variables for several engines [16, 47, 48]. Figure 1.16 depicts the trends in the number of control variables for military and commercial engines through late 1980s [17]. As we look back at the period from the time when these trends were

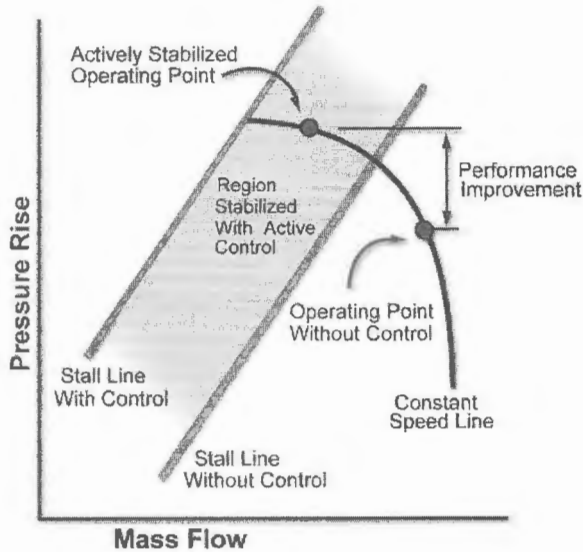


Fig. 1.14 Active compressor stability control extends operating region for engines (from [40]).

first projected in the mid-1980s to the mid-2000s, these trends have stayed almost flat at their relative levels since the mid-1990s (for the engines of conventional design). Additional increase in the control complexity beyond the current levels will most likely come from three drivers: efficiency optimization, emission reduction, and noise reduction. Note that the engine configurations used in the comparison of Fig. 1.7 are 1) a two-spool, afterburning turbofan engine with thrust vectoring nozzle for military engines and 2) a two-spool, high-bypass turbofan engine with thrust reverser for commercial engines.

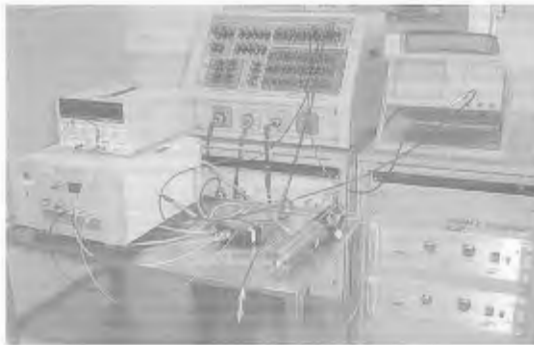
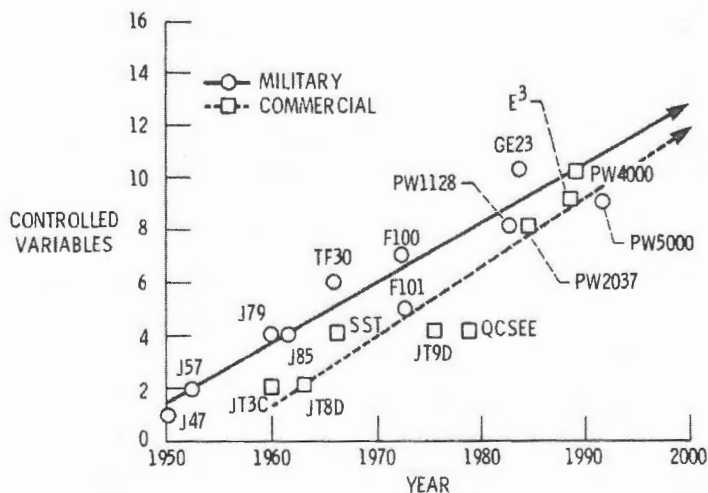


Fig. 1.15 High-response fuel valve for active combustion control (from [44]).

Table 1.1 Control variables of some selected engines (courtesy of General Electric for GE90 data)

Variable	TFE731	F100	F110	PW2037	GE90
Compressor inlet variable vane position command	—	CIVV	IGV	—	—
Rear compressor variable vane position command	—	RCVV	VSV	VSV	VSV
Compressor tip clearance control valve	—	—	—	CCC	—
Cowl cooling control valve	—	—	—	—	CC
Main burner fuel valve	WFM	WFM	WFM	WFM	WFM
Combustor mode valve	—	—	—	—	Comb
Surge/variable bleed command	Wb25	—	—	—	Wb
(HP and LP) Turbine tip clearance control valve	—	—	—	TCChp TCClp	TCChp TCClp
Augmentor fuel flow command #1	—	WFAC (Core)	WFAB	—	—
Augmentor fuel flow command #2	—	WFAD (Duct)	—	—	—
Sequencing valve position	—	SVP	—	—	—
Exhaust nozzle area command	—	A8	A8TM	—	—

**Fig. 1.16** Trends of engine control complexity through 1990 (from [17], including discrete control variables).

1.3.6.2 Control complexity in the number of measured variables.

Because measured variables are used to provide feedback to the control system, as well as to provide awareness of the health of the engine, the number of measured variables is also a good measure of control complexity. Table 1.2 lists the measured gas path variables for selected engines [16, 47]. Note that the control (actuator) feedback variables are not included in the list. The number of measured variables is expected to increase rapidly in future engines as active control strategies and additional health management capabilities are added to the engine control system. Figure 1.17 shows the first commercial engine to be rated at 100,000-lb thrust.

1.3.6.3 Evolution of engine controllers. The word “controls” is frequently used to mean the components of a control system. However, in most cases, it means the computational unit or the “brain” that is responsible for the control law and control logic. This latter meaning helps us view the evolution of engine controllers on a historical timeline from the 1940s HMC to the modern, dual-channel FADEC. Figure 1.18 shows this evolution of engine controls of historically important engines. Figure 1.19 illustrates the evolution of electronic controls in the last 30 years at Honeywell Engines.

1.3.7 Historically Interesting Engines

A description of some control-historical aircraft engines is given in Tables 1.3–1.6. These tables are organized by the engine model, the year when the engine was first flight tested (or first run on the ground when the flight-test date was not confirmed), the aircraft model, and the technology advancements made by the engine [49–52].

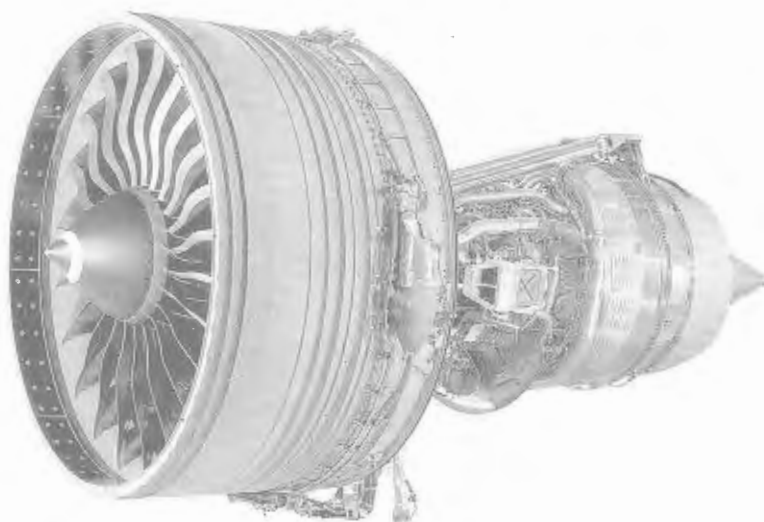
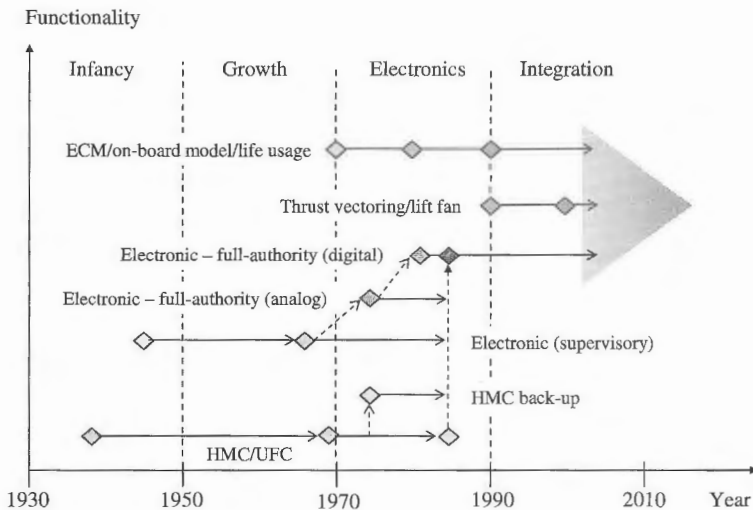


Fig. 1.17 GE 90 engine and its composite material fan blade (courtesy of General Electric).

Table 1.2 Measured gas-path variables of selected engines (courtesy of General Electric for GE90 data)

Variable	TFE731	F100	F110	GE90
Fan inlet total temperature	T_{t2}	T_{t2}	T_{t2}	T_{t2}
Fan inlet or ambient pressure	P_{t2}	P_{s2}	—	P_{s0}
Compressor inlet temperature	—	—	—	T_{t25}
Compressor discharge temperature	—	—	—	T_{t3}
Burner pressure or compressor discharge pressure	—	P_b (or P_{s3})	P_{s3C}	P_{s3}
Compressor bleed pressure	—	—	P_{3B} (or CDPCBP)	—
High pressure turbine blade metal temperature	—	—	T_{4B}	—
Inter-turbine temperature	T_{t45}	FTIT	—	—
Exhaust temperature	—	—	—	T_{t5}
Turbine exit total pressure (fan & core mixed stream)	—	P_{t6m}	—	—
LP spool speed	N_1	N_1	N_F	N_1
HP spool speed	N_2	N_2	N_G	N_2
	—	LOD	—	—

**Fig. 1.18** Evolution of engine controls.

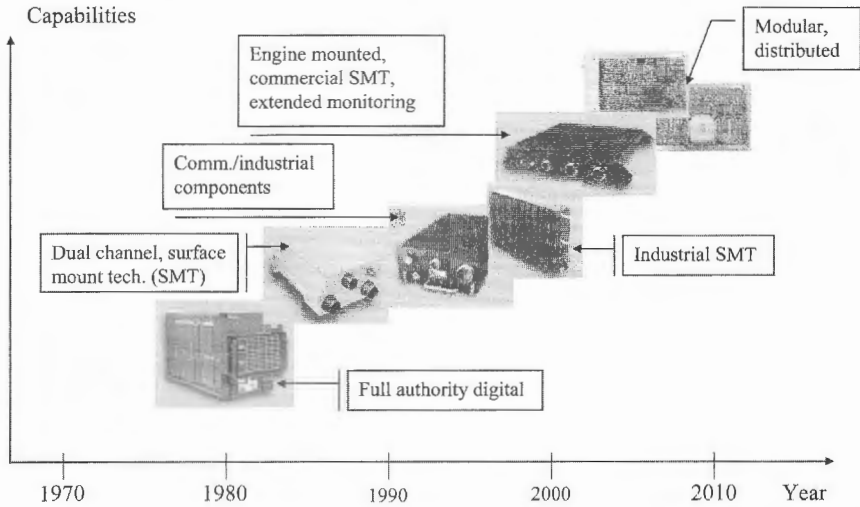


Fig. 1.19 Evolution of electronic engine controls at Honeywell (courtesy of Honeywell International).

Table 1.3 Major milestones for propulsion controls in the infancy period (before 1950)

Year	Engine model	Aircraft model	Technology advancements
1942	GE I-A	Bell XP-59A (1st ft)	Design based on British Whittle engine (single-stage centrifugal compressor with canned combustors) Hydromechanical governor Flameout protection (min. flow stop) Maximum-flow schedule
1943	GE TG-100/T31		World's first turboprop engine tested First two-spool turbine engine with the free turbine driving the propeller at the front First axial-flow compressor aircraft gas turbine
1943	GE I-16	Bell P-59 (1st ft)	First afterburner (AB) experiment on this engine
1944	GE J-33	Lockheed XP-80 (1st ft)	A direct descendant of the British Whittle engine of the early 1940s

(Continued)

Table 1.3 Major milestones for propulsion controls in the infancy period (before 1950) (Continued)

Year	Engine model	Aircraft model	Technology advancements
			November 1945, the Allison Division of General Motors assumed complete responsibility for the development and production of J33 series engines
1945	Westinghouse J30	McDonnell FH-1 (1st flt)	First pure jet fighter for the Navy
1946	GE J35	Republic XP-84 (1st flt)	First axial-flow compressor turbojet Late in 1947, complete responsibility for the production of the engine was transferred to the Allison Division of General Motors
1948	GE J47	F-86, F-86D (1st flt with AB)	Production started in 1948, the most produced aircraft turbine engine in history First electronically controlled afterburner (AB), AB control was linked to main engine control.
1949	PW J52	A-4, A-6	First anti-ice control system First two-spool, aerodynamically connected turbine engine design; J52-P3 production started in 1959. Predecessor of JT8D

Table 1.4 Major milestones for propulsion controls in the growth period (1950–1969)

Year	Engine model	Aircraft model	Technology advancements
1951	PW J57	B-50 (1st flt)	First two-spool, aerodynamically connected turbine engine in production; production started in 1953
1955	GE J79	NA B-45 (1st flt) XF-4D, F-104	First high-compression-ratio engine flight tested First variable-geometry engine, variable-stator compressor
1956	GE J85	GAM-72 drone	First engine to reach Mach 2 in late 1955 First small turbojet engine

(Continued)

Table 1.4 Major milestones for propulsion controls in the growth period (1950–1969) (Continued)

Year	Engine model	Aircraft model	Technology advancements
1959		T-38	Afterburning version first flown in T-38
1957	PW JT3	B-707, DC-8	Commercial version of J57 in production First U.S. commercial turbine engine
1958	PW J58	SR-71 (1st flt)	First engine to reach Mach 3 Variable intake and nozzle controls First variable cycle engine (var. comp. bypass) First electronic temperature trim, predecessor of supervisory control First engine with a controlled bleed valve (during engine start)
1958	GE CJ805	Convair 880 (1st flt.) Convair 990	Commercial version of J79 CJ805-23 was the first U.S. turbofan (free turbine driven aft fan); it first flew on the 990
1958	PW TF30	F-111, F-14	First afterburning <i>turbofan</i> engine HMFC and separate controls for AB and compressor stators
1959	GE J85/ Lift fan	XV-5A	First lift fan system powered by two J85s tested Twin lift fans with controllable louvers First fixed-wing VTOL flight on Republic XV-5A
1960	GE CJ610	Learjet	Business jet/general aviation version of J85
1961	PW JT8D	DC-9, B-727, B737	First engine run of commercial version of J52 in 1961 Production started in 1963 Annular full-length bypass duct
1964	GE J93	NA XB-70 (1st flt)	Predecessor of GE's Supersonic transport engine, the GE4 2300°F turbine inlet temp. with turbine cooling Mach 3 cruise for a large bomber/transport demonstrated in a 1965 flight
1966	GE TF39	C-5, B-52	First derivative of the core engine concept from the GE1 demonstrator engine First GE high-bypass-ratio (8:1) turbofan Improved foreign object damage (FOD) rejection
1968	PW JT9D	B-747 (first flight)	First PW high-bypass-ratio (5:1) turbofan
1968	GE CF6	DC-10	First ground run of the commercial version of TF39 in 1968 Significant increase in fuel efficiency

Table 1.5 Major milestones for propulsion controls in the electronics period (1970–1989)

Year	Engine model	Aircraft model	Technology advancements
1970	PW F100	F-15	First modular concept engine tested on ground First UFC First supervisory EEC—digital First engine with event history recorder, predecessor of engine diagnostic unit (EDU)
1972	Garrett TFE731	Falcon, Learjet	First gas turbine engine with full-authority electronic control—analogue, single channel Reduced capability HMC as a backup. First modulated interstage bleed valve
1972	GE F101	Rockwell B-1	HMC with full-authority electronic AB control—analogue
1974	CFMI CFM56	DC-8 Series 70, KC-135, B-737	GE F101 core and SNECMA M56 fan, turbine, and gearbox Better fuel efficiency, lower noise and emissions
1981	PW F100	F-15, F-16	First flight test of a full-authority digital electronic engine control (DEEC) Single-channel DEEC with a backup HMC First FADEC-controlled military engine
1982	PW JT9D-7R4	B-747-400	Entry into service in 1982 Supervisory electronic control with HMC
1984	GE F110-100	F-16	Entry into service Set single-engine reliability and safety record
1984	PW2037	B-757	First commercial service of advanced turbofan Dual-channel FADECs Active tip clearance controls for high-pressure compressor, high- and low-pressure turbines Multiple bleed valve controls Engine program plug in the FADEC to select multiple engine models

(Continued)

Table 1.5 Major milestones for propulsion controls in the electronics period (1970–1989) (Continued)

Year	Engine model	Aircraft model	Technology advancements
1989	IAE V2500	A320, MD-90	Integrated thermal management for oil and fuel Advanced thrust reverser Predecessor of F117 for C-17 Entry into service First engine with autostart (ignition and restart) Model-based tip clearance control Central maintenance features built in the FADEC

Table 1.6 Major milestones for propulsion controls in the integration period (1990–)

Year	Engine model	Aircraft model	Technology advancements
1990	PW F119	F-22	First flight in 1990 Dual-channel, <i>active</i> FADECs Two-dimensional thrust vectoring nozzle Self-tuning onboard real-time model (STORM) for diagnostics Analytical redundancy for gas-path sensors Separate diagnostic unit (CEDU) from FADEC
1990	GE YF120		First flight in 1990 Variable cycle engine (variable bypass) Flight-qualified engine that has the highest number of control variables
1992	GE F110-129	F-16, F-14	Entry into service on F-16 in 1992 FADEC
1992	PW F100-220	F-15, F-16	First life extension control implementation by limiting rotor speed excursions with varying nozzle area and inlet guide vanes
1995	PW F100-229	F-15 Active	First supersonic-yaw thrust vectoring nozzle flight demonstration (pitch yaw balanced beam nozzle)
1995	GE90	B777	Lighter and more durable fan blades by using a composite material technology (see Fig. 1.19) Model-based control for better fuel efficiency
2000	PW F135	JSF	First flight in September 2000 Increased integration of flight-propulsion controls (for STOVL)

(Continued)

Table 1.6 Major milestones for propulsion controls in the integration period (1990–) (Continued)

Year	Engine model	Aircraft model	Technology advancements
2002	PW 6000	A-318	Integrated lift fan and jet propulsion system
			First flight-critical propulsion system
			First implementation of multivariable control
			Three-bearing swivel duct nozzle
			First flight in January 2002
2007	Trent 1000	B787	First commercial application of PW's self-tuning onboard real-time model
			Three-shaft, high-bypass turbofan with bleedless design
			Power takeoff from the intermediate-pressure spool

References

- [1] Mattingly, J. D., *Elements of Propulsion: Gas Turbines and Rockets*, AIAA, Reston, VA, 2006.
- [2] Mattingly, J. D., Heiser, W. H., and Pratt, D. T., *Aircraft Engine Design*, 2nd ed., AIAA Education Series, AIAA, Reston, VA, 2002.
- [3] Jaw, L. C., and Garg, S., "Aircraft Propulsion Control Technology Development in the U. S. – a Historical Perspective," International Symposium on Air Breathing Engines, Paper NASA/TM 2005-213978, Sept. 2003.
- [4] Jaw, L. C., and Guo, T.-H., "Recent Advancements in Aircraft Engine Health Management Technologies and Recommendations for the Next Step," NASA Technical Report, Nov. 2004.
- [5] Sobey, A. J., and Suggs, A. M., *Control of Aircraft and Missile Powerplants*, Wiley, New York, 1963.
- [6] Spang, H. A., III, and Brown, H., "Control of Jet Engines," *Control Engineering Practice*, Vol. 7, No. 9, Sept. 1999, pp. 1043–1059.
- [7] Dorf, R. C., and Bishop, R. H., *Modern Control Systems*, 10th ed., Prentice–Hall, Upper Saddle River, NJ, 2004.
- [8] Brogan, W. L., *Modern Control Theory*, 3rd ed., Prentice–Hall, Upper Saddle River, NJ, 1991.
- [9] Boyne, W. J., and Lopez, D. S., *The Jet Age – Forty Years of Jet Aviation*, Smithsonian Inst., The John D. Lucas Printing Co., 1981.
- [10] Otto, E. W., and Taylor, B. L., III, "Dynamics of a Turbojet Engine Considered as a Quasi-Static System," NACA Rept. 1011, 1951.
- [11] Athans, M., and Falb, P. L., *Optimal Control – an Introduction to the Theory and its Applications*, McGraw–Hill, New York, 1965.
- [12] Bryson, A. E., Jr., and Ho, Y. C., *Applied Optimal Control*, Hemisphere, New York, 1975.
- [13] Ketchum, J. R., and Craig, R. T., "Simulation of Linearized Dynamics of Gas-Turbine Engines," NACA-TN-2826, Lewis Flight Propulsion Laboratory, Cleveland, OH Nov. 1952.

- [14] McKinney, J. S., "Simulation of Turbofan Engine – Part I: Description of Method and Balancing Technique," Air Force Aero Propulsion Lab., Air Force Systems Command, AFAPL-TR-67-125, Wright-Patterson AFB, Nov. 1967.
- [15] Caine, D. A., "Recent Developments in Digital Control for Helicopter Powerplants," American Helicopter Society (AHS) Rotary Wing Propulsion Specialists Meeting Paper, Nov. 1982.
- [16] Keck, M. F., Fredlake, J. J., and Schwent, G. V., "A Control Concept Combining the Best of the Current Hydromechanical and Electronic Technologies," Society of Automotive Engineers, Paper 740380, Feb. 1974.
- [17] Merrill, W., Lehtinen, B., and Zeller, J., "The Role of Modern Control Theory in the Design of Controls for Aircraft Turbine Engines," *Journal of Guidance, Control, and Dynamics*, Vol. 7, No. 6, Nov.–Dec. 1984, pp. 652–661.
- [18] Rosenbrock, H. H., and Munro, N., "The Inverse Nyquist Array Method," Paper #3.1, *Alternative for Linear Multivariable Control*, National Engineering Consortium, Inc., Chicago, IL, Jan. 1978, pp. 101–137.
- [19] McMorran, P. D., "Design of Gas Turbine Controllers Using Inverse Nyquist Array Method," *Proceedings of Institute of Electrical Engineers*, Vol. 117, London, U.K., 1970, pp. 2050–2056.
- [20] MacFarlane, A. G. J., et al., "Applications of Multivariable Control Techniques to Aircraft Gas Turbine Engines," *Proceedings of Multivariable Control Systems Design and Applications*, IEE, London, U.K., Sept. 1971.
- [21] Michael, G. J., and Farrar, F. A., "Development of Optimal Control Modes for Advanced Technology Propulsion Systems," United Aircraft Research Lab., Internal Rept., East Hartford, CT, Jan.–Feb. 1974.
- [22] Stone, C. R., Miller, N. E., Ward, M. D., and Schmidt, R. D., "Turbine Engine Control Synthesis—Vol. I, II" Air Force Aero Propulsion Lab., AFAPL-TR-75-14, March 1975.
- [23] DeHoff, R. L., Hall, W. E., Adams, R. J., and Gupta, N. K., "F100 Multivariable Control Synthesis Program—Volume I: Development of F100 Control System," Air Force Aero Propulsion Lab., AFAPL-TR-77-35, 1977.
- [24] Szuch, J. R., Soeder, J. F., Seldner, K. and Cwynar, D. S., "F100 Multivariable Control Synthesis Program—Evaluation of Multivariable Control Using a Real-Time Engine Simulation," NASA Technical Paper 1056, Oct. 1977.
- [25] Lehtinen, B., DeHoff, R. L., and Hackney, R. D., "Multivariable Control Altitude Demonstration on the F100 Turbofan Engine," AIAA Paper 79-1204, 1979.
- [26] Smith, K. L., "Design Methods for Integrated Control Systems," AFWAL-TR-86-2103, Wright Patterson AFB, Dec. 1986.
- [27] Shaw, P. D., Rock, S. M., and Fisk, W. S., "Design Methods for Integrated Control Systems," AFWAL-TR-88-2061, Wright Patterson AFB, June 1988.
- [28] Garg, S., and Mattern, D., "Integrated Flight/Propulsion Control System Design Based on a Centralize Approach," AIAA Paper 89-3520, 1989.
- [29] Mattern, D., and Garg, S., "Integrated Flight/Propulsion Control System Design Based on a Decentralize, Hierarchical Approach," AIAA Paper 89-3519, Aug. 1989.
- [30] Przybylko, S. J., Hutcheson, L., Suder, B., and Warwick, T.R., "Advanced Integrated Digital Engine Simulation—a Step Forward for Propulsion System Testing," AIAA Paper 70-633, June 1970.
- [31] Mihalow, J. R., and Drummond, C. K., "STOVL Aircraft Simulation for Integrated Flight and Propulsion Controls Research," NASA TM 102419, June 1989.

- [32] Adibhatla, S., "Propulsion Issues in Design of Integrated Flight and Propulsion Control Systems—Lessons Learned from a STOVL Design Study," AIAA Paper 94-3610-CP, 1994.
- [33] Garg, S., and Mattern, D., "Application of an Integrated Methodology for Propulsion and Airframe Control Design to a STOVL Aircraft," AIAA Paper 94-3611-CP, 1994.
- [34] Adibhatla, S., and Lewis, T. J., "Model-Based Intelligent Digital Engine Control (MoBIDEC)," AIAA Paper 1997-3192, July 1997.
- [35] Military Air Systems, "Joint Strike Fighter (JSF) Shaft-Driven Lift-Fan System with Three-Bearing Swivel Main Engine Nozzle," Federation of American Scientists (FAS), <http://www.fas.org/man/dod-101/sys/ac/jsf.htm>, retrieved Dec. 2008.
- [36] Burcham, F. W., Jr., Ray, R. J., et al., "Propulsion Flight Research at NASA-Dryden from 1967 to 1997," NASA/TP-1998-206554, July 1998.
- [37] Greitzer, E. M., "The Stability of Pumping Systems—The 1980 Freeman Scholar Lecture," *Journal of Fluids Engineering*, Vol. 103, June 1981, pp. 193–242.
- [38] Lorenzo, C. F., Chiaramonte, F. P., and Mehalic, C. M., "Determination of Compressor In-Store Characteristics from Engine Surge Transients," AIAA Paper 84-1206, June 1984.
- [39] Luppold, R. H., Roman, J. R., Gallops, G. W., and Kerr, L. J. "Estimating In-Flight Engine Performance Variations Using Kalman Filter Concepts," AIAA Paper 89-2584, July 1989.
- [40] Maine, T. A., Gilyard, G., and Lambert, H., "A Preliminary Evaluation of an F100 Engine Parameter Estimation Process Using Flight Data," NASA TM-4216, 1990; also AIAA Paper 90-1921, July 1990.
- [41] Garg, S., "Propulsion Controls and Health Management Research at NASA Glenn Research Center," JANNAF Inter-agency Propulsion Committee Meeting, NASA/TM—2002-211590, July 2002.
- [42] Lorenzo, C. F., and Merrill, W. C., "Life Extending Control—A Concept Paper," NASA-TM-104391, June 1991.
- [43] Wiseman, M. W., and Guo, T-H., "An Investigation of Life Extending Control Techniques for Gas Turbine Engines," *Proceedings of the 2001 American Control Conference*, Vol. 5, IEEE, June 2001, pp. 3706–3707.
- [44] Jaw, L. C., Wu, D. N., and Bryg, D., "Tracking and Control of Gas Turbine Component Damage/Life," *Proceedings of NATO AVT Panel Symposium on Aging Mechanisms and Control*, Paper MP-69-P-01, Manchester, U.K., Oct. 2001.
- [45] Tran Van, H., et al., "Design, Simulation, and Test of a High Response, High Modulation Fuel Valve for Active Thermo-Acoustic Instabilities Control in Combustors and Afterburners," *Proceedings of the 2003 International Society of Air Breathing Engines Symposium*, Cleveland, OH, Sept. 2003.
- [46] Drain, D. I., "Propulsion Control and Control Theory—A New Research Focus," NASA Paper, 1979.
- [47] Myers, L. P., "F-15 Digital Electronic Engine Control System Description," NASA CP 2298, March 1984.
- [48] Kulikov, G. G., and Thompson, H. A., *Dynamic Modeling of Gas Turbines – Identification, Simulation, Condition Monitoring and Optimal Control*, Springer-Verlag, London, 2004.
- [49] General Electric Co., *Seven Decades of Progress—a Heritage of Aircraft Turbine Technology*, Aero Publishers, Inc., 1979.

- [50] Garvin, R. V., *Starting Something Big – The Commercial Emergence of GE Aircraft Engines*, AIAA, Reston, VA, 1998.
- [51] St. Peter, J., *The History of Aircraft Gas Turbine Engine Development in the United States ... a Tradition of Excellence*, American Society of Mechanical Engineers, IGTL, 1999.
- [52] Mattingly, J. D., "Pictures of Aircraft Gas Turbine Engines," <http://www.aircraftenginedesign.com/enginepics.html>, retrieved Feb. 2007.

Engine Modeling and Simulation

This chapter discusses various forms of engine models from a control system engineering perspective. The control engineering perspective treats each engine module as a whole, that is, the thermodynamic and fluid mechanic properties of the air/gas are assumed the same within the entire module like a fan, a compressor, or a turbine, rather than dividing the module into finite, smaller parts or stages. This lumped-parameter approach allows us to simplify the model and reduce the complexity of the control law. A gas turbine engine model often consists of both a steady-state model and a dynamic model. Both types of models are derived based on the first principles of physics. Both the steady-state and dynamic models will be discussed in this chapter.

A gas turbine engine is inherently nonlinear as a result of the complex interactions of thermal, fluid, and mechanical effects. Naturally, these nonlinear characteristics also appear in engine models. However, nonlinear models are typically not used in control law design because the characteristics of nonlinear models (or systems) change with the initial and the current operating conditions [1], making nonlinear design techniques too generalized for meeting specific performance requirements at a specific operating condition. Also, simpler linear control design techniques have been successfully applied to real-world problems; hence, linear control techniques are still being used in practice. In a practical control design approach, after the linear control law is designed, nonlinear control features, like scheduling and switching, are added, and this “augmented” system is evaluated in computer simulations or even hardware-in-the-loop simulations. Nonlinear simulation and hardware-in-the-loop simulations will be discussed in more detail in Chapter 6. In this chapter, only the topic of extracting linear models from nonlinear engine simulations will be discussed to facilitate control law design using established linear control design techniques. At the end of this chapter, we will discuss data-driven engine models.

2.1 Steady-State Engine Models

When the internal state of an engine reaches a thermodynamic equilibrium, that is, there is no change in the values of state variables, the engine is said to be in a steady state. Steady-state models of jet engines are traditionally called *cycle decks*. Cycle deck derives its name from the fact that it is a computer program to analyze the thermodynamic engine cycle and performance at any operating condition

in the flight envelope. Also, these programs were first coded in FORTRAN language onto a “deck” of computer punch cards. Hence, a cycle deck is the engine model that represents the steady-state characteristics of *each* engine module (or component) over its entire range of operation. The industry practice is to use data or empirical-analytical correlations in the form of component “maps” to give greater accuracy (see Chapter 8 of [2]). These maps are converted into a set of multivariate functions of corrected speed, corrected flow rate, pressure ratio, and efficiency for each module. These functions are then implemented in the form of look-up tables for efficient digital computation. The map functions are commonly adjusted for Reynolds-Number effects and variable geometry settings.

A cycle deck is also called a component level model and is a high-fidelity, physics-based simulation of an engine’s steady-state performance. The generic components of a turbofan engine include fan, compressor, combustor, turbine, mixer, and nozzle. Depending on the design, for example, a two-spool turbofan engine, an engine can have a separate low-pressure compressor (LPC) and a high-pressure compressor (HPC), as well as a low-pressure turbine (LPT) and a high-pressure turbine (HPT). For some engines, for example, the Garrett ATF3 [3] and the Rolls Royce RB211, a third spool is added to provide additional flexibility in shaping the performance of the engine at different operating points in the flight envelope. Figure 2.1 illustrates the modules of a two-spool turbofan engine. The fan, the LPC, and the LPT modules are connected on one shaft, while the HPC and the HPT modules are connected on another shaft. The two shafts are concentric and are primarily linked by aerodynamic forces.

Typically there are several model versions that represent an engine’s performance. The “customer deck” is the model that is given to an engine customer that represents the average performance of new engines. This model does not account for variations of engines in the fleet, nor for changes in operating characteristics over the lifetime of a specific engine. The model that represents the current performance of an engine (by considering engine-to-engine variation and deterioration) is called an “engine-specific deck.” Most engine models emphasize accuracy and completeness, and they execute in non-real time. However, some engine models that are used for control analysis and diagnostic evaluation can run in real time and can be implemented in a FADEC. Appendix B, Sec. B.6.9 provides some background on engine performance and operability analysis; specifically, one set of equations that represents engine steady-state performance can be found in Sec. B.6.7.

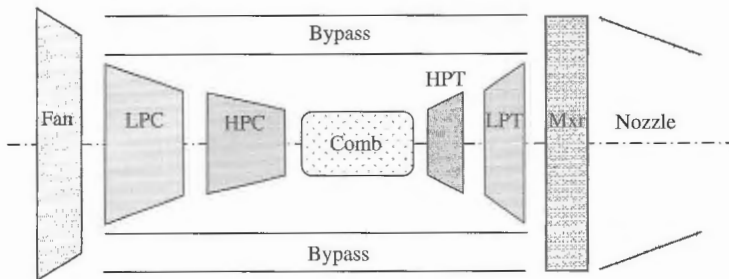


Fig. 2.1 Modules of a two-spool turbofan engine.

2.2 Dynamic Engine Models

Control design fundamentally deals with the dynamics of a system. Because of the complexity of the gas turbine, the analysis of engine dynamics is multidisciplinary, that is, it includes physical phenomena in mechanical, fluid mechanic, and thermodynamic fields. In this section we will derive three basic types of dynamics in gas turbine engines by using the first principles of physics. These types are shaft dynamics, pressure dynamics, and temperature dynamics. We will also derive simplified forms, that is, the linear approximations, of these basic dynamics. The basic dynamics derived in this chapter are applicable to all forms and derivatives of gas turbine engines.

2.2.1 Shaft Dynamics—Effect of Inertia

Shaft dynamics represent the simplest form and yet the most important dynamic behavior of a gas turbine engine. Shaft dynamics have the same characteristics as the two-disk system shown in Fig. 2.2, where two round disks are connected by a rigid shaft. The acceleration of the combined rigid body (consisting of both disks and the shaft) based on the principle of Newtonian mechanics is

$$\dot{\omega} = \frac{\Delta Q}{I} \quad (2.1)$$

where $\dot{\omega}$ is the angular acceleration of the body, ΔQ is the differential torque exerted on the disks, and I is the mass moment of inertia of the combined body.

In the context of a gas turbine engine, the angular velocity ω is usually substituted by the shaft rotational speed N . The differential torque ΔQ can be expressed by a function of shaft speed and fuel flow rate W_f . Substituting these in the torque function, we write the shaft dynamic equation as

$$\dot{N} = \frac{f(N, W_f)}{I} \quad (2.2)$$

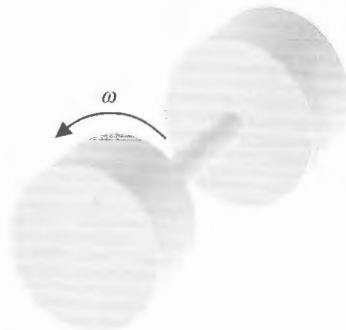


Fig. 2.2 Analogy of shaft dynamics by a two-disk system.

If we take the Taylor's series expansion of the function f at a (steady-state) nominal operating point and retain only the first-order terms, we get the following linearized shaft dynamic equation:

$$\dot{N} = a \cdot \Delta N + b \cdot \Delta W_f \quad (2.3)$$

where

$$a \equiv \frac{1}{I} \frac{\partial f}{\partial N}, \quad b \equiv \frac{1}{I} \frac{\partial f}{\partial W_f} \quad (2.4)$$

2.2.1.1 Shaft dynamics of one-spool engines. The linear model of shaft dynamics for a one-spool engine [4], expressed in the state equation form of Eq. (2.3), is

$$\dot{N} = \frac{1}{I} \frac{\partial Q}{\partial N} \cdot \Delta N + \frac{1}{I} \frac{\partial Q}{\partial W_f} \cdot \Delta W_f \quad (2.5)$$

The output equation for any engine variable y can be expressed as a function of speed and fuel flow as well, so that a small variation of the output variable from its nominal value is expressed as

$$\Delta y = \frac{\partial y}{\partial N} \cdot \Delta N + \frac{\partial y}{\partial W_f} \cdot \Delta W_f \quad (2.6)$$

For jet engines, the second term in the output equation $\partial y / \partial W_f$ is nearly zero for most output variables except those that can be changed directly by a control inputs. An example of nonzero D-term is the high-pressure (HP) turbine gas temperature, which can be affected directly by the fuel flow input. Another example is the engine thrust, which can be affected directly by nozzle area or bypass ratio. Note that the first-order partial derivatives, similar to stability derivatives in airplane dynamics, are evaluated at the specific nominal condition, about which the linear model is expected to apply.

Now let us assume there are four output variables, and they are the shaft speed, the compressor discharge pressure, the turbine exit temperature [or the exhaust gas temperature (EGT)], and the turbine exit pressure, respectively. The output variable, expressed in vector form, is $y = [N \ P_{s3} \ T_{t5} \ P_{t5}]^T$. The partial derivatives for each output variable, therefore, represent the slopes of a small change in the output variable with respect to a small change in the state variable N and with respect to a small change in the input variable W_f . Any small change in torque (or differential torque) represents the difference between the torque produced by the turbine and the torque required by the compressor, that is,

$$\Delta Q = Q_t - Q_c \quad (2.7)$$

If we express the torques in terms of mass flow rate and temperature rise (or drop) across the compressor (or turbine), we get

$$\Delta Q = \dot{m}_t c_{p_t} (T_{t4} - T_{t5}) - \dot{m}_c c_{p_c} (T_{t3} - T_{t2}) \quad (2.8)$$

At steady state, that is, when the engine is running at a constant speed with a constant fuel flow rate, the compressor required torque is the same as the turbine produced torque; hence, the differential torque ΔQ is zero. However, as the fuel flow is increased (or decreased) from the steady-state value, the torque balance no longer holds, and the differential torque becomes the driving force for the engine to accelerate (or decelerate).

The derivatives can be calculated from a steady-state engine cycle deck by conducting a series of small perturbation “experiments.” Each of these experiments allows the model to run to a slightly “perturbed” value for a state variable or an input variable *one at a time*. Each perturbed value includes a small variation from its nominal value. One thing to note is that these small perturbation experiments must be run with the torques between the compressor and the turbine intentionally unbalance. This approach is called the *unbalanced torque performance analysis* or *perturbation analysis*. After the perturbed model simulations reach a “pseudo” steady state, we can calculate the difference in shaft torque from each experiment and divide the difference by the variation in the state variable or the input variable that has been perturbed. As an example, the partial derivative of torque with respect to fuel flow is calculated as $(\Delta Q / \Delta W_f)$, where Δ represents a small variation in one of these perturbed variables.

Equations (2.5) and (2.6) constitute the *time-domain representation* of the shaft dynamics in a jet engine. The *frequency-domain representation* of the same dynamic system is the transfer function that “transmits” from the input variable to an output variable (see Appendix A, Sec. A.2.1). Hence, the transfer function from the input variable fuel flow to the output variable y is expressed as

$$\frac{Y(s)}{W_f(s)} = \frac{cb}{s-a} + d \quad (2.9)$$

where

$$a \equiv \frac{1}{I} \frac{\partial Q}{\partial N}, \quad b \equiv \frac{1}{I} \frac{\partial Q}{\partial W_f}, \quad c \equiv \frac{\partial y}{\partial N}, \quad d \equiv \frac{\partial y}{\partial W_f}$$

where all partial derivatives are evaluated at a nominal condition. For a gas turbine engine, the coefficient a is always less than zero (i.e., < 0) in the control envelope (see Fig. 1.4); this means that a gas turbine engine is inherently stable within this region. This stability property is confirmed by the negative real part of the pole of the denominator polynomial of Eq. (2.9) (or the eigenvalue of the engine model). Equation (2.9) represents a first-order lag, which means that the speed response behaves like a lag function after the fuel flow is changed. The time constant of the engine is defined by $\tau = 1/-a$. The results from [4] are summarized in Fig. 2.3, which

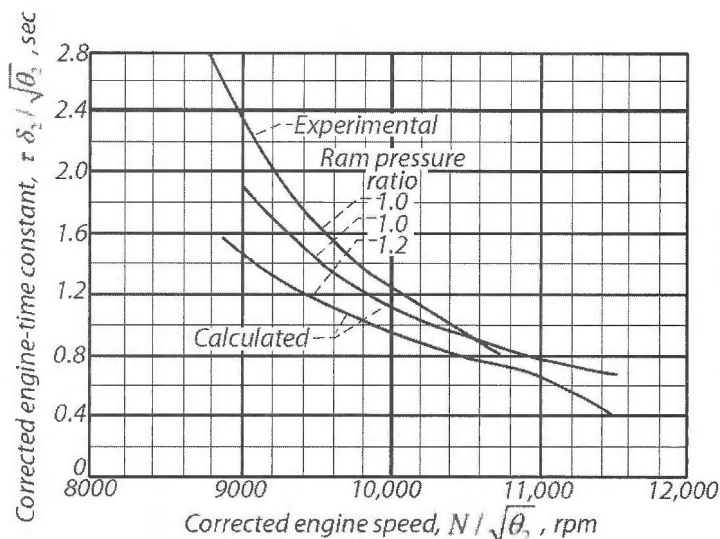


Fig. 2.3 Corrected engine time constant as a function of corrected engine speed (from [4], courtesy of Sanjay Garg of NASA).

shows the experimental and calculated results for corrected engine time constant ($\tau_c = \tau \delta_2 / \sqrt{\theta_2}$) as a function of the corrected engine shaft speed ($N_c = N / \sqrt{\theta_2}$). This figure shows that the time constant of the engine at low speed is three to four times as long as that at high speed.

2.2.1.2 Shaft dynamics of two-spool engines. The linear model of shaft dynamics for a two-spool jet engine can be derived by extending the one-spool model of Eqs. (2.5) and (2.6) with the dynamics of the second shaft [5]. For a two-spool engine, we have

$$\begin{aligned} \dot{N}_1 &= \frac{1}{I_1} \left(\frac{\partial Q_1}{\partial N_1} \cdot \Delta N_1 + \frac{\partial Q_1}{\partial N_2} \cdot \Delta N_2 + \frac{\partial Q_1}{\partial W_f} \cdot \Delta W_f \right) \\ \dot{N}_2 &= \frac{1}{I_2} \left(\frac{\partial Q_2}{\partial N_1} \cdot \Delta N_1 + \frac{\partial Q_2}{\partial N_2} \cdot \Delta N_2 + \frac{\partial Q_2}{\partial W_f} \cdot \Delta W_f \right) \end{aligned} \quad (2.10)$$

where

$$\Delta Q_1 = \dot{m}_{t1} c_{p_{t1}} (T_{t4.5} - T_{t5}) - \dot{m}_{c1} c_{p_{c1}} (T_{t2.5} - T_{t2}) \quad (2.11)$$

and

$$\Delta Q_2 = \dot{m}_{t2} c_{p_{t2}} (T_{t4} - T_{t4.5}) - \dot{m}_{c2} c_{p_{c2}} (T_{t3} - T_{t2.5}) \quad (2.12)$$

Similarly, the output equation is given by

$$y = \frac{\partial y}{\partial N_1} \cdot \Delta N_1 + \frac{\partial y}{\partial N_2} \cdot \Delta N_2 + \frac{\partial y}{\partial W_f} \cdot \Delta W_f \quad (2.13)$$

In a matrix notation, the shaft dynamics for a two-spool engine are expressed as

$$\begin{bmatrix} \dot{N}_1 \\ \dot{N}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} W_f \quad (2.14)$$

$$y = [c_1 \quad c_2] \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} + d W_f \quad (2.15)$$

Note that we have dropped the Δ symbol without loss of generality for a linear system. We have also used a , b , c , and d in the matrices to denote partial derivatives.

The frequency-domain representation of the two-spool engine dynamics expressed in transfer function form for the output variable y is (see Appendix A)

$$\frac{Y(s)}{W_f(s)} = C(sI - A)^{-1} B + D = \frac{k(s + z_1)}{(s + r_1)(s + r_2)} \quad (2.16)$$

where I is the identity matrix.

This transfer function represents a second-order dynamic system. We have substituted the matrices in Eqs. (2.14) and (2.15) with A , B , C , and D , respectively, where k is a constant, z_1 is the zero of the transfer function, and r_1 and r_2 are the two poles (or roots of denominators) of the transfer function. Depending on the output variable y selected, the zero and the gain constant will change; however, the poles and the characteristic equation will not change. This means that the transfer functions from fuel flow to shaft speed (N_1 or N_2) or to engine pressure ratio (P_{15}/P_{12}) are different only in the numerator of their respective transfer functions.

For most engines, Eq. (2.16) can be approximated by a first-order system (or a first-order lag). The frequency responses of the LP and the HP shaft speeds with respect to sinusoidal inputs in fuel flow were measured and compared to theoretical first-order lag results [5]. These results are summarized in Fig. 2.4, where the inner spool refers to the HP spool and the outer spool refers to the LP spool.

2.2.2 Pressure Dynamics—Effect of Mass Storage

A jet engine consists of numerous small chambers called (air or gas) *volumes*. Each volume is capable of storing thermal energy and air/gas masses. Because the air and combustion gases are continuous media, the dynamics of energy and mass in a volume behave like a distributed system, and the size of the volume determines how good the lumped parameter approximation for the volume dynamics

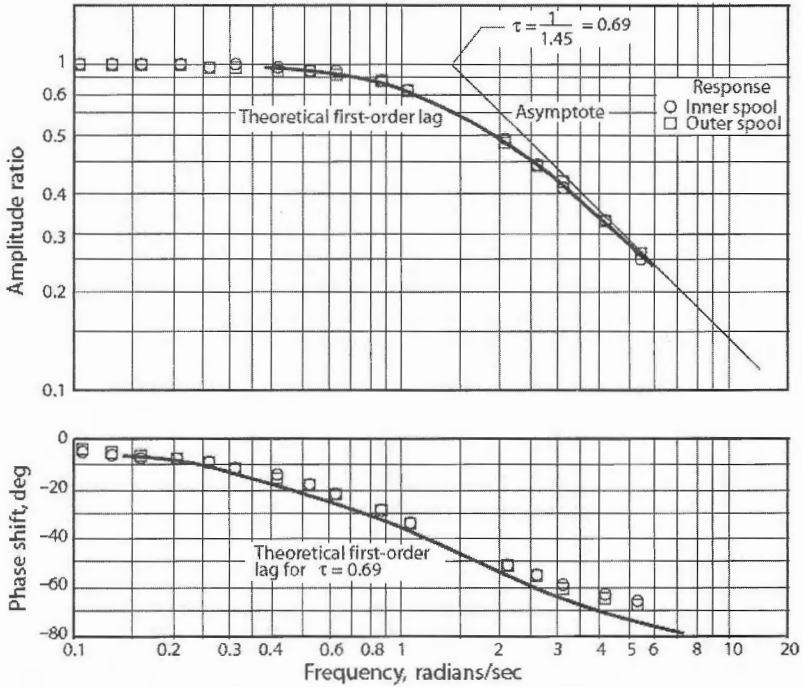


Fig. 2.4 Frequency responses of the LP and HP shaft speeds to fuel input (from [5]).

is. This approximation is sufficiently accurate if we break the air/gas passages in an engine into smaller chambers. For instance, we can consider the combination of the air passage in the LPC module and the space between the LPC and the HPC modules as one volume. A suggested breakdown of volumes in a two-spool turbofan engine is illustrated in Fig. 2.5 by the gray, dotted regions.

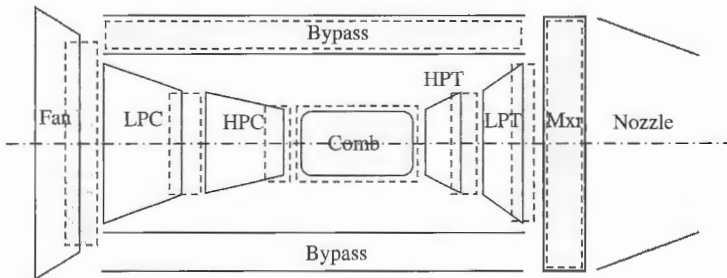


Fig. 2.5 Air and gas volumes in a two-spool turbofan engine.

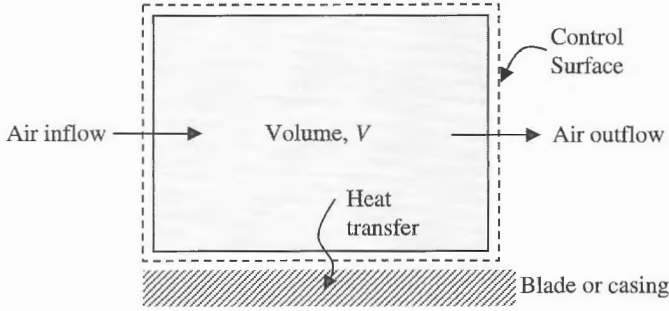


Fig. 2.6 An air/gas volume.

The effect of mass storage in a volume causes the pressure in the volume to change with a corresponding change in the temperature. These changes are results of the thermodynamic relationship between the pressure and temperature of the air or gas in the volume. We will derive the pressure change, or the pressure dynamics, in this subsection while leaving the derivation of temperature dynamics in the next subsection. Now, let us consider an air volume as shown in Fig. 2.6 [6].

The rate of change of air (or gas) masses within the *control surface* is

$$\dot{m} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \quad (2.17)$$

The air in the volume is characterized by the temperature T , the pressure P , and the density ρ . Assuming that the air behaves like a perfect gas near a nominal operating condition, we can write a small change of the pressure in the volume as

$$dP = (\rho R)_0 dT + (RT)_0 d\rho = (\rho R)_0 dT + \left(\frac{RT}{V} \right)_0 dm \quad (2.18)$$

where V is the volume within the control surface, which is constant, and the subscript 0 denotes a nominal operating condition. After differentiating the preceding equation with respect to time, we get

$$\dot{P} = (\rho R)_0 \dot{T} + \left(\frac{RT}{V} \right)_0 \dot{m} \quad (2.19)$$

So the rate of change of the pressure in the volume is related to the rates of change of the temperature and of the mass, respectively. Because the first term in the preceding equation is a lot smaller than the second term [7, 8], we can drop the first term; hence, the pressure term in Eq. (2.19) is approximately proportional to the rate of change of mass within the volume as

$$\dot{P} \approx \left(\frac{RT}{V} \right)_0 \dot{m} \quad (2.20)$$

This means that the change in pressure ΔP is a time integral of the right-hand side of Eq. (2.20). The frequency-domain representation of the linearized pressure dynamics is

$$\frac{P(s)}{\Delta M(s)} = \frac{(RT/V)_0}{s} \quad (2.21)$$

where $\Delta M(s)$ is defined as the Laplace transform of $\dot{m}(t)$, which is a rate of change of the mass flow rate in the volume, and it can also be expressed as $M_{\text{in}}(s) - M_{\text{out}}(s)$.

2.2.3 Temperature Dynamics—Effect of Energy Storage

A jet engine has two different types of temperature dynamics: one is the change of temperature associated with a direct change of the thermodynamic state of the air or gas in a volume, and the other is the change of temperature associated with heat transfer between the surrounding metal parts of the engine hot section and the gas flow. A direct change of thermodynamic state can happen when the pressure or the temperature of the air or gas changes as a result of turbomachinery work input or extraction. A change in thermodynamic state can also happen when the air or gas is heated by chemical reactions. The dynamics caused by these direct thermodynamic effects are very fast compared to those caused by metal-heating (or heat-soak) effects; consequently, the characteristic time of the temperature change caused by thermodynamic effects can be an order of magnitude faster than that caused by metal-heating effects. Thus, thermodynamic temperature changes are sometimes called the *fast* temperature dynamics in a module or a volume, and the metal-heating temperature changes are called the *slow* temperature dynamics. Temperature dynamics caused by the metal-heating effect in a gas turbine are particularly of interest to engine control and monitoring system designs because they are slow, visible, and yet large. This large temperature change must be captured in a model so that its transient response matches well with that of the actual engine.

From the law of energy conservation, we know that the change of temperature in a module, including both the fast and the slow temperature dynamics, can be expressed by [9]

$$\dot{T} = \left(\frac{RT}{c_v PV} \right) (\dot{m}_{\text{in}} h_{\text{in}} - \dot{m}_{\text{out}} h_{\text{out}} + \dot{q}) - \left(\frac{RT^2}{PV} \right) \dot{m} \quad (2.22)$$

where c_v is the constant-volume specific heat of the gas and $\dot{q}$ is the heat-transfer rate, mainly from the metal surroundings *into* the gas within a volume, that is, the slow temperature dynamics.

Let us now focus on the slow temperature change in a gas turbine engine. This is possible by applying the concept of frequency separation caused by the large

disparity in time constant between the fast and the slow temperature changes. Suppose the temperature of the gas is higher than the temperature of the metal, and then the heat-transfer rate from the gas to the metal can be approximated by a one-dimensional heat convection equation [10] as

$$\dot{q} = \bar{h}A_M(T - T_M) = c_M m_M \dot{T}_M \quad (2.23)$$

where $\bar{h}$ is the normalized convective heat-transfer coefficient (or film conductance); A_M is the area of the metal wall that is exposed to the air or gas, that is, the wetted area; T is the gas temperature; T_M is the metal temperature; c_M is the specific heat of the metal; and m_M is the mass of the metal. Because the energy (or heat) lost from the gas is the same as the energy gained by the metal, we can express the relationship between the gas temperatures before and after the heat transfer as

$$\dot{q} = c_p \dot{m}_t (T_{\text{in}} - T_{\text{out}}) = \bar{h}A_M(T - T_M) \quad (2.24)$$

where $\dot{m}_t$ is the mass flow rate of the gas through the turbine (control volume), T_{in} is the gas temperature at the inlet, and T_{out} is the gas temperature at the outlet (or exit). Equations (2.23) and (2.24) are the dynamic equations for the slow temperature change in the volume.

Substituting Eq. (2.23) into Eq. (2.24) with $T = T_{\text{in}}$ and taking the Laplace transform of both sides of the resultant equation, we get the transfer function (TF) from the turbine inlet temperature to the turbine exit temperature as

$$\frac{T_{\text{out}}(s)}{T_{\text{in}}(s)} = \frac{[(c_M m_M / h A_M) - (c_M m_M / c_p \dot{m}_t)]s + 1}{(c_M m_M / h A_M)s + 1} \quad (2.25)$$

where the time constant is a function of the mass flow rate through the turbine and the exit gas temperature [11], that is, for the HP turbine, the time constant is

$$\tau_{\text{slow}} = \frac{c_M m_M}{h A_M} = f(\dot{m}_{41}, T_{41}) = c_1 \frac{\sqrt{T_{41}}}{\dot{m}_{41}^{c_2}} \quad (2.26)$$

where the subscript 41 stands for the HP turbine inlet station and c_1 (the multiplier) and c_2 (the power of the mass flow rate) are empirically determined constants.

2.3 Modeling of Complete Engine Dynamics

In Sec. 2.2, we explained three basic types of dynamic effects in gas turbine engines, namely, shaft dynamics caused by the inertial effect, pressure dynamics

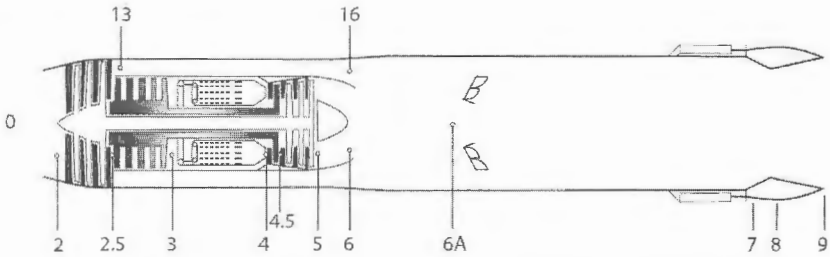


Fig. 2.7 Generic two-spool afterburning turbofan engine (adapted from [12]).

caused by the mass storage effect, and temperature dynamics caused by the energy storage as well as the heat transfer between the gas and the surrounding “walls.” Now we will show how to apply these effects to a complete engine from the inlet to the exhaust. We will use an afterburning turbofan engine as an example because it represents the most complex control design problem among all forms of turbine engines. Figure 2.7 is a cutaway drawing of a generic two-spool, afterburning turbofan engine with the key engine stations identified (adapted from Fig. SM7.3a of [12]). Figure 2.8 shows the detailed internal components of a real-world example of the two-spool turbofan engine, the GE F110.

To build a complete dynamic model, we first identify each and every element in the engine that contributes to one of these dynamic effects. We do this by taking the schematic diagram of Fig. 2.7 and converting it into a computation flow diagram as shown in Fig. 2.9 [13]. The model in this figure represents a full engine model that is capable of estimating both the transient and the steady-state performance of an engine. Hence the model is called a *transient* cycle deck or a transient component-level model.

Note that a volume element (filled in light gray color) is inserted between each pair of engine modules to allow for the mass and energy storages, which give rise to pressure and temperature dynamics, respectively. Without the volume elements, the model in Fig. 2.9 reduces to a steady-state model, and the performance of each

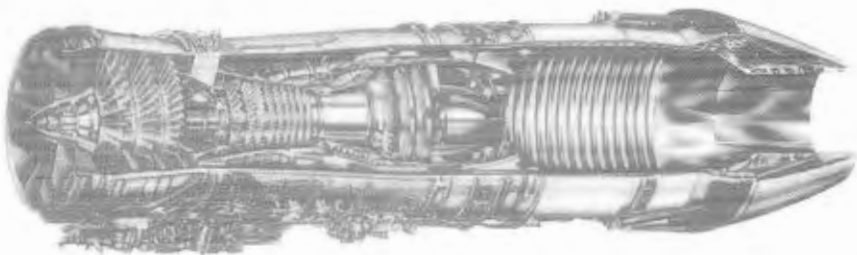


Fig. 2.8 GE F110 engine (courtesy of General Electric Company).

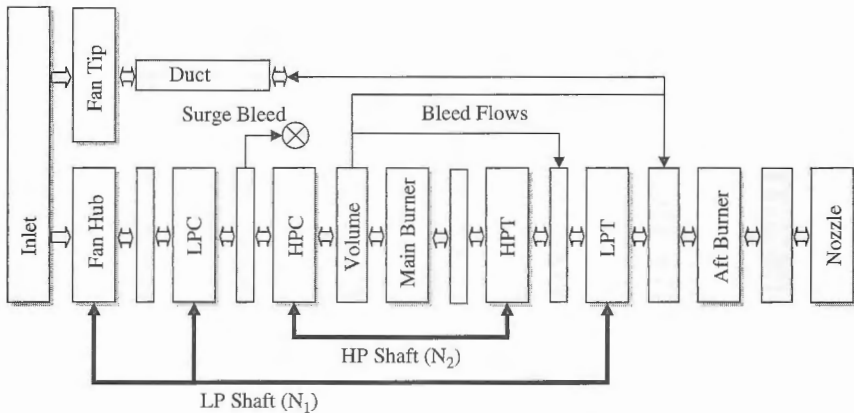


Fig. 2.9 Computational flow diagram of a two-spool afterburning turbofan. (See also color figure section at the back of the book.)

engine module, such as the fan module, the LPC module, the HPC module, etc., can be determined by using the component maps representing each module as described in Sec. 2.1. Also note that the “yellow” arrows indicate data (or information) flows through the engine.

Because the dynamics of each volume can be represented by two state variables—the pressure [expressed by Eq. (2.20)]—and the temperature [expressed by Eq. (2.22)]—so the total number of state variables for the eight volumes of the engine is 16. Adding to this number the two state variables representing the LP and the HP shaft speeds, we have a total of 18 state variables for this afterburning turbofan engine. The number of state variables for the engine will increase further if we consider the relatively slow changing temperature as a result of the metal heating or cooling effect [see Eq. (2.23)]. These slow temperature state variables have fairly long time constants, and they are significant only for the HP and the LP turbine modules. So the number of state variables for a two-spool, mixed-flow, afterburning turbofan engine has now increased to 20.

Moreover, each computational block (either a turbomachinery module or a volume) in the flow diagram of Fig. 2.9 represents a lumped-parameter system for that element. This means that the internal air or gas flow properties of each block are uniform throughout the block regardless of the size of the block. In reality, this assumption is generally not true because the internal properties can depend on the spatial location within the block. This is especially true when the block is a volume element and it is relatively large compared to the characteristic length traveled by air/gas particles. Take the volume of the afterburner as an example. This volume is large compared to the volume between the LP and the HP compressors, or the volume between the HP and the LP turbines; therefore, the properties within the afterburner are expected to vary depending on the actual location in the afterburner where the properties are calculated. Another example is a multistage axial compressor or turbine module, which consists of a series of compression or expansion stages with a small volume between two adjacent stages. In this kind of module,

the flow properties are different at each stage. Hence, they depend on the spatial location within the module.

This phenomenon of varying internal properties with respect to spatial location makes an element a distributed dynamic system. One approach to converting a distributed system into a lumped-parameter system is to divide the distributed system into a series of smaller systems, each of which is small enough so that its internal properties are approximately uniform within the system boundary. This approach is similar to the finite difference method in computational fluid mechanics or structural mechanics. By dividing a large volume, or a multistage compressor or turbine module, into a series of smaller modeling elements, the total number of computational elements in Fig. 2.9 can increase significantly; hence, the total number of state variables for a gas turbine engine can easily exceed 50.

The general nonlinear form of the state and the output equations of an engine can be expressed by Eqs. (A.83) and (A.84). If the engine can be approximated as a linear-time-invariant (LTI) system, then the nonlinear equations can be simplified to the following equations:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

where x is an $n \times 1$ vector, u is an $m \times 1$ vector, y is an $l \times 1$ vector; the dimensions of the matrices A , B , C , and D are $n \times n$, $n \times m$, $l \times n$, and $l \times m$, respectively; and all of the matrix elements, $A = \{a_{ij}\}$, $B = \{b_{ik}\}$, $C = \{c_{ij}\}$, and $D = \{d_{ik}\}$ are constants. The preceding equations are the most common time-domain representation for the *linearized* dynamics of a gas turbine engine. The corresponding frequency-domain representation for the input-output relationship of the engine (or system) is the TF $G(s)$, which is derived from Eq. (A.3) and Eq. (A.4)

$$G(s) = \frac{Y(s)}{U(s)} = \frac{k_e(s^m + b_1s^{m-1} + \cdots + b_{m-1}s + b_m)}{s^n + a_1s^{n-1} + \cdots + a_{n-1}s + a_n} \quad (2.27)$$

where k_e is an engine-specific constant and $m \leq n$.

2.4 Modeling of Actuator and Sensor Dynamics

Engine actuators are used primarily to control gas path performance and operability. These actuators fall into three categories: 1) fuel control actuators, typically called fuel metering valves; 2) position control actuators, which set inlet guide vane (or stator vane) angles, the exhaust nozzle area (for an afterburning engine), as well as certain discrete positions like the fuel shut-off solenoid; and 3) air- or gas-flow control actuators, typically called bleed valves.

In the simplest form, the dynamics of an engine actuator can be modeled by a first-order lag as

$$\frac{u_a(s)}{u_{\text{cmd}}(s)} = \frac{b}{s + a} \quad (2.28)$$

where u_{cmd} is the actuator command calculated by the controller and u_a is the actuator output (or the actuation). More complex models of common engine actuators are described later in this section.

Engine sensors are used to measure specific physical properties to accomplish two goals: 1) providing feedback for the control system and 2) obtaining an awareness of the health and the usage of the engine. Engine sensors fall into five categories as follows:

1) *Gas-path sensors* typically measure pressures, temperatures, airflow rates, fuel flow rates, and shaft rotational speeds.

2) *Vibration sensors* measure the displacement or acceleration of a rotating part or a casing component in the engine.

3) *Oil and debris sensors* count the number of oil chips and sizes; some of them also measure specific particles going through the engine inlet or exhaust duct.

4) *Actuator feedback sensors* measure the position, angle, or speed of an actuator for feedback purposes in control loops; some of these measurements provide “local” loop closure around the actuators for faster closed-loop response.

5) *Damage sensors* measure cracks or delamination of structures, although most of these sensors are ground-based and considered nondestructive-inspection equipment.

In the simplest form, the dynamics of an engine sensor can be represented by a “unity” transfer function because the dynamic response of the sensor is typically much faster than the dynamic response of the engine or the actuator. In most cases, a first-order lag with a fast pole (i.e., a large negative root of the denominator polynomial) will be adequate to model sensor dynamics. So we have

$$\frac{z(s)}{y(s)} = 1 \quad \text{or} \quad \frac{z(s)}{y(s)} = \frac{b}{s + a} \quad (2.29)$$

where y is the output variable being measured and z is the measured output variable.

2.4.1 Hydraulic Actuators

Hydraulic actuators (or hydraulic valves) are the most common actuation devices in hydromechanical controls on gas turbine engines. The “working” medium of a hydraulic actuator is a fluid, which can be a hydraulic fluid or engine fuel (to reduce the weight of carrying hydraulic fluid in the air). Typical hydraulic actuators are shown in Fig. 2.10 [14].

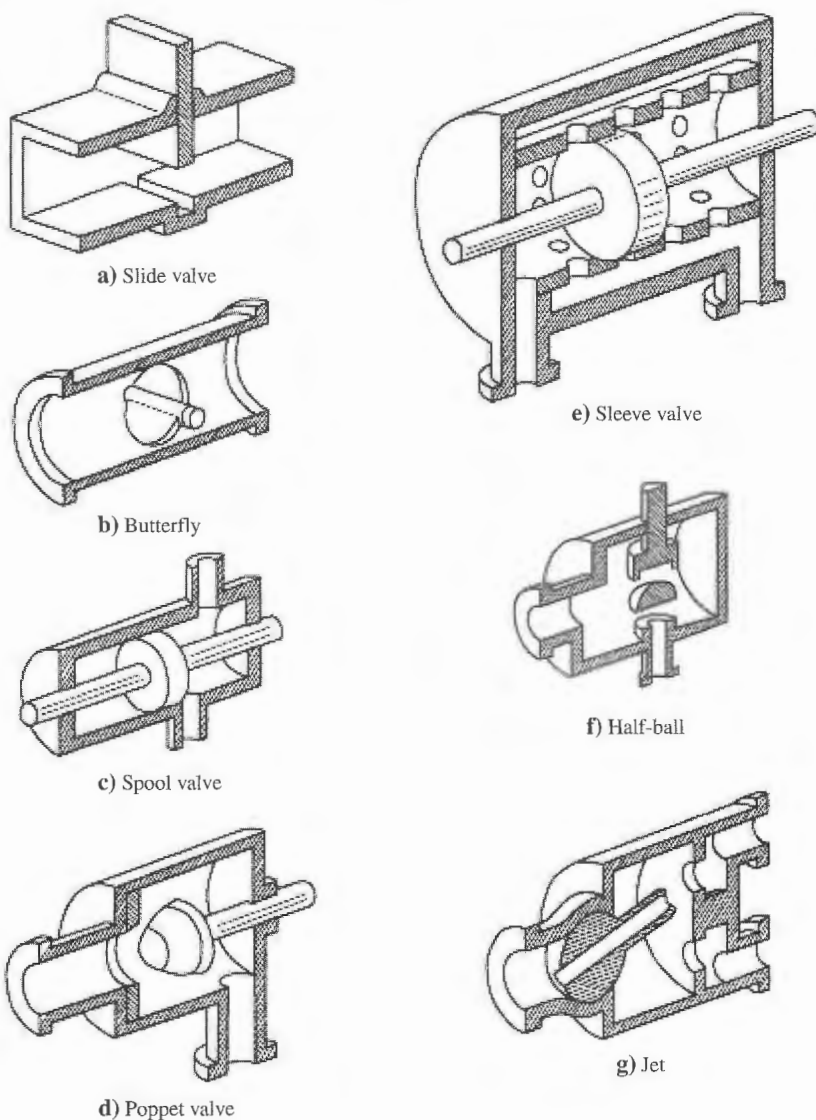


Fig. 2.10 Typical hydraulic actuators (from [14], reprinted with permission).

Generally, these actuators can be divided into two types: 1) a load-bearing hydraulic actuator, like a rod-and-cylinder (or a piston-rod-cylinder combination) assembly; 2) a servovalve to control the hydraulic fluid in and out of the load-bearing actuator. Typically, the servovalve is a combination of an electrohydraulic servovalve (EHSV) and a spool valve. Figure 2.11 is an example of a multistage hydraulic actuator, which consists of three stages: the rod-and-cylinder component on the bottom of the figure to carry the load, the spool valve in the middle

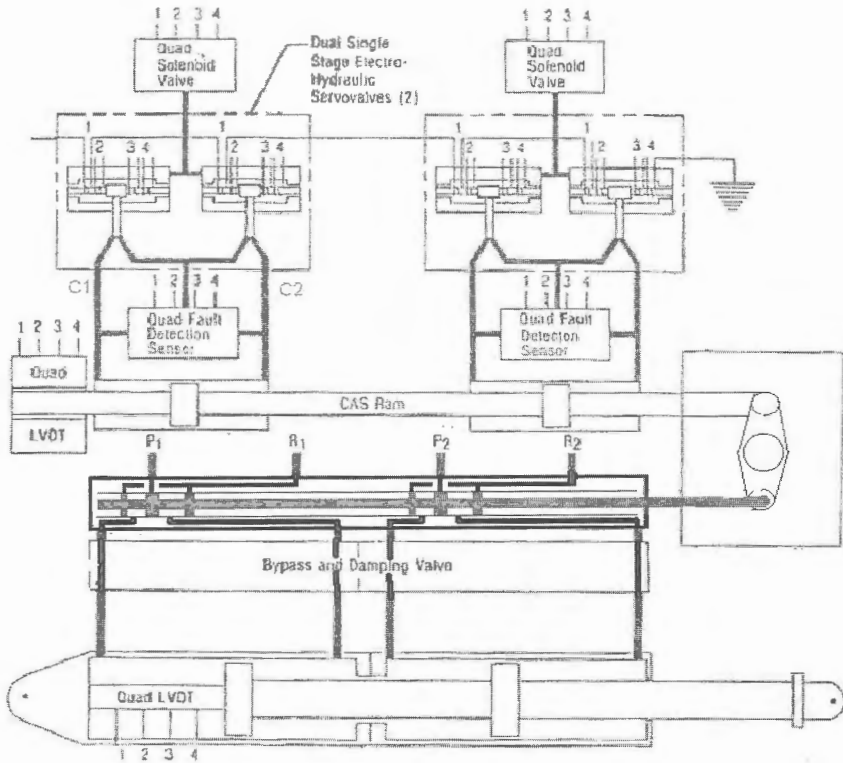


Fig. 2.11 Example hydraulic actuator (courtesy of Boeing Phantom Works). (See also color figure section at the back of the book.)

(with the rod colored in blue) acting as a boost valve, and the four identical torque motors on the top (in two separate packages for redundancy purposes).

2.4.1.1 First-order model of a hydraulic actuator. The simplest form of model for a hydraulic actuator is an integrator as shown in Fig. 2.12. This first-order model converts the volumetric flow rate of the fluid (i.e., $Q = dVol/dt$) into the movement of the piston by assuming that the hydraulic fluid is incompressible.

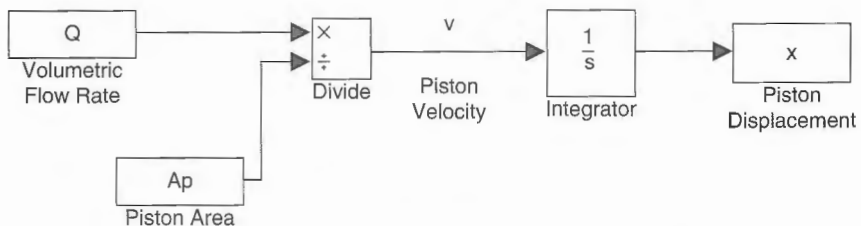


Fig. 2.12 First-order model of a hydraulic actuator.

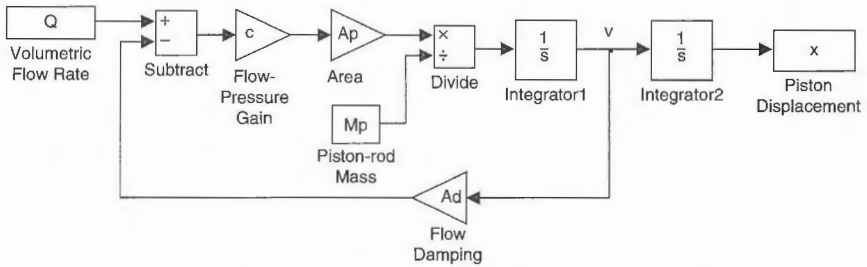


Fig. 2.13 Second-order model of a hydraulic actuator.

In the TF form, this model is given by

$$\frac{X(s)}{Q(s)} = \frac{(1/A_p)}{s} \quad (2.30)$$

2.4.1.2 Second-order model of a hydraulic actuator. A more complex form of a hydraulic actuator model is second order, and it behaves like a mass-spring-damper (mechanical) system as shown in Fig. 2.13. This model considers the inertia of the moving parts, that is, the piston-and-rod assembly.

In the TF form, this model is given by

$$\frac{X(s)}{Q(s)} = \frac{cA_p}{M_p s^2 + cA_p A_d s} \quad (2.31)$$

2.4.1.3 Third-order model of a hydraulic actuator. The most general form of a hydraulic actuator model is third order because it includes the effects of both the piston inertia and fluid compressibility as shown in Fig. 2.14, where c is a gain constant and T_c is the time constant of the compressibility effect, which is a function of the fluid's bulk modulus B and the fluid volume of the actuator.

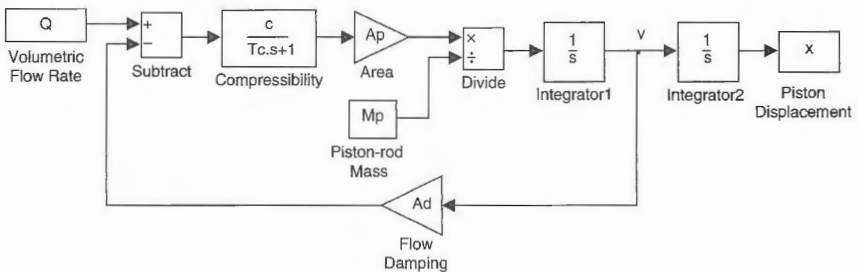


Fig. 2.14 Third-order model of a hydraulic actuator.

In the TF form, this model is given by

$$\frac{X(s)}{Q(s)} = \frac{cA_p}{M_p T_c s^3 + M_p s^2 + cA_p A_d s} \quad (2.32)$$

Note that the models just given have not included the load exerted on the piston-rod assembly, nor have they included fluid leakages between the piston and the cylinder. Leakage essentially increases the damping of the actuator. Furthermore, these models are open-loop models for a generic actuator because the actuator position (or displacement) feedback has not been included.

2.4.1.4 Model of a servovalve. In the preceding subsections, the volumetric flow rate Q has been used as the input to the actuator model. You might ask where this flow rate comes from. A simple answer is from an upper-stage flow control device, which is often called a servovalve or an electro-hydraulic servovalve (EHSV). The primary purpose of the servovalve is to supply the right amount of hydraulic fluid (in terms of the volumetric flow rate) to the downstream valve or actuator. Commonly used servovalves for engine controls fall into two categories: torque motor and jet pipe; both types are electrohydraulic devices. Although the moving mechanisms of these two types of servovalves are different, their dynamic models are similar.

An EHSV can be modeled as a combination of an electrical motor and a flow control device, usually called an orifice, in series as shown in Fig. 2.15. The motor is modeled as a second-order system, and the orifice is modeled as a servogain, at a nominal operating condition.

In the TF form, this model is given by

$$\frac{Q(s)}{I(s)} = \frac{c_1 c_2}{s^2 + a_1 s + a_2} \quad (2.33)$$

Note that a gain constant has been used to represent the relationship between the volumetric flow rate and the motor displacement. By doing so, the following approximation has been made:

$$\frac{Q(s)}{X_m(s)} = c_2 \quad (2.34)$$

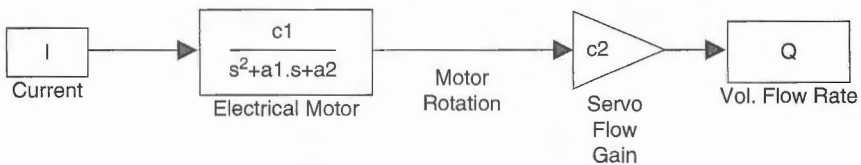


Fig. 2.15 Servovalve model.

where $X_m(s)$ is the Laplace transform of motor displacement $x_m(t)$. This approximation is true only near a nominal operating point of the servovalve because the flow rate through the orifice is approximated by

$$Q \approx 38.034 \cdot c_d A_0 \sqrt{\frac{\Delta P}{sp}} \quad (2.35)$$

where Q is the volumetric flow rate, A_0 is the orifice area in square inches, sp is the specific gravity of the fluid, and ΔP is the pressure drop across the orifice in pounds per square inch. For most fuel metering valves, the pressure drop across the valve is regulated at a constant value; hence, the volumetric flow rate through the valve is proportional to the motor displacement as

$$\delta Q \approx 38.034 \cdot c_d \left(\frac{\partial A_0}{\partial x_m} \right)_0 \cdot \delta x_m \equiv c_2 \cdot \delta x_m \quad (2.36)$$

where the nominal valve operating point is denoted by the subscript 0. The reader is referred to [15] and [16] for additional background information about hydraulic actuators and fluid power components.

2.4.2 Pneumatic Actuators

Unlike a hydraulic actuator, a pneumatic actuator uses a compressed air or gas as the working medium. The primary difference in dynamics between a pneumatic actuator and a hydraulic actuator is the effect of compressibility because the air or gas is essentially compressible compared to a hydraulic fluid.

The model for a pneumatic actuator is similar to the third-order model of a hydraulic actuator, except that the volumetric flow rate is replaced by the mass flow rate of the air or gas. Figure 2.16 shows a model of the pneumatic actuator pushing a piston rod. Note that the actuator load has been added to calculate the net force acting on the piston.

The effect of compressibility also creates a unique phenomenon in pneumatic actuators called “choking.” Choking is the condition where the amount of air or gas flow through the valve reaches a maximum value and cannot be increased by increasing the upstream (or supply) pressure. From a control design standpoint, choking

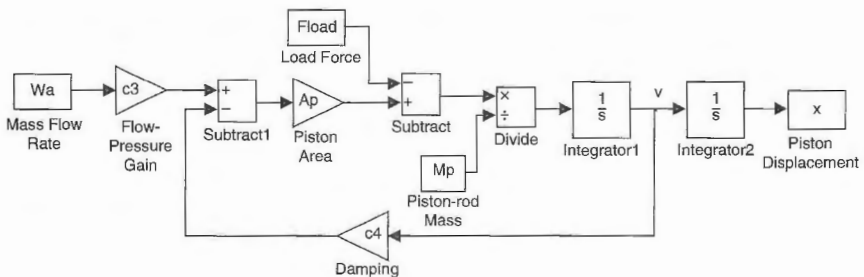


Fig. 2.16 Model of a pneumatic actuator.

introduces saturation nonlinearity (see Appendix A), which is a common cause of limit cycles in closed-loop systems, and it makes the design more challenging.

The mass flow rate in Fig. 2.16 is given by [2, 17]

$$W_a = \dot{m}_a = \frac{c_d A_n P_{tu}}{\sqrt{T_{tu}}} \sqrt{\frac{2g_c}{R} \frac{\gamma}{\gamma-1} \left[\left(\frac{P_d}{P_{tu}} \right)^{\frac{2}{\gamma}} - \left(\frac{P_d}{P_{tu}} \right)^{\frac{\gamma+1}{\gamma}} \right]} \quad (2.37)$$

where A_n is the actuator inlet flow nozzle area, P_{tu} is the total pressure of the flow upstream of the nozzle, P_d is the static pressure downstream of the nozzle (in the actuator housing), T_{tu} is the total temperature of the flow upstream of the nozzle, and g_c is the gravity constant. Note that a discharge coefficient c_d has been added to account for nonisentropic flows.

If the flow going into the actuator volume is choked, that is, the ratio of the actuator pressure to the supply pressure (P_d/P_{tu}) is less than the constant 0.528 (for the airflow), then the mass flow rate in Eq. (2.37) becomes

$$W_a = \dot{m}_a = 0.532 \cdot C_d \frac{A_n P_{tu}}{\sqrt{T_{tu}}}, \quad \text{for air} \quad (2.38)$$

where the constant on the right side of the equation corresponds to $\gamma = 1.4$; it takes on a value of 0.518 when $\gamma = 1.3$ (see Fig. B.8 and the explanation of the mass flow parameter in Appendix B). Note that this equation is valid when the unit of the flow rate is pounds per second, the unit of area is square inches, the unit of pressure is pounds per square inch, and the unit of temperature is degrees Rankine.

2.5 High-Fidelity Engine Simulations

High-fidelity engine simulations, such as the transient cycle deck based on the model in Fig. 2.9, have been developed by engine manufacturers for engine and control system designs. These simulations perform nonlinear estimation of the performance of every engine module from the inlet to the exhaust. Different nonlinear simulations are developed for different engine types (or models) to represent the specific configuration and characteristics of each engine type. High-fidelity engine simulations have also been developed by various U. S. government laboratories. However, most of these simulations are implemented in programming languages that require compilation and linking to run on a specified computing hardware. In addition, these simulations are usually proprietary because of the sensitive nature of module performance characteristics.

Recently, there has been an increasing interest in using graphical modeling tools, like the Simulink® from The Mathworks, to build high-fidelity simulations of complex systems. These graphical modeling tools provide an efficient design interface for building models as well as for analyzing control system performance with a wide range of control design tools. Two of these graphical-model-based simulations are described in this section: The first is the U.S. Air Force Research Laboratory's generic engine simulation for a two-spool, low-bypass, afterburning

turbofan engine, and the second is NASA's generic, two-spool, high-bypass turbofan engine. Both generic simulations are constructed in the Simulink® modeling and simulation environment.

2.5.1 U.S. Air Force Generic Turbofan Engine Model

The U.S. Air Force operates a wide variety of aircraft turbine engine types. These engine types are produced by multiple vendors, with some variations among different engine models within each engine type. High-fidelity simulation models exist for most engine types, but a standard approach to the development of these models among all vendors does not exist.

The Air Force Research Laboratory (AFRL) at the Wright-Patterson Air Force Base contracted Scientific Monitoring, Inc., to develop a generic turbofan engine model. This model (or the AFRL model) provides a basic framework for both the development of engine component models and the analysis of component interactions at the system level within the engine. These intentions are consistent with the models previously discussed by Adibhatla and Gastineau [18], Gastineau [19], Mink and Behbahani [20], and the Numerical Propulsion System Simulation (NPSS) [21]. The AFRL model can also serve as a tool for engine control and health-management-related research.

The AFRL model can capture both steady-state and transient behavior of the engine. In the generic engine model framework, model development and enhancement are standardized, and they can be shared among multiple organizations, as required. Component modules developed for one engine can be reused in subsequent models, and the time and the cost required to produce analytical results are significantly reduced.

The generic, two-spool, turbofan engine model represents a modern, high-bypass, nonaugmented, turbofan engine model. The model conforms to the following requirements [22]:

- 1) The model accurately represents engine steady-state behavior, especially at the design point.
- 2) The model credibly represents transient engine behavior.
- 3) The model outputs simulate engine measured variables.
- 4) The model operates over the entire flight envelope of the engine.
- 5) The model simulates engine behavior in real time.
- 6) The model is modifiable to simulate other turbine engine cycles.
- 7) The model is capable of interfacing with hardware for testing and validation.
- 8) The model provides user inputs for environment conditions, power lever angle, and load demands for bleed and power extraction.

The capabilities that the AFRL model can provide are as follows: 1) a tool to investigate the engine performance and operability interactions with aircraft subsystems, 2) a tool to predict the consequences of specific operational scenarios and modifications to existing engine or control systems, 3) a tool to integrate test results from test facilities and incorporate them into existing or new models, 4) a testbed for the development of advanced engine control and health management system architectures and algorithms, and 5) a tool to archive prior design experience and field-obtained knowledge.

Many simplified models leave out some critical features necessary to match model results with manufacturers' cycle deck output values. The AFRL model begins with simplified approaches, such as using the constant-pressure specific heat c_p and the ideal-gas equation to calculate component work performance. However, the model does include the following more sophisticated effects:

1) *In the area of thermodynamics*: The enthalpy h and the ideal enthalpy h_{id} tables with dissociations are adapted from available thermodynamics tables, including the JANAF (Joint Army Navy Air Force) tables. These values are in agreement with those used in NPSS. Dissociations are assumed for equilibrium conditions, and the ability to "freeze" the constituents was not implemented.

2) *In the area of fluid mechanics*: The Reynolds numbers at local conditions are compared with reference Reynolds numbers to get the effects of flow fields on gas-path component work and efficiencies. The reference Reynolds number is meant to represent the rig test conditions where the component map was developed. Application of these effects allows the correct performance adjustments with altitude. All engine gas-path components have Reynolds-number calculations, and their effects can be adjusted.

3) *In the area of heat transfer*: Heat absorption of the metal parts is included for the HPC and each subsequent component. The primary driver for heat absorption is the change in temperature during a transient. Inputs for this calculation are the mass of the metal parts and the user-assignable reference time constant. The effective time constant is recalculated from the reference time constant with Reynolds effects.

The Simulink® block diagram view of the AFRL generic engine model is shown in Fig. 2.17. The major blocks (representing key engine components) are inlet, fan tip, fan hub, compressor, combustor, HPT, LPT, bypass duct, mixer, nozzle, and cooling flow paths.

An example of transient simulation using this model is shown in Fig. 2.18, where a rapid deceleration is followed by a rapid acceleration. The amber line marks the time when the deceleration is commanded, the green line marks the acceleration, and the red line marks the start of fuel flow cutback to keep the turbine temperature below limit.

2.5.2 NASA Generic Turbofan Engine Model

NASA's generic, two-spool, high-bypass turbofan engine, called C-MAPSS (Commercial Modular Aero Propulsion System Simulation) represents a realistic, large commercial engine rated at 90,000-lb thrust [23]. C-MAPSS is based on the MAPSS technology developed for a generic, component-level model of a modern, high-performance turbofan simulation [24]. A graphical user interfaces has been developed to facilitate the use of the engine simulation and to design regulator and limit controllers. Figure 2.19 shows the main elements of the engine model.

C-MAPSS allows the user to simulate the performance of a deteriorated engine by varying 13 health parameter variables through a convenient user interface. These 13 health parameters are the efficiency, the flow, and the pressure-ratio modifiers for each of the fan, the LPC, and the HPC modules (total number is 3 times 3 = 9), and the efficiency and the flow modifiers for each of the HPT and the LPT modules. An example of the simulated fan speed response to a health

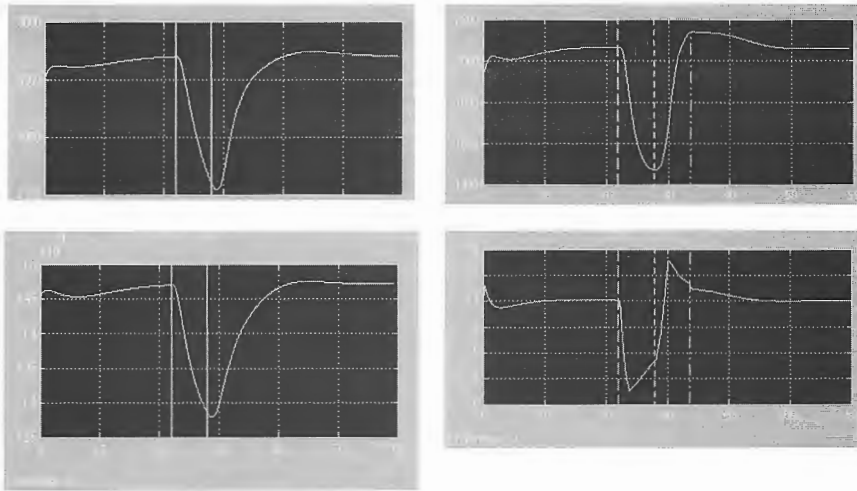


Fig. 2.18 Transient simulation of the AFRL generic engine model (N_1 in top left, N_2 in bottom left, T_{45} in top right, W_f/P_3 in bottom right). (See also color figure section at the back of the book.)

parameter degradation at a constant throttle position (or PLA) is shown in Fig. 2.20. The figure shows that the fan speed initially goes up when the degradation is first added (or injected), and then the fan speed control loop brings the speed back to the commanded value.

2.6 Derivation of Linear Engine Models

Linear models are needed for control system analysis and design because most control design tools apply only to linear models. To derive a linear model around a nominal operating point for an engine, we usually resort to one of two methods: 1) the small perturbation method based on the Taylor's series expanded around the nominal operating point and 2) the system identification method

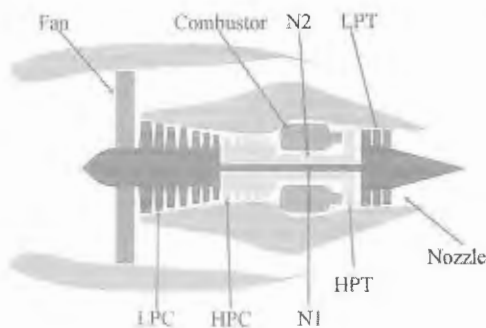


Fig. 2.19 Main elements of C-MAPSS model [23].

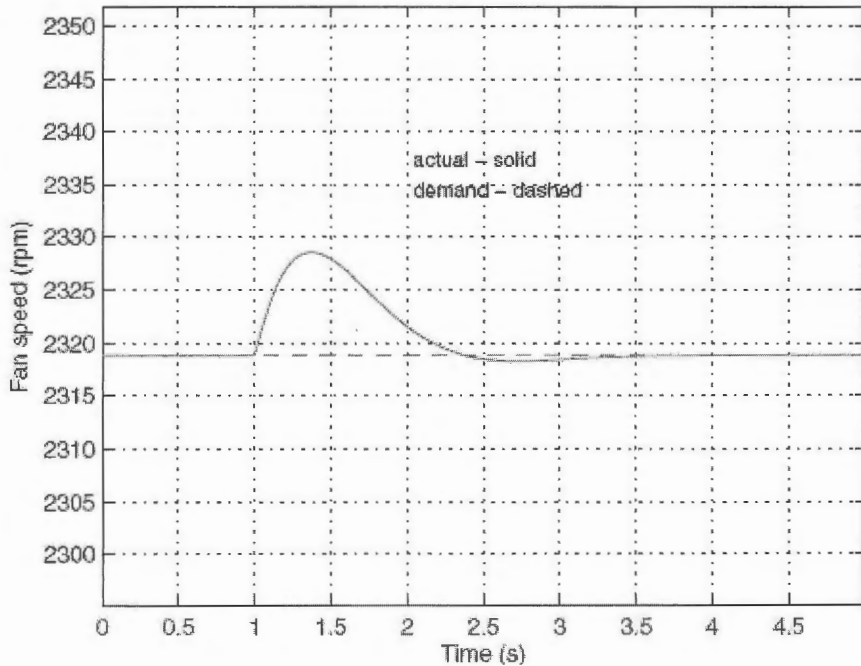


Fig. 2.20 Fan speed transient after degradation is injected in the model [23].

based on minimizing the error between the output of an assumed model structure and the actual output treated as the reference value. The small perturbation method is almost always applied to a nonlinear simulation of the engine, whereas the system identification method can be applied to either a nonlinear simulation or a data set containing the time histories of input and output variables of the system.

2.6.1 Extraction of Linear Models from Nonlinear Simulations

The nonlinear state and output equations for a dynamic system are expressed by Eqs. (A.83) and (A.84) in Appendix A and are repeated here

$$\dot{x}(t) = f[x(t), u(t), t] \quad (2.39)$$

$$y(t) = g[x(t), u(t), t] \quad (2.40)$$

where f and g are nonlinear functions of the state variable, the input variable, and time (in general form). For turbine engines, f and g are smooth enough, within the engine's operating envelope, to have a Taylor-series approximation around the nominal operating condition x_0 and u_0 , so that

$$f[x(t), u(t), t] = f[x_0(t), u_0(t), t] + \left. \frac{\partial f}{\partial x} \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial f}{\partial u} \right|_{x_0, u_0} \delta u(t) + o[\delta^2] \quad (2.41)$$

$$g[x(t), u(t), t] = g[x_0(t), u_0(t), t] + \left. \frac{\partial g}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial g}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) + o[\delta^2] \quad (2.42)$$

where $o[\delta^2]$ represents a small-order term and is the collection of the remaining terms in the Taylor-series expansion. This small term is in the order of the square of the “deltas,” that is, $o[\delta x^2, \delta u^2]$. We can make this term arbitrarily small by choosing small enough deltas in state variables, input variables, and in time, so that $o[\delta^2]$ is negligible because

$$x(t) = x_0(t) + \delta x(t) \quad \text{and} \quad u(t) = u_0(t) + \delta u(t) \quad (2.43)$$

Equations (2.42) and (2.43) can be written as

$$\dot{x}(t) \approx f[x_0(t), u_0(t), t] + \left. \frac{\partial f}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial f}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) \quad (2.44)$$

$$y(t) \approx g[x_0(t), u_0(t), t] + \left. \frac{\partial g}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial g}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) \quad (2.45)$$

Because the first term on the right-hand side of Eq. (2.44) is zero at the nominal (equilibrium) condition, the preceding equations can be further simplified to

$$\dot{x}(t) \approx \left. \frac{\partial f}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial f}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) \quad (2.46)$$

$$y(t) - y_0(t) \approx \left. \frac{\partial g}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial g}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) \quad (2.47)$$

For a multi-input and multi-output system, as in the case of a modern gas turbine engine, the linearized equations of (2.46) and (2.47) can be expressed in matrix form as

$$\dot{x}(t) \approx \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \bigg|_{x_0, u_0} \delta x(t) + \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \cdots & \frac{\partial f_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix} \bigg|_{x_0, u_0} \delta u(t) \quad (2.48)$$

$$y(t) - y_0(t) \approx \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \dots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \dots & \frac{\partial g_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_l}{\partial x_1} & \frac{\partial g_l}{\partial x_2} & \dots & \frac{\partial g_l}{\partial x_n} \end{bmatrix} \bigg|_{x_0, u_0} \delta x(t) + \begin{bmatrix} \frac{\partial g_1}{\partial u_1} & \frac{\partial g_1}{\partial u_2} & \dots & \frac{\partial g_1}{\partial u_m} \\ \frac{\partial g_2}{\partial u_1} & \frac{\partial g_2}{\partial u_2} & \dots & \frac{\partial g_2}{\partial u_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial g_l}{\partial u_1} & \frac{\partial g_l}{\partial u_2} & \dots & \frac{\partial g_l}{\partial u_m} \end{bmatrix} \bigg|_{x_0, u_0} \delta u(t) \quad (2.49)$$

These equations represent the extracted linear model of the engine from a nonlinear simulation.

During the extraction process, we need to change (or perturb) one state or input variable at a time. After each state or input variable is perturbed, we calculate the differences in the functional values (f and g) as a result of this perturbed variable. Then we can calculate the partial derivative in the columns corresponding to this perturbed variable. For example, suppose that the j th state variable is perturbed while all other state and input variables are held at their nominal condition values. Then the following matrix elements can be calculated: $\partial f_i / \partial x_j = \Delta \dot{x}_i / \Delta x_j$, for $i = 1, 2, \dots, n$ and $\partial g_k / \partial x_j = \Delta y_k / \Delta x_j$, for $k = 1, 2, \dots, l$. To minimize the effect of asymmetry in the functional value around the nominal point, a two-step perturbation approach is suggested. In this approach, each state or input variable is perturbed backward first (i.e., $-\Delta$) and then perturbed forward with the same step size (i.e., $+\Delta$), before the total differences and the partial derivatives are calculated [6].

2.6.2 Identification of Models from Experimental/Operational Data

Another approach to deriving the linear model of an engine is through *system identification* (or system ID). System ID is a data-driven modeling approach by which input and output data of a (physical) system are collected from either a test/simulation or a real operation. It treats the system as a “black box” and requires limited knowledge of its internal working principles. It tries to find the best match (or fit) between the model, usually with an assumed model structure or format, and the input-output data set. The input-output data set is also called the observed data, although simulated data are sometimes used as a substitute for measured data.

We are usually interested in open-loop models of a physical system in the system ID process. The models derived from the process can be categorized into *parametric* and *nonparametric* models, depending on whether the method used in the identification process assumes a finite dimensional set of parameters in the search of the best fit between the data and the model. Examples of parametric models are regression models, autoregression moving average models, equation-error models, output-error models, and state-space models, etc. Examples of nonparametric models are impulse-response functions, step-response functions, frequency-response functions, empirical-transfer functions, etc. To obtain a good understanding of the system ID theory and its practices, the reader is referred to [25].

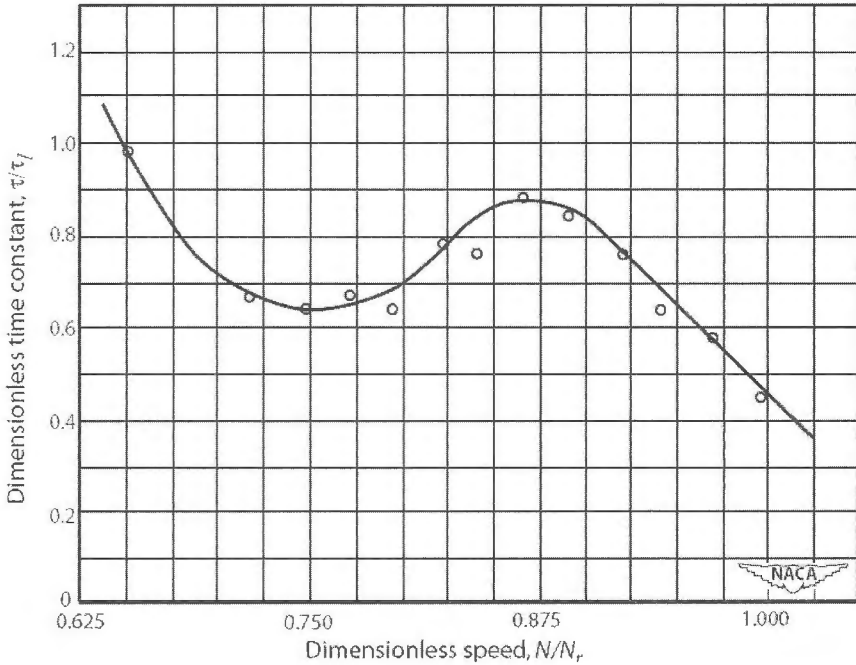


Fig. 2.21 Variation of engine time constant with shaft speed (from [27]; courtesy of NASA).

In the gas turbine industry, system ID concept was used in the early 1950s to derive engine models from observed data by using both step-response and frequency-response methods [26, 27]. Figure 2.21 shows the variation of engine time constant vs shaft speed, determined from the time-response analysis of engine transient data.

Development of system ID theories and methods continued in the 1970s [28]. The parameters of a multivariable gas turbine engine model were identified from stochastic input-output data by using a least-squares algorithm [29].

Most system ID methods assume a model structure and a model order before the parameters of this model are calculated by “fitting” the model to the data. If the structure or order is wrong to begin with, the resultant model might not be correct. To overcome this deficiency of a presumption of model order, a system ID method that yields a linear, time-invariant, multivariable, and stochastic model was developed [30]. The identified model is expressed in the discrete form as

$$x(k+1) = Fx(k) + Gu(k) + w(k) \quad (2.50)$$

$$y(k) = Hx(k) + Ju(k) + v(k) \quad (2.51)$$

where k stands for the discrete time step; w and v are discrete, Gaussian, white noises with zero mean and unknown covariances; and F , G , H , J are discrete

model matrices. The dimensions of these matrices are $n \times n$, $n \times m$, $k \times n$, and $k \times m$, respectively. First, the model order (n) needs to be determined based on a time-series analysis of the output data. Then the parameters to be identified are grouped in a vector and designated as θ . The parameter vector is defined as

$$\theta^T = [f_{ij} \quad g_{ij} \quad h_{ij} \quad j_{ij}]^T \quad (2.52)$$

where subscript ij represents a matrix element and f, g, h, j are the corresponding matrix elements of F, G, H, J .

This method was applied to the identification problem of a single-spool turbojet engine with constant nozzle area [31]. The resultant second-order model takes the form of Eqs. (2.50) and (2.51), and the numerical values of corresponding model matrices at the nominal condition of 100% shaft speed are given here:

$$F = \begin{bmatrix} 0 & 1 \\ -0.258 & 1.06 \end{bmatrix}, \quad G = \begin{bmatrix} 45,947 \\ 19,978 \end{bmatrix} \quad (2.53)$$

$$H^T = \begin{bmatrix} 1 & 0.0153 & 0.0025 & -0.02 & 0.00051 & -0.017 & 0.0023 & 0.026 \\ 0 & -0.00064 & -0.0002 & 0.00366 & -0.00006 & 0.0038 & -0.00023 & -0.0034 \end{bmatrix}^T$$

$$J = \begin{bmatrix} 687,485 \\ 291,423 \end{bmatrix} \quad (2.54)$$

where the output variables are $y^T = [N \ T_{13} \ P_3 \ T_{15} \ P_{15} \ T_{14} \ P_{14} \ F_N]^T$ and the input variable is fuel flow. The unit of shaft speed is in percentage, the unit of temperature is in degrees Rankine, the unit of pressure is in pounds per square inch absolute (psia), the unit of thrust is in pounds, and the unit of fuel flow is in pounds per second (pps).

Both time-domain and frequency-domain system ID methods have been applied comfortably to engine modeling by control engineers. For more system ID results for gas turbine engines, the reader is referred to [32]. While the optimal model structure and model order can be iterated through a well-designed modeling process, one thing that is often overlooked is the excitation function (or the time history of input data) that is used to produce the outputs from the engine. A good excitation function should cover both the positive and the negative sides of the nominal value of an input variable. Preferably, there should be at least two crossovers in the excitation function.

References

- [1] Graham, D., and McRuer, D., *Analysis of Nonlinear Control Systems*, Wiley, New York, 1961, pp. 44–45.
- [2] Mattingly, J. D., *Elements of Propulsion: Gas Turbines and Rockets*, AIAA, Reston, VA, 2006.

- [3] St. Peter, J., *The History of Gas Turbine Engine Development in the United States . . . A Tradition of Excellence*, American Society of Mechanical Engineers, IGTI, 1999, pp. 430–431.
- [4] Otto, E. W., and Taylor, B. L., III, “Dynamics of a Turbojet Engine Considered as a Quasi-Static System,” NACA TN-2091, May 1950.
- [5] Novik, D., “Some Linear Dynamics of Two-Spool Turbojet Engines,” NACA-TN-3274, June 1956.
- [6] Jaw, L. C., “Control of Helicopter Engines in Low Altitude Flight,” Ph. D. Dissertation, Stanford Univ., Stanford, CA, Aug. 1991, pp. 87–92.
- [7] Saravanamuttoo, H. I. H., and Fawke, A. J., “Simulation of Gas Turbine Dynamic Performance,” American Society of Mechanical Engineers, Paper 70-GT-23, May 1970.
- [8] Szuch, J. R., and Seldner, K., “Real-Time Simulation of F100-PW-100 Turbofan Engine Using the Hybrid Computer,” NASA-TM-X-3261, 1975.
- [9] Khalid, S. J., and Hearne, R. E., “Enhancing Dynamic Model Fidelity for Improved Prediction of Turbofan Engine Transient Performance,” AIAA Paper 80-1083, June 1980.
- [10] Kreith, F., *Principles of Heat Transfer*, 3rd ed., Intext Educational Publishers, 1976.
- [11] Ballin, M. G., “A High Fidelity Real-Time Simulation of a Small Turboshaft Engine,” NASA TM 100991, July 1988.
- [12] Mattingly, J. D., *Solutions Manual, Elements of Propulsion: Gas Turbines and Rockets*, AIAA Education Series, AIAA, Reston, VA, 2006.
- [13] Szuch, J. R., “Models for Jet Engine Systems, Part I—Techniques for Jet Engine Systems Modeling,” *Controls and Dynamic Systems—Advances in Theory and Applications*, edited by C. T. Leondes, Vol. 14, Academic Press, 1978, pp. 213–253.
- [14] Sobey, A. J., and Suggs, A. M., *Control of Aircraft and Missile Powerplants*, Wiley, New York, 1963.
- [15] Merritt, H. E., *Hydraulic Control Systems*, Wiley, New York, 1967.
- [16] Viersma, T., *Analysis, Synthesis, and Design of Hydraulic Servosystems and Pipelines*, Elsevier, New York, 1980.
- [17] Anderson, B., “Pneumatic Controls Analysis,” AiResearch Manufacturing Co. of Arizona, Rept. CA-5023-R, Ch. 1, Jan. 1963.
- [18] Adibhatla, S., and Gastineau, Z., “Tracking Filter Selection and Control Mode Selection for Model Based Control,” AIAA Paper 94-3204, June 1994.
- [19] Gastineau, Z., “Robust Internal Model Turbomachinery Control,” Ph. D. Dissertation, Mechanical Engineering Dept., Southern Methodist Univ., Dallas, TX, 1998.
- [20] Mink, G., and Behbahani, A., “The AFRL ICF Generic Gas Turbine Engine Model,” AIAA Paper 2005-4538, July 2005.
- [21] Evans, A. L., Follen, G., Naiman, C., and Lopez, I., “Numerical Propulsion System Simulation’s National Cycle Program,” AIAA Paper 98-3113, July 1998.
- [22] Mink, G., “Turbine Engine Dynamic Simulator—Final Report,” Air Force Research Lab., Rept. AFRL-PR-WP-TR-2005-2124, Wright-Patterson AFB, OH, June 2006.
- [23] Frederick, D. K., DeCastro, J. A., and Litt, J. S., “User’s Guide for the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS),” NASA/TM-2007-215026, Oct. 2007.
- [24] Parker, K. I., and Guo, T.-H., “Development of a Turbofan Engine Simulation in a Graphical Simulation Environment,” NASA TM 2003-212543, Aug. 2003.
- [25] Ljung, L., *System Identification: Theory for the User*, Prentice-Hall, Upper Saddle River, NJ, 1987.

- [26] Otto, E. W., and Taylor, B. L., III, "Dynamics of a Turbojet Engine Considered as a Quasi-Static System," NACA Rept. 1011, 1951.
- [27] Delio, G. J., "Evaluation of Three Methods for Determining Dynamic Characteristics of a Turbojet Engine," NACA Technical Note 2634, Feb. 1952.
- [28] Sage, A., and Melsa, J., *System Identification*, Academic International Press, New York, 1971.
- [29] Michael, G., and Farrar, F., "Identification of Multivariable Gas Turbine Dynamics from Stochastic Input-Output Data," United Aircraft Research Lab., Rept. R941620-3.
- [30] Tse, E., and Weinert, H., "Structure Determination and Parameter Identification for Multivariable Stochastic Linear Systems," *Proceedings of the 1973 Joint American Control Conference*, IEEE Transactions on Automatic Control, Vol. 20, No. 5, Oct. 1975, pp. 603–613.
- [31] Merrill, W. C., "An Application of Modern Control Theory to Jet Propulsion Systems," Ph. D. Dissertation, Unive. of Toledo, OH, June 1975.
- [32] Kulikov, G. G., and Thompson, H. A., *Dynamic Modeling of Gas Turbines – Identification, Simulation, Condition Monitoring and Optimal Control*, Springer–Verlag, London, 2004 Part II, pp. 65–155.

Model Reduction and Dynamic Analysis

A physical system is considered *dynamic* if at least one of its state variables changes with time. The rate of change of the state variable is directly related to how fast it responds to an external input. The faster the response is, the shorter it takes to change from one state (or operating condition) to another. A generalized measure of system response is *time constant*. Time constant is the characteristic time of the system. For a first-order system, this characteristic time is equal to the time it takes for the system to reach 63.2% of the desired change in a step input command [1]; so the shorter the time constant is, the faster the system responds to an external input or excitation, and the more dynamic the system is.

The time constant and the pole [in radians per second (r/s)] are reciprocal of each other, that is, $\tau = 1/a$, where a is the *corner* frequency; it is also the pole of the characteristic equation $s + a = 0$. In linear algebra terminology, a is the eigenvalue of the first-order system. For a second- or higher-order system, the time constant is not as clearly defined because there are more than one time constants. In fact, the number of time constants is the same as the number of poles (or eigenvalues) of the system. Hence, the response of this higher-order system is the linear combination of all the individual first-order subsystems.

When we talk about the dynamics of a system, we are always interested in how they compare against the dynamic capability of the control system. The dynamic capability of a control system is often expressed as the control bandwidth in the frequency domain. The bandwidth is the frequency range (or spectrum), within which a desired closed-loop response can be achieved. By comparison, the system dynamics that fall inside the control bandwidth are called *dominant* dynamics. On the other hand, the dynamics that fall outside the control bandwidth are often treated as disturbances or noise and need to be filtered out or suppressed.

For a two-spool gas turbine engine, the order of the system can easily exceed five. These time constants are the characteristic times associated with the dynamics of the three physical effects described in Chapter 2: 1) the inertias of rotating shafts; 2) mass storages in various volumes in the air and gas paths; and 3) heat storages in rotating blades, disks, and casing (see Chapter 2). The characteristic time for the shaft dynamics is typically 0.1 s, and the characteristic time for pressure dynamics is typically 0.01 s.

3.1 Example of Engine Dynamics

In Sec. 2.3, we explained that the total number of state variables for a two-spool afterburning turbofan engine is approximately 20. This number increases if the distributed nature of some engine modules is considered, and it decreases if the number of module or volume is reduced. Now let us focus on the lumped-parameter model of an engine and use a real-world example to illustrate the need for dynamic analysis. A 16th-order linear model of the Pratt and Whitney F100 engine is our example. The open-loop poles of the engine model are listed in Table 3.1 [2] for the “dry” operating mode (i.e., without afterburning) at sea-level, static, intermediate power.

From experience we know that the bandwidth of the primary fuel control loop for most traditional engine control systems is 0 to 2 Hz (or approximately 10 r/s). By *traditional* control, we mean the control systems on all modern and historical “production” turbine engines. On the other hand, *nontraditional* control systems correspond to the high-response, active stability, and tip clearance controls that are still in research or development stage. Nontraditional control systems have much higher bandwidth and will be discussed in Chapter 7.) The engine dynamic characteristics that fall within this 2 Hz bandwidth are usually called the dominant dynamics. For the F100 engine, the dominant dynamics are those five poles listed on the bottom of Table 3.1 (in boldfaced numbers). The exact locations of these poles will vary based on a specific engine design because the size and the configuration of each engine module determines the rate of change of pressure or temperature within each module or volume. The number of dominant dynamics (or state variables) for all types of turbine engines typically ranges from three to five.

There are at least four actuators to control the gas-path performance of an afterburning turbofan engine: main burner fuel metering valve, afterburner fuel metering valve, inlet guide vane actuator, and nozzle actuator. Furthermore, all aircraft engines have “custom bleed” valves to extract airflows from compressors for cabin pressurization and air-conditioning purposes. Some engines even have bleed

Table 3.1 Open-loop pole locations of F100 engine at sea-level, static, intermediate power (from [2])

Pole location, r/s	State variable	Remark
-501	P_{t45}	—
-181	P_{t3}	—
-56	T_{t45}	—
$-47 \pm 4.8 i$	T_{t4}, T_{t7}	—
-42	T_{t4Hi}	Fast response T_{t4}
-21	T_{t25}	Fan hub exit
-22	T_{t13}	Fan tip exit
$-20 \pm 4.2 i$	T_{t3}, T_{t5}	—
-16	T_{t16}	Fan duct mixer
$-4 \pm 3 i$	N_1, P_{t7}	—
-2.7	N_2	—
-1.9	T_{t45Lo}	Slow response T_{t45}
-0.67	T_{t4Lo}	Slow response T_{t4}

Table 3.2 Actuator dynamics—pole locations of F100 engine control actuators

Pole location, r/s	State variable	Definition
-50	X_{CDP}	Valve position—comp. discharge pressure
-50	X_{FIGV}	Actuator position—fan inlet guide vane
-50	X_{CVS}	Actuator position—comp. variable stator
-20	X_{A8}	Actuator position—nozzle
-10	X_{WFM}	Valve position—main burner fuel metering

valves to avoid compressor surge and stall. More sophisticated engines have two or more stages of controllable guide vanes or stators. For the F100 engine, the pole locations of the actuator transfer functions are listed in Table 3.2. Note that the afterburner fuel metering valve is not included because the numerical example is for nonafterburning mode of operation as is the case for Table 3.1. Also note that these engine control actuators have been approximated by a first-order system, that is, the actuators act like an integrator in the open-loop sense. These first-order poles are smaller in absolute values compared to the natural frequencies of the actuators if they had been modeled as a second-order system (see Chapter 2 for more details).

The fuel flow metering valve controls the engine's primary loop, which is to regulate the engine's power (or thrust). The remaining actuators listed in Table 3.2 are for secondary control loops, such as the guide vane or stator position control and the bleed flow control. The bandwidth of these secondary control loops is from 0 to 10 Hz (or approximately 60 r/s). The control bandwidth for most traditional engine control systems, including the primary and the secondary control loops, rarely exceeds 10 Hz.

One thing worth noting is that we shall not confuse the control bandwidth with the sampling frequency of a digital control system. The bandwidth of a control loop is the frequency spectrum of the closed-loop system within which the system's response to an external input, in continuous time, is stable with adequate stability margins. On the other hand, the sampling frequency, or the control update rate, is the number of repeated calculations in a second that the control command is updated. The sampling frequency for a digital engine control system typically ranges from 30 to 100 Hz (cycles per second), although for some limit protection control loops, such as the overspeed protection control loop, the sampling frequency can be as high as 200 Hz, that is, the control command is updated in every 0.005 s. The reason that the digital control update rate is so much higher than the actual control bandwidth is to reduce the quantization effect, which is the primary nonlinearity that erodes the stability of the closed-loop system when it is transformed from a continuous-time control system to a discrete-time control system [3].

3.2 Frequency Spectrums of Interest

The key to control system design is to know the right *frequency spectrum of interest* for the closed-loop system, and the key to selecting the right frequency spectrum of interest for design is to know the dynamic characteristics of the system.

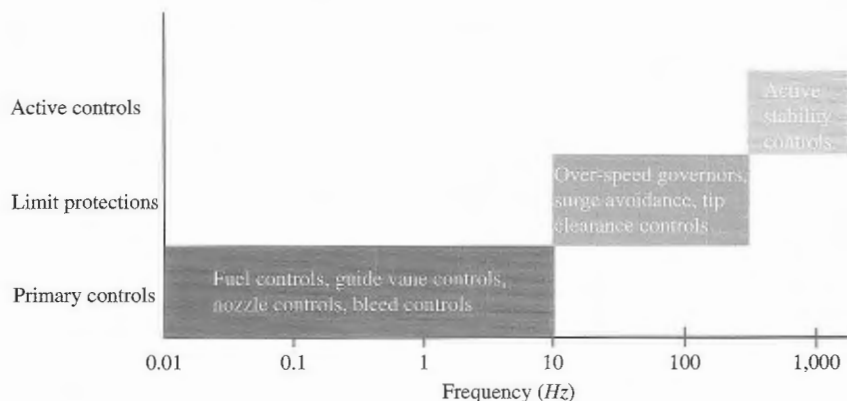


Fig. 3.1 Frequency spectrums of interest for turbine engine control designs.

The frequency spectrum of interest is the frequency range, within which the closed-loop control system is expected to perform with satisfactory performance, even in the presence of disturbances and uncertainties. This frequency spectrum is a function of the dynamics of the system, the actuator(s) selected to control the system, and the control law(s) designed to give satisfactory performance, both in steady state and transient. As indicated in Tables 3.1 and 3.2, the frequency spectrum of the F100 engine ranges from 0.5 to 500 r/s (or from 0.08 to 80 Hz). This is a large range from a control design standpoint because the maximum frequency of the spectrum is 1000 times, or three orders of magnitude, higher than the minimum frequency. The large range poses a challenge for control design because different sensing and actuation technologies are most likely needed to control various dynamic characteristics at different frequency orders of magnitude. To overcome this problem, we must first determine the primary frequency band within which the dynamics are most important to us and are responsible for meeting the performance requirements of the closed-loop system. This primary frequency band is the frequency spectrum of interest for control system design, and the dynamics within the band are the dominant dynamics of the system to be controlled.

We can divide the dynamics of an engine into three frequency spectrums of interest: 1) the *primary control* bandwidth, which ranges from 0 to 10 Hz, 2) the *limit protection control* bandwidth, which ranges from 10 to 200 Hz, and 3) the *active control* bandwidth, which is above 200 Hz, and it can go well beyond 1000 Hz. Figure 3.1 shows the three frequency spectrums of interest and their associated control objectives for gas turbine engines.

We will analyze the dominant dynamics in the primary control bandwidth in the remainder of this chapter. The analysis of the dominant dynamics in the active control bandwidth will be treated in Chapter 7.

3.3 Dominant Dynamics in the Primary Control Bandwidth

The equation of motion (EOM) is the starting point for all dynamic analysis. The EOMs for the shaft dynamics, the pressure dynamics, and the temperature

dynamics can be analytically derived as discussed in Chapter 2, if the physical dimensions and the thermodynamic properties of the air (or gas) within each engine module are known. A much more efficient way of deriving the EOMs is to extract the linear model at a specified, nominal operating point from a numerical simulation of the engine, and this procedure was described in Chapter 2.

Once a linear model is derived, we can analyze the dynamics of the system and derive a reduced-order model of the engine. The purpose of the reduced-order model is to include only the dynamic elements that fall in the primary control bandwidth. The reduced-order model is more suited for control law designs because, without such simplification, the engine model can result in overly complex (and higher-order) control laws that are sensitive to parameter variations and modeling uncertainties. We will show how to apply two well-known model reduction methods to arrive at a reduced-order model for an engine.

3.3.1 Model Reduction by Spectral Decomposition

We recall from Chapter 2 that the dynamics of an engine can be represented by a linear, time-invariant model near a nominal operating point within a finite time interval. This model, in the state equation form, can be transformed into a diagonal form, if the dynamic matrix A in Eq. (2.28) has no repeated eigenvalues (as it is the case for the F100 engine) [4]. The diagonal form of the model is expressed by

$$\dot{\tilde{x}} = \Lambda \tilde{x} + \tilde{B}u \quad (3.1)$$

$$y = \tilde{C}\tilde{x} + \tilde{D}u \quad (3.2)$$

where $\tilde{x}$ is the vector that contains *modal* state variables, and it is related to the original state variables by the canonical transformation matrix T , such that $x = T\tilde{x}$; and where $\Lambda = T^{-1}AT$ is a $n \times n$ diagonal matrix with the eigenvalues of A as the diagonal elements. We recall from Appendix A that the eigenvalues of A are the roots of the characteristic equation of the transfer function from input u to output y .

The model of Eqs. (3.1) and (3.2) can be partitioned into a set of r reduced states $\tilde{x}_1$ and $(n - r)$ remaining states $\tilde{x}_2$ as [2]

$$\begin{bmatrix} \dot{\tilde{x}}_1 \\ \dots \\ \dot{\tilde{x}}_2 \end{bmatrix} = \begin{bmatrix} \Lambda_1 & \vdots & 0 \\ \dots & \dots & \dots \\ 0 & \vdots & \Lambda_2 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \dots \\ \tilde{x}_2 \end{bmatrix} + \begin{bmatrix} \tilde{B}_1 \\ \dots \\ \tilde{B}_2 \end{bmatrix} u \quad (3.3)$$

$$y = \begin{bmatrix} \tilde{C}_1 & \vdots & \tilde{C}_2 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \dots \\ \tilde{x}_2 \end{bmatrix} + \tilde{D}u \quad (3.4)$$

where Λ_1 is an $r \times r$ diagonal matrix and Λ_2 is an $(n-r) \times (n-r)$ diagonal matrix. The corresponding partition of the transformation matrix is given by

$$\begin{bmatrix} x_1 \\ \dots \\ x_2 \end{bmatrix} = \begin{bmatrix} T_{11} & \vdots & T_{12} \\ \dots & \dots & \dots \\ T_{21} & \vdots & T_{22} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \dots \\ \tilde{x}_2 \end{bmatrix} \quad (3.5)$$

Now let us assume that the dynamics associated with the $(n-r)$ states $\tilde{x}_2$ are much faster than those of the r reduced states $\tilde{x}_1$; this means that the time constants of the states $\tilde{x}_2$ are much shorter. Hence, the corresponding dynamics will have settled in a steady state much sooner than the reduced states $\tilde{x}_1$. If this assumption is true, we have

$$\dot{\tilde{x}}_2 \approx 0 \quad (3.6)$$

then Eq. (3.3) becomes

$$\dot{\tilde{x}}_1 = \Lambda_1 \tilde{x}_1 + \tilde{B}_1 u \quad (3.7)$$

and the reduced-order model, with an order of r , for the original state variables, after substituting Eqs. (3.5) and (3.6) into Eqs. (3.3) and (3.4), becomes

$$\dot{x}_1 = A_r x_1 + B_r u \quad (3.8)$$

$$\begin{bmatrix} x_2 \\ y \end{bmatrix} = \begin{bmatrix} \tilde{C}_r \\ C_r \end{bmatrix} x_1 + \begin{bmatrix} \tilde{D}_r \\ D_r \end{bmatrix} u \quad (3.9)$$

where

$$A_r = T_{11} \Lambda_1 T_{11}^{-1} \quad (3.10)$$

$$B_r = T_{11} (\Lambda_1 T_{11}^{-1} T_{12} \Lambda_2^{-1} \tilde{B}_2 + \tilde{B}_1) \quad (3.11)$$

$$\tilde{C}_r = T_{21} T_{11}^{-1} \quad (3.12)$$

$$C_r = C_1 + C_2 T_{21} T_{11}^{-1} = C_1 + C_2 \tilde{C}_r \quad (3.13)$$

$$\tilde{D}_r = (T_{21} T_{11}^{-1} T_{12} - T_{22}) \Lambda_2^{-1} \tilde{B}_2 \quad (3.14)$$

$$D_r = D + C_2 [(T_{21} T_{11}^{-1} T_{12} - T_{22}) \Lambda_2^{-1} \tilde{B}] = D + C_2 \tilde{D}_r \quad (3.15)$$

and we have used the following partitioned output equation [based on Eq. (2.29)] in the derivation of Eqs. (3.13) and (3.15):

$$y = [C_1 \quad \vdots \quad C_2] \begin{bmatrix} x_1 \\ \vdots \\ x_2 \end{bmatrix} + Du \quad (3.16)$$

The solution of Eq. (3.8) is expressed in terms of state transition matrix, and the reader can refer to Appendix A, Sec. A.4.1 for more details. Note that as long as Eq. (3.6) holds, the reduced-order model, expressed by Eqs. (3.8) and (3.9), will have the same dynamic characteristics in the primary control bandwidth as the original full-order model; therefore, spectral decomposition essentially acts like a frequency separator, which treats high-frequency dynamics independent of low-frequency dynamics. The high-frequency dynamics are relative to the frequency spectrum of interest, and in this case, it is the primary control bandwidth.

A numerical example of the three-state linear model, which was reduced from a 17th-order linear model extracted from a nonlinear simulation of an F100-like two-spool, afterburning turbofan engine at sea-level, idle power, is presented here [5]:

$$\dot{x}(t) \approx \begin{bmatrix} -1.56 & 1.61 & -1.94 \\ 0.08 & -1.05 & -0.32 \\ 2.02 & 0.21 & -14.7 \end{bmatrix} \begin{bmatrix} \Delta N_1 \\ \Delta N_2 \\ \Delta P_{t7} \end{bmatrix} + \begin{bmatrix} 0 & 0 & -0.33 & 1.95 \\ 0 & 0 & 0.05 & 1.57 \\ -0.94 & 0 & -0.01 & 1.21 \end{bmatrix} \begin{bmatrix} \Delta A_8 \\ \Delta IGV \\ \Delta RCVV \\ \Delta W_{f_M} \end{bmatrix} \quad (3.17)$$

$$\begin{bmatrix} \Delta F_N \\ \Delta T_{t4} \\ \Delta W_{a2} \\ \Delta SM_F \\ \Delta SM_C \end{bmatrix} \approx \begin{bmatrix} 0 & 0 & 2.21 \\ -0.08 & -0.21 & -0.11 \\ 0.8 & -0.05 & -0.25 \\ -0.01 & 0.32 & -1.56 \\ 0.08 & 1.08 & 0.02 \end{bmatrix} \begin{bmatrix} \Delta N_1 \\ \Delta N_2 \\ \Delta P_{t7} \end{bmatrix} + \begin{bmatrix} 0.07 & 0 & 0 & 0 \\ 0 & 0 & 0.12 & 2.31 \\ 0 & 0 & -0.01 & 0 \\ 0 & 0.07 & -0.08 & -0.02 \\ 0 & 0 & 0 & -1.21 \end{bmatrix} \begin{bmatrix} \Delta A_8 \\ \Delta IGV \\ \Delta RCVV \\ \Delta W_{f_M} \end{bmatrix} \quad (3.18)$$

where the state variables are the LP shaft speed N_1 in the unit of revolutions per minute (rpm), the HP shaft speed N_2 in the unit of revolutions per minute, and the augmentor exit pressure or the exhaust nozzle pressure P_{t7} in the unit of pounds per square inch; the input (or control) variables are the exhaust nozzle area A_8 in the unit of square feet, the fan inlet guide vane position (IGV) in the unit of degrees, the compressor stator vane position (RCVV) in the unit of degrees, and the main burner fuel flow rate W_{f_M} in the unit of pounds per hour; and the output variables are the net thrust F_N in the unit of pounds, the HP turbine inlet gas temperature T_{t4} in the unit of degrees Rankine, the airflow rate through the fan w_{a2} in the unit of pounds per second, the surge margin of the fan SM_F , and the surge

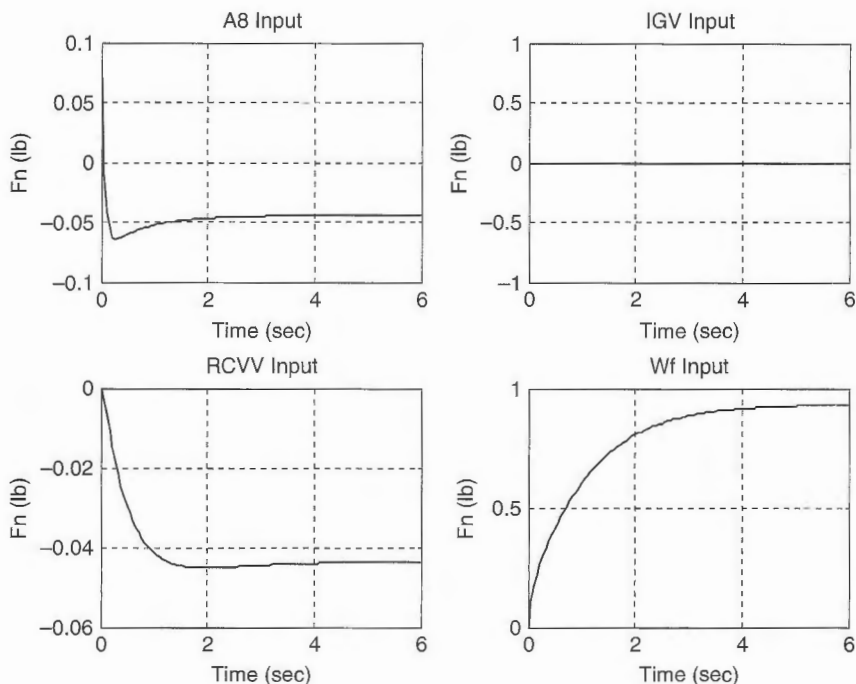


Fig. 3.2 Step responses of the net thrust to the four input variables.

margin of the compressor SM_C . The poles of this third-order model are -0.99 , -1.92 , and -14.4 .

The step responses of the net thrust to each of the four input variables are shown in Fig. 3.2. The step responses of the turbine temperature to each of the four input variables are shown in Fig. 3.3. The step responses of the compressor surge margin to each of the four input variables are shown in Fig. 3.4.

3.3.2 Model Reduction by Singular Value Decomposition (Balanced Realization)

Another useful model reduction technique is the (internally) *balanced realization* [6]. It is called a balanced realization because the reduced model has the same properties as the original model in terms of controllability and observability, or, in plainer language, the state variables are as controllable as they are observable. One of the greatest advantages of this technique is its ability to reject noise, that is, the reduced model is more robust in the presence of noise.

In a balanced realization, the reduced model is obtained by truncating the less important parts of the original model. This truncation is based on the relative values of a set of *factors* in some transformed coordinates. These factors are

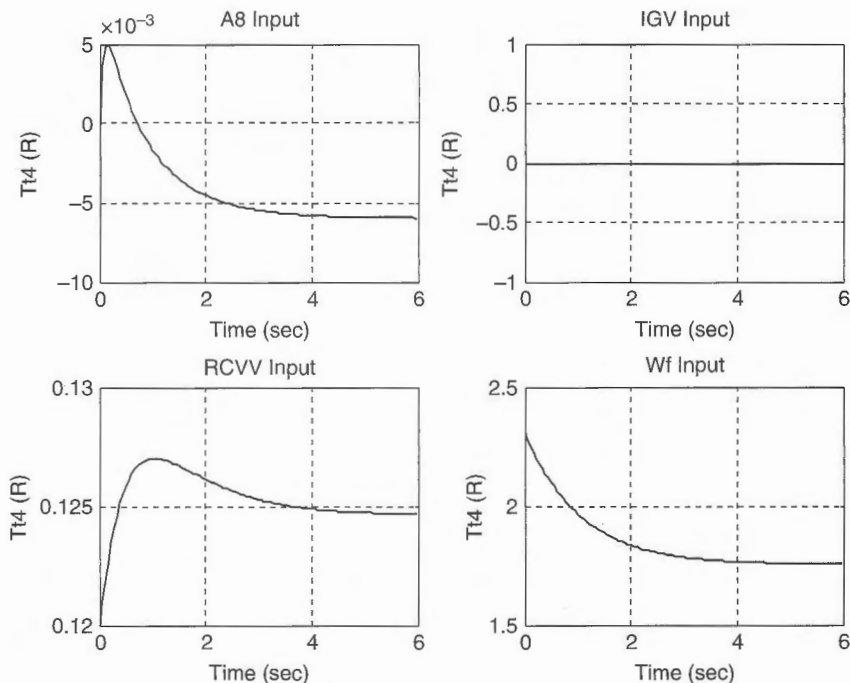


Fig. 3.3 Step responses of the turbine temperature to the four input variables.

similar to the eigenvalues (or poles) of the dynamic matrix A in a state equation, but they are called *singular values* and are expressed as σ . Now suppose the reduced-order model is also defined by Eq. (3.8); this model with order r is a good approximation of the original, full-order model, if the r th singular value σ_r is quite large compared to the $(r+1)$ th singular value σ_{r+1} .

A modification of the balanced realization technique is to apply a frequency-dependent weighting to selected input-output relationships (or TFs) during the model reduction process [7]. A simplification of this weighting concept is to apply only scalar weights [8]. An advantage of the scalar weighting is that it does not require special computation routine; therefore, standard balanced realization tool can be used to find the reduced model.

Suppose $G(s)$ is the TF matrix defining the input-output relationships of a multi-input/multi-output (MIMO) linear system, that is, $Y_{l \times 1}(s) = G_{l \times m}(s)U_{m \times 1}(s)$; the TF matrix can be realized into a state equation form and reduced to a lower order by using the balanced realization method. However, before the balanced realization is performed, we can multiply each TF in the matrix by a weighing factor to emphasize the specific TF as a way to influence the result, and these weighing factors can be either frequency-dependent or scalar values. Although not all of the TFs in the matrix can be weighted freely, a few selectively weighted TFs can effectively lower the model reduction error. The resultant reduced-order

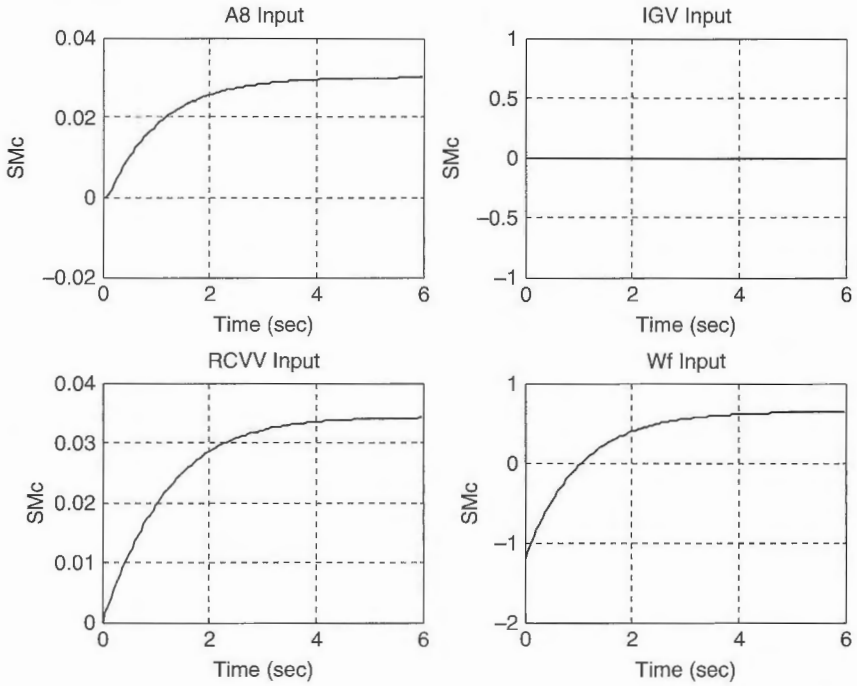


Fig. 3.4 Step responses of the compressor surge margin to the four input variables.

model is a *weighted* balanced realization. Let us consider a 3×3 TF matrix, where each entry represents an input-output relationship in the frequency domain. The weighted TF matrix is given by

$$G(s) \approx \begin{bmatrix} w_{11}G_{11}(s) & w_{12}G_{12}(s) & w_{13}G_{13}(s) \\ w_{21}G_{21}(s) & w_{22}G_{22}(s) & w_{23}G_{23}(s) \\ w_{31}G_{31}(s) & w_{32}G_{32}(s) & w_{33}G_{33}(s) \end{bmatrix} \quad (3.19)$$

Applying this technique, a weighted balanced realization of a turboshaft engine model was derived. This third-order model was reduced from a ninth-order model. The “original” ninth-order model was identified from a set of nonlinear simulation data by using the output-error model format.

The state and the output equations of this reduced, third-order model are shown here:

$$\dot{x}(t) \approx \begin{bmatrix} -2.78 & 0.62 & 6.60 \\ 0 & -2.05 & -2.49 \\ 5.41 & -4.17 & -44.91 \end{bmatrix} x(t) + \begin{bmatrix} 102 & 1.9 \\ -40 & 41.5 \\ -298 & -3.7 \end{bmatrix} \begin{bmatrix} \Delta W_f \\ \Delta P_t \end{bmatrix} \quad (3.20)$$

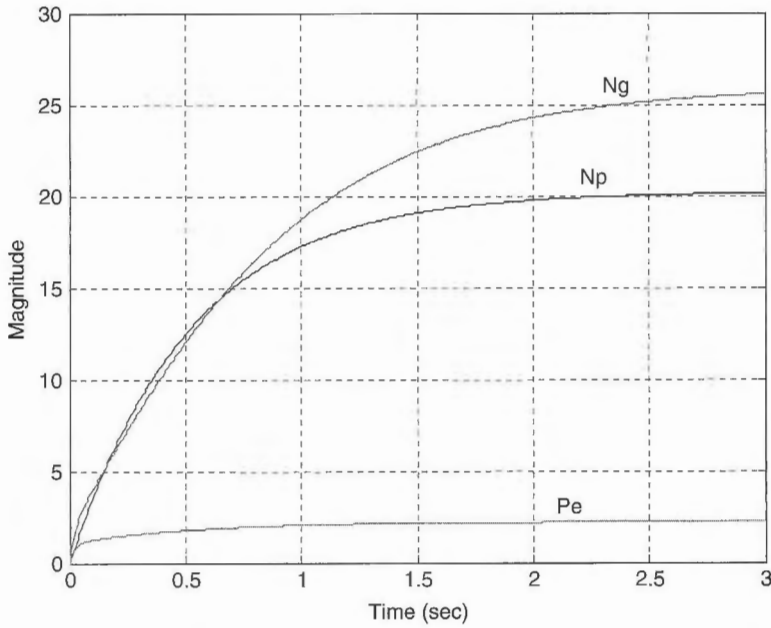


Fig. 3.5 Responses of the three output variables to a unit step input in fuel flow.

$$\begin{bmatrix} \Delta N_g \\ \Delta N_p \\ \Delta P_e \end{bmatrix} \approx \begin{bmatrix} 0.688 & -0.014 & 0.118 \\ -0.458 & -2.2 & -0.202 \\ 0.058 & -0.017 & -0.149 \end{bmatrix} x(t) \quad (3.21)$$

where the output variables are the gas generator shaft speed N_g , the power turbine shaft speed N_p , and the engine shaft power delivered by the power turbine shaft P_e ; and the input (or control) variables are the fuel flow rate W_f and the load power transmitted from the helicopter rotor system to the power turbine through the drive train P_l . The units of the outputs are revolutions per minute, revolutions per minute, and horsepower, respectively; the units of the inputs are pounds per hour and horsepower, respectively. The poles of this third-order model are -45.98 , -1.99 , and -1.77 .

Figure 3.5 shows the responses of the three engine output variables N_g , N_p , and P_e to a unit step input in the fuel flow.

3.4 Integrated Engine and Vehicle Dynamics

The dynamics of a turbine engine can be approximated by a low-order (typically less than five), linear model around a nominal operating point. This model should be integrated with the vehicle model to study the dynamic characteristics

of the combined vehicle-engine system. Using a conventional air vehicle as an example, the aircraft has six degrees of freedom in motion, which include three translational motions along the aircraft's body axes x , y , and z , and three rotational motions about each of the body axes ϕ , θ , and ψ [9]. In steady rectilinear flight, that is, all of the linear and angular velocities in the six degrees of freedom are zero, then a linear model of the aircraft near a nominal flight condition can be expressed as [10]

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \bar{X}_u & \bar{X}_w & -u_0 \sin \theta_0 & -g \cos \theta_0 \\ \bar{Z}_u & \bar{Z}_w & u_0 \cos \theta_0 & -g \sin \theta_0 \\ \bar{M}_u & \bar{M}_w & \bar{M}_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \end{bmatrix} + \begin{bmatrix} \bar{X}_{\delta_e} & T_{\delta_e} \cos \varepsilon \\ \bar{Z}_{\delta_e} & -T_{\delta_e} \sin \varepsilon \\ \bar{M}_{\delta_e} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_e \\ \delta_t \end{bmatrix} \quad (3.22)$$

$$\begin{bmatrix} \dot{v} \\ \dot{r} \\ \dot{p} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \bar{Y}_v & -u_0 \cos \theta_0 & u_0 \sin \theta_0 & g \cos \theta_0 \\ \bar{N}_v & \bar{N}_r & \bar{N}_p & 0 \\ \bar{L}_v & \bar{L}_r & \bar{L}_p & 0 \\ 0 & \tan \theta_0 & 1 & 0 \end{bmatrix} \begin{bmatrix} v \\ r \\ p \\ \phi \end{bmatrix} + \begin{bmatrix} \bar{Y}_{\delta_a} & \bar{Y}_{\delta_r} \\ \bar{N}_{\delta_a} & \bar{N}_{\delta_r} \\ \bar{L}_{\delta_a} & \bar{L}_{\delta_r} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \delta_a \\ \delta_r \end{bmatrix} \quad (3.23)$$

where Eq. (3.22) is called the *longitudinal* equation of motion and Eq. (3.23) is the *lateral* equation of motion, and these two equations are decoupled, that is, the longitudinal and the lateral motions can be controlled separately. The symbols in these equations follow the conventional definition of aircraft dynamics, succinctly: X , Y , Z are the forces in the body axes x , y , z , respectively; T is the propulsive (or engine) thrust; L , M , N are the moments about the three body axes; the subscripted forces and moments (e.g., X_u and L_r) stand for the partial derivatives (or stability derivatives) of the forces and moments with respect to the subscript variable; the overbar stands for a derived variable (with other terms included) in the same axis as the original variable; θ is the pitch angle; ε is the thrust offset angle from the body axis x ; the subscript 0 stands for the steady rectilinear condition; and δ stands for a control input variable with the subscript e for elevator angle, t for (engine) throttle angle, a for aileron angle, and r for rudder angle.

In a conventional aircraft, that is, without any thrust vectoring, the engine exhaust thrust is the only input from the engine to the aircraft model. Suppose that the engine has two spools and its dynamics can be approximated by a second-order linear model as described in Eqs. (2.14) and (2.15), so we can express the thrust as

$$F_N = [c_1 \quad c_2] \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} + d_1 W_f \quad (3.24)$$

where the two shaft speeds are the state variables that satisfy Eq. (2.14). In an extremely simplified case, the engine thrust can be modeled as a first-order lag, that is, $F_N(s)/W_f(s) = k_e/(s + a_1)$, or in the state equation form, it is written as

$$\begin{bmatrix} \dot{N}_1 \\ \dot{N}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} W_f = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} + \begin{bmatrix} \bar{b}_1 \\ \bar{b}_2 \end{bmatrix} \delta_t \quad (3.25)$$

$$F_N = c_1 N_1 + c_2 N_2 + d_1 W_f = c_1 N_1 + c_2 N_2 + \bar{d}_1 \delta_t \quad (3.26)$$

where it has been assumed that the engine fuel flow is a linear function of the engine throttle angle δ_t near the nominal operating point, that is, the rectilinear flight condition.

Combining Eqs. (3.22), (3.25), and (3.26) and substituting the trust T_{δ_t} with F_N , we get the following integrated aircraft-engine model for the longitudinal motion:

$$\begin{bmatrix} \dot{u} \\ \dot{w} \\ \dot{q} \\ \dot{\theta} \\ \dot{N}_1 \\ \dot{N}_2 \end{bmatrix} = \begin{bmatrix} \bar{X}_u & \bar{X}_w & -u_0 \sin \theta_0 & -g \cos \theta_0 & c_1 \cos \varepsilon & c_2 \cos \varepsilon \\ \bar{Z}_u & \bar{Z}_w & u_0 \cos \theta_0 & -g \sin \theta_0 & -c_1 \sin \varepsilon & -c_2 \sin \varepsilon \\ \bar{M}_u & \bar{M}_w & \bar{M}_q & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{11} & a_{12} \\ 0 & 0 & 0 & 0 & a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} u \\ w \\ q \\ \theta \\ N_1 \\ N_2 \end{bmatrix} + \begin{bmatrix} X_{\delta_e} & \bar{d}_1 \cos \varepsilon \\ Z_{\delta_e} & -\bar{d}_1 \sin \varepsilon \\ \bar{M}_{\delta_e} & 0 \\ 0 & 0 \\ 0 & \bar{b}_1 \\ 0 & \bar{b}_2 \end{bmatrix} \begin{bmatrix} \delta_e \\ \delta_t \end{bmatrix} \quad (3.27)$$

Note that this model is applicable for a single engine airplane, or for a multi-engine airplane without any differential thrust produced between the left-hand side and the right-hand side of engines. For a multi-engine airplane with wing-mounted engines, the differential thrust between the engines on the two sides of the fuselage can be used to enhance the controllability of the aircraft. This scenario is of particular interest for the situation where the flight control system is severely damaged and the engines are used to modulate the lateral motion of the aircraft to attempt safe landing. In this situation, the differential thrust must be considered in the lateral equation of motion in (3.23). This concept of propulsion-controlled aircraft will be discussed further in Chapter 9.

The primary goal of a turbojet or turbofan engine is to produce thrust, but the primary goal of a turboshaft or turboprop engine is to produce torque to drive helicopter rotors, aircraft propellers, and electric power generators. Just as engine thrust is an input to the aircraft model, engine torque is an input to the helicopter model. However, the helicopter model should also include the dynamics of the rotor system in addition to the motion of the helicopter body. Rotor dynamics are a major source of vibrations on a helicopter. Furthermore, rotor dynamics directly affect the response of the engine, because the engine and the rotor are mechanically connected. Thus, the integrated rotor-engine system is of great interest to the engine control design, and it will be discussed next.

3.4.1 Dynamics of Turboshaft Engine

Most turboshaft and turboprop engines incorporate the *free turbine* configuration, where the gas-generator section of the engine is decoupled from the power turbine (or the LP spool) section. Figure 3.6 is a drawing of the GE T700 engine [11] showing the power turbine in the downstream of the flow path of the gas generator.

Technically speaking, a propeller mounted at the inlet of a turboprop engine acts like a fan or LP compressor to the gas generator; hence, the performance of the gas generator is affected by the power turbine-propeller dynamics. However, because the propeller speed governor keeps the speed nearly constant all the time in the prop governing mode, the power turbine-propeller dynamics are only loosely coupled to the gas-generator performance.

On a helicopter, the flow paths of the turboshaft engine and the rotor system are typically not aligned; hence, the power turbine is decoupled from the gas-generator section. Because of the large load consumed by helicopter (main and tail) rotors, the integrated rotor-engine dynamics are strongly influenced by the rotor. Such dynamics are manifested in torsional oscillations of the rotor-engine drive train. These torsional oscillations are in the frequency spectrum of the engine's primary fuel control loop; hence, they are also dominant dynamics from the perspective of engine control design. Drive train torsional dynamics are most prominent when a helicopter undergoes an agile maneuver, such as a nap-of-the-earth flight or a dive-and-wave-off flight [8].

The frequency spectrum of interest in fuel control system designs for turboshaft engines is illustrated in Fig. 3.7 [11], where most of the dynamic interactions between the engine and the rotor/drive train take place below the one-per-revolution (or 1/rev) frequency of the main rotor. These dynamics should be considered in the design of engine controls.

A numerical example of a linear state equation of the T700 turboshaft engine in hover is given by [11]

$$\dot{x}(t) \approx \begin{bmatrix} -4.474 & 0 & -692.8 & 1462 & -2942 \\ -0.024 & -0.565 & -41.28 & 33.02 & 232.1 \\ 0.752 & 0 & -423 & 407.1 & 0 \\ 0 & 0 & 4115 & -4526 & 0 \\ 1.063 & 0 & -269.5 & 810.1 & -3063 \end{bmatrix} x + \begin{bmatrix} 22.88 \\ 1.715 \\ 0 \\ 52.53 \\ 11.169 \end{bmatrix} W_f \quad (3.28)$$

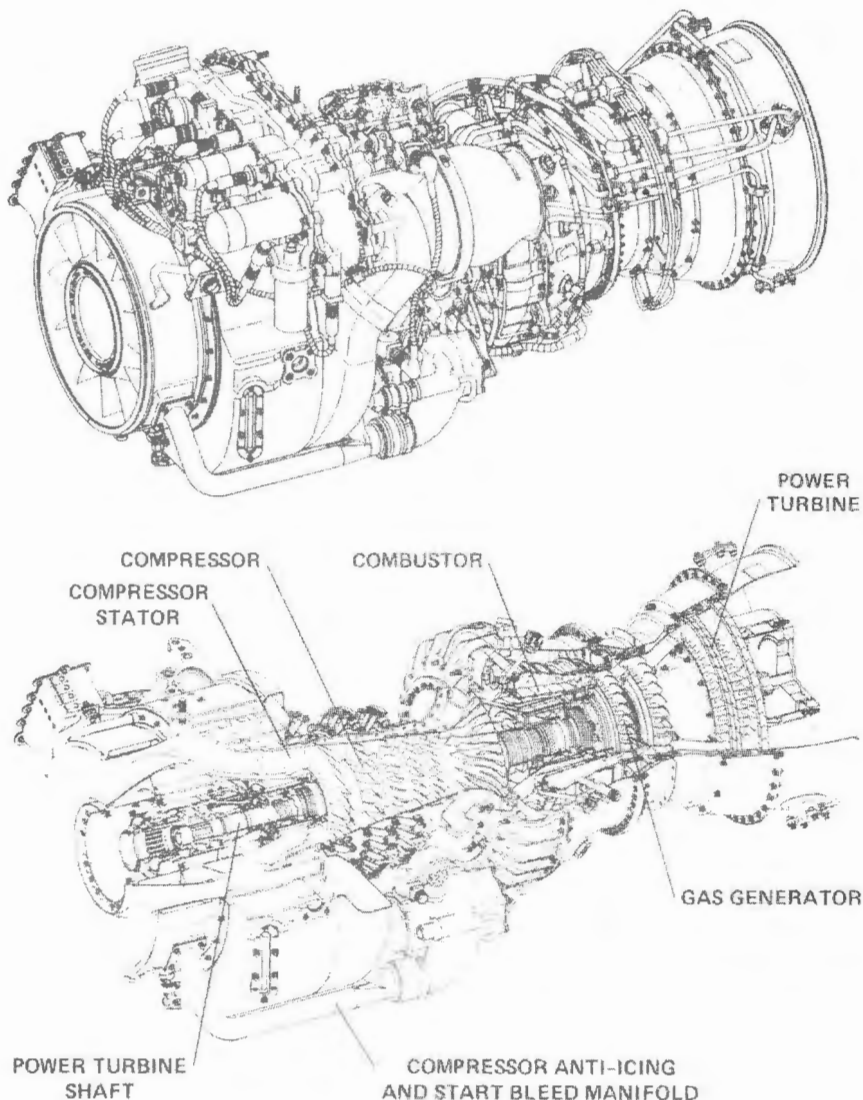


Fig. 3.6 GE T700 turboshaft engine with free power turbine stages (from [11]).

where the state variables are the gas-generator shaft speed N_1 or N_G , the power turbine shaft speed N_2 or N_P , the compressor discharge pressure P_3 , the gas-generator turbine inlet pressure P_{141} , and power turbine inlet pressure P_{145} , that is, $x^T = [N_G \ N_P \ P_3 \ P_{141} \ P_{145}]^T$. The units of the state variables are revolutions per minute for the first two and pounds per square inch absolute for the next three; and the unit of the input variable fuel flow rate is pounds per hour.

The poles of this fifth-order, open-loop model of Eq. (3.28) are -2.66 , -0.565 , -51.6 , -4900 , and -3060 for the five state variables, respectively. The step

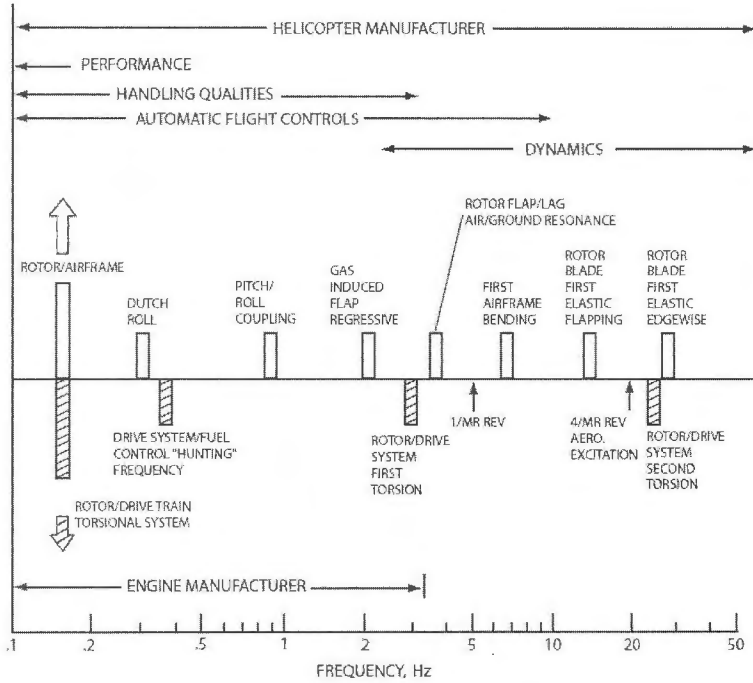


Fig. 3.7 Frequency spectrum of interest in helicopter engine control design (from [11]).

responses of four selected output variables, that is, N_g , N_p , P_3 , and P_{t45} , to the fuel flow input variable are shown in Fig. 3.8.

This five-state model was extracted from a nonlinear simulation of the engine by the small perturbation method. During the model extraction, it was assumed that the pressure and the mass flow in each volume were in equilibrium at all times, that is, in a quasi-steady state. The rationale behind this assumption is that the modal frequencies of these neglected pressure and temperature dynamics are so high compared to the dominant dynamics of the engine. Also note that the metal heating (or heat-sink) effect had not been included in nonlinear simulation when the linear model was extracted; hence, Eq. (3.28) does not include the slow temperature dynamics. However, a slow temperature dynamic can be added after the fact, that is, after the linear model is extracted. We can define a new state variable for the gas-generator turbine inlet temperature *with* the metal heating effect as T_{41L0} , so that the TF of this state variable with respect to the fast temperature dynamic at the turbine inlet can be expressed by [see Eq. (2.25)]

$$\frac{T_{41L0}(s)}{T_{41}(s)} = \frac{\tau_2 s + 1}{\tau_1 s + 1} \quad (3.29)$$

Because the fast temperature state T_{41} can be treated as an output variable, we can then combine the fast temperature output equation with Eqs. (3.28) and (3.29) to

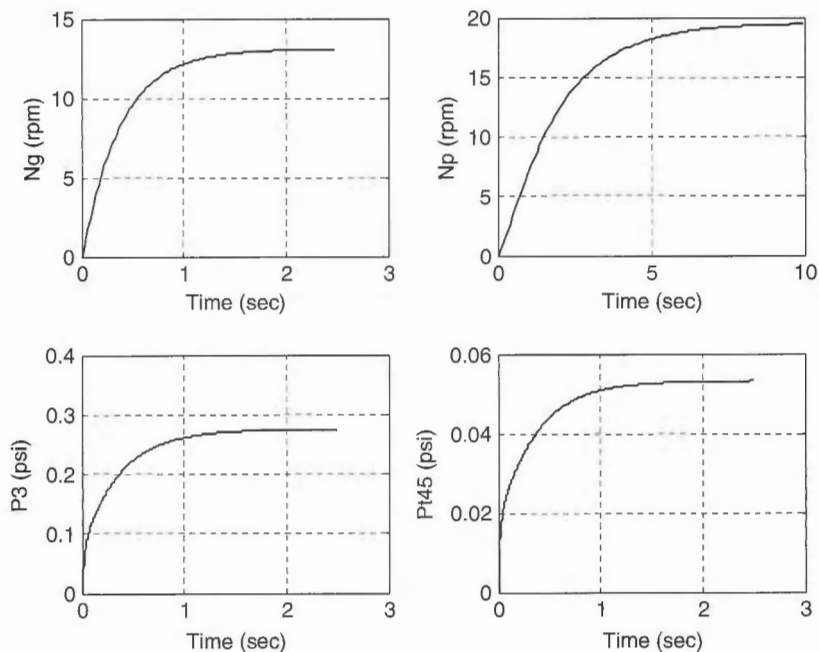


Fig. 3.8 Responses of a fifth-order model of T700 to a unit step input in fuel flow.

get an augmented linear model with the metal heating effect as the sixth state. An example of this six-state engine model for the nominal condition of level flight at 80 knots is given in Eqs. (3.30) and (3.31) [11]:

$$\dot{x}(t) \approx \begin{bmatrix} -3.659 & 0 & -118.5 & 612.7 & -1548 & 8.968 \\ 0.0128 & -0.446 & 0 & -3.099 & 182.5 & 0.5212 \\ 0.634 & 0 & -409 & 389.9 & 0 & 0 \\ 0 & 0 & 4909 & -5287 & 0 & 18.68 \\ 0.846 & 0 & -4.315 & 701 & -4038 & 4.501 \\ -22.79 & 0 & 188,900 & -201,700 & 0 & 663 \end{bmatrix} x + \begin{bmatrix} 15.069 \\ 0.876 \\ 0 \\ 35 \\ 7.561 \\ 1242.5 \end{bmatrix} W_f \quad (3.30)$$

$$y(t) \approx I_{6 \times 6} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1.68 \end{bmatrix} W_f \quad (3.31)$$

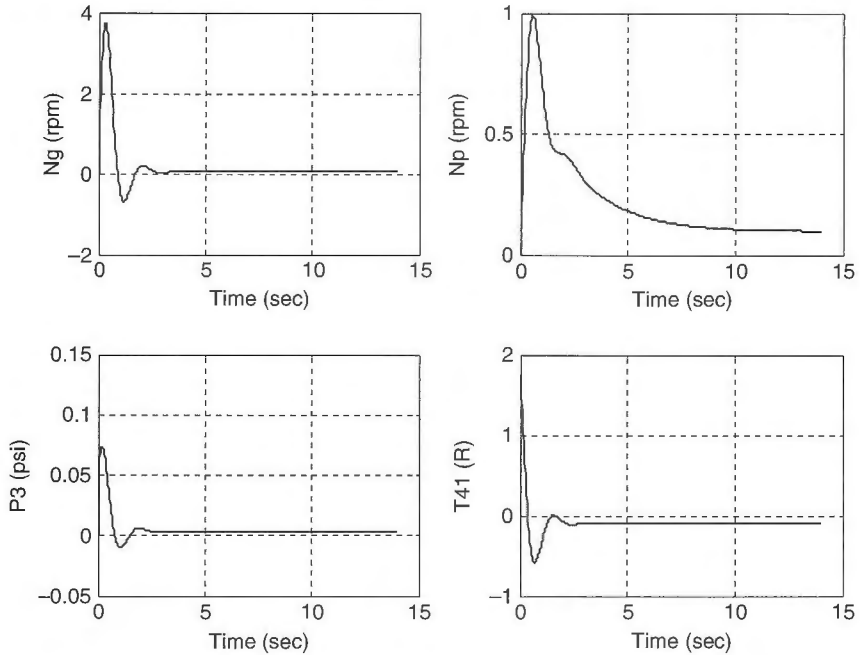


Fig. 3.9 Responses of a sixth-order model of T700 to a unit step input in fuel flow.

where the state variable vector is $\mathbf{x}^T = [N_G \ N_P \ P_3 \ P_{t41} \ P_{t45} \ T_{41Lo}]^T$ and $I_{6 \times 6}$ is an identity matrix with order six. The unit of the temperature state variable is degrees Rankine. The poles of this sixth-order model are -0.446 , -48.4 , -4985 , -4038 , -361.4 , and $-1.844 \pm 3.592i$. The step responses of four selected output variables, that is, N_g , N_p , P_3 , and T_{41Lo} , to the fuel flow input variable are shown in Fig. 3.9. Note that the responses are oscillatory at the beginning of the transient. The oscillations are apparent because the metal heating effect.

3.4.2 Engine and Airframe Compatibility Issues

Design compatibility between the engine and the rotor-driven train is not so easily achieved because different engineering disciplines are required for the designs of these two systems. Traditionally, airframe designers assume a constant rotor speed during the operation, while the engine designers assume a constant load from the rotor. In reality, the turbine shaft speed and the rotor speed vary all the time as a result of vehicle maneuvers and disturbances of the ambient air. Commonly encountered compatibility problems in rotorcraft include [12–15]: 1) the drive-train torsional stability problems characterized by unacceptable shaft torque oscillations with a high-gain fuel control system; 2) large rotor speed variations during maneuvers, such as a 3–5% speed droop after a sudden pull of the collective stick, and a 10% variation in a quick-turn or

wave-off maneuver [16,17]; and 3) excessive vibrations that cause hardware failure or breakage.

Of these problems, the torsional stability problem and the rotor speed variation problem are particularly relevant to engine control designs; therefore, we want to study them further.

To analyze the torsional stability of the engine-rotor-drive system, we can use a simplified, linear, dynamic model recommended by the Society of Automotive Engineers [18]. This mass-spring-damper model, as shown in Fig. 3.10, provides a basis for predicting torsional flexural modes of the system and is often used by engine control engineers to gain insight of potential compatibility problems. This model consists of three masses and two springs to represent the torsional flexibility of the drive train between the power turbine and the rotor. The torsional flexibility is so progressive that the power turbine can be "twisted" by two revolutions with respect to the rotor blades. In the figure, the parameters that affect the dynamics of the coupled system are as follows: rotor moment of inertia (denoted by I_r); rotor hub and gearbox moment of inertia (I_h); power turbine shaft moment of inertia (I_p); effective lag hinge spring stiffness (K_l), which is caused mainly by the centrifugal force on the rotor blade; b_l is the effective damping coefficient of the blade lead-lag damper (b_l); and drive shaft stiffness (k_d).

The dynamic model in Fig. 3.10 can be further simplified by assuming that all of the shafts are rigid [19], that is, the shaft between the power turbine and the gearbox as well as the shaft between the gearbox and the rotor hub are rigid by comparison with the stiffness of the rotor lead-lag damper.

Although this type of dynamic models has been used to analyze the torsional compatibility between the engine and the rotor, it has two drawbacks: 1) the damping ratio and the stiffness are crude approximations of the lead-lag dampers on rotor blades, and 2) the effects of engine-rotor coupling on high-frequency

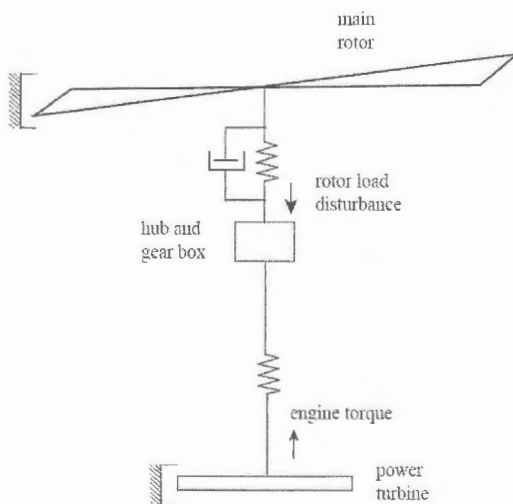


Fig. 3.10 Simplified linear dynamic model for analyzing the engine-rotor torsional compatibility problem.

progressive and regressive lead-lag modes is not considered. A more accurate model that considers the individual dynamics of rotor-blade elements will be discussed in the next subsection.

3.4.3 Helicopter Rotor and Turboshaft Engine

A simplified, linear rotor-engine model, derived by considering only the blade lead-lag motion (or degree of freedom) for a three-bladed rotor, is given by [8]

$$\begin{bmatrix} \dot{\xi}_0 \\ \ddot{\xi}_0 \\ \dot{\Omega}_r \\ \dot{N}_G \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -a_0\omega_l^2 & -a_0d_l & -\sqrt{3}\frac{b_0}{I_{eq}} & \sqrt{3}\frac{h_1}{I_{eq}} \\ -c_0\omega_l^2 & -c_0d_l & -\frac{b_0}{I_{eq}} & \frac{h_1}{I_{eq}} \\ 0 & 0 & 0 & -f_1 \end{bmatrix} \begin{bmatrix} \xi_0 \\ \dot{\xi}_0 \\ \Omega_r \\ N_G \end{bmatrix} + \begin{bmatrix} 0 \\ \sqrt{3}\frac{l_1}{I_{eq}} \\ \frac{l_1}{I_{eq}} \\ g_1 \end{bmatrix} W_f \quad (3.32)$$

where ξ_0 is the equilibrium (or collective) lead-lag angle, Ω_r is the rotor speed, N_G is the gas-generator speed of the engine, I_{eq} is the equivalent moment of inertia of the turbine-drive train, and

$$Q_e = [h_1 \quad h_2] \begin{bmatrix} N_G \\ N_P \end{bmatrix} + l_1 W_f \quad (3.33)$$

$$a_0 \equiv 1 + \frac{3I_{z_0}}{I_{eq}} \quad (3.34)$$

$$b_0 \equiv 3h_w + h_2 \quad (3.35)$$

$$c_0 \equiv \frac{\sqrt{3}I_{z_0}}{I_{eq}} \quad (3.36)$$

where h_w is the rotor speed damping, which is a function of the rotor speed and its geometry.

A general expression of the TF from the fuel flow to the rotor speed for this rotor-engine model is

$$\frac{\Omega_r(s)}{W_f(s)} = \frac{l_1/I_{eq} (s+z_e)(s+z_\Omega)(s+\bar{z}_\Omega)}{(s+f_1)(s+p_\Omega)(s+\bar{p}_\Omega)(s+p_\xi)} \quad (3.37)$$

where z_e is the engine zero, z_Ω is the rotor speed zero, $\bar{z}_\Omega$ is the conjugate of the rotor zero, f_1 is the engine gas-generator mode, p_Ω is the rotor speed mode, $\bar{p}_\Omega$ is the conjugate of the rotor speed mode, and p_ξ is the collective lead-lag mode.

A numerical example of Eq. (3.32) is given here:

$$\begin{bmatrix} \dot{\xi}_0 \\ \dot{\xi}_0 \\ \dot{\Omega}_r \\ \dot{N}_G \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -289.7 & -9.41 & 248.4 & -154.4 \\ 1.881 & -0.0565 & -2.091 & 1.224 \\ 0 & 0 & 0 & -1.94 \end{bmatrix} \begin{bmatrix} \xi_0 \\ \dot{\xi}_0 \\ \Omega_r \\ N_G \end{bmatrix} + \begin{bmatrix} 0 \\ -31.05 \\ 0.246 \\ 0.391 \end{bmatrix} W_f \quad (3.38)$$

The transfer function of Eq. (3.36) is

$$\frac{N_p(s)}{W_f(s)} = \frac{l_1/I_{eq} (s + 12.29)(s + 14.09 - 0.06i)(s + 14.09 + 0.06i)}{(s + 1.94)(s + -5.51 + 16.11i)(s + -5.51 - 16.11i)(s + 0.47)} \quad (3.39)$$

Note that we have substituted the power turbine speed for the rotor speed based on the earlier assumption of a rigid drive train. The step responses of three selected output variables, that is, the equilibrium lead-lag angle of the rotor blades, the gas generator speed N_g , and the power turbine speed N_p , to the fuel flow input variable are shown in Fig. 3.11. The Bode plot of the TF in Eq. (3.39) is shown in Fig. 3.12. As the damping ratio of the lead-lag damper is decreased, the notch-filter-like behavior will be increasingly apparent.

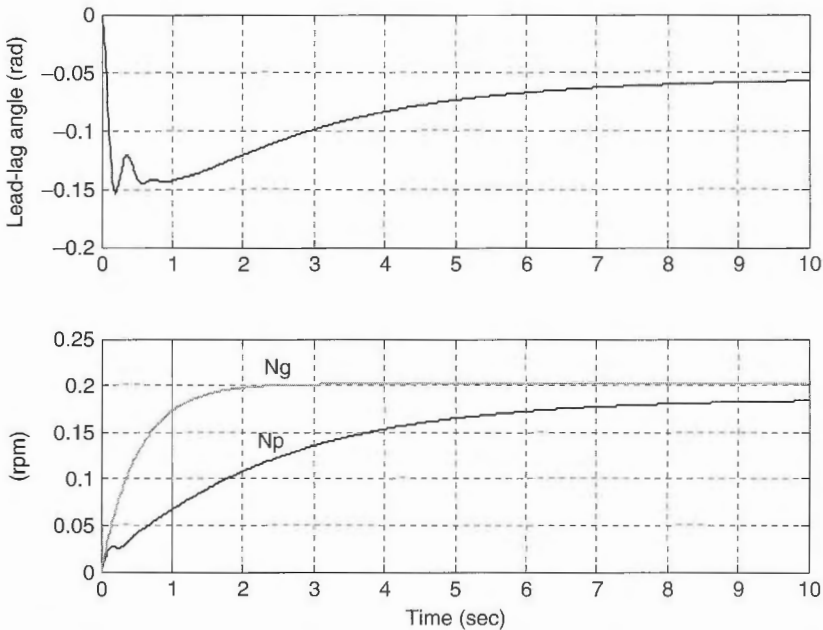


Fig. 3.11 Responses of an engine-rotor model to a unit step input in fuel flow.

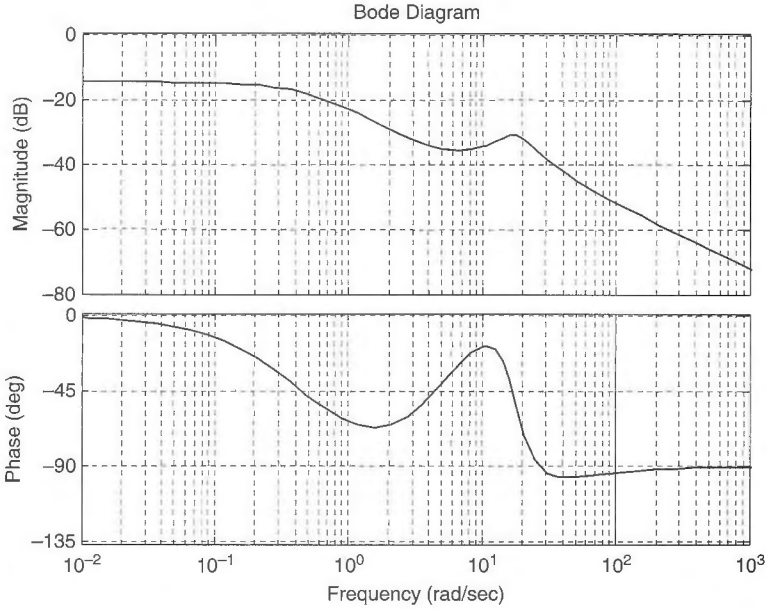


Fig. 3.12 Bode plot of the TF from engine fuel flow to power turbine speed.

3.5 Variations in Engine Dynamics

In Chapter 2, we derived the transfer functions for engine dynamics, and in the preceding section we explained how to analyze the open-loop poles to determine the dominant dynamics in the frequency spectrum of interest. In this subsection, we will explain how the dynamics of an engine, that is, the pole locations and the coefficients of the transfer functions, change with engine power.

Moving the engine throttle at different power settings is the most obvious way to see the changes in engine dynamic response. Depending on the nonlinearity of the engine, an engine at takeoff (TO) power responds three to five times as fast as idle power to a control command. This means that the poles of the open-loop transfer functions in Eqs. (2.9) and (2.16) of Chapter 2, for one-spool and two-spool engines, respectively, are several times larger at TO power than at idle power. This large variation in engine dynamics suggests that the coefficients of a control law also change from idle to TO power to ensure proper stability margins. We will discuss this with more details in Chapter 4.

Using the F100 engine as an example, a linear model valid near the intermediate power (which is equivalent to the TO power of a commercial engine) is given next [5]:

$$\dot{x}(t) \approx \begin{bmatrix} -2.02 & 3.75 & -1.81 \\ 0.05 & -3.5 & -0.25 \\ 2.76 & 0.35 & -4.53 \end{bmatrix} \begin{bmatrix} \Delta N_1 \\ \Delta N_2 \\ \Delta P_{t7} \end{bmatrix} + \begin{bmatrix} 0 & 0.35 & -0.81 & 1.53 \\ 0 & 0 & 0.42 & 0.94 \\ -4.01 & -0.49 & 0.01 & 1.09 \end{bmatrix} \begin{bmatrix} \Delta A_g \\ \Delta IG V \\ \Delta RC V V \\ \Delta W_{f_M} \end{bmatrix} \quad (3.40)$$

Table 3.3 Comparison of open-loop poles of the F100 turbofan engine

Pole (mode)	Idle power	Intermediate power
N_1	-0.99	$-2.99 + 1.92i$
N_2	-1.92	-4.08
P_{r6}	-14.4	$-2.99 - 1.92i$

$$\begin{bmatrix} \Delta F_N \\ \Delta T_{i4} \\ \Delta W_{a2} \\ \Delta SM_F \\ \Delta SM_C \end{bmatrix} \approx \begin{bmatrix} 0 & 0 & 1.48 \\ -0.06 & -0.12 & -0.11 \\ 0.31 & 0.01 & -0.04 \\ 0.29 & 0.17 & -0.5 \\ -0.08 & 1.17 & 0.02 \end{bmatrix} \begin{bmatrix} \Delta N_1 \\ \Delta N_2 \\ \Delta P_{i7} \end{bmatrix} + \begin{bmatrix} 0.74 & 0 & 0 & 0 \\ 0 & 0 & 0.09 & 0.57 \\ 0 & -0.14 & 0 & 0 \\ 0 & 0 & -0.04 & 0 \\ 0 & 0 & 0.09 & -0.19 \end{bmatrix} \begin{bmatrix} \Delta A_g \\ \Delta IGV \\ \Delta RCVV \\ \Delta W_{fM} \end{bmatrix}$$

(3.41)

The open-loop poles of this model derived at the intermediate power are compared to those at the idle power [in Eqs. (3.17) and (3.18)] in Table 3.3. The step responses of the engine thrust to a unit of fuel flow are compared between idle and the intermediate power in Fig. 3.13. Note that the settling time of the idle transient is approximately three times as long as that of the intermediate power transient.

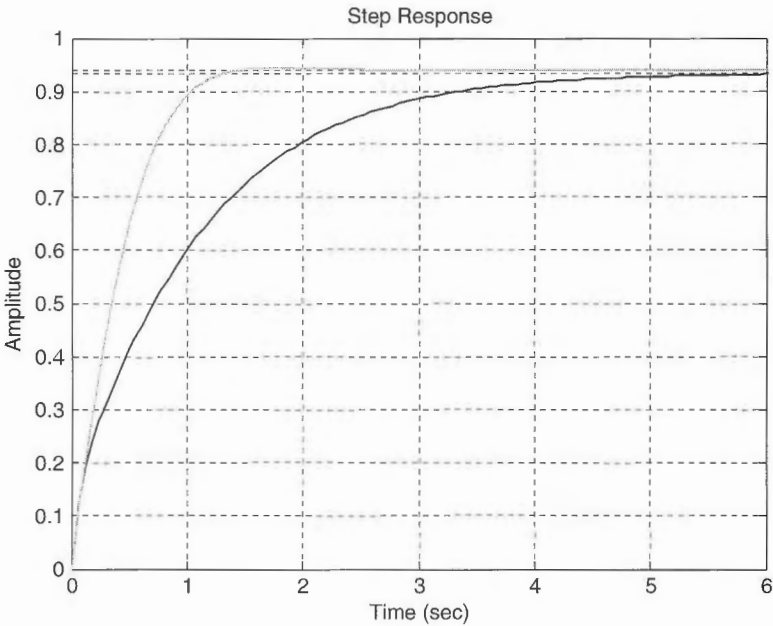


Fig. 3.13 Comparison of thrust responses of the F100 engine at idle (blue line) and intermediate power (green line). (See also color figure section at the back of the book.)

Table 3.4 Comparison of the dc gains of the LP and the HP spool responses of a turbofan engine (courtesy of [20])

Response TF	Low power	High power
N_1	0.195	0.056
N_2	0.262	0.075

Although the engine response, measured by the rate of change of the LP and the HP spool speeds increases with the increasing power level, the dc gain, that is, the TF gain when $s = 0$, typically decreases with the increasing power level. An example is listed in Table 3.4 [20]. The results were derived from test cell data by using system-identification techniques.

References

- [1] Dorf, R. C., and Bishop, R. H., *Modern Control Systems*, 10th ed., Pearson Prentice-Hall, Upper Saddle River, NJ, 2005, Ch. 2.
- [2] DeHoff, R. L., and Hall, W. E., Jr., "Jet Engine Systems Models, Part II—State Space Techniques and Modeling for Control," *Controls and Dynamic Systems—Advances in Theory and Applications*, edited by C. T. Leondes, Vol. 14, Academic Press, 1978, pp. 259–299.
- [3] Franklin, G. F., and Powell, J. D., *Digital Control of Dynamic Systems*, Addison Wesley Longman, Reading, MA, 1980, Chs. 1 and 10.
- [4] Kailath, T., *Linear Systems*, Prentice-Hall, Upper Saddle River, NJ, 1980, Ch. 2.
- [5] Weinberg, M. S., and Adams, G. R., "Low Order Linearized Models of Turbine Engines," Air Force Systems Command, Technical Rept. ASD-TR-74-24, Wright-Patterson AFB, OH, July 1975.
- [6] Moore, B. C., "Principal Component Analysis in Linear Systems: Controllability, Observability, and Model Reduction," *IEEE Transactions on Automatic Control*, Vol. AC-26, No. 1, IEEE, Feb. 1981, pp. 17–32.
- [7] Enns, D., "Model Reduction for Control System Design," Ph. D. Dissertation, Stanford Univ., Stanford, CA, 1984.
- [8] Jaw, L. C., "Control of Helicopter Engines in Low Altitude Flight," Ph. D. Dissertation, Stanford Univ., Stanford, CA, Aug. 1991, Appen. D.
- [9] McRuer, D., Ashkenas, I., and Graham, D., *Aircraft Dynamics and Automatic Control*, Princeton Univ. Press, Princeton, NJ, 1973, Ch. 4.
- [10] Bryson, A. E., Jr., *Control of Spacecraft and Aircraft*, Princeton Univ. Press, Princeton, NJ, 1994, Ch. 10.
- [11] Ballin, M. G., "A High Fidelity Real-time Simulation of a Small Turboshift Engine," NASA TM 100991, July 1988.
- [12] Hanson, H. W., et al., "Engine/Airframe/Drive Train Dynamic Interface Documentation—Bell Helicopter Investigation Report," USARTL-TR-78-15, 1978.
- [13] Richardson, D. A., and Alwang, J. R., "Engine/Airframe/Drive Train Dynamic Interface Documentation—Boeing Vertol Investigation Report," USARTL-TR-78-11, 1978.

- [14] Needham, J. F., and Banerjee, D., "Engine/Airframe/Drive Train Dynamic Interface Documentation—Hughes Helicopter Investigation Report," USARTL-TR-78-12, 1978.
- [15] Twomey, W. J., and Ham, E. H., "Engine/Airframe/Drive Train Dynamic Interface Documentation—United Technologies Sikorsky Aircraft Investigation Report," USARTL-TR-78-13, 1978.
- [16] Bolton, R. L., "Adaptive Fuel Control Feasibility Investigations," *American Helicopter Society 38th Annual Forum Proceedings*, Anaheim, CA, May 1982.
- [17] Howlett, J. J., Morrison, T., and Zagaranski, R. D., "Adaptive Fuel Control for Helicopter Applications," *Journal of American Helicopter Society*, Oct. 1984, pp. 43–54.
- [18] Society of Automotive Engineers, "Helicopter Engine-Rotor System Compatibility," Aerospace Recommended Practice (ARP) 704, June 1962.
- [19] Alwang, J. R., "Engine-Airframe Transient Compatibility: Analysis and Test Requirements," American Helicopter Society, Nov. 1979.
- [20] Kulikov, G. G., and Thompson, H. A., *Dynamic Modeling of Gas Turbines – Identification, Simulation, Condition Monitoring and Optimal Control*, Springer-Verlag, London, U.K., 2004.

Design of Set-Point Controllers

Chapter 1 describes the three basic functions of an automatic control system as set-point control, transient control, and limit protection. We will discuss the first of these functions in this chapter. The purpose of set-point control is to regulate the engine's performance near a desired operating condition for a relatively long period of time. Set-point control is also called steady-state control, and a steady-state operating condition is sometimes called a set point, and it corresponds to a fixed power level produced by the engine, and this fixed power is controlled by engine speed or pressure ratio. Commonly known set points for engines are idle, cruise, and takeoff (TO).

The controller designed for a steady-state operating condition is called a set-point controller, because it tries to control the engine from anywhere in the neighborhood of this operating condition back to the center or the "set-point." This concept is illustrated by the dotted circle around each of the three commonly known set points in Fig. 4.1. The figure shows three set points on the (steady-state) operating line of a typical compressor map: idle, cruise, and TO. You may recall from Chapter 3 that the TO power is the "military" power or the "intermediate" power (without afterburning) for a fighter engine, and it is the "maximum" power for a commercial engine. A set-point controller has been traditionally called a *governor*, and in modern control terminology it is called a *regulator*.

The operating line can shift up or down on the compressor map depending on the amount of bleed air and shaft power extracted from the engine. Consequently, a set-point controller should accommodate such shifts in the operating line. Even if the operating line stays unchanged on the compressor map, a set point can move to the left or right along the operating line for the same throttle position, because the idle and TO power set points change with the altitude and inlet temperature, and because the exact fan speed or pressure ratio command for each set point is proportional to the "difference" between the corresponding commands at idle and TO. The exact value of speed or pressure ratio command is calculated by the "power-setting logic," which will be covered in Chapter 6. Hence the goal of the set-point controller is two-fold: 1) to regulate engine performance in the neighborhood of a desired operating condition and 2) to determine what this operating condition is based on the pilot's (or ground operator's) command. This command is usually a throttle position or a power lever angle.

The design of set-point controllers, both in the time-domain and frequency-domain, will be discussed in this chapter. Set-point controller design for one-spool

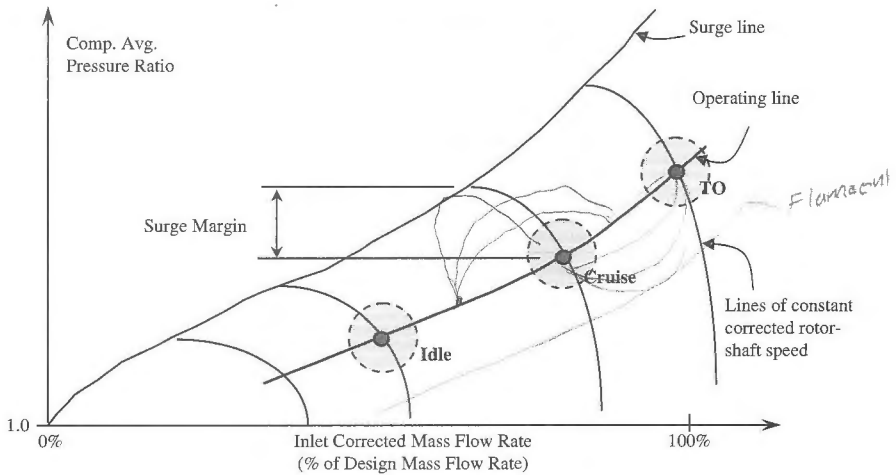


Fig. 4.1 Operating line on a compressor map.

engines is covered in Sec. 4.1. Controller design for two-spool engines is covered in Sec. 4.2. The design techniques selected in this chapter are classical control design methodologies; hence, they apply to single-input single-output (SISO) systems primarily. Multi-input multi-output (MIMO) design methodologies will be discussed in Chapter 7. In Sec. 4.3, controller designs for turboshaft engines are discussed. Lastly, in Sec. 4.4, some practical considerations in engine set-point controls are discussed.

4.1 Controller Design for One-Spool Engines

Although the one-spool engine represents the simplest form of gas turbine engines, the control design concept for a one-spool engine is essentially the same for a multispool (two-spool or three-spool) engine. From the classical control design perspective, all set-point controllers have a primary control loop, and the primary control loop tries to produce the steady-state and transient characteristics that are representative of a specific control mode. In this section, two classical control design techniques, the root-locus method and the frequency response method, are applied to the controller design of one-spool engines. For a review of these design methods, the reader can refer to Appendix A.

As explained earlier, because the engine thrust cannot be measured reliably in flight, another engine variable must be selected as the output variable to be regulated in feedback control. One essential requirement for the selection of this output variable is that it must correlate well with the thrust in all engine operating conditions. Two such variables have been used in the engine industry and are briefly mentioned in Chapter 1, and they are the shaft speed N and the engine pressure ratio EPR.

From Chapter 2 we know that the engine transfer function (TF), from fuel flow to shaft speed, is a first-order lag. If we use a proportional-plus-integral (PI) control law to control the engine speed, then the block diagram for this control loop

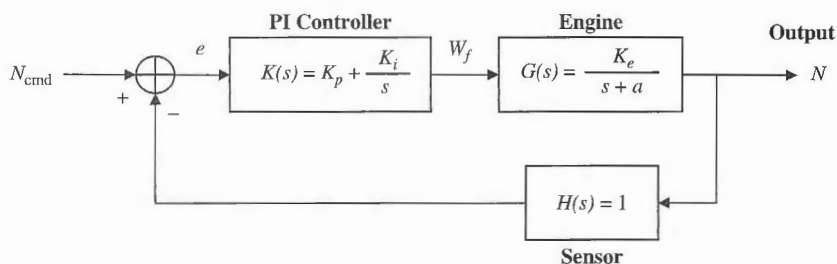


Fig. 4.2 Block diagram of the speed control loop for a set-point controller.

is shown in Fig. 4.2, where $1/a$ is the time constant for the engine, K_e is the engine TF gain, K_p is the proportional control gain, and K_i is the integral control gain. The output variable is the shaft speed N , and the controller will regulate the speed at the command value N_{cmd} .

Note that we have assumed that the dynamics of the speed sensor is so much faster than that of the shaft speed that its TF can be approximated by 1. From Appendix A, the open-loop transfer function (OLTF) is

$$\text{OLTF} = \frac{K_e K_p (s + K_i/K_p)}{s(s + a)} \quad (4.1)$$

This is a second-order system with poles located at 0 and $-a$ and a zero located at $-K_i/K_p$. The constant $K_e K_p$ is called the *loop gain*.

Note also that we have neglected the actuator TF for the same reason that the dynamics of the actuator is fast compared to that of the engine. However, if we want to include the actuator dynamics in the OLTF, and we may as well use a more generalized control law, that is, a proportional-plus-integral-plus-derivative (PID) control law, then the OLTF of Eq. (4.1) is expanded to

$$\text{OLTF} = \frac{K_e K_a K_d (s^2 + K_p/K_d s + K_i/K_d)}{s(s + a)(s + b)} \quad (4.2)$$

where b is the time constant of the actuator, K_a is the transfer function gain of the actuator, and the loop gain is $K_e K_a K_d$. This TF has a third-order denominator polynomial with its roots (or poles) located at 0, $-a$, and $-b$. These roots can take on one of the following three forms depending on the relative values of the coefficients of the polynomial: 1) two distinct real roots, 2) two identical real roots, and 3) two complex conjugate roots.

To explain and illustrate design results, we define the nominal operating condition of a one-spool engine as follows: the steady-state speed is 33,000 rpm; the engine time constant = $1/0.6$ s, or $a = 0.6$ r/s; and the fuel flow to speed TF [i.e., $K_e/(s + a)$] gain is $K_e = 33$ rpm/pph.

Design of a set-point controller for this operating condition with the root-locus method is discussed in Sec. 2.1.1 and design with the frequency-response method is discussed in Sec. 2.1.2.

4.1.1 Root-Locus Design for One-Spool Engine

As explained in Appendix A, the root-locus method is a graphical design technique for determining the closed-loop pole locations of a control system by choosing the value of the control gain. An objective of the root-locus design is to make sure that all closed-loop poles stay on the left-hand side of the s plane to ensure stability of the closed-loop system. Because the poles of the denominator of the OLTF of Eq. (4.1) are located at the 0 (origin) and $-a$ and because the zero of the OLTF is located at $-K_i/K_p$; therefore, from one of the rules of the root-locus method, we know that both closed-loop poles will be on the left-hand side of the s plane, no matter what the value of the loop gain is selected. Furthermore, the location of the zero can have two possibilities:

Case 1—conservative integrator: A smaller value of the integrator gain will result in a zero location between 0 and $-a$.

Case 2—aggressive integrator: A larger value of the integrator gain will result in a zero location on the left-hand side of $-a$ (or more negative).

In the first case, one of the closed-loop poles will be located between 0 and $-K_i/K_p$, and the other will be located further to the left of $-a$. Both poles will remain on the real axis so that the closed-loop response of the output variable is a combination of the two first-order lag responses; hence, there will be no oscillation in the output response. In the second case, the closed-loop poles can take on either two real values or two complex conjugate values. The reason is that two initial branches of the root loci will meet up between 0 and $-a$ on the real axis, and then the branches will split and depart from the real axis until they meet again on the real axis somewhere to the left of the zero, $-K_i/K_p$. In most circumstances, the design approach in case 1 is preferred.

In each case, the root loci are symmetric about the real axis and only the branches on the upper part of the s plane need to be shown. The arrow on each branch indicates the migration of the closed-loop pole on that branch as the loop gain is increased from 0 to ∞ . Because we already know the value of the engine gain K_e in the TF, the loop gain is simply a function of the proportional control gain K_p ; this means that the closed-loop pole locations are directly related to the value of the proportional control gain. In summary, for the design of a PI-control law, the problem amounts to selecting two design parameters: the ratio of control gains K_i/K_p and the proportional control gain K_p .

Using the Matlab® technical computing software, we can draw the root loci for the one-spool engine with a PI control law for the first case just described. Let us choose the zero location to be at -0.1 and the resultant root loci of the one-spool engine are shown in Fig. 4.3a, and an expanded view of the right-hand branch is shown in Fig. 4.3b.

Similarly for the second case, choose the zero location to be at -1 , that is, K_i/K_p is 10 times as large in the preceding case. The resultant root loci of the one-spool engine are shown in Fig. 4.4.

Now let us look at the time response of the closed-loop control system. Suppose the value of the proportional control gain K_p is chosen to be 0.04 pph/rpm, that is, the loop gain is $K_p K_e = 1.32$, then the step responses of the output in the two design cases are compared in Figs. 4.5a and 4.5b, respectively.

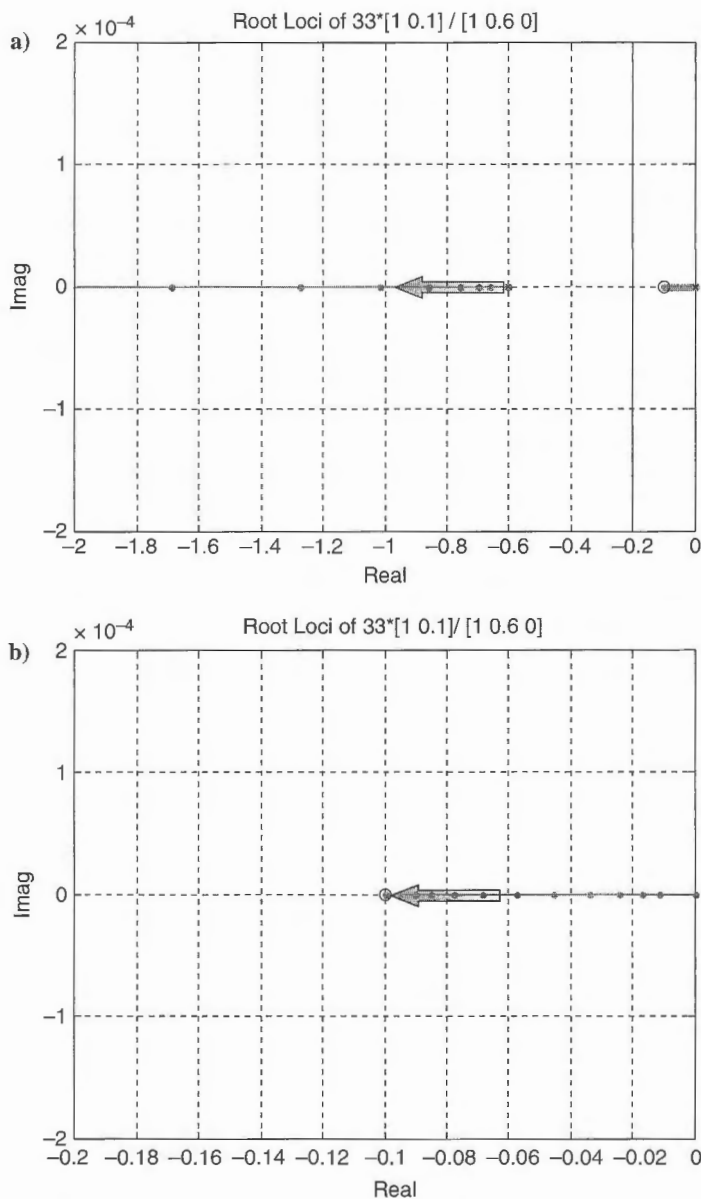


Fig. 4.3 (a) Root-locus plot of a PI control law for one-spool engine with a conservative integrator. (b) Expanded view of the right-hand-side branch of the root-locus plot in Fig. 4.3a.

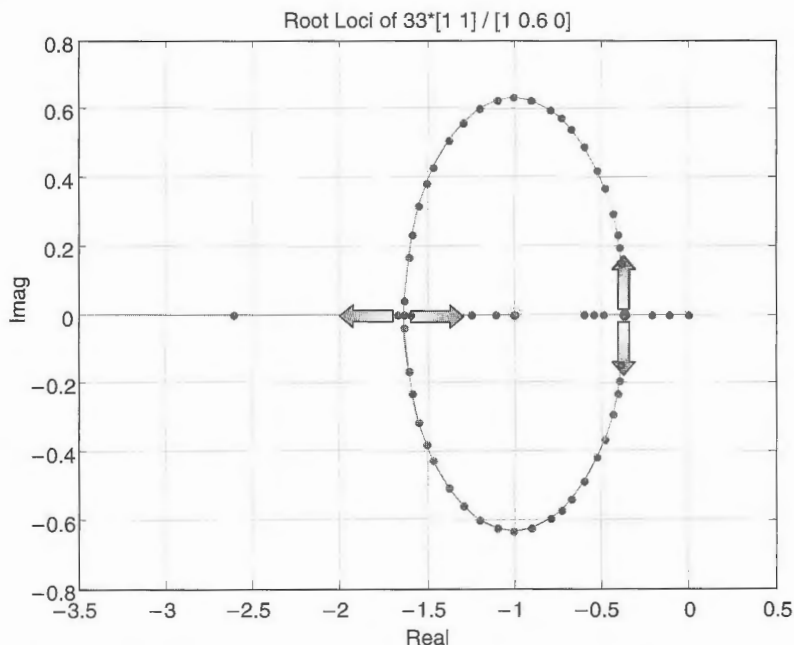


Fig. 4.4 Root-locus plot of a PI control law for one-spool engine with an aggressive integrator.

Now, if the proportional control gain is increased by 10 times, that is, $K_p = 0.4$ pph/rpm, then the unit step responses for the two cases are compared in Fig. 4.6.

The figures show that the output response is still underdamped when we choose a more aggressive gain ratio K_i/K_p , even though the proportional control gain is relatively small. This underdamped response is caused by the two imaginary closed-loop poles, which are located on the top and the bottom branches of the root loci and are formed after the two branches “meet” each other and “jump off” the real axis as shown in Fig. 4.4. To avoid the underdamped response, we can either reduce the proportional control gain or increase the gain so that the two closed-loop poles remain on the real axis. If we choose to increase the proportional gain by 10 times, then the closed-loop poles in case 2 become two real values. Hence, by choosing both a small and a large proportional gain for each of the two design cases, we can grasp the concept of the changes of closed-loop pole locations as the design parameter, proportional control gain, is varied. The results of this parametric analysis are listed in Table 4.1.

As the table shows, a more aggressive integrator produces a faster output response. To avoid undesirable oscillations, we need to increase the proportional control gain. However, by increasing the proportional gain, the closed-loop system becomes more sensitive to disturbances and excitation sources that appear at “higher” frequencies. One example of higher-frequency disturbances is sensor noise. Because sensing noise always exists in the measurement and data-collection

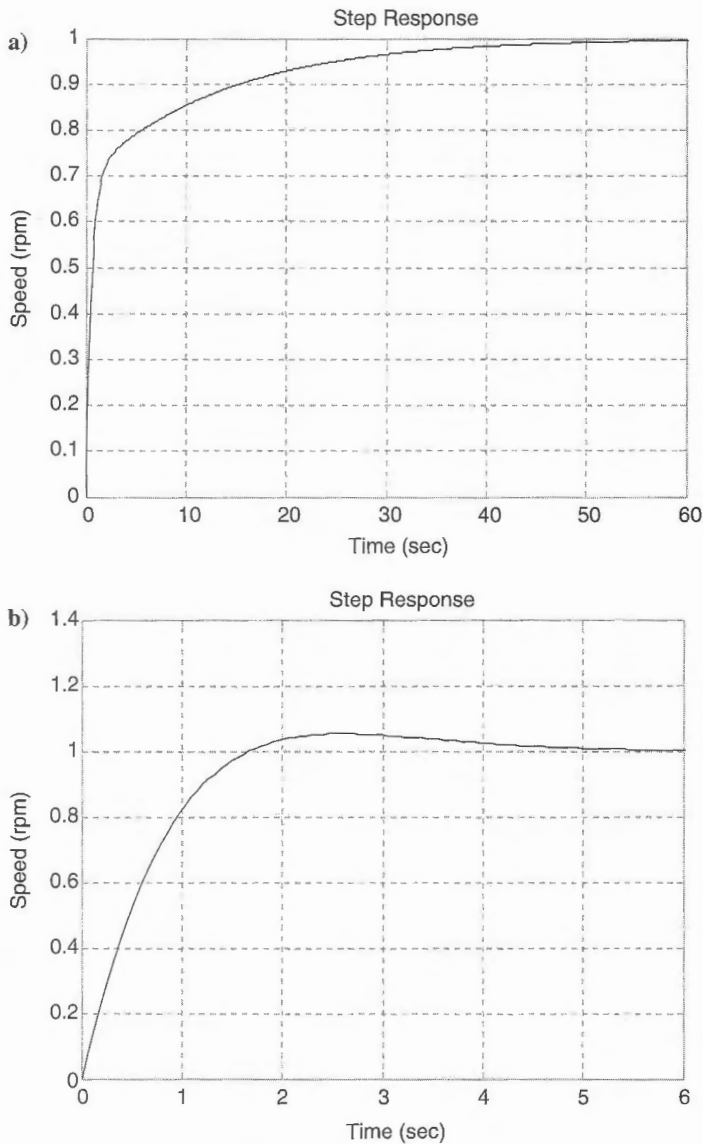


Fig. 4.5 (a) Closed-loop response of the shaft speed to a unit step input for design case 1. (b) Closed-loop response of the shaft speed to a unit step input for design case 2.

process, the noise shows up in the feedback error (i.e., $N_{\text{cmd}} - N$), and consequently, it is magnified by a high-gain controller. This means that the engine fuel flow, which is the controlled variable and is the output of the controller, will experience larger swings. Larger swings in fuel flow are undesirable for two reasons: 1) the fuel control actuator experiences larger peak values, and 2) the output variable (shaft

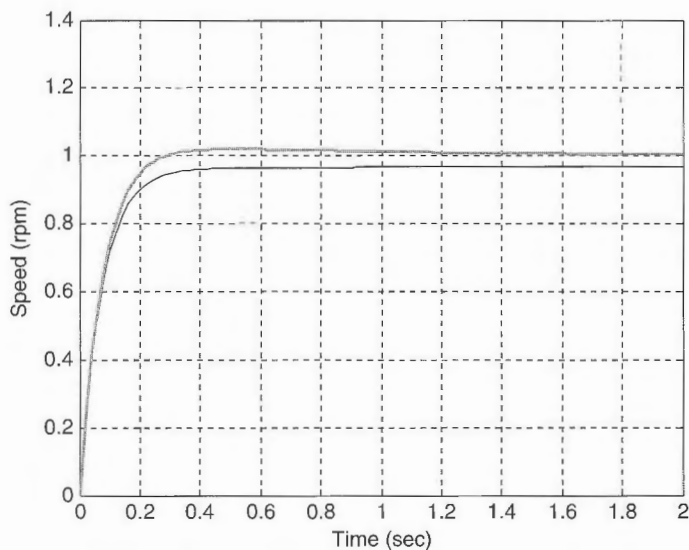


Fig. 4.6 Comparison of the closed-loop responses of the speed with a high-gain controller for both design cases (case 1 in blue, case 2 in green). (See also color figure section at the back of the book.)

speed in this case) might behave like a limit cycle, that is, a steady-state error that is cyclic between an upper and a lower bounds. (The reader can refer to Appendix A for more explanation of the characteristics of nonlinear systems.)

Note that an underdamped response is not necessarily bad because it usually comes with a shorter rise time and a shorter setting time than those of an overdamped response. An underdamped response with excessive oscillations, especially near the end (around the set point), is what we want to avoid in the design. In general, control law design, like other design disciplines, is a *tradeoff* among many conflicting objectives. Different operating conditions and mission scenarios often result in different control laws. A practical example of this design tradeoff is the much smaller tolerance for speed oscillations at the TO power than that at the idle or part-power operating conditions. At TO power, we want the engine to settle as fast as it can, so that the engine thrust can stabilize quickly during the TO run and

Table 4.1 Comparison of closed-loop poles between high and low proportional gains

Gain	Conservative integrator ($K_i/K_p = 0.1$)	Aggressive integrator ($K_i/K_p = 1$)
Small ($K_p = 0.04$)	-1.8486 and -0.0714	$-0.9600 \pm 0.6312i$
Large ($K_p = 0.4$)	-13.7037 and -0.0963	-12.766 and -1.034

in the initial phase of the climb. As a result, the design of the set-point controller at TO power should strive to limit the speed overshoot to 5%.

The closed-loop response of the shaft speed to a unit step input with $K_p = 0.4$ and $K_i/K_p = -1$ is shown in Fig. 4.5b. Note that the overshoot is close to 5%. The speed response of the same design to a unit impulse response is shown in Fig. 4.7. The step response represents a transient characteristic of the control system when the speed command is changed, whereas the impulse response represents a transient characteristic when a disturbance is introduced in the control system. The peak value in the impulse response indicates the sensitivity of the control loop to a disturbance, and the slope of the response indicates the equivalent time constant of the system.

4.1.2 Frequency-Response Design for One-Spool Engine

From Appendix A, we know that the design philosophy of the frequency-response method is to select controller TF so that the overall OLTf satisfies the desired response characteristics, which are represented by a “corridor” on the magnitude vs frequency diagram (see Fig. A.19). In summary, the frequency-response method guides us to *shape* the frequency response of the OLTf. A successful design often behaves in two ways: 1) the OLTf acts like an integrator at very low frequency, and 2) it acts like a first-order lag at very high frequency (i.e., a low-pass filter). Instead of going through the details of choosing the controller TF to fit in the corridor, we will take the controller design results from the root-locus method in the preceding subsection and examine their corresponding frequency responses for compliance with the desired characteristics.

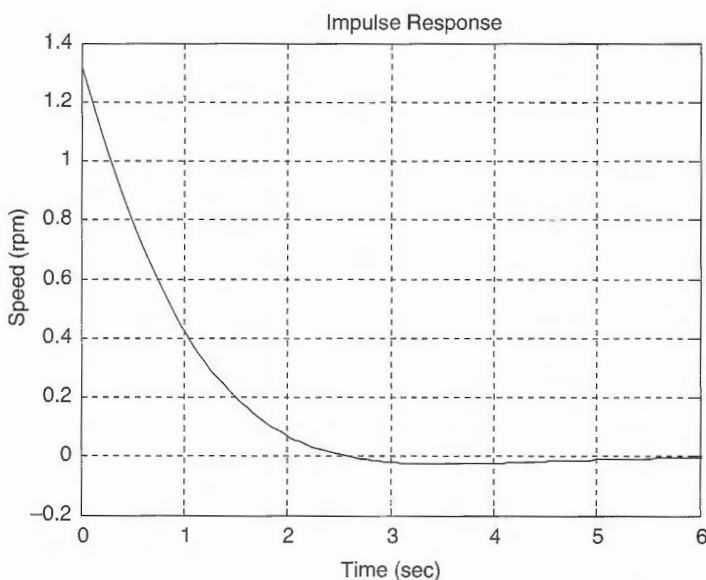


Fig. 4.7 Closed-loop speed response to an impulse input for a one-spool engine.

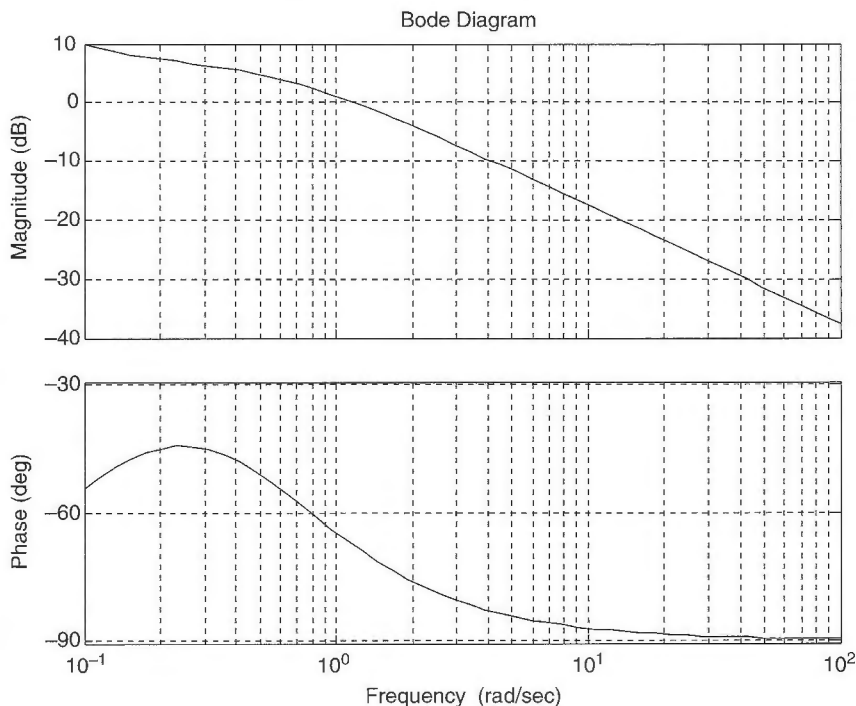


Fig. 4.8 Bode plot of the speed control loop with a conservative integrator.

Figure 4.8 shows the frequency response in the form of Bode plot for the speed control loop from case 1 of the root-locus design with a small proportional gain. Figure 4.9 shows the frequency response from case 2 of the root-locus design with a high control gain. The OLTF for case 1 is $1.32(s + 0.1)/(s^2 + 0.6s)$, and the OLTF for case 2 is $1.32(s + 1)/(s^2 + 0.6s)$. In each of the cases, the magnitude diagram satisfies the design corridor described in Appendix A; furthermore, the phase shift approaches -90 deg at high frequencies while the phase margin (PM) is more than 45 deg in both cases at the gain crossover frequency.

4.2 Controller Design for Two-Spool Engines

Two-spool engines represent the most common form of aircraft propulsion systems in use to date. As explained earlier, the primary goal of a set-point controller is to regulate engine performance at a fixed operating condition, and this operating condition is defined by a fixed N_1 or EPR. To illustrate an alternative in engine control design, we will use EPR as the power setting variable for the two-spool design problem and will design control laws for the EPR control loop as opposed to the speed control loop. Similarity in the design approach between the EPR control loop and the N_1 control loop should be evident in design examples to be discussed in this section.

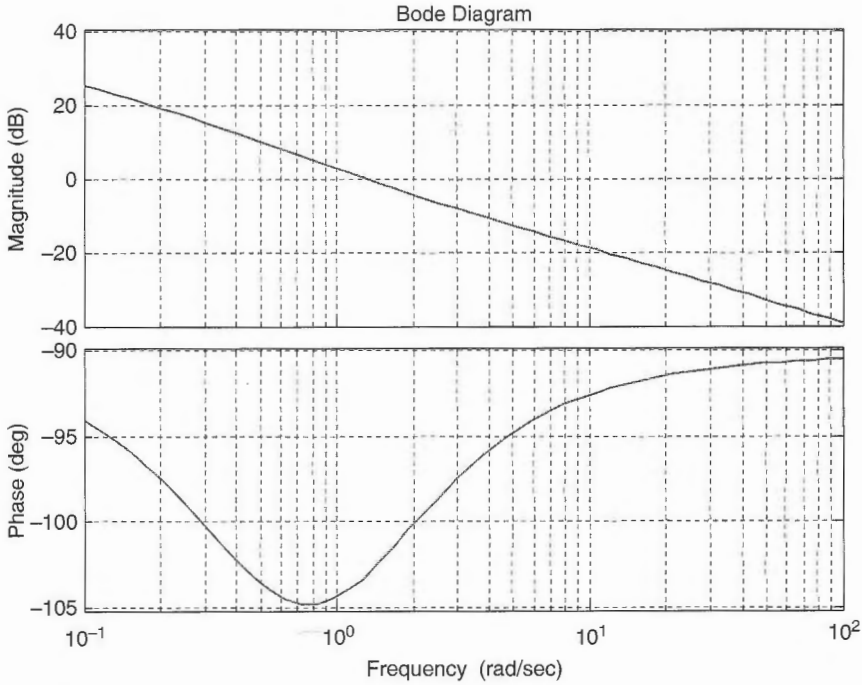


Fig. 4.9 Bode plot of the speed control loop with an aggressive integrator.

Again from Chapter 2, we know that the engine TF from fuel flow to engine pressure ratio for a two-spool engine is a second-order system. The block diagram for the EPR control loop with a PI control law is shown in Fig. 4.10, where z_1 is the zero of the TF and r_1 and r_2 are the two poles.

From Fig. 4.10, the OLTf is

$$\text{OLTf} = \frac{K_e K_p (s + (K_i/K_p)) (s + z_1)}{s(s + r_1)(s + r_2)} \quad (4.3)$$

This is a third-order system with poles located at 0, $-r_1$, and $-r_2$ and two zeros located at $-K_i/K_p$ and $-z_1$. As before, the constant $K_e K_p$ is the loop gain. If we add the actuator dynamics in the transfer function, and if we use a PID control law, then the OLTf becomes

$$\text{OLTf} = \frac{K_e K_a K_d (s^2 + (K_p/K_d)s + (K_i/K_d)) (s + z_1)}{s(s + b)(s + r_1)(s + r_2)} \quad (4.4)$$

where, as defined earlier, b is the time constant of the actuator, K_a is the transfer function gain of the actuator, and the loop gain is $K_e K_a K_d$. Equation (4.4) is the general form (in the context of classical control design methodologies) of the OLTf for a two-spool engine. The corresponding block diagram is shown in Fig. 4.11.

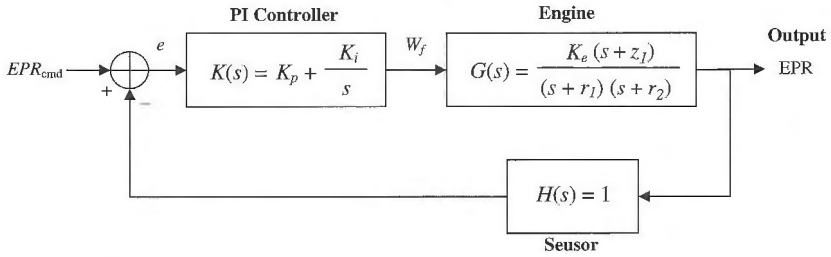


Fig. 4.10 Block diagram of the EPR control loop for a two-spool engine.

By using the same steady-state operating condition, as in the case of one-spool engine control design, a numerical example of the TFs for three different output variables—EPR, N_1 , and EGT—of a two-spool engine are given as follows:

$$G_{\text{EPR}}(s) = \frac{\text{EPR}(s)}{W_f(s)} = \frac{1.24(110s + 550)}{s^2 + 460s + 550} \quad (4.5)$$

$$G_{N_1}(s) = \frac{N_1(s)}{W_f(s)} = \frac{330.6(-1.35s + 10)}{s^2 + 12.3s + 10} \quad (4.6)$$

$$G_{\text{EGT}}(s) = \frac{\text{EGT}(s)}{W_f(s)} = \frac{60.38(1000s + 600)}{s^2 + 345s + 600} \quad (4.7)$$

where the unit of fuel flow W_f is pounds per second, the unit of N_1 is revolutions per minute, and the unit of EGT is degrees Fahrenheit. The open-loop responses of these output variables to a unit step input in fuel flow are shown in Fig. 4.12. Note that the 0 in Eq. (4.6) falls on the right-hand side of the s plane, and it causes

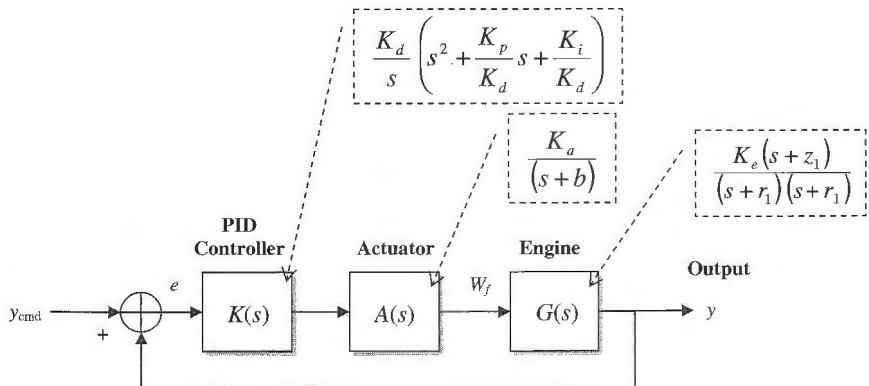


Fig. 4.11 Block diagram of a general control loop for a two-spool engine.

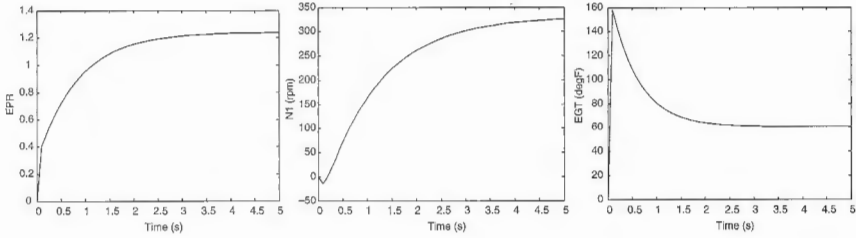


Fig. 4.12 Open-loop responses of three engine variables to a unit step input in fuel flow.

the response of the transfer function to “droop” initially. This phenomenon is typical of a high-bypass turbofan engine.

4.2.1 PI Control Law Design for Two-Spool Engine Without Actuator Dynamics

By choosing $K_i/K_p = -2$, the OLTF for this control loop from Eq. (4.3) becomes

$$\text{OLTF}_1 = \frac{K_e K_p (s + 2)(110s + 550)}{s(s^2 + 460s + 550)} \quad (4.8)$$

The root-locus plot of this OLTF for the EPR control loop, by using the SISO Design Tool in the Matlab® Control System Design Toolbox, is shown in Fig. 4.13. Similarly, the Bode plot of the same OLTF is shown in Fig. 4.14. Note the pink squares marked on the plots show the closed-loop pole locations if the loop gain

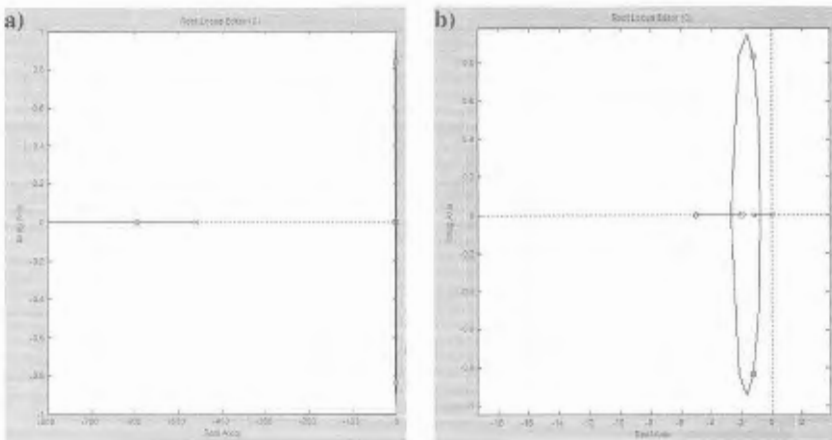


Fig. 4.13 (a) Overall root-locus plot for a PI control loop. (b) Right side of root-locus plot for a PI control loop.

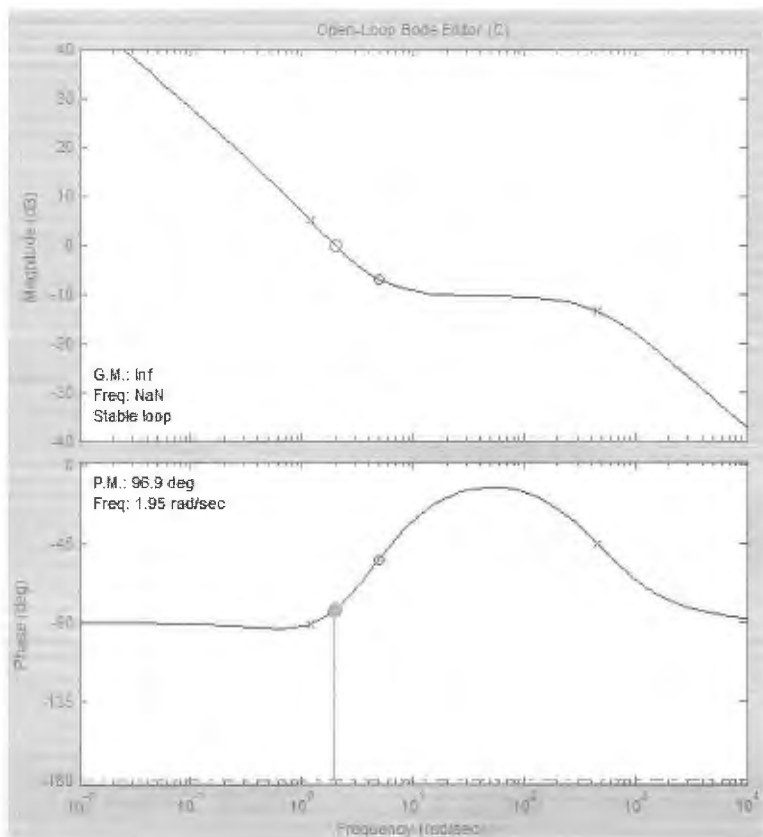


Fig. 4.14 Bode plot of a PI control loop for a two-spool engine. (See also color figure section at the back of the book.)

$K_e K_p$ is chosen to be 2. Also note from the Bode plot that the gain margin is infinite while the PM is 96.9 deg.

The closed-loop step response from EPR command to the final EPR is shown in Fig. 4.15.

4.2.2 PID Control Law Design for Two-Spool Engine with Actuator Dynamics

If we include the actuator transfer function with $b = 50$ and the derivative control in the control law, that is, the PID control law, then the OLTF of Eq. (4.4) becomes

$$\text{OLTF} = \frac{1.24 K_a K_d (s^2 + 8s + 16)(110s + 550)}{s(s + 50)(s^2 + 460s + 550)} \quad (4.9)$$

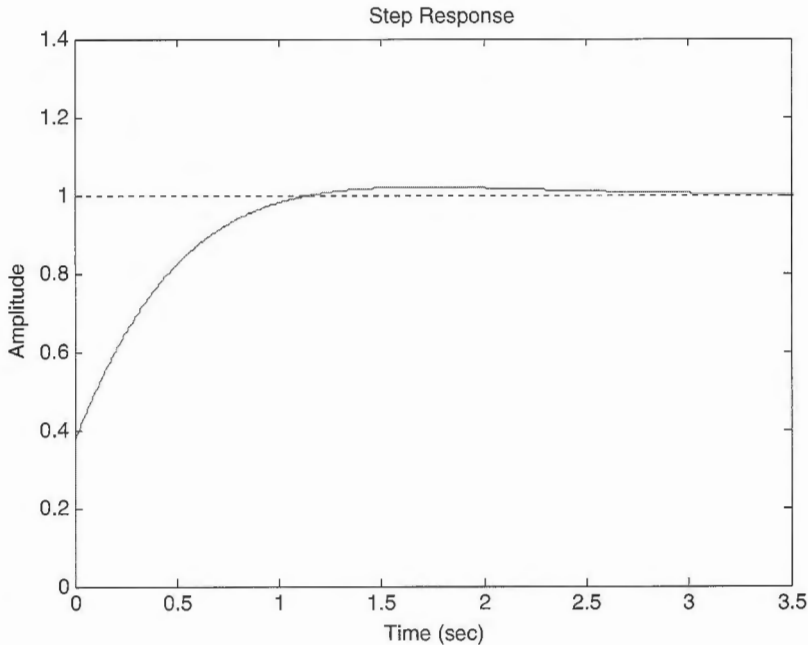


Fig. 4.15 Unit step response of a two-spool engine with a PI control law.

where $K_p/K_d = 8$ and $K_i/K_d = 16$. The corresponding root-locus plot and Bode plot are shown in Figs. 4.16 and 4.17, respectively.

The closed-loop step response from the EPR command to the EPR output is shown in Fig. 4.18 with a loop gain ($K_a K_d$) of 100. By comparing the step response of the PID controller in this figure to the response of the PI controller in Fig. 4.15, we see that the response rises much faster with the derivative control action; however, the derivative control also comes with a drop after the initial rise.

4.3 Control Design for Turboshaft Engines

Turbine engines are used in rotorcraft propulsion and electric power generation. These engines fall in the category of turboshaft because the shaft power is the primary output variable of these engines instead of the thrust. The majority of turboshaft engines contain two spools, where the gas-generator (GG) spool connects the compressor module with the HP turbine module and where the other spool connects the LP turbine module with the drive train of the rotor system of a helicopter or to an electric generator. The LP turbine module in this configuration is called a *free* turbine or a *free* power-turbine (PT).

In both the rotorcraft and the power generation applications, a constant rotational speed for the drive shaft is desired to ensure flight safety, handling quality, and generator stability. Hence, maintaining a constant LP shaft speed is the primary objective of the control system for a turboshaft engine. As in turbojet and

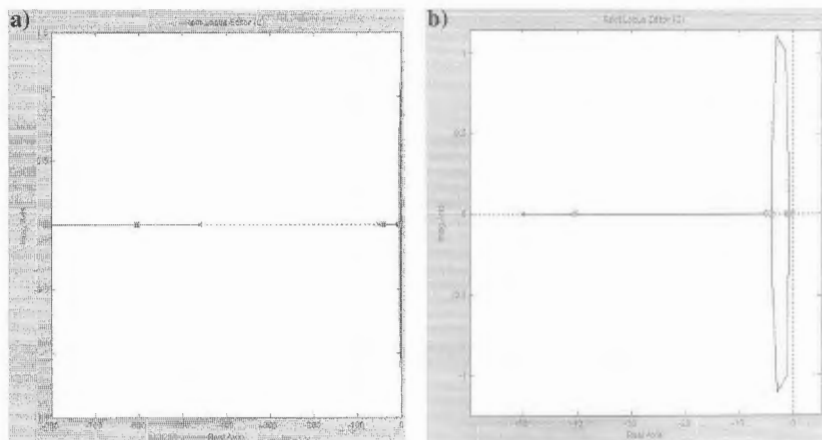


Fig. 4.16 (a) Overall root-locus plot for a PID control loop. (b) Right side of root-locus plot for a PID control loop.

turbofan engines, this objective needs to be accomplished by both the set-point controller and the transient controller; however, the set-point controller is more simplified in a turboshaft engine because the range of engine speed variations across all steady-state operating conditions is much smaller than that of a turbojet or turbofan engine. On the other hand, the transient controller for a turboshaft engine is more complicated than a turbojet engine because it has to deal with the possibility of large rotor load variations during different maneuvers, and the range of engine speed variations during these maneuvers can cover the entire engine power spectrum from idle to maximum power.

4.3.1 Feedback Control for Power Turbine Speed

In a rotorcraft, the rotor speed is presumed to be constant at all times; hence, the pilot controls lift by controlling the collective pitch angle. This presumed *decoupling* of flight control from engine control dictates the engine control requirement to be maintaining a constant turbine output shaft speed. Hence, the primary control loop in a turboshaft engine is the power turbine shaft speed N_p . The reference command of this control loop is a near 100% speed. Because the power turbine is one-way coupled to the gas generator, a cascade inner loop for controlling the gas-generator speed N_g is usually incorporated to improve the transient response of the engine. An example of the cascaded N_p and N_g control loops is shown in Fig. 4.19. Note we have assumed that the sensor dynamics for both the gas-generator speed and the power turbine speed are instant (i.e., the TF is 1).

The design methods described earlier can be applied to the design of the two controllers $K_1(s)$ and $K_2(s)$ (for the PT and the GG spools, respectively), except that the inner-loop controller (or the GG controller) must be designed first, and then the outer-loop controller (or the PT controller) can be designed by substituting the inner loop by its closed-loop transfer function in the figure.

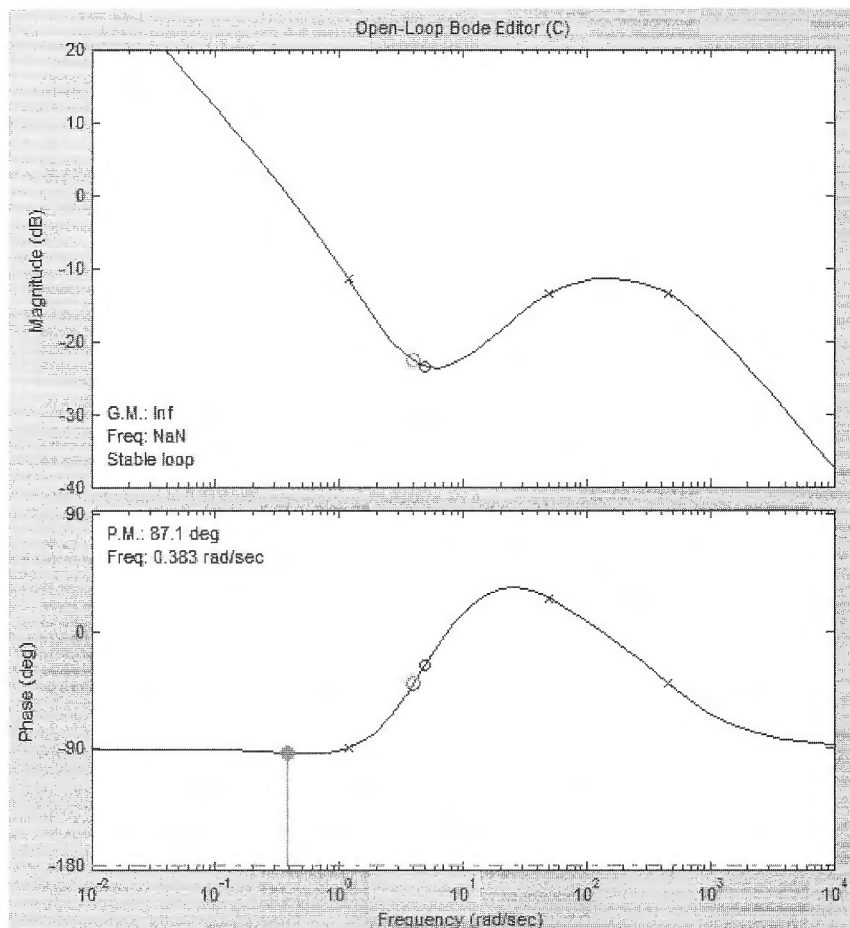


Fig. 4.17 Bode plot of a PID control loop for a two-spool engine.

4.3.2 Feed-Forward Control for Rotor Load Anticipation

For helicopter engines, the rotor load can change widely during certain flight maneuvers. This large change in load often causes unacceptable delays in the engine's response to increase or decrease power to keep the rotor speed at the desired (constant) value. To reduce the delays, we usually add a feedforward control loop to the feedback control loops. The feedforward loop essentially anticipates a change in rotor load and sends a command signal to the fuel control unit to increase or decrease the fuel flow, even before the load change shows up in the PT speed. The advantage of feedforward is that it increases the control bandwidth while preserving the stability margins of the closed-loop control system. Figure 4.20 illustrates the combined feed-forward and feedback control loops for

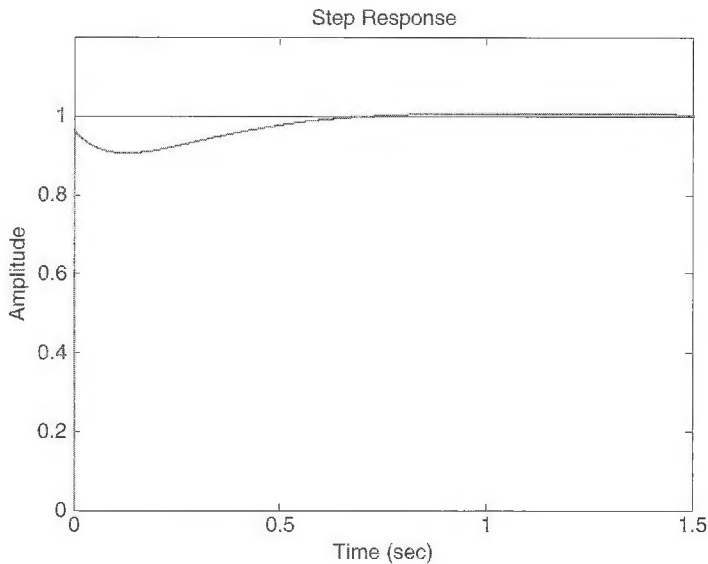


Fig. 4.18 Unit step response of a two-spool engine with the PID control law.

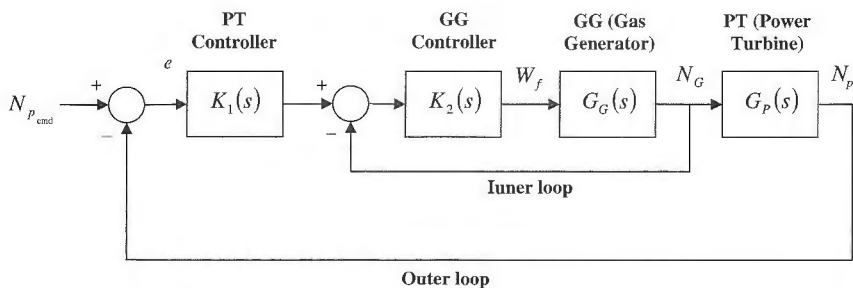


Fig. 4.19 Cascaded control loops for a turboshaft engine.

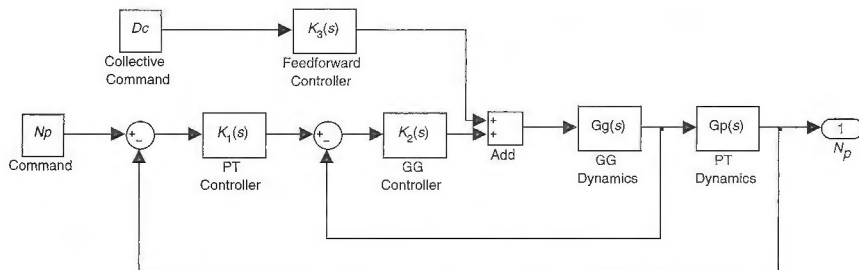


Fig. 4.20 Combined feedforward and feedback control loops for helicopter engines.

a helicopter engine, where the input of the feed-forward loop is the rotor-blade collective pitch angle δ_c command, represented by D_c .

4.3.3 Adaptive Control for Optimal Fuel Consumption

At cruise condition, the fuel consumption varies with the rotor speed for a given rotor load, air speed, and altitude. In some cases, a 5% variation in rotor speed can reduce the engine fuel consumption appreciably. Figure 4.21 illustrates the benefit in the reduction of fuel consumption for a Sikorsky S-76 helicopter if the rotor speed is allowed to change slightly to achieve optimal cruise fuel consumption [1]. The figure indicates that 5–10% less fuel is consumed in a 130-knot, 5000-ft, 10,000-lb-weight cruise by operating the rotor in the 104–107% speed range, instead of a fixed 100%.

To realize the benefit of reduced fuel consumption in cruise, the engine control must be able to *seek* the minimum fuel consumption rotor speed during the flight depending on the weight, altitude, and air speed of the rotorcraft. This rotor speed corresponds to the minimum power condition, which is a tradeoff between blade angle of attack and blade tip Mach number [1]. An adaptive control logic that is activated by the pilot can be designed. When activated, it modifies the engine control functions during flight. Basically, the adaptive logic induces an incremental change in rotor speed and monitors the fuel flow rate of the engine. This process is repeated over a predetermined rotor-speed range until the minimum fuel consumption condition is identified [2]. The predetermined

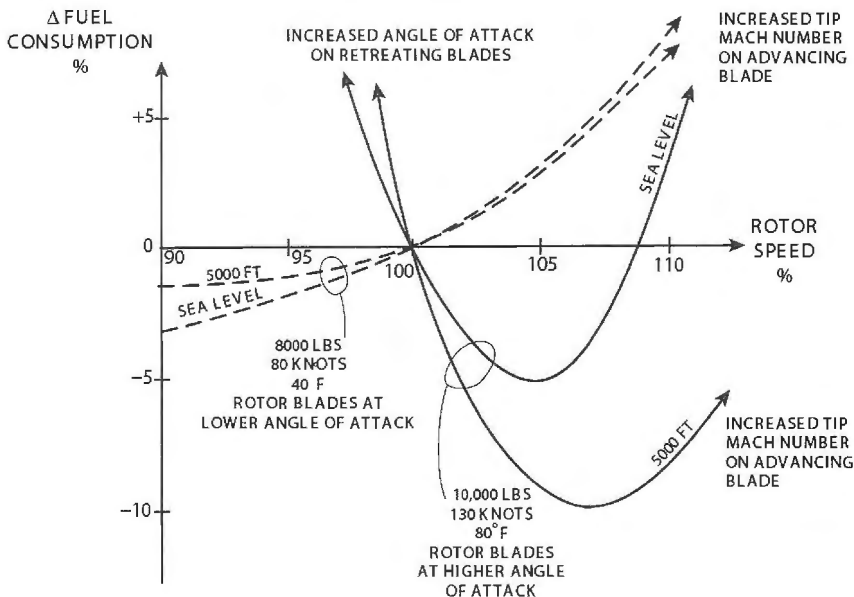


Fig. 4.21 Fuel consumption variations in cruise with respect to rotor speed (from [1], reprinted with permission).

speed range is typically narrow for the reasons of acceptable vibration, noise, and ride comfort.

4.4 Some Practical Considerations for Set-Point Controls

We have discussed linear control design methods for both SISO and MIMO systems, but before we leave this chapter let us spend some more time to emphasize a few practical considerations in the design of control laws for gas turbine engines. These considerations include, but are not limited to, control authorities under different control laws, engine responses under performance drift (or shift), and controlling the engine using fuel ratio instead of the fuel flow directly.

4.4.1 Engine Responses Under Different Control Laws

We are interested in the effects of three basic control laws on the closed-loop engine performance. Let us use a design example for the single-spool engine described in Sec. 2.1 to explain these effects. Recall that the closed-loop speed response to a step input command is shown in Fig. 4.5b. In addition to the speed response, we are particularly interested in the time histories of the fuel control commands corresponding to both the proportional and the integral control laws. These time histories are shown in Fig. 4.22, where the step input size is 100 rpm,

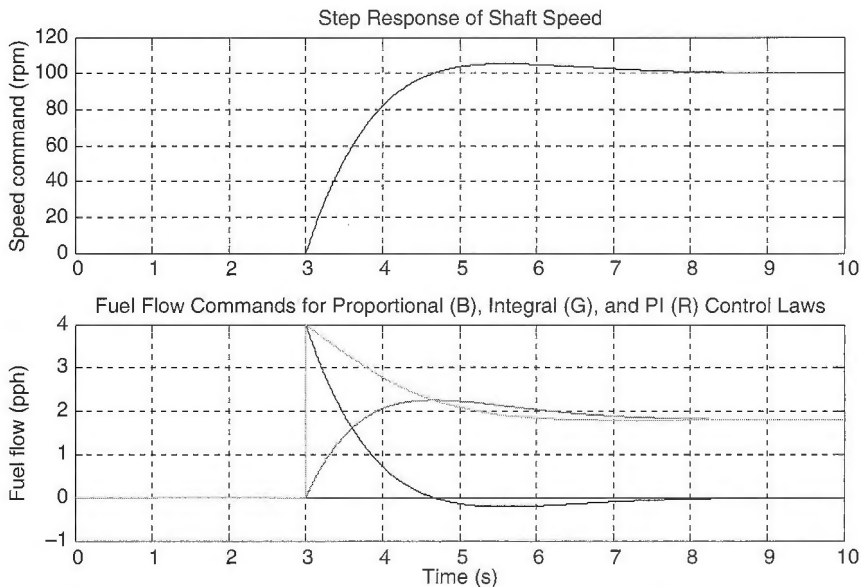


Fig. 4.22 Fuel control commands for proportional, integral, and PI control laws in design case 2 of the single-spool engine. (See also color figure section at the back of the book.)

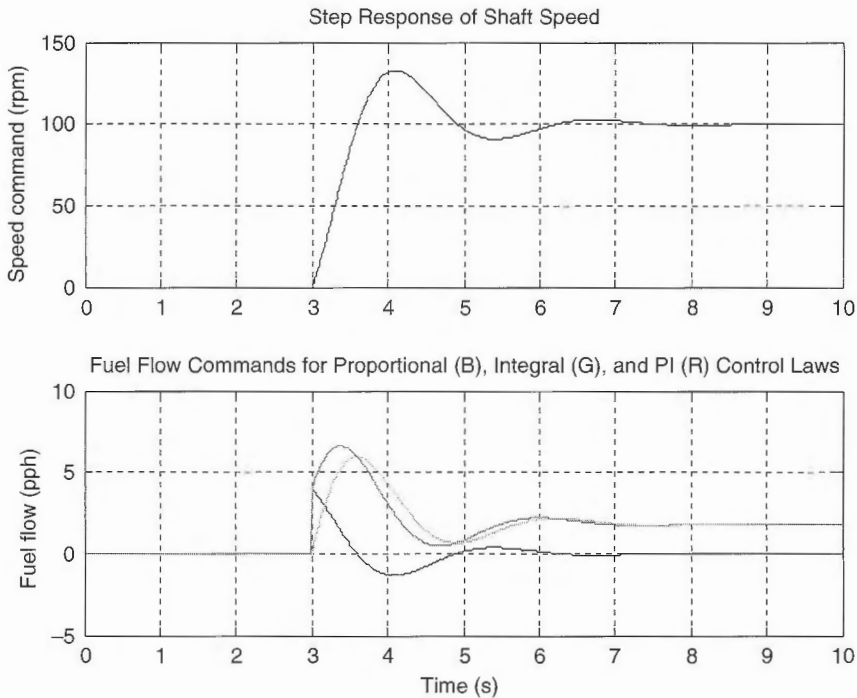


Fig. 4.23 Fuel control commands for proportional, integral, and PI control laws with the integral control gain increased by five times from design case 2.

the proportional control command is in blue, the integral control is in green, and the combined PI control is in red.

Suppose we increase the integral control gain by five times, the corresponding results of the closed-loop controlled performance are shown in Fig. 4.23. The responses of the speed and the fuel commands are both more oscillatory.

For the case of increased integral control gain, if we plot the fuel commands for the proportional and the integral control laws, respectively, vs the shaft speed, then we see in Fig. 4.24 that the control-vs-output relationships are different for the two control laws. Notably, the proportional control command during this step-response transient falls on a straight line with a slope being the *negative* of the proportional gain (-0.04 pph/rpm); the integral control command during the transient follows a spiral pattern from zero to the final steady-state condition. Note that the integral control command is not zero at the end of the transient (as marked by a red circle).

If we add the derivative control law in the design, then the time response of the speed and the relationship between the derivative control command and the speed are shown in Fig. 4.25. The derivative control gain used in the analysis value is 0.0001 pph/rpm-s. The derivative control command is directly proportional to the rate of change of speed; hence, it is the largest where the rate of change is the largest (e.g., at the beginning of the acceleration).

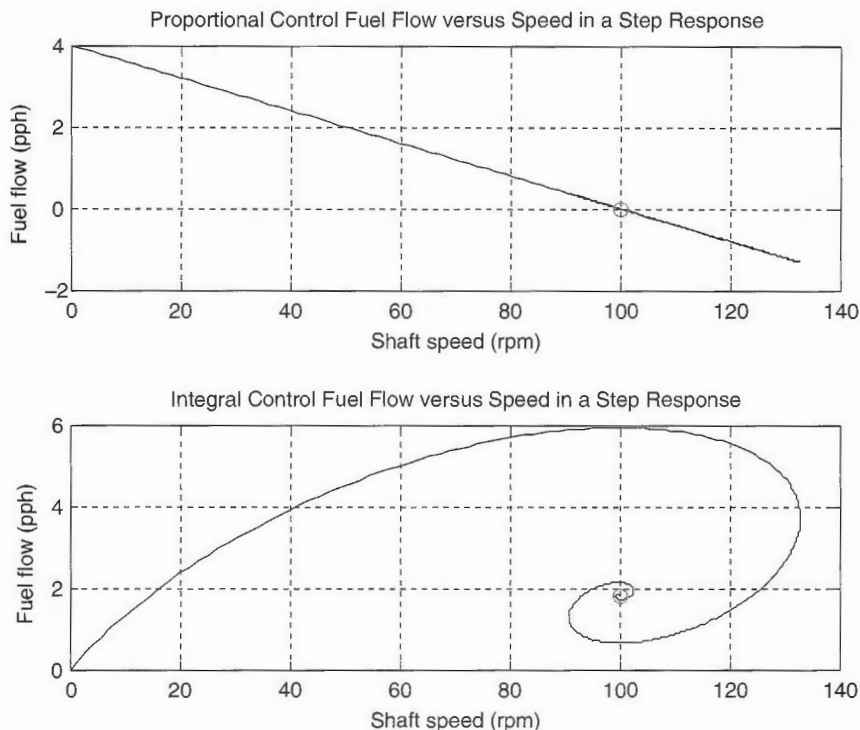


Fig. 4.24 Comparison of the parametric relationship of fuel control command vs engine speed between a proportional control and an integral control. (See also color figure section at the back of the book.)

4.4.2 Using Fuel Ratio Unit as Control Variable

In all of the design examples discussed so far, we have used the fuel flow (rate) as the control variable; however, in reality, the fuel ratio (unit), $RU \equiv W_f/P_3$, has almost been universally accepted as the fuel control variable in the engine control industry. There are three reasons for using the fuel ratio instead of the simpler fuel flow rate as the primary fuel control variable: 1) the fuel ratio provides a good control of turbine gas temperature T_{t4} because of its proportional relationship to the fuel-to-air ratio of the main burner [3], 2) it provides a self-recovery feature when the engine surges, and 3) it simplifies the control law by reducing the variability (or smaller range) of the controller gain constants, just as how corrected variables have been used to reduce the variability of engine performance variables.

We recall, from Appendix B [Eq. (B.33)], that the corrected fuel flow for the main burner is $\dot{m}_{fc} = W_{fc} = W_f/(\delta_2\sqrt{\theta_2})$, which is “corrected” by both the total pressure and the total temperature at the inlet. By choosing the compressor exit (or discharge) static pressure as the correction parameter, we have greatly reduced the difference in fuel ratios between the TO power and the idle power, that is, the ratio of the TO fuel ratio to the idle fuel ratio (RU_{TO}/RU_{idle}). This is significant from a historical perspective because reducing this ratio means to reduce the size

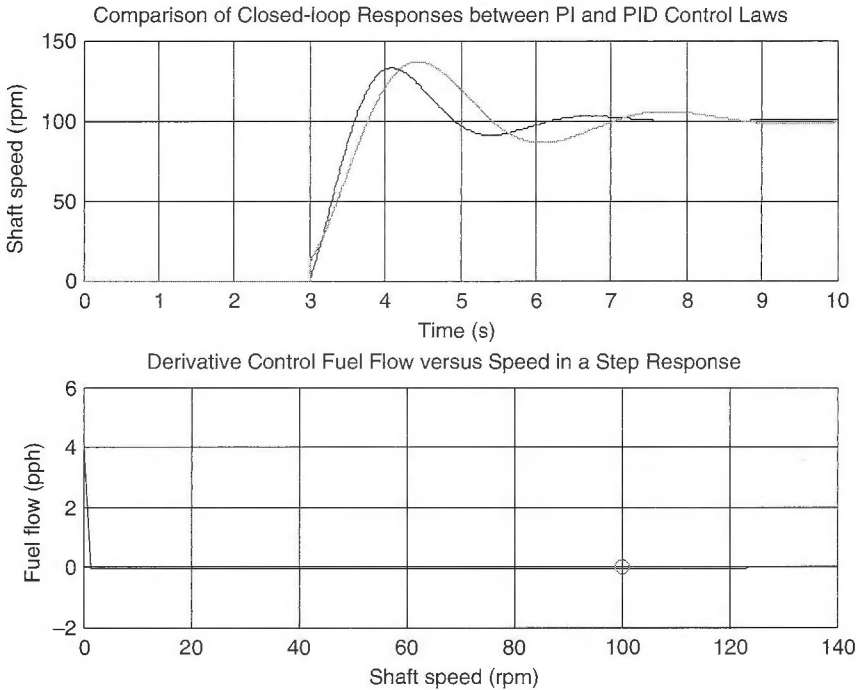


Fig. 4.25 Time response of shaft speed and the parametric relationship of a derivative fuel control command vs speed.

of the fuel scheduling cam in the hydromechanical control (HMC) unit. (The cam functions as a mechanical calculator to schedule the desired fuel flow ratio unit from idle to TO power.) Reducing this ratio also makes the engine control more “linear” and more predictable, based on the classical control design methodology, over the entire operating line.

When the fuel ratio is used as the control variable, it needs to be multiplied by the compressor exit pressure to get the actual fuel flow (rate) that is injected in the burner. This logic is illustrated in Fig. 4.26. This multiplication step ensures that

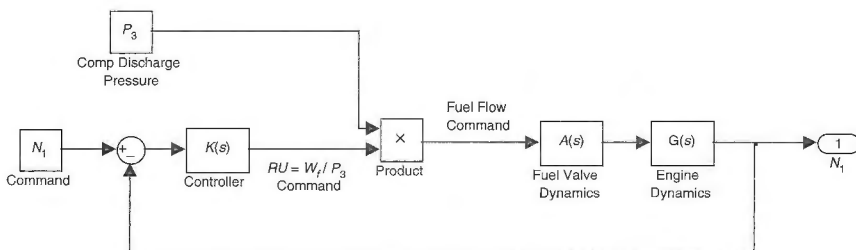


Fig. 4.26 Control logic relating controller output to main burner fuel flow rate.

the fuel flow rate going into the main burner is reduced when the engine starts to stall or surge because the compressor exit pressure will drop at the instant of stall/surge. The multiplication operation is easily performed in a modern digital electronic control; however, in the HMC it is accomplished by a hinged mechanical bar, where one end of the bar is the output quantity of fuel flow and the other end is the input quantity of the pressure. The pressure input has been converted from the pneumatic quantity to a mechanical movement, typically through a bellows.

References

- [1] Howlett, J. J., Morrison, T. and Zagaranski, R. D., "Adaptive Fuel Control for Helicopter Applications," *Journal of the American Helicopter Society*, Oct. 1984, pp. 43–54.
- [2] Bolton, R. L., "Adaptive Fuel Control Feasibility Investigation," *American Helicopter Society 38th Annual Forum Proceedings*, Anaheim, CA, May 1982.
- [3] Spang, H. A., III, and Brown, H., "Control of Jet Engines," *Control Engineering Practice* 7, Elsevier Science Ltd., 1999, pp. 1043–1059.

Design of Transient and Limit Controllers

An engine (or any physical system) is in transient operation when some or all of its performance variables are changing with time. Strictly speaking, an engine never reaches a *true* steady state, where all performance variables stay constant over time, because, even in a *stable* cruise condition, the engine's control variable fuel flow varies constantly to counter the disturbances that are inherent in any control system. Common disturbances include variations in the ambient condition such as the wind speed and the air temperature, measurement noises, actuator nonlinearity and inconsistency. Compared to the stable cruise condition, a transient condition experiences much larger and more observable performance changes in a relatively short time period.

Transient control design accounts for approximately three-quarters (or 75%) of the total control law design and development effort. Why? Because transient control by nature has to cover the speed range of several different operating points (or set points) that, by nature, makes it nonlinear, and because it has to protect the engine from running beyond operating limits during transient operations. Some examples of these operating limits include: the physical limit of a shaft speed, the maximum temperature for turbine blades, the maximum limit of the burner (or combustor) pressure, and the surge (or stall) limit of the compression system. Succinctly, transient control design is concerned with two things: 1) how to change the engine's operating condition from one state to another and 2) how to keep the engine from exceeding operating limits while making these state changes. An engine's operating state is commonly represented by its power level, which is directly related to net thrust for a turbofan or turbojet engine.

To illustrate the concept of transient control, Fig. 5.1 has expanded the set-point control diagram in Fig. 4.1 with two notional transient trajectories on the compressor map; the dotted line on the upper side of the operating line is a trajectory for engine acceleration and the dotted line on the lower side of the operating line is a trajectory for deceleration. In addition to the surge line, two operating limits have been added on this compressor map: a turbine temperature limit and a burner blowout limit. Note that the acceleration and the deceleration trajectories have stayed within all operating limits in the figure. This means that engine operating limits bound all transient operations, and these limits form the basis of transient schedules. Developing transient schedules is one of the most important tasks in transient control design and it will be discussed in this chapter.

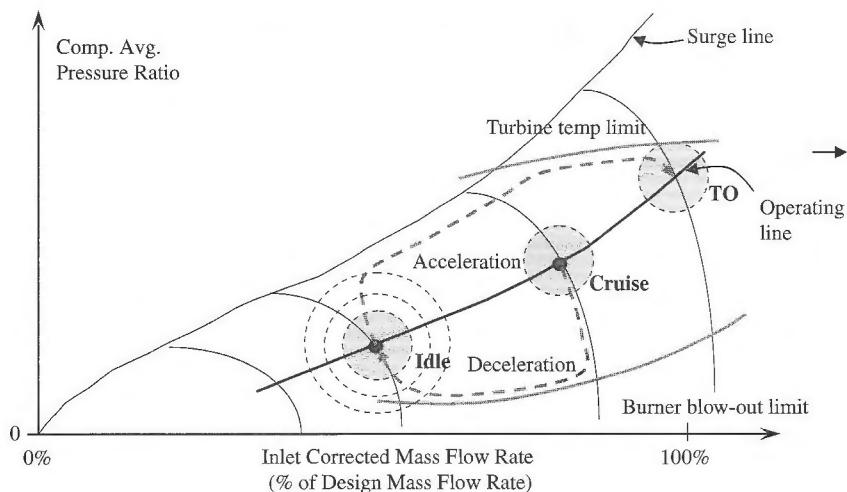


Fig. 5.1 Transient operating trajectories and operating limits on a compressor map.

In addition to making the engine stay within all operating limits while changing set points, a transient controller faces another challenge, that is, it must provide acceptable transient performance in terms of the rise time, the settling time, and the overshoot/undershoot of each controlled variable (like N_1 or EGT). This challenge is further complicated by the need to switch the control mode between set-point control and transient schedules. Hence, the goal of transient (and limit protection) control design is to achieve a “graceful” transition in every mode switching, that is, a smooth transition in engine performance during any control mode transfer throughout the entire transient operation. This absence of performance “jump” or sudden change is more easily understood than accomplished.

The objective of this chapter is two-fold: 1) to explain the two most common transient control approaches, namely, the *schedule-based* control and the *acceleration-based* (or $\dot{N}$) control and 2) to introduce the common operating limits and how to design controllers to avoid these limits. In Sec. 5.1, we will discuss the design of schedule-based transient controllers. The schedule-based approach to transient control has been adopted by the majority of gas turbine engines. In Sec. 5.2, some design considerations for nonlinearity will be covered, specifically, two techniques that are commonly used in transient control designs—gain-scheduling and integrator wind-up protection. In Sec. 5.3, we will discuss how to design an acceleration-based transient controller. Section 5.4 will cover the design of limit-protection controllers; which are traditionally called *topping governors*. Lastly in Sec. 5.5, some considerations of transient control design for turboshaft engines will be discussed.

5.1 Schedule-Based Transient Controller Design

For the set-point controller of a turbofan engine, the fan speed N_1 or the engine pressure ratio (EPR) is the feedback variable for the primary control loop. For

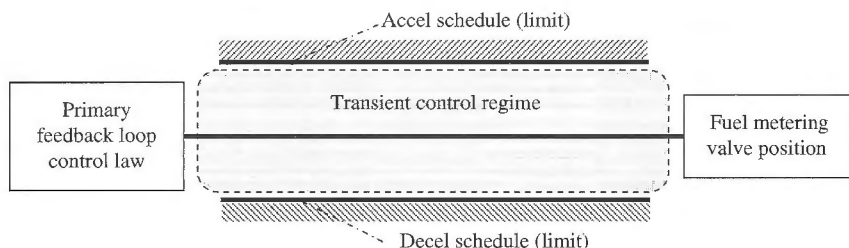


Fig. 5.2 Concept of saturation nonlinearities for transient schedules.

transient controls, we still have the primary control loop; however, this control loop is bounded by the acceleration and the deceleration schedules, that is, the control variable fuel flow will not be allowed to exceed the values on the acceleration schedule on the upper side and the deceleration schedule on the lower side. To understand this concept better, let us consider an acceleration transient starting from idle and ending at TO power. If we let the set-point control neighborhood at idle (the dotted circle in Fig. 5.1) expand its circle of “influence” because of the proportional control action, until it touches the acceleration schedule, then the fuel flow will be limited to the scheduled value. Similarly, as the engine accelerates close to the TO power, the set-point control neighborhood around the TO power will shrink from where it barely touches the acceleration schedule to the center of the neighborhood, while the transient controller hands over the control of fuel flow to the set-point controller. Figure 5.2 illustrates this transient control concept by representing the accel (short for acceleration) schedule and decel (short for deceleration) schedule as saturation nonlinearities around the primary control loop. The acceleration and the deceleration schedules are also called transient schedules.

So how do we develop (or derive) the transient schedules? This question will be addressed in Chapter 6; however, one thing worth nothing here is that the unit of transient schedules should be the same as that of the control variable in the primary control loop. In Chapter 4, we see that the fuel ratio unit ($RU \equiv W_f/P_3$) is the preferred control variable rather than the fuel flow itself. Hence, most of the transient schedules for turbine engines are expressed in RU.

5.1.1 Concept of Control Schedules

If we express the operating limits of Fig. 5.1 in a fuel flow (or fuel ratio unit) vs shaft speed diagram, we get the *control envelope* for the engine as shown in Fig. 5.3. This control envelope illustrates the safe operating region of the control variable vs the output variable. Similarly, we can express the control envelope in terms of the engine pressure ratio if the primary control loop is for EPR.

After the engine starts up, the control envelope, although is not a rectangle, is still bounded by the maximum (or max) flow limit, the minimum (or min) flow limit, the idle power speed limit, and the TO power speed limit. The maximum flow limit prevents the engine from surge (or stall) as well as from overtemperature,

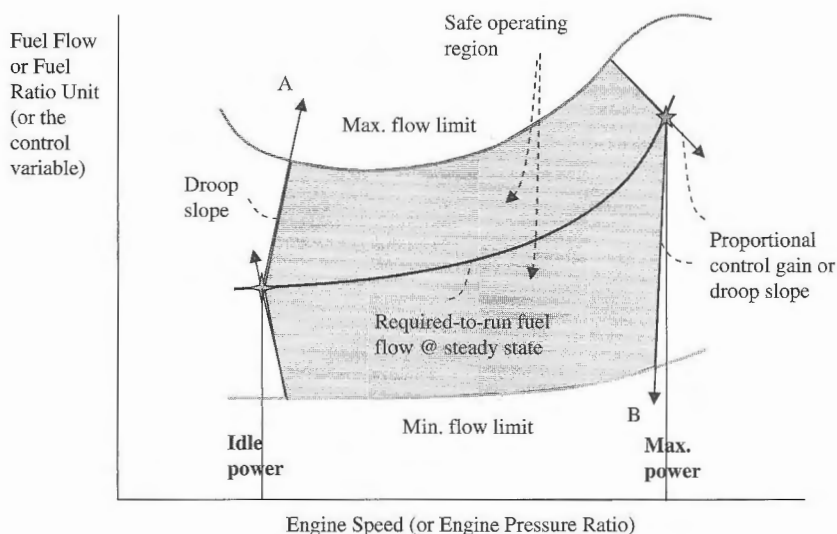


Fig. 5.3 Control envelope for gas turbine engines.

whereas the minimum flow limit prevents the engine from flame-out. The idle power is typically set to produce a desirable ground thrust for taxiing the airplane, and the TO power produces the rated thrust for TO. The max and the min flow limits are the accel and the decel schedules, respectively. These transient schedules as well as the idle and TO speed settings will change with the engine's inlet conditions, and will be explained in Chapter 6. One thing to note is that the droop slopes from idle to point A and from TO power to point B are almost vertical in practice as the result of a large initial speed error.

Figure 5.3 also shows two pairs of straight lines intersecting the idle power point and the TO power point on the steady-state operating curve. Each pair of lines consists of a line that “emanates” from the set point and a line that “approaches” the set point. These lines are called *droop lines*, and the slope of each droop line represents the proportional control gain ($\Delta W_f / \Delta N_{\text{err}}$) at the set point. Actually, the trajectory between a set-point (e.g., idle, TO, or any other point in between) and a transient schedule does not really follow the droop line, because the trajectory includes the effects of both the proportional control and integral control. Many modern engine controllers also add the derivative control and other types of compensation logics to make the trajectory more sophisticated than a simple straight line.

Protections for other operational limits are handled by engine transient controls. Examples of these limits include turbine and compressor temperature limits, the compressor discharge pressure limit, and shaft overspeed and underspeed limits. These additional limit protection logics built into the basic accel-decel transient controller, allow a pilot *unrestricted* throttle movement throughout the operating envelope, hence significantly reducing the pilot's workload.

5.1.2 Acceleration Control

Because every engine acceleration or deceleration begins and ends with the set-point controller, the time it takes for the set-point controller to hand over the control “authority” to the accel schedule depends on the difference between the beginning and the ending speeds defined by the corresponding set points. If the difference is small, the accel schedule may not be activated at all, that is, the acceleration is totally controlled by the set-point controllers at both ends.

From Chapter 4 we know that a set-point control law $K_{sp}(s)$ should consist of at least two basic control laws, that is, the proportional control and the integral control, as expressed by

$$\frac{\Delta W_f}{\Delta N_{err}} = K_{sp}(s) = K_p + \frac{K_i}{s} \quad (5.1)$$

where $\Delta N_{err} = N_{cmd} - N_{ls}$, that is, the difference between the N_l command and the N_l sensed. From this equation, we can express the acceleration control logic as follows:

$$\Delta W_{f_{cmd}} = K_{sp} \cdot \Delta N_{err} \quad \text{if} \quad K_{sp} \cdot \Delta N_{err} < (W_f|^{accel} - W_{f_{ss}}) \quad (5.2)$$

$$\Delta W_{f_{cmd}} = W_f|^{accel} - W_{f_{ss}} \quad \text{if} \quad K_{sp} \cdot \Delta N_{err} \geq (W_f|^{accel} - W_{f_{ss}}) \quad (5.3)$$

where $\Delta W_{f_{cmd}}$ is the commanded “delta” fuel flow sent to the fuel metering valve, $W_f|^{accel}$ is a value on the accel schedule, $W_{f_{ss}}$ is the steady-state (or required-to-run) fuel flow at the setpoint, and the Laplace operator s has been omitted without loss of generality. Figure 5.4 illustrates the acceleration control logic graphically. Note that the steady-state fuel flow corresponding to the “current” engine speed has been added to the output of the set-point controller in the “Select the smaller” block before the sum is compared to the accel schedule to determine whether the control authority of the set-point controller should yield to the accel schedule.

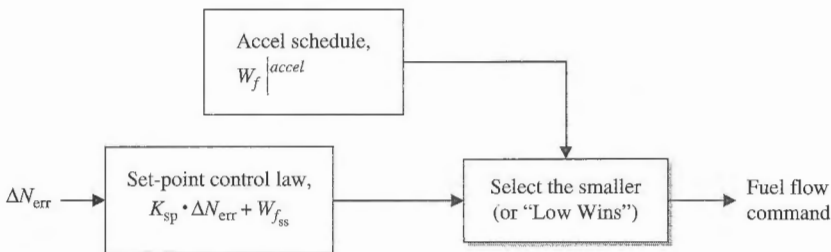


Fig. 5.4 Concept of acceleration control logic.

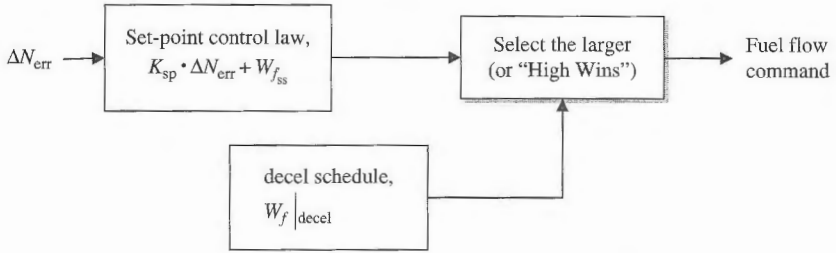


Fig. 5.5 Concept of deceleration control logic.

5.1.3 Deceleration Control

The deceleration control logic is similar to the acceleration control logic, except that the decel limit acts as the “lowest limit” for the control authority instead of the “highest limit” as does for the accel limit. If the scale of deceleration is moderately large, then the decel schedule will become active during the deceleration meaning that the decel limit is larger than the output of the set-point controller.

The deceleration control logic is expressed by

$$\Delta W_{f_{cmd}} = K_{sp} \cdot \Delta N_{err}, \quad \text{if } |K_{sp} \cdot \Delta N_{err}| < |(W_f|_{decel} - W_{f_{ss}})| \quad (5.4)$$

$$\Delta W_{f_{cmd}} = (W_{f_{ss}} - W_f|_{decel}), \quad \text{if } |K_{sp} \cdot \Delta N_{err}| \geq |(W_f|_{decel} - W_{f_{ss}})| \quad (5.5)$$

where $W_f|_{decel}$ is the value on the decel schedule. Figure 5.5 illustrates the deceleration control logic graphically. Note also that the steady-state fuel flow corresponding to the “current” engine speed has been added to the output of the set-point controller in the “Select the larger” block before the sum is compared to the decel schedule.

5.2 Nonlinear Considerations in Transient Controls

Gas turbine engines are highly complex and nonlinear systems. Nonlinear characteristics are exhibited in both the steady-state performance and the *dynamics* of the engine (see Chapter 2). Typically, the time response of the engine near TO power is three to five times as fast as the response near idle. This indicates that a control law designed for TO power will be inadequate to control the engine at idle, and vice versa, because one set of controller gains is only valid for a relatively small neighborhood around the set point for which the control law is designed. To overcome this problem, we often resort to the method of small perturbation (or piecewise linearization), which divides the engine operating line into multiple, yet finite, number of nominal points (or set points). We then design a set-point controller for each nominal point, knowing that the controller is only valid in the vicinity of the nominal point.

The result of this piecewise linearization is a set of linear set-point controllers. Then a methodology is needed to switch between two neighboring set-point

controllers without introducing discontinuities in engine output variables. The most commonly used technique for switching between two neighboring controllers is the method of gain scheduling, which will be discussed in the next subsection.

In addition to the nonlinearities of the engine, there is another form of nonlinearity that is common for all nonlinear control systems. This type of nonlinearity is caused by switching between a linear controller and a nonlinear schedule; it is called the *saturation nonlinearity* (see Appendix A for more explanation). Because all linear controllers incorporate some form of integral control to improve performance and robustness, the integral control law will continue to compute a desired input (or actuator) command as long as the feedback error is not zero, even though the linear controller has yielded the control authority to the nonlinear controller. This phenomenon is also called *integrator wind-up*. Integrator wind-up protection will be covered in Sec. 5.2.2.

5.2.1 Gain Scheduling

Gain scheduling is an effective technique used in controlling nonlinear dynamic systems like an aircraft engine or a high-performance airplane, whose performance varies widely over the operating envelope. The technique effectively “glues” a family of linear controllers into one “super” controller, which has the same structure (or format) as each individual, linear, set-point controller. However, the parameters in this super controller are changed frequently depending on the system’s operating condition and its relationship to nearby set-point conditions. This super controller is called a *gain-scheduled* controller.

To schedule the gains (or the parameters) of the controller, we start by choosing one or more state or output variables as the scheduling variables. A scheduling variable is an independent variable capable of discerning which of the set-point controllers are most adequate for a given operating condition. Another way to explain the scheduling variables is to consider all set-point controllers or linear models as functions of these variables, that is, the linear controllers and models corresponding to different nominal operating conditions are parameterized by these scheduling variables [1]. An example of scheduling variables is the engine shaft speed. Let us assume that the dynamics of an engine can be represented by three linear models: idle, cruise, and TO power. From Chapter 2, we can write the state equations for the three linear models as

$$\begin{bmatrix} \dot{N}_1 \\ \dot{N}_2 \end{bmatrix} = A_k \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} + B_k W_f \quad (5.6)$$

$$y = C_k \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} \quad (5.7)$$

where the subscript k stands for idle, cruise, or TO depending on the specific operating point we are referring to. Note that D in Eq. (5.7) has been dropped for simplicity. Suppose a linear controller for each of the three linear models has been

designed and these controllers are expressed as a second-order system in the following equation:

$$\begin{bmatrix} \dot{x}_{c_1} \\ \dot{x}_{c_2} \end{bmatrix} = A_{c_k} \begin{bmatrix} x_{c_1} \\ x_{c_2} \end{bmatrix} + B_{c_k} e \quad (5.8)$$

$$W_f = C_{c_k} \begin{bmatrix} x_{c_1} \\ x_{c_2} \end{bmatrix} + D_{c_k} e \quad (5.9)$$

where x_{c_1} and x_{c_2} are the state variables of the controller; e is the feedback error; A_c , B_c , C_c , and D_c are the controller parameter matrices, or the state variable feedback gain matrix; and the subscript k again denotes the k th operating point. Let us further assume that N_1 is the controlled variable, that is, the primary control loop is to control N_1 , and so we have

$$e = N_{1_{\text{cmd}}} - N_1 \quad (5.10)$$

The closed-loop system near the k th operating point, expressed in the state equation form, can be simplified to

$$\begin{bmatrix} \dot{N}_1 \\ \dot{N}_2 \\ \cdots \\ \dot{x}_{c_1} \\ \dot{x}_{c_2} \end{bmatrix} = \begin{bmatrix} A_k - [B_k D_{c_k} & 0] & \vdots & B_k C_{c_k} \\ \cdots & \cdots & \cdots & \cdots \\ [-B_{c_k} & 0] & \vdots & A_{c_k} \end{bmatrix} \begin{bmatrix} N_1 \\ N_2 \\ \cdots \\ x_{c_1} \\ x_{c_2} \end{bmatrix} + \begin{bmatrix} B_k D_{c_k} \\ \cdots \\ B_{c_k} \end{bmatrix} N_{1_{\text{cmd}}} \quad (5.11)$$

where all matrix elements are valid near the k th nominal point. For any operating condition located between the two adjacent nominal points k and $k + 1$, the values of the parameters in the controller, that is, Eqs. (5.8) and (5.9), can be interpolated between the corresponding values of the two controllers at the k th and the $(k + 1)$ th nominal points.

Suppose we choose N_2 as the scheduling variable, then the matrices in Eq. (5.11) become functions of the scheduling variable and can be written as

$$\begin{bmatrix} \dot{N}_1 \\ \dot{N}_2 \\ \cdots \\ \dot{x}_{c_1} \\ \dot{x}_{c_2} \end{bmatrix} = A_{\text{GS}}(N_2) \begin{bmatrix} N_1 \\ N_2 \\ \cdots \\ x_{c_1} \\ x_{c_2} \end{bmatrix} + B_{\text{GS}}(N_2) \cdot N_{1_{\text{cmd}}} \quad (5.12)$$

where A_{GS} and B_{GS} are the gain-scheduled matrices for the closed-loop system, which is valid for the family of nominal operating points.

Similarly, we can use the transfer function representation to describe gain scheduling. Let us first express the linear model of the k th nominal point as

$$G_{1_k}(s) \equiv \left. \frac{N_1(s)}{W_f(s)} \right|_k = \frac{k_{1_k}(s + b_{1_k})}{(s^2 + a_{1_k}s + a_{2_k})} \quad (5.13)$$

$$G_{2_k}(s) \equiv \left. \frac{N_2(s)}{W_f(s)} \right|_k = \frac{k_{2_k}(s + b_{2_k})}{(s^2 + a_{1_k}s + a_{2_k})} \quad (5.14)$$

We also express the PID control law designed for the k th nominal point as

$$K_k(s) \equiv \left. \frac{W_f(s)}{E(s)} \right|_k = \frac{k_{d_k}s^2 + k_{p_k}s + k_{i_k}}{s} \quad (5.15)$$

where $E(s)|_k = N_{1_{cmd}}(s) - N_1(s)$. After substitution, we get the transfer function of N_1 over $N_{1_{cmd}}$ as

$$\left. \frac{N_1(s)}{N_{1_{cmd}}(s)} \right|_k = \frac{G_k(s)K_k(s)}{1 + G_k(s)K_k(s)} \quad (5.16)$$

Equation (5.16) is the CLTF for a two-spool engine with gain-scheduling. Each of the coefficients' k , a , and b in the equation is a function of N_2 and is interpolated between the corresponding model values of two adjacent nominal points. Further simplification of Eq. (5.16) will involve a lot of algebraic manipulation, even for this simple design problem. They are presented here to give the reader some appreciation of what a control engineer had to go through to design an engine control law without using a modern, affordable control design tool like Matlab®.

5.2.2 Integrator Wind-up Protection

The accel and decel controls just described incorporate “hard” limits on the control variable calculated by the set-point controller. These hard limits force *instant* switching between the set-point controller and the accel (or decel) schedule without taking into account of the integrator wind-up (IWU) issue in the set-point controller. IWU occurs when the linear controller output is limited by a transient schedule, an actuator magnitude limit, or an actuator rate limit. Each of these IWU scenarios causes the integral control value to keep increasing as a result of a non-zero and same-signed feedback error. To avoid IWU, the integral control has to be frozen or trimmed “down” when an IWU scenario is active. Different techniques of trimming the integral control have been studied and applied. This concept has

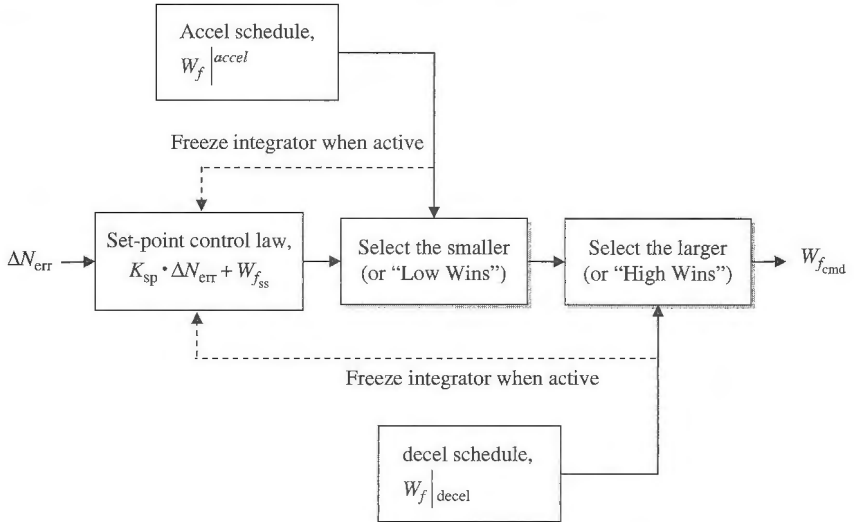


Fig. 5.6 Concept of the complete transient control logic with wind-up protection.

even been investigated for multivariable control problems [2]. However, the simplest approach to avoiding the IWU problem is to “freeze” the integrator (or the integral control) when the linear controller output is constrained. Although this might not be the most elegant approach, it has proven effective in numerous engine control applications because the initial buildup of the integral control value before it is frozen is relatively small and not large enough to cause sudden jumps.

Applying this integrator-freeze logic to IWU protection, we get a more robust transient control logic as shown in Fig. 5.6. This more robust logic combines IWU protection with accel/decel controls.

5.2.3 Deceleration Surge

Although this concept of deceleration surge seems “funny,” it actually happens in some aircraft engines at high altitudes. You might ask why the decel surge occurs. A simple answer is spool *mismatch*. Just as in an accel surge, a decel surge can occur when the pumping capacities between the LP compressor and the HP compressor are different, hence causing a sudden pressure rise at the exit of the compressor. Typically on a two-spool aircraft engine, the LP spool surges in accel, and the HP spool surges in decel as illustrated in Fig. 5.7 [3].

A common approach to avoiding the decel surge is to open the interstage bleed valve during a decel, where the rotor speed mismatch is large enough to have exceeded a safe operating limit.

5.3 Acceleration-Based Transient Controller Design

The second approach to controlling engine transients is to modulate engine fuel flow so as to produce a desired acceleration (or deceleration) in engine speed.

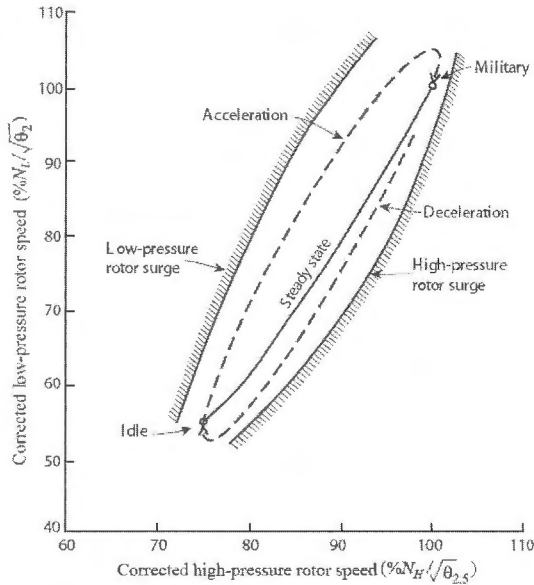


Fig. 5.7 Dynamic rotor matching limits for a typical turbojet engine (from [3], reprinted with permission).

This acceleration-based control is commonly referred to as the *N-dot control* because it tries to control the rate of change of the engine speed to the desired value during an accel or decel. The motivation for N-dot control is two-fold: 1) consistent accel and decel performance in shaft speed, temperature, pressure, and power for an engine even when it has aged or deteriorated and 2) consistent accel and decel performance for all engines in the fleet even with engine-to-engine variations caused by manufacturing and material tolerances. These are strong arguments for the proponents of N-dot controls.

The concept of N-dot control is described in Sec. 5.3.1. This is followed by a transient control design example for a two-spool engine in Sec. 5.3.2. Then in Sec. 5.3.3, the advantages of the N-dot control over the scheduled control will be summarized.

5.3.1 N-Dot Control Concept

N-dot control, as the name implies, uses the rotor acceleration (or $\dot{N}$) as the controlled variable in the feedback. The reference command of this feedback control loop is a desired rotor acceleration. N-dot is defined as the rate of change of the shaft speed of an engine, and typically it is the speed of the gas generator or the HP shaft. Hence, for a turbofan engine, N-dot is the rate of change of N_2 , and for a turboshaft engine, it is the rate of change of N_g or N_1 . Figure 5.8 illustrates this concept of N-dot control loop for an engine.

Note that N-dot is not directly measured on an engine, so it is “calculated” from the shaft speed N by differentiating the speed over time. The differentiator

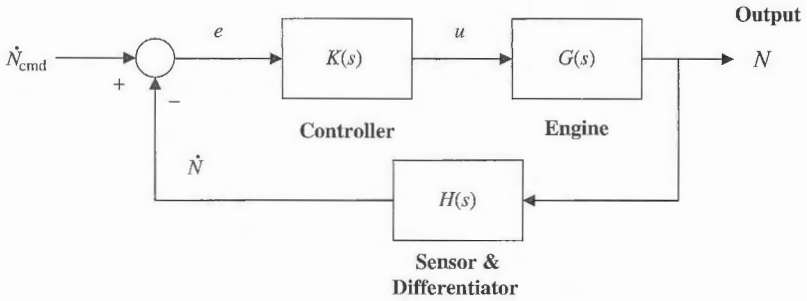


Fig. 5.8 Concept of $\dot{N}$ control.

in the feedback loop is usually implemented as a digital filter to smooth out the jitter in the calculated $\dot{N}$ from each time step.

To eliminate the steady-state or tracking error, we can use an integral control (in addition to the proportional control) in the $\dot{N}$ control loop. This PI control law for the $\dot{N}$ control loop is shown in Fig. 5.9. Note also that the PI control law in this figure can be replaced by other control laws with no significant degradation in the performance of $\dot{N}$ control. The two commonly used alternative $\dot{N}$ -dot control laws are 1) a pure integral control and 2) a compensator followed by an integral control.

The OLTF of the $\dot{N}$ -dot control loop is

$$\left. \frac{\dot{N}(s)}{E(s)} \right| = sG(s) \left(\frac{k_p s + k_i}{s} \right) \quad (5.17)$$

After canceling the s term in the numerator and the denominator, we get

$$\left. \frac{\dot{N}(s)}{E(s)} \right| = G(s)(k_p s + k_i) \quad (5.18)$$

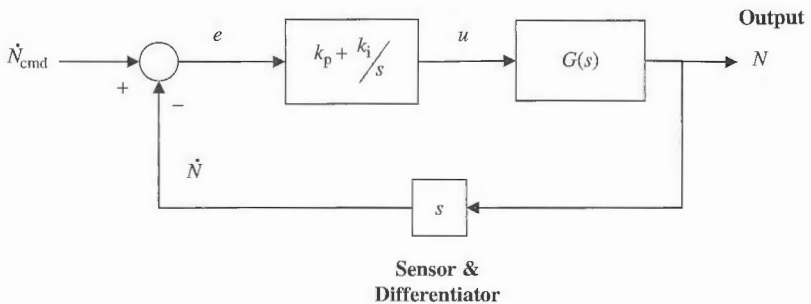


Fig. 5.9 $\dot{N}$ -dot control loop with a PI control law.

From this equation we know that the N -dot control loop has a guaranteed stability because the phase lag of the OLTf at high frequencies is less than 90 deg and the phase margin of the loop is at least 90 deg. Well, this is true theoretically, but practically, N -dot control can become unstable when there is significant computational delay; furthermore, Eq. (5.18) does not consider fuel control and actuator dynamics, which can increase the closed-loop system order and phase lag.

The desired rotor acceleration $\dot{N}_{cmd}$ in Fig. 5.9 is usually not a constant value; instead, it is scheduled as a function of the rotor speed N in a staircase function to account for nonlinearities and to provide a greater flexibility in scheduling engine acceleration. Transition between a set-point controller and an N -dot controller deserves careful consideration. The transition is usually handled by one of two approaches: 1) switching between the set-point controller and the N -dot controller in a way similar to the schedule-based control for engine transients as described in the preceding section and 2) cascading (nesting) the set-point control loop with the N -dot control loop to form inner- and outer-loop controls. These two approaches are shown in Fig. 5.10.

The switching controller approach is preferred for two reasons: 1) the two controllers, the set-point controller and the N -dot controller, can be designed separately; 2) the (linear-system) stability properties of the two controllers can be preserved, unlike in the cascade controller approach where the stability of the inner loop will be affected by the stability of the outer loop and vice versa.

We should note that N -dot control alone cannot prevent the engine from surge, overtemperature, or burner blowout. In fact, N -dot control has a stronger tendency to drive the engine into these operational limits because it is singly focused on attaining the desired rotor acceleration regardless what the state of the engine is in, that is, it does not “care” whether the engine is in a deteriorated or even damaged state. Figure 5.11 shows the maximum acceleration potential of a turbojet (or turbofan) engine, and it shows the various operational limits acting as the bounds of the safely achievable acceleration and deceleration rates [3].

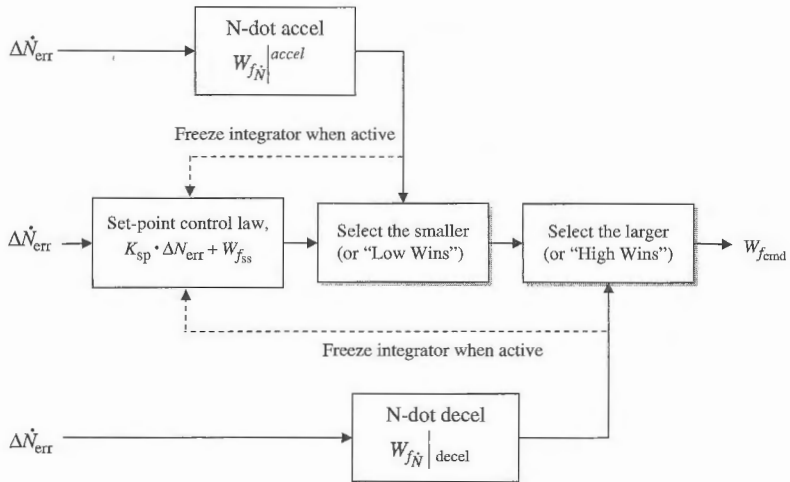
To prevent N -dot control from driving the engine into these problems, accel and decel schedules are put in place to bound the N -dot control authority. These schedules basically work in the same way as in the scheduled-based transient controllers (see Fig. 5.6). The schedule-based limit protection concept for N -dot control is shown in Fig. 5.12 for the case of a cascade controller.

5.3.2 N -Dot Control Design for a Two-Spool Turbofan Engine

Using the two-spool engine example described in Chapter 4, we can design a PI control law for the N -dot control loop shown in Fig. 5.9 for the engine TF from fuel flow to N_f . Although N -dot control is usually applied to the HP shaft speed, here we are using the LP shaft (or fan) speed with the same TF that we used in Chapter 4 to illustrate a design approach of N -dot control. Substituting the engine-transfer function of Eq. (4.6) into Eq. (5.18) we get the following OLTf:

$$\frac{\dot{N}(s)}{E(s)} = \frac{2.777(0.07s + 1)(-1.35s + 10)}{(s^2 + 12.3s + 10)} \quad (5.19)$$

Switching Controllers



Cascade (or Nested) Controllers

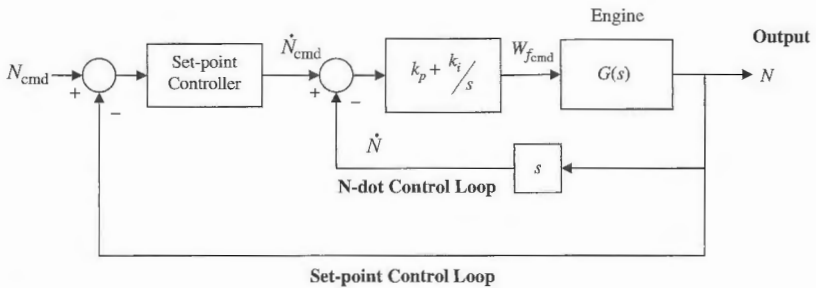


Fig. 5.10 Two approaches to transitioning between a set-point controller and a N-dot controller.

where the zero of the PI control law is located at -14.29 r/s. The Root Locus plot and the Bode plot of Eq. (5.19), generated by the Matlab® "sisotool" is shown in Fig. 5.13. The closed-loop step response of this N-dot control loop is shown in Fig. 5.14. Note that the magnitude of the output variable N_1 increases linearly with time because the N-dot reference command value stays constant.

If we implement the differentiating filter as $s/(0.002s + 1)$, that is, the calculated derivative value is slightly smoothed by a low-pass filter, then the calculated N-dot is compared with the reference command in Fig. 5.15. Note that the calculated value is still jittery, although it has been improved a lot. The oscillations in the calculated N-dot are caused by numerical instabilities in the digital computer simulation. This result points out the need for a more sophisticated filter to produce a smoothed N-dot value in the feedback loop.

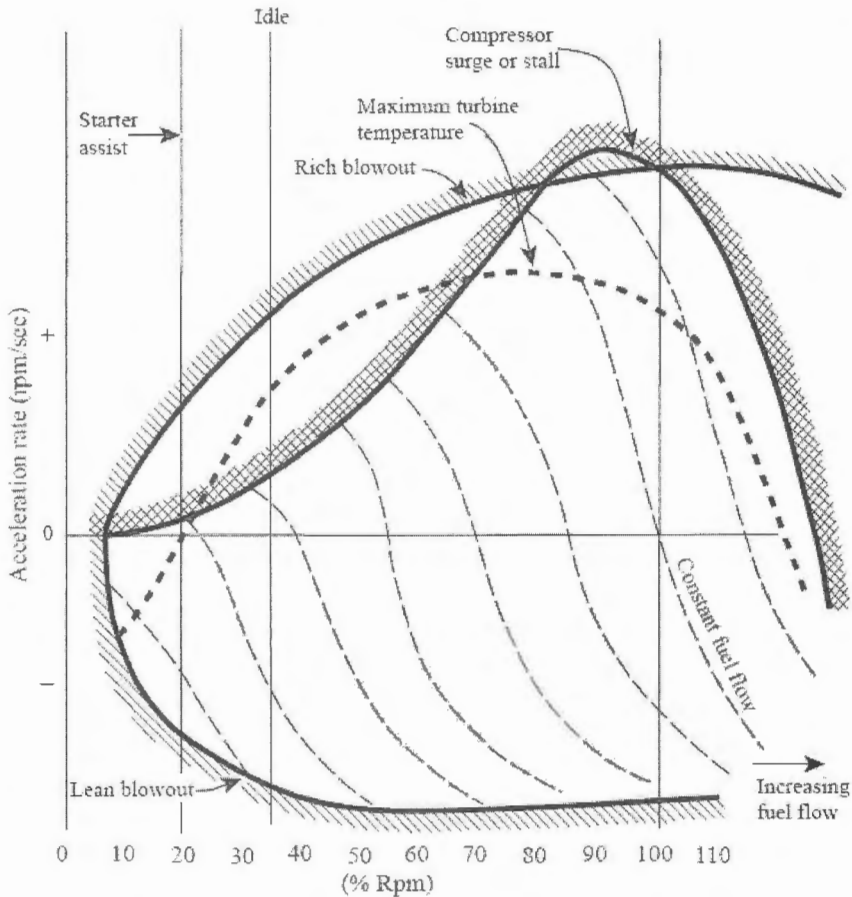


Fig. 5.11 Maximum acceleration potential and limits for a turbine engine (from [3], reprinted with permission).

5.3.3 Comparison Between Scheduled and N-Dot Controls

N-dot controls have been applied to turbofan and turboshaft engines. The advantages of N-dot control are summarized as follows: 1) consistent transient performance for all of the engines in the fleet regardless of their manufacturing variations, ages, and deteriorations; 2) direct control of the rotor acceleration produces more responsive engine transients; and 3) good stability property for the control loop, hence allowing higher-gain control laws. Knowing these advantages, however, we need to know its limitations as well. The most serious limitation of N-dot control is its tendency to drive the engine into stall, surge, or flame-out; therefore, some kind of accel and decel schedules, like those in the schedule-based controls, are almost necessary to protect the engine from these potentially catastrophic events. The need for transient schedules adds time and cost in the development of N-dot controllers.

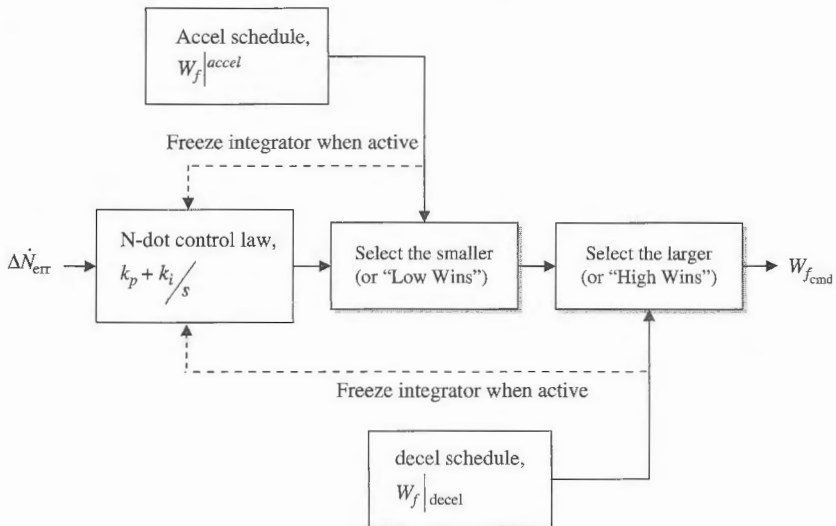


Fig. 5.12 Accel and decel schedules superimposed on N-dot controller.

A second limitation of N-dot control is that the accel time in engine transients is not minimized but scheduled. In a schedule-based control, the acceleration time and the deceleration time are minimized because these transients behave almost like a bang-bang control, which switches between the maximum and the minimum control limits similar to the accel and the decel schedules. Minimum-time accel or

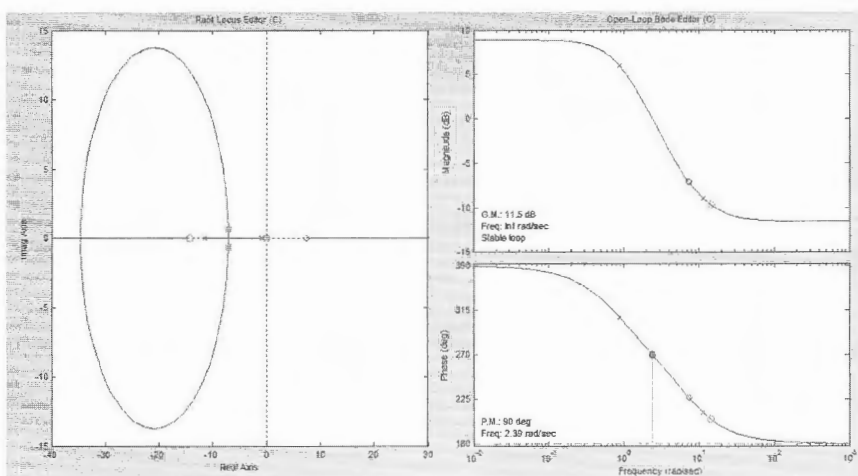


Fig. 5.13 Root Locus and Bode plots of an N_1 -dot control design for a two-spool engine.

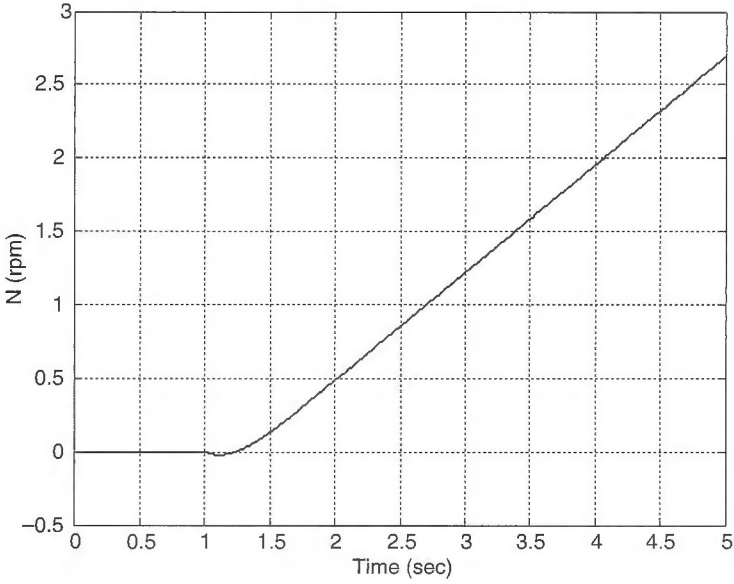


Fig. 5.14 Simulated response of N_1 with an N -dot control.

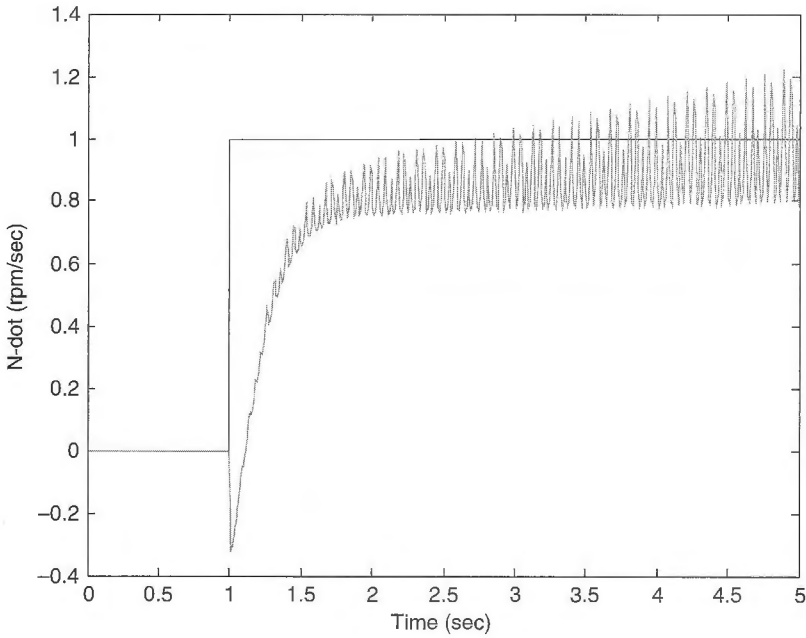


Fig. 5.15 Comparison between the calculated and low-pass-filtered $\dot{N}_1$ (in green) and the $\dot{N}_1$ reference command (in blue). (See also color figure section at the back of the book.)

decel might not be the most desirable transient operating scenario for a commercial engine because of the high mechanical and thermal stresses imparted on the hot-section components whose service lives are reduced. We will learn in Chapter 9 that minimum-time accel increases turbine temperature beyond what is necessary to meet aviation regulatory requirements for acceleration; hence, it reduces the lives of engine hot-section components. On the other hand, minimum-time acceleration is almost a design requirement for military fighter engines and attack helicopter engines for fast response, agility, and survivability. In fact, when flight safety concern becomes a high priority, even commercial aviation engines have a need to accelerate from idle to TO power in minimal time. This need is most apparent in a go-around attempt during a landing approach. Another example is a situation where an emergency power boost is needed when the aircraft encounters a downdraft (or wind shear) during TO or landing.

5.4 Controller Design for Limit Protections

In this chapter, we have learned the technique of using accel and decel schedules to protect engines from undesirable operating conditions like stall, surge, exceeding the turbine temperature limit, and combustion flame-out. Besides these normal transient limits, a gas turbine engine can run into other potentially damaging conditions, and it is the responsibility of the controller to prevent such conditions from occurring by imposing additional limits. Unlike the accel and decel schedules, which are activated in almost all moderate-to-large transients, these additional limits are only activated when the engine is operating in certain regions of envelope, or when there is an excessive change in engine performance that was not anticipated during design, or when the engine is under a severe abnormal or damaged state. Among these undesirable conditions, the following four pose most frequent concerns in engine control design:

1) *Exceeding the maximum LP-shaft physical speed N_1 limit:* Overspeed protection generally addresses the concern of disk burst or blade break-off. The most common cause of this condition is when the afterburner nozzle area opens too fast or too much, while the speed governor (or the set-point controller at TO power) does not act fast enough to reduce the speed by reducing the fuel flow. Another possible condition where overspeed can occur is when the afterburner flames out. For nonafterburning engines, this condition can occur when the engine is deteriorated or damaged such that the speed match between the LP and the HP shafts has deviated far from the design relationship.

2) *Exceeding the maximum HP-shaft physical speed N_2 limit:* This condition can occur when the LP-shaft goes into overspeed for whatever reason. Typically, when the exhaust nozzle is open too widely, overspeed conditions are possible. Other conditions for this to occur are failed speed sensor and failed fuel control unit. One thing needs to be said about the overspeed protection: This is more critical for the engines designed before mid-1980s because these engines are controlled either with a pure hydromechanical control (HMC) unit or with a combination of a single-channel electronic control unit (ECU) and a hydromechanical backup control unit. To protect the HMC-equipped engine from overspeed, we usually put a physical stop in the fuel metering valve stroke within the HMC so that the

fuel flow would not exceed an allowable value. For modern, dual-channel electronic controlled engines, the likelihood of the overspeed condition resulted from single sensor or actuator failure has been drastically reduced.

3) *Exceeding the maximum compressor exit static pressure P_3 limit*: This protection addresses the concern of combustor liner burst. This condition can occur when the engine is in overspeed or when the inlet total pressure exceeds the design limit (see Fig. B.2).

4) *Exceeding the maximum EGT limit in turbines*: This protection addresses the concern of the erosion of turbine blades caused by high temperature. This condition can occur when the engine is overfueled, or when there is excessive extraction of bleed air or mechanical power.

For the engines operating at high pressure ratios, typically those of military fighters, other operational limits might need to be considered. One of these limits is the maximum compressor exit temperature T_3 limit. This limit protects the engine from excessive thermal differential stress in the rim of the latter compressor stages. This condition can occur when the engine is in overspeed or when the inlet total temperature exceeds the design limit. Other operational limits are afterburner light-off limit, engine airstart limit, and structural limits.

Common practices in the design of these limit-protection controllers are to apply a PID control law to the feedback error, which is calculated by subtracting the limiting value from the engine output variable. The output variable being protected is called a control-limiting variable. The derivative control in the PID control law provides anticipatory actions based on the rate of change of the limiting variable as it approaches the limit. Limit-protection controllers are traditionally called *topping governors*.

Again, we are confronted with the integration problem of making the topping governors work seamlessly with the primary-loop controller. The primary controller can be either a set-point controller or an N-dot controller. We resolve this integration problem by incorporating the topping governor output as a "trim" or "correction" to the command value of the primary control loop. The reason for doing so is to avoid the limit-cycle phenomenon when the two controllers "fight" against each other to win the ultimate authority in providing a value for the control variable. Hence, for the N_1 primary control loop, the block diagram in Fig. 5.16 illustrates the control logic for a limit-protected primary control loop. Similarly, for a cascaded primary control design, the limit protection logic is shown in Fig. 5.17, where the inner loop is for N-dot control.

We normally make the topping governor active only on one side of the limit. Another way to describe it is that the control authority of the topping governor (subscript TG) is limited to positive values, or $\Delta N_{TG} \geq 0$.

Additional limit-protection topping governors can be handled in the same way, that is, using the topping governors to trim the command value of the primary control loop (N_1 or EPR). However, one subtle detail about the integration of these various governors must be pointed out: Because multiple topping governors can be active at the same time (which would correspond to a situation where multiple limits have been violated), we can either let all of the active governors make their own contributions to trim the command value (we call this approach "equal voting"), or we can let one of the governors

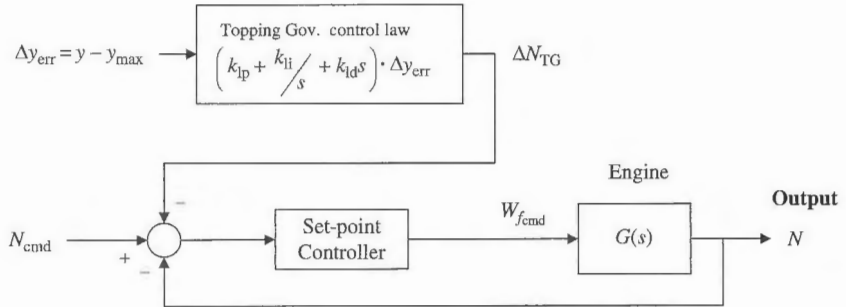


Fig. 5.16 Integrating a topping governor with the primary control loop.

take over the lead role and provide a single trim to the command value (we call this approach “leader-follower”). These two approaches are illustrated in Fig. 5.18.

Both approaches have the same effect of having only one topping governor being active *eventually* because as soon as the engine is cleared of an operating limit the output of the corresponding topping governor is reset to zero in the equal-voting approach. The equal-voting approach has the ability to move the engine away from operating limits faster as multiple governors can work together to achieve this purpose. On the other hand, the leader-follower approach selects only one topping governor to be active at any given time, and this governor automatically ensures that all other limits are not being violated.

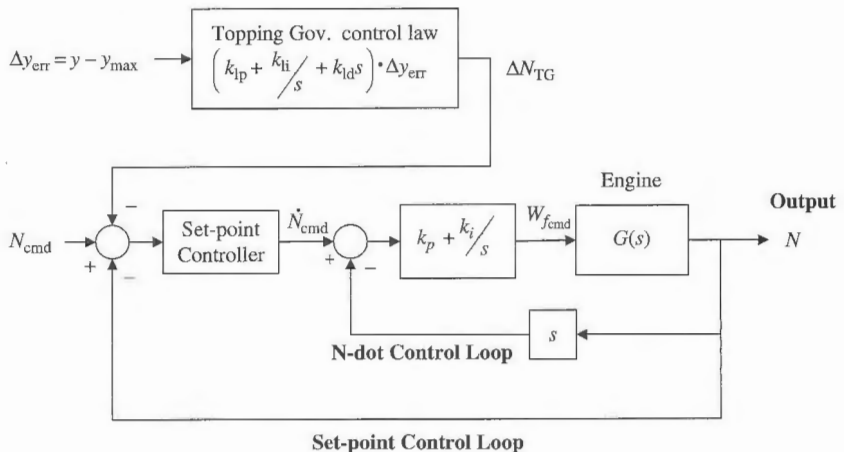


Fig. 5.17 Limit protection logic for a cascaded primary control loop.

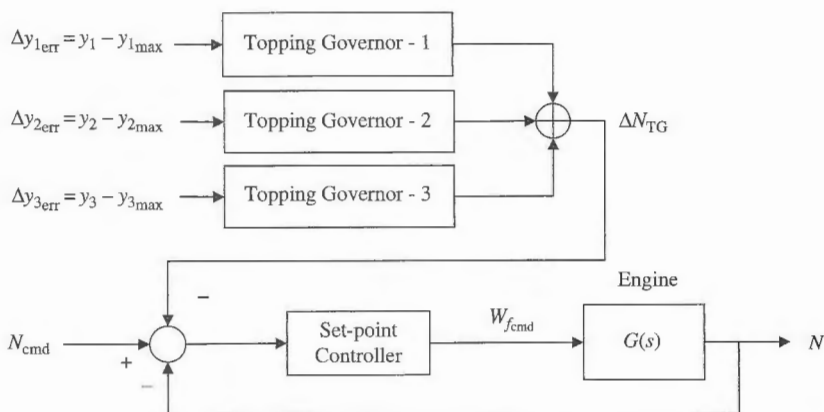
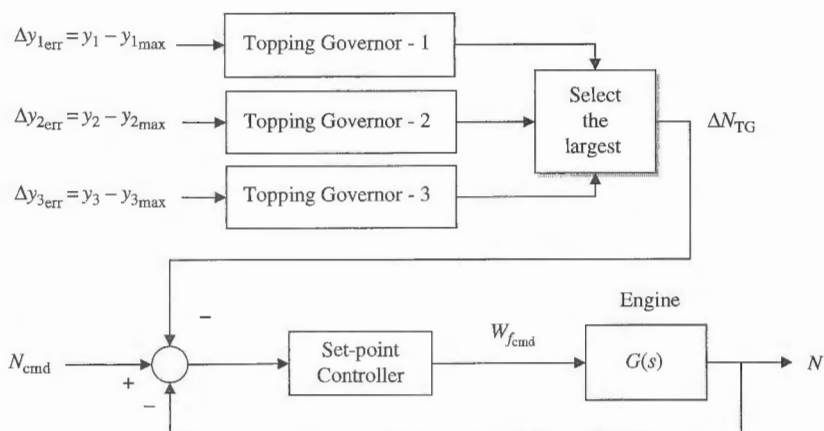
Equal VotingLeader-Follower

Fig. 5.18 Two approaches to integrating multiple topping governors.

5.5 Rotorcraft Engine Considerations

5.5.1 Power Management

Power management is an important control function for a multi-engine rotorcraft. In a multi-engine helicopter, power can be provided from one, two, or all engines; however, the major benefit of multiple engines is fail-safe operation. When all of the engines are operating normally (i.e., not deteriorated or damaged), sharing rotor load among all of the engines equally is desirable because all engines will respond roughly at the same time after a disturbance or during a transient. Another reason for sharing the load among all engines is

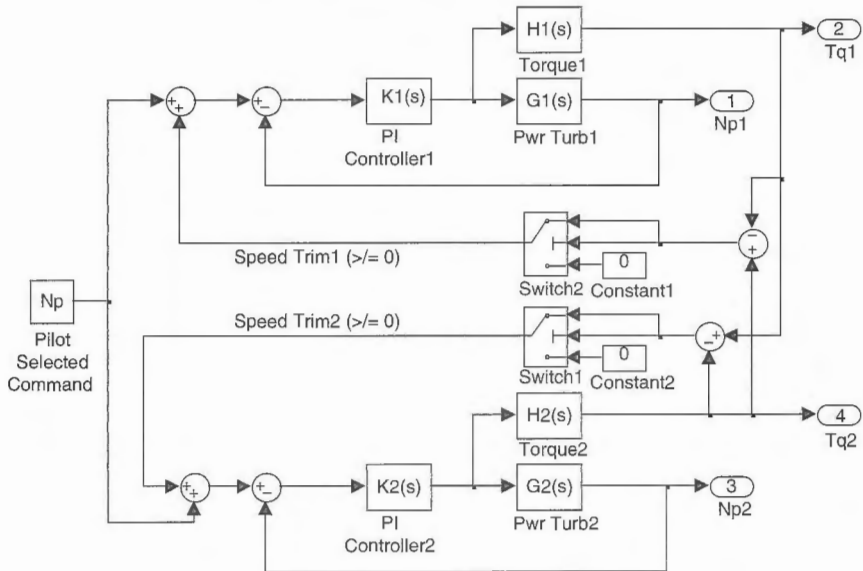


Fig. 5.19 Torque-matching logic for load sharing.

to recover the power promptly when one or more engines fail to deliver the expected power. This is particularly important when a failed engine is deteriorated or damaged.

An effective way of sharing the load is to match the torques measured at the turbine shafts of all engines. For a two-engine helicopter, this torque-matching logic for load sharing is shown in Fig. 5.19. Note that an *upward-only* trim is added to each of the power turbine feedback loops to prevent the failure of one engine from reducing the power of the good engine.

5.5.2 Adaptive Acceleration Control and Surge Recovery

In Sec. 5.3.1, the potential danger of $\dot{N}$ control driving the engine into surge was discussed. This adverse effect can be marginalized if the $\dot{N}$ command (or $\dot{N}$ -dot schedule) is corrected by the compressor discharge pressure (P_{CD} or P_3), that is, $\dot{N}/P_3$. This rationale is the same as the use of fuel ratio unit ($RU \equiv \dot{W}_f/P_3$) as the control variable for surge recovery as explained in Sec. 4.4.2.

When surge occurs, the engine control has a high priority to recover from the surge smoothly; furthermore, when surge margin is lost as a result of engine deterioration or damage, the engine control wants to compensate for the lost margin by taking a more conservative acceleration. Both of these have been accomplished in an adaptive fuel control study conducted in the 1980s [4]. The adaptive control employed a closed-loop control for $\dot{N}/P_3$. In addition, a surge detector was implemented in the ECU. The surge detector logic is based on discriminating an abnormally large negative rate of change of the compressor

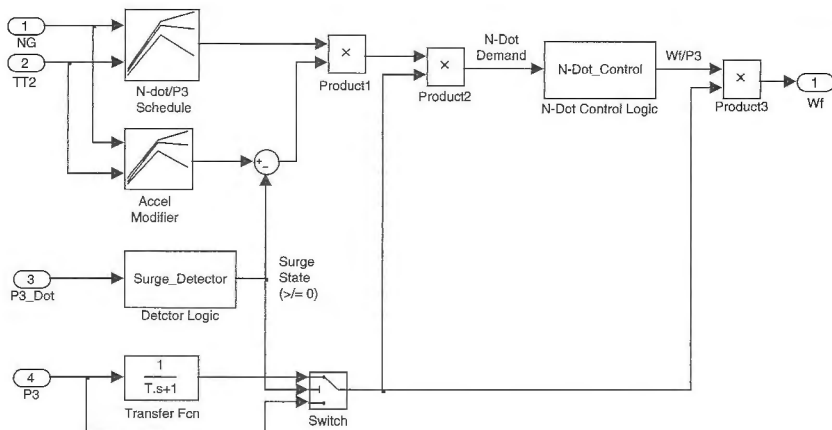


Fig. 5.20 Adaptive acceleration control logic.

discharge pressure, that is, $-\dot{P}_3 \geq c > 0$. When this excessive decrease in the pressure is detected, a flag to indicate surge margin loss is set, and the fuel flow rate is suppressed even when the pressure starts to recover from the bottom value [5].

To avoid repeating the surge condition, the surge flag also decreases an accel modifier. The modifier is multiplied to the N-dot schedule, hence effectively reducing the N-dot command where the surge was detected. This adaptive acceleration control logic is shown in Fig. 5.20 [5].

References

- [1] Khalil, H. K., *Nonlinear Systems*, 3rd ed., Prentice-Hall, Upper Saddle River, NJ, 2002, Ch. 12.
- [2] Mattern, D., "A Comparison of Two Multi-Variable Integrator Windup Protection Schemes," AIAA Paper 93-3812, Aug. 1993.
- [3] Sobey, A. J., and Suggs, A. M., *Control of Aircraft and Missile Powerplants*, Wiley, New York, 1963.
- [4] Howlett, J. J., Morrison, T., and Zagaranski, R. D., "Adaptive Fuel Control for Helicopter Applications," *Journal of the American Helicopter Society*, Oct. 1984, pp. 43–54.
- [5] Achgill, D. M., and Zagaranski, R. D., "Adaptive Fuel Control Testing," *American Helicopter Society 41st Annual Forum Proceedings*, Ft. Worth, TX, May 1985.

6 Control System Integration

In preceding chapters, we have discussed the dynamic interactions of complex aero-thermal-mechanical effects in a gas turbine engine; we have also discussed different approaches to the design of engine controllers for the three primary control functions, that is, set-point, transient, and limit-protection controls. In this chapter, we will cover the integration issues facing engine control designs. These issues include power setting, transient schedules, control modes, and engine accessories. Near the end of the chapter, we will use two simplified engine models (a one-spool engine and a two-spool engine) to illustrate how to synthesize multiple set-point control laws and transient schedules into a coherent controller that is capable of rapid accel and decel between idle and TO power at a selected engine ambient (or flight) condition. The design examples are intended for the reader to gain some appreciation of real-world design challenges. The examples can be readily extended to the entire engine operating envelope covering a wide range of altitude and Mach number.

6.1 Quick Review

We are now familiar with the basic elements of engine control logics. These elements include the following: 1) a primary control loop (or set-point governor) to regulate engine thrust through the controlled variable N_1 or EPR, 2) a reference command representing the pilot's power request for the primary control loop, 3) an acceleration schedule and corresponding transient control logics, 4) a deceleration schedule and corresponding transient control logics, 5) several limit-protection control loops (or topping governors) for critical engine performance variables like EGT, P_3 , N_1 , and N_2 , etc. These elements are illustrated in Fig. 6.1 as different blocks in the control logic diagram.

In the figure, the pilot's power request is the reference command for the primary control loop. The reference command is calculated based on the throttle position set by the pilot. This calculation is called *power setting logic*, and it is divided into three basic logic paths: TO power, idle power, and part power. Power setting is one of the most important engine control logics because it affects both flight safety and operational characteristics. The concern about flight safety is usually with the accel time from a low power to the TO power in a go-around (or aborted landing), while the concern about operations is with the pilot's ability to

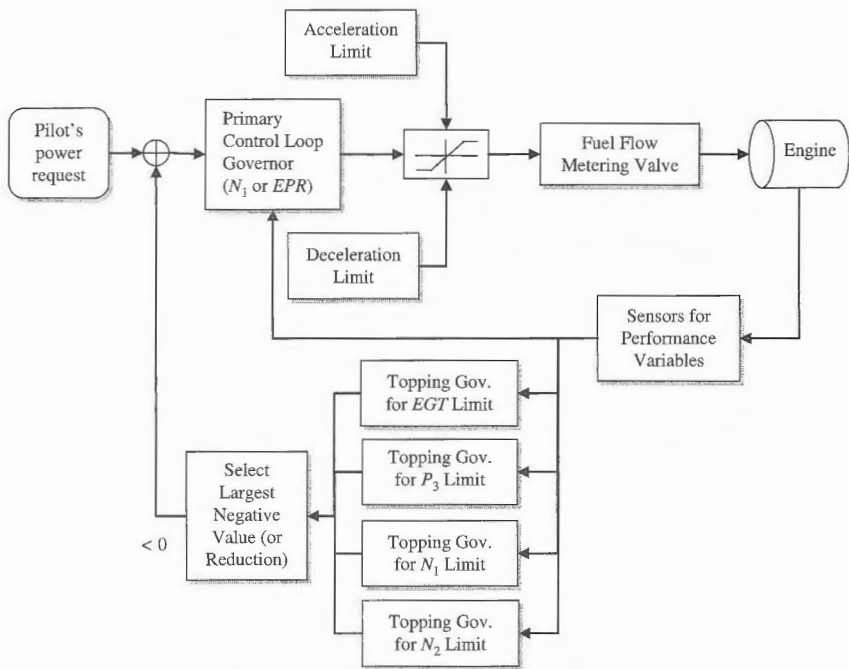


Fig. 6.1 Basic elements of a gas turbine engine control logic.

modulate thrust fast and accurately during TO and landing. Power setting logics are discussed in the next subsection.

6.2 Power Setting

Modern aircraft engines are designed to give the pilot an ability of unrestricted throttle movement over the entire engine operating envelope. Unrestricted throttle movement means that the pilot, after engine start, can move the throttle to any desired position at any time without causing engine surge, overtemperature, overspeed, or any other operational limit exceedance. This is partly accomplished by what we now know as the transient and the limit-protection controllers with the help of the accel schedule, the decel schedule, and various operational limits. The primary purpose of a throttle is to allow the pilot to change the power or thrust level from one set point to another; hence, we must know how to calculate the reference command value corresponding to each throttle position.

The reference value for the set point of the primary control loop at any given throttle position is sensitive to the inlet condition of the engine. The need for varying the reference value with the inlet condition can be understood by considering the following requirement: the same aircraft with the same gross weight needs to run its engine(s) at a higher speed (or pressure ratio) on a hot day than on a cold day to produce the same amount of thrust TO. Because of this dependency of the inlet condition, we must make the power setting reference value at each

throttle position as a function of the ambient pressure, the ambient temperature, and the aircraft Mach number. These functions have historically been implemented as look-up tables in an electronic control, or as the surface contour of a mechanical device called cam in a hydromechanical control. The functions are traditionally called *power setting schedules*. Typical power setting schedules for aircraft gas turbines will be discussed in the following subsections.

6.2.1 Takeoff Power Schedule

The TO power schedule is most commonly expressed as a function of the ambient temperature and the altitude. The TO schedule is made up of two separate schedules: the flat-rate schedule and the temperature schedule as illustrated in Fig. 6.2. As the name implies, the flat-rate schedule produces a constant thrust (or power rating) regardless of the ambient condition. The temperature schedule, on the other hand, is provided to prevent the engine from exceeding the temperature limit of the high-pressure turbine. Note that the flat-rate schedule can be applied to either the physical speed N_1 or the corrected speed N_{1c} . In the case of the corrected speed, it is indeed a “flat” line and corresponds to a constant corrected thrust. Also note that the TO schedule for the EPR control loop is effectively the same as the corrected N_{1c} schedule as indicated by the red lines in Fig. 6.2a.

To generate the flat-rate schedule, we usually run the steady-state cycle deck of the minimum-performing engine to a desired, constant, maximum thrust while varying the ambient temperature and altitude (or ambient pressure) independently. To generate the temperature schedule, we again run the steady-state cycle deck to match to a constant, maximum turbine inlet blade temperature while varying the ambient temperature and altitude independently.

To combine the flat-rate schedule and the temperature schedule, we select the lower value of the two, in a way similar to how the accel schedule is selected between the surge schedule and the turbine temperature schedule, to ensure that no limit will be violated. The combined TO schedule is illustrated in Fig. 6.3 (see also Fig. B.15 and Fig. B.16). It is the basis of the TO power setting table in the pilot’s flight manual.

The TO power for a military fighter engine is commonly called the *military* power, to differentiate it from the *maximum* power, which corresponds to the sum of the thrust produced at the military power plus the extra thrust produced by the full afterburner (that is, the afterburner is running with the maximum fuel flow).

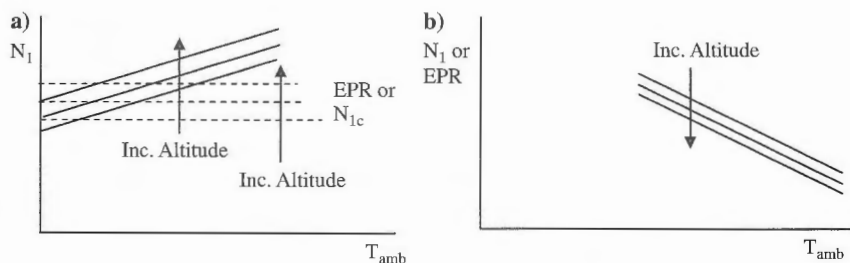


Fig. 6.2 (a) Flat-rate schedule. (b) Temperature schedule. (See also color figure section at the back of the book.)

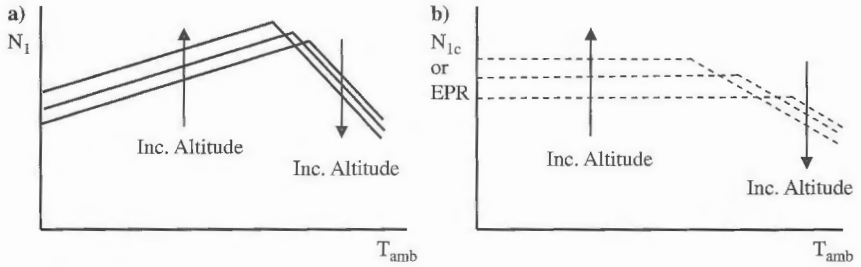


Fig. 6.3 TO schedules after combining the flat-rate and the temperature schedules.

6.2.2 Idle Power Schedule

It is common for an aircraft engine to have two idle schedules: one is the *ground idle* schedule, and the other is the *flight idle* schedule. The ground idle schedule sets the reference value of the minimum steady-state power for an engine. This reference value is a compromise among three conflicting objectives: 1) to reduce the acceleration time in a snap acceleration from idle to TO power, 2) to reduce the wear of aircraft brakes during ground maneuver like taxiing, and 3) to maintain minimal bleed air and accessory load. The result is a schedule that provides a near constant thrust at all altitudes (or, more precisely, for all engine inlet pressures). This means that the engine fan speed or pressure ratio increases with increasing altitude (or decreasing inlet pressure). The idle power schedule for a turbofan or turbojet engine is commonly expressed as a function of the ambient pressure as illustrated in Fig. 6.4.

Similar for the rationale behind the ground idle, the flight idle is a compromise among the following: 1) reducing the acceleration time from idle to TO power, 2) reducing the engine thrust during descent and landing approach, and 3) keeping the power high enough to maintain minimal bleed and accessory load. A typical flight idle schedule is set (or located) slightly higher than the ground idle.

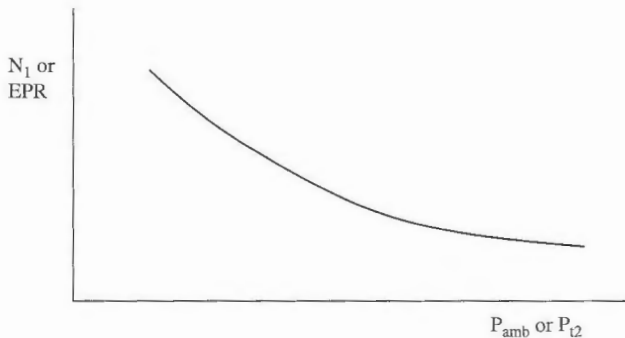


Fig. 6.4 Idle power schedule for a turbofan or turbojet engine.

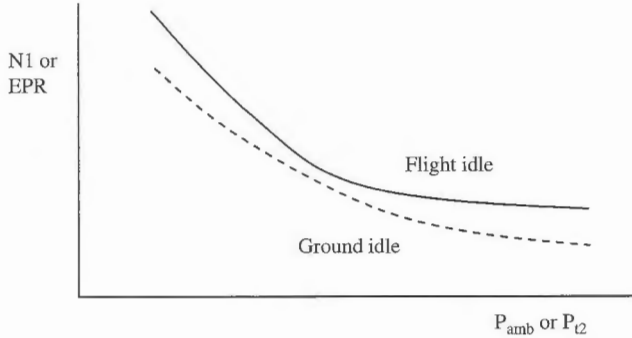


Fig. 6.5 Flight idle schedule for a turbofan or turbojet engine.

Switching between the ground idle and the flight idle schedules is determined by the aircraft's "weight on wheel" switch, which is triggered when the aircraft is on the ground. An example of the flight idle schedule, in comparison to the ground idle schedule, is shown in Fig. 6.5.

To generate the ground or flight idle schedule, we again run the steady-state cycle deck of the engine to match to a constant thrust value while varying the inlet pressure. Note that both idle schedules need to satisfy the partial requirements for the engine-airframe integration.

6.2.3 Part Power

Part power refers to the engine power that corresponds to any throttle position between the idle and the TO power. The throttle position is most often represented by power level angle (PLA). Now let us assume, without loss of generality, that the throttle position at idle corresponds to a PLA of 20 deg and that the throttle position at TO corresponds to a PLA of 100 deg, and then the simplest approach to calculating the part-power reference command value is to interpolate between the idle and the TO reference values with respect to PLA. This approach is expressed by

$$N_{1_{ref}} = N_{1_{idle}} + \frac{(PLA - 20) \cdot (N_{1_{TO}} - N_{1_{idle}})}{(100 - 20)} \quad (6.1)$$

where the fan speed N_1 has been used as the power setting variable, and the subscripts "idle" and "TO" represent the idle and the TO power, respectively.

A more complex approach to part-power setting is to "schedule" the power setting variable so that the engine performance follows a desired functional relationship of the throttle position (or PLA). The most desirable relationship for a turbofan or turbojet engine is to schedule N_1 or EPR so that the engine thrust is linearly proportional to PLA between idle and TO power. Similarly, the most desirable relationship for a turboprop or turboshaft engine is to schedule the gas-generator speed or pressure ratio so that the torque of the output shaft is linearly proportional to PLA between idle and maximum power.

For a turbofan engine, this relationship is expressed by

$$N_{1_{\text{ref}}} = f(\text{PLA}) \quad (6.2)$$

where f is a function of PLA, and the function causes the engine net thrust to be linearly related to PLA as

$$F_{N_{\text{PLA}}} = F_{N_{\text{idle}}} + \frac{(\text{PLA} - 20) \cdot (F_{N_{\text{TO}}} - F_{N_{\text{idle}}})}{(100 - 20)} \quad (6.3)$$

Because the net thrust is also a function of the power setting variable (speed or pressure ratio), it can be represented by a continuous, finite function g as

$$F_{N_c} = g(N_{1_c}) \quad (6.4)$$

where the subscript c stands for “corrected” or “referred” variable. After combining Eqs. (6.2–6.4) and simplifying the integrated equation, we get

$$N_{1_{c_{\text{ref}}}} = g^{-1}(F_{N_{\text{idle}}}) + g^{-1} \left\{ \frac{(\text{PLA} - 20) \cdot [g^{-1}(F_{N_{\text{TO}}}) - g^{-1}(F_{N_{\text{idle}}})]}{(100 - 20)} \right\} \quad (6.5)$$

where $g^{-1}()$ is the “inverse” function of g , that is, $N_{1_c} = g^{-1}(F_{N_c})$. Equation (6.5) can be graphically illustrated in Fig. 6.6.

Other forms of functional relationships can be defined, and the part-power setting reference value can be derived accordingly. For instance, we might want a *segmented* (or piecewise linear) schedule so that the power setting reference value of the cruise and climb segment is less sensitive to PLA for ease of manual throttle adjustment by the crew or operator. This kind of desensitized part-power schedule is particularly desirable for commercial engines. An example of this segmented part-power setting schedule is shown in Fig. 6.7 [1].

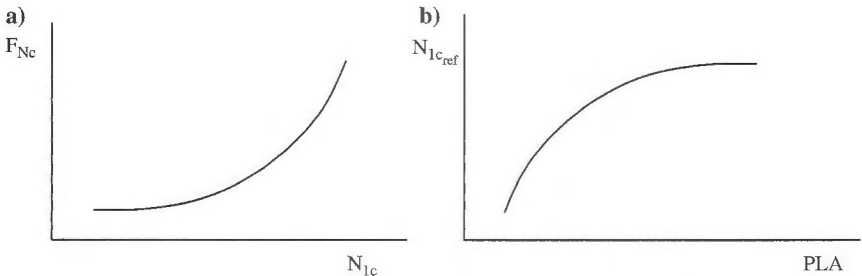


Fig. 6.6 (a) Thrust vs speed. (b) Speed reference vs PLA.

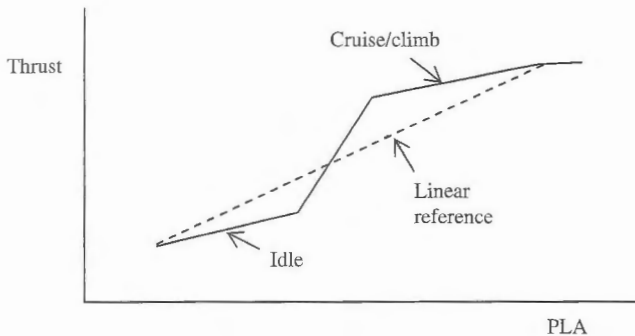


Fig. 6.7 Segmented part-power setting schedule.

6.2.4 Trimming Function and Autotrim

An automatic control system relies on sensors and actuators to accomplish the task of closed-loop control, in addition to the controller and control logics. For a turbine engine, these sensors and actuators are considered accessories, and they can be removed from one engine and installed on another engine. Installing a different serial-numbered accessory component on the same engine means that the new combination of the engine and the accessory component needs to be checked on a test stand (or test cell) to ensure that the overall system performance is acceptable, even though each individual accessory component has passed its own bench test. When an unacceptable performance discrepancy is found, we need a quick and reliable way of adjusting the control system so that the discrepancy is reduced or eliminated. This way of changing the control system characteristics is called *trim* or *trimming function*.

Historically, the trimming function was accomplished by turning a screw, a knob, or a rigging pin on the fuel control (or an inlet guide vane control) to compensate for unacceptable performance discrepancies. The screw or pin effectively changes the spring constant or the mechanical stop of a servocomponent, which, in turn, changes the performance of the closed-loop controlled engine. This process is manual, and the trimming function is called a *manual trim*. In addition to trimming a specific actuator or servocomponent, manual trim can also be applied to the power setting reference value of the primary control loop to account for engine-to-engine variations and engine deteriorations. The most common form of this trim is to increase or decrease the TO power so that all engines will produce the same amount of power or thrust when they are new (or just overhauled).

For a modern engine equipped with an electronic control, the trimming function is automatic when a different control unit is installed on the engine because the control has an inherent ability to compensate for actuator degradations and sensor drifts; however, most of these degradations and drifts can be detected by the built-in-test (BIT) logics in the electronic control. Furthermore, more advanced electronic controls incorporate self-calibration logics to adjust the engine power setting reference values when engine deteriorations are detected by an on-line monitoring capability.

We should not confuse trim with *de-rate*. De-rate is a deliberate reduction of engine power or thrust below the normal TO power to take advantage of a lighter aircraft TO weight. The magnitude of a de-rate is typically an order of magnitude larger than that of a trim. De-rate can be implemented as a fixed adjustment or a variable adjustment. In the case of a fixed adjustment, the TO flat-rate schedule is lowered, and this lowered schedule is stored in the nonvolatile memory of the electronic control unit; while in a variable adjustment, the flat rate is reduced by (or multiplied by) a factor entered by a pilot or operator through an interface with the electronic control. Some modern engines have even simplified the process of changing the engine rating at the TO power by using an electronic equivalent of mechanical bias called *power plug* or *data entry plug*.

6.3 Transient Schedules

Unlike the power-setting logic, the accel and the decel transient schedules are not directly related to the inlet condition of the engine, but are more dependent on the relationship between the LP and the HP shaft speeds, or more precisely, the mismatch between the two shaft speeds.

Work on engine fuel control parametric analysis at NASA established the basic concept of transient schedules [2]. The concept is illustrated in Fig. 6.8, where the accel fuel flow limit and the decel limit are simply expressed as linear functions of engine corrected parameters $(W_f/N)_c \equiv (W_f/\delta\sqrt{\theta})/(N/\sqrt{\theta})$ and $PR \equiv (P_{13}/P_{12})$, with minor adjustments for inlet condition variations. Note that these limits did not consider engine surge and blowout because the engine used in the analysis and simulation is J85, a single-spool turbojet engine.

This simplified, and yet intuitive, parametric relationship can be correlated to traditional fuel ratio unit and corrected shaft speed, and the resultant parametric relationship is shown in Fig. 6.9.

6.3.1 Acceleration Schedule

The acceleration schedule shown in Fig. 5.3 is a combination of two separate schedules: 1) the surge schedule and 2) the turbine temperature schedule. As

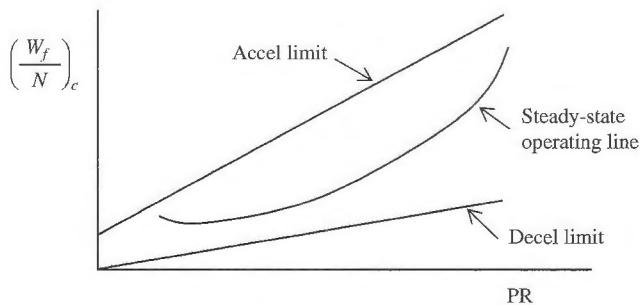


Fig. 6.8 Parametric relationships for accel and decel limits for a single-spool engine.

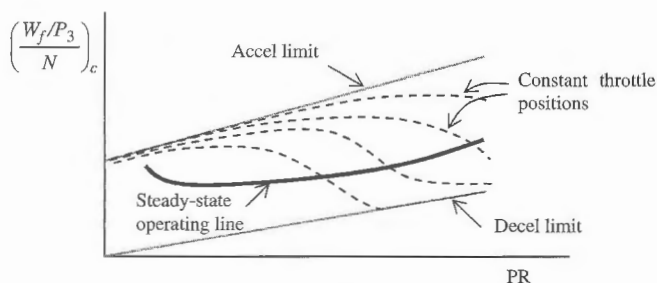


Fig. 6.9 Traditional parametric relationships for accel and decel limits.

illustrated in Fig. 5.1, the surge schedule is usually activated during the low-to-medium speed range, while the temperature schedule is activated in high-speeds (near the TO power).

The surge schedule is derived empirically by lowering the surge line with some safety margins. These safety margins are also called the surge margin *stacking*, which corresponds to a series of potential surge margin reductions caused by variations in engine operating conditions and or manufacturing tolerances. Figure 6.10 illustrates the concept of surge margin stacking. The acceleration schedule is hence set below all of these surge margin stacks. Also shown in the figure are the potential shifts in the engine's steady-state operating line. With all of these uncertainties and variations, the surge line has a tendency to move downward, and the operating line has the tendency to move upward. The net effect is a reduced operating region for acceleration (as indicated by the green region in the figure).

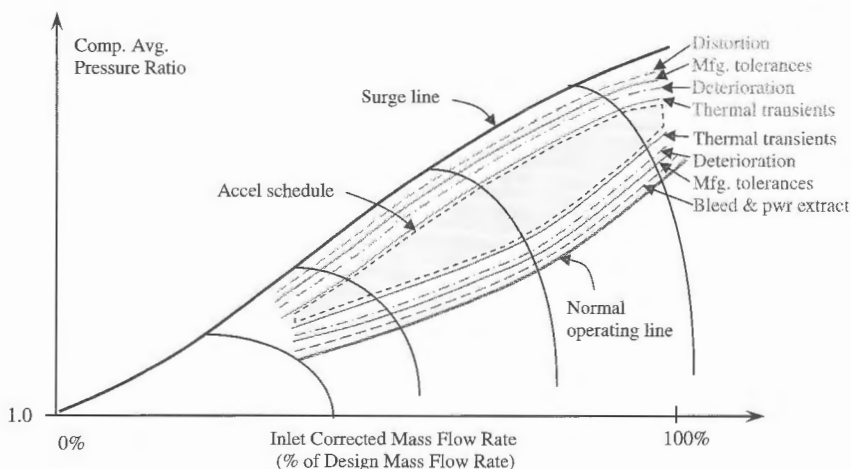


Fig. 6.10 Surge margin stacking and the reduced acceleration region (in green). (See also color figure section at the back of the book.)

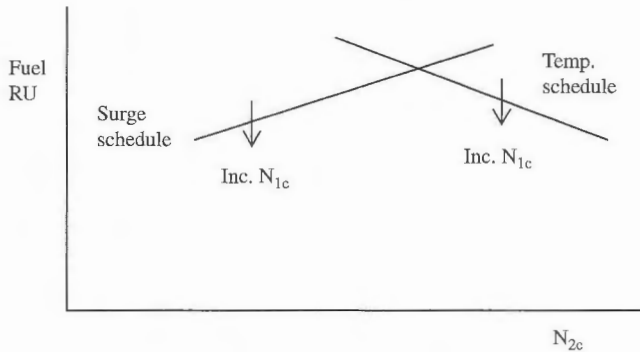


Fig. 6.11 Typical acceleration schedule.

The temperature schedule is usually derived by running the steady-state cycle deck to the constant, maximum value of HP turbine temperature T_{t4} while varying the gas-generator speed N_2 . Figure 6.11 shows a typical shape of the accel schedule.

6.3.2 Deceleration Schedule

The deceleration schedule shown in Fig. 5.3 is also a combination of two separate schedules: 1) the (lean) blowout schedule and 2) the minimum flow schedule. Figure 6.12 shows a typical shape of the decel schedule.

6.4 Control Modes

A control mode is often defined by the control variable in a control loop, for example, N_1 or EPR is the control variable in the primary fuel control loop, hence the N_1 or the EPR control mode. In this section, we will introduce a more general definition for control modes. These control modes, the so-called *system control modes*, can be thought of as a “super” set of the variable-specific definition of control modes.

In Chapter 2, the concept of *functional control modes*, which provide the three basic functions of an automatic control system, that is, set-point control, transient control, and limit-protection control, were discussed. On the other hand, *systems control modes* are related to the capability of a propulsion or power system as a whole. The capability goes beyond engine performance and considers engine operability and durability (or the usage and extension aspect of the engine’s life cycle). By this definition, there are at least three system control modes: *performance*, *operability*, and *durability*.

The performance control mode is interested in maximizing (or optimizing) the performance of an engine, and it includes all of the functional control modes that have been described earlier, such as N_1 , EPR, schedule-based control, and N-dot control. For the primary control loop, most gas turbine engines use the N_1 control mode; however, both the N_1 and the EPR control modes have proven effective in regulating engine power, although a potential disadvantage of EPR control is its

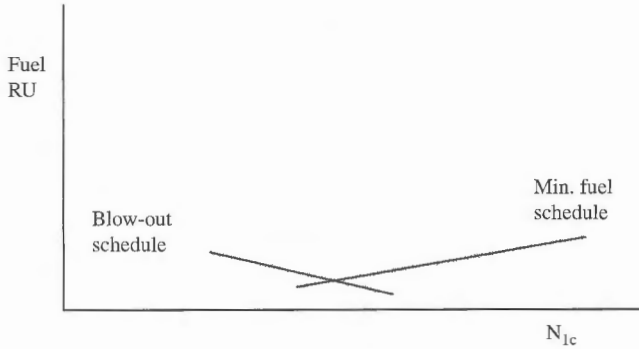


Fig. 6.12 Typical deceleration schedule.

higher susceptibility to erroneous reading from the inlet (total) pressure probe caused by FOD or icing. Because N_1 and EPR correlate very well with the engine thrust, theoretically, the calculated net thrust F_N itself can be a performance control mode, that is, the controlled variable for the primary control mode.

The operability control mode is interested in maintaining the stability of an engine during all kinds of aircraft maneuver and throttle excursions. We are typically interested in three types of stability in an engine: fan stability, compressor stability, and combustion stability. Operability engineers have used *margin* as an indicator for the relative stability between the operating condition of the engine and the stability limit. So the fan surge margin SM_f , the compressor surge margin SM_c , and the burner lean blowout margin BM_b are possible measures for these types of stability. They can be used as the controlled variables in the operability control mode.

The durability control mode is interested in preserving or extending the life of an engine; that is, it is concerned with the life usage, life remaining, and life extension of engine components, especially those components that are rotating or exposed to hot air or gases. The controlled variables in the durability control mode can be the cycle, the equivalent cycle, the total accumulated cycle (TAC), the maximum speed N_{max} , the dwell time at a certain temperature $t_{@T}$, the peak temperature T_{max} , or any other form of measure that accurately reflects the usage and the damage of an engine component. Table 6.1 summarizes the relationships among system control modes, functional control modes, and controlled variables.

Note that the controlled variables in the operability control mode (SM_f , SM_c , or BM_b) cannot be measured directly. These variables are the *unmeasured* output variables. In a state-equation formulation (see Appendix B), we can express these unmeasured, but controlled, variables by y_c in the following rearranged equations:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (6.6)$$

$$y_m(t) = C_m(t)x(t) \quad (6.7)$$

$$y_c(t) = C_c(t)x(t) \quad (6.8)$$

Table 6.1 Engine control modes and controlled variables

System control mode	Functional control mode	Controlled variable (or control loop)	Type
Performance	Set point	N_1 , EPR, or F_N	Linear
	Transient	N_1 , EPR, or $\dot{N}$	Nonlinear
	Limit protection	EGT (T_{t45} or T_{t5}), N_1 or N_2 , P_3 , or T_3	Linear with nonlinear switching
Operability	Set point	None	None
	Transient	SM_f , SM_c , or BM_b	Nonlinear
	Limit protection	SM_f , SM_c , or BM_b	Nonlinear
Durability	Set point	None	None
	Transient	TAC, $N_{\max}$, $t_{@T}$ or $T_{\max}$	Nonlinear
	Limit protection	TAC, $N_{\max}$, $t_{@T}$ or $T_{\max}$	Nonlinear

where the output variable has been split into two (types of) variables, the measured output y_m and the unmeasured and controlled output y_c . Note also that the engine thrust, the stability margins of critical engine components, and the component damage (or remaining life) all fall in this category of unmeasured but controlled outputs depending on which system control mode is active in the control system.

Before we leave this section on engine control modes, an important concept about the control mode is worth noting, that is, the controlled variable in a control mode can be substituted by another variable if this second variable provides a better correlation of the first variable. The second variable is particularly of interest to us if it is measurable. Regardless which controlled variable is used in the control loop, the system control mode or the functional control mode should remain the same as those listed in Table 6.1. An example of this substitution is the N_1 or the EPR for the thrust in the fuel control loop. Another example is the use of compressor exit Mach number M_{Nc} as the controlled variable for the compressor surge margin SM_c , because the exit Mach number provides a good correlation to the surge margin as shown in Fig. 6.13 [3].

6.5 Engine Accessories

6.5.1 Categories of Accessories

Engine accessories (or sometimes called accessory systems) are those components that are required to support an engine's operation throughout its operating envelope from start to shutdown. They are most often attached to the engine case and are external to engine air and gas paths. Engine accessories are divided into the following seven categories:

1) *Sensing components or sensors*: Examples of this category are speed, pressure, temperature, flow, and position sensing systems including probes and signal conditioning circuitry, etc.

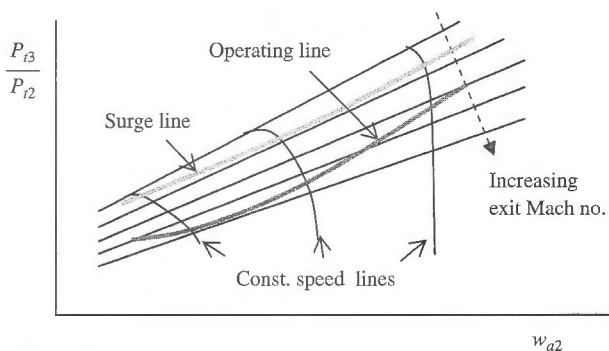


Fig. 6.13 Correlation of compressor exit Mach number to surge margin on a compressor map.

2) *Actuation components or actuators*: Examples of this category are valves, motors, fuel actuators, stator or vane position actuators, bleed actuators, area control actuators, solenoids, starters, igniters, and throttles, etc.

3) *Control components*: Examples of this category are fuel controls (for main engine combustor and augmentor), electronic control units (ECU), or full-authority digital engine control (FADEC).

4) *Fuel and oil flow delivery components*: Examples of this category are tanks (or reservoirs), pumps, valves, regulators, flow dividers, filters, chip detector, for both the fuel and the oil.

5) *Thermal management components*: Examples of this category are heat exchangers between air and fuel, also between fuel and oil.

6) *Power extraction or management components*: Examples of this category are shafts, gearboxes, and generators.

7) *Connecting components*: Examples of this category are wires, cables, connectors, harnesses, plumbing, fittings, brackets, and air or fluid lines.

Although we have mainly discussed one type of actuator in Chapter 3, that is, the fuel metering valve, a modern gas turbine engine usually has two other types of actuators: the bleed flow actuator and the inlet guide vane (IGV) actuator. Here IGV is meant to be a generic name for fan or compressor guide vanes, stator vanes, and variable-geometry flow controls. For modern commercial aircraft or military transport engines, active clearance controls have become yet another type of actuation mechanism to improve engine efficiency and fuel economy (e.g., reducing engine specific fuel consumption during cruise). Additional actuators are provided for military engines, such as the augmentor fuel control actuator and the nozzle-area control actuator.

6.5.2 Electronic Control Unit

The ECU is a generic name for supervisory and full-authority electronic controls. An ECU greatly simplifies the hardware used in an engine control system by eliminating cams, linkages, valves, and other computational components found in an hydro-mechanical unit (HMU). All modern gas turbine engines (from the early 1980s) are controlled by FADEC units.

The first FADEC-controlled engine is the Pratt and Whitney's F100 engine. The digital electronic engine control (DEEC), as it is called on the F100 engine, is a single-channel unit with a backup HMU. The DEEC has two basic control modes: the N_1 control mode for nonafterburning (or *dry*) operations and the EPR (defined as the ratio of P_{16} to P_{12}) control model for the afterburning (or *wet*) operation. Note that both control modes are required in afterburning operations. The logics behind these control models are quite complex, but the concept is simple, that is, to keep the nozzle area constant when the power setting is below the military power, and to open the nozzle area while maintaining the same EPR when the power setting is above the military power in afterburning mode.

The benefits of the DEEC, assessed in a NASA flight-test program [4], include increased thrust, faster transient response, increased air-start envelope, improved augmentor performance, elimination of ground trimming, and the fail-operate capability. In addition to the control functions, the DEEC incorporates extensive fault-detection and accommodation functions.

An interface diagram of the DEEC is shown in Fig. 6.14 [5], where the three major control components are the DEEC computer, the gas-generator fuel metering unit, and the augmentor fuel metering unit. The engine sensors and actuators are listed in Table 6.2.

An interesting fact about the F100 DEEC is that it does not measure inlet total pressure P_{12} as other engines do. Instead, it uses a single static-pressure probe P_{s2} located at the hub of the inlet spinner to calculate the inlet total pressure. One explanation is that the static-pressure probe provides a better estimate of the inlet

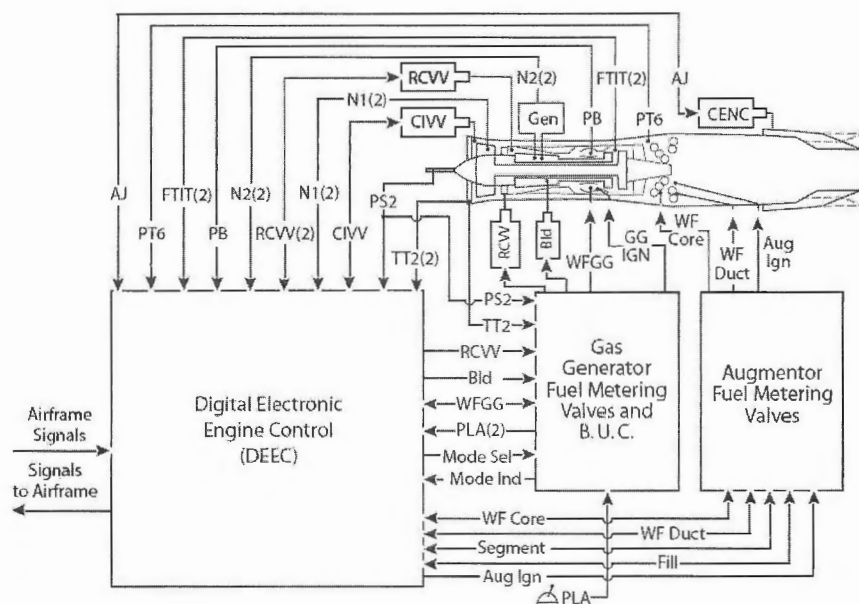


Fig. 6.14 F100 DEEC interface diagram.

Table 6.2 F100 DEEC sensors and their numbers

Name	Symbol	Number
Fan speed	N_1	2
Gas-generator speed	N_2	2
Inlet static pressure	P_2 (or P_{s2})	1
Burner pressure	P_b	1
LP turbine exit total pressure	P_{t6}	1
Inlet total temperature	T_{t2}	2
Fan turbine inlet temperature	FTIT	2
Throttle position	PLA	2
Rear-compressor variable vane position	RCVV	2
Compressor-inlet variable vane position	CIVV	1
Gas-generator fuel metering valve position	WFGG	1
Augmentor-core fuel metering valve position	WFAC	1
Augmentor-duct fuel metering valve position	WFDC	1
Exhaust nozzle-area actuator position	AJ	1
Start bleed valve position	SVP or BLD	1
Light-off detector	LOD	1

total pressure than what can be measured by a single total-pressure probe, when the airflow at the inlet is distorted [6].

In comparison, the GE F414 engine (see Fig. 6.15) is equipped with dual-channel FADECs, which require two identical sensors for each measurement. Although there is only one actuator for each control variable, each actuator has two excitation coils to be energized by each of the FADEC channels. The FADEC sensors for this engine are listed in Table 6.3.

The F414 engine also has two control modes: the CPR (core pressure ratio, which is defined as the ratio of P_{21} to P_{t56}) control mode for nonafterburning operations and the T_{t5} control mode for the afterburning operation. The reason

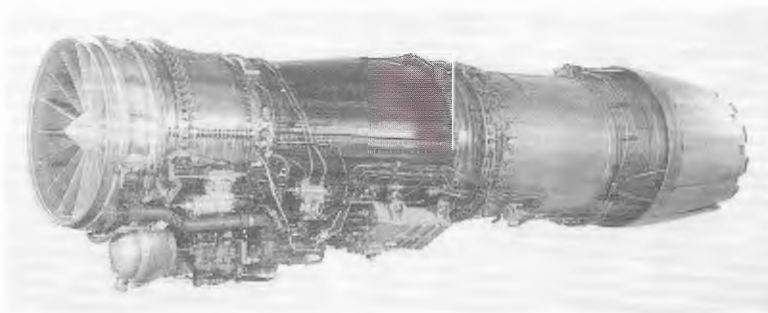
**Fig. 6.15 GE F414 engine (courtesy of General Electric Company).**

Table 6.3 F414 FADEC sensors and their numbers

Name	Symbol	Number
Fan speed	N_1	2
Gas-generator speed	N_2	2
Inlet total pressure	P_{t2}	2
Fan tip exit static pressure	P_{21}	2
Compressor discharge static pressure	P_3	2
LP turbine exit total pressure	P_{t56}	2
Inlet total temperature	T_{t1}	2
HP compressor inlet total temperature	T_{t25}	2
Exhaust gas temperature	T_{t5}	2
Throttle position	PLA	2
Light-off detector	LOD	2
Fan variable geometry position	FVG	2
Compressor variable geometry position	CVG	2
Main burner fuel metering valve position	WFM	2
Augmentor fuel metering valve position	WFA	2
Exhaust nozzle-area actuator position	A_8	2

for choosing CPR instead of N_1 is because it correlates better with dry thrust, hence giving a smoother relationship between the thrust and throttle position. The control variable in the CPR control mode is the main burner fuel flow, and the control variable in the T_{t5} control mode is the exhaust nozzle area A_8 .

For turboshaft engines, the Lycoming LTS101 is the first helicopter engine that received an Federal Aviation Administration (FAA) certification for electronic control. The electronic control computer (ECC) on the engine is a single-channel FADEC with a backup control. The ECC features adaptive control techniques, fault-tolerant design, and self-diagnostics. The basic control mode is N_p , the power turbine speed, which is regulated at constant 100%. The logic diagram for the "basic" control mode [7] is shown in Fig. 6.16. Note that the control logic consists of two transient control features: a lead-follower type of limit protection control (see Fig. 5.18) and a switching type of N-dot control (see Fig. 5.10).

The ECC receives input signals from five engine sensors: the power turbine speed N_p , the gas-generator speed N_G , the inlet total temperature T_{t1} , the measured gas temperature (between the gas generator turbine and the power turbine), and the rotor drive shaft torque Q . The only control actuator is the torque motor, which drives a flapper valve, which in turn modulates the bypass flow of the mechanical fuel metering unit.

6.6 Controller Synthesis Examples

The objective of engine controller design often boils down to a synthesis of various linear set-point control laws, nonlinear transient schedules, limit protection,

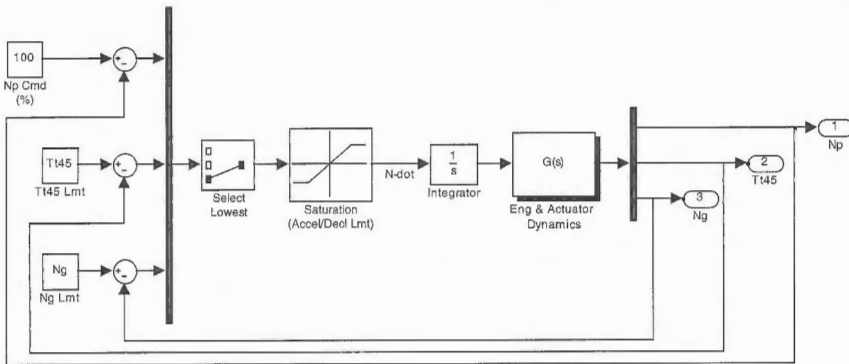


Fig. 6.16 LTS101 electronic engine control logic diagram (data from [7]).

and switching logics. The most effective way to accomplish this objective is through computer simulations of engine large transients at all extreme (or corner) operating conditions. In this section, the design synthesis concept will be illustrated by two numerical simulation examples. These examples conceptually cover a reduced engine control envelope.

6.6.1 Design and Synthesis of Controllers for a One-Spool Engine

To meet the requirements of rapid and large transients between idle and TO power, the engine controller must change the engine shaft speed (or pressure ratio) over a wide operating range. Large transients require frequent switching from one set-point control law to another as a result of the nonlinearity of the engine; they also require periodic transitions between set-point controls and transient schedules to avoid physical and operational limits. For controller design, we are mostly interested in large transients because satisfactory performance in large transients is a sufficient condition for satisfactory performance in small (or part-power) transients. Actually, controller design for large transients involves synthesis of various control laws and decision logics. This control synthesis approach will be explained next by using a simplified one-spool engine model as an example.

A simplified, piecewise-linear model of a hypothetical one-spool engine has been prepared. The model is a collection of linear models at various nominal operating points from idle to TO power. Each linear model has the form of the one-spool engine dynamics of Eqs. (2.5) and (2.6) for a given set point. This piecewise linear engine model is shown in Fig. 6.17.

The model has two input variables: the delta fuel flow DW_f and the shaft speed N . The delta fuel flow is defined as the difference between the actual fuel flow rate injected into the engine combustion chamber and the steady-state fuel flow rate.

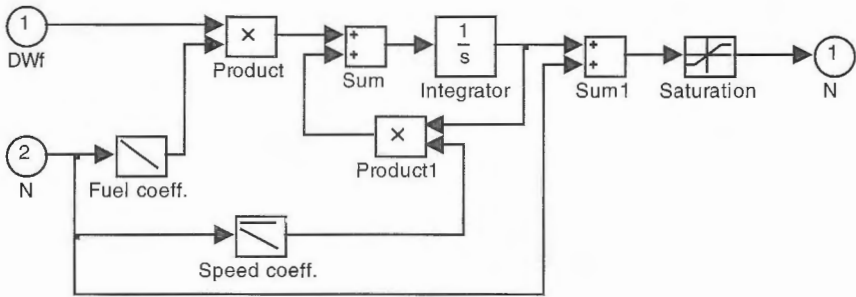


Fig. 6.17 Piecewise linear model of a simplified one-spool engine.

From Fig. 4.1 we know that the steady-state operating line is a collection of steady-state operating points. In a (large) transient, the steady-state operating point corresponding to the instantaneous shaft speed changes constantly along the operating line, consequently, the state-state fuel flow rate also changes constantly during the transient. The delta fuel flow is calculated outside the engine model (block) in the control model. This will become apparent when the engine model is integrated with the control model later.

Furthermore, the fuel coefficient and the speed coefficient in Fig. 6.15 correspond to the partial derivatives a and b in Eq. (2.4), respectively. These coefficients are modeled as piecewise linear functions and implanted as look-up tables. The values of these coefficients, as functions of engine shaft speed, are listed in Table 6.4.

6.6.1.1 Design of set-point controllers. A PI controller structure is assumed for the set-point controller. A number of set-point controllers have been designed, and each design contains a pair of control gains K_p and K_i , which are functions of the engine shaft speed (see Sec. 5.2.1). The controller block diagram is shown in Fig. 6.18. Note that the shaft speed has been chosen as the controlled variable, and the block diagram represents the primary control loop. The speed-scheduled set-point control gains for this hypothetical one-spool engine are given in Table 6.5.

Table 6.4 Piecewise linear model coefficients for a hypothetical one-spool engine

Speed N , rpm	Speed coeff. a , rpm / rpm-s	Fuel coeff. b , rpm / pph-s
25,000	- 0.5	35
40,000	- 1.1	30
55,000	- 1.8	24

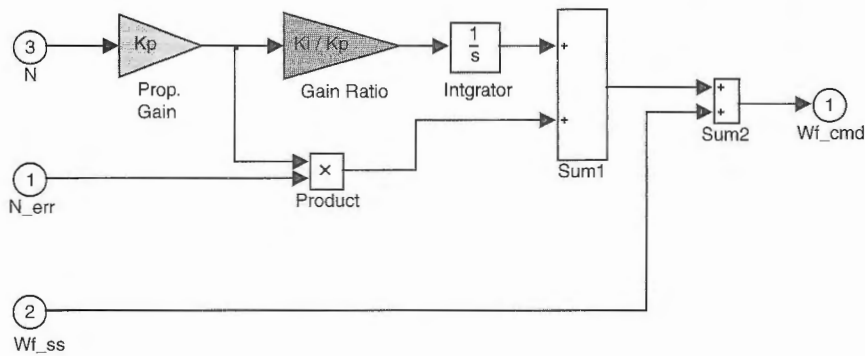


Fig. 6.18 PI control law is used for set-point control.

6.6.1.2 Design of a transient controller. The simplest form of a transient controller contains only two schedules: the accel schedule and the decel schedule. From Chapter 5 we know that these schedules are incorporated in the controller to prevent the engine from surge, overtemperature, and blowout. The accel and the decel schedules for this simplified engine are again assumed to be functions of the shaft speed and are defined in Table 6.6. Also listed in the table are the values of the steady-state fuel flow rate at various shaft speeds. In this simple example, no operational limit, like the turbine temperature limit or the shaft speed limit, is considered.

As explained in Chapter 5, an important part of a transient controller is to handle the transition between a linear (set-point) control law and a transient schedule. Figures 5.4 and 5.5 provide the basis for this transition logic.

6.6.1.3 Synthesis of set-point and transient controllers for one-spool engine. The process of bringing together a number of set-point controllers and transient (and limit protection) schedules to ensure satisfactory performance over a wide range of engine operating conditions is called *synthesis*. Just as how synthesis is applied to the *selection* of the most suitable actuation and sensing capabilities to support the optimal design of a controller, so it is used to *adjust* linear control gains and transition logics to achieve fast and well-damped transient

Table 6.5 Speed-scheduled set-point control gains for a hypothetical one-spool engine

Speed N	Prop. gain K_p	Integ. gain K_i
25,000	0.0005	0.0001
40,000	0.0003	0.00004
55,000	0.0001	0.00001

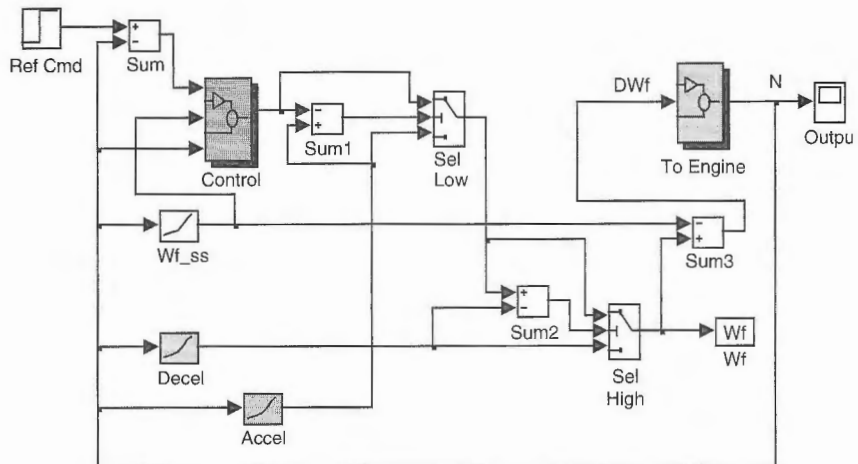
Table 6.6 The accel and decel schedules for a hypothetical one-spool engine

Speed, rpm	Accel schedule, pph	Decel schedule, pph	S. S. fuel flow, pph
10,000	120	50	100
30,000	150	60	120
35,000	200	75	150
40,000	250	100	200
45,000	350	130	320
50,000	500	190	450
55,000	600	210	550

performance in engines. Typically, the most difficult adjustment happens near the terminal set-point condition of a large transient, that is, the TO power operating condition in a rapid accel or the idle operating condition in a rapid decel.

After some tuning of the control gains, a synthesized engine controller block diagram for large transient operations is shown in Fig. 6.19. The values of the control gains are listed in Table 6.5. Note that the linear control gains in the table already represent the adjusted values resulted from the synthesis process.

Also note that this controller model has neglected the dynamics of fuel actuator and speed sensor because they are assumed fast compared to the engine shaft dynamics; hence, their transfer functions (TF) are set to unity (or 1).

**Fig. 6.19** Synthesized large-transient controller for a hypothetical one-spool engine.

6.6.1.4 Simulation of closed-loop performance during large transients of one-spool engine. Simulations of the closed-loop response of the engine during a rapid accel from idle to TO power, as well a rapid decel from TO power to idle, are shown in Figs. 6.20 and 6.21, respectively. In the real world, the overshoot near the TO power in a rapid accel should be kept under 10% to avoid excessive thrust variation during TO. On the other hand, the overshoot in Fig. 6.20 is acceptable because the engine model in this simulation does not include the turbine metal heating effect, which exists in real engines and tends to “wash out” the speed overshoot.

The block diagram of the piecewise linear model used in the preceding simulations is shown in Fig. 6.22. This diagram is similar to Fig. 6.19, except that the engine speed is fed back to the engine model with a unit-step time delay and used as a scheduling variable for the model parameters.

To illustrate a case of unacceptable transient performance, the integral control gain in Table 6.5 is increased until excessive oscillations occur. An example of this controller-induced oscillation is shown in Fig. 6.23, where the commanded fuel flow W_f is overlaid on top of the S. S. fuel flow W_{fss} on the lower-right-hand corner to show the magnitude of difference and the phase lag between the two fuel flows.

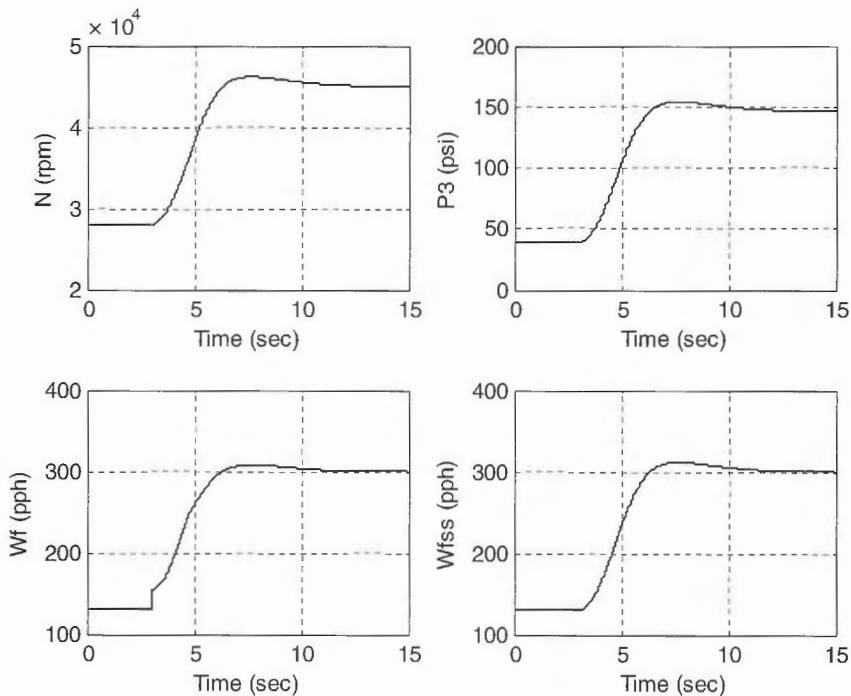


Fig. 6.20 Rapid accel from idle to TO power for a hypothetical one-spool engine.

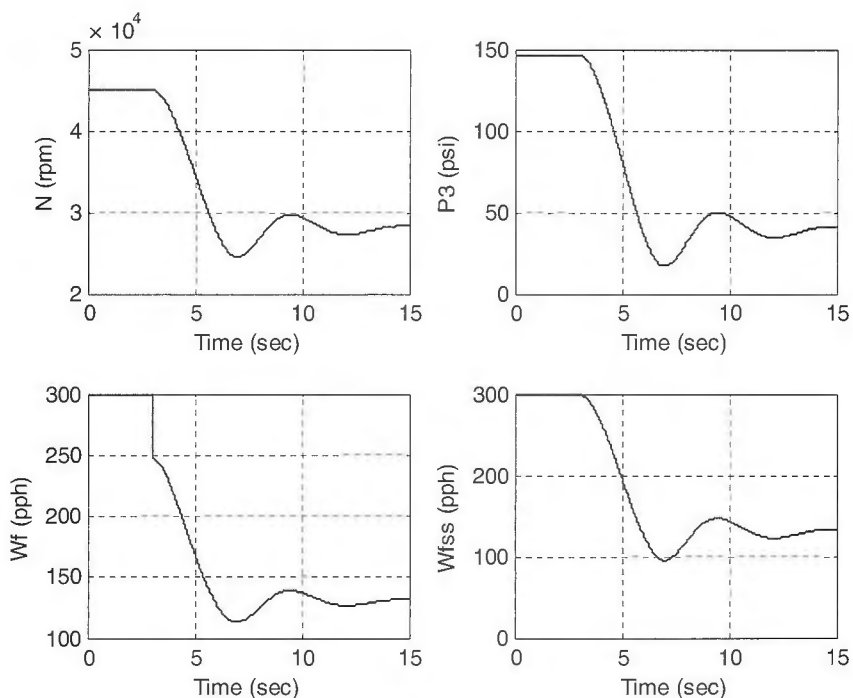


Fig. 6.21 Rapid decel from TO power to idle for a hypothetical one-spool engine.

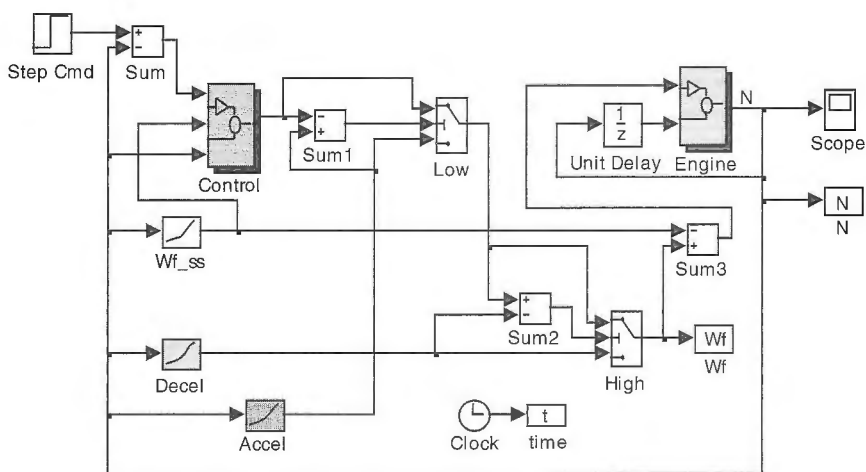


Fig. 6.22 Block diagram for closed-loop simulations of a hypothetical one-spool engine.

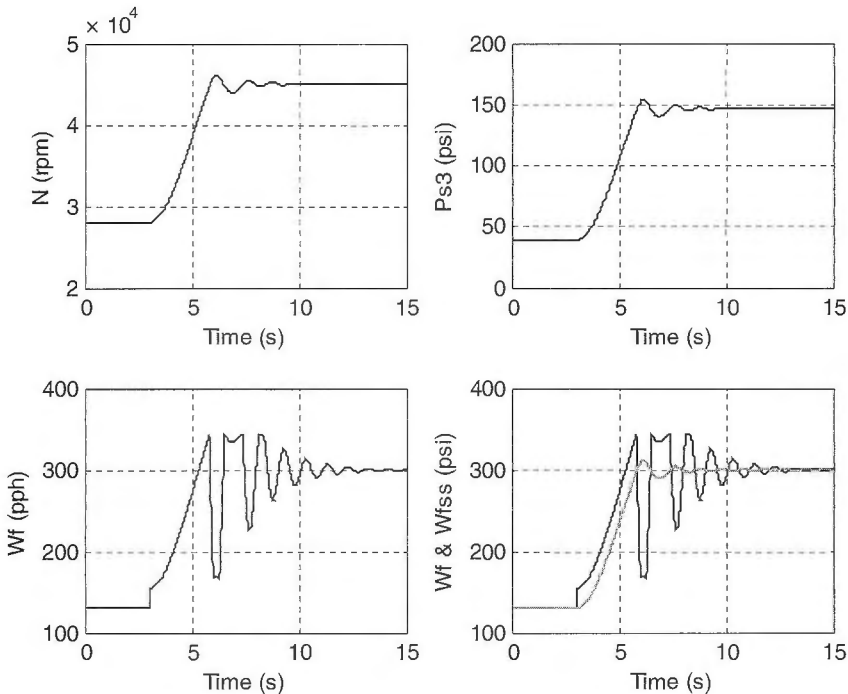


Fig. 6.23 Controller-induced oscillations in a rapid accel from idle to TO power for a hypothetical one-spool engine.

6.6.2 Design and Synthesis of Controllers for a Two-Spool Engine

A simplified, linear model of a hypothetical two-spool engine has also been prepared. The engine is similar to a generic, high-bypass turbofan commonly seen on commercial transport airplanes. The model is represented by a set of three TFs relating three engine output variables of interest to the input variable fuel flow. The three output variables are EPR , N_1 , and T_{15} (or EGT). This TF model of the two-spool engine is shown in Fig. 6.24. The input to the model is again the delta fuel flow. Each output from the model is the sum of the individual variation from the nominal operating point and the steady-state value of the output at the nominal point, which is designated as the initial value (or “ini”) in the figure.

The open-loop responses of the three output variables to a unit step input in fuel flow are shown in Fig. 6.25. Note a slight reversal of the fan speed N_1 response at the beginning of the transient. This is typical for a large-bypass turbofan engine.

The TF model in Fig. 6.24 is valid for the engine operating conditions near the TO power. To represent engine dynamics over the entire operating line, we need to connect a number of TF models at various engine power levels to form a piecewise linear model as described in Sec. 6.6.1 (for the one-spool engine). For the set of TF models, the coefficients of the numerator and the denominator polynomials

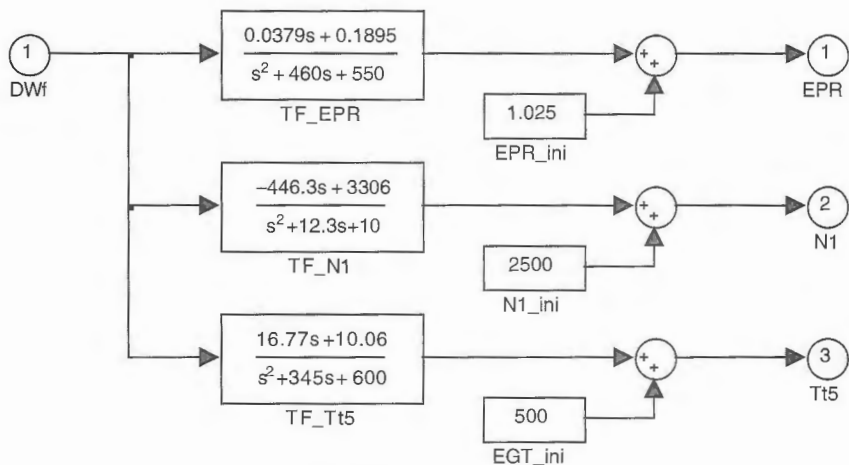


Fig. 6.24 TF model of a hypothetical two-spool engine.

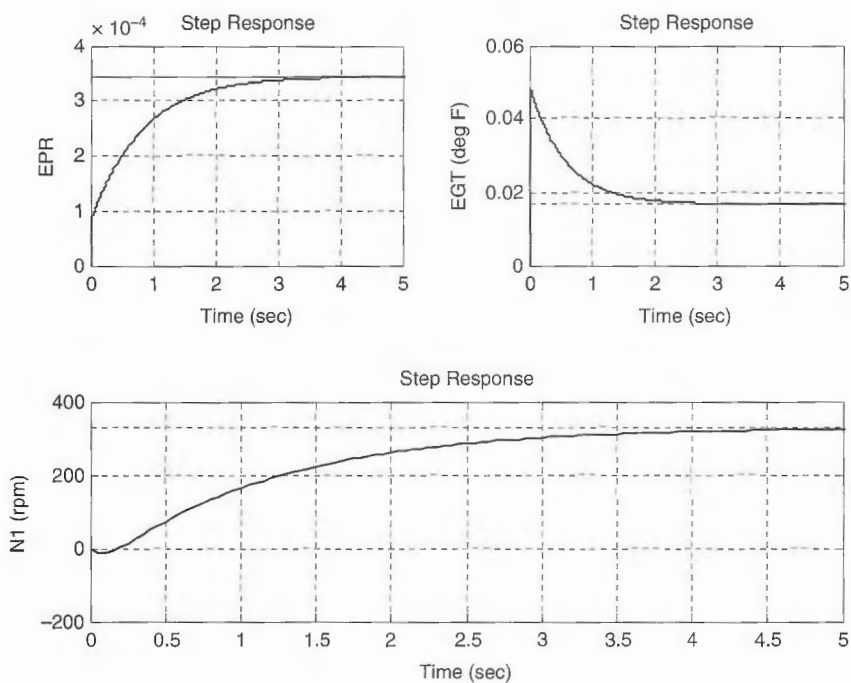


Fig. 6.25 Open-loop responses to a unit step input in fuel for a hypothetical two-spool engine.

change with the power level; even the order of the polynomials can change with the power level, although we often force the same model order and structure for all of the linear models.

Rather than spending time to synthesize a piecewise linear model for the two-spool engine and follow the design approach of the one-spool engine, let us assume that the linear model in Fig. 6.24 is valid for the entire operating line from idle to TO power and focus the design effort on an alternative design approach, such as the N-dot control law as opposed to the traditional scheduled-based transient control.

6.6.2.1 Design of an EPR controller. Suppose we choose EPR as the primary control variable for the two-spool engine, we can design a set-point controller using the traditional PI control law to control the EPR as shown in Fig. 6.26. The responses of the three engine output variables to a step command in the EPR reference command input are shown in Fig. 6.27.

In Fig. 6.27, the time to accelerate the engine from idle to TO power is approximately 10 s. This time is about twice as long as the desired. To speed up the acceleration, we can increase the proportional and the integral control gains; however, as these gains are increased, the engine transient is pushed closer to compressor surge and the turbine temperature limit. Hence an accel schedule is needed, just as the design example of the one-spool engine, to prevent engine from operating beyond the safe control region.

6.6.2.2 Design of an N-dot controller. Now let us apply an alternative approach to controller design for the two-spool engine. This approach uses an N-dot (or rate) control loop in addition to the primary EPR control loop. You might recall from Chapter 6 that the purpose of N-dot control is to provide a self-compensating feature for accel and decel transients, so that both the accel and the decel time as well as the time histories of the controlled variable are consistent for all of the engines in the fleet in all operating environments.

Figure 6.28 shows the block diagram of this alternative design approach. The EPR control loop (in blue) and the N-dot control loop (in red) are arranged in switching positions (see Fig. 5.10) to select the smaller of the two fuel flow commands as the output of the controller DW_f . T_f is the N-dot filter time constant.

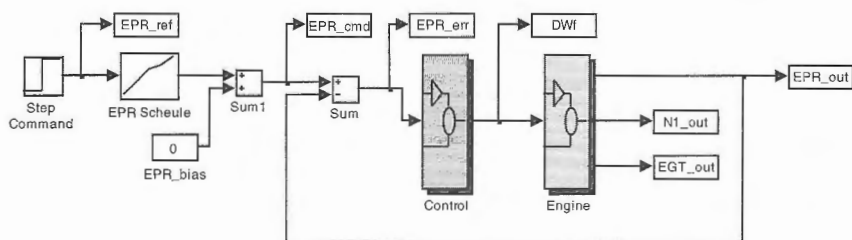


Fig. 6.26 EPR control loop with traditional PI control law for the hypothetical two-spool engine.

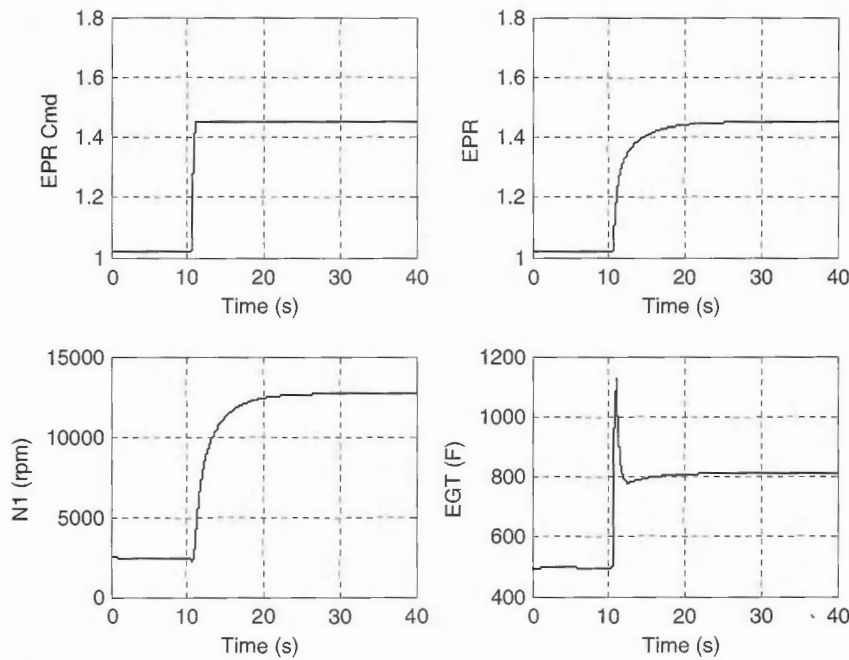


Fig. 6.27 Rapid accel from idle to TO power with a PI control law for a hypothetical two-spool engine.

After substituting this control model into Fig. 6.28 and choosing the value of the filter time constant and the integral control gain in the N-dot control loop to be 0.002 and 0.5, respectively, we can plot the simulated transient response to the same step command in EPR reference input in Fig. 6.29. Note that the fan speed response shows a constant acceleration rate (or N-dot) between 8,500 and 13,000 rpm. This acceleration rate corresponds to the value of 700 rpm/s in the N-dot schedule.

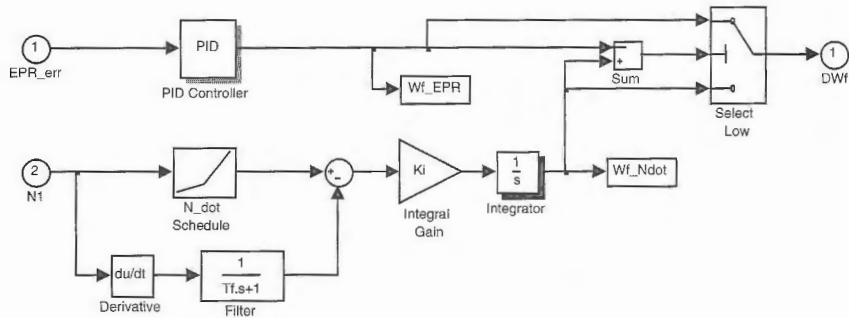


Fig. 6.28 Combined EPR control and N-dot control for a hypothetical two-spool engine. (See also color figure section at the back of the book.)

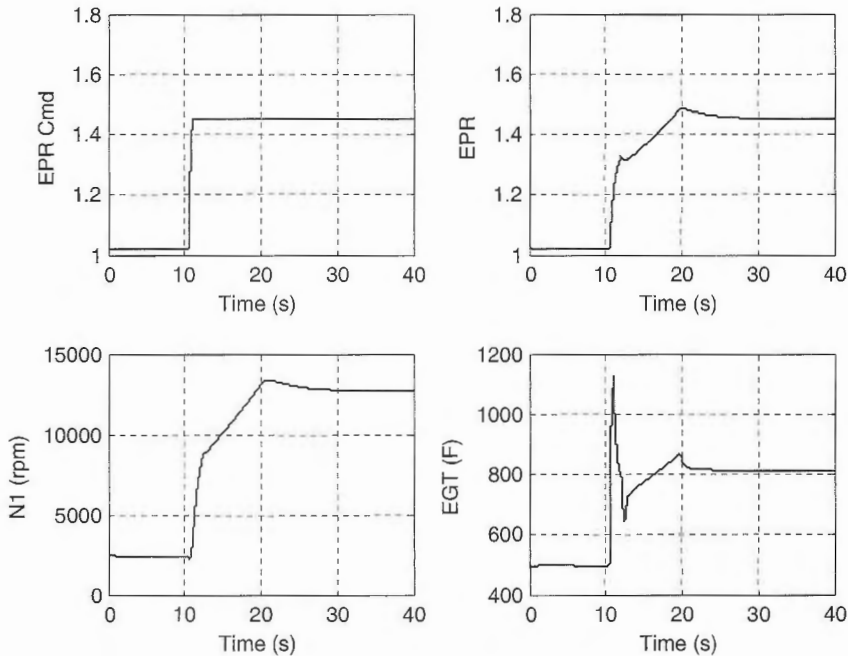


Fig. 6.29 Rapid accel from idle to TO power with both EPR and N-dot control laws for a hypothetical two-spool engine.

Designing individual controllers and integrating these controllers into a robust control system for the entire operating envelope of a turbine engine represent one of the greatest challenges in control engineering. To meet this challenge, we need an in-depth understanding of engine dynamics, actuator and sensor characteristics, modeling techniques, and control design methods. We also need practical experience in handling various types of system nonlinearities and design tradeoffs. Most importantly, we need to know the design requirements including system, performance, stability, and life usage requirements.

Engine control requirements usually start with a desired control mode on the top level. This desired control mode determines the primary control loop, which then determines the feedback controlled variable. Each control loop must rely on one of the four types of actuators to accomplish its control objective: the fuel control actuator, the vane positioning actuator, the bleed flow actuator, and the nozzle control actuator. The controlled variable in each control loop specifies the engine output variables to be measured and their sampling rates. Now multiplying this requirement flow-down process by the total number of control modes covering all possible steady-state, transient, and limit-protection control scenarios, we easily come up with more than 20 individual controllers that need to be designed and integrated. This quick analysis gives a feel of the scale of engine control design effort for a modern, two-spool engine, and it provides some justification for the materials covered in Chapters 2 through 6.

References

- [1] Fredlake, J. J., and Adams, M. R., "Electronic Fuel Controls for Executive Jet Aircraft," *Advances in Aerospace Propulsion*, P-131, 1983.
- [2] Seldner, K., and Gold, H., "Computer and Engine Performance Study of a Generalized Parameter Fuel Control for Jet Engines," NASA TN D-5871, June 1970.
- [3] Wagner, R. P., "Modern Control Systems of Turbine Engines," Aero Propulsion Short Course, Univ. of Tennessee Space Inst., Tullahoma, TN, May 1985.
- [4] Kock, B., "Digital Electronic Engine Control F-15 Overview," *Proceedings of the NASA Symposium on Digital Electronic Engine Control (DEEC) Flight Evaluation in an F-15 Airplane*, May 1983.
- [5] Myers, L. P., "F-15 Digital Electronic Control System Description," *Proceedings of the NASA Symposium on Digital Electronic Engine Control (DEEC) Flight Evaluation in an F-15 Airplane*, May 1983.
- [6] Hughes, D. L., and Mackall, K. G., "Effects of Inlet Distortion on a Static Pressure Probe Mounted on the Engine Hub in an F-15 Airplane," *Proceedings of the NASA Symposium on Digital Electronic Engine Control (DEEC) Flight Evaluation in an F-15 Airplane*, May 1983.
- [7] Janik, S., "Full Authority Digital Control for a Small Turbine Helicopter Engine," *Proceedings of SAE Aerospace Congress & Exposition*, Oct. 1983.

Advanced Control Concepts

In preceding chapters, we have learned some fundamental techniques for engine modeling and control law design. These modeling and design techniques have been used by the engine control community to solve traditional control problems. Nontraditional control problems, such as multivariable controls, active controls, and life-extending controls, have been studied in the last 30 years. These problems can be grouped in a category called *advanced controls* (as opposed to the traditional set-point and transient controls). The field of advanced controls is largely in the research stage. However, the control concepts introduced in this chapter offer some interesting insight into both the direction that the research has taken and the technologies that are being explored by the control community.

This chapter starts with an introduction of multivariable controls for turbine engines. Then the concept of active controls will be described. Active controls use high-frequency actuators to modulate the stability characteristics of engine modules. Specifically, we are interested in three applications of active controls: the active control of compressor surge/stall, the active control of compressor or turbine tip clearance, and the active control of combustion instability. These applications all point out the need for high-response actuators that operate at the frequency spectrum that is an order of magnitude higher than conventional fuel and IGV actuators. Hence, this chapter spends considerable efforts in discussing high-response actuator designs.

7.1 Multivariable Control

In the context of this chapter, multivariable control (MVC) is another name for multi-input/multi-output (MIMO) control and is a modern control design technique as opposed to traditional single-input/single-output (SISO) control design techniques. MVC is based on an optimization approach [1], which achieves a closed-loop control design that maximizes (or minimizes) an objective function of key performance indices of the engine. Through the optimization process, a control law that specifies the control action for each and every control input of the engine is derived. The resultant control law is called an optimal control law because the resultant control actions among all input variables optimize the performance objective.

In 1977, the National Engineering Consortium (NEC) organized an international forum in Chicago, Illinois, by bringing together the world's leading authorities,

to discuss the then-emerging research area of linear MVC. Most of the papers presented at the international forum were compiled into a book [2]. A unique feature of the papers presented at this forum is the use of a unifying theme problem based on a modern turbofan engine. This theme problem will be briefly described in Sec. 7.1.1 and the results of two representative papers, one using the frequency-domain method and the other using the time-domain method, will be discussed in Secs. 7.1.2 and 7.1.3, respectively.

7.1.1 MVC Design Theme Problem

The criteria and the models for the MVC design theme problem are described in a paper at the NEC international forum [3]. The Pratt and Whitney F100 engine, a high-performance turbine engine with several effectors, was used as the theme problem. The control requirements and specification categories for the F100 engine are outlined as follows: 1) engine protection against temperature, speed, pressure, and structural limits; 2) steady-state performance and accuracy for thrust modulation, fuel consumption, control sensitivity, deterioration, and repeatability; 3) transient performance thrust response and accel/decel times; 4) operability or stability for fan and compressor stalls, combustion stability, augmentor spikes, and flow fluctuations; 5) inlet/aircraft compatibility; 6) trim capability; and 7) start and transition. Transient time requirements (in the unit of seconds) for large power (or thrust) changes are summarized in Table 7.1. Large power changes are defined by larger than 70% of the intermediate (or Intrm) power, where the intermediate power is also called the military power, which is the maximum engine thrust without afterburning. Maximum (or Max) thrust refers to full, augmented engine thrust.

A total of 16 linear-time-invariant (LTI) models of the F100 engine were extracted from a nonlinear simulation of the F100 engine (see Chapter 2 for a description of the derivation of linear models). These 16 models cover the full range of operating conditions as defined by the altitude-Mach number flight envelope of the engine (see Chapter 6 about the piecewise linear model). Each linear model has 16 state variables, five (5) input (or control) variables and seven (7) (unmeasured) output variables. Typically there are five to ten (10) measured output variables on a turbine engine, and these measured outputs are also the state variables. The definitions of these model variables are given in Tables 7.2–7.4.

The model for the operating condition of sea level, static (Mach zero), near-intermediate power (PLA = 83deg) was chosen as the theme problem because every engine has to pass through this operating condition during TO [4]. An alternative, fifth-order model was also provided for those design approaches that

Table 7.1 Power change response requirements [data from 3]

Thrust change, %	Idle to Intrm	30% Intrm to Intrm	Idle to Max	Intrm to Max	Max to Intrm	Intrm to Idle
90	4	3.5	8	4.5	2	3
98	15	15	15	12	7	20

Table 7.2 Definition of the state variables of the linear model in the theme problem [data from 3]

Variable	Symbol	Description
x_1	N_1 , rpm	Fan speed
x_2	N_2 , rpm	Compressor speed
x_3	P_{t3} , psi(a)	Compressor discharge pressure
x_4	P_{t45} , psi(a)	Interturbine volume pressure
x_5	P_{t7m} , psi(a)	Augmentor pressure (or P_{t7})
x_6	T_{t25h} , °R	Fan inside diameter discharge temperature (or T_{t13})
x_7	T_{t25c} , °R	Duct temperature (or T_{t25})
x_8	T_{t3} , °R	Compressor discharge temperature
x_9	T_{t4Hi} , °R	Burner-exit fast response temperature
x_{10}	T_{t4Lo} , °R	Burner-exit slow response temperature
x_{11}	T_{t4} , °R	Burner-exit total temperature
x_{12}	T_{t45Hi} , °R	Fan turbine inlet fast response temperature
x_{13}	T_{t45Lo} , °R	Fan turbine inlet slow response temperature
x_{14}	T_{t5} , °R	Fan turbine exit temperature
x_{15}	T_{t6c} , °R	Duct exit temperature (or T_{t16})
x_{16}	T_{t7m} , °R	Duct exit temperature

Table 7.3 Definition of the input variables of the linear model in the theme problem [data from 3]

Variable	Symbol	Description
u_1	W_{fm} , pph	Main burner fuel flow
u_2	A_j , ft ²	Nozzle jet area (or A_8)
u_3	CIVV, deg	Compressor inlet guide vane position
u_4	RCVV, deg	High variable stator position
u_5	BLC, %	Custom compressor bleed flow

Table 7.4 Definition of the output variables of the linear model in the theme problem [data from 3]

Variable	Symbol	Description
y_1	F_N , lb	Engine net thrust
y_2	w_{a2} , pps	Total engine airflow
y_3	T_{t4} , °R	Turbine inlet temperature
y_4	SM_F	Fan stall margin
y_5	SM_C	Compressor stall margin
y_6	$\Delta P/P_{2d}$	Data-based fan exit pressure ratio $(P_{t25} - P_{25}) / P_{25}$
y_7	$\Delta P/P_{2t}$	Theory-based fan exit pressure ratio

Table 7.5 Elements of A matrix in the reduced-order model

-0.3245e1	-0.2158e1	-0.9155e3	0.5731e0	0.1342e3
0.1642e1	-0.5941e1	-0.2816e3	0.1897e0	0.5705e2
0.1685e-1	-0.2554e-1	-0.1003e2	0.7994e-2	0.5807e0
0	0	0	-0.1000e2	0
-0.2163e1	0.6862e1	0.7405e3	0.1195e1	-0.1715e3

would experience computational difficulty when using the full-order (16th-order) model. The reduced, fifth-order model in state equation form is

$$\dot{x}(t) = A \begin{bmatrix} \Delta N_1 \\ \Delta N_2 \\ \Delta P_{t7} \\ \Delta X_{WFM} \\ \Delta P_{t3} \end{bmatrix} + B \begin{bmatrix} \Delta W_{FM} \\ \Delta A_j \\ \Delta CIVV \\ \Delta RCVV \\ \Delta BLC \end{bmatrix} \quad (7.1)$$

$$\begin{bmatrix} \Delta F_N \\ \Delta W_{a2} \\ \Delta T_{t4} \\ \Delta SM_F \\ \Delta SM_C \end{bmatrix} = Cx(t) + Du(t) \quad (7.2)$$

where X_{WFM} is the main burner fuel metering actuator position. The model matrices A , B , C , and D are listed in Tables 7.5–7.8.

Actuator models for the four relevant (nonafterburning) control variables are shown in Fig. 7.1. A sensor model for the fan turbine inlet temperature is also shown in Fig. 7.1. In the case of the full-order model, these actuator and sensor dynamics would have been added to the engine dynamic model for a complete dynamic representation. In the case of the reduced-order model, only the dominant mode of the fuel metering actuator was added to complete the reduced-order dynamic representation (see Fig. 7.1).

Table 7.6 Elements of B matrix in the reduced-order model

0.1432e-1	-0.3553e3	-0.9906e2	-0.1549e2	0.2220e5
0.2871e0	0.7286e3	0.2514e2	-0.6487e2	0.8122e4
-0.2469e-2	-0.1030e3	0.6333e0	-0.3213e0	-0.7418e2
0.1000e2	0	0	0	0
-0.1311e0	0.3295e3	-0.2500e2	0.6257e2	-0.6445e5

Table 7.7 Elements of C matrix in the reduced-order model

0.1662e1	-0.1768e1	0.7999e2	-0.1890e0	0.3771e2
0.1383e-1	0.3142e-5	-0.1060e-1	0.1289e-3	-0.1839e-6
0.1694e0	-0.1129e0	-0.4959e1	0.7386e-1	-0.1835e0
0.7590e-4	0.3269e-5	-0.1477e-1	0.2284e-5	0.4315e-4
-0.4859e-4	0.1381e-3	0.1140e-1	0.1951e-4	-0.2688e-2

7.1.2 Frequency-Domain Methods

MVC design methods in the frequency domain naturally extend the classical control design methods, such as the root-locus method and the Bode plot method, that use transfer functions (TFs) to synthesize the controller. The greatest advantage of frequency-domain design is its insensitivity to model order and modeling errors. Frequency-domain MVC methods require intensive algebraic manipulations, and because of a lack of sustained development in computerized design tools, these methods have not been as widely adopted as their time-domain counterparts. Nevertheless, we will review the inverse Nyquist array (INA) [5] method to get an idea of multivariable frequency design techniques. The goal of INA is to use matrix transformation to achieve an approximately decoupled dynamic system so as to apply classical control methods to each of the “diagonally” dominant TF of the engine.

As explained in Appendix A, the TF for a MIMO system is a TF matrix, whose elements are individual TFs, and each of which represents a specific input-output relationship. Suppose the block diagram of a MVC system is shown in Fig. 7.2, where $G(s)$ is the engine TF, $K(s)$ is the input compensator TF, $L(s)$ is the output compensator TF, and $F(s)$ is the feedback compensator TF. The dimensions of these TFs are $m \times l$, $l \times k$, $k \times m$, and $k \times k$, respectively.

To achieve diagonal dominance, we first need to select (controlled) output variables so that the number of the controlled outputs is the same as the number of inputs. Generally speaking, there are more measured output variables than input variables, hence selecting a smaller set of measured output variables as the controlled output variables conceptually is not a problem. However, selecting the correct variables, which enable important control design goals, is a key practical design concern that requires a thorough understanding of the system to be controlled and should not be taken lightly. Next, we need to select a

Table 7.8 Elements of D matrix in the reduced-order model

0.1302e0	0.1992e3	0.4802e2	-0.1503e2	0.1083e5
0.1449e-6	0.3395e0	0.6806e0	0.2812e-3	0.3204e-3
0.2967e-1	0.7927e2	0.2567e1	-0.7631e0	0.2066e4
0.1046e-5	-0.7720e-2	-0.5814e-2	0.1157e-3	0.6605e-1
-0.8395e-5	-0.7897e-2	-0.6841e-3	-0.9643e-3	-0.2815e0

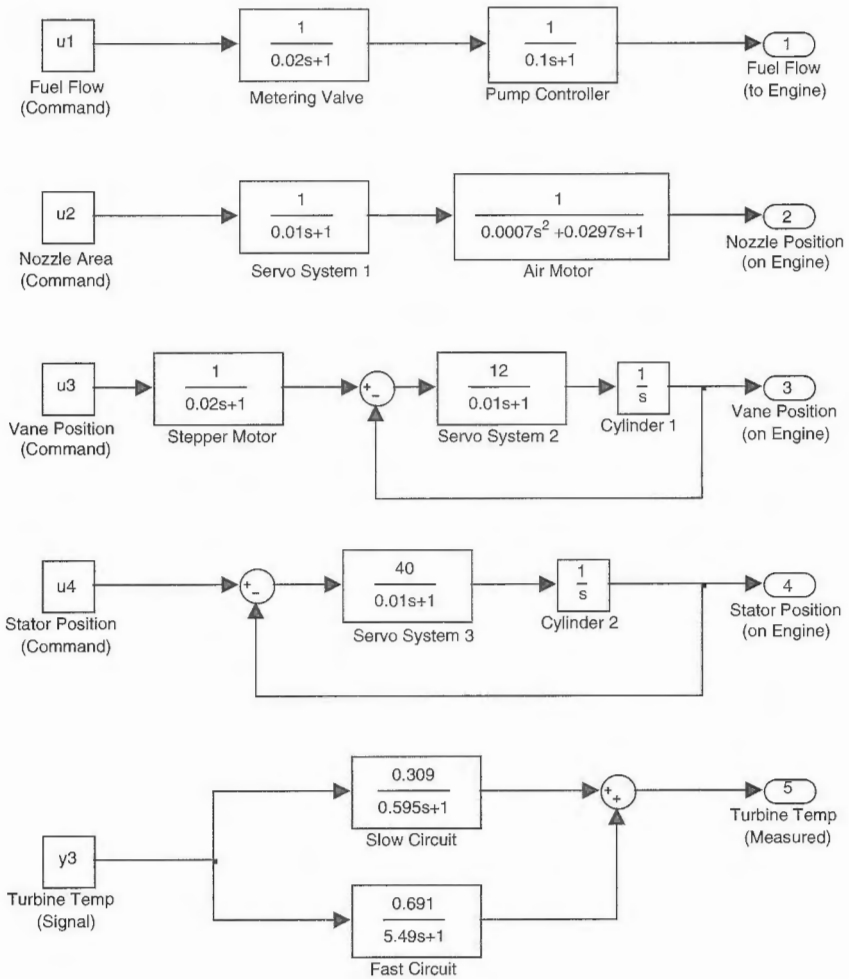


Fig. 7.1 Actuator and sensor models for the theme problem.

nonsquare matrix TF $M(s)$ so that the original (nonsquare) plant TF $G(s)$ can be approximated by a square TF $G_d(s)$ for the frequency spectrum of interest, that is, $G_d(s) \cong M(s)G(s)$.

An example of applying the INA method to the theme problem yields the following design for the compensators [6]:

$$K(s) = \begin{bmatrix} 0 & 0 & 5.0 \\ -5.33e-4 & 2.4e-2 & 3.2e-3 \\ 0 & 5.35 & 0 \end{bmatrix} \quad (7.3)$$

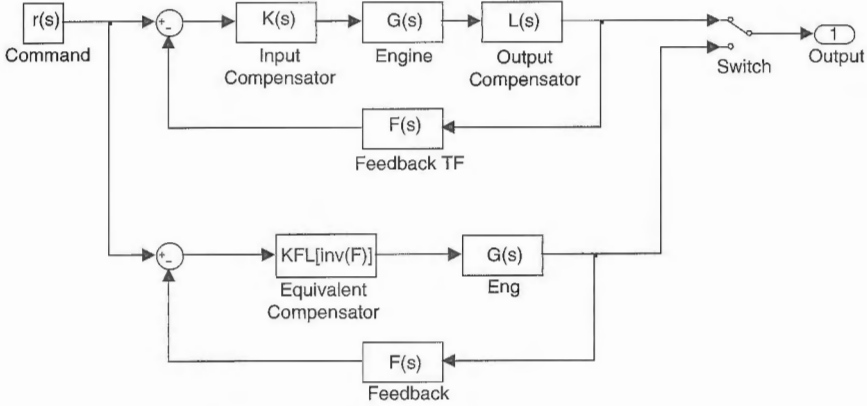


Fig. 7.2 TF block diagrams of a MVC system.

$$L(s) = \begin{bmatrix} 1 & 0 & 0 \\ -0.8 & 1.33 & -0.6 \\ 0 & 0 & 1.43 \end{bmatrix} \quad (7.4)$$

$$F(s) = \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 75 \end{bmatrix} \quad (7.5)$$

Note that this particular design has all scalar elements in the compensators. However, nonscalar elements (i.e., a rational function of s) can appear in the compensators depending on design choices. In this design, the number of input variables is three and the number of controlled variables is also three. The input variables are the main burner fuel flow u_1 , the nozzle jet area u_2 , and the compressor inlet guide vane position u_3 . The measured output variables are the fan speed x_1 , the compressor speed x_2 , the compressor discharge pressure x_3 , the augmentor pressure x_5 , and the thermocouple measurement of the (averaged) fan turbine inlet temperature $\tilde{x}_{12} \equiv (x_{12} + x_{13})/2$. Three of the unmeasured output variables (in Table 7.4) have been selected as the controlled output variables, and they are the engine net thrust y_1 , the total engine airflow y_2 , and the turbine inlet temperature y_3 . Hence the plant TFs are

$$G_d(s) = \begin{bmatrix} \frac{y_1(s)}{u_1(s)} & \frac{y_1(s)}{u_2(s)} & \frac{y_1(s)}{u_3(s)} \\ \frac{y_2(s)}{u_1(s)} & \frac{y_2(s)}{u_2(s)} & \frac{y_2(s)}{u_3(s)} \\ \frac{y_3(s)}{u_1(s)} & \frac{y_3(s)}{u_2(s)} & \frac{y_3(s)}{u_3(s)} \end{bmatrix}_{3 \times 3} = C_d(sI - A)^{-1}B + D + D_d \quad (7.6)$$

and

$$G(s) = \begin{bmatrix} \frac{x_1(s)}{u_1(s)} & \frac{x_1(s)}{u_2(s)} & \frac{x_1(s)}{u_3(s)} \\ \frac{x_2(s)}{u_1(s)} & \frac{x_2(s)}{u_2(s)} & \frac{x_2(s)}{u_3(s)} \\ \frac{x_3(s)}{u_1(s)} & \frac{x_3(s)}{u_2(s)} & \frac{x_3(s)}{u_3(s)} \\ \frac{x_5(s)}{u_1(s)} & \frac{x_5(s)}{u_2(s)} & \frac{x_5(s)}{u_3(s)} \\ \frac{\tilde{x}_{12}(s)}{u_1(s)} & \frac{\tilde{x}_{12}(s)}{u_2(s)} & \frac{\tilde{x}_{12}(s)}{u_3(s)} \end{bmatrix}_{5 \times 3} = C(sI - A)^{-1}B + D \quad (7.7)$$

where A , B , C , C_d , D , and D_d are engine model matrices as in Eqs. (A.3) and (A.4). Note that the full 16th-order model was used in the design and the dynamics of the corresponding actuators and sensors are also included.

7.1.3 Time-Domain Methods

MVC design methods in the time domain are centered around the linear-quadratic-regulator (LQR) method and the model reference (or model-based) control method, with the LQR method being the predominant choice. The LQR method uses a set of weights distributed in two weighting matrices, one for the state variables and the other for the input variables, to derive the optimal control law. Its stochastic companion, which includes state dynamic and measurement noises, is called LQG (linear quadratic Gaussian) method. In the general case of a time-varying linear system, the state and the output equations are given by Eqs. (A.1) and (A.2), and are repeated here to facilitate the discussion:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (7.8)$$

$$y(t) = C(t)x(t) + D(t)u(t) \quad (7.9)$$

We are interested in designing a *continuous* state variable feedback control law expressed as

$$u(t) = -K(t)x(t) \quad (7.10)$$

so that the state variables can reach the origin, that is, the nominal condition, while minimizing the following quadratic objective function:

$$J = \frac{1}{2} (x^T S_f x)_{t=t_f} + \frac{1}{2} \int_{t_0}^{t_f} (x^T Q x + u^T R u) dt \quad (7.11)$$

where t_f is the terminal time (or time at the end of the control), t_0 is the start time, S_f and Q are positive-semidefinite matrices, and R is a positive-definite matrix. The solution of this optimization problem is [1]

$$u(t) = -K(t)x(t) = -R^{-1}(t)B^T(t)S(t) \quad (7.12)$$

and S satisfies the following matrix Riccati equation:

$$\dot{S} = -SA - A^T S + SBR^{-1}B^T S - Q \quad (7.13)$$

In case of an LTI system, all matrices in Eqs (7.12) and (7.13) have scalar elements, that is, the feedback control law of Eq. (7.12) reduces to a constant state-variable feedback gain matrix K .

A particularly interesting approach to applying the LQR method is the *output* (variable) feedback regulator (OFR) for a LTI system. The control law in an OFR is

$$u(t) = -K_y z(t) \quad (7.14)$$

where the measured output variable z is a linear combination of the state variable superimposed with a white noise v , that is,

$$z(t) = Cx(t) + v(t) \quad (7.15)$$

and the number of measured outputs is less than the number of states.

Let the objective function in optimization be defined as

$$J = \frac{1}{2} \int_0^{\infty} (x^T Q x + u^T R u) dt \quad (7.16)$$

then the solution of the feedback gain matrix K_y satisfies the following set of equations [7]:

$$\begin{cases} KA_0 + A_0^T K + Q + C^T F^T R K_y C = 0 \\ LA_0^T + A_0 L + X_0 = 0 \\ K_y = R^{-1} B^T K L C^T (C L C^T)^{-1} \end{cases} \quad (7.17)$$

where X_0 is the covariance matrix of the state variable at $t = 0$ and $A_0 = A - BK_y C$.

7.1.4 Comparison of MVC Methods

MVC methods include more techniques than just the INA, the LQR, and the OFR. Most modern control theory books [8,9] cover a wide range of methods, including some of the most common ones: LQR, OFR, LQG, frequency-weighted LQG, H^∞ (H-infinity), model following control, eigenstructure assignment, INA, and characteristic locus. A comparison of the advantages and disadvantages of these MVC methods is listed in Table 7.9 [10].

Table 7.9 Comparison of MVC Methods (data from [10], courtesy of Stephen Rock)

MVC method	Advantages	Disadvantages
LQR	Guaranteed stability margins Robustness can be improved through frequency shaping Easy to compute	Need full state feedback Need accurate model
OFR	Use available outputs Low-order dynamic feedback	Possibly many design iterations Convergence problem Some poles might not be moved No guaranteed margins Possibly many design iterations
LQG	Use available measurements Easy to compute	No guaranteed margins Need accurate model Possibly many design iterations
Frequency-weighted LQG	Use available measurements Singular values can be adjusted through cost and noise matrices Easy to compute	Possibly many design iterations Frequency-dependent cost and noise matrices may be needed leading to high-order controllers
H^∞	Simultaneously addressing stability margins, sensitivity, robustness, response, and (different types of) uncertainties Maximum achievable performance over the whole frequency spectrum of interest Direct one-step design for shaping singular values	High-order controllers Conservative controller (worst-case scenario)
MFC	Control over transient responses Incorporation of nonlinear characteristics Applicable to many synthesis techniques Decreasing regulator activities through feed-forward	Depending on synthesis methods used Implementation complexity
Eigenstructure assignment	Arbitrary pole placement Maximizing a lower bound of stability margins Relatively easy to compute	Need accurate model, especially for unstable systems Selection of eigenvalues might not be optimal. Possibly many iterations
INA	Design reduced to classical single-loop compensator problem Extension of well known classical control theory	Difficult to apply to systems with more than five inputs/outputs No guaranteed robustness Possibly many iterations
CL	Generalization of the Nyquist stability criterion to MIMO	Difficult to apply to systems with more than five inputs/outputs No guaranteed robustness Possibly many iterations

7.2 Active Clearance Control

Engine capability is an important metric for the performance of a turbine engine. One of the definitions of *capability* is given by

$$\text{Capability} \equiv \frac{F_N/W}{W_f/F_N} \quad (7.18)$$

where W is the weight of the engine, the numerator is the thrust-to-weight ratio, and the denominator is the thrust specific fuel consumption. To increase the capability with the same engine weight, we want to increase the thrust and reduce the fuel consumption, which are synonymous to increasing engine pressure ratio in the compressor and reducing fuel-to-air ratio in the combustor.

Active controls can potentially increase engine capability because they can expand the engine's stability margins through high-frequency modulations of aero-thermal characteristics. In Fig. 3.1, the difference in frequency spectrum between active controls and conventional controls was illustrated. The frequency spectrum for active controls is in the order of 10^2 to 10^3 Hz, and the frequency spectrum of conventional controls is up to 10 Hz. The most promising active control technologies to achieve this goal increase the capability are 1) active tip clearance control to improve compressor or turbine efficiency; 2) active stall and surge control to increase engine pressure ratio and stability margin; and 3) active combustion control to suppress lean-combustion instability, reduce emissions, reduce liner fatigue, and improve turbine inlet temperature distribution (or the so-called combustor *pattern factor*).

Tip-clearance control and pattern factor control are performance control modes, while stall/surge control and combustion stability control are operability control modes. The basic concept of each of the three active controls is discussed in separate subsections below; specifically, actuation technologies for active controls will be emphasized because actuation has been identified as the most important enabling technology for active controls.

Tip clearance is the distance between the tip of a rotating blade and the shroud. It is usually calculated as the average distance over all of the blades on the same rotating stage. Active clearance control (ACC) can improve component efficiency and operability by reducing air- or gas-flow losses. ACC can also reduce tip rubs, which are often a cause for engine removals. ACC can be applied to compressors or turbines. Turbine ACC is more challenging than compressor ACC because of the harsher environment that the control system has to deal with, that is, high temperature and high pressure environment. All modern commercial engines incorporate some form of heating or cooling for ACC of turbines. This technique uses bleed air from the engine to control the thermal growth of the shroud and stator structures. Although it can reduce fuel consumption in the cruise operating condition, it is inherently slow and can compromise the overall cycle efficiency because of the loss of the bleed flow. ACC with a high-response actuation mechanism is desirable for the engines that experience extreme and sudden loads on their shafts and bearings.

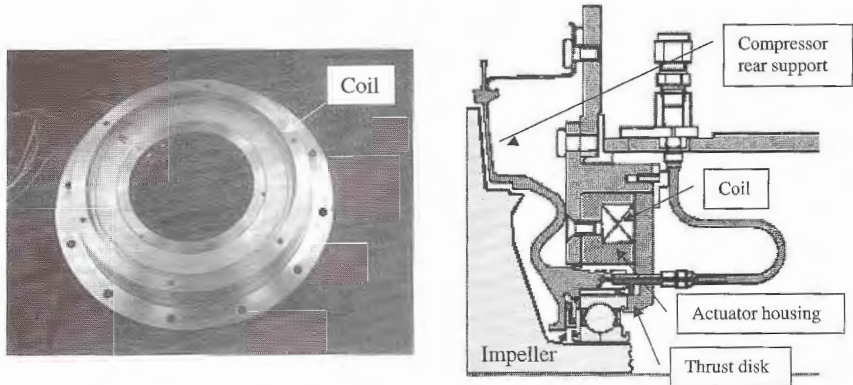


Fig. 7.3 Electromagnetic actuator for centrifugal compressor tip-clearance control (courtesy of Scientific Monitoring, Inc.).

We will use an example to symbolize the challenges facing a high-response ACC system. The example is about an experimental investigation of clearance control for a centrifugal compressor [11]. The control concept is to modulate the axial tip clearance by moving the impeller-shaft assembly in and out of the shroud [12]. Controlling the axial tip clearance of a centrifugal compressor is beneficial because it is the dominant performance-determining part of the clearance starting from the “knee” of the shroud to the exit of impeller.

The actuation system is an electromagnetic actuator located in the rear housing of the impeller. The actuator moves the impeller-shaft assembly in the axial direction to control the axial tip clearance; therefore, it acts like a thrust bearing for the compressor shaft. Figure 7.3 shows the circular magnetic thrust bearing. The inner diameter of the magnetic coil is 6.5 in.

7.2.1 ACC Modeling

This ACC system can be modeled as a second-order mass-spring-damper system. The mass is the impeller, which moves in the axial direction. The spring is in the form of a Belleville spring. Figure 7.4 shows a simplified model of the ACC system. The variables and their values are defined Table 7.10.

The equation of motion for this model is given by

$$m\ddot{x} = F_a + F_b - k(x + x_i) - b\dot{x} \quad (7.19)$$

After substituting the numerical values of the constants, the equation becomes

$$\frac{1}{24,000}\ddot{x} = F_a + F_b - 84.8 \cdot (x + 5.425) - 6\dot{x} \quad (7.20)$$

where $x_i = 460/84.8 = 5.425$ mil is the initial spring deflection, which is designed to pull the impeller back from compressor casing in case that the actuator fails.

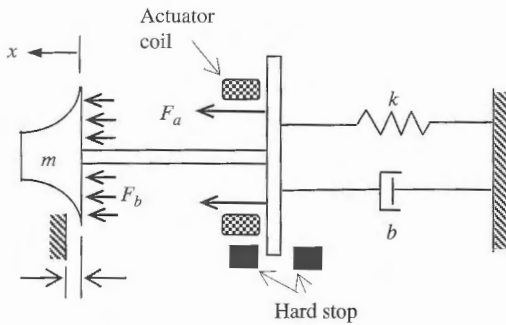


Fig. 7.4 Simplified model of the ACC system for centrifugal compressor.

In Eq. (7.20), the unit of force is in pounds, displacement is in mil (0.001 in.), and time is in seconds. Figure 7.5 shows a block diagram model of this ACC system in Simulink®. A limit integrator is used to simulate the effect of the impeller resting on the mechanical stop. The initial tip clearance was measured to be 12.17 mil, which is less than the “cold” clearance C_{xc} measured at 15 mil because of the aerodynamic force acting on the impeller.

The open-loop TF from the actuator force to the clearance is

$$\frac{X(s)}{F_a(s)} = \frac{1}{ms^2 + bs + k} = \frac{1}{(1/24,000)s^2 + 6s + 84.8} \quad (7.21)$$

and the open-loop poles (also called the modal frequencies) of this TF are located at $s_1 = -14.14$ and $s_2 = -143,990$ rad/sec, respectively. Note that the second pole is so large that it can practically be ignored in the analysis. Indeed, the TF for the inner loop with feedback damping in low frequencies can be written as

$$\frac{24,000}{s + 24,000 \cdot 6} \approx \frac{1}{6} \quad (7.22)$$

Table 7.10 Definition and values of ACC model variables

Symbol	Description	Value
m	Total mass of moving components	0.5 Slug
k	Spring constant	84.8 lb/mil (mil = 0.001 in.)
b	Viscous damping coefficient	6 lb-s/mil
x	Rotor displacement in the axial direction	Time varying, mil
c_x	Tip clearance ($c_x = c_{xc} - x$), $C_{xc} = 15$	Time varying, mil
F_a	Force generated by electromagnetic actuator	Time varying, lb
F_b	Aerodynamic force acting on the impeller	Time varying, lb

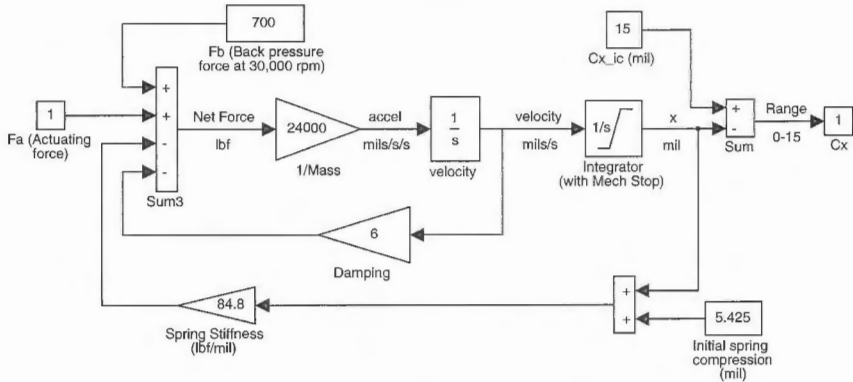


Fig. 7.5 Simulation model for the simplified ACC system.

The electromagnet model relates the magnetic force to the electric current in the coil and the gap between the electromagnet and the thrust plate. The ideal force is proportional to the square of the coil current and inversely proportional to the square of the gap and is given by

$$F_a \sim \frac{N^2 i^2 A}{h^2} \quad (7.23)$$

where F_a is the electromagnetic force in pounds; N is the number of turns of wire that make up the coil; I is the current in amperes; A is the area of the air gap in square inches; and h is the gap between the coil and the thrust plate, in mil, and $h = h_0 - x$.

The proportional constant to complete Eq. (7.23) depends on the permeability, which in turn depends on the shape of the electromagnet. The estimated force of the actuator vs the current vs the gap is depicted in Fig. 7.6. It is clearly seen that the electromagnet force is proportional to the current and inversely proportional to the gap.

The power of the actuator during the experiment came from a pulse-width-modulated (PWM) servo amplifier. The PWM excitation frequency was approximately 27 kHz, and it was high enough that the high-frequency content of the amplifier was filtered out by the inductance of the coil. To eliminate the variations in force with changes in electrical resistance and inductance (from the coil), the servo amplifier was configured as a voltage-controlled current source. This allowed the actuator to provide a force that is proportional to the input voltage regardless of the variations in temperature and signal frequency. The input of the amplifier is the control voltage, and the output is the current going into the actuator coil. Thus the amplifier and the coil could be modeled as a typical electrical circuit with internal feedbacks to regulate the output current. Figure 7.7 shows the block diagram of this amplifier model.

Capacitance position sensors were used to measure the tip clearance. The tip clearance was an averaged value over the individual tip clearances of 36 blades

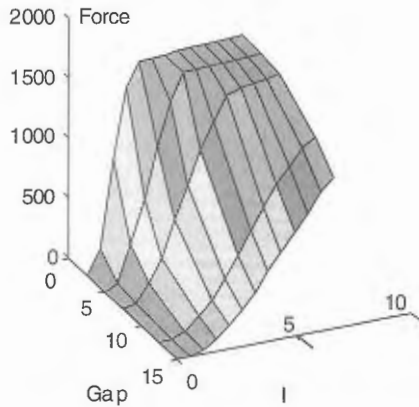


Fig. 7.6 Electromagnet force vs current I vs gap.

passing by the probes. This allowed the measurements to be updated once per revolution of the rotor speed. The sensor had a bandwidth of 20kHz, which was so much faster than the rest of system dynamics; hence, the sensor dynamics were neglected.

7.2.2 Linear Control Law Design

A linear model of the complete ACC system was obtained by first linearizing the actuator force around a nominal point with respect to the current i and the clearance x . After substituting the linearized actuator force from Eq. (7.23) into Eq. (7.19), we get the combined impeller-shaft-actuator model as

$$m\ddot{x} + b\dot{x} + \left(k + \frac{\partial F_a}{\partial h} \bigg|_0 \right) \delta x = \frac{\partial F_a}{\partial i} \bigg|_0 \delta i \quad (7.24)$$

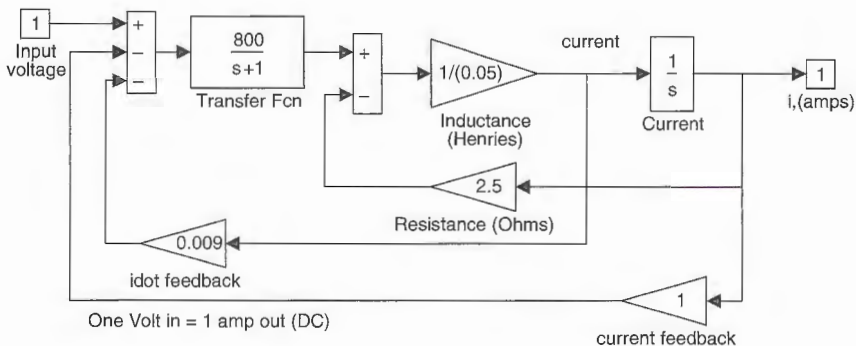


Fig. 7.7 Simplified model of the amplifier and the coil.

Because the partial derivatives of the force with respect to the gap is a negative number, the system in Eq. (7.24) can become open-loop *unstable* if

$$k < -\left. \frac{\partial F_a}{\partial h} \right|_0 \quad (7.25)$$

A good design guideline is to make the system stable in the open-loop sense by making sure that Eq. (7.25) does not happen near the nominal operating condition.

Linear control laws were designed at three nominal operating points, and the controller gains were scheduled between any two adjacent points [13]. The control laws were designed by the H-infinity design method with built-in robustness to account for the uncertainties in the electromagnetic force and the mechanical damping. Subsequent model reduction resulted in a first-order PI control law for all three points. Standard frequency-response phase and gain margins were used to check the closed-loop robustness of the reduced-order controllers at the three nominal points. An example of the control laws is

$$K(s) = \frac{-2.67(s + 9.3)}{s} \quad (7.26)$$

The minus sign is required because the gain from electrical current to the gap in tip clearance is a negative number. The control gains [-2.67 and 9.3 in Eq. (7.26)] were scheduled with two variables: the tip clearance and the electric current.

The gain-scheduled controller can be evaluated by using the closed-loop system shown in Fig. 7.8.

Note that a prefilter of 25 rps was used to offset the zero in the PI controller. Without this prefilter the zero would cause an overshoot in the system step response. To evaluate the nonlinear, simulated response, a series of step commands were given starting from 12 mil, decreasing by 2 mil every second for a period of 4 s and ending at a clearance of 4 mil. The resulting simulated response is shown in Fig. 7.9.

An interesting result in the experiment is that reducing the tip clearance of the compressor delays surge onset. This phenomenon is shown by two curves in

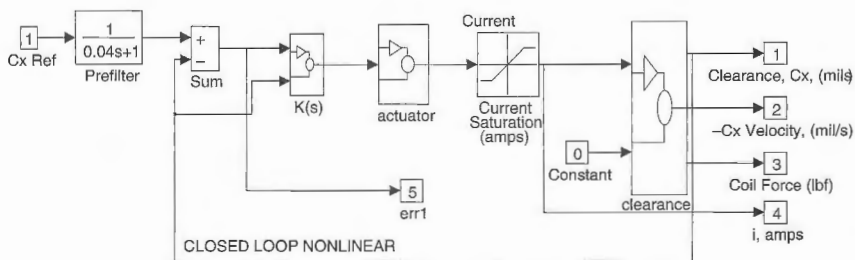


Fig. 7.8 Block diagram of the closed-loop simulation model of the ACC.

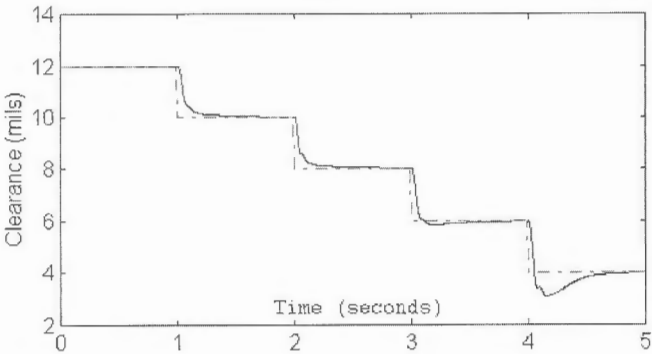


Fig. 7.9 Simulated closed-loop tip-clearance response to a series of step commands.

Fig. 7.10—the measured tip clearance on the top and the actuator current on the bottom. It can be explained on the basis of blade aerodynamics, because decreasing the tip clearance reduces the leakage flow that spills from the pressure side of the blade to the suction side, hence reducing the loss in the compressor.

7.3 Active Stall and Surge Control

Air and gas flows in turbine engines are susceptible to instabilities. These flow instabilities are characterized by rapid, unsteady fluctuations in velocity, temperature, and pressure. The unsteady fluctuations, associated with compressor surge and stall, are often deterministic and repeatable. They disrupt surrounding flow-fields and cause severe stresses on engine components; they also degrade engine efficiency and even cause flow reversals or flameout.

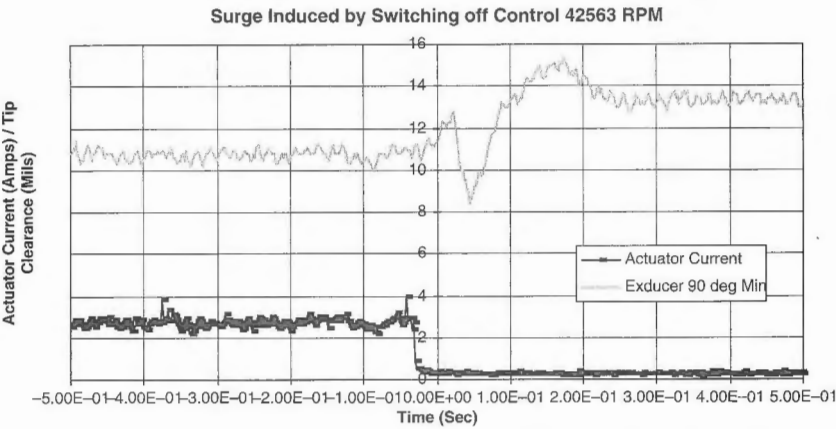


Fig. 7.10 Experimental result of compressor surge induced by widening tip clearance.

Rotating stall and surge represent two most notable flow instabilities in a turbine engine compression system [14–16]. Because engine components are being exposed to increasingly high thermal and aerodynamic loads, and because the components are operating more closely to their physical limits, stall and surge have become important design and operability issues. To address these issues, design engineers have first tried to avoid stall and surge and then to control them actively. Much work has been done on the control of compressor surge and rotating stall in the 1990s [17]. Most of the work relies on modulating flow properties in the compressor at a high speed (or fast response). Fast modulation eliminates flow instabilities while they are still in the beginning stage of stall/surge development. In the beginning stage, disturbances are small, the control authorities (or energy) required to suppress such disturbances are also small, and the growth of the disturbances can be slow enough to be effectively suppressed. For active stall/surge controls, the speed of modulation could be as high as half, or even three-quarters (75%), of the rotational speed of the compressor (or fan depending on the source of the instability).

Two approaches to controlling stall/surge have evolved. One approach involves the flow injection at the inlet of a compressor [18, 19]; the other approach involves the bleed air from an interstage or the exit of a compressor [20, 21]. While inlet injection requires an airflow source, the operating environment of injection actuators is more favorable than the high-temperature and high-pressure environment of bleed air. On the other hand, inlet injection requires multiple actuators to be sequenced properly to suppress rotating stall “cells,” but exit bleed only requires one or a far fewer number of lower-response actuators, although these bleed actuators can be bigger.

Regardless of the approach taken, the actuation system has consistently been identified as the most critical technology to enable active stall/surge control. In early 1990s, the bandwidth of surge control actuation systems was in the range of 400 to 500 Hz [22, 23]. By late 1990s, the bandwidth was extended beyond 1000 Hz [24]. So rather than discussing the unsteady aerodynamics of compressor stall/surge, let us devote the rest of this section to some practical considerations in the design of an active surge control actuator.

7.3.1 Actuation Challenges

In active flow controls, actuation response should be measured for the entire flow control system including the prime mover, flow supply and delivery components, and the flow control device (typically a valve). Flow supply and delivery components involve ducting and plenums; therefore, they inevitably introduce transport delays. These delays can significantly reduce the response of the actuation system.

Another challenge for the actuation system is the diminishing control authority at high frequency. This means that the actuator must be capable of controlling a relatively large amount of flow rate at high frequency for it to be effective in suppressing stall/surge.

Yet another challenge is packaging of the actuation system. Packaging must consider the following implementation-related issues: number of actuators and the optimal placement of each actuator on the engine; size and weight of each

actuator; electrical power for the high-response actuators; supporting components like the manifold and ducting for the storage and delivery of the air or gas flow to the actuators; and electronic control unit with required control laws.

7.3.2 Actuator Design

A high-response actuator for active stall/surge control has been designed. The actuator consists of a prime mover, a valve, and an injector. The prime mover is a piezoelectric translator, which consists of a stack of piezoelectric laminates housed in a cylindrical container. Two obvious reasons for selecting the piezoelectric translator are high actuation bandwidth and large force output. The major drawback of the translator is its relatively small displacement, typically in the range of a few milli-inches for a reasonably sized translator. In order to use the translator to modulate enough amount of the airflow for active stall/surge control, we need to magnify its displacement where it is connected to the flow control valve. A question of interest is whether we can design the magnification mechanism and the valve in a conformal manner so that the complete actuation system is effective and compact.

An answer to the preceding question is a circular-shaped valve. The valve consists of a number of pie-shaped moving parts, which together form the circumference of a circle on the outer edge. Each moving part is hinged near its “waist” on both sides to provide a fulcrum for the magnification of the *vertical* displacement of its outer circular edge, when the center is pushed downward. The piezoelectric prime mover is positioned vertically to the valve, so when it pushes down at the center of each circular-shaped valve, the outer edge of the valve deflects upward and opens a flow passage allowing the air to flow from the top of the valve to the air volume beneath the valve. The air is subsequently injected into the engine inlet flow stream through a closely coupled injector to reduce transport delays. This valve concept is shown in Fig. 7.11, where each of the four circular valves is shaped like an anchor to reduce weight and to increase mechanical strength.

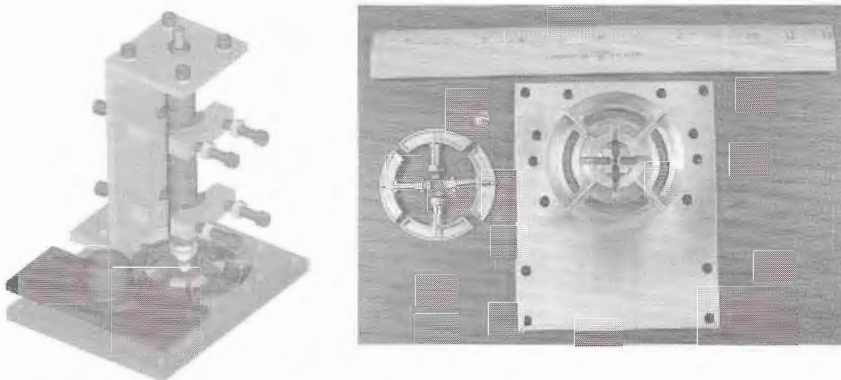


Fig. 7.11 Design model and a fabricated prototype of the high-response air-gas-flow control valve.

7.3.3 Actuator Model

A simplified model of the actuator can be developed to estimate valve performance and flow characteristics. First, we can express the mass flow rate through a valve as follows (see Sec. 2.4.2):

For choked airflow,

$$w_v = 0.532 \frac{A_v C_d P_{t1}}{\sqrt{T_{t1}}} \quad (7.27)$$

For unchoked airflow, Eq. (2.37) can be simplified according to the Fliegner's approximation, which suggests the following equation as a good "fit" for Eq. (2.37):

$$w_v \approx 1.06 \frac{A_v C_d}{\sqrt{T_{t1}}} \sqrt{P_2(P_1 - P_2)} \quad (7.28)$$

where w_v is the mass flow rate (in the unit of pounds per second), A_v is the valve opening area (square inches), C_d is the valve discharge coefficient, P_1 is the valve upstream pressure (pounds per square inch), P_2 is the valve downstream pressure (pounds per square inch or pounds per square inch absolute), and T_1 and T_2 are the flow temperatures in the upstream and downstream of the valve, respectively (in degrees Rankine). This equation is valid for most of the unchoked flow conditions where $\gamma = 1.4$, except when the pressure drop across the valve or injector is very small ($< 5\%$).

The valve opening (i.e., the orifice) area is proportional to the displacement of the piezoelectric translator, which is, in turn, proportional to the translator position command from the control computer. Let the control variable for the position command be V (or V_{cmd}), and then the valve area A_v can be expressed as

$$A_v = K_v V \quad (7.29)$$

where K_v is a conversion coefficient. Substituting Eq. (7.29) into Eqs. (7.27) and (7.28), we can represent the valve flow rate as

$$w_v = 0.532 \frac{K_v C_d V P_{t1}}{\sqrt{T_{t1}}}, \quad \text{if } (P_2/P_{t1}) < 0.528 \quad (7.30)$$

$$w_v \approx 1.06 \frac{K_v C_d V}{\sqrt{T_{t1}}} \sqrt{P_2(P_1 - P_2)}, \quad \text{otherwise} \quad (7.31)$$

Also because the flow going through the circular-shaped valve must be collected in the downstream and discharged either to the fan (or compressor) flow stream in an injection control mode or to the outside of the engine in an overboard bleed control mode, this requires a second opening (or orifice) to complete the

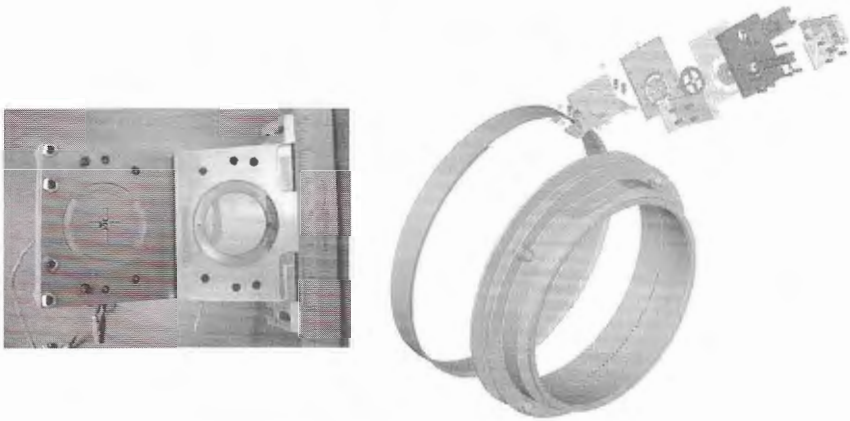


Fig. 7.12 Two serial flow restrictions in the high-response actuator.

discharge function. This second opening essentially forms a second flow restriction in series with the upstream (circular) flow control valve. The second opening is called an injector and is shown in Fig. 7.12. The photo on the left-hand side shows the downstream, annular flow collector and the injector (near the right-hand side of the collector). The drawing on the right-hand side illustrates how the actuator is installed in a slanted way on a ring in front of the compressor. The reason for the angled installation of the actuator is to minimize the distance of travel for the injected air between the valve and the injector, hence reducing the transport delay of the actuation system.

We can analyze the actuation system by applying Eqs. (7.27) and (7.28) to the two serial flow restrictions of the valve and the injector. We are primarily interested in estimating the flow rate and the “control” pressure. The control pressure is the static pressure between the valve and the injector. Because the air flow can choke either at the valve or at the injector, the number of possible combinations (or cases) of flow characteristics in the actuator is four. The flow rates and the control pressures for the four cases are summarized in Table 7.11. A dimensionless coefficient k has been defined as

$$k = \frac{K_v C_{dv} V}{A_i C_{di}} \quad (7.32)$$

Note that the coefficient k is a function of control command V . Also note that making the flow choke at the valve is a more desirable condition because then the flow will be linearly proportional to the control command (voltage).

The variables referenced in Table 7.11 are defined as follows:

w_v = mass flow rate through the valve opening, pps

w_i = mass flow rate through the injector, pps

A_v = valve opening area, in.²

A_i = injector area, in.²

C_{dv} = valve discharge coefficient

Table 7.11 Possible flow conditions in the high-response stall/surge actuator

Case no.	Choking conditions	Pressure and mass flow rate
1	Valve choked Injector unchoked	$w_v = 0.532 \frac{K_v C_{dv} V P_0}{\sqrt{T_1}}$ $P_1 = P_2 + k^2 \frac{P_0^2}{P_2}$
2	Valve choked Injector choked	$w_v = 0.532 \frac{K_v C_{dv} V P_0}{\sqrt{T_1}}$ $P_1 = k P_0$
3	Valve unchoked Injector unchoked	$w_v = 1.06 \frac{K_v C_{dv} V}{\sqrt{T_2}} \sqrt{P_1(P_0 - P_1)}$ $P_1 = \frac{(k^2 P_0 - P_2) + \sqrt{(k^2 P_0 - P_2)^2 + 2k^2 P_2^2}}{2k^2}$
4	Valve unchoked Injector choked	$w_v = 1.06 \frac{K_v C_{dv} V}{\sqrt{T_2}} \sqrt{P_1(P_0 - P_1)}$ $P_1 = \frac{4k^2 P_0}{1 + 4k^2}$

C_{di} = injector discharge coefficient (assumed equal to C_{dv})

P_0 = air supply pressure, psi

P_1 = valve downstream pressure or control pressure, psi

P_2 = injector downstream pressure, psi

T = temperature (assumed the same for all three locations), °R.

7.3.4 Actuator Experimental Results

The high-response air valve was tested first in a low-speed wind tunnel and then in a compressor rig. Figure 7.13 shows the experimental setup of the valve mounted on the top of the wind tunnel. The photo on the right-hand side shows the hot-wire anemometer used to measure the velocity profile of the air coming out of the injector by traversing across the width of the injector opening.

Figure 7.14 shows the experimental setup of the valve mounted on a Honeywell compressor rig during testing. The drawing on the right-hand side shows the assembled actuator as it sits on top of a flange in front of the compressor revealing the slanted mount.

Table 7.12 summarizes the results of one of the wind-tunnel tests to characterize the static performance of the actuator as the position command X_{cmd} of the piezoelectric translator is changed from fully closed to fully open. For this test, the supply pressure P_0 of the air is 40 psia. The static performance is measured

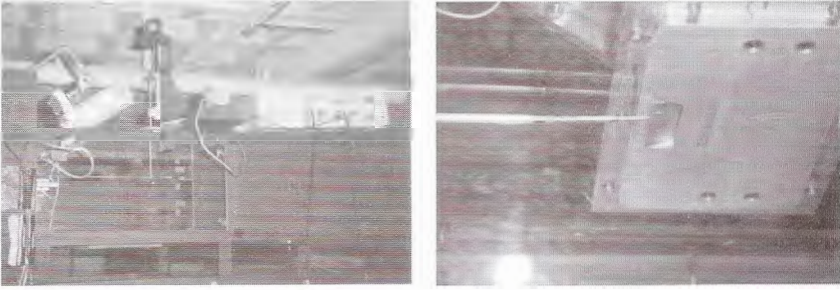


Fig. 7.13 Wind tunnel (left) test of the high-response actuator (courtesy of Scientific Monitoring, Inc.).

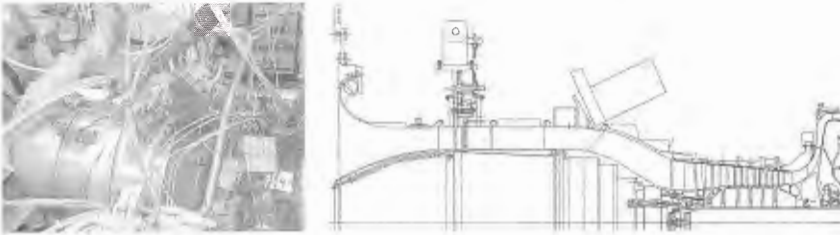


Fig. 7.14 Compressor rig (left) test of the high-response actuator (courtesy of Honeywell).

Table 7.12 Summary of a wind-tunnel test of the high-response actuator

Test	$X_{\text{cmd}}, \mu\text{m}$	V_j, fps	w_v, pps
1	0	73	—
2	20	85	—
3	40	87	0.018
4	60	98	0.021
5	80	108	0.024
6	100	117	0.026
7	120	124	0.026
8	140	161	0.035
9	160	215	0.047
10	180	268	0.051

by the jet velocity V_j of the air coming out of the injector and the mass flow rate of the air w_{jv} .

The flow rate measured on the compressor rig, as a function of translator command and supply pressure, is shown in Fig. 7.15. Because the valve opening is proportional to the translator displacement, which in turn is proportional to the voltage command, we have expressed the flow rate in terms of the voltage command. Note that a slight leakage (about 0.04 pps) exists in the actuator.

The dynamic test of the high-response actuator with the aerodynamic load was conducted by using the frequency response method, that is, sinusoidal excitations in the frequency range between 10 and 1000 Hz (with a total of 18 test frequencies) were used. At each test frequency, three variables were measured at the rate of 5000 samples per second, and two variables were measured at the rate of 12,500 samples per second. The three lower-rate measured variables are the translator command from valve controller V_{cmd} , the translator electrical potential from power amplifiers V_{amp} , and the translator displacement V_{pzt} . The remaining two measured variables are the static pressure in the upstream of the valve P_1 and the static pressure in the downstream of the valve P_2 .

Figure 7.16 shows the time histories of the three lower-rate measurements at the excitation frequency of 10 Hz. The signal of V_{cmd} is in blue, the signal of V_{amp} multiplied by 20 is in green, and the signal of V_{pzt} is in red. Figure 7.17 shows the two higher-rate signals at the excitation frequency of 1000 Hz in the upper-left-hand chart; also plotted in the figure are the power spectral density of the flow rate and the mass flow rate itself.

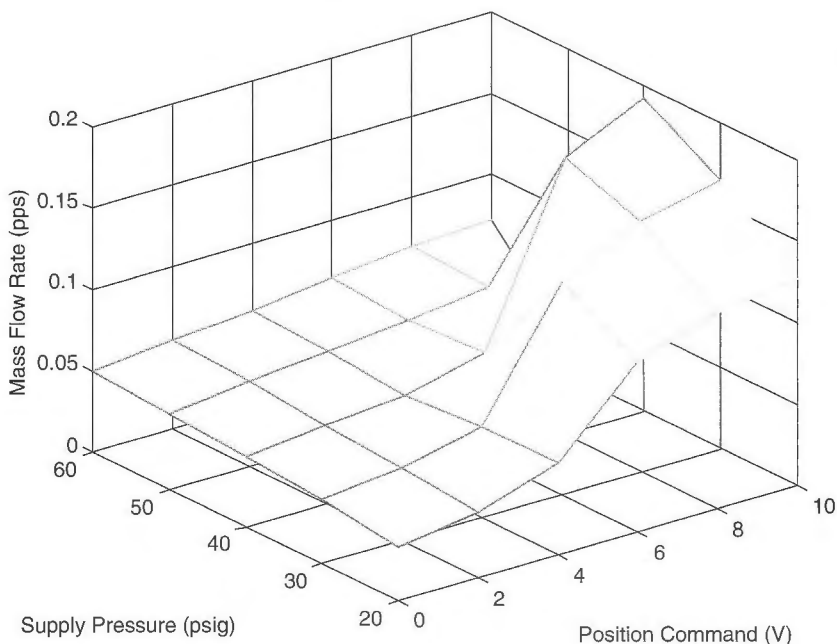


Fig. 7.15 Measured steady-state flow rates through the stall/surge control actuator.

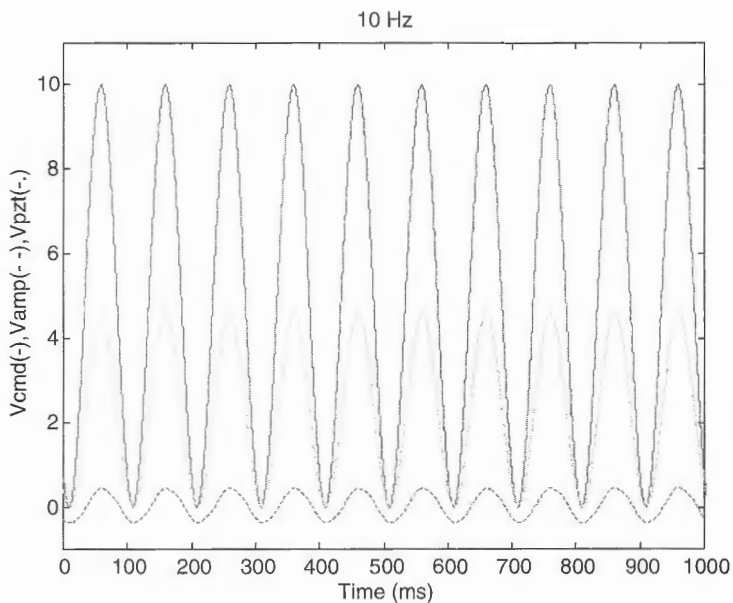


Fig. 7.16 Translator command and displacement signals at 10 Hz. (See also color figure section at the back of the book.)

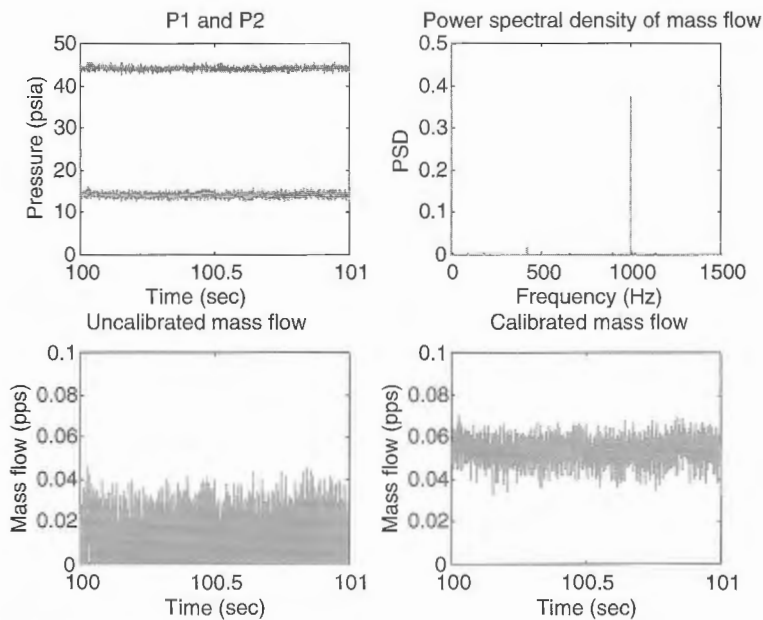


Fig. 7.17 Valve pressures and estimated flow rates at 1000 Hz.

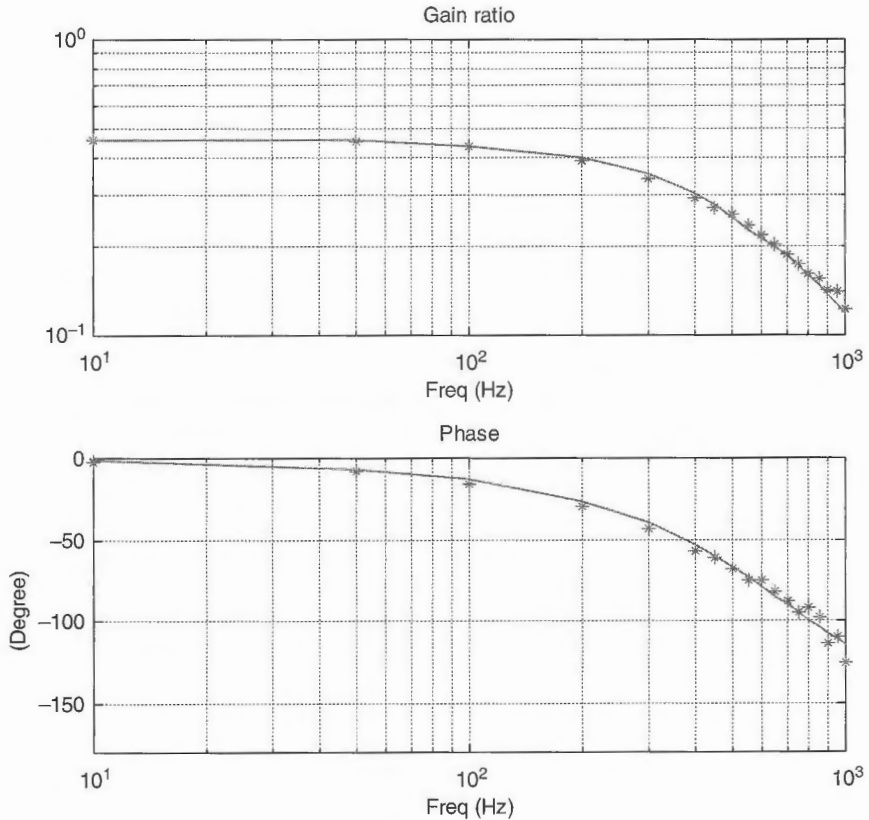


Fig. 7.18 Frequency response of the actuator mass flow rate.

The frequency response of the valve was also computed for both the dry (or unloaded) and the wet (or loaded) test conditions. Figure 7.18 shows the flow rate at the wet condition on the compressor rig test.

The rig test confirmed the large-flow high-bandwidth capability of the actuator. The maximum static flow measured on the rig was 0.202 pps. This flow rate was measured at the piezoelectric translator potential of 100 V and the air supply pressure of 50 psig. The mechanical integrity of the valve with multiple moving parts was also proven. Furthermore, the heat dissipation of the piezoelectric translator was found to be moderate even when it operated at frequencies above 500 Hz continuously for approximately 15 s.

7.4 Active Combustion Control

Active combustion controls can be divided into two categories: active instability control and active combustion enhancement, with the instability control being

more widely studied [25, 26]. Combustion instabilities are characterized by rapid thermo-acoustic oscillations. These oscillations (sometimes called *thermo-acoustic instabilities*) are primarily caused by the unsteady interaction between the pressure and the heat release in the combustion chamber [27, 28]. This interaction is particularly strong in gas turbine engines with a lean combustion technology and can contribute to a resonant growth of the oscillations. Lean combustion has the advantages of reducing fuel consumption and carbon emissions, although it can cause flame-out and sonic fatigue in structural components, or even increase the production of oxides of nitrogen (NO_x).

Active instability control aims at reducing thermo-acoustic instabilities resulting from the coupled pressure and heat oscillations, either in the main burner or the afterburner. For active instability controls, actuation technology has again been identified as an enabling technology.

Two approaches have been considered for active controls: 1) controlling the total fuel flow directly and 2) controlling a fraction of the total fuel flow. In the second approach, a modulated, secondary fuel flow is superimposed on the primary fuel flow in order to eliminate the pressure oscillations, which typically fall in the frequency range of 300 to 1000 Hz. The advantage of the secondary flow control is that the stability control system can be designed to accommodate large variations in the combustion response that change with engine operating conditions [29]. Regardless of which approach is taken, the control effectiveness increases with decreasing distance between the valve and the fuel nozzle because this distance is a measure of transport delay and damping [30, 31]. Depending on the type of instability and the size of the combustion zone that the secondary fuel flow is expected to control, the modulation magnitude typically requires 10 to 20% of the primary fuel flow rate. Therefore, a high-response actuator that can modulate secondary fuel streams up to $\pm 20\%$ of the total fuel flow rate at frequencies above 500 Hz is of high interest to active instability controls.

Several candidate technologies for the high-response fuel actuator have been investigated by various researchers. The three most promising types of actuation technology are 1) smart materials such as piezoelectric and magnetostrictive translators, 2) solenoids, and 3) high-response servovalves. Each of these types has bandwidth and force limitations [32]. However, the actuation technology that is described here falls in the same category of smart materials as the magnetostrictive; actually, it is similar to the piezoelectric technology used in the high-response air valve described in the preceding section for stall/surge control.

In this section, design and test of an active combustion control actuator to suppress instabilities are discussed. The actuator is capable of modulating the fuel flow rate of $\pm 90\%$ at the *valve* exit over the test frequency range of 300–550 Hz. The modulation range of the actuator is $\pm 50\%$ of the flow rate at the *nozzle exit*, after a 6-in. fuel line between the valve and the nozzle (hence introducing transport delays).

7.4.1 Actuator Design

A family of high-response fuel actuators for active instability controls has been designed [33]. These actuators consist of a prime mover and a valve. The

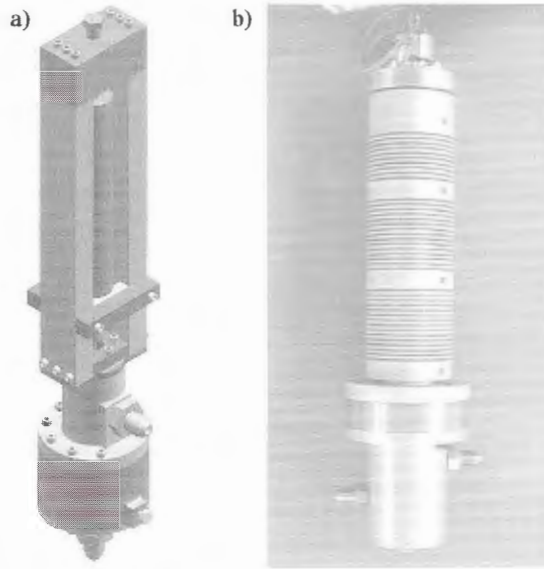


Fig. 7.19 Design model and a prototype of the high-response fuel actuator.

prime mover is a piezoelectric translator, and the valve is a flapper-controlled port. A three-dimensional mechanical model of the valve is shown in Fig. 7.19a; a prototype actuator with improved cooling capability around the piezoelectric translator and with optimized flow path in the valve is shown in Fig. 7.19b.

7.4.2 Actuator Modeling

A closed-loop control system model of the fuel actuator has also been developed to analyze the performance of the actuator and different control options. Figure 7.20a shows the closed-loop simulation model of the fuel actuator, represented by the high-response, high-modulation fuel valve block in the Simulink® diagram. Figure 7.20b shows the model of the entire fuel control system including the fuel delivery and injection system.

By measuring pressure oscillations and modulating the fuel flow at the dominant frequency of these pressure oscillations, we can achieve the purpose of eliminating or significantly reducing thermo-acoustic instability. As shown in the model, pressure oscillations are caused by the instantaneous heat release from the fuel (represented by DQ) and are measured by a sensor as the instantaneous pressure change (DP). The control law computes a position command for the prime mover, which modulates the fuel flow through the valve (DW_{fuel}) to remove the heat oscillations. The heat-release DQ and the resultant pressure oscillations DP cover a wide frequency band. However, the frequencies at which self-excitation occurs are the resonant frequencies of combustion instabilities.

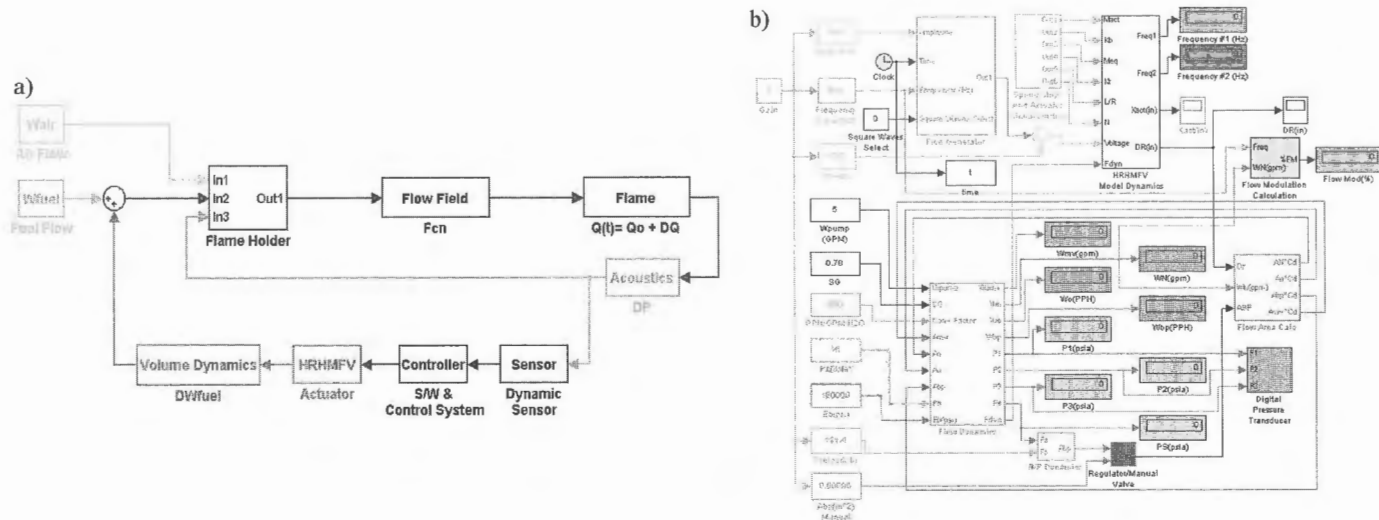


Fig. 7.20 Dynamic simulation model of the high-response fuel actuator.

Assuming that the flame can be approximated by a one-dimensional unsteady flow, then the volumetric heat-release rate q can be expressed as a function of both spatial location x and time t ,

$$q(x, t) = q(x_0, t) = \delta(x - x_0) \cdot q_0(t) \quad (7.33)$$

where x_0 is the location of the flame and $\delta()$ is the delta function. If x_0 remains fairly steady with time (assuming that the flame front is close to the fuel injector), the small-perturbation heat-release rate becomes a function of time only; hence, the preceding equation can be further simplified to

$$q(x, t) = q_0(t) \approx \bar{q}_0 + \tilde{q}(x_0, t) \quad (7.34)$$

where the “bar” represents the mean value and the “tilde” represents a small perturbation around the mean value. Equation (7.34) provides the basis for combustion instability control, and a simplified control law can be expressed as

$$\Delta W_f = -k_1 \cdot \Delta q \approx -k_2 \cdot \Delta p \quad (7.35)$$

where Δp represents the instantaneous pressure change, as a result of the instantaneous heat release, in the chamber.

7.4.3 Actuator Experimental Results

Because of the potential danger involved in testing the high-response actuator with flammable fuel, the fuel actuator was subject to two types of testing: one is a dry test (without fluid in the valve), and the other is a wet test. In the dry test, the valve housing was removed exposing the flapper so that the movement of the flapper, when it was “actuated” by the prime mover, could be measured. The dry test was conducted to verify the actuation concept and mechanical integrity of the prototype actuator. Specifically, the test was designed to verify that the actuator can operate, in a no-load condition, up to 500 Hz, although higher actuation frequencies could be tested also. A simple optical sensor was built to detect the direction and frequency of the displacement of the flapper. The sensor consists of a laser, a lens attached to the flapper, and a photocell as the position detector. The laser projects a beam on the lens, and the photocell is placed in the direction of the reflected laser beam. When the flapper is pushed by the prime mover, the reflected laser beam changes its direction and points to a different position on the photocell. The voltage change from the photocell is used to determine position change of the laser beam and thus the movement of the flapper. The experimental setup of this dry test is shown in Fig. 7.21.

Figure 7.22 is a blown-up view of the oscilloscope showing the 500-Hz waveform of the signal received by the photocell. The waveform is consistent with the square wave command sent to the prime mover from the desktop control computer.

The wet test was conducted at a fuel bench. Figure 7.23 shows the setup of the fuel bench with the improved actuator prototype.

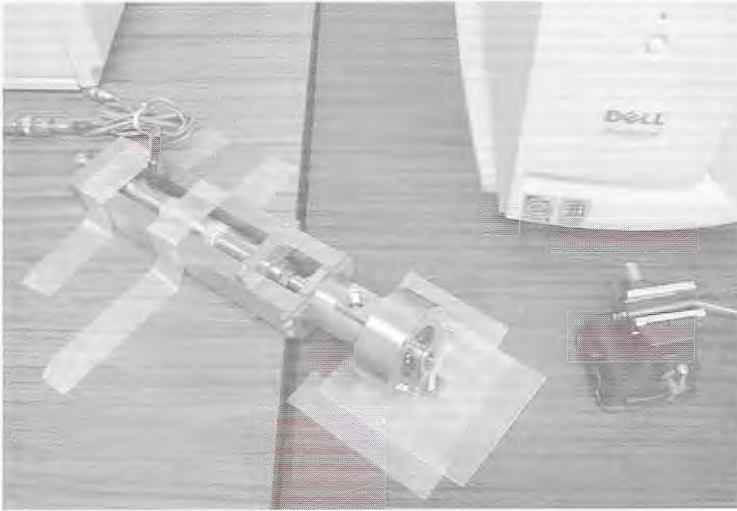


Fig. 7.21 Dry-test setup showing the laser pointing at valve.

Two actuator prototypes were built and tested on the fuel bench. The fuel supply (or valve inlet) pressures for the dynamic test ranged between 300 and 615 psig. For each supply pressure, three actuation frequencies were tested, and they are 300, 450, and 550 Hz. The actuators were calibrated so that the total fuel flow through each actuator was 2000 pph when the valve was fully open and the supply pressure was 600 psig. The wet test demonstrated a modulation capability of $\pm 90\%$ of the total flow at the valve exit over the frequency range of 300–550 Hz.

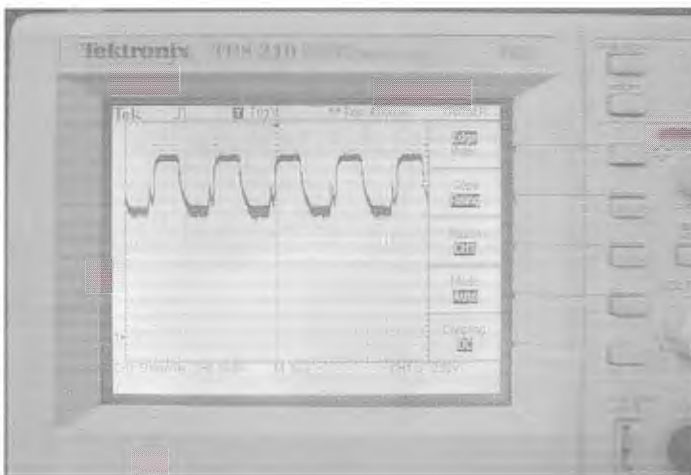


Fig. 7.22 Photocell output waveform showing the valve moving at 500 Hz.



Fig. 7.23 Wet-test setup of the actuator on a fuel bench.

and the supply pressure range of 300–615 psig. The test was repeated after a 30-in. long pipe between the valve and the fuel nozzle was installed to simulate the transport delay on an engine. The modulation capability of the actuators reached $\pm 50\%$ even with the transport delay. Table 7.13 summarizes the results of the wet test of the two actuators [34].

Figure 7.24 shows the measured steady-state fuel flow rate through the two prototype actuators at the supply pressure of 300 psig. Figure 7.25 shows three measured dynamic pressures (relative to their respective nominal values) and their corresponding power-spectral-density (or power spectrum) diagrams. The tests were conducted at the supply pressure of 615 psig and actuation frequency of 300 Hz. The three measured pressures are the valve upstream pressure P_1 , the valve downstream pressure P_2 , and the nozzle inlet pressure P_{2N} , at the end of the 30-in. pipe.

Table 7.13 Summary of results of fuel actuator wet tests

Test	Supply pressure, psig	Operating frequency, Hz	Actuator 1		Actuator 2	
			Mean flow, pph	Flow modulation, %	Mean flow, pph	Flow modulation, %
1	300	300	550	± 45	417	± 54
2	300	450	720	± 23	540	± 57
3	300	550	700	± 41	475	± 34
4	615	300	1300	± 17	900	± 11
5	615	450	1368	± 25	1143	± 13
6	615	550	1368	± 6	1095	± 7

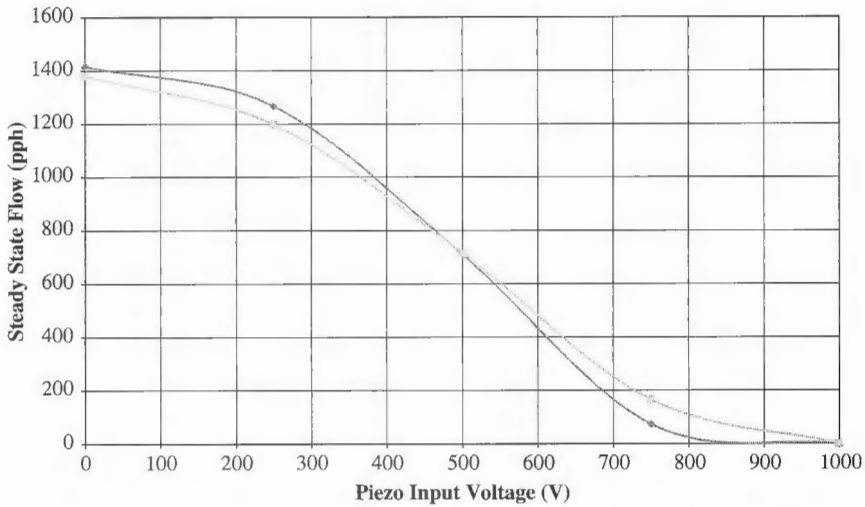


Fig. 7.24 Steady-state fuel flows for valve 1 (blue) and valve 2 (red) vs prime-mover input voltage at 300 psig. (See also color figure section at the back of the book.)

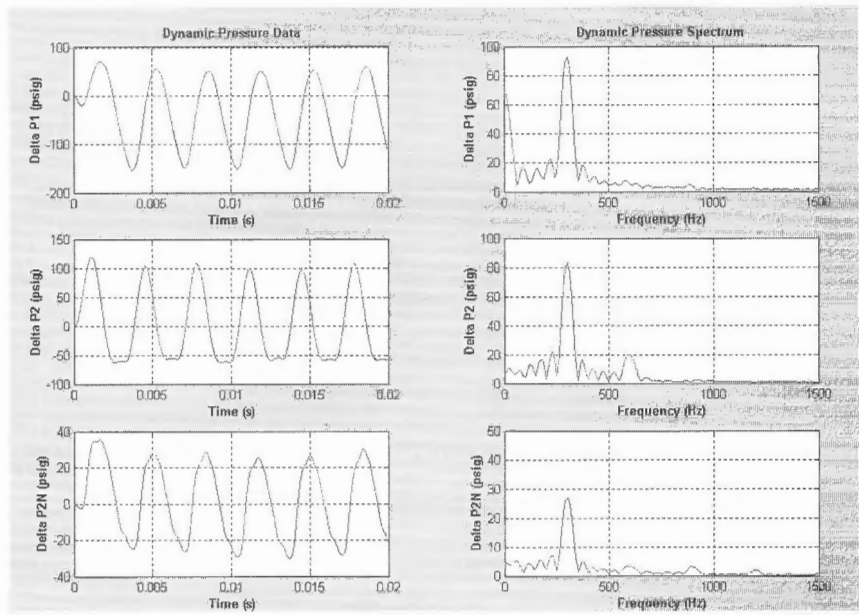


Fig. 7.25 Dynamic pressure measurements and the corresponding power spectrums of one actuator at 615-psig supply pressure and 300-Hz modulation frequency.

References

- [1] Bryson, A. C., Jr., and Ho, Y.-C., *Applied Optimal Control—Optimization, Estimation, and Control*, Hemisphere, New York, 1975.
- [2] Sain, M. K., Peczkowski, J. L., and Melsa, J. L. (eds.), *Alternative for Linear Multivariable Control*, National Engineering Consortium, Inc., 1978.
- [3] Hackney, R. D., Miller, R. J., and Small, L. L., "Engine Criteria and Models for Multivariable Control System Design," Paper #1.1, *Alternative for Linear Multivariable Control*, National Engineering Consortium, Inc., 1978.
- [4] Sain, M. D., "The Theme Problem," Paper #1.2, *Alternative for Linear Multivariable Control*, National Engineering Consortium, Inc., 1978.
- [5] Rosenbrock, H. H., *Computer-Aided Control System Design*, Academic International Press, New York, 1974.
- [6] Rosenbrock, H. H., and Munro, N., "The Inverse Nyquist Array Method," Paper #3.1, *Alternative for Linear Multivariable Control*, National Engineering Consortium, Inc., 1978.
- [7] Merrill, W., "Output Feedback Regulator Design for Jet Engine Control System," Paper #6.2, *Alternative for Linear Multivariable Control*, National Engineering Consortium, Inc., 1978.
- [8] Dorf, R. C., and Bishop, R. H., *Modern Control Systems*, 10th ed., Pearson Prentice-Hall, Upper Saddle River, NJ, 2005.
- [9] Franklin, G. F., Powell, J. D., and Emami-Naeini, A., *Feedback Control of Dynamic Systems*, 5th ed., Pearson Prentice-Hall, Upper Saddle River, NJ, 2006.
- [10] "Comparison of Major Control Design/Synthesis Techniques," System Controls Technologies, Rept., 1988 (courtesy of Stephen Rock, Stanford Univ., Stanford, CA).
- [11] Jaw, L. C., et al., "Active Tip Clearance Control for Small Turbine Engines," U.S. Army Aviation and Troop Command, Rept. USAATCOM TR 96-D-43, Dec. 1996.
- [12] Weimer, M., "Design and Test of an Active Tip Clearance System for Centrifugal Compressors," *Proceedings of the 28th Joint Propulsion Conference*, AIAA, Washington, D.C., 1992; also AIAA Paper 92-3189, 1992.
- [13] Mattern, D., Jaw, L. C., et al., "Experimental Results of an Active Tip Clearance Control System for a Centrifugal Compressor," *Proceedings of the International Gas Turbine Congress and Exposition*, 1997.
- [14] Emmons, H. W., Pearson, C. E., and Grant, H. P., "Compressor Surge and Stall Propagation," *Transactions of the ASME*, May 1955, pp. 455–469.
- [15] Greitzer, E. M., "Surge and Rotating Stall in Axial Flow Compressors, Part I: Theoretical Compression System Model; and Part II: Experimental Results and Comparison with Theory," *Journal of Engineering for Power*, Vol. 98, No. 2, April 1976, pp. 190–216.
- [16] Moore, F. K., "A Theory of Rotating Stall of Multistage Compressors, Part I, II, III," *Journal of Engineering for Power*, Vol. 106, No. 2, April 1984, pp. 313–336.
- [17] Paduano, J. D., "Recent Developments in Compressor Stability and Control," Paper, Feb. 1998.
- [18] Day, I. J., "Active Suppression of Rotating Stall and Surge in Axial Compressors," American Society of Mechanical Engineers, Paper 91-GT-87.
- [19] Epstein, A. H., Ffowcs Williams, J. E., and Greitzer, E. M., "Active Suppression of Aerodynamic Instabilities in Turbomachines," AIAA Paper 86-1994, 1986; also *Journal of Propulsion*, Vol. 5, No. 2, 1989, pp. 204–211.

- [20] Adomaitis, R. A., and Abed, E. H., "Local Nonlinear Control of Stall Inception in Axial Flow Compressors," AIAA Paper 93-2230, June 1993.
- [21] Badmus, O. O., Chowdhury, S., Eveker, K. M., and Nett, C. N., "Control-Oriented High-Frequency Turbomachinery Modeling: Single-Stage Compression System 1D Model," American Society of Mechanical Engineers, Paper 93-GT-18, May 1993.
- [22] Berndt, R. G., Weigl, H. J., Paduano, J. D., and Epstein, A. H. "Experimental Techniques for Actuation, Sensing and Measurement of Rotating Stall Dynamics in High Speed Compressors," *Society of Photo-Optical Instrumentation Engineers (SPIE) Proceedings for Sensing, Actuation, and Control in Aeropropulsion*, edited by J. D. Paduano, Vol. 2494, SPIE, Bellingham WA, April 1995, pp. 166–185.
- [23] Mattern, D., and Owen, A. K., "A Voice Coil Actuated Air Valve for Use in Compressor Forced Response Testing," *Society of Photo-Optical Instrumentation Engineers (SPIE) Proceedings for Sensing, Actuation, and Control in Aeropropulsion*, edited by J. D. Paduano, Vol. 2494, SPIE, Bellingham WA, April 1995, pp. 215–223.
- [24] Jaw, L. C., Wu, D. N., Zhou, G. A., and Bryg, D. J., "A High-Response High-Gain Actuator for Active Flow Control," AIAA Paper 99-2128, June 1999.
- [25] McManus, K. R., Poinso, T., and Candel, S. M., "A Review of Active Control of Combustion Instabilities," *Progress in Energy and Combustion Sciences*, Vol. 19, No.1, 1993, pp. 1–29.
- [26] Docquier, N., and Candel, S., "Combustion Control and Sensors: a Review," *Progress in Energy and Combustion Science*, Vol. 28, No.2, 2002, pp. 107–150.
- [27] Annaswamy, A. M., and Ghoniem, A. F., "Active Control in Combustion Systems," *IEEE Control Systems Magazine*, Vol.15, No. 6, Dec. 1995, pp. 49–63.
- [28] Zinn, B. T., "An Overview of Active Control of Combustion Instabilities," AIAA Paper 1997-461, 1997.
- [29] Richards, G. A., and Robey, E. H., "Dynamic Response of a Premix Fuel Injector," American Society of Mechanical Engineers, Paper 2001-GT-0036, 2001.
- [30] Ziada, S., and Graf, H., "Feedback Control of Combustion Oscillations," *Journal of Fluids and Structures*, Vol. 12, No.4, May 1998, pp. 491–507.
- [31] Johnson, C. E., Neumeier, Y., Darling, D., Sattinger, S., Neumaier, M., and Zinn, B., "Demonstration of Active Control of Combustion Instabilities on a Full-Scale Gas Turbine Combustor," American Society of Mechanical Engineers, Paper 2001-GT-0519, 2001.
- [32] Richards, J. S., "An Exploration of Secondary Fuel Injection as Actuation for Control of Combustion Instabilities in a Laminar Premixed Tube Combustor," MSME Thesis, Virginia Polytechnic Inst. and State Univ., Blacksburg, VA, 2000.
- [33] Tran Van, H., et al., "Design, Simulation, and Test of a High Response, High Modulation Fuel Valve for Active Thermo-Acoustic Instabilities Control in Combustors and Afterburners," *Proceedings of the 2003 International Society of Air Breathing Engines Symposium*, 2003.
- [34] Tran Van, H., and Jaw, L. C., "Smart Material Actuators for High-Response Flow Control—Final Report," U.S. Air Force Research Lab., Contract No. F33615-00-C-2015, May 2004.

Engine Monitoring and Health Management

Because the aircraft propulsion system is critical to flight safety and dispatch reliability, engine/propulsion monitoring has led the way in the maturation of monitoring and health management capabilities for aircraft. Historically, engine condition monitoring and engine diagnostics are handled by control systems because the data collected from onboard sensors are primarily used to control the engines. From the early monitoring capabilities of the 1970s, such as event history recorders and engine health monitors, engine monitoring has evolved into a highly sophisticated and necessary capability to ensure safety, reliability, and availability for modern gas turbine engines.

Indeed, *safety*, *reliability*, and *availability* are the primary goals of any monitoring and health management system. These goals are interrelated; for example, an aircraft might be grounded (or unavailable) for a safety concern of the engine part that is about to fail because of poor reliability. To achieve these goals, an engine health management system must possess three essential things: 1) onboard data collection and analysis, 2) off-board data trending and maintenance decision support, and 3) support resources to keep the engine operating (i.e., sustainment), such as maintainers, ground-support personnel and equipment, parts, and supplies.

In this chapter, some basic concepts of engine monitoring and health management will be introduced. Then, the architecture and algorithmic framework of engine monitoring systems will be discussed. Lastly, the mathematical foundations of monitoring algorithms, with an emphasis on model-based algorithms, will also be discussed.

8.1 Basic Concepts

8.1.1 Safety, Reliability, and Availability

Safety is the degree of assurance that a system will not cause human or property damage in case of a failure. By an aviation industry's de facto safety standard [1], airborne software can be certified, according to their criticality of safety, in five different levels: A, B, C, D, or E. The criticality is defined as the consequence of a potential failure, and there are five different criticality levels: catastrophic, hazardous-severe, major, minor, and no-effect. In the engine design and development phase, safety is addressed by both sound product designs and stringent certification processes. After the engine is certified and

introduced into service, safety is enforced by disciplined audits with periodic checks and tests.

Reliability is a measure of the probability of nonfailure of a product or system, that is, the probability that the product will not fail at any given time; hence, reliability is typically decreasing with the (service) life of a product. The level of reliability can be measured in two different contexts: 1) inherent reliability, which is used to define, measure, and evaluate a *design* program or process; and 2) operational reliability, which is used to describe reliability *performance* when operated in the expected environment. A typical metric for inherent reliability is mean time between failure, and typical metrics for operational reliability are mean time between maintenance (MTBM) and mean time between removals [2].

Engine removals fall into two types: scheduled and unscheduled. Scheduled removals can be planned so as to avoid undesirable delays and cancellations. Unscheduled removals are not planned and have a major impact on system reliability. Hence, the mean time between unscheduled removals (MTBUR) is a more important metric for operational reliability.

A similar metric as MTBUR is time on wing (TOW), which is typically measured by hours. TOW is an indication of the availability of the engine (fleet) to support (aircraft) operations. We will review the concept of availability in a short while, but let us first focus on the concept of maintainability. Maintainability is typically measured by mean time to repair (MTTR), maintenance man-hour (MMH) per maintenance event, MMH per flying/operating hour, downtime, logistics delay time, and the number of no fault found.

Availability is the ability of a product (or service) to be ready for use when the customer wants to use it. For a product to be ready for use, it must be able to provide desired functions with a sufficient reserve for every consumable item, such as the fuel or the lubrication oil. Availability is generally defined as

$$A \equiv \frac{\text{Uptime}}{\text{Uptime} + \text{Downtime}} \quad (8.1)$$

where Uptime is equal to the hours that the product works and Downtime is equal to the total number of hours that the product is inoperable. Thus, the operational availability can be defined as

$$A_o \equiv \frac{\text{MTBM}}{\text{MTBM} + \text{MDT}} \quad (8.2)$$

where MDT is the mean downtime.

8.1.2 Anomaly, Fault, and Failure

There are three basic types of abnormal conditions: 1) a *fault*, 2) an *anomaly*, and 3) a *failure*, although these conditions are often used interchangeably in monitoring and health management literatures, hence creating confusions for a reader. First, an abnormal condition is defined as a "sizable" deviation of the actual operating condition from the normal (or expected) condition. The deviation can apply to a single measured variable or a group of variables that together represent a distinguishable pattern. A sizable deviation means that the difference is larger than a

predetermined threshold of tolerance. A deviation is typically called a *residual*. Abnormal conditions generally result from two common causes: 1) usage-induced degradation and 2) external factors that exert an abrupt change or an accelerated rate of change (e.g., foreign object damage, or FOD, in gas turbine engines).

Secondly, a fault is defined as a *known* and *repeatable* abnormal condition, an anomaly is an *unknown* (that is, not seen before, not fully understood, or intermittent) abnormal condition, and a failure is the abnormal condition where the performance of a product has degraded to the point that it can no longer provide intended functionality. Figure 8.1 shows four different quadrants (or sets) that divide the operating condition space and the knowledge space. Specifically, the region of faults and the region of anomalies are divided by our knowledge about the abnormal conditions.

Because an abnormal condition degrades continuously over time, the *severity* of a fault increases with the usage of a system. The time history of degradation is depicted as trajectory in a performance-vs-time diagram in Fig. 8.2.

This trajectory is a degradation path, or more technically, a failure progression path, because the degradation will reach a functionally unacceptable level defined as a failure, given enough time of usage. The figure also shows the relative locations of an anomaly, a fault, and the point of failure in the degradation process. At first, a *precursor* of performance degradation can be observed as the earliest detectable sign of an abnormal condition. This precursor reflects the sensitivity of detection of a monitoring system, although it depends largely on the sensing and modeling capabilities used in system. As the degradation increases, a condition is reached where the degradation satisfies the criteria to alarm the operator or the maintainer about the situation, and we call this condition a fault. A fault is usually associated with a specific *root cause*. We consider a fault less severe than a failure, and in most cases a product can continue to operate with one or more faults present (typically up to a predetermined time period or usage count). A fault can eventually develop into a failure, and the product is said to have reached its service life limit. If it continues to operate beyond this point, the product will break down and can cause a catastrophic scenario. Furthermore, a fault *symptom* is a collection

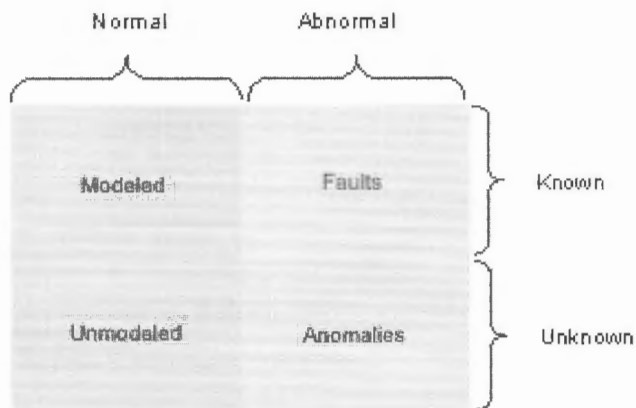


Fig. 8.1 Recommended definitions for faults and anomalies.

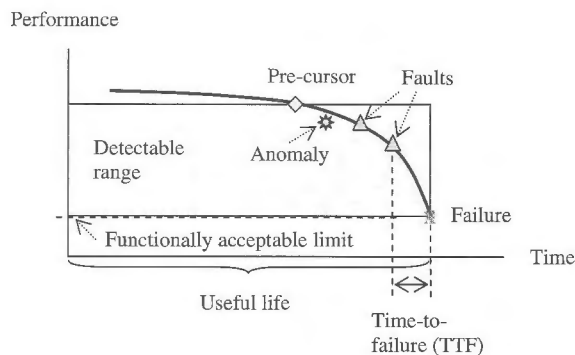


Fig. 8.2 Performance vs time diagram and degradation path.

of individual evidences of an abnormal condition that occur in the same time frame of interest. A fault symptom can include one or more *failure modes*, and each failure mode describes a failure mechanism that decreases the functionality of a product. If we can measure the functionality quantitatively and replace the word “performance” in Fig. 8.2 with “functionality” or “capability,” then the figure represents a failure progression path of a specific failure mode.

8.1.3 Diagnostics and Prognostics

Two basic monitoring concepts are developed around the abnormal condition, and they are diagnostics and prognostics. In a simple description, diagnostics refers to the process of finding *current* and *past* abnormal conditions (often called events), whereas prognostics refers to the process of estimating the *future* performance, functionality, or capability of a product based on the currently detected faults.

Modern development of monitoring capabilities for machines or industrial equipment (collectively called systems) started in the 1960s. Initially, the interest was in the detection and isolation of failures in a system, and it was primarily based on our knowledge of the failure modes and effects of the system [3]. Then the interest was expanded to accommodate the failure so that the system can continue to operate for a period of time until proper maintenance can be performed. Failure accommodation can be realized by having redundant hardware; however, hardware redundancy increases cost and weight. A more preferred approach is *software* (or *analytical*) redundancy, which relies on a model to provide the redundant information needed to meet monitoring and safety requirements [4, 5].

Since the 1990s, prognostics has become the primary focus for engine monitoring systems, largely because of the new requirements of autonomic logistics for the Joint Strike Fighter (JSF) [6]. With this new focus, monitoring functions have been increasingly integrated with maintenance planning, maintenance management, and logistics functions. Hence, a new generation of systems has evolved to provide a complete range of *health management* functions. These “total” health management functions are often called prognostics and health management (PHM) [7, 8]. Figure 8.3 illustrates the evolution of monitoring and health

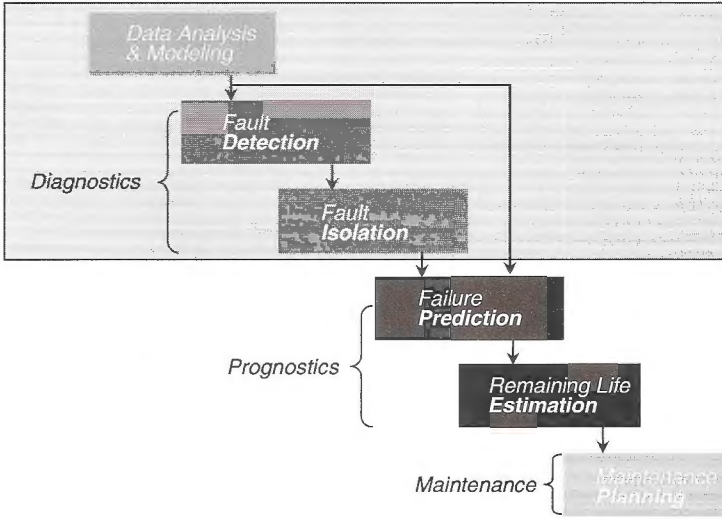


Fig. 8.3 Evolution of monitoring and health management functions.

management functions for aircraft engines. In the figure, failure prediction and life estimation are both related to the time to failure (TTF) of a part or system, when it is subject to a defined sequence of operational scenarios.

From Appendix A, a general linear system can be represented by Eqs. (A.1) and (A.2). As the system (or product) deteriorates, its performance and functionality degrade. Performance degradation can be manifested in time-varying changes in the parameters of the system equations, that is, time-varying parameters. Because time-varying linear systems are difficult to analyze, an alternative approach to degradation modeling is to treat any time-varying parameter as an additional state variable. This is achieved by augmenting the original state and output equations with an additional state variable (or state vector), called the *damage state variable*, $x_d(t)$, that is,

$$\dot{x}_A(t) = A_A x_A(t) + B_A u(t) \quad (8.3)$$

$$y(t) = C_A x_A(t) + Du(t) \quad (8.4)$$

where $B_A = [B \ 0]^T$, $x_A(t)$ is the augmented state variable and is defined as

$$x_A(t) = [x(t) \ x_d(t)]^T \quad (8.5)$$

Note that Eqs. (8.3) and (8.4) can be used as the general form of performance degradation (or damage) model by defining the state variable as consisting of the normal state variable x and the damage state variable x_d . Also note that these equations represent a system fault model in continuous time.

The goal of diagnostics is to detect faults and to isolate the faults to one or more root causes. To detect a fault, some performance degradation must be identified

by observing a larger-than-normal change in one or more output variables. Mathematically, the fault-detection problem is to check whether a measured output variable z has exceeded its alarm *threshold* γ as expressed here:

$$z(t) = y(t) + v(t) > \gamma \quad (8.6)$$

where the measured output variable is the sum of the true output variable y and a measurement noise v . Note that we have assumed that a higher output value corresponds to a more degraded engine. This assumption works obviously for turbine temperature. However, for compressor discharge pressure, the opposite is true, that is, the lower the pressure, the more degraded the engine is. In the latter case, the $>$ sign in Eq. (8.6) needs to be replaced with the $<$ sign to represent the fault-detection criterion.

To isolate the root cause of a fault, we need to check whether a specific damage state variable has exceeded the alarm threshold ε , that is,

$$\hat{x}_d(t) > \varepsilon \quad (8.7)$$

where the caret stands for an estimated value of the damage state variable. By identifying the specific damage state variable, we know the root cause (or damage) of the performance degradation.

Similarly, the goal of prognostics is to predict when (and how) a detected degradation will deteriorate to an unacceptable level. This goal can be expressed as

$$z(t+d) = y(t+d) + v(t+d) > \zeta \quad (8.8)$$

$$\hat{x}_d(t+d) > \xi \quad (8.9)$$

where ζ is the acceptable performance limit for useful life, ξ is the corresponding state variable limit, and d is the TTF (measured from the current time t) or *remaining useful life* (RUL) of the product [9].

8.1.4 Maintenance Concepts

Maintenance practices for engines and other industrial equipment have evolved from the *reactive*, or failure-driven, maintenance concept to the *preventive* maintenance concept, and more recently, to the *condition-based* maintenance (CBM) concept. Whereas preventive maintenance relies on periodic inspections to find failures in their early stages, CBM replaces time-based inspections with condition-based repairs so that maintenance is performed only upon evidence of need, in other words, when the condition of an engine or a subsystem has degraded to the point where loss of the system's function required to complete a desired mission is probable. In modern maintenance terminologies, CBM is synonymous to *predictive* maintenance, *proactive* maintenance and PHM [10].

The reliability-centered-maintenance (RCM) concept defines a process to determine the maintenance requirements of any physical asset (or system) in its operating context [11]. In this context, RCM is an enabling step of CBM. The goal of RCM is to maximize the reliability of an engine or a system while optimizing

the maintenance cost. It achieves this goal by a detailed analysis of failure modes and by providing early warning of potential failures. RCM practices for gas turbine engines are most effectively applied to the removals of life-limited parts (LLPs) like rotating disks. These practices have resulted in efficient engine build policies with an optimal planning of work scopes.

Engine removal (or rejection) caused by module life limits is the second largest cause for lower-than-expected TOW in military engines. (The number one cause is FOD.) Modern gas turbine engines are designed under the modular concept for improved maintainability. A typical two-spool engine has 5 to 10 modules, such as fan, compressor, combustor, turbine, nozzle, casing, etc. The RCM-motivated module matching (or alignment) concept is an example of extending the TOW or reducing the engine removal rate. Module matching is an optimization technique that identifies the most appropriate module to be put in an engine from the pool of available (uninstalled) modules of the same kind, so that the remaining lives of all modules in the engine are approximately the same, after the engine is rebuilt in the shop. The objective function of a general module matching algorithm that minimizes the discrepancy among all possible combinations of the remaining lives of the modules to be included in the engine build process is

$$\sum_{i=1}^N \sum_{j=1}^N (r_{1i} - r_{1j})^2 \quad \forall i \neq j, j > i \quad (8.10)$$

where N denotes the number of modules in an engine, r is the remaining life, r_{1i} is the remaining life of engine 1 for module i , and r_{1j} is the remaining life of engine 1 for module j , with the assumption that engine 1 is to be built. What complicates this optimization problem is that each module to be included in the build process is selected from a pool of modules that are available at the time of this engine build, and these available modules come either from the engines that have been removed or from spare engines (or parts). So Eq. (8.10) needs to be expanded to include the possibility of multiple "choices" for each kind of module (like the compressor or the turbine module). This need can be addressed by adding the following equation to the optimization process of Eq. (8.10):

$$r_{1i} \in \{r_{1i}, r_{2i}, \dots, r_{ki}\} \quad (8.11)$$

where k is the number of available choices in the pool of module i .

A module matching tool, based on the preceding optimization concept and simulated on a fleet of in-service fighter engines of the Royal Air Force of United Kingdom, shows that the tool is able to increase the TOW by more than 100% for the selected engine fleet in a period of just 12 months. The potential benefit in cost reduction as a result of decreasing engine removals and recovering some of the residual life from LLPs is measured in tens of million pounds [12].

The U.S. Department of Defense has developed guidelines for CBM+ by combining the practices of CBM and RCM with a goal to perform equipment maintenance based only on the evidence of a need [13]. CBM+ relies on embedded diagnostic and prognostic capabilities as well as ground-based trending and decision support software to accomplish its goal.

Table 8.1 Comparison of various maintenance concepts

Concept	Philosophy	Terminology
Event driven or failure driven	Repair or fix problems when they arise	Reactive maintenance
Time based	Perform periodic/routine inspections and maintenance	Preventive maintenance
Usage based	Perform maintenance based on the estimated usage of a part or a system, primarily for LLP	Scheduled maintenance
Condition based	Perform maintenance based on evidence of need; the evidence is gathered by analyzing failure modes, effects, root causes, and corrective actions	Predictive maintenance, PHM, RCM, proactive maintenance, anticipatory maintenance
Condition-based plus	Enhance CBM with embedded diagnostics and prognostics	CBM+

Table 8.1 summarizes various maintenance concepts that are being practices in various industries utilizing gas turbine engines.

8.1.5 Support Concepts

In the beginning of this chapter, it is mentioned that support resources (including people, tools, and parts) are an essential part of a comprehensive health management system. There are two fundamentally different concepts in providing support (or sustainment) for engines: 1) bill-and-material and 2) power-by-the-hour (PBH). The former is the traditional approach of keeping an engine on the aircraft wing as long as possible by the operators themselves. In this approach, an operator owns its own engine shop and buys parts from original equipment manufacturers (OEMs) to compete repairs. The latter is an innovative approach that became popular in the 1990s in the commercial engine market. In this “newer” approach, a third-party company has a contract to provide all of the parts and repairs needed to keep the engine operational for the operator at a fixed cost per flight hour (CPFH).

This third-party support contractor is often the engine OEM itself. An example of the PBH concept is Rolls-Royce’s TotalCare®, which helps airline customers concentrate on their core business of flying passengers for a fixed CPHH. Another example is Lufthansa Technik’s Total Engine Support, TES®, under which a maintenance, repair, and overhaul company is responsible for engine maintenance and support for the operators. Regardless what it is called and who is providing the support, the *total support concept* has become increasingly the preferred choice for all types of turbine engines in air, land, and sea applications.

This total support concept has grown to the airplane or vehicle (or system) level. Both Boeing and Airbus offer similar concept under the name of GoldCare® and Tailored Support Package (TSP), respectively. The reason is that the trend in

total support is driving the following two goals: 1) *guaranteed performance* in terms of equipment availability and reliability and 2) *predictable financial risks* in the form of a fixed cost per usage of the equipment.

The total support concept is not limited to the commercial world. In the military and defense arena, the mission-capable rate [14] and the “ready-for-tasking at reduced cost (from a baseline)” are examples of the trend towards using a single metric to drive operations and support. Regardless what it is called, *the total support concept is increasingly recognized as a good justification for better monitoring and health management capabilities.*

8.2 Monitoring System Design

The purpose of monitoring is to support the business objectives of high performance and a predictable, as well as manageable, cost. To accomplish these objectives, we need the engine to meet the goals of safety, reliability, availability, operability, efficiency, and supportability at a reasonable cost. Because safety should not be compromised and usually is treated as an absolute goal, the remaining goals are all related and should be addressed simultaneously, hence the emphasis of a comprehensive health management process instead of just the onboard or the off-board monitoring piece. However, covering the entire spectrum of health management system design is not feasible in one or two chapters; therefore, only the monitoring part of the health management system design will be covered in this book, because monitoring is a core capability of health management and it is closely tied to the control system, which is the primary subject of this book.

Generally speaking, a monitoring system has two objectives: 1) when a system or subsystem is going to fail and 2) what to do about it. The former is related to data collection, trending, diagnostics, and prognostics, whereas the latter is more or less related to maintenance. Monitoring is not just about knowing system failures; it is also about knowing the need for servicing (or resupply of) various consumable items that keep an engine running (or airplane flying), such as engine oil, tire pressure, and oxygen bottles, etc. Again in this book, the main focus is to monitor system failures and performance degradations.

A review of engine monitoring systems for commercial aviation engines [15] revealed that these systems had proven effective in providing early warning for impending failures; however, the high number of false alarms had caused some reluctance among commercial users to rely on the results. For helicopters, although the cost of implementing a health and usage monitoring system (HUMS) was still high, the benefits had been steadily increasing.

The greatest challenge facing engine monitoring system design is how to reduce false alarms. Reducing false alarms can be addressed in two separate aspects: 1) improving the effectiveness of signal condition and data manipulation for measured data and 2) increasing the accuracy of models used as the baseline for diagnostics and prognostics. Both of these are related to monitoring algorithms. Before we discuss algorithm designs, let us first review some system-level design concepts such as system requirements, system architecture, monitoring capabilities, and monitoring functions.

8.2.1 System Requirements

The generic system requirements of an engine monitoring system are to provide the following: 1) automated analysis, alerting, and maintenance decision support; 2) accurate results with high confidence; 3) robust monitoring against noise and corrupt information; 4) a broad coverage of possible fault symptoms; and 5) an ability to predictive failures [16]. Furthermore, these requirements need to be satisfied with existing, or as few as possible, sensing instrument, and they should be implemented with flexibility, modularity, and user friendliness.

These top-level system requirements are translated (or broken down) into lower-level engineering requirements depending on business objectives and design constraints. Some examples of the engineering requirements are functional requirements, architectural requirements, software or hardware requirements, and interface requirements.

Some of these requirements will be described briefly in the following subsections, but emphasis will be placed on functional requirements and algorithmic approaches in the rest of this chapter.

8.2.2 System Architecture

A widely accepted partition for an engine monitoring system consists of two subsystems: 1) an onboard system and 2) an off-board (or ground-based) system. Typically, the onboard system collects data, performs built-in tests and other monitoring functions, such as detection, assessment, prediction, and alarm. It can also provide health-assessment results to the control system for adaptive or fault-tolerant controls. Today, the majority of engine trending and prognostic functions are performed on the ground.

Ground-based monitoring usually starts when the in-flight data and pilot reports are transmitted (or downloaded) to the ground system. The system performs additional analysis, correlations, and assessments. The results are then used by maintenance crew and operations personnel to support decision making timely and accurately. Typically engine maintenance is done at three levels: 1) at the flight line (or sometimes called the line maintenance or the operational-level maintenance), 2) at the engine shop (or the intermediate-level maintenance), and 3) at the overhaul facility (or depot-level maintenance). Each level has a different “timescale of urgency” in completing the maintenance task so as to support operations. This concept of urgency is illustrated in Fig. 8.4.

8.2.3 Monitoring Capabilities

Engine monitoring capabilities historically started with the interest in engine performance. Engine performance is defined by the thermodynamic characteristics of the gas-path (GP) modules, such as the fan, the compressor, the burner, and the turbine (see Appendix B). So GP performance has been the primary focus of engine control and monitoring activities as evidenced by a large number of temperature and pressure sensors, which make up the majority of engine sensors.

If we analyze common engine failures, in addition to the GP problems such as efficiency degradation and flow capacity shift, we find that mechanical and

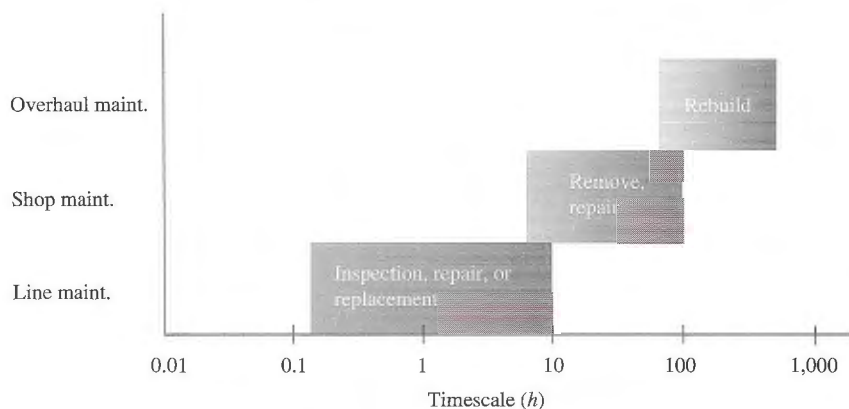


Fig. 8.4 Maintenance timescale of urgency (in hours).

lubrication problems, such as chips in the oil, tip rub, rotor imbalance, bearing crack, and seal leak, constitute the majority of the causes for in-flight diversion or delay and cancellations. To detect these failures before a schedule interruption or catastrophic failure occurs, we need to monitor more “things” about the engine, and these additional monitoring capabilities, together with the GP monitoring, form a complete range of monitoring capabilities that cover from gas-path performance to oil quality and to vibration level.

A comprehensive engine monitoring system includes the following four major monitoring capabilities: 1) GP performance monitoring, 2) oil contamination and debris (OD) monitoring, 3) mechanical vibration (MV) monitoring, and 4) usage and consumption (UC) monitoring, which is particularly important for LLPs and controls and accessory (C&A) components. Each monitoring capability deals with a set of measurements giving a “perspective” of the condition of an engine; hence, each set represents a *dimension* of the information that is available for diagnosis and prognosis. An effective monitoring system combines information from different dimensions to make the best decision. This approach is commonly known as information *fusion*.

8.2.4 Monitoring Functions

According to the recommendation of the International Organization for Standardization (ISO) Technical Committee 108 for Mechanical Vibration, Shock, and Condition Monitoring [17], or commonly referred to as the ISO-13374, the functionality of a comprehensive monitoring system should include six functional blocks (or layers): 1) data acquisition (DA), 2) data manipulation (DM), 3) state detection (SD), 4) health assessment (HA), 5) prognostic assessment (PA), and 6) advisory generation (AG). These functional blocks are shown as the middle section in Fig. 8.5.

An implementation example of the ISO condition monitoring guideline is the Open System Architecture for Condition Based Maintenance (OSA-CBM) [18].

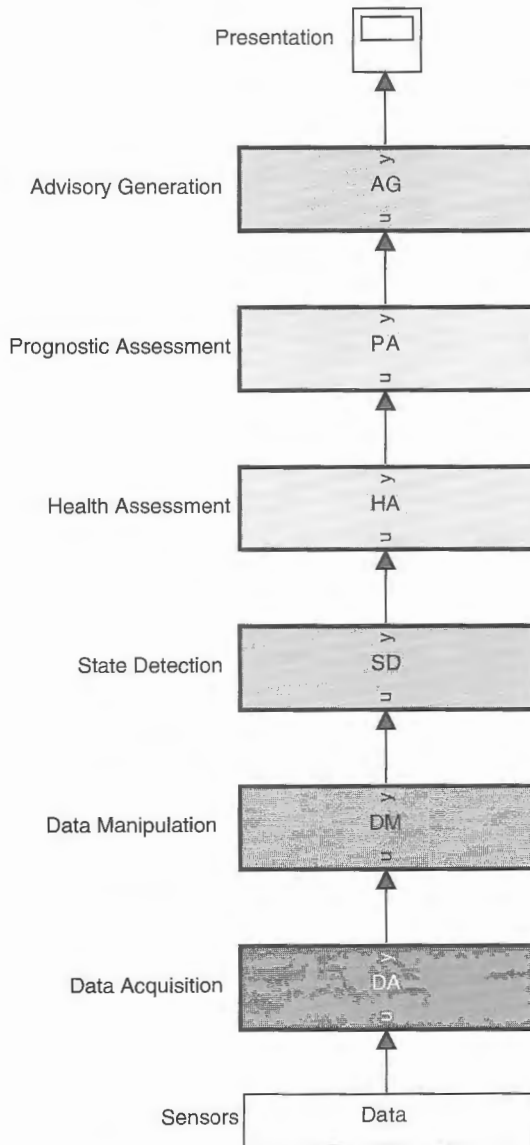


Fig. 8.5 Standard condition monitoring functions.

OSA-CBM is an attempt to simplify the process of integrating a wide variety of software and hardware components by specifying a standard architecture and framework. It describes the six ISO functional blocks of a monitoring system to facilitate the capabilities of CBM; it also defines the interfaces between these blocks. The OSA-CBM standard is managed by a trade organization, called MIMOSA (<http://www.mimosa.org>).

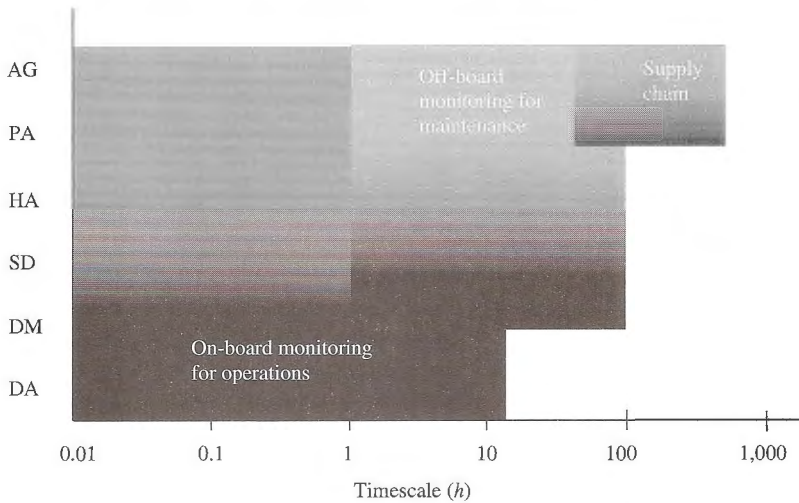


Fig. 8.6 Timescale of urgency (in hours) for monitoring functions.

Depending on the system partition, the concept of timescale of urgency is also applicable to the monitoring functions. For onboard monitoring functions, DA, DM, and SD are more urgent than PA and AG, whereas for off-board monitoring functions, trending, PA, and AG are more important. Supply chain or logistics, on the other hand, has a longer timescale than the off-board monitoring. Figure 8.6 illustrates this concept.

8.2.5 Software Architecture

A robust software architecture for engine monitoring systems must be able to support the various functions described in the ISO guideline; it must also handle all monitoring capabilities to provide a coherent “picture” of the condition of an engine. A software architecture that meets these requirements is illustrated in Fig. 8.7 [19]. Each vertical line represents a monitoring capability, which provides one perspective of the condition of the engine. Each row represents a monitoring function. Each intersection of a column and a row represents an information processing *node*, in which one or more monitoring and health management algorithms reside.

8.2.6 Algorithmic Approaches

Algorithms are used in a monitoring system to analyze data, detect and isolate faults, assess damages and health, predict failures and RUL, and recommend actions. In general, they play the role of extracting (or aggregating) information to support decision making at different functional levels. A certain type of algorithms can be applied to all of the monitoring capabilities and functions as illustrated in Fig. 8.5. However, in most cases, one type of algorithm can be more effective than others for a given monitoring capability and for a given monitoring function.

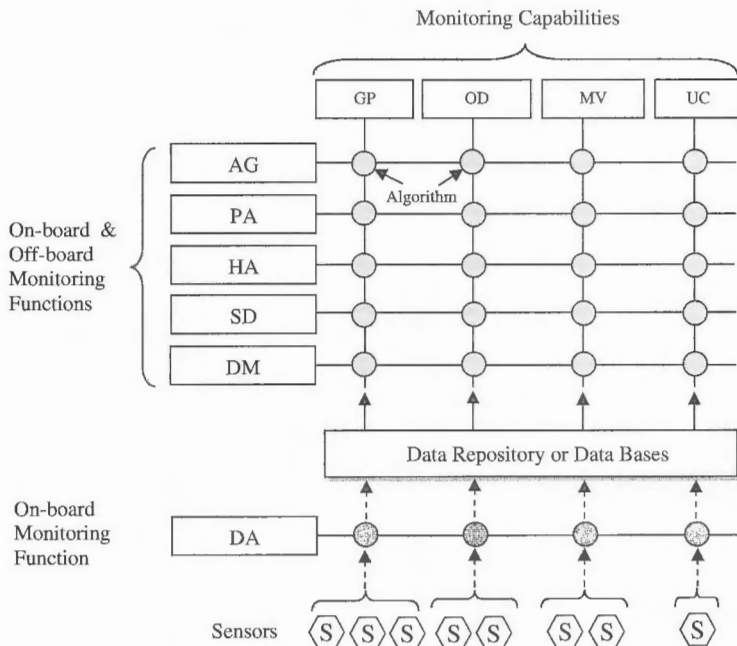


Fig. 8.7 General software architecture for engine monitoring systems.

Algorithms fall into two broad classes: *data-driven* algorithms and *physics-based* algorithms. Examples of data-driven algorithms are statistical and machine-learning methods. Examples of model-based algorithms are physics-of-failure models and state equations. Some algorithms combine the attributes of both classes, and they are called hybrid algorithms. Examples of hybrid algorithms include data-driven corrections to physics-based models, algorithms that combine direct measurements with (hidden) features, and algorithms that combine symbolic information processing (e.g., expert system) with numeric information processing (e.g., artificial neural network). Hybrid algorithms have proven effective in achieving the goal of reducing false alarms. A graphic illustration of the fidelity and the complexity of hybrid algorithms, as opposed to data-driven and physics-based algorithms, is shown in Fig. 8.8.

Engine gas-path analysis (GPA) is most matured among all four monitoring capabilities. Two comprehensive reviews of diagnostic algorithms used in GPA suggest a qualitative assessment of the computation speed vs the complexity of various algorithms in Fig. 8.9 [19, 20].

8.3 Monitoring Algorithm Design

Data-driven algorithms, including statistical and empirical approaches, have been widely published. In this section, we will discuss some of the most common

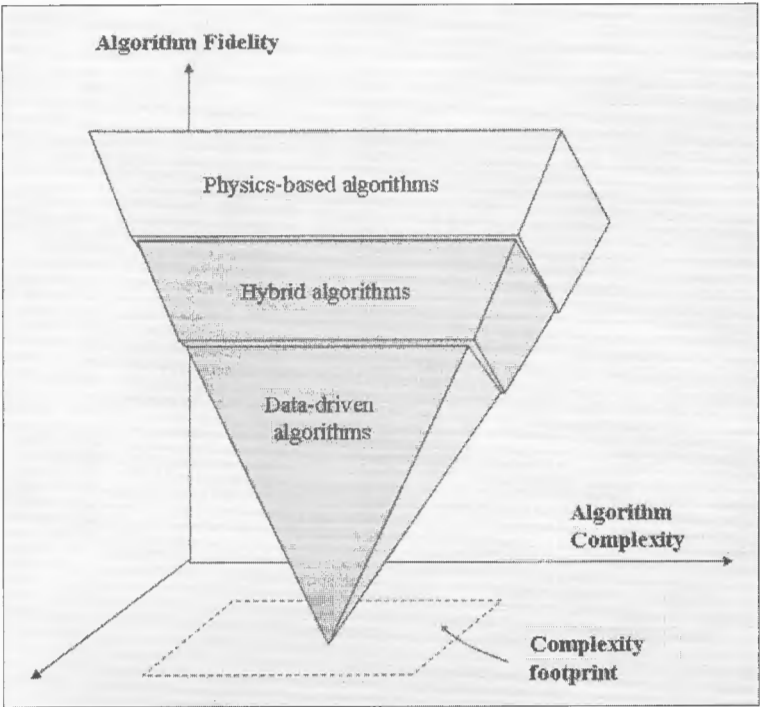


Fig. 8.8 Illustration of the fidelity and the complexity of different types of algorithms.

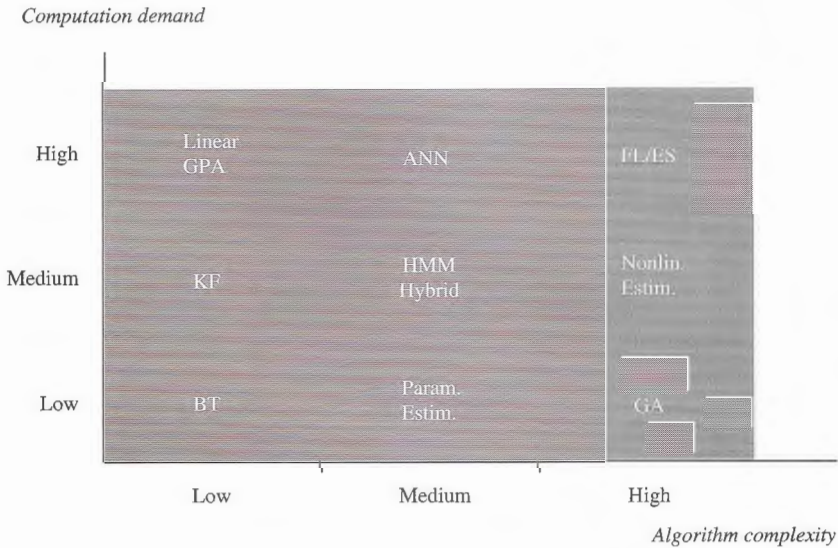


Fig. 8.9 Computational performance of GPA diagnostic algorithms. (See also color figure section at the back of the book.)

physics-based approaches. Physics-based approaches are increasingly favored because they can provide more accurate estimates of expected operating states and because they are naturally extensible to the physics-of-failure models, which are considered essential for producing high-confidence prognostics.

Physics-based algorithms are essentially physics-based models. These models are expressed in the form of the state and the output equations, and they are used in monitoring systems to calculate the expected values of the state and output (both measured and unmeasured) variables for past, present, and future times. These model-expected values are then compared to the actual measurements to provide the evidence of a fault and a failure. In Sec. 8.1.3, we used a unified mathematical formulation to represent both the diagnostic and the prognostic problems. This formulation, however, does not show explicitly how the state and the output variables are calculated, nor does it show how the alarm thresholds are determined.

The process of calculation of the state and the output variables from models is called *estimation*. Estimation, in general, is an optimization process that minimizes the error between the estimated value and the true value of a variable. The two most common methods used in estimation are the *least-square* estimation and the *Kalman filter*. We will briefly describe these two methods in the following subsections [9, 21].

8.3.1 Least-Square Estimation

To estimate the state variable x_A using the measurement variable z , containing random error v , the least-square (LS, or more precisely, the weighted-least-square) estimate is

$$\hat{x}_A = \bar{x}_A + PC_A^T R^{-1} (z - C_A \bar{x}_A) \quad (8.12)$$

where $\hat{x}_A$ is the estimated value of the state variable, $\bar{x}_A$ is the (estimated) value of the state variable before the “new” measurement is made, and we have assumed, without loss of generality,

$$z = C_A x_A + v \quad (8.13)$$

and

$$R = E(vv^T) \quad (8.14)$$

$$M = E[(x_A - \bar{x}_A)(x_A - \bar{x}_A)^T] \quad (8.15)$$

$$P^{-1} = M^{-1} + C_A^T R^{-1} C_A \quad (8.16)$$

R and M are known matrices of the errors (for measurement error and prior-estimate error, respectively) and are called the covariance matrices. R^{-1} and M^{-1} are the weighting matrices in the LS objective function. The objective function can be thought of as one to minimize the *sum of the squares of errors* for both x_A

and v . If we assume Gaussian distributions for x_A and v , then the LS estimate is identical to the maximum-likelihood or the minimum-variance estimates. It can be shown that P is the covariance matrix of the error in the new estimate.

8.3.2 Kalman Filter

Suppose the measurements can be made repeatedly, and the system can be represented by a multistage stochastic process (or stochastic state equation in discrete time), then the state and the output equations can be expressed by [21]

$$x_{k+1} = \Phi_k x_k + \Gamma_k w_k \quad (8.17)$$

$$y_k = H_k x_k + v_k \quad (8.18)$$

After applying the LS estimation solution to this multistage system, we get

$$\hat{x}_i = \bar{x}_i + K(z_i - H_i \bar{x}_i) \quad (8.19)$$

where

$$\begin{aligned} K_i &= P_i H_i^T R_i^{-1} \\ P_i &= M_i - M_i H_i^T (H_i M_i H_i^T + R_i)^{-1} H_i M_i \\ M_{i+1} &= \Phi_i P_i \Phi_i^T + \Gamma_i Q_i \Gamma_i^T \end{aligned} \quad (8.20)$$

and where Q_i is the covariance matrix of the disturbance w .

Equations (8.19) and (8.20) represent the Kalman filter (KF) for a linear multistage system. Note that R_i and Q_i depend on the size of the time step between measurements. Also note that we have dropped the subscript A in favor of the standard (or nonaugmented) state equation.

KF is a recursive estimator for the state variable based on measured output variables as well as the system model. For a continuous linear dynamic system with continuous measurements, assuming that both the disturbance and the measurement are white noises, then the KF (or Kalman-Bucy filter) is expressed as

$$\dot{\hat{x}} = A\hat{x} + Bu + K(z - C\hat{x}), \quad \hat{x}(0) = 0 \quad (8.21)$$

where

$$K = PC^T R^{-1} \quad (8.22)$$

$$\dot{P} = AP + PA^T + BQB^T - PC^T R^{-1} CP \quad (8.23)$$

and Q is the covariance matrix of the disturbance w , that is,

$$Q = E[(w - \bar{w})(w - \bar{w})^T] \quad (8.24)$$

KF is the most widely used model-based method for estimating the state variables of a system. It treats the engine (or any physical system) as a linear dynamic system with finite measurements, and it assumes that the uncertainties associated with the dynamic process w and the measurement v can be characterized by white noises. Different variations of the KF have been studied, including the *filter bank* [22, 23]; the *constrained filter* [24, 25]; the *Bayesian-enhanced filter* [26]; the *dual-estimation filter* [27, 28]; and the *baseline-resetting filter* [29].

State estimation has traditionally been used by control engineers to provide the feedback (of estimated states or outputs) in closed-loop control systems, especially in MIMO systems. State estimation can also be used for detection, isolation, and prediction of failures in any system for health management purposes. By applying the KF concept to the state detection and prediction problems described in Sec. 8.1.3, we get

$$\hat{\hat{x}}_A = \begin{bmatrix} \hat{\hat{x}} \\ \hat{\hat{x}}_d \end{bmatrix} = A_A \hat{\hat{x}}_A + B_A u + K \left[z - \left(C_A \begin{bmatrix} \hat{\hat{x}} \\ \hat{\hat{x}}_d \end{bmatrix} + Du \right) \right] \quad (8.25)$$

For prognostics, the KF can not be used as a “feedback” observer because the future values of measurements are not available at the current time; hence, prognostics is inherently an open-loop state transition problem by using the best current estimate, that is, the transition of the current state estimate to a future estimate is shown here:

$$\dot{\hat{x}}_A = \begin{bmatrix} \dot{\hat{x}} \\ \dot{\hat{x}}_d \end{bmatrix} = A_A \hat{x}_A + B_A u, \quad \text{with } \hat{x}_A(t) = \begin{bmatrix} \hat{x}(t) \\ \hat{x}_d(t) \end{bmatrix} \quad (8.26)$$

8.3.3 Probability Estimation – Bayesian Method

As stated earlier, the goal of diagnostics and prognostics is to estimate the system state variable and determine whether one or more individual state variables exceed thresholds. To put it differently, we are trying to find a set of state variables at time step k when the values of the variables exceed their corresponding thresholds, that is, for $\{x_k, k \geq 0\} \in \Omega = \Omega_N \cup \Omega_F$, where Ω_N is the set of normal state variables and Ω_F represents the set of degraded (or faulty) state variables, the set of faulty state variables Ω_F consists of the estimated state variables that have violated the degradation thresholds. This fault set is expressed, at time k , as

$$\Omega_F = \{\hat{x}_k, k \geq 0 : \|\hat{x}_k\| > \varepsilon\} \quad (8.27)$$

and

$$\Omega_F = \Omega - \Omega_N$$

where ε (or ξ in the case of prognostics) is the upper bound of the acceptable degradation.

Alternatively, we can express the set of faulty state variables in probabilistic terms; hence, for the case of prognostics, the fault set is

$$p(x_{k+d} \in \Omega_F | U_k) = \int_{\Omega_F} p(x_{k+d} | U_k) dx_{k+d}, \quad d = 1, 2, \dots \quad (8.28)$$

where $U_k = \{U_{k-1}, y_k\}$ is the information of the system up to time k and $p(\cdot)$ is the probability density function (PDF). Actually, the probability is a conditional probability (given the information up to time step k). This generalized framework lets us “bridge” the estimation problem, or the prognostic problem, from deterministic computation to stochastic estimation.

For the noise-affected, measured variable z , the output equation can be expressed as [see Eq. (8.6)]

$$z = g(x, v) \quad (8.29)$$

Now suppose that the following quantities can be calculated (or obtained) from Monte Carlo simulation or actual data: 1) the PDF of z , that is, $p(z)$, and 2) the PDF of x and z , that is, $p(x, z)$. Then the conditional probability of $p(x | z)$, that is, the PDF of x given z , can be calculated by the Bayes rule:

$$p(x | z) = \frac{p(x, z)}{p(z)} \quad (8.30)$$

Note that the conditional probability of $p(z | x)$ can always be obtained from the preceding equation and $p(x, z)$. Hence, an alternative calculation is

$$p(x | z) = \frac{p(z | x)p(x)}{p(z)} \quad (8.31)$$

where $p(x | z)$ is also called the *posterior* density function of x . From this density function, we can estimate the state variable, that is, $\hat{x}$.

Depending on the estimation objective functions, the estimated state variable varies. Three most common objective functions are maximum likelihood, minimum variance, and minimum error. When Eq. (8.29) is linear [i.e., Eq. (8.29) reduces to Eq. (8.6)] and the noise is Gaussian, it can be shown that the three common Bayesian estimation criteria all yield the same KF.

Bayesian methods use conditional probability to identify the most likely faults or fault states. Although elegant in the mathematical formulation, they are often difficult to apply in practice because of the uncertainties associated with the conditional probabilities, especially when there is a large number of conditional probabilities to be calculated.

8.3.4 Hybrid Model Estimation

Estimation using hybrid models is primarily used to provide expected values for the output variables of a system in a specified operating condition. These expected values are then compared to the actual, measured values to provide “evidence” of an anomaly or a fault. There are almost countless ways of coming up with hybrid

models, but in this section we will only describe one form of hybrid model that has been used effectively in anomaly and fault detections [30]. Succinctly, this type of hybrid model uses the first principles of physics to provide an approximate, first order-of-magnitude estimate for the output variable of interest, and it uses data-driven correction coefficients to improve estimation accuracy.

This hybrid model can be called an approximate physical model or a *heuristic* physical model, which is different from an exact physical model. In an exact model, the dependent variable y is a function of the independent variable x , while x can be a function of time t . This model is expressed in the output equation form as

$$y = f[x(t), t] \quad (8.32)$$

A common form of approximate model is the Taylor-series expansion of the preceding equation at a nominal operating condition (see Chapter 2 for more explanation), that is,

$$\tilde{y} \approx f(x_0, t) + \left. \frac{\partial f}{\partial x} \right|_0 (x - x_0) \quad (8.33)$$

However, this approximation model is only valid near the nominal operating point and requires some knowledge of the functional expression (or numerical values) in the neighborhood of the operating point. If the functional expression is not well known, then the modeling error, manifested in the higher-order terms in the Taylor's expansion, will be excessive. Consequently, a type of hybrid models is to multiply the physical model with a data-driven function η . It is expressed as

$$y_H \cong \tilde{y} \cdot \eta(x, t) \quad (8.34)$$

In Eq. (8.34), η can be thought of as a correction factor derived by comparing the data to the physical model, and it is a function of the independent variable and time. This correction factor can be in a parametric form or a nonparametric form. For instance, it can be substituted by an artificial neural network (ANN) in a nonparametric form. An example of this is the *cooperating* neural network [31], where an ANN runs side by side with a physical model to “learn” the difference between the model and the actual data. This concept is illustrated in Fig. 8.10. The data-driven neural network model is added to the simplified physics-based model to learn the difference (or sometimes called the innovation sequence) between the simplified, physical model and the actual engine data.

The cooperating neural-network type of hybrid models is particularly interesting to us in constructing individualized engine models for trending, diagnostics, and prognostics because it has an inherent ability to adapt to engine-to-engine variations (caused by manufacturing tolerances) and time-varying component deteriorations (caused by aging and operational damage). This type of model can also be implemented onboard a monitoring system [32].

Even with a hybrid model, unique identification of the failure mode or root cause associated with a fault symptom often cannot be achieved because of the

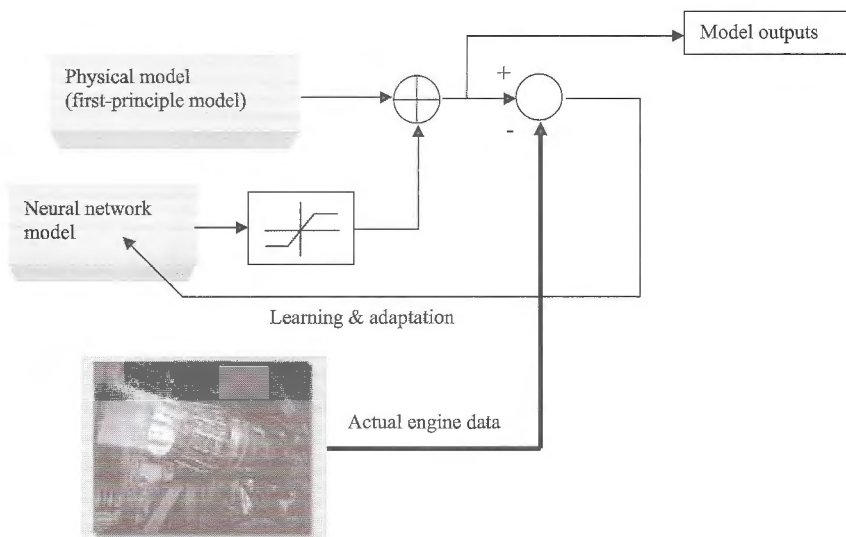


Fig. 8.10 Cooperating neural-network concept as a nonparametric example of hybrid models.

lack of information. This lack of information is primarily a result of 1) insufficient sensors or measurements and 2) similar fault symptoms for multiple failure modes or root causes. The end result is the presence of ambiguity in diagnostics. The collection of possible failure modes corresponding to the same fault symptom is usually called the *ambiguity group*. To reduce the ambiguity group, the diagnostic results from different algorithmic approaches should be compared; furthermore, the monitoring results from different information dimensions (see Fig. 8.7) should be consolidated (or fused).

8.4 Trend Monitoring from Periodically Recorded Data

Trend monitoring or trending is a charting technique for detecting variations and anomalies in a physical or business process. It is the most widely used technique for fault detection and isolation of a machine's health condition. It can also be used as a prognostic tool to predict the future performance or an impending failure of the machine by extending the trends discovered in the trending process. In this section, a numerical example to describe the basic concept of trending will be introduced. This example provides a foundation to explain the difference between data-driven and physics-based algorithms.

8.4.1 A Numerical Example of Trend Monitoring

Trend monitoring is a systematic way of discovering variations and trends in the data by using consistent charting rules. Effective trend monitoring requires same kinds of data to be collected over a period of time. Same kinds of data refer

to both the same *set* of data and the same *operating* condition. For aircraft engines, TO and cruise conditions (for commercial engines) are the typical conditions where data are collected for trend-monitoring purposes.

Trend monitoring is most effectively accomplished graphically by presenting the data in a visual picture to allow easy identification of variations and trends. An example of this graphical presentation of engine trend data is shown in Fig. 8.11. As expected, the data are highly correlated in the three-dimensional space formed by the X-Y-Z coordinates corresponding to three measured variables: N_1 , W_f , and EGT, respectively. Furthermore, we see a trend of increasing EGT with increasing N_1 or W_f . The question is whether the increase in EGT is a normal trend or an *abnormal* trend? A normal trend is one that represents an inherent physical relationship. For instance, the turbine temperature increases with the engine fuel flow. On the other hand, an abnormal trend is one that reveals an underlying change or degradation of the engine. For instance, the turbine temperature increases *faster* than what it normally does for the same amount of fuel increase.

In trend monitoring, we are primarily interested in abnormal variations and trends in the data points; however, we generally cannot discern an abnormal trend until we know what the normal trend is. Knowing what the normal trend is can be a difficult task because we are usually limited in the number of measurements available for trending due to cost, weight, and reliability concerns. Various statistical methods and data-analysis techniques have been suggested to help us build normal trends and use them to discover abnormal trends. For instance, Fig. 8.12

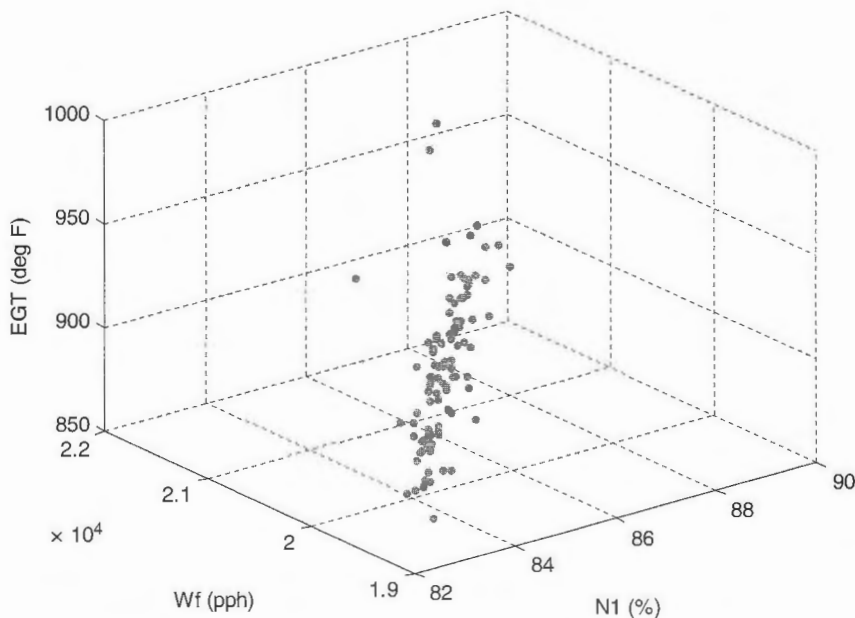


Fig. 8.11 Three-dimensional view of the data collected for three engine variables.

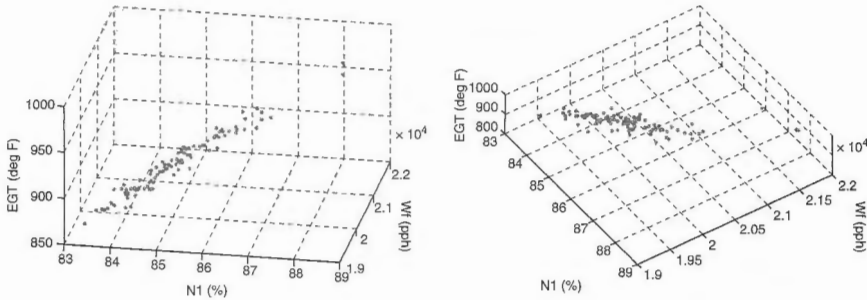


Fig. 8.12 Graphical presentation of the same data from two different viewpoints.

shows two different views of the same data in the three-dimensional space after some coordinate transformations. Through these transformations, we hope to identify certain data *components* that are more distinctive than others so as to identify similar and dissimilar trends more easily.

When the number of measured variables exceeds three, graphical presentation of the data becomes impractical. Mathematically we can throw all of the time-elapsing data of any dimension (i.e., any number of measured variables) in a hyperspace, R^N , where N is the number of dimensions or variables. Statistically we can even calculate the mean value, the covariance matrix, and other properties for the data that fall in the same cluster in the hyperspace and use these properties to define normal trends. This concept is the basic premise of the data-driven approach for monitoring.

8.4.2 Trend Charts and Thresholds

For each measured variable, we can calculate the statistical mean value and standard deviation of the variable from a historical database, which represents sufficiently large data about the variable in the operating conditions of interest. The mean value, the standard deviation, and the measurements form the basis of a trend chart for the variable to be monitored.

Figure 8.13 shows the trend charts of the three measured variables described in the preceding subsection. More data points have been included in these charts to show trends more prominently over a longer time period. Note that these data points on the trend charts are distributed around the centerline, which has been shifted to coincide with the horizontal axis by subtracting the mean value from the actual value at each data point. Note also that two red lines have been added on each trend chart. These lines represent the thresholds, or the *control limits*, for the chart. The line on the top is called the upper control limit (UCL), and the line on the bottom is the lower control limit (LCL). These control limits trigger alarms for any point that is located outside the limits.

The trend charts shown in Fig. 8.13 are called *control charts* [33]. They are originated from the Shewhart solution, which focuses on the behavior of data

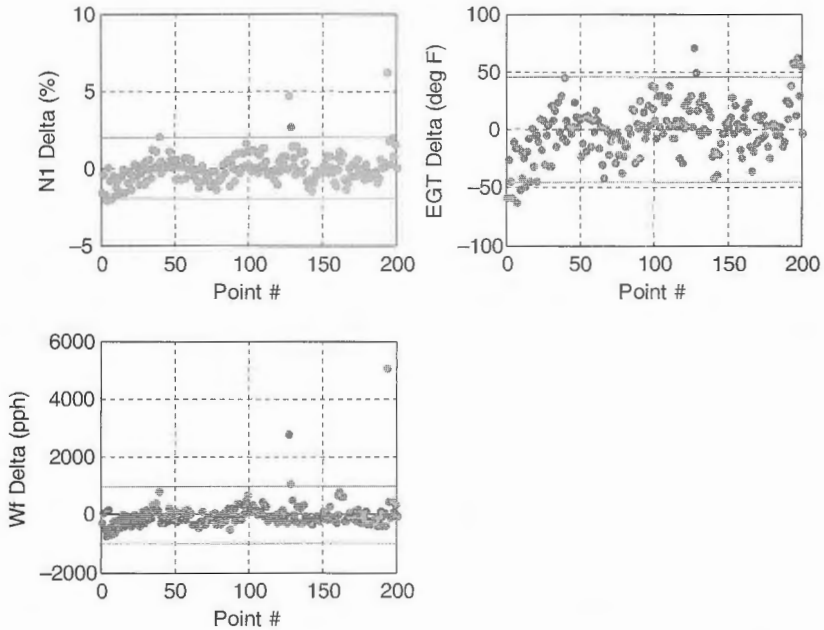


Fig. 8.13 Trend charts for the fan speed, the turbine exhaust temperature, and the fuel flow. (See also color figure section at the back of the book.)

locally rather than globally [34], so they are also called the Shewhart charts for the mean. A mean-value control chart consists of three parts:

- 1) The first part is set of data (or measurements of a variable) sampled at different time instances.
- 2) A center-line representing the statistical mean value of the variable; it is often approximated by the average value of a *large enough* data set.
- 3) A UCL and an LCL are traditionally represented as some multiples of the standard deviation. The standard deviation (denoted by σ) is a measure of the variability in the variable being monitored. The number three is used as the multipliers in the Shewhart chart.

Although the primary goal of a control chart is to detect any point falling outside the control limits (or the control band), the chart has two secondary goals: 1) detecting changes in the mean value and standard deviation and 2) revealing patterns in the data, such as an upward or downward trend, a biased distribution in one area of the chart or on one side of the center line, and a cyclical pattern. These goals are typically accomplished by modifying the mean-value control chart and incorporating additional detection logic and rules.

Some examples of modified control charts used in statistical process controls are [35] the standard-deviation control chart, the range control chart, and the cumulative sum control chart. The control charts in Fig. 8.13 actually represent another form of modified control chart, where the mean value has been removed

from each data point in the chart causing the centerline coincide with the horizontal axis. The modified chart effectively treats the mean value as the *baseline* and plots the residuals (i.e., the difference between an actual measured value and the baseline) on the chart. This residual control chart provides a foundation for introducing physics-based models in the otherwise data-driven trend monitoring technique. This hybrid data-and-physics monitoring technique will be discussed in the next subsection.

Because the centerline is supposed to represent the mean value of a large data set, it is not sensitive enough to reveal a shift in the mean value, which might have been caused by an underlying change in the process or a failure in the system. However, such a shift in the mean can be represented by a trend line representing the movement of the mean. This trend line is formed by a series of “short-term” mean values, each of which is calculated from a relatively small data set (or sample size). The slope of the trend line gives an indication (or a forecast) of how fast the monitored variable will reach a control limit. This predictive feature in trend monitoring is a data-driven approach to prognostics. Figure 8.14 is an example of the trend line on the EGT trend chart. The trend line predicts a “near-future” intersection of the UCL.

Note that several points have fallen outside the control band in Fig. 8.14. Although some of these out-of-control points are a result of the shift in the mean value as indicted by the upward-moving trend line, some other points are a result of *false alarms*. False alarms (or type I errors) are an inherent problem of a finite-valued control limit because the “theoretical” distribution of the variable is infinite. For example, if the sampled data of a variable behave like a normal

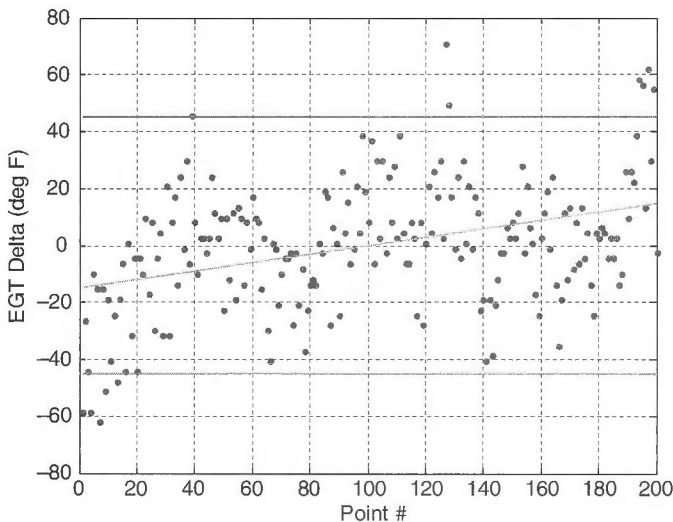


Fig. 8.14 Control limits (in red) and the trend line (in green) on EGT trend chart. (See also color figure section at the back of the book.)

distribution, then the probability of a point falling outside the $\pm 3\sigma$ control band is 0.0027, although there is nothing wrong with this point in terms of process quality or system health.

8.4.3 Increased Accuracy Through Physics-Based Modeling

For most turbine engines, the number of measured variables typically ranges between five to 10; hence, graphical presentation of the measured data for trend monitoring becomes impractical. Moreover, the wide range of inlet conditions, especially for aircraft engines, causes the accuracy of the data-driven approach to decrease with increasing data dimensions.

The most widely known physics-based approach to reducing the variability in engine gas-path data is the use of *corrected* variables, such as $N_{1c} \equiv N_1/\sqrt{\theta_2}$ and $T_{15c} \equiv T_{15}/\theta_2$, throughout the operating envelope [36]. These corrected variables are calculated by referring the actual measured value to the sea-level, standard-day condition by multiplying or dividing some correction factors. The appropriate correction factors for each engine variable are determined from dimensional analyses.

Further use of physical principles can yield additional benefit of removing variability and discerning trends. To accomplish this, we often deduce our knowledge about an engine or accessory component into some form of mathematical relationships among measured variables. Indeed, one of the most effective ways of applying physical principles to modeling is to identify one or a few independent variables among all of the measured variables. After the independent variables are identified, the rest of the dependent variables can be expressed as functions of these independent variables. By doing so, we have effectively reduced the dimension of the hyperspace, and consequently, breaking the original high-dimensional space into several lower-dimensional spaces, which might allow us to identify trends more easily.

From our knowledge of engine performance (see Appendix B), we know that the fan speed is directly related to engine thrust and can be selected as an independent variable. Actually, the corrected fan speed is a more desirable independent variable than the fan speed, and the dependent variables should be corrected to the sea-level, standard-day condition as well. However, to simplify the discussion, the original, uncorrected variables will be used to illustrate the physics-based monitoring concept here.

By choosing the fan speed as an independent variable, we can now make the other two variables functions of the fan speed. The functional relationships of these two dependent variables with respect to the fan speed are depicted in Figs. 8.15 and 8.16, and these relationships are the projections of the data points onto the N_1 - W_f plane and the N_1 -EGT plane, respectively. The green lines on top of the data points in the figures are the linear regression lines excluding the outliers.

Once we have established the functional relationships between the independent and the dependent variables, we can modify the concept of the data-driven control chart described in the preceding subsection to create a hybrid, data-and-physics-based control chart. This hybrid control chart also contains three parts:

- 1) A data set of a given *dependent* variable. The data set may contain either raw measurements of the variable or some calculated values involving the variable. An example of the calculated variable is the corrected variable described earlier.

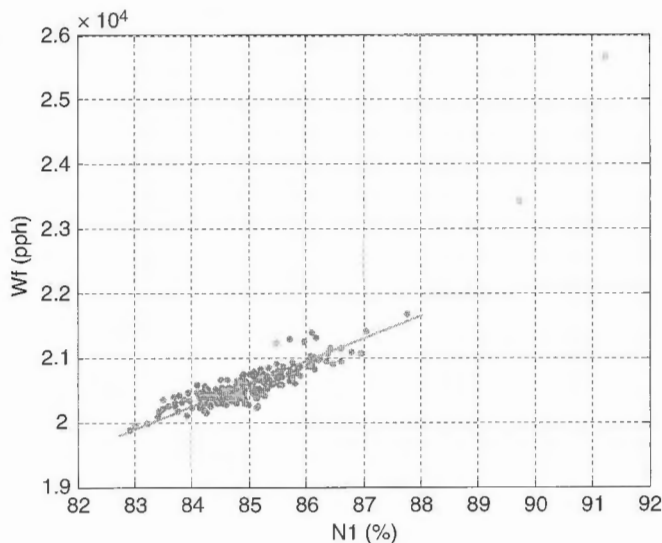


Fig. 8.15 Fuel flow rate vs N_1 . (See also color figure section at the back of the book.)

2) A baseline that represents the expected functional relationship between the dependent variable and its independent variable. A linear regression line is usually sufficient to cover all of the variations in operating conditions. After the baseline is determined, the expected value for each data point can be found. The expected value

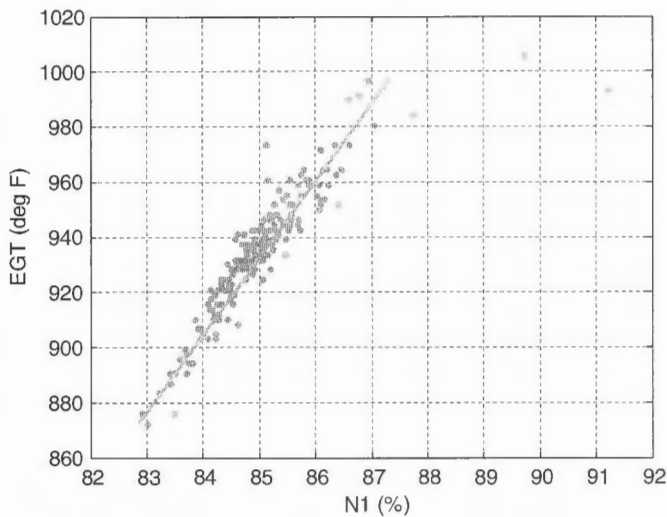


Fig. 8.16 Turbine temperature vs N_1 . (See also color figure section at the back of the book.)

is then subtracted from the absolute value of the data point to form a residual. The residuals corresponding to all of the data points are plotted vs time.

3) A set of control limits, which are multiples of the standard deviation of the data.

Note that the statistical mean is generally *not* the expected value because the latter is based on a physical model of the system being monitored, while the former is a mathematical average of all of the observed data. In other words, an expected value is the *truth* value that the system is expected to produce if the uncontrollable variability is removed.

8.4.4 Alarm Rules

Alarm rules are derived to detect any data point that represents a nonrandom, unusual pattern in the data. Some of these unusual patterns are defined as follows:

- 1) *Instability pattern*: the presence of points outside the control limits.
- 2) *Mixture pattern*: the absence of points near the centerline.
- 3) *Stratification pattern*: the absence of points near the control limits.
- 4) *Trend pattern*: the presence of a upward or downward trend.

Of these patterns, the instability pattern is the most essential one to be identified on a control chart. The goal of identifying this pattern is accomplished by a set of rules, and the most commonly used rules are the Western Electric rules (or the zone rules) [37]. These rules divide the control band into three different zones: zone A is defined as the region enclosed between the 2σ and the 3σ control limits, zone B corresponds to the region between the 1σ and the 2σ control limits, and zone C corresponds to the region between $\pm 1\sigma$, as illustrated in Fig. 8.17. The four basic zone rules for the mean-value control chart are as follows: 1) any single data point falls outside the UCL or LCL (i.e., $\pm 3\sigma$), 2) two out of three consecutive points fall beyond a 2σ limit, 3) four out of five consecutive points fall beyond a 1σ limit, and 4) eight consecutive points fall on the same side of the centerline.

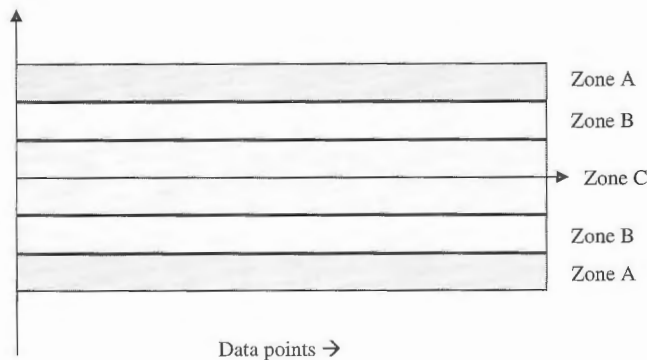


Fig. 8.17 Zone definition in the Western Electric zone alarm rules.

Additional alarm rules have been suggested to supplement the original zone rules. These combined rules are sometimes called modified Western Electric rules. These modified rules are intended to increase the sensitivity of the control chart to a small shift so that proper actions can be taken more quickly. An example of the modified Western Electric rules is described as follows [35]: 1) rules 1–4 in the Western Electric rules, 2) six points in a row steadily increasing or decreasing, 3) 15 points in a row in zone C, 4) 14 points in a row alternating up and down, 5) eight points in a row on both sides of the centerline with none in zone C, and 6) one or more points near a control limit.

References

- [1] *Software Consideration in Airborne Systems and Equipment Certification*, Radio Technical Commission for Aeronautics, Inc., DO-178B, 1992.
- [2] *Reliability Toolkit: Commercial Practices Edition*, Reliability Analysis Center, Griffiss AFB, Rome, NY, 1997.
- [3] Willsky, A. S., "A Survey of Design Methods for Failure Detection in Dynamic Systems," *Automatica*, Vol. 12, No. 4, Pergamon Press, Tarrytown, NY, 1976, pp. 601–611.
- [4] Merrill, W. C., DeLaat, J. C., and Bruton, W. M., "Advanced Detection, Isolation, and Accommodation of Sensor Failures—Real-time Evaluation," *Journal of Guidance, Control, and Dynamics*, Vol. 11, No. 6, 1988, pp. 517–526.
- [5] Frank, P. M., "Fault Diagnosis in Dynamic Systems Using Analytical and Knowledge-Based Redundancy—A Survey and New Results," *Automatica*, Vol. 26, No. 3, Pergamon Press, Tarrytown, NY, May 1990, pp. 459–474.
- [6] Henley, S., Currer, R., Scheuren, W., Hess, A., and Goodman, G., "Autonomic Logistics—the Support Concept for the 21st Century," *Proceedings of IEEE Aerospace Conference*, IEEE Paper ZF11_0701, March 2000.
- [7] Smith, G., Schroeder, J. B., Navarro, S., and Haldeman, D., "Development of a Prognostics and Health Management Capability for the Joint Strike Fighter," *Proceedings of IEEE AUTOTESTCON*, Sept. 1997, pp. 676–682.
- [8] Scheuren, W., "Joint Strike Fighter Prognostics & Health Management," AIAA Paper 98-3710, July 1998.
- [9] Jaw, L. C., and Wang, W., "Mathematical Formulation of Model-Based Methods for Diagnostics and Prognostics," *Proceedings of ASME Turbo Expo 2006: Power for Land, Sea and Air*, ASME Paper GT2006-90655, May 2006.
- [10] Jaw, L. C., "Putting CBM and EHM in Perspective," *Maintenance Technology*, Applied Technology Publications, Inc., Nov. 2001, pp. 14–18.
- [11] Moubray, J., *Reliability-Centered Maintenance*, 2nd ed., Industrial Press, Inc., 1997, p. 7.
- [12] Wang, Y., Jaw, L., Rendek, P., Moses, E., Robinson, M., Driver, S., and Senior, K., "Demonstration of a Reliability Centered Maintenance (RCM) Tool to Extend Engine's Time-on-Wing (TOW)," *Proceedings of the 2007 IEEE Aerospace Conference*, IEEE Paper 10-1109, March 2007.
- [13] "Condition-Based Maintenance Plus Organizational Charter," U.S. Dept. of Defense Instruction, No. 4151-22, DOD, Washington, D.C., Dec. 2007.
- [14] Office of Aviation Management (OAM), "Implementation Guide—Aviation Programs Performance Indicators (Metrics) for Use with DOE O 440.2B, Aviation Management and Safety," U.S. Dept. of Energy, DOE G 440.2B-1A, 19 Sept. 2005.

- [15] Tumer, I., and Bajwa, A., "A Survey of Aircraft Engine Health Monitoring Systems," AIAA Paper 99-2528, June 1999.
- [16] Tsalavoutas, A., et al., "Combining Advanced Data Analysis Methods for the Constitution of an Integrated Gas Turbine Condition Monitoring and Diagnostic System," *Proceedings of ASME Turbo Expo 2000*, ASME Paper 2000-GT-0034, May 2000.
- [17] "Condition Monitoring and Diagnostics of Machines—Data Processing, Communication and Presentation—Part 1: General Guidelines," International Organization for Standardization, Standard No. ISO 13374-1, 2003.
- [18] "Building an OSA-CBM (Open Systems Architecture for Condition-based Maintenance) System—a Guide for Developers," OSA-CBM Ver. 3.1, Machinery Information Management Open Standards Alliance (MIMOSA), <http://www.mimosa.org/downloads/39/index.aspx>, May 2009.
- [19] Jaw, L. C., "Recent Advancements in Aircraft Engine Health Management Technologies and Recommendations for the Next Step," *Proceedings of Turbo Expo 2005: 50th ASME International Turbine & Aeroengine Technical Congress*, ASME Paper GT2005-68625, June 2005.
- [20] Li, Y. G., "Performance-Analysis-Based Gas Turbine Diagnostics: a Review," *Journal of Power and Energy*, Proceedings of the Institution of Mechanical Engineers, Vol. 216, No. 5A, 2002, pp. 363–377.
- [21] Bryson, A. E., Jr., and Ho, Y. C., *Applied Optimal Control*, Hemisphere, New York, 1975, Ch. 12.
- [22] Merrill, W. C., DeLaat, J. C., and Burton, W. M., "Advanced Detection, Isolation, and Accommodation of Sensor Failures—Real-Time Evaluation," *Journal of Guidance, Control, and Dynamics*, Vol. 11, No. 6, 1988, pp. 517–526.
- [23] Kobayashi, T., and Simon, D. L., "Application of a Bank of Kalman Filters for Aircraft Engine Fault Diagnostics," NASA/TM-2003-212526, Aug. 2003.
- [24] Simon, D., and Simon, D. L., "Aircraft Turbofan Engine Health Estimation Using Constrained Kalman Filtering," *Journal of Engineering for Gas Turbines and Power*, Vol. 127, No. 2, April 2005, pp. 323–328.
- [25] Dewallef, P., Leonard, O., and Mathioudakis, K., "On-Line Aircraft Engine Diagnostic Using a Soft-Constrained Kalman Filter," *Journal of Engineering for Gas Turbines and Power*, Vol. 129, No. 4, Oct. 2007, pp. 970–976.
- [26] Dewallef, P., Romessis, C., Leonard, O., and Mathioudakis, K., "Combining Classification Techniques with Kalman Filters for Aircraft Engine Diagnostics," *Journal of Engineering for Gas Turbines and Power*, Vol. 128, No. 2, April 2006, pp. 281–288.
- [27] Borguet, S., Dewallef, P., and Leonard, O., "On-Line Transient Engine Diagnostics in a Kalman Filtering Framework," *Proceedings of 2005 ASME Turbo Expo*, ASME Paper GT2005-68013, June 2005.
- [28] Borguet, S., Dewallef, P., and Leonard, O., "A Way to Deal with Model-Plant Mismatch for a Reliable Diagnosis in Transient Operation," *Journal of Engineering for Gas Turbines and Power*, Vol. 130, No. 3, May 2008, pp. 601–608.
- [29] Kobayashi, T., and Simon, D. L., "Hybrid Kalman Filter Approach for Aircraft Engine In-Flight Diagnostics: Sensor Fault Detection Case," NASA/TM-2006-214418, Oct. 2006.
- [30] Jaw, L. C., and Wu, D. N., "Anomaly Detection and Reasoning with an Embedded Physical Model," *Proceedings of the 2002 IEEE Aerospace Conference*, IEEE, Vol. 6, March 2002, pp. 3073–3081.

- [31] Jaw, L. C., "Neural Network Modeling of Engine Tip Clearances," *Proceedings of the 1997 AIAA/ASME/SAE/ASEE Joint Propulsion Conference*, AIAA, Reston, VA.
- [32] Brotherton, T., Volponi, T., Luppold, R., and Simon, D. L., "eSTORM: Enhanced Self Tuning On-Board Real-Time Engine Model," *Proceedings of 2003 IEEE Aerospace Conference*, IEEE, Vol. 7, March 2003, pp. 3075–3086.
- [33] Deming, W. E., *Out of the Crisis: Quality, Productivity and Competitive Position*, MIT Press, Cambridge, MA, 1982, Ch. 11.
- [34] Wheeler, D. J., *Normality and the Process Behavior Chart*, SPC Press, 2000, Ch. 4.
- [35] Montgomery, D. C., *Introduction to Statistical Quality Control*, 6th ed., Wiley, New York, 2008, Chs. 6 and 9.
- [36] Mattingly, J. D., Heiser, W. H., and Pratt, D. T., *Aircraft Engine Design*, 2nd ed., AIAA Education Series, AIAA, Reston, VA, 2002.
- [37] *Statistical Quality Control Handbook*, Western Electric Corp., 1956.

Integrated Control and Health Monitoring

In Chapter 8, we discussed how to use the Kalman filter (KF) to estimate state variables and “unmeasured” output variables. These estimated variables can be used in feedback control and health-monitoring. Because KF is a special case of the class of observers or estimators in system theories, it is intuitive to think that the same estimator can be used by the control system for *state feedback* and by the monitoring system for *state awareness*. This idea is a motivation for integrated control and monitoring of the engine. In fact, integrated control and health management has increasingly become a design requirement for any high-performance, high-readiness physical system. These systems are aware of their own functional and physical capabilities, and some of them are even capable of reconfiguring their own control resources to complete a desired mission while in coordination with other assets.

Integrated control and health-monitoring (or management) systems, by conventional wisdom, are *intelligent* systems. An intelligent engine is self-aware and can modify itself in response to internal and external changes, and this vision was first conceived for the space shuttle main engine. The ultimate goal of an intelligent engine is to operate more safely and reliably with a reduced life-cycle cost [1]. The subject of integrated control and health management encompasses a large scope, and it can form its own volume. To make the contents in this chapter manageable and interesting, we will discuss three unconventional control problems as examples to introduce the need and challenges of this subject. The discussions in this chapter are more control oriented than monitoring oriented. Again, sensor and related technologies are not covered in this chapter. Interested readers are referred to [2] for a summary of various sensor needs for integrated and intelligent engines.

In Sec. 9.1, some basic integration concepts including the architecture, the measure for affordable capabilities, and the requirements for monitoring will be described. In Sec. 9.2, a concept of life-extending control (LEC) will be introduced; specifically, an approach that optimizes engine performance while extending the service life of critical (hot-section) engine components will be discussed. In Sec. 9.3, a concept of safety assurance control will be introduced; specifically, an approach to run an engine with reduced (or less-than-normal) operability margins during emergency landing maneuvers will be discussed. Then in Sec. 9.4, an approach to fast power control of a helicopter engine to support nap-of-the-earth flights will be discussed.

9.1 Integration Architecture, Capabilities, and Requirements

9.1.1 Integration Architecture

Many different kinds of architectures for control and health management integration can be defined, for instance, an architecture that replaces the sensor block in the closed-loop control system diagram of Fig. 1.1 with the monitoring system, which acts like a virtual sensor/estimator, or an architecture that augments the controller block in the Fig. 1.1 with health management and control reconfiguration capabilities, essentially making it an adaptive control system. Either architecture is acceptable from the design standpoint. However, both architectures might be unacceptable from the operational standpoint because the control system typically has a higher *criticality* level for safety than the health management system; hence, they have different certification requirements. Furthermore, there are deliberate barriers (or mediators) implemented in the onboard control and monitoring systems to prevent adverse interactions from happening as a result of malfunctions or ill assessment of the health management system.

To mitigate the risk of the health management system “corrupting” the control system, the integration architecture of Fig. 9.1 has been suggested [3, 4]. In this architecture, the control system performs its normal functions, while the health management system continuously assesses the current health condition and future capabilities of the engine. The results of these assessments are used to answer the following three questions:

- 1) What is the *current* level of performance?
- 2) How much longer [i.e., remaining useful life (RUL)] can the engine continue to produce acceptable power and response in all of the desired operational scenarios?
- 3) How confident is the *current* prediction of this RUL?

The answers to these questions are passed to the control system, which then determines the best control mode and control laws, based on how much risk the control system is willing to take with a degraded engine, to accomplish the goals

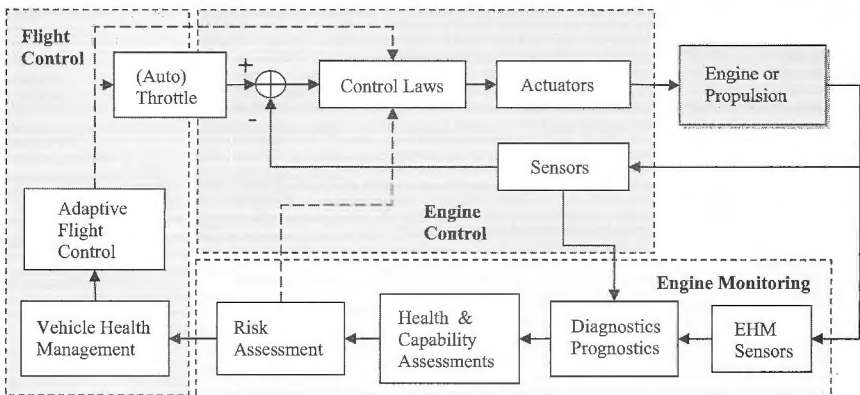


Fig. 9.1 Block diagram of a suggested integration architecture for engine control and health management.

of control. Note that, in the effort to answer the three preceding awareness-related questions, the health management system can rely on some special sensors, so-called the dedicated health management sensors, which are not used for feedback control directly.

This integration architecture can be applied to air vehicles, sea and ground vehicles, as well as other complex systems that can benefit from a tighter integration between the control and monitoring functions.

9.1.2 Affordable Capabilities

In Sec. 7.2, we discussed that engine performance capability can be defined as the ratio of thrust to weight to fuel consumption, that is, $(F_N/W)/\text{SFC}$. This definition considers the engine capability only from a performance perspective. However, engine performance alone is not sufficient to address the operation and support (O&S) requirements to keep an engine available and ready for missions throughout its life cycle. This O&S or life-cycle aspect must consider *cost* in any measure of capabilities because cost enforces a balance between engine performance and mission readiness. The inclusion of life-cycle cost in a performance measure constitutes an *affordable capability* (AC).

One way to define an AC for engines is to divide the performance capability described above by the total cost of owning and operating the engine. This AC measure is also called a cost-capability index (CCI) [5] in the United States, and its generic definition is expressed as

$$\text{CCI} = \frac{\text{Perf Capability}}{\text{Cost}} = \frac{(F_N/W)/(\text{SFC})}{\text{Cost}} \quad (9.1)$$

where cost is the total cost of ownership of an engine type, and it is a weighted sum of the development, production, and maintenance costs.

In Chapter 8, we discussed the total support concept and introduced the concept of “RFT at a reduced cost.” This concept suggests an AC measure for a fleet of aircraft of the same type, just as CCI is for a fleet of engine of the same type. However, MCR (see Sec. 8.1) alone is not a measure of AC for aircraft, just as TOW alone is not an AC measure for engines. The reason that MCR and TOW are not AC measures can be explained by considering the following scenario: a high level of inventory of spare aircraft and engine parts can result in high MCR and TOW. However, excess inventory comes at the expense of an increased life-cycle cost. Indeed, increased life-cycle cost is a direct result of overstocked parts or inefficient supply chain. Another not-so-apparent cause for increased life-cycle cost is replacing parts wrongfully or prematurely. Replacing parts wrongfully can be a result of wrong diagnosis, and replacing parts prematurely can be a result of inaccurate prognosis. The latter is particularly relevant for performance-based support contracts because the “temptation” to pull a degraded part prematurely so as to avoid the penalties associated with lower-than-guaranteed life can be high enough for a contractor to make more conservative repair/replacement decisions for a suspected failure. By considering cost in an AC measure, the conservative decisions can be neutralized by higher incurred costs for the new parts to replace the existing parts that still have a fair amount of residual life.

All of these factors, that is, high inventory, inefficient supply chain, wrong diagnosis, and inaccurate prognosis, are related to the functions of health management; therefore, *health management has a direct influence in the life-cycle cost, hence the affordable capability, of engines, airplanes, ships, and other safety-critical and uptime-sensitive systems.* Of these factors, accurate diagnosis and prognosis is the key to reducing the life-cycle cost or increasing affordable capabilities.

9.1.3 System Requirements for Monitoring

Monitoring requirements here specifically refer to diagnostic and prognostic requirements for affordable capabilities. It is well known that diagnostic requirements are often defined by the false-alarm rate (or the false-positive rate and commonly denoted by α) and the missed-detection rate (or the false-negative rate and commonly denoted by β); furthermore, the performance of a diagnostic system is commonly evaluated graphically in a receiver operating characteristic curve, which is a diagram of the true detection rate (or sensitivity, $1 - \beta$) vs the false-alarm rate (or α).

Given a diagnostic capability of an engine, in terms of the false-alarm rate and the missed-detection rate, we are interested in analyzing how good this capability is in completing scheduled flights or combat missions by a fleet of aircraft. If we use the MCR as a performance measure for this analysis, and if we assume there is no spare engine available to replace a failed engine, then the MCR can be parameterized as functions of α and β , as shown in Figs. 9.2 and 9.3 [6]. Figure 9.2 shows that MCR is a function of false-alarm rate for varying levels of sensitivity

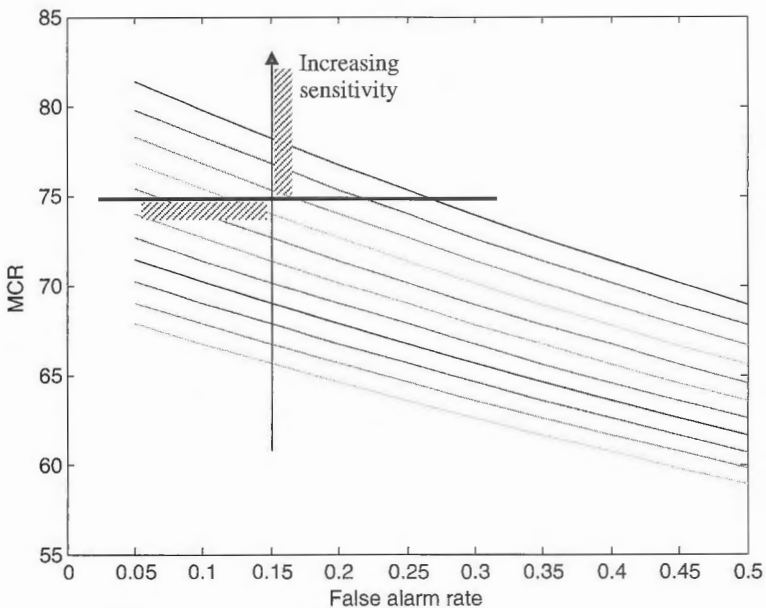


Fig. 9.2 MCR as a function of false-alarm rate α with varying sensitivity ($1 - \beta$).

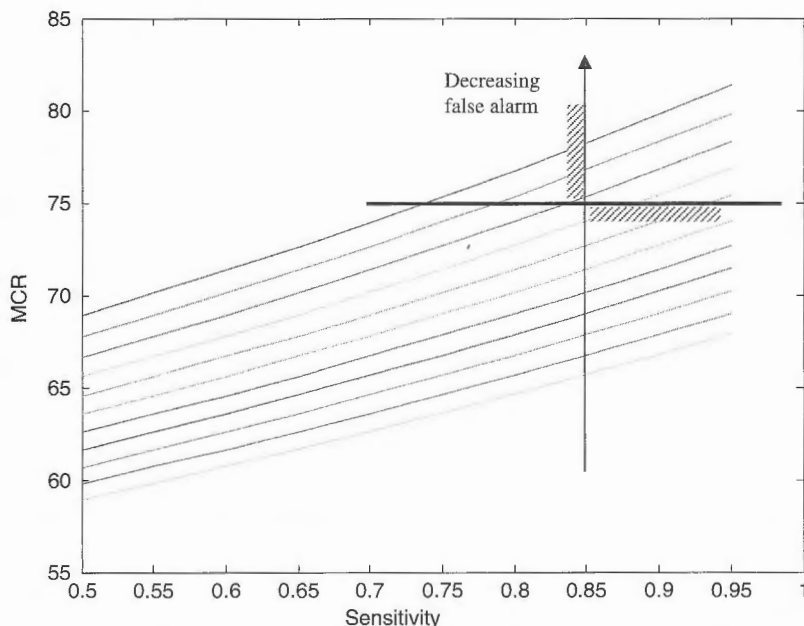


Fig. 9.3 MCR as a function of sensitivity with varying false-alarm rate.

$(1 - \beta)$. So hypothetically, suppose that a desired minimum MCR is 75% and that diagnostic capability of less than 15% false-alarm rate, we can then define the region in the upper-left-hand corner of the MCR vs α diagram where we want the diagnostic system to perform in order to support a desired MCR.

Similarly, Fig. 9.3 shows that MCR is a function of sensitivity $(1 - \beta)$ for varying false-alarm rates. So again, suppose that a desired minimum MCR is 75% and that a diagnostic capability of less than 15% missed detection rate, we can then define the region in the upper-right-hand corner of the MCR vs $(1 - \beta)$ diagram where the diagnostic system must perform in order to meet the desired MCR.

The purpose of prognostics is to predict the RUL of a system or subsystem with acceptable confidence. Life forecasting (or prediction) must be done periodically to ensure *convergence* in predicted RUL because variations in operating conditions, mission scenarios, and data can result in different outcomes from one prognosis to another. As one can expect, the predicted RUL and associated confidence level at the beginning of the service life of a system are not as accurate as those predicted near the end of the service life. Hence, as the failure trend is further developed in time, the prediction accuracy and confidence will both increase. This phenomenon in prognostics can be illustrated in Fig. 9.4 [7], where the predicted, instantaneous RUL curve of a control actuator is shown in red, and the smoothed confidence band with an upper limit (in blue) and a lower limit (in green) corresponding to the confidence levels of $\pm 3\sigma$, respectively, are also shown.

The RUL can be thought of as the prognostic lead time (PLT), which is the lead time for the logistic system to respond to anticipated maintenance actions required

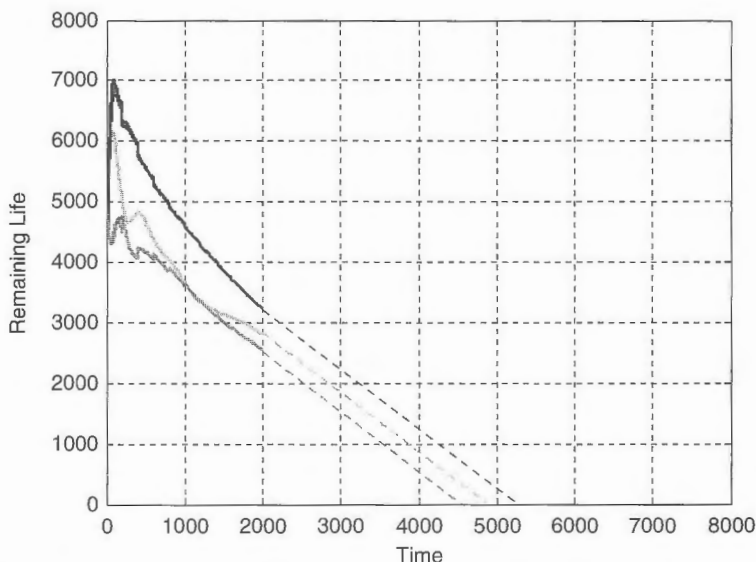


Fig. 9.4 RUL forecast with the confidence band. (See also color figure section at the back of the book.)

for the system that is predicted to fail. The lead time is used to *alert* the supply chain about certain parts that are needed for anticipated repair (or replacement). It is also used to alert the planning staff about certain human and facility resources that should be scheduled to perform the repair. The time that is required for the supply chain and the resource planning and scheduling functions to complete is called the *logistic response time* (LRT). To reduce the LRT, we desire a lean supply chain [8] so that an optimal level of parts can be maintained, and they are stocked at the most efficient locations [9].

Regardless of the LRT, an effective prognostic capability should minimize any logistic delay, which lengthens the downtime of the system to be repaired or replaced. Based on this principle, we can define the “minimal” requirements for an acceptable prognostic capability as follows:

1) The predicted RUL of an engine (or any of its subsystems) shall be greater than the LRT of the engine (or subsystem). This requirement can be stated differently as follows: when the predicted RUL of an anticipated failure is equal to the LRT, then replacement parts shall be ordered, and sufficiently skilled maintainers shall be scheduled. If this requirement is met, then there is essentially no logistic delay, and the downtime for repair is exactly the same as the time needed to perform the repair, given available parts, maintainers, facilities, and tools.

2) The prediction confidence [or confidence factor (CF)], when RUL is equal to LRT, must be greater than or equal to a minimally acceptable value $CF_{\min}$, that is, this requirement can be expressed as

$$CF|_{RUL=LRT} \geq CF_{\min} \quad (9.2)$$

3) The confidence of a predicted RUL can be relaxed with increasing value of RUL, that is, further upstream of a predicted failure, and the confidence is inversely proportional to RUL. This is a “weak” requirement for prognostics, and it can be expressed as

$$CF|_{RUL > LRT} = f\left(\frac{1}{RUL}\right) < CF_{\min} \quad (9.3)$$

9.2 Life-extending Control

The discussion of affordable capabilities earlier has motivated us to change our mindset about how we control engines. The traditional engine control philosophy has been to maximize the performance, but now we have a new opportunity to control engines differently, that is, to optimize the performance and cost consequences together, or to maximize the AC measure. Moreover, in certain situations compromises must be made, not between the performance and cost, but between performance and flight safety.

The primary objective of a traditional aircraft engine control system is to produce thrust or shaft power as fast as possible in response to the pilot’s command. Fast response is desirable in certain situations, for instance, takeoff, go-around, air combat, or emergency landing; nevertheless, it comes at a price in some other situations. The most notable price to pay for fast response (performance) is reduced life for engine components because of increased thermal and mechanical stresses. For commercial aircraft engines and industrial gas turbines, extending the service life of components so as to reduce the operational and support cost is a strong interest for the owners and operators of these engines.

In this section, a nontraditional concept of designing engine control logics are described. This concept is called life-extending control (LEC) [10], and as the name implies, the purpose of the control is to prolong the life of gas-path components, especially for the disks, blades, and stators (or vanes) in the hot section of an engine. LEC is a new paradigm for engine control design. It can be thought of as an additional control logic that modifies a traditional engine control to reduce component damages while maintaining (flight) safety [11]. This concept is illustrated in Fig. 9.5.

9.2.1 Component Damages

The most common damage modes for gas-path components are turbine-blade creep/rupture, uncooled turbine stator creep, turbine-cooled stator thermal-mechanical fatigue (TMF), and low-cycle fatigue of disks. Damage is interpreted as the accumulation of strain at the microstructural level. When the cumulative strain exceeds a certain limit, cracks can form and grow (or propagate). The unit for damage accumulation is typically a time interval or a cycle of speed or temperature. The damage incurred within the accumulation unit is called the damage rate d . Different damage rates are observed for different damage modes and for different components. Damage rate is typically a geometric function of one or

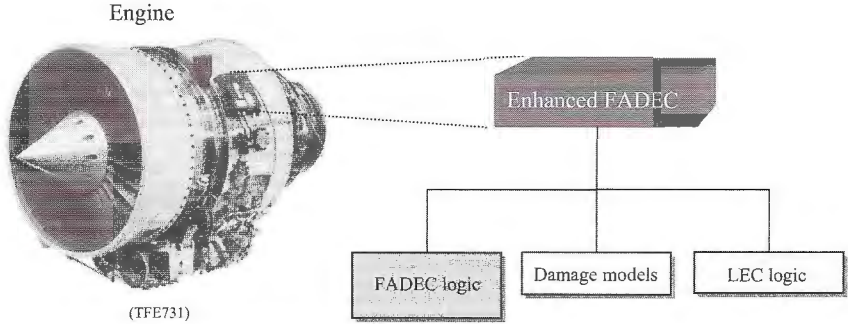


Fig. 9.5 LEC concept demonstrated for a commercial turbofan engine.

more variables that reflect the operating or environmental condition, which are called the damage drivers.

The time to failure, or the remaining life, of an engine component, subject to a standard usage profile, is expressed by

$$L(T_f) = T_{f_0} - \sum_{i=1}^{T_f} d_i \quad (9.4)$$

where d_i is the life decrement per cycle (or time interval), L is the remaining life, T_f is the time or cycle when the component fails to provide useful service, and T_{f_0} is the predicted life of the component when it is new. Figure 9.6 shows two examples of the normalized damage rates.

While T_{f_0} depends on material properties and operating conditions, it is more convenient for us to set it as a constant for a standardized material and operating condition and then to adjust the equivalent cycle count T_{fe} for actual operational usage [12].

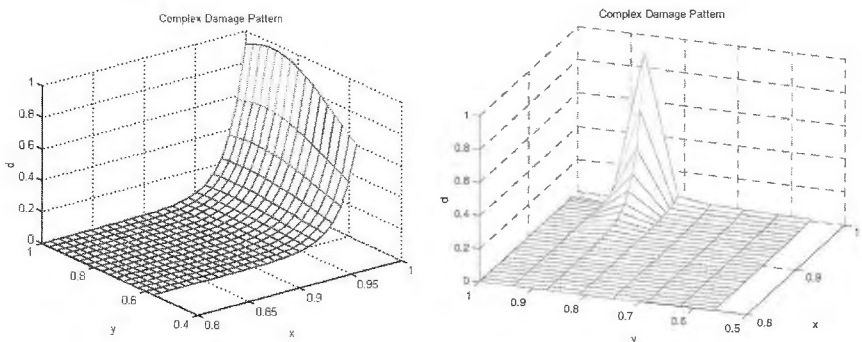


Fig. 9.6 Damage patterns that depend on two operating parameters (x and y).

The damage (rate) algorithm, or the lifing algorithm as it is called sometimes, for a given mode and for a given component is developed through an analytical-empirical process. For turbine blade creep/rupture damage, the damage algorithm can be represented by

$$d_{c/r} = [c_1^{(T_{45})}][c_2 + c_3 N_H^2] \quad (9.5)$$

where T_{45} is the turbine gas temperature, N_H is the high-pressure turbine speed, and c_1, c_2, c_3 are constants (in simplified form). Basically, the damage rate increases in a square function of the high-pressure turbine speed, and it increases in a power function of the turbine temperature. Therefore, forecasting of the remaining life of the turbine blade takes the form of

$$L(T_{fe}) = T_{fo} - \sum_{i=1}^{T_{fe}} [c_1^{(T_{45})}]_i [c_2 + c_3 N_H^2]_i \quad (9.6)$$

Lifing algorithms, like the one in Eq. (9.6), will most likely be modified with different coefficients through the life cycle of a component (or system) as a result of the new knowledge gained from operational experience. With the new knowledge gained from analyzing field data, c_1, c_2, c_3 can be made a function of the rate of change, the time duration, and the peak value of T_{45} and N_H . This updated information takes consideration of the level of operation such as severe vs normal missions [13].

Life-forecasting (or prediction) results can be presented in a remaining useful life (RUL)-vs-usage diagram shown in Fig. 9.7, which is similar to Fig. 9.4 but

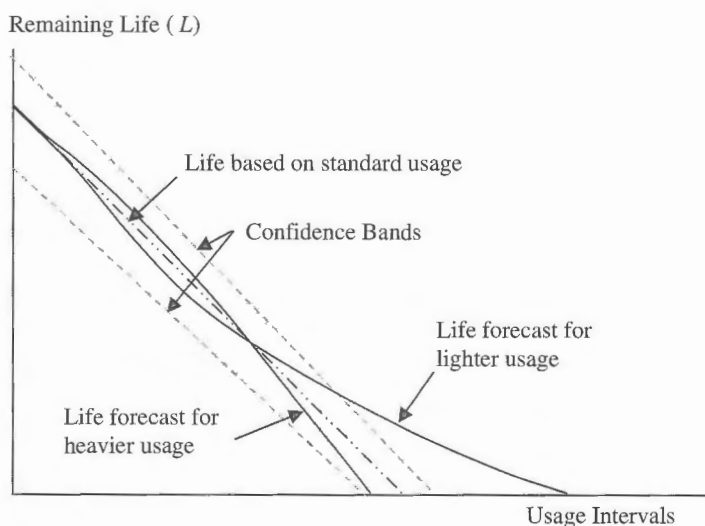


Fig. 9.7 Life-forecasting chart showing RUL vs usage (time or cycle).

has two differences: 1) the horizontal axis is measured in usage intervals rather than in time directly, and 2) two additional life-forecast curves have been added to represent the predicted RULs for a lighter-than-normal usage scenario and for a heavier-than-normal usage scenario, respectively. Note that a confidence band for the normal (or standard) usage scenario has been added to indicate the probabilities associated with the prediction.

9.2.2 Control Design

Design of LEC logic was applied to reducing TMF damage for a hot-section component of a small commercial turbofan engine. The LEC logic trims the traditional transient control logic of the engine during snap acceleration from idle to the TO power. Because the response time of a snap acceleration is a key performance index for all aviation engines, the design goal was to reduce TMF damage while not adversely affecting the transient performance of the engine. Several design options were investigated, including resetting the target (set-point) speed, increasing or decreasing control gains, and reducing the accel schedule. It was found from engine simulations that reducing the accel schedule is the most effective option.

To reduce TMF damage, the LEC logic lowered the accel schedule by a certain percentage as soon as the difference between the actual speed and the target (set-point) speed, DN, is less than a predefined threshold, that is, the N-dot acceleration schedule is lowered by a certain percentage when the speed difference is within a preselected range of the target speed. This logic concept is illustrated in Fig. 9.8. Note that the power setting variable is the corrected HP spool speed N_{2C2} .

Three different speed error (DN) thresholds were selected (800, 1000, and 1200 rpm), and the reductions in TMF damage (in percentage) and the increase of rise time of fan speed N_1 (an indicator of engine thrust) during the engine acceleration from ground idle to maximum power are shown in Tables 9.1–9.3. Figure 9.9

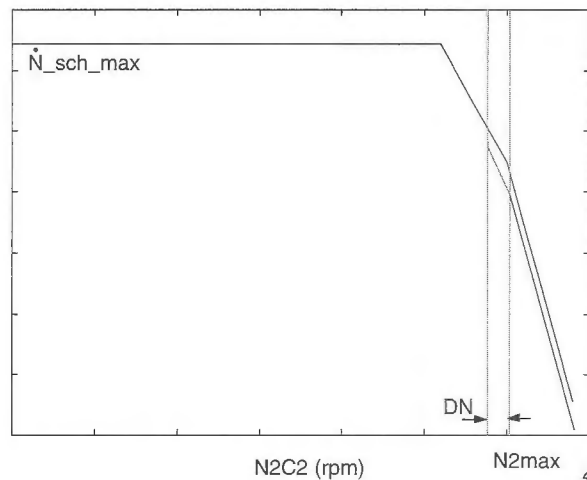


Fig. 9.8 Illustration of acceleration schedule reduction logic.

Table 9.1 TMF damage reduction for speed error threshold of DN = 800 rpm

Accel fuel reduction, %	TMF reduction, %	Extra rise time, s
10	13.7	0.06
20	24.5	0.12
30	35.3	0.22
40	45.6	0.32
50	49.0	0.58

Table 9.2 TMF damage reduction for speed error threshold of DN = 1000 rpm

Accel fuel reduction, %	TMF reduction, %	Extra rise time, s
10	14.7	0.06
20	26.4	0.16
30	37.7	0.28
40	47.5	0.40
50	54.3	0.74

Table 9.3 TMF damage reduction for speed error threshold of DN = 1200 rpm

Accel fuel reduction, %	TMF reduction, %	Extra rise time, s
10	14.7	0.08
20	27.5	0.18
30	39.2	0.32
40	49.0	0.50
50	56.9	0.86

shows the reduction in TMF damages vs reductions in acceleration fuel flow for different thresholds of speed error. Figure 9.10 shows the reduction in TMF damages vs changes in rise time for different thresholds of speed error. It can be seen that the greater the reduction in TMF damage is, the greater the increase in rise time is. It is also found that the relationship between the TMF damage reduction and increase in rise time is not sensitive to the threshold values. For all three cases, significant reductions in TMF damage can be achieved with only a very small increase in rise time for N_1 and thrust.

9.3 Safety Assurance Control

An engine control system needs not only to provide desired performance with adequate safety margins, but it also needs to provide satisfactory performance by accommodating engine component deteriorations. Conventional deteriorations, such

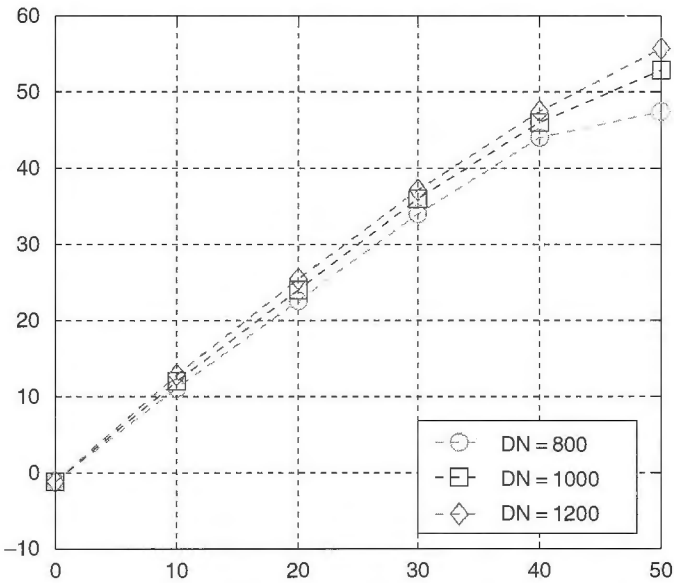


Fig. 9.9 Reduction in TMF damage (in percentages) vs reduction in acceleration schedule (in percentages) vs speed threshold (in revolutions per minute).

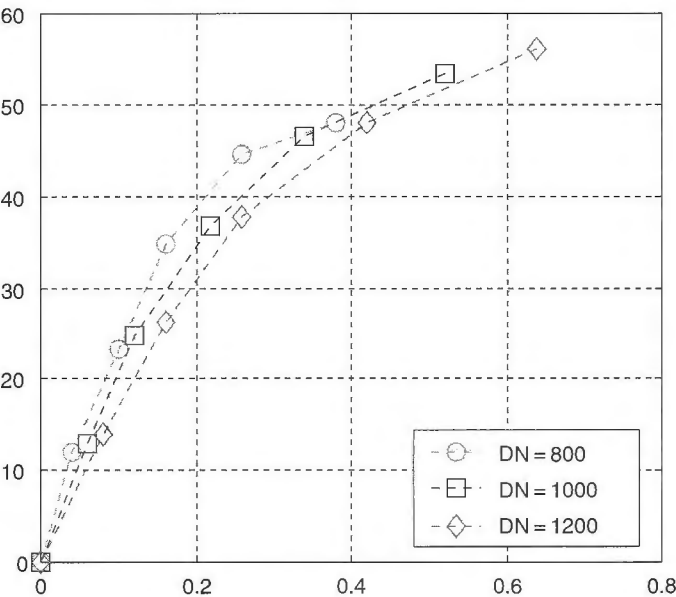


Fig. 9.10 Reduction in TMF damage (in percentages) vs increase in rise time (in percentages) vs speed threshold (in revolutions per minute).

as failures and aging, have routinely been included in control design considerations. Techniques such as feedback controls, built-in-tests, embedded models, and engine health management have been employed in engine controls to deal with conventional deteriorations. However, these techniques alone are not adequate to handle some unconventional operating scenarios, such as an uncontained blade failure, a loss of hydraulic power for flight controls, a loss of a section of airplane structures, a loss of control surfaces caused by an air collision or explosion, or any combat-related damage.

The propulsion system contributed to approximately 12% of all jet transport accidents. Of these propulsion-related accidents, the top two causes are uncontained failures (approximately 18%) and a combination of propulsion system malfunction and crew errors (15%) [14]. In the event of an engine or system failure, the (integrated) engine control and monitoring system plays an important role in providing continued and safe operation of the aircraft. Even if a failure is not caused by the propulsion system, the engine control and monitoring system is expected to provide "extra" capabilities to ensure a safe emergency landing of the aircraft.

The need for this kind of more intelligent engine controls in the event of an unconventional failure is reinforced by the following two real-world cases [15]:

Case 1: On 19 July 1989, the United Airlines Flight 232 suffered a failure in the center engine of the DC-10 aircraft. The failure was uncontained, and the fragments from the engine severed all hydraulic power to the flight control system, hence losing the controllability of the aircraft. The pilots made an emergency landing in Sioux City, Iowa, by adjusting the thrusts of the two remaining, wing-mounted engines. Almost two-thirds of the passengers and crew onboard were saved.

Case 2: On 24 November 2003, an A300 freighter operated by the international courier DHL was hit by a surface-to-air missile. The damage caused loss of hydraulic power and, consequently, the aircraft's control surfaces. The crew made an emergency landing in Baghdad, Iraq, airport by coordinating the engine thrusts by recalling the lessons learned from the Sioux City accident.

In such circumstances, the primary objective for the air vehicle is to land safely as soon as possible. The concept of propulsion-controlled aircraft (PCA) is the most effective method to accomplish this objective. Flight tests of the PCA concept by NASA Dryden Flight Research Center with the F-15 concluded with a successful landing on 21 April 1993, using only the engine power to turn, climb, and descend. A successful follow-on program with an MD-11 transport aircraft conclusively demonstrated the success of the concept in 1995 [16]. These research works concluded that: 1) the pitch motion of an aircraft can be controlled by modulating engine thrusts, and 2) the pilot workload increases significantly with the propulsion-controlled aircraft, and it is more feasible when the control is automated.

Whether the PCA is used as the only means for flight-path control or it is used to augment a partially damaged flight control system, the engine is expected to change thrust faster and over a larger range between the min and the max levels. Increasing the thrust and its response time can improve the controllability of a damaged aircraft by modulating the differential thrust between the two sides of the aircraft. To increase the thrust beyond its normally scheduled value, the engine controller must *reset* (or *relax*) some of the operational limits (or constraints). To increase the thrust response, the engine controller must modify the control laws or actuation mechanisms. However, in doing so, the controller should avoid

compressor surge or turbine blade meltdown. Relaxing constraints and modifying control laws are two practical approaches to *damage-adaptive controls* under unconventional operating conditions.

In the first approach, we are interested in relaxing some or all of the performance and operational limits in the controller. These limits include power setting variable (N_1 or EPR), accel limit, decel limit, shaft mechanical speeds, EGT, and P_3 . By resetting these limits “out of the way,” the engine is able to produce the thrust that is unattainable under normal engine control logic to increase the thrust, or to recover some of the thrust lost as a result of the damage. This approach, however, does not consider the remaining life of the engine, nor does it try to extend the life of the engine by shaping the control law as described in the preceding section. Basically this approach assumes that the engine can withstand the increased mechanical and thermal loads over the damage mitigating period of approximately 20 minutes until the aircraft makes an emergency landing.

9.3.1 Limit-reset Control

An example of the limit-reset controller is described in this subsection. However, before a damage-adaptive control for increased thrust level can be designed, a large decrease in engine thrust, caused by a presumed, unconventional damage situation, must be simulated to help us understand the problem. The damage scenario is simulated by a combination of several large (i.e., >10%) degradations in engine component parameters, such as the efficiency and the flow capacity of a specific component.

The combination of component degradations include fan flow reduction, LPC efficiency reduction, HPC flow leakage, and tail nozzle-area increase. The initial values of these degradations are then adjusted, together with the adjustment of each bias used by the controller to relax one or more corresponding operational constraints. This procedure is adopted because it gives the flexibility to select a set of parameter degradations and a set of limit biases to best show the effect of the limit-reset control.

The simulation results are shown in Figs. 9.11 [17]. The closed-loop response of a generic commercial engine with a normal controller is shown in Fig. 9.11a. Note that the damage is introduced at 20 s. The observed effect of the damage is a sudden drop in EPR, which is the presumed power-setting variable with a simultaneous, sudden jump in EGT. The engine controller tries to increase fuel flow to maintain the desired (or commanded) value of EPR. However, the actual EPR increases only slightly before the fuel flow runs into the accel limit.

Figure 9.11b shows the activation diagram for each of the five operational limits: the max fuel limit ($W_f/P_3|_{\max}$, or the variable name of “Wf/P3 Max” in the figure), the min fuel limit ($W_f/P_3|_{\min}$, or the variable name of “Wf/P3 Min”), the max low-pressure shaft speed ($N_1|_{\max}$, or “N1max”), the max high-pressure shaft speed ($N_2|_{\max}$, or “N2max”), and the max compressor exist static pressure ($P_3|_{\max}$, or “P30max”). A value of one means that the limit is active; a value of zero means the limit is inactive.

The closed-loop response of a limit-reset controller, after the simulated damage scenario is introduced, is shown in Figs. 9.12a and 9.12b, in which all operational

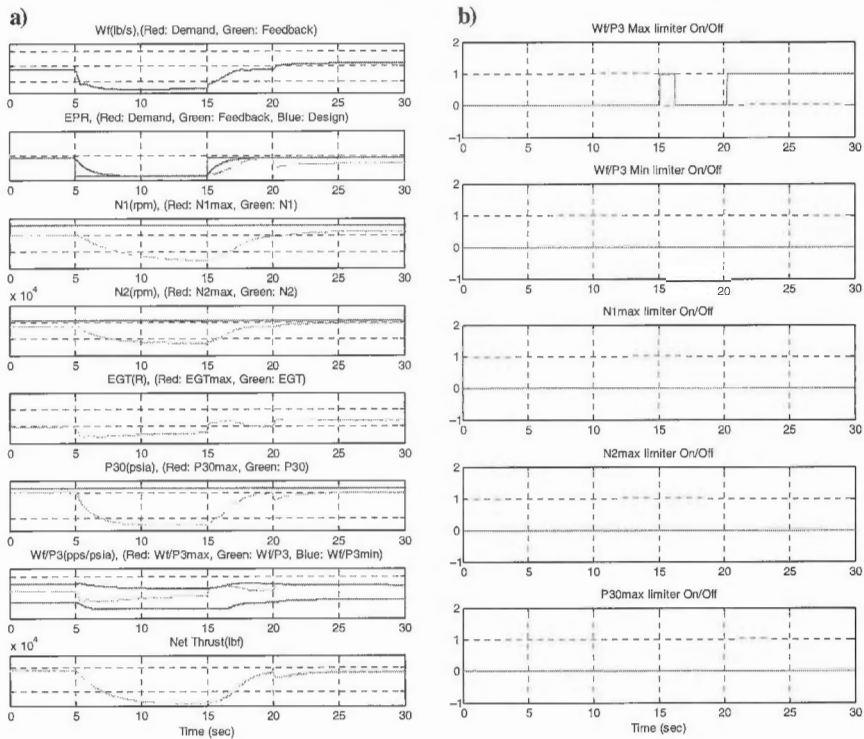


Fig. 9.11 (a) Engine response with a normal controller. (See also color figure section at the back of the book.) (b) Operational limit activation diagram.

limits are reset so as to not interfere with the engine accel and decel. The time for the EPR to recover the damage-induced decrease is about 3.5 s.

9.3.2 High-Gain Control

An example of faster response control is shown in Fig. 9.13, in which the controller gain constants are magnified. Note that the acceleration time for the EPR to recover to the before-damage value is reduced by half (or 50%).

Damage-adaptive control tries to increase the engine response and thrust level. This control capability is important to assure safe and secure flights under severe, adverse conditions. This capability is currently an active area of research in engine controls at NASA. The capability can only be realized by integrating control and health management functions.

9.4 Fast-Response Control for Helicopter Engines

In Chapter 2, we discussed helicopter engine-airframe compatibility issues and how they can cause large rotor speed variations. We also discussed a reduced,

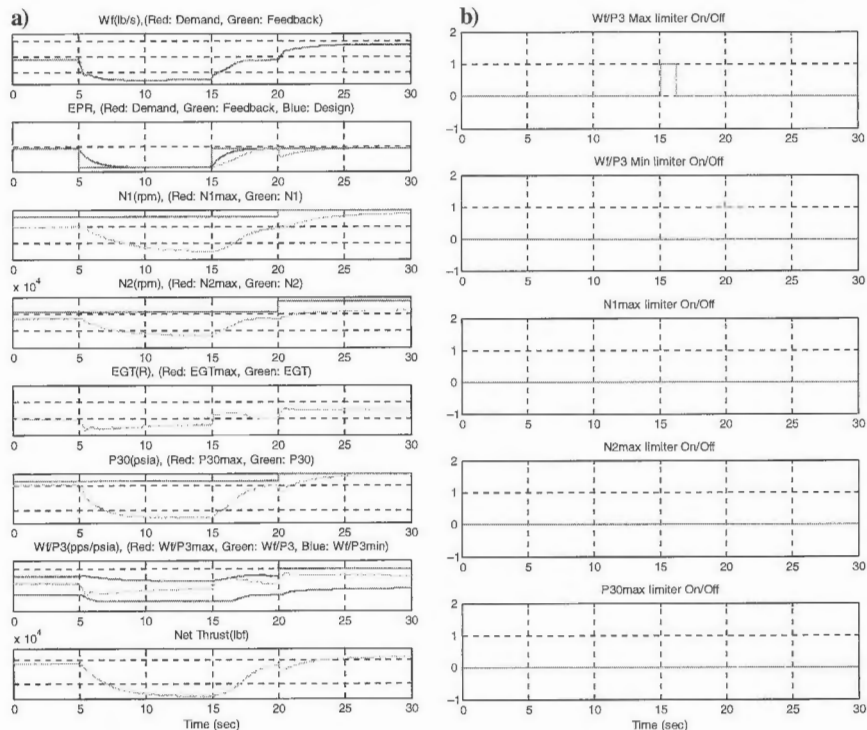


Fig. 9.12 (a) Engine response with a limit-reset controller. (See also color figure section at the back of the book.) (b) Operational limit activation diagram.

third-order model [given in Eqs. (3.20) and (3.21)] of a turboshaft engine to illustrate the torsional flexibility of the engine-drive train-rotor system [as expressed in Eq. (3.31)]. If this torsional flexibility is not properly suppressed, it can compromise the controllability of the helicopter.

Large turbine speed variations caused by large changes in rotor loads can occur in forward flight and hover. Some of the forward-flight maneuvers that can cause large load changes include quick-turn, quick stop, wave-off, and nap-of-the-earth (or terrain following). On the other hand, some of the hover maneuvers that can cause large load changes include unmasking (or bob-up behind an obstacle) and quick heading change. During these “extreme-load” maneuvers, engines are being commanded to maximum acceleration or deceleration; however, from Chapter 5, we know that these maximum transients are limited by the accel and decel schedules for surge and flame-out protections. Hence, we are interested in nonconventional ways to increase the engine’s transient response in these extreme maneuvers.

In this section, an innovative control concept that has the potential to increase engine transient responses (or the rate of change of shaft output power) by an *order*

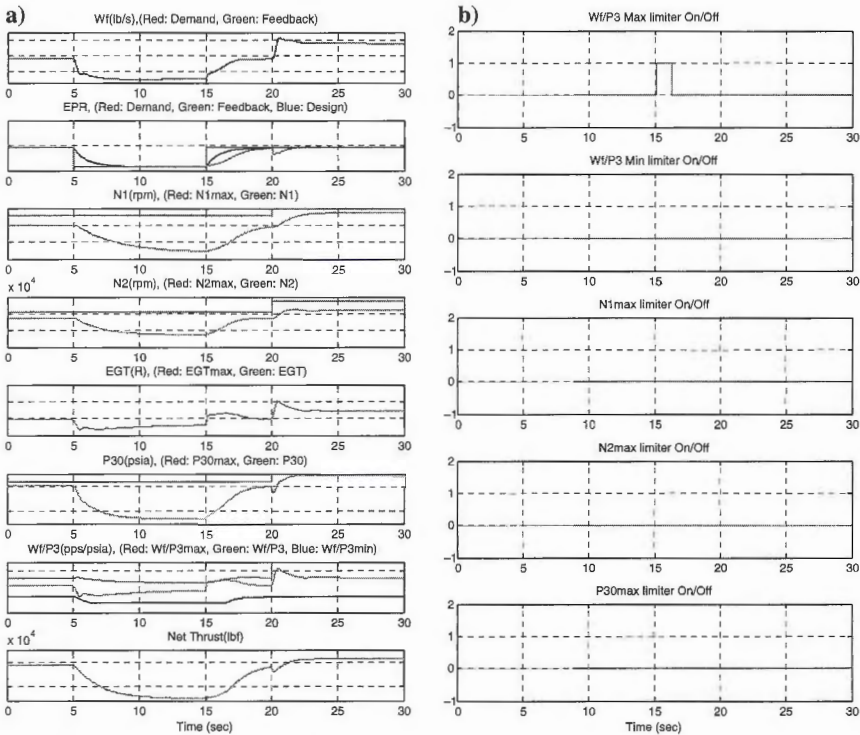


Fig. 9.13 (a) Engine response with a gain-magnified controller. (See also color figure section at the back of the book.) (b) Operational limit activation diagram.

of magnitude compared to traditional approaches is introduced. Increased engine responses potentially reduce rotor speed droops and improve the handling quality and survivability of helicopters.

9.4.1 Turbine Bleed Control Mode

In the preceding section, we discussed two approaches to faster control of engine thrust: the limit-reset control and the gain-magnifying control. The first approach is not acceptable here because the helicopter engine has not been damaged; hence, all normal control limits should be observed. The second approach has a very limited applicability, because the linear control part of the acceleration control where the gain-magnifying control is expected to work will quickly be overtaken by the accel and decel schedules in the transient operation. Hence, the second approach essentially has little effect on reducing undesirable speed variations.

Although the approach that incorporates a feed-forward control loop from the pilot's collective command to the fuel flow command (as described in

Chapter 5) has demonstrated merit in reducing speed variations by anticipating rotor load changes, this approach will still be limited by the accel and decel fuel limits as described before. In fact, the amount of fuel flow available to the engine will be further limited by the magnitude and rate of change of the fuel actuator.

An innovative bleed flow control concept for faster power responses in turboshaft engines was suggested. In this concept, a “turbine” bleed flow is regulated simultaneously with the traditional fuel control. The turbine bleed flow is located between the gas-generator turbine and the power turbine. Unlike the conventional compressor bleed control, the turbine bleed control modulates a small portion of the high-energy combustion gases between the two turbine modules [18]. Modulating the combustion gases produces a larger scale of change in engine power. This concept is illustrated in Fig. 9.14.

Based on a series of computer simulations of this turbine bleed control concept on a modern turboshaft engine, the following results can be concluded: 1) the turbine speed responds much faster in a turbine bleed control than in a compressor bleed control, and 2) the magnitude of change in power turbine shaft output power, for the same amount of bleed flow, is much larger in a turbine bleed control than in a compressor bleed control. However, the benefit of faster response and larger magnitude of power change comes at an extra cost. The extra cost is in reduced propulsive efficiency because a portion of the aerothermal energy is wasted by throwing away some of the combustion gases, even though the amount of bleed flow is small. Hence, the turbine bleed control mode is recommended only for certain types of helicopter maneuvers that experience fast and large load changes while maintaining agility and handling quality.

From an operational standpoint, the turbine bleed control would be activated shortly before the helicopter starts an agile maneuver, and it would be turned off right after the maneuver is finished. When it is activated, the bleed valve opens to a nominal position, and the valve is modulated around this nominal position during the entire maneuver. In this operating scenario, the engine is operating with a

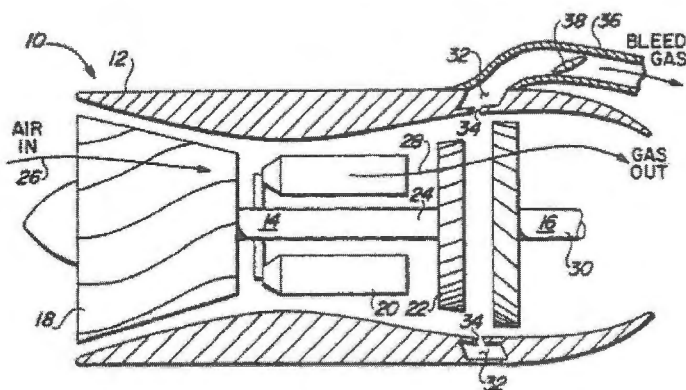


Fig. 9.14 Turbine bleed control concept (from [18]).

leaky (bleed) valve, but the gain in transient performance and controllability of the helicopter can outweigh the loss in fuel consumption and propulsive efficiency. This concept is similar to the use of wing-mounted spoilers on fixed-wing aircraft. These spoilers are modulated during landing to achieve a fast control of altitude to get the aircraft back on the glide slope.

9.4.2 Integrated Engine-Helicopter Model

A reduced-order model for a modern, midpower-class turboshaft engine with both the fuel flow and turbine bleed flow controls is presented here [19]:

$$\dot{x}(t) \approx \begin{bmatrix} -2.78 & 0.62 & 6.60 \\ 0 & -2.05 & -2.49 \\ 5.41 & -4.17 & -44.91 \end{bmatrix} x(t) + \begin{bmatrix} 102 & 453 & 1.9 \\ -40 & 85.9 & 41.5 \\ -298 & 46.1 & -3.7 \end{bmatrix} \begin{bmatrix} \Delta W_f \\ \Delta w_{45} \\ \Delta P_l \end{bmatrix} \quad (9.7)$$

$$\begin{bmatrix} \Delta N_g \\ \Delta N_p \\ \Delta P_e \end{bmatrix} \approx \begin{bmatrix} 0.688 & -0.014 & 0.118 \\ -0.458 & -2.2 & -0.202 \\ 0.058 & -0.017 & -0.149 \end{bmatrix} x(t) + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -18.77 & 0 \end{bmatrix} \begin{bmatrix} \Delta W_f \\ \Delta w_{45} \\ \Delta P_l \end{bmatrix} \quad (9.8)$$

where w_{45} denotes the bleed control input, W_f is the fuel flow input, and P_l is the rotor load input, which is added to ensure mechanical compatibility with the drive train. The unit of fuel flow rate is in pounds per hour, the unit of the bleed flow is in percentages (engine airflow rate), the unit of rotor speeds is in revolutions per minute, and the unit of shaft powers is in horsepower. The open-loop step responses of the engine output power with respect to the fuel flow input and the bleed flow input are compared in Fig. 9.15. The scales of the vertical axes are different to provide better resolution for each output response.

Note that a positive bleed flow input means to increase the bleed valve opening. Opening bleed valve corresponds to taking more flow out of the engine, hence reducing the engine output power. Note also that the engine output power drops immediately when the bleed valve is opened to dump a small amount of turbine gases overboard. The initial power drop is then washed out by about 50%. It is this large, fast, initial change in power that is most beneficial in countering large rotor load changes.

Because the model has been scaled, a unit step input in fuel flow is 1 pph, and a unit step input in bleed flow is 1%. Figure 9.15 suggests that the steady-state change in engine output power is approximately the same for -1% of bleed input and $+3.5$ -pph fuel input. Nevertheless, we are more interested in the initial change in power than the steady-state change. Hence, it is the large, initial change in power affected by the bleed control that is needed to overcome the instantaneous change in rotor load.

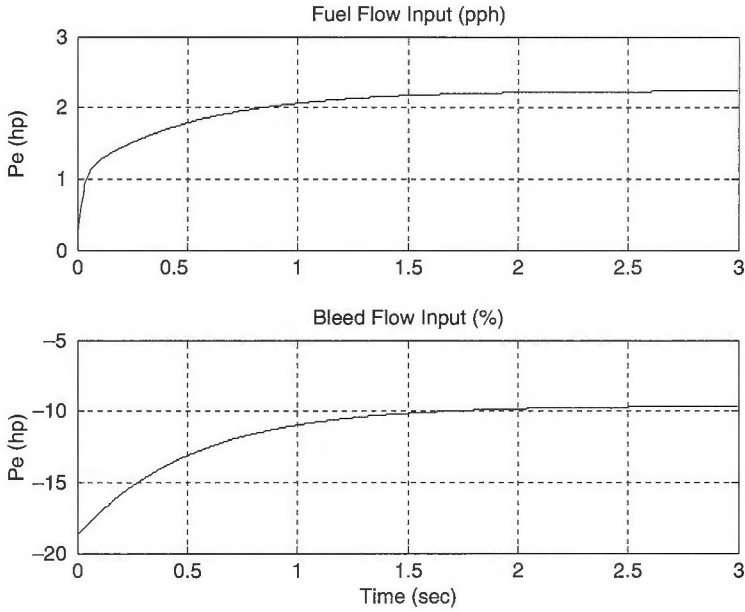


Fig. 9.15 Comparison of engine output power responses between fuel flow and bleed flow controls.

After integrating the preceding engine model with a helicopter model in vertical flight, we get a seventh-order, state-space model [19] of the engine-drive train-rotor system for a Blackhawk (UH-60) type of utility helicopter as

$$\dot{x} = \begin{bmatrix} -0.0193 & -5.73 & -0.015 & 0 & -0.02 & -2.564 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 140.6 & -815.8 & -26.76 & 2.8 & 2.326 & 293.9 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -2.78 & -2.94 & 3.18 & -289.7 & -9.41 & 248.4 & -154.4 \\ 0 & 0.0269 & 0 & 1.881 & 0.0565 & -2.091 & 1.224 \\ 0 & 0 & 0 & -0.0068 & 0 & 0.0249 & -1.94 \end{bmatrix} x(t) + \begin{bmatrix} 0.387 & 0 & 0 \\ 0 & 0 & 0 \\ 777.9 & 0 & 0 \\ 0 & 0 & 0 \\ -55.92 & -31.05 & 53.76 \\ 0 & 0.246 & -0.426 \\ 0 & 0.391 & 0.318 \end{bmatrix} u(t) \quad (9.9)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -7.597 & 12.29 \end{bmatrix} x(t) + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 2.471 & -4.279 \end{bmatrix} u(t) \quad (9.10)$$

where the seven state variables are $x^T = [w \ \delta\beta_0 \ \delta\dot{\beta}_0 \ \delta\xi_0 \ \delta\dot{\xi}_0 \ \delta N_p \ \delta N_G]$; the three input (or control) variables are: $u^T = [\delta_c \ \delta W_f \ \delta w_{45}]$, where δ_c is the flight control input representing the collective command (typically an angle for the position of the collective spindle) to change the rotor pitch angle; and the four output variables are $y^T = [w \ \delta N_p \ \delta N_G \ \delta Q_e]$; and where w is the (vertical) velocity of the helicopter in the Z-body axis, β_0 is the equilibrium (or collective) flap angle of the main rotor, ξ_0 is the equilibrium (or collective) lead-lag angle of the main rotor, Q_e is the engine shaft output torque. Note that this model has been scaled according to the importance of each variable, as such, the unit of angles and collective command is in centi-radian (or crad), the unit of shaft speeds is in 1000 rpm, the unit of fuel flow is in 10 pph, and the units for other variables remain the same as their corresponding unscaled units, that is, percentages, feet per pound, and seconds.

The open-loop step response of the turbine speed N_p with respect to each of the three input variables is plotted in Fig. 9.16. Because the model is again scaled, the unit step inputs in the three control variables are 1 crad, 10 pph, and 1%, respectively. Also because the main rotor speed is physically connected to the power turbine speed through a fixed gear ratio, so the axis of N_p can be

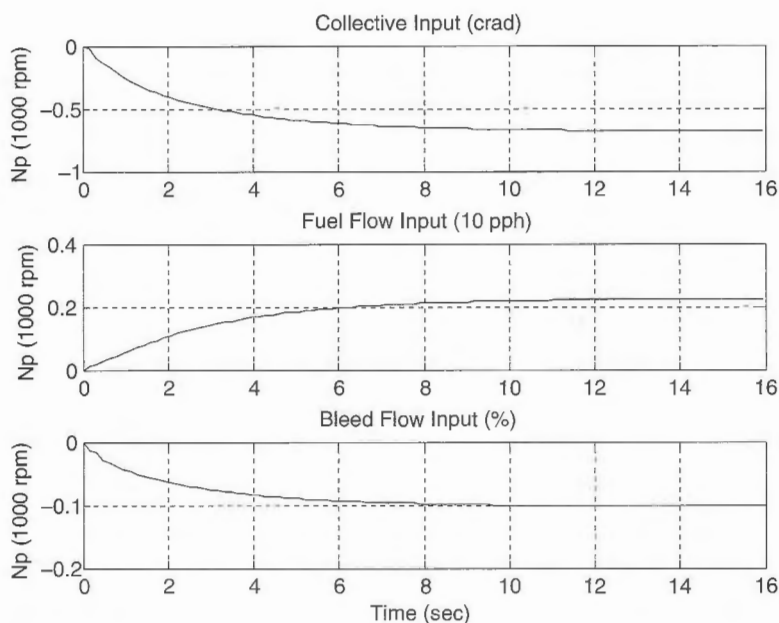


Fig. 9.16 Comparison of the power turbine speed with respect to three control inputs.

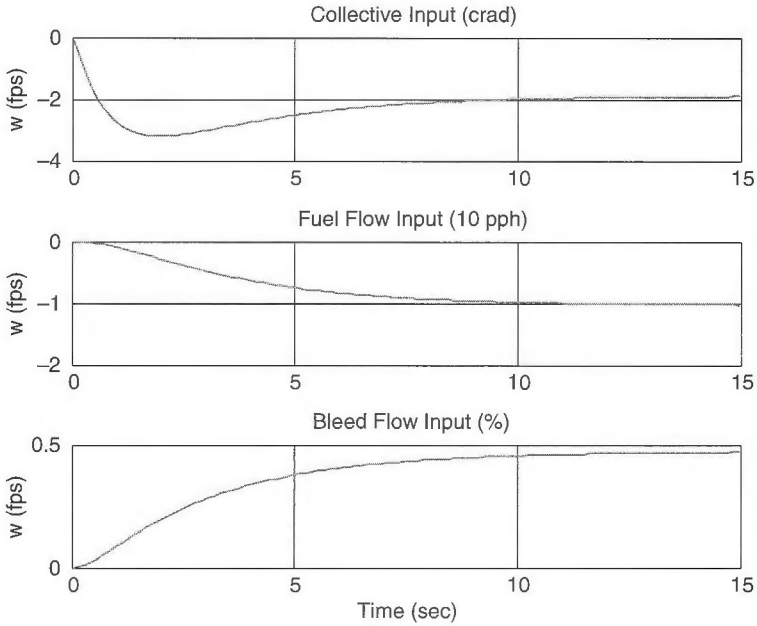


Fig. 9.17 Comparison of the vertical velocity with respect to three control inputs.

scaled down to represent the response of the rotor speed. Similarly, a comparison of the responses of the body vertical velocity w to the three input variables is plotted in Fig. 9.17. Note that a negative vertical velocity in the body axis means a positive climb rate.

9.4.3 Proportional Control Law Design

We can design a proportional control law for the integrated helicopter-engine system using a multivariable control design approach. Multivariable design provides the advantage of coordinating all three control inputs simultaneously (see Chapter 7). Design coordination is based on an objective function that considers both the performance and the control “energy” of the closed-loop control system in the optimization process. The optimization objective function can be a function of the state variable and the output variable. The resultant design is a compromise between better performance and lower control energy. One form of the objective function that balances between the effects of state and input variables is in a quadratic form and is expressed as

$$J = \frac{1}{2} \int_0^{\infty} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt \quad (9.11)$$

where $\mathbf{Q}$ is a weighting matrix for the state variable vector $\mathbf{x}$ and $\mathbf{R}$ is a weighting matrix for the input (or control) variable vector $\mathbf{u}$.

The design technique that uses the objective function in Eq. (9.11) is called the linear quadratic regulator (LQR). LQR is one of the most widely studied methods

in multivariable control design. Let's pick the state and the input weighting matrices Q and R to be simple diagonal matrices as

$$Q = \text{diag}([1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 10 \quad 1]) \quad (9.12)$$

$$R = \text{diag}(0.1 \quad 0.1 \quad 0.1) \quad (9.13)$$

The optimal state-variable feedback (SVFB) gain K_b can be calculated by using the Matlab's® "lqr" command in the Control System Design Toolbox as

$$K_b = \begin{bmatrix} -2.6004 & 9.2252 & 3.1281 & 0.4149 & -0.1781 & 0.3506 & 0.072 \\ -0.2576 & -0.1665 & -0.0986 & -0.7231 & -1.4738 & 4.8182 & 2.3297 \\ 0.3125 & 0.7161 & 0.1688 & 3.4334 & 2.5232 & -8.1145 & -0.5152 \end{bmatrix} \quad (9.14)$$

The closed-loop system dynamic matrix, after including the SVFB, is

$$A_{cl} = A - BK_b \quad (9.15)$$

The closed-loop responses of the climb rate and the turbine speed with respect to a step input in collective are shown in Fig. 9.18. Similarly, the closed-loop

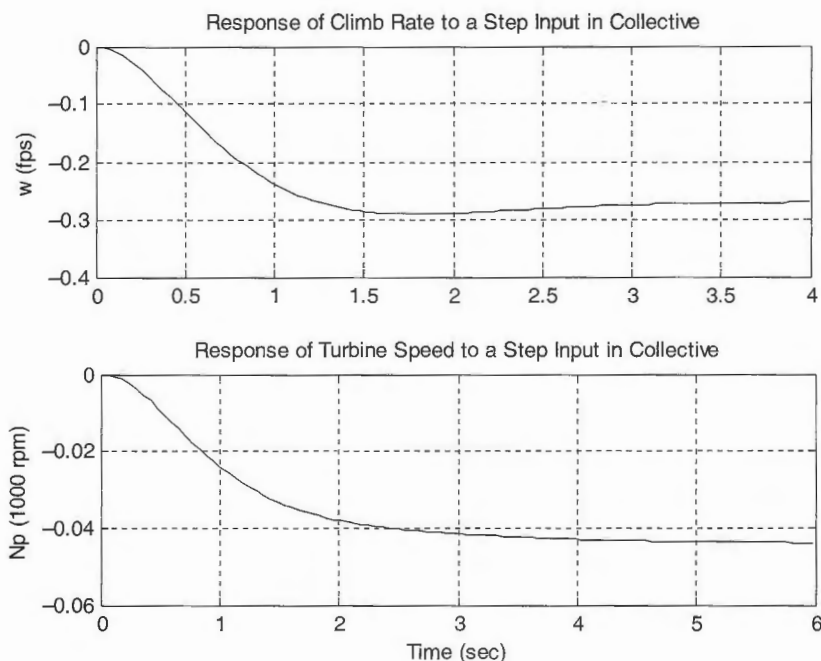


Fig. 9.18 Closed-loop responses of climb rate and turbine/rotor speed to a unit step input in collective.

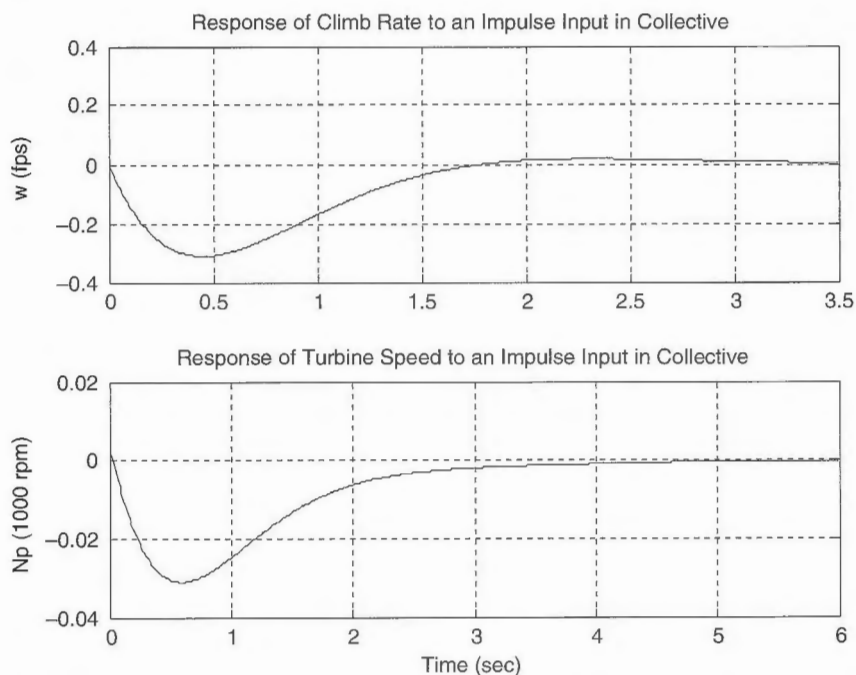


Fig. 9.19 Closed-loop responses of climb rate and turbine/rotor speed to a unit impulse input in collective.

responses with respect to a unit impulse input (analogous to a gust disturbance on the rotor) in collective are shown in Fig. 9.19.

The LQR design just described is merely an example to illustrate how to derive a control law for the helicopter-engine model in Eqs. (9.9) and (9.10). There can be many extensions to this example, for instance, tuning the closed-loop performance and responses by adjusting the weighting matrices, designing a closed-loop control law to track a nonzero set point of climb rate, and designing a control law by using other multivariable design techniques. Interested readers are referred to the textbooks that are specifically written about multivariable control design methods (e.g., [20, 21]).

Even without the turbine bleed control, a conventional control system can still benefit from the integrated approach to modeling the rotor and engine system as well as from MVC design techniques. An example of applying both the LQR and the Kalman filter techniques to the turboshaft engine control design problem is found in [22]. The combined LQR and KF design technique is called LQG synthesis (see Chapter 7). In an LQG synthesis, the KF acts like an estimator (see Chapter 8) for the state variables, which are used in the optimal feedback control law. The design problem is, again, to increase the closed-loop response of the rotor-turbine speed governor. The optimal governor includes a proportional control law and an integral control law. The proportional control is derived from the LQG design technique, and the integral control is a single integrator with the power turbine speed error (between

the commanded and the actual speeds) as its input. Furthermore, the optimal governor is augmented with an anti-alias filter to attenuate high-frequency noise. The results of this LQG synthesis showed that the bandwidth of the speed governor was increased to about 6 rad/s from 3 rad/s in the baseline, non-LQG design, and that the power turbine speed overshoot was reduced to about 0.5% from 6%.

References

- [1] Litt, J. S., Simon, D.L., Garg, S., Guo, T-H., Mercer, C., Millar, R., Behbahani, A., Bajwa, A., and Jensen, D., "A Survey of Intelligent Control and Health Management Technologies for Aircraft Propulsion Systems," NASA TM 2005-213622, ARL TR 3413, 2005; also published on *Journal of Aerospace Computing Information and Communication*, Vol. 1, AIAA, Dec. 2004.
- [2] Simon, D. L., Garg, S., Hunter, G. W., Guo, T-H., and Semega, K. J., "Sensor Needs for Control and Health Management of Intelligent Aircraft Engines," ASME Paper GT2004-54324, Aug. 2004; also NASA TM 2004-213202, ARL-TR-3251.
- [3] Buderath, M., "Effects of Diagnostic Capabilities on the Mission Capable Rate," *EADS Military Air Systems Technical Forum on Condition Based Maintenance*, Munich, Germany, Dec. 2008.
- [4] Jaw, L. C., Van, H. T., et al., "Model-based Approach to Validation and Verification of Flight Critical Software," *Proceedings of IEEE Aerospace Conference*, Track 11, Paper 407, MT, Mar. 2008.
- [5] AIAA Air Breathing Propulsion Technical Committee, "Versatile Affordable Advanced Turbine Engine (VAATE) Initiative—an AIAA Position Paper," Jan. 2006; www.aiaa.org/Participate/Uploads/VAATE.pdf.
- [6] Buderath, M., et al., "Effects of Diagnostic Capabilities on the Mission-Capable Rate," *EADS Military Air Systems Technical Forum on Condition Based Maintenance*, Munich, Germany, Dec. 2008.
- [7] Jaw, L. C., and Van, H. T., "Real-time Simulation of Actuator Prognostics Using a V&V Test Bench," 2006 Integrated System Health Management (ISHM) Conference, Paper, Aug. 2006.
- [8] Khalak, A., and Tierno, J., "Influence of Prognostic Health Management on Logistic Supply Chain," *Proceedings of the 2006 American Control Conference*, 1-4244-0210-7, IEEE, June 2006, pp. 3737–3742.
- [9] Jaw, L. C., Bryg, D., and Khalak, A., "Concept and Prototype of GCSS-AF Research Environment for Condition Based and Reliability Centered Maintenance," 2007 Integrated System Health Management (ISHM) Conference, Poster Session, Aug. 2007.
- [10] Guo, T-H., "A Roadmap for Aircraft Engine Life Extending Control," *Proceedings of the American Control Conference*, Vol. 5, June 2001, pp. 3702–3705.
- [11] Wu, D. N., and Jaw, L. C., "Intelligent Engine Life Usage Control (ILEC Program) Demonstration – Final Report," submitted to NASA Glenn Research Center under Contract NAS3-98112, Mar. 2002.
- [12] Guo, T-H, Chen, P., and Link, J., "Intelligent Life-Extending Controls for Aircraft Engines," *Proceedings of the First Intelligent Systems Conference*, AIAA paper 2004-6468, Sept. 2004.
- [13] Suarez, E. L., Duffy, M. J., Gamache, R. N., Morris, R., and Hess, A. J., "Jet Engine Life Prediction System Integrated with Prognostics Health Management,"

- Proceedings of the 2004 IEEE Aerospace Conference*, Vol. 6, IEEE, March 2004, pp. 3596–3602.
- [14] Simon, D. L., “An Overview of the NASA Aviation Safety Program Propulsion Health Monitoring Element,” NASA TM 2000-210348; also AIAA Paper 2000-3624, July 2000.
 - [15] Lutz, T., and Greeves, B., “Throttle Only Control (TOC)—10 Steps to a Survivable Landing Following Loss of Normal Flight Control,” INTERpilot, International Federation of Air Line Pilots’ Associations (IFALPA), 2004, pp. 32–34.
 - [16] Burcham, F. W., Jr., Ray, R. J., Conners, T. R., and Walsh, K. R., “Propulsion Flight Research at NASA Dryden from 1967 to 1997,” NASA TP-1998-206554, 1998; also AIAA Paper 1998-3712, 1998.
 - [17] Jaw, L. C., Van, H. T., and Mink G. R., “Simulation of an Intelligent Engine Control System for Aircraft under Adverse Conditions,” AIAA Paper 2007-2810, May 2007.
 - [18] Jaw, L. C., “Method for Rapidly Changing the Power Output of a Turbine Engine,” U.S. Patent, #5,280,702, 25 Jan. 1994.
 - [19] Jaw, L. C., “Control of Helicopter Engines in Low Altitude Flight,” Ph. D. Dissertation, Stanford Univ., Stanford, CA, Aug. 1991, Appendix A.
 - [20] Bryson, A. E., Jr., *Control of Spacecraft and Aircraft*, Princeton Univ. Press, Princeton, NJ, 1994.
 - [21] Kwakernaak, H., and Sivan, R., *Linear Optimal Control Systems*, Wiley, New York, 1972.
 - [22] Pfeil, W. H., de los Reyes, G., and Bobula, G. A., “The Application of LQR Synthesis Techniques to the Turboshift Engine Control Problem,” *20th Joint Propulsion Conference*, AIAA Paper 1984-1455, June 1984.

Appendix A

Fundamentals of Automatic Control Systems

Some fundamentals of automatic control systems are reviewed in this appendix. Although this book is not an introduction to control systems, the contents of this appendix can provide a foundation for the readers of all levels to understand better the terminologies and symbols used in the book.

This foundational material will be presented in such a way that different pockets of knowledge are consolidated into a coherent understanding about the subject of automatic control. This approach is believed to reflect the “thinking” of a control engineer as he/she tries to solve real-world control problems. This kind of thinking takes a multidimensional view of the solution space, regardless if it is in the frequency domain or the time domain, or if it is a classical control or a modern control technique. The material can be found in most introductory textbooks on control system analysis and design, and some of these books are referenced at the end of the appendix.

A brief explanation and comparison between open-loop and closed-loop controls is given first. Then the most commonly used mathematical representations for a linear dynamic system (in both the time domain and the frequency domain) is discussed. We focus on linear dynamic systems because of two reasons: 1) this class of systems can be modeled and analyzed by the basic theories of calculus and algebra, and 2) there are well-established control design methods for these systems.

In Sec. A.3, three basic control laws, namely, the proportional, the integral, and the derivative controls, are described. This is followed by a quick review of the performance and the stability of linear systems in Sec. A.4. Then in Sec. A.5, two most commonly used classical control design methods, the root-locus method and the frequency response method, are described. Lastly, some basic nonlinear systems and their effects are discussed in Sec. A.6.

A.1 Open-Loop and Closed-Loop Controls

Automatic control systems fall into two types: *open-loop* control and *closed-loop* (or feedback) control. The basic difference between these two types is whether any output variable of the system is used to determine new control actions. In a closed-loop control system, at least one output variable is measured, and its value is fed back to the controller to calculate an updated control actions. A closed-loop control system consists of four primary components: 1) the physical system to be controlled (also called the process or the plant), 2) the controller, 3) the actuator,

and 4) the sensor. Figure 1.1 of Chapter 1 shows the block diagrams of these components in an open-loop as well as a closed-loop sense. Because the open-loop control system does not use process output to update the control action, hence the feedback line from the output block to the controller block is absent. The term automatic control system is commonly used in place of the closed-loop control system.

A closed-loop control system provides three basic *functions*:

1) *Set-point control*: This function is responsible for keeping the physical system near a stable, controlled operating condition over time. Set-point control applies to the time-varying reference command, if the rate of change of the reference command is slow compared to the system dynamics to be controlled. This type of control is called tracking. Set-point control is also called steady-state control or a regulator.

2) *Transient control*: This function is responsible for changing the system between two operating conditions from one to another while satisfying a desired response.

3) *Limit protection control*: This function is responsible for protecting the system from an unsafe or undesirable operating regime like compressor surge or turbine overtemperature.

The advantages and the disadvantages of closed-loop vs open-loop controls are compared in Table A.1. The advantages of closed-loop control far out-weigh its disadvantages.

Examples of open-loop and closed-loop control systems are abundant, such as a toaster oven, an electric motor, an elevator, a car cruise control, a water level float regulator, a robotic manipulator, etc. [1, 2]. The benefit of closed-loop control can be demonstrated easily as follows: Ask a person to hold a ball and stand in front of you by about 10 ft away. Then ask the person to close both eyes, turn 360 deg twice, and throw the ball at you. The most likely outcome of this demonstration is that the ball will not reach you because the sense of direction is lost without seeing

Table A.1 Comparison between closed-loop and open-loop control systems

Advantages of closed-loop control	Disadvantages of closed-loop control
Performance	—
Near zero steady-state tracking error	
Tighter tracking or regulation of desired performance	
Robustness	Stability
Less sensitive to system variations	Instability can happen with ill-designed control law
Less sensitive to disturbances	
Expanded region of stable operations	
Safety	Cost
Increased reliability	More expensive to implement initially because of the costs associated with sensing and controller design
Improved maintainability	

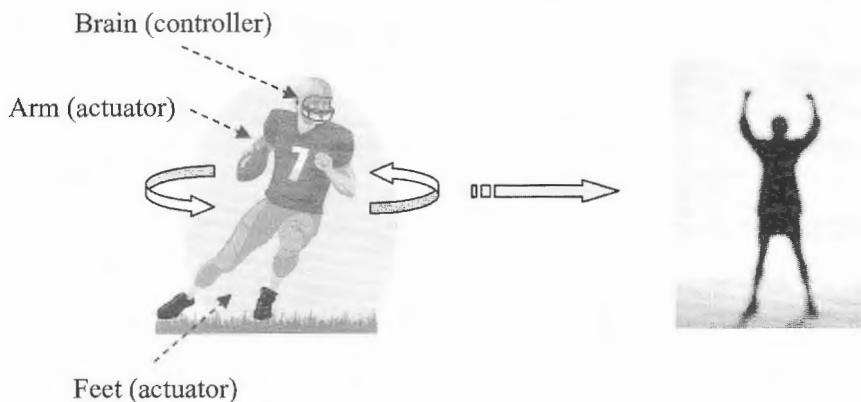


Fig. A.1 Simple demonstration of open- and closed-loop control systems.

the target. Seeing the target is the sensory feedback in this human closed-loop control system. Rotating the body until the target is lined up is a control action that combines the controller (brain) and the actuator (feet). Throwing the ball at the target is also a control action. Suppose a second demonstration is conducted with the person's eyes open, you will notice how quickly and accurately the ball will reach you. Figure A.1 illustrates this human control system demonstration.

A.2 Representations of Linear Dynamic Systems

Almost all real-world dynamic systems are nonlinear, so why are we studying linear systems? The reason is two-fold: 1) linear systems have analytical solutions, whereas only a few of the special types of nonlinear systems have such kind of solutions; and 2) most control design methods and computer-aided design tools work only for linear systems. Therefore, our goal as a control engineer is to 1) find proper linear approximations of the nonlinear system that we are trying to control, 2) apply linear control design techniques to the approximate linear systems, and then 3) adjust the control design by considering the effects of nonlinearity and uncertainties. This is the design process for any automatic control system.

The greatest advantage of working with linear systems is that they satisfy the Principle of Superposition by possessing the *additivity* and the *homogeneity* properties [3]. This means that a linear combination of a system's inputs results in a linear combination of the system's outputs for those individual inputs.

A linear dynamic system can be expressed in the state equation and the output equation form as follows:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (\text{A.1})$$

$$y(t) = C(t)x(t) + D(t)u(t) \quad (\text{A.2})$$

where x , u , and y stand for the state, the input, and the output variables, respectively; where $A(t)$ and $B(t)$ are the dynamic and the input-distribution matrices for

the linear state equation, respectively; and where $C(t)$ and $D(t)$ are the state-to-output distribution and the input-to-output distribution matrices for the linear output equation, respectively.

The same symbols x , u , and y are used for both *scalar* and *vector* forms. A scalar form has only one variable, whereas a vector form has at least two variables. For example, the number of state variables in a turbine engine can exceed 20. In the vector form, we denote the number of state variables as n , and it is also the *order* of the system. Similarly, we denote the number of input variables as m and the number of output variables as l . Hence, x is an $n \times 1$ vector, u is an $m \times 1$ vector, and y is an $l \times 1$ vector. Furthermore, the dimensions of the matrices A , B , C , and D are $n \times n$, $n \times m$, $l \times n$, and $l \times m$, respectively.

The order of the system is n , regardless of the dimension of u or y . When both u and y are vector variables (i.e., more than one element in each vector), the system described by Eqs. (A.1) and (A.2) is called a *multi-input, multi-output* (MIMO) system. For MIMO systems, the matrix elements in Eqs. (A.1) and (A.2) are the *parameters* of the system (model). Because these parameters are functions of time, Eqs. (A.1) and (A.2) represent the *linear, time-varying* model of a system.

A special case of the MIMO system is when both the input u and the output y are scalar variables, and then the system is called a *single-input, single-output* (SISO) system, and it is represented by the following equations (where d is a scalar):

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (\text{A.1a})$$

$$y(t) = C(t)x(t) + d(t)u(t) \quad (\text{A.2a})$$

Furthermore, a special case of the SISO system is when the state x is also a scalar variable, and then the system is called a *first-order* system, and Eqs. (A.1) and (A.2) can be further simplified to

$$\dot{x}(t) = a(t)x(t) + b(t)u(t) \quad (\text{A.1b})$$

$$y(t) = c(t)x(t) + d(t)u(t) \quad (\text{A.2b})$$

where all parameters a , b , c , and d are scalars.

When the model parameters in Eqs. (A.1) and (A.2) are not functions of time, or when they can be approximated as constants during a finite time interval of interest, the equations can be further simplified to represent a *linear, time-invariant* (LTI) model as

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (\text{A.3})$$

$$y(t) = Cx(t) + Du(t) \quad (\text{A.4})$$

where all of the matrix elements a_{ij} , b_{ik} , c_{lj} , and d_{lk} are constant values. The LTI model is the most widely used form for control system designs. We sometimes see a simplified form of the output equation where the feedthrough (or feedforward)

term in the output equation is removed to focus only on the *dynamics* of the system, that is,

$$y = Cx \quad (\text{A.5})$$

Equations (A.3) and (A.4) are the most frequently used *time-domain* representation of a linear time-invariant system. An alternative representation of the linear system is in the *frequency domain*, and is more commonly known as the transfer function (TF). We will express the linear system in the frequency domain in the next subsection.

A.2.1 Frequency-Domain Representation of a Control System

TF describes the relationship between a pair of input and output variables in the frequency domain. It is defined as the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable [4]. We can use a TF to represent the input-output relationship of any LTI system. An example is the open-loop control system shown in Fig. A.2, where the input-output relationships of the four components of the control system, namely, the command shaping block, the controller block, the actuator block, and the process block, have been expressed by TFs $F(s)$, $K(s)$, $A(s)$, and $G(s)$, respectively.

These individual TFs are defined as

$$F(s) \equiv \frac{C(s)}{R(s)}, \quad K(s) \equiv \frac{U(s)}{C(s)}, \quad A(s) \equiv \frac{U_a(s)}{U(s)}, \quad G(s) \equiv \frac{Y(s)}{U_a(s)} \quad (\text{A.6})$$

where r is the reference input variable, c is the command variable, and u_a is the actuated control variable (i.e., the actuator output). This chain of relationships is what we have called an open-loop control system (from reference input to the process output without feedback). An important thing to remember about the TF is that it applies only to the specific pair of input and output variables. A different input-output pair will result in a different transfer function.

By feeding back the output variable to the command variable, we get a closed-loop control system, and we show this closed-loop system in Fig. A.3, where the “sensed” output variable z is subtracted from the command and e is the error of this feedback, so

$$E(s) = C(s) - Z(s), \quad H(s) \equiv \frac{Z(s)}{Y(s)} \quad (\text{A.7})$$

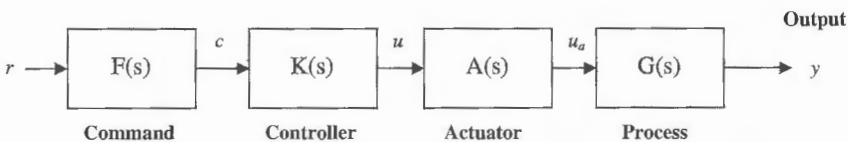


Fig. A.2 Block diagram of an open-loop control system in transfer-function form.

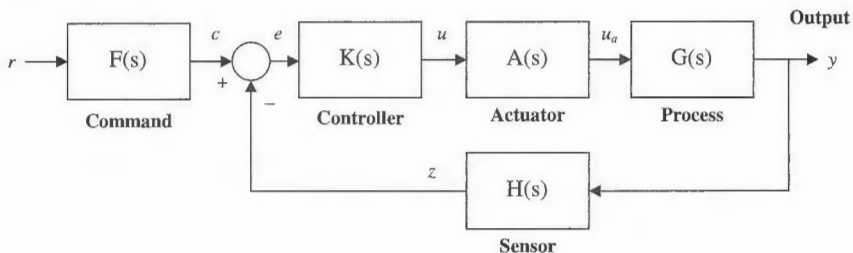


Fig. A.3 Block diagram of a closed-loop control system in TF form.

The TF for the open-loop system (shown in Fig. A.2) can be derived by substituting the output of a component by the input of its successive component in the block diagram. After doing so, we get

$$\frac{Y(s)}{R(s)} = G(s)A(s)K(s)F(s) \quad (\text{A.8})$$

Similarly, the TF for the closed-loop system is given by

$$\frac{Y(s)}{R(s)} = \frac{G(s)A(s)K(s)F(s)}{1 + H(s)G(s)A(s)K(s)} \quad (\text{A.9})$$

In a simpler form, the closed-loop control system can be represented by just three blocks and a comparator as shown in Fig. A.4. Note that, in this simplified form, we have included the actuator transfer function in the process transfer function $G(s)$, and we have removed the command shaping filter $F(s)$.

Note that the TF of the process (or the system) to be controlled $G(s)$ is the frequency-domain counterpart of the state and the output equations of Eqs. (A.3) and (A.4), respectively. This TF defines the relationship between *one* input variable $u(t)$ to *one* output variable $y(t)$, so it represents a SISO system. To represent a MIMO system, we must define a TF for each and every possible input-output combination. Hence, for a system with m input variables and l output variables, we replace the TF $G(s)$ in the block diagram with a *TF matrix* containing $l \times m$ elements of individual TFs, that is, the matrix of individual TFs contains $\{Y_i(s)/U_j(s); i = 1, \dots, l; j = 1, \dots, m\}$.

Based on the simplified closed-loop system diagram in Fig. A.4, we can write the closed-loop transfer function (CLTF) as

$$\text{CLTF} = \frac{Y(s)}{R(s)} = \frac{G(s)K(s)}{1 + H(s)G(s)K(s)} \quad (\text{A.10})$$

Note that the denominator of Eq. (A.10) is the characteristic polynomial of the closed-loop TF. By equating this polynomial to zero, we get the characteristic

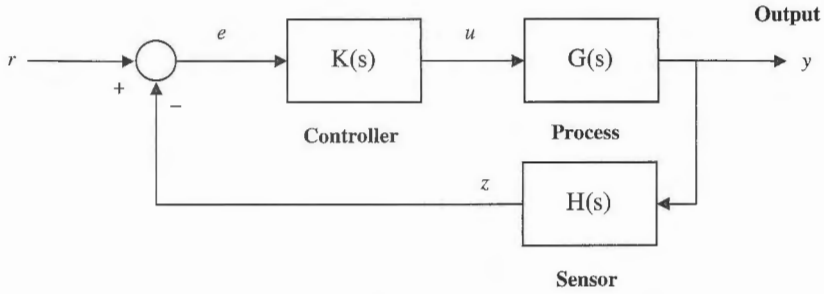


Fig. A.4 Block diagram of the simplest closed-loop control system.

equation of the closed-loop system. The roots of the characteristic equation are called the *poles* of the closed-loop system. These poles provide the most important insight into the stability characteristics of the system. The number of poles is equal to the highest power of the denominator polynomial; hence, it is the *order* of the system (or TF). Similarly, the roots of the numerator polynomial of Eq. (A.10) are called the *zeros* of the closed-loop system. In fact, the concept of poles and zeros is valid for any TF, not just for the CLTF just shown.

In addition to the CLTF, we can define two other *loop* transfer functions that are used frequently in control system design. The first one is the open-loop transfer function (OLTF), which is the TF from e to z (see Fig. A.4) and is expressed in Eq. (A.11). The second is the forward-loop transfer function (FLTF), which is the TF from e to y and is expressed in Eq. (A.12).

$$\text{OLTF} = \frac{Z(s)}{E(s)} = H(s)G(s)K(s) \quad (\text{A.11})$$

$$\text{FLTF} = \frac{Y(s)}{E(s)} = G(s)K(s) \quad (\text{A.12})$$

Because TFs constitute the essential part of our understanding of the classical control theory, we should spend more effort analyzing them. In general, a TF is a rational function of two polynomials: the numerator polynomial, designated by $N(s)$, and the denominator polynomial, designated by $D(s)$; both polynomials have real coefficients. Let us assume that the order of the numerator polynomial is m and the order of the denominator polynomial is n , which is also the *order* of the LTI system represented by the TF; then the *polynomial form* of the TF, or the polynomial form of the frequency-domain representation, can be expressed as

$$\frac{N(s)}{D(s)} = \frac{k(s^m + b_1s^{m-1} + \cdots + b_{m-1}s + b_m)}{s^n + a_1s^{n-1} + \cdots + a_{n-1}s + a_n} \quad (\text{A.13})$$

where k is a constant and called the *loop gain*, and $m \leq n$.

Factoring Eq. (A.13) into individual roots for both the numerator and the denominator polynomials, we get the *pole-zero form* of the TF

$$\frac{N(s)}{D(s)} = \frac{k(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)} = \frac{k \prod_{j=1}^m (s - z_j)}{\prod_{i=1}^n (s - p_i)} \quad (\text{A.14})$$

where p_i are the poles of the TF and z_j are the zeros of the TF. Note that Eq. (A.14) applies to any TF, whether it is OLTF or CLTF.

The poles and the zeros of a TF are traditionally represented by the “x” and the “o” symbols on the s plane, respectively. Figure A.5 illustrates the pole-zero configuration of a hypothetical TF. Because the TF has real-valued coefficients, all of the poles and zeros that are not located on the real axis must appear in complex conjugate pairs as illustrated in the figure. Note that the poles and zeros can be on either side of the imaginary axis depending on the sign of the real part. We will see later that the location of a pole relative to the imaginary axis determines the stability characteristic of the TF (or process or system).

Equation (A.14) can be expanded into a sum of partial fractions. If the poles of the TF are all distinct, then the number of fractions is equal to n , and the *partial-fraction form* (or *residue-pole form*) of the TF is

$$\frac{N(s)}{D(s)} = d_0 + \frac{c_1}{s - p_1} + \frac{c_2}{s - p_2} + \cdots + \frac{c_n}{s - p_n} \quad (\text{A.15})$$

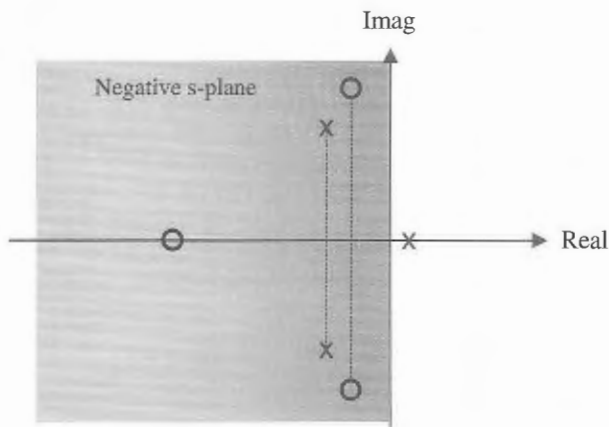


Fig. A.5 Graphical representations of poles (x) and zeros (o).

where c_i are the constant coefficients and are sometimes called the residues of the TF. If some of the poles of the TF are repeated, then the partial-fraction form becomes

$$\begin{aligned} \frac{N(s)}{D(s)} = & d_0 + \frac{c_1}{s-p_1} + \frac{c_2}{s-p_2} + \cdots + \left[\frac{c_{lj}}{(s-p_l)^j} + \frac{c_{l(j-1)}}{(s-p_l)^{j-1}} + \cdots + \frac{c_{l1}}{s-p_l} \right] \\ & + \cdots + \frac{c_{(n-j)}}{s-p_{(n-j)}} \end{aligned} \quad (\text{A.16})$$

where p_l is the multiple pole, j is the number of times the multiple pole is repeated; and for multiple complex conjugate poles, j is an even number. Interested readers are referred to Appendix A of [5] for more discussions on linear system representations.

A.2.2 Time-Domain Representation of a Control System

The state and the output equations of the entire closed-loop control system can be derived by combining the individual state and output equations of various components of the control system. To illustrate this concept and to prevent being confused by the complexity of the algebra, we will derive the overall system equations for the simplified closed-loop system shown in Fig. A.4.

First, we write down the state and the output equations for each of the three control components in the figure. The equations for the physical system under control (or the process) can be expressed by

$$\dot{x}_p = A_p x_p + B_p u \quad (\text{A.17})$$

$$y = C_p x_p + D_p u \quad (\text{A.18})$$

where subscript p stands for the process.

Similarly, the equations for the controller and the sensor components are

$$\dot{x}_c = A_c x_c + B_c e \quad (\text{A.19})$$

$$u = C_c x_c + D_c e \quad (\text{A.20})$$

and

$$\dot{x}_s = A_s x_s + B_s y \quad (\text{A.21})$$

$$z = C_s x_s + D_s y \quad (\text{A.22})$$

where subscript c stands for the controller and subscript s stands for the sensor.

Combining the preceding three state equations and introducing the relationship between e and z , that is, $e = r - z$, we get an augmented state equation for the entire closed-loop control system as

$$\begin{bmatrix} \dot{x}_p \\ \dot{x}_c \\ \dot{x}_s \end{bmatrix} = \begin{bmatrix} A_p - B_p D_c Q D_s C_p & B_p C_c - B_p D_c Q D_s D_p C_c & -B_p D_c Q C_s \\ -B_c Q D_s C_p & A_c - B_c Q D_s D_p C_c & -B_c Q C_s \\ B_s C_p - B_s D_p D_c Q D_s C_p & B_s D_p C_c - B_s D_p D_c Q D_s D_p C_c & A_s - B_s D_p D_c Q C_s \end{bmatrix} \begin{bmatrix} x_p \\ x_c \\ x_s \end{bmatrix} + \begin{bmatrix} B_p D_c Q \\ B_c Q \\ B_s D_p D_c Q \end{bmatrix} r \quad (\text{A.23})$$

where $Q \equiv (1 + D_s D_p D_c)^{-1}$. Similarly, the output equation is given by

$$y = \begin{bmatrix} C_p - D_p D_c Q D_s C_p & D_p C_c - D_p D_c Q D_s D_p C_c & -D_p D_c Q C_s \end{bmatrix} \begin{bmatrix} x_p \\ x_c \\ x_s \end{bmatrix} + D_p D_c Q r \quad (\text{A.24})$$

Although Eqs. (A.23) and (A.24) look complex and intimidating, they can be computed easily by a standard linear algebra or control design software tool. Putting the equations in a shorthand notation, we get

$$\dot{x}_A = A_A x_A + B_A r \quad (\text{A.25})$$

$$y = C_A x_A + D_A r \quad (\text{A.26})$$

where subscript A stands for “augmented” variables in the state and output equations for the closed-loop control system. If we assume that the orders of the controller, the process, and the sensor are n_c , n_p , n_s , respectively, then the order of the combined closed-loop system of Eq. (A.25) is $n_c + n_p + n_s$. Equations (A.25) and (A.26) can be represented in block diagram form as illustrated in Fig. A.6.

Taking the Laplace transform of both Eqs. (A.25) and (A.26) and rearranging the terms, we arrive at an important relationship between the time-domain representation and the frequency-domain representation of the same system, that is, the Laplace transform of the output variable $y(t)$ is related to the Laplace transform of the input variable $r(t)$ according to the following equation:

$$Y(s) = [C_A(sI - A_A)^{-1}B_A + D_A]R(s) \quad (\text{A.27})$$

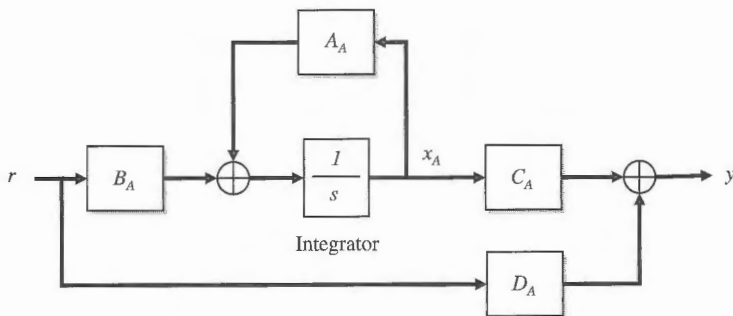


Fig. A.6 Block diagram of a control system in time-domain representation.

In the case of a SISO system, D_A is a scalar (having a dimension of 1×1), B_A is a column vector, C_A is a row vector, and Eq. (A.27) represents the classical TF. Note that Eq. (A.27) is valid for any LTI system defined by Eqs. (A.3) and (A.4). For simplicity, we will drop the subscript A in the remainder of this appendix as we continue to review the time-domain representation of linear systems.

The roots of the characteristic equation of the TF of Eq. (A.27), that is, poles of the denominator polynomial of the closed-loop system, are the *eigenvalues* of the dynamic matrix A (or A_A). Just as the poles are important to the characteristic equation of a TF, the eigenvalues are important to any square matrix A . We can decompose any square matrix of order n that has distinctive eigenvalues λ_i into the following form [6]:

$$A = \sum_i \lambda_i p_i q_i^T = P \Lambda Q \quad (\text{A.28})$$

where $\Lambda = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ is the diagonal matrix containing all of the eigenvalues of A ; P is a matrix that contains n number of *column* (or *right*) eigenvectors p_i such that $A p_i = \lambda_i p_i$; and Q is the matrix inverse of P and contains n number of the *row* (or *left*) eigenvectors q_i^T , that is,

$$A p_i = \lambda_i p_i, \quad P = [p_1, p_2, \dots, p_n], \quad P^{-1} = Q = \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix} \quad (\text{A.29})$$

Substituting Eq. (A.28) into Eq. (A.3) and multiplying both sides of the equality sign by P^{-1} on the left-hand side of the matrix terms, we get

$$Q \dot{x}(t) = \Lambda Q x(t) + Q B u(t) \quad (\text{A.30})$$

Now let us define a new state variable $x_m \equiv Qx$, called the *modal state variable*, then Eq. (A.30) can be written as

$$\dot{x}_m(t) = \Lambda x_m(t) + B_m u(t) \quad (\text{A.31})$$

where $B_m \equiv QB$. Equation (A.31) is the *diagonal form* of the state equation. Similarly, we can change the state variable in the output equation of (A.4) and get the corresponding modal output as

$$y(t) = CPx_m(t) + Du(t) \quad (\text{A.32})$$

The preceding discussion leads into the concept of *state-space realization* of a system. A realization is a set of state and output equations that represent the system in the time domain, hence, Eqs. (A.3) and (A.4) are one form of the realization, whereas Eqs. (A.31) and (A.32) are another form of the realization for the same system. In fact, a linear dynamic system can have many more realizations, and any two of these realizations are related by a matrix transformation called *similarity transformation*, which is represented by T .

Suppose T is an invertible matrix of order n ; a new realization (or similarity transformation) of Eqs. (A.3) and (A.4) is

$$\dot{x}_s(t) = (T^{-1}AT)x_s(t) + (T^{-1}B)u(t) \quad (\text{A.33})$$

$$y(t) = (CT)x_s(t) + Du(t) \quad (\text{A.34})$$

where we have transformed the original state variable to the new state variable $x_s \equiv T^{-1}x$.

Because the realization of a system in the time-domain is not unique, we are interested in defining some standards for comparing and analyzing different realizations of the same system. These standards are called the *canonical forms* of state-space realizations.

In addition to the diagonal form of realization just described, other frequently used canonical forms are the *controllability* (or the controllable) form, the *observability* (or the observable) form, the *controller* form, and the *observer* form. Also, there is the *Jordan form* to express the dynamic matrix A in a block diagonal form for a system with repeated eigenvalues. Interested readers are referred to textbooks on the analysis of linear systems.

Before we move on to the next subject, we are reminded of a fact about the state-space representation: even though a system can have multiple time-domain realizations, it can only have one TF (or frequency-domain realization).

A.3 Basic Control Laws

Controllers are traditionally called *control laws*. So far we have not discussed the detailed structure of the controller $K(s)$, and have only represented it as a TF

in the frequency-domain and as a matrix equation in the time-domain. From Eq. (A.13), a control law TF can be represented in the polynomial form as

$$\frac{U(s)}{E(s)} \equiv K(s) = \frac{K_c(s^m + b_1s^{m-1} + \cdots + b_{m-1}s + b_m)}{s^n + a_1s^{n-1} + \cdots + a_{n-1}s + a_n} \quad (\text{A.35})$$

where K_c is the controller gain. The same control law can be represented in the state equation form as in Eqs. (A.19) and (A.20). Because both representations are for the identical control law, we can write

$$\frac{K_c(s^m + b_1s^{m-1} + \cdots + b_{m-1}s + b_m)}{s^n + a_1s^{n-1} + \cdots + a_{n-1}s + a_n} = C_c(sI - A_c)^{-1}B_c + D_c \quad (\text{A.36})$$

which relates the frequency-domain control law to its time-domain counterpart. We will now use this equation to explain the three basic control laws used in automatic control systems.

A.3.1 Proportional Control Law

A proportional control law causes the output of the controller, that is, the computed control variable u , to be linearly proportional to the input of the controller, that is, the feedback error e . The proportional control law is expressed in the frequency domain as

$$K_p(s) = K_p \quad (\text{A.37})$$

where K_p is the same as the *feedthrough* term D_c on the right-hand side of Eq. (A.36) for a SISO system, with A_c , B_c , and C_c being all zero. The equivalent time-domain representation of the proportional control law is

$$u(t) = K_p e(t) \quad (\text{A.38})$$

A.3.2 Integral Control Law

An integral control law integrates the feedback error over time and keeps increasing the value of the control variable to eliminate the error. The integral control law is expressed in the frequency domain as

$$K_i(s) = \frac{K_i}{s} \quad (\text{A.39})$$

where K_i is the *integral gain* and is the product of C_c and B_c in Eq. (A.36) for a SISO system, with A_c and D_c being zero. The equivalent time-domain representation is

$$\dot{u}(t) = K_i e(t) \quad (\text{A.40})$$

Putting Eq. (A.40) in the state equation form, we get

$$\dot{v} = K_i e \quad (\text{A.41})$$

$$u = v \quad (\text{A.42})$$

where we have changed the variable u to the variable v to denote the state variable of the controller and retained u as the output variable of the controller.

A.3.3 Derivative Control Law

A derivative control law causes the computed control variable to be the first-order derivative of the feedback error with respect to time. This control law is able to anticipate “future” value of the feedback error, based on the rate of change of the current error, so as to increase or decrease the control variable before the actual error appears. The derivative control law is expressed in the frequency domain as

$$K_d(s) = K_d s \quad (\text{A.43})$$

The time-domain representation of this control law is

$$u = K_d \dot{e} \quad (\text{A.44})$$

and the state equation form of this controller is given by

$$\dot{e} = w (= \dot{r} - \dot{z}) \quad (\text{A.45})$$

$$u = K_d w \quad (\text{A.46})$$

where w is calculated as the difference between the rate of change of the reference command $\dot{r}$ and the rate of change of the sensed output $\dot{z}$.

A.3.4 Proportional-Integral-Derivative Control Law

A proportional-integral-derivative (PID) control law combines the preceding three individual control laws to provide a general-purpose control law for most systems under all operating conditions. The PID control law is expressed in the frequency domain as

$$K_{\text{PID}}(s) = K_p + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_p s + K_i}{s} \quad (\text{A.47})$$

We can express the PID control law in the state equation form as

$$\dot{v} = K_i e \quad (\text{A.41a})$$

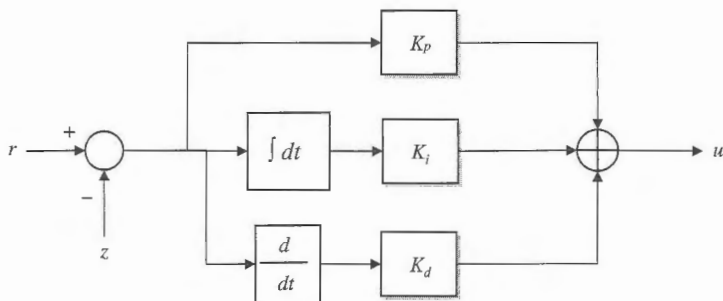


Fig. A.7 Block diagram of the PID control law.

$$\dot{e} = w \quad (\text{A.45a})$$

$$u = \begin{bmatrix} 1 & K_p \end{bmatrix} \begin{bmatrix} v \\ e \end{bmatrix} + K_d w \quad (\text{A.48})$$

This PID control law is illustrated graphically in Fig. A.7.

A.4 Analysis of Linear Dynamic Systems

We are interested in the general solution of the state equation of an LTI system. Because the solution is applicable to all linear systems, we can use it to analyze the properties and performance of any LTI system. Specifically, we are interested in the following characteristics of a system: controllability, observability, steady-state accuracy, transient response, and stability.

A.4.1 Solution of the State Equation

The state equation of an LTI system is expressed in Eq. (A.3). The solution of this first-order, ordinary, *matrix* differential equation is similar to that of a scalar differential equation. The (unique) solution consists of two parts: 1) a homogeneous solution that is an exponential function of time and 2) a particular solution that satisfies the entire state equation (with the nonhomogeneous input term). The solution of Eq. (A.3) is given by [3]

$$x(t) = e^{A(t-t_0)} x(t_0) + \int_{t_0}^t e^{A(t-\tau)} B u(\tau) d\tau \quad (\text{A.49})$$

where e^{At} is a $n \times n$ matrix, and we have assumed that the initial time is t_0 .

A special case of the preceding solution is when the system is a first-order, SISO system, and then Eq. (A.49) can be simplified to

$$x(t) = e^{a(t-t_0)} x(t_0) + \int_{t_0}^t e^{a(t-\tau)} b \cdot u(\tau) d\tau \quad (\text{A.49a})$$

where a is a scalar number and is the eigenvalue (or the pole) of the first-order system.

An alternative expression for Eq. (A.49) is

$$\phi(t; t_0, x_0, u) = e^{A(t-t_0)} x(t_0) + \int_{t_0}^t e^{A(t-\tau)} B u(\tau) d\tau \quad (\text{A.50})$$

where $x_0 \equiv x(t_0)$, and $\phi(t; t_0, x_0, u) \equiv x(t)$ is called the *state transition matrix*, which is a clearer notation than $x(t)$ in Eq. (A.49).

A more compact form of Eq. (A.50) can be derived if we define a more *restrictive* state transition matrix $\phi(t, t_0)$, which is the solution of the homogeneous part of the state equation (A.3), that is,

$$\frac{\partial}{\partial t} \phi(t, \tau) = A \phi(t, \tau), \quad \text{with } \phi(\tau, \tau) = I \quad (\text{A.51})$$

and Eq. (A.50) can be written as

$$\phi(t; t_0, x_0, u) = \phi(t, t_0) x(t_0) + \int_{t_0}^t \phi(t, \tau) B u(\tau) d\tau \quad (\text{A.52})$$

This equation can also be written as

$$\phi(t; t_0, x_0, u) = \phi(t; t_0, x_0, 0) + \phi(t; t_0, 0, u) \quad (\text{A.53})$$

This means that the solution of a linear state equation consists of two parts: one is a zero-input ($u = 0$) solution, and the other is a zero-state ($x_0 = 0$) solution. This property holds for both the time-varying and the time-invariant linear systems.

If we substitute Eq. (A.49) into Eq. (A.4), we get the output time history (or the output response) to an input function $u(t)$ as

$$y(t) = C \left[\phi(t, t_0) x(t_0) + \int_{t_0}^t \phi(t, \tau) B u(\tau) d\tau \right] + D u(t) \quad (\text{A.54})$$

or

$$y(t) = C \phi(t; t_0, x_0, u) + D u(t) \quad (\text{A.55})$$

Now let us look at the output response of the system for the case $x_0 = 0$; Eq. (A.54) reduces to

$$y(t) = C \int_{t_0}^t \phi(t, \tau) B u(\tau) d\tau + D u(t) = \int_{t_0}^t [C \phi(t, \tau) B + D \delta(t - \tau)] u(\tau) d\tau \quad (\text{A.56})$$

where the matrix inside the bracket, that is, $C\phi(t, \tau)B + D\delta(t - \tau)$, is called the *impulse response matrix*, which is the output response of the linear system at time t to a unit delta function at time τ , or $\delta(t - \tau)$; the second term of Eq. (A.56) is also called a *convolution integral*.

The impulse response matrix is analogous to the impulse response of a first-order SISO system, which is expressed as $g(t, \tau)$, and the output response of the system is

$$y(t) = \int_{-\infty}^{\infty} g(t, \tau)u(\tau)d\tau \quad (\text{A.57})$$

Note that Eqs. (A.49–A.57) for the solution of an LTI system are also applicable to the solution of a time-varying linear system [defined by Eqs. (A.1) and (A.2)]. When we apply these equations to time-varying systems, we need to remember that all of the matrices A , B , C , and D in these equations are functions of time.

Because for an LTI system, $g(t, \tau) = g(t - \tau, 0)$, Eq. (A.57) can be written, by shifting the timescale and changing the variables, as

$$y(t) = \int_{-\infty}^{\infty} g(t - \tau, 0)u(\tau)d\tau = - \int_{-\infty}^{\infty} g(\zeta, 0)u(t - \zeta)d\zeta \quad (\text{A.58})$$

We are familiar with this equation from the classical control theory because it can be obtained by solving the inverse Laplace transform of the TF of the system $G(s)$ as follows:

$$y(t) = L^{-1}\{G(s)\} \quad (\text{A.59})$$

where $L^{-1}\{\cdot\}$ stands for inverse Laplace transform. This leads naturally to the alternative approach to finding the solution of an LTI system by using the Laplace transform method.

If we substitute Eq. (A.28) into Eq. (A.50) for A , then the solution of the state equation (A.3) becomes

$$\phi(t; t_0, x_0, u) = \sum_i^n p_i \Lambda q_i^T e^{At} x(t_0) + \int_0^t e^{A(t-\tau)} Bu(\tau)d\tau \quad (\text{A.60})$$

A.4.2 Finding the Solution by the Laplace Transform

The solution of an LTI system can also be found by using the Laplace transform method, which has the advantage of solving an algebraic equation rather than a differential equation as in Eq. (A.4). From Eq. (A.10), the output time history for a SISO system is computed by

$$y(t) = L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{G(s)K(s)}{1 + H(s)G(s)K(s)}R(s)\right\} \quad (\text{A.61})$$

Because the Laplace transform of the output $Y(s)$ can be rearranged into the partial-fraction form of Eq. (A.15) (assuming that there is no repeated pole), then Eq. (A.61) can be expressed as a sum of individual inverse transforms by

$$y(t) = L^{-1}\{d_0\} + L^{-1}\left\{\frac{c_1}{s-p_1}\right\} + L^{-1}\left\{\frac{c_2}{s-p_2}\right\} + \cdots + L^{-1}\left\{\frac{c_n}{s-p_n}\right\} \quad (\text{A.62})$$

Also because the Laplace transforms and the inverse Laplace transforms of commonly encountered dynamic functions have been tabulated in mathematical handbooks, computing Eq. (A.15) is relatively fast and straightforward compared to the state transition method of Eq. (A.49) or Eq. (A.50), which involves numerical integration. Table A.2 lists some of the most frequently encountered functions in control and monitoring systems and their respective Laplace transforms.

Equation (A.56) can be extended to a MIMO control system by repeating it ($l \times m$) times, if the output variable is an $l \times 1$ vector and the input variable is an $m \times 1$ vector, that is, the set of individual TFs $\{Y_i(s)/U_j(s)\}$.

A.4.3 Controllability and Observability

A linear system, expressed by Eqs. (A.1) and (A.2), is said to be *controllable* at time t_0 , if there exists a finite time interval $t_f - t_0 < \infty$ and an input $u(t)$ over this time interval $[t_0, t_f]$, such that the system can be transferred from any initial state $x(t_0)$ to any final state $x(t_f)$ within the state space of the system. The necessary and sufficient condition for a system to possess the property of controllability is the linear independence of the *rows* of the matrix function $\phi(t, \tau)B(\tau)$ in Eq. (A.54). Note that we have allowed matrix B to be a function of time because the same equation holds for time-varying systems.

For an LTI system, controllability can be tested by the following condition:

$$\text{Rank}([A \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B]_{n \times nm}) = n \quad (\text{A.63})$$

Table A.2 Laplace transform of commonly seen functions

Functions in time (or inverse transforms)	Laplace transforms in frequency
Impulse: $f(t) = \delta(t - a)$	1
Step: $f(t) = 1$, for $t > 0$	$\frac{1}{s}$
Ramp: $f(t) = at$	$\frac{a}{s^2}$
Exponential decay: $f(t) = e^{-at}$	$\frac{1}{s + a}$
Exponential reach: $f(t) = 1 - e^{-at}$	$\frac{a}{s(s + a)}$
Sine: $\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
Cosine: $\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$

Or alternatively, by the property of row independence of the matrix

$$(sI - A)^{-1}B = \begin{bmatrix} r_1^T \\ r_2^T \\ \vdots \\ r_n^T \end{bmatrix}_{n \times m}, \quad \text{where } (ar_i^T + br_j^T) \neq 0 \quad (\text{A.64})$$

where a and b are arbitrary real numbers. Controllability is a property of the state equation only, and so it does not involve the output equation.

A linear system, expressed by Eqs. (A.1) and (A.2), is said to be (state) *observable* at time t_0 if there exists a finite time interval $t_f - t_0 < \infty$ such that the system's initial state $x(t_0)$, for any point within the state space, can be determined from the input $u(t)$ and the output $y(t)$ values over this time interval $[t_0, t_f]$. Like the property of controllability, the necessary and sufficient condition for a system to possess the property of observability is the linear independence of the *columns* of the matrix function $C(t)\phi(t, t_0)$ in Eq. (A.54). Note that we have again allowed matrix C to be a function of time because the same equation holds for time-varying systems.

For an LTI system, observability can be tested by the following condition:

$$\text{Rank} \begin{pmatrix} \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}_{nl \times n} \end{pmatrix} = n \quad (\text{A.65})$$

Or alternatively, by the property of column independence of the matrix

$$C(sI - A)^{-1} = [c_1 \quad c_2 \quad \cdots \quad c_n]_{l \times n}, \quad \text{where } (ac_i + bc_j) \neq 0 \quad (\text{A.66})$$

Observability is a property of both the state and the output equations.

A.4.4 Performance

Accuracy and response are the two most important aspects of the performance of a control system. Accuracy is the ability to maintain (or regulate) the controlled variable at a desired value when the system reaches a steady state. Response is the "speed" of change of the controlled variable when the system is forced (or excited) to move away from its current operating condition (whether intentionally or unintentionally). These two properties are discussed further in the following subsections.

A.4.4.1 Accuracy. Accuracy is a measure of the closeness of a system's output variable to a desired reference value (see Figs. A.2 and A.3). Accuracy is typically measured at, or near, steady state; hence, it is often quantified by the

steady-state error of the controlled variable. However, when the reference value continues to change with time and the system does not appear to reach a steady-state condition, the word *accuracy* can still be used to measure the system's ability to track the reference variable, and in this case, we call the feedback error a *tracking error*.

Before we analyze the accuracy of a closed-loop control system, we want to introduce the concept of system *type*. Succinctly, the type of a closed-loop control system is equal to the number of integrators in the FLTF [i.e., $G(s)K(s)$]. Suppose the FLTF has d number of integrators, that is, there are d number of poles located at the origin of the s plane; we can express the FLTF in the polynomial form as

$$\text{FLTF} = \frac{k(s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m)}{s^n + a_1 s^{n-1} + \cdots + a_{n-d} s^d} \quad (\text{A.67})$$

This equation can be rearranged to show the dimension of the integrator explicitly as

$$\text{FLTF} = \frac{k(s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m)}{s^d (s^{n-d} + a_1 s^{n-d-1} + \cdots + a_{n-d})} \quad (\text{A.68})$$

and the closed-loop control system is defined as type d . The type of a system determines the characteristics of the output variable. For example, a type 0 system will produce a constant output, when the system is forced by a constant external input; a type 1 system will produce a constant rate of change of the output when it is forced by the same constant external input [1].

Phrases like *external input*, *forcing function*, *driving function*, and *excitation* are used interchangeably in the control's field to mean a reference input $r(t)$. The three most common forms of external inputs in the analysis of control system performance are *step* function, *ramp* function, and *impulse* function. They are illustrated in Fig. A.8.

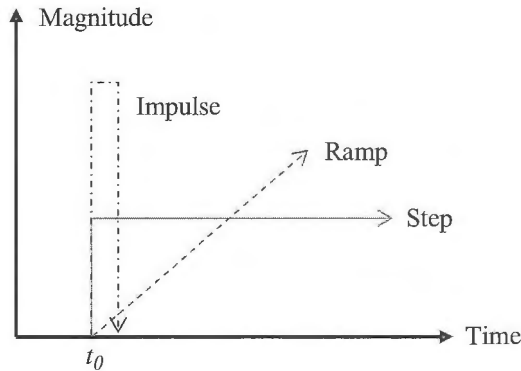


Fig. A.8 Time histories of three common input functions: step, ramp, and impulse.

The Final-Value Theorem of the Laplace transform provides the theoretical basis for evaluating the steady-state error. According to the theorem, the final (or steady-state) value of a function f is the same as the initial value of the Laplace transform of the same function multiplied by the Laplace operator s , that is, $s F(s)$, and it is written as

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s) \quad (\text{A.69})$$

By applying this theorem, the final value of the error function $e(t)$ can be evaluated as

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} \left[s \frac{1}{1 + H(s)G(s)K(s)} R(s) \right] \quad (\text{A.70})$$

So the final value of the controlled variable depends on both the reference input $r(t)$ and the *closed-loop sensitivity function*, which is defined as $1/(1+H(s)G(s)K(s))$. Now suppose the TF of the sensor component $H(s)$ is unity (or 1) then Eq. (A.70) reduces to

$$\lim_{t \rightarrow \infty} e(t) = \lim_{t \rightarrow \infty} [r(t) - y(t)] = \lim_{s \rightarrow 0} \left[s \frac{1}{1 + G(s)K(s)} R(s) \right] \quad (\text{A.71})$$

The steady-state errors corresponding to the three most common reference inputs are given by the following three equations:

For a step input,

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} \left[s \frac{1}{1 + G(s)K(s)} \left(\frac{r_0}{s} \right) \right] \quad (\text{A.72})$$

For a ramp input,

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} \left[s \frac{1}{1 + G(s)K(s)} \left(\frac{l_0}{s^2} \right) \right] \quad (\text{A.73})$$

For an impulse input,

$$\lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} \left[s \frac{1}{1 + G(s)K(s)} (I_0) \right] \quad (\text{A.74})$$

where r_0 is the final value of the step function, l_0 is the slope of the ramp function, and I_0 is the impulse energy, which is the area under the impulse function.

A.4.4.2 Response. Response is a measure of the speed of change of a system's output variable as it is going through a transient. A transient is a temporary condition where the state variable of the system changes with time. In most cases, transients are caused by changes in the reference input, such as a change in engine's throttle position by a pilot. The response of a system can be interpreted in either the time domain or the frequency domain. The characteristics used to determine a system's response in each domain are described in the following subsections.

A.4.4.2.1 Time-domain response. Time-domain response deals with the *time history* of the output variable of a control system. The time history of the output is the time-domain solution of the output equations, that is, $y(t)$. We are mostly interested in three basic characteristics of a system's response: *rise time*, *settling time*, and *overshoot* (or undershoot). These characteristics are illustrated in Fig. A.9 for an output variable of a hypothetical system to a *step* input function $r(t)$. Hence, the rise time, $t_r = t_1 - t_0$, is the time duration for the output variable to reach a desired rise threshold y_r . The rise threshold is usually defined as a percentage (e.g., 90%) of the desired *change* in the output from the start of a transient. On the other hand, the settling time, $t_s = t_2 - t_0$, is the time duration for the output to reach and stay inside a settling band. The settling band is a set of upper and lower limits from the desired final value (e.g., $\pm 2\%$) of the output variable. Maximum overshoot is the peak value that the output variable has attained before it settles in its final value. Overshoot is usually expressed in percentages of the desired change, for example, 10% of the reference value r_0 .

A.4.4.2.2 Frequency-domain response. Frequency-domain response deals with the *frequency response* of the output TF of a control system. Strictly speaking, frequency response can be applied to any of the TFs in Eq. (A.6); however, from a control design's standpoint, we are mostly interested in the frequency response of the OLTF [Eq. (A.11)], because we can use it to infer the performance and stability properties of the closed-loop system. A frequency response consists of two basic

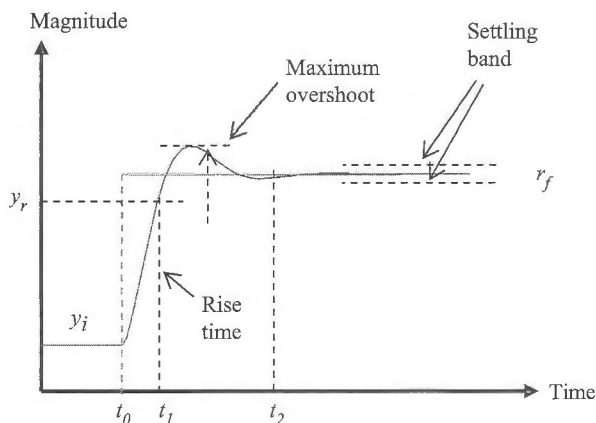


Fig. A.9 Time-domain response properties.

characteristics of the TF: 1) magnitude of the TF as a function of frequency and 2) phase angle of the TF as a function of frequency. An example of the frequency response for the transfer function of $1/(s + 10)$ is shown in Fig. A.10. The magnitude and the phase diagrams of a frequency response are called the Bode diagram.

Note that the x axis in both the magnitude and the phase diagrams as well as the y axis in the magnitude diagram are in logarithmic scale; however, the y axis in the phase diagram is in linear scale. Also note that the unit of the magnitude is in decibels, which is defined as $1 \text{ db} = 20 \log_{10}(M)$, where M is the magnitude of the TF at any given frequency ω , that is,

$$M(\omega) = |\text{OLTF}(s = j\omega)| \quad (\text{A.75})$$

and

$$\phi(\omega) = \angle[\text{OLTF}(s = j\omega)] \quad (\text{A.76})$$

The frequency ω in the preceding equations is the excitation frequency of a sinusoidal input function, for instance, the sine function $\sin(\omega t)$. So the frequency response in Fig. A.10 is a collection of all of the magnitude-phase pairs calculated by Eqs. (A.75) and (A.76) for all of the excitation frequencies specified in the calculation. The excitation frequencies usually span the frequency spectrum of interest within which the control system is expected to operate.

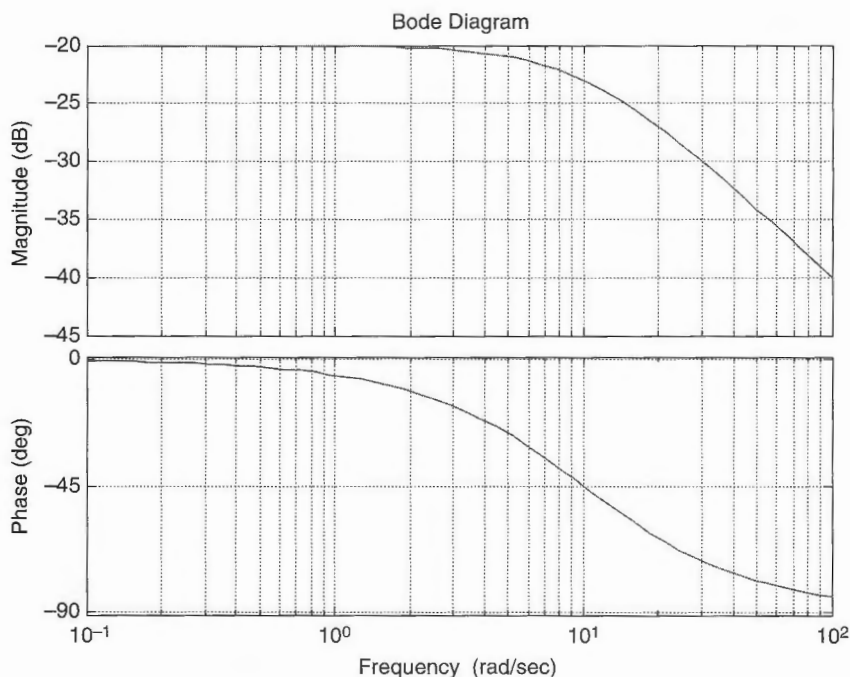


Fig. A.10 Example of frequency response—Bode diagram.

A.4.5 Stability

Stability deals with the controllability aspect of a control system, and it is closely related to the safety of the system. Stability can be described as the ability of a control system to keep its feedback error within an acceptable bound. Stability, in a broader sense, means that the error function $e(t)$ always stays within a fixed region in the state space. This kind of stability is called *bounded-input* and *bounded-output stability*; it is also called *stability in the sense of Lyapunov* or *Lyapunov stability*. In most cases, a controlled system with Lyapunov stability is not sufficient to meet the performance and the stability requirements because the output variable is oscillatory (or cyclic) even though the oscillation does not grow with time. Figure A.11 shows the concept of Lyapunov stability with a sustained oscillation in the output variable. To make the output variable *converge* to a final value, that is, to make the steady-state error approach zero, we will need a more stringent criterion for stability.

A more stringent stability criterion would require that the output variable approaches the desired final value asymptotically. This *asymptotic stability* is represented by a step response of the output variable that gradually reaches the desired reference value (see Fig. A.9). Depending on how fast the output response converges to the desired value, the step response of an asymptotically stable system can fall into three types: *underdamped*, *critically damped*, and *overdamped*. The damped system responses are typical characteristics of second-order linear dynamic systems. From Eq. (A.14) we can express a general second-order system in the pole-zero form as

$$\frac{N(s)}{D(s)} = \frac{k(s - z_1)}{(s - p_1)(s - p_2)} = \frac{k(s - z_1)}{s^2 + (p_1 + p_2)s + p_1p_2} \equiv \frac{k(s - z_1)}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{A.77})$$

where we have defined two new parameters on the far right-hand side of the equation to represent the denominator polynomial: the *damping ratio* ζ and the *natural frequency* ω_n . When $\zeta < 1$, the system represented by the TF in Eq. (A.77) is called underdamped, and the response of the system will exhibit some oscillations before it approaches the desired value. When $\zeta = 1$, the system is called critically damped,

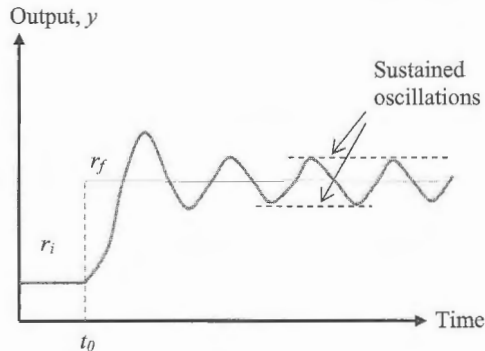


Fig. A.11 Example of bounded stability in a step response.

Table A.3 Summary of the characteristics of a general second-order system

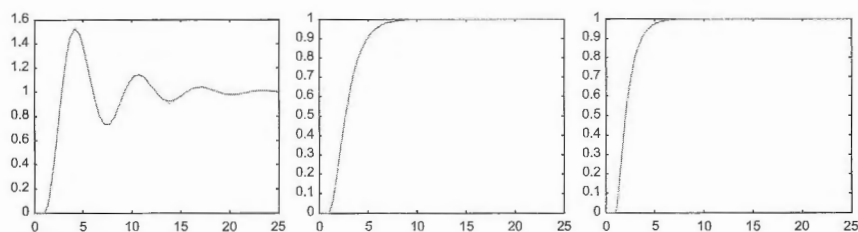
Type	Damping ratio	Poles	Response
Underdamped	$\zeta < 1$	Two complex conjugate roots	Oscillatory
Critically damped	$\zeta = 1$	Two equal real roots	Exponential
Overdamped	$\zeta > 1$	Two different real roots	Exponential

and the response of the system will act like one that consists of two identical first-order systems. When $\zeta > 1$, the system is called overdamped, and the response of the system will act like one that consists of two different first-order systems. These characteristics are summarized in Table A.3. Examples of these characteristics are shown in Fig. A.12.

Note that Eq. (A.77) and Table A.3 are valid for all second-order, linear dynamic systems, whether they are stable or not. So, how do we know if a system is stable? And how can we generalize the stability criterion to a system whose order is higher than two? To answer these questions, we know from Eq. (A.49) that the solution of the state equation is asymptotically stable if the eigenvalues of the dynamic matrix A all have negative real parts, that is, $\text{Re}(\lambda_i) < 0$. This is more easily understood with the help of the simplified, first-order system response in Eq. (A.49a) because it explicitly shows an exponential decay function when a has a negative real part. Similarly, in the frequency-domain representation, a system is asymptotically stable if, and only if, the real part of every pole of the characteristic equation falls on the left-hand side of the s plane.

When the real part of an imaginary pole is zero, that is, the pole is located on the imaginary axis but not at the origin, then the system is said to be *marginally stable* because the output response exhibits oscillatory behavior as in a sinusoidal function. This kind of marginal stability is the same as the Lyapunov stability described at the beginning of this subsection.

Stability can also be inferred from the Bode diagram of the OLTF. When the *phase lag* exceeds 180 deg (i.e., phase angle is < -180 deg), the control system becomes unstable with increasing error because the feedback loop becomes additive instead of subtractive. An example of an unstable output response is shown in Fig. A.13. The corresponding simulation model is shown in Fig. A.14, and the

**Fig. A.12** Three examples of second-order responses ($\zeta = 0.2$, left; $\zeta = 1$, center; $\zeta > 1.15$, right).

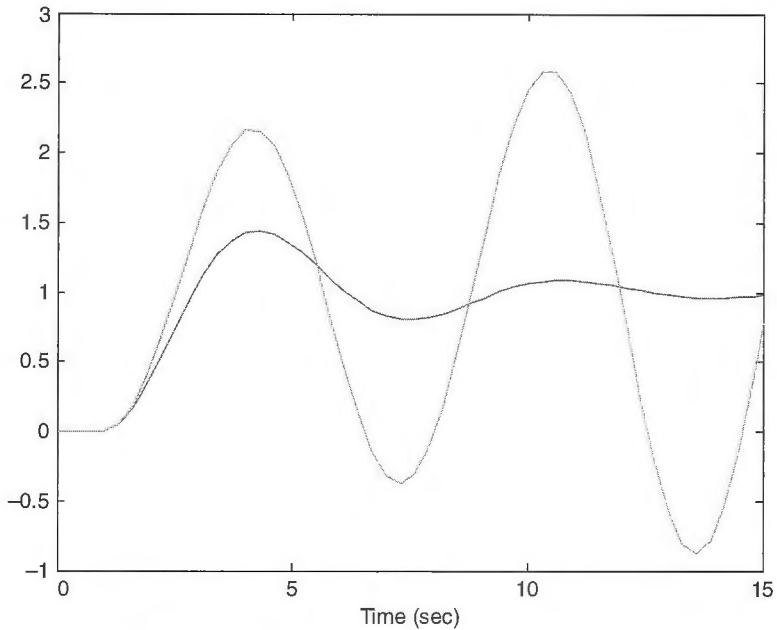


Fig. A.13 Example of an unstable system—output response (red, stable; blue, unstable). (See also color figure section at the back of the book.)

Bode diagram is shown in Fig. A.15. Because the roots of the unstable system have a small positive real part (+0.05), the response of the system is slowly diverging. Note that the phase angle in Fig. A.15 is already < -180 deg at low frequency, indicating an unstable system.

The *relative* stability of an automatic control system is a quantitative description of the “distance” between the current operating condition and the “nearest” unstable operating condition. Although the real part of any pole of an OLTF pole is a measure of the relative stability of the system, the traditional way of measuring relative

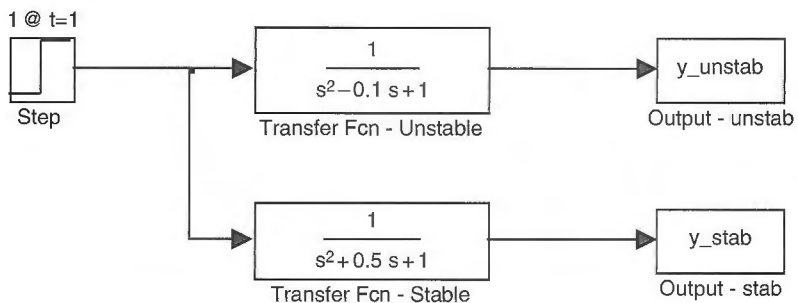


Fig. A.14 Simulation model of the unstable system.

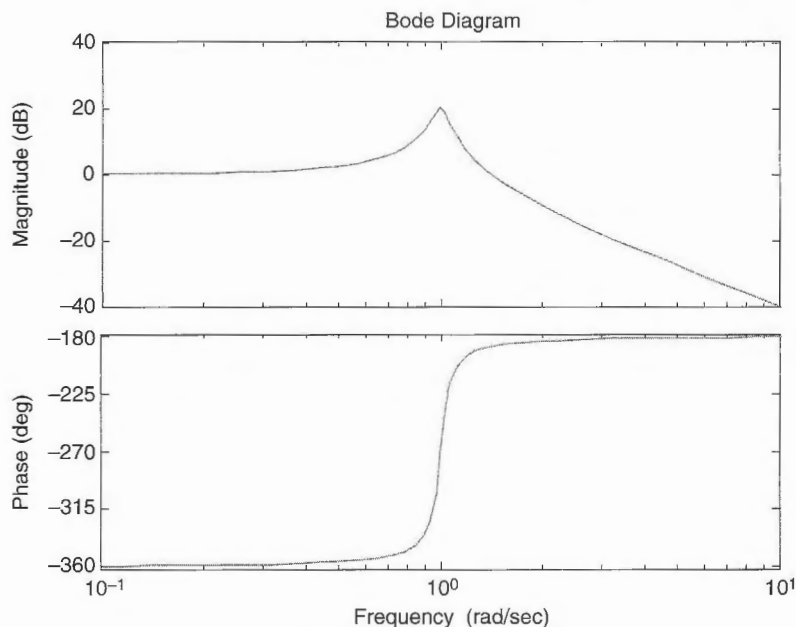


Fig. A.15 Bode diagram of an unstable system.

stability has been to use the *phase margin* (PM) and the *gain margin* (GM) in the frequency domain. Simply put, PM is the additional phase shift that a control system can tolerate before becoming unstable while keeping the loop gain unchanged [see Eq. (A.14)]. Similarly, GM is the additional loop gain that the system can accept before becoming unstable. Because of the uncertainties in measurement processes and models, experience suggests that a minimum PM of 45 deg for the entire open-loop transfer function be necessary to ensure stability.

A.5 Two Linear Control Design Methods

Control system design formally should include the following activities: 1) development of control system and functional requirements, 2) analysis of the system to be controlled, 3) sensor selection and placement, 4) actuator selection and placement, 5) control law design, and 6) design assurance through simulation-based validation and verification. However, in most organizations these design activities are divided into three basic functional groups: system engineering, design engineering, and test engineering; moreover, the design engineering is divided into hardware design and software design. For the purpose of this book, we are mostly interested in step 5 and design engineering, that is, control law design and actuator/sensor analysis.

To design the control law for a linear system, we need to understand the performance and stability requirements imposed on the control system from the system engineering activities. These requirements must be met even in the presence of

Table A.4 Classification of common linear control design methods

Domain	Classical control (single variable control)	Modern control (multivariable control)
Time domain	Root locus	Linear quadratic
Frequency domain	Bode plot	Inverse Nyquist array, characteristic loci, H-infinity

operational uncertainties. Operational uncertainties can be grouped into three categories: *process* (including actuator, plant, and sensor) uncertainty, *measurement* uncertainty, and *environment* uncertainty. The ability of a control system to maintain acceptable performance and stability under these uncertainties is called *robustness*.

Commonly used linear control design methods can be classified in Table A.4. The two classical control methods are reviewed in this appendix, whereas the modern control design methods are covered in Chapter 7 in the specific context of gas turbine engines.

A.5.1 Root-Locus Design Method

The *root-locus* method is a graphical method of predicting the movement of the roots of the characteristic equation of the CLTF as the control gain is increased. The method infers the locations of closed-loop poles from the locations of the open-loop poles and zeros. It uses a set of rules to construct the trajectories of all closed-loop poles on the s plane by varying the (open) loop gain. The trajectories are also called the branches of the root loci. The number of branches depends on the orders of the denominator and the numerator polynomials of the OLTF. Once a value is selected for the loop gain, the locations of closed-loop poles are determined, and the characteristics of the closed-loop system can be estimated.

Recall from Eqs. (A.10) and (A.14) that the characteristic equation of the CLTF is a sum of the denominator and the numerator polynomials of the OLTF, and it can be expressed as

$$\frac{-1}{k} = \frac{\text{OLTF}}{k} = \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)} = \frac{(s^m + b_1 s^{m-1} + \cdots + b_{m-1} s + b_m)}{s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \quad (\text{A.78})$$

where the p_i and z_j are the open-loop poles and zeros, respectively. From Eq. (A.78), we can derive the basic rules of the root-locus method. They are summarized in Table A.5.

An example of the root-locus plot is shown in Fig. A.16 for the OLTF $K(s+1)/s(s-1)(s^2+4s+16)$ [7] by using the Matlab® computing tool from the Mathworks.

Table A.5 Summary of basic rules of the root-locus plot

Rule name	Rule description
$k = 0$	Closed-loop poles are at the locations of open-loop poles.
$k = \pm\infty$	Closed-loop poles are at the locations of open-loop zeros.
Branch	The number of branches is n , the order of the denominator polynomial of OLTF.
Asymptote	The number of asymptotes is $n - m$, the difference between the orders of the denominator polynomial and the numerator polynomial of OLTF.
Symmetry	The root loci are symmetric about the real axis.
Real roots	Root loci on the real axis, that is, real-valued roots, always lie on the left-hand side of an <i>odd</i> number of the sum of real-valued poles and zeros.
Centroid	All asymptotes of the complete root loci are centered on the real axis at $b_1 - a_1/n - m$.

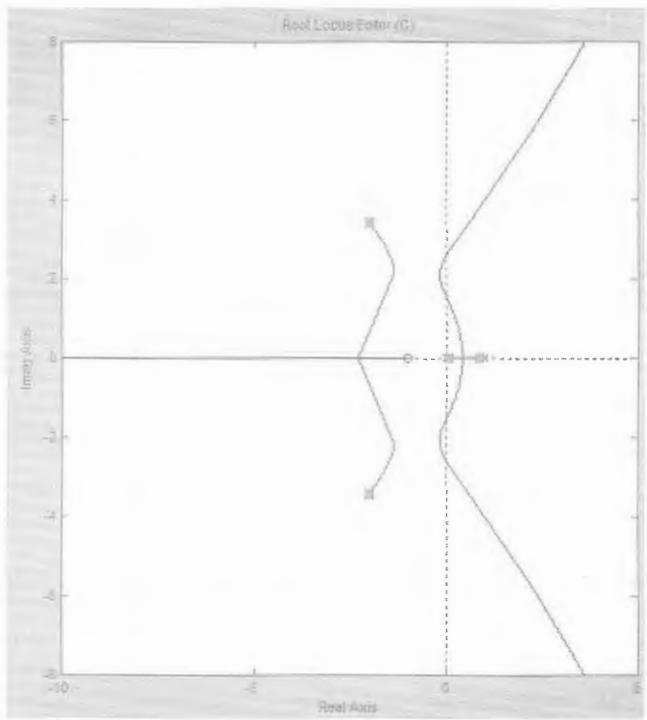


Fig. A.16 Example of root-locus diagram.

A.5.2 Frequency-Response Design Method

The *frequency-response* method is a systematic way of constructing the controller TF so that the resultant CLTF satisfies some desired characteristics in the frequency spectrum. The method uses the frequency response of the OLTF to infer the relative stability of the closed-loop system. The three strongest advantages of the frequency-response method are 1) existence of convenient sinusoidal test signals at various magnitudes and frequencies to “measure” the frequency response of a system or TF, 2) stability of the closed-loop system can be inferred from the phase shift diagram directly, and 3) the controller design is inherently robust against noise and disturbance, if it satisfies certain desired characteristics in the frequency domain.

To calculate the magnitude and the phase angle of the OLTF, we first rearrange Eq. (A.78) and then substitute it in Eqs. (A.75) and (A.76) to get

$$M(\text{OLTF}) = \left| \frac{k(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)} \right|_{s=i\omega} = k \frac{\prod_{j=1}^m (s - z_j)}{\prod_{i=1}^n (s - p_i)} \Big|_{s=i\omega} \quad (\text{A.79})$$

$$\phi(\text{OLTF}) = \angle \left[\frac{k(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)} \right]_{s=i\omega} = \angle \left[\frac{\prod_{j=1}^m (s - z_j)}{\prod_{i=1}^n (s - p_i)} \right]_{s=i\omega} \quad (\text{A.80})$$

If we plot the magnitude of Eq. (A.79) in a log-log scale, that is, both the magnitude and the frequency are in logarithmic scales, then the equation can be simplified to

$$\log(M) = \log \left(k \frac{\prod_{j=1}^m (s - z_j)}{\prod_{i=1}^n (s - p_i)} \right)_{s=i\omega} = \log(k) + \sum_{j=1}^m \log(|s - z_j|_{s=i\omega}) - \sum_{i=1}^n \log(|s - p_i|_{s=i\omega}) \quad (\text{A.81})$$

So the multiplicative operation of individual (first or second order) TFs in Eq. (A.81) has been converted into additive operation by taking the logarithm of the equation. This is another advantage of the frequency-response method, that is, the ease of constructing the overall frequency response of the OLTF by adding or subtracting the responses of individual TFs.

Now let us look at the frequency responses of three basic TFs: an integrator $1/s$, a first-order lag $1/(s + a)$, and a second-order, underdamped system $1/(s^2 + as + b)$. Suppose that all of the coefficients a and b in the TFs take on the

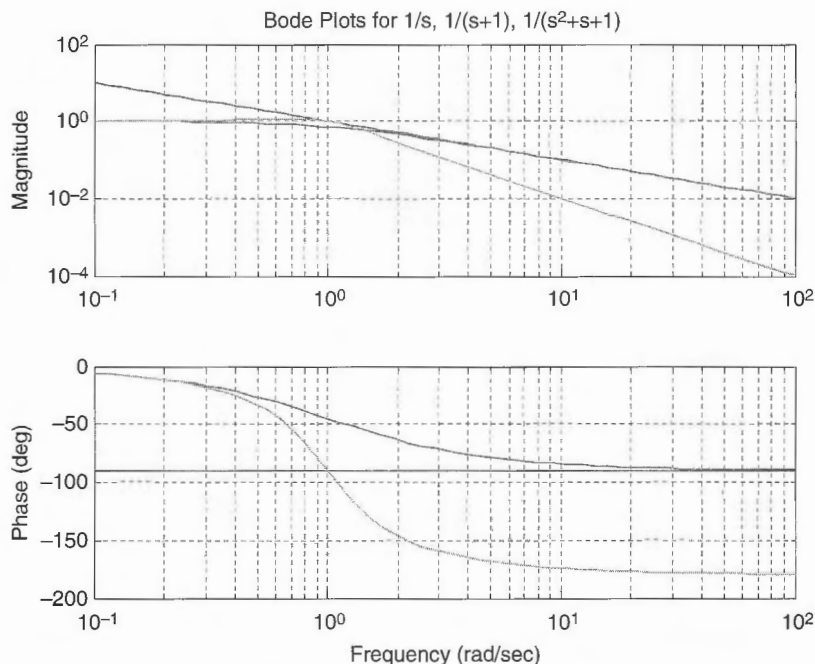


Fig. A.17 Comparison of the frequency responses of three basic TFs. (The integrator is in blue, the first-order lag is in green, and the second-order TF is in red.) (See also color figure section at the back of the book.)

value of one; the frequency responses of these TFs are compared in Fig. A.17, where the magnitude plot is on the top and the phase plot on the bottom. The combination of magnitude and phase plots is called the Bode diagram. From the figure, we note the following: 1) the maximum phase shift for the integrator and the first-order lag is -90° ; 2) the maximum phase shift for the second-order system is -180° ; 3) the magnitude of the integrator decreases linearly with increasing frequency, and this decrease is called *attenuation* or *roll-off*; 4) the rate of attenuation for the first-order lag approaches that of the integrator at high frequencies; and 5) the rate of attenuation of the second-order system is twice as large as that of the integrator at high frequencies.

It has been a tradition to express the magnitude in decibels by the following formula:

$$\text{db} \equiv 20 \log_{10} (M) \quad (\text{A.82})$$

where $\log_{10}$ is a logarithm function of base 10. After applying this formula to the magnitude results of Fig. A.17, the same Bode plot is expressed in the unit of decibels as shown in Fig. A.18. From this figure, we see that the rate of attenuation of the integrator is -20 db per *decade* of frequency, that is, -20 db/decade. A decade corresponds to a frequency range where the maximum frequency is 10 times that of the minimum frequency.

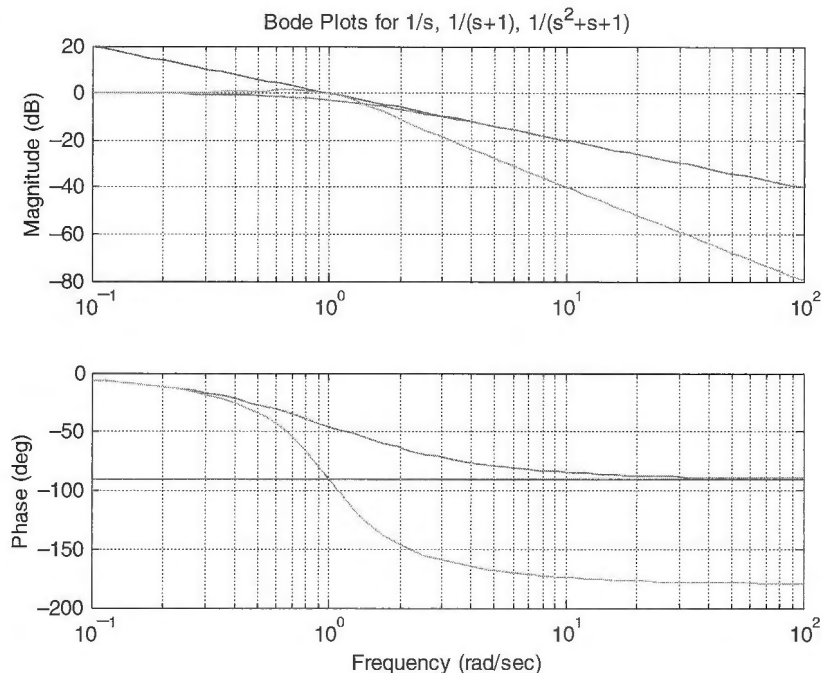


Fig. A.18 Comparison of the frequency responses of three basic TFs with magnitudes in the unit of decibels. (The integrator is in blue, the first-order lag is in green, and the second-order TF is in red.) (See also color figure section at the back of the book.)

The design philosophy of the frequency-response method is to select controller TF so that the overall OLF satisfies the following rules of thumb: 1) high gain (or magnitude) at low frequencies for disturbance rejection, 2) low gain at high frequencies for noise attenuation, 3) greater than ($>$) 45-deg PM (measured from -180 deg) at gain crossover frequency, and 4) less than ($<$) 6-dB reduction in magnitude at the frequency where the PM is at least 45 deg. These rules suggest that the frequency response of the resultant OLF fit in the decaying “corridor” as illustrated in Fig. A.19.

A.6 Nonlinear Dynamic Systems

Unlike its linear system counterpart in Eqs. (A.1) and (A.2), a nonlinear, dynamic system is expressed as nonlinear functions of time, state, and input variables as follows:

$$\dot{x}(t) = f[x(t), u(t), t] \quad (\text{A.83})$$

$$y(t) = g[x(t), u(t), t] \quad (\text{A.84})$$

where x , u , and y again stand for the state, the input, and the output variables, respectively.

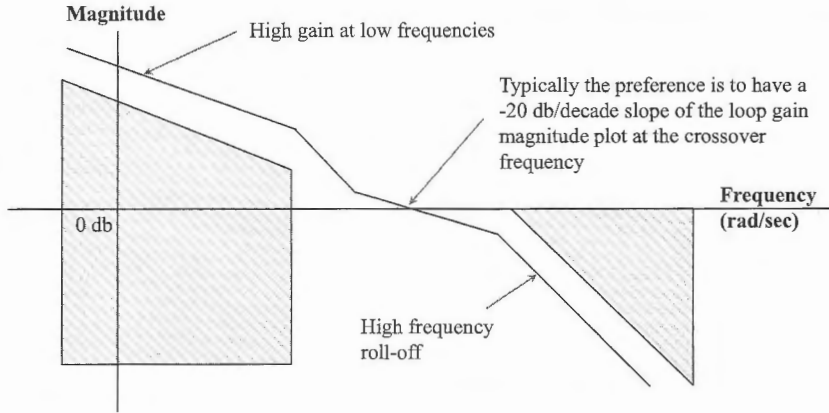


Fig. A.19 Response corridor of the frequency-response design method.

Nonlinear systems are difficult to analyze because there does not exist a general, analytical method for the exact solution of the state equation (A.83), except for a few special cases, such as the van der Pol equation, $\ddot{x} + \mu(1 - x^2)\dot{x} + kx = q$, when its nonlinear damping coefficient μ is zero. To analyze nonlinear systems, we usually resort to the method of numerical solution, which is an *approximate* solution of the nonlinear system. Numerical solutions have become popular because of the availability of powerful digital computers and cost-effective computational tools. Historically, there have been other forms of approximate solutions, such as the method of *linearization* and the method of *averaging* (for periodic nonlinear systems), both of which fall in the category of *perturbation* theory.

Linearization is the most widely used method for the analysis of nonlinear systems. It is based on the Taylor-series expansion near a *nominal* condition. A Taylor series can approximate a continuous function $f(x)$ with an arbitrary degree of accuracy depending on the number of terms in the series. A nominal condition is any value of the independent variable and is usually written as x_0 . For a function of multiple (independent) variables, for example, $f(x, u)$, it can still be represented by a Taylor series; however, the ordinary derivatives in the series are replaced by the partial derivatives of f with respect to each individual variable.

Let a nominal condition of the nonlinear equation (A.83) be defined as $x_0 = x(t_0)$ and $u_0 = u(t_0)$ when $t = t_0$; suppose that the partial derivatives of f with respect to x and u are continuous in a neighborhood of x_0 as well as in a neighborhood of u_0 , and then the Taylor-series expansion around the nominal condition, by retaining only the terms up to the second-order derivatives, is given by

$$\begin{aligned} \dot{x}(t) \approx & f(x_0, u_0) + \left. \frac{\partial f}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial f}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) \\ & + \frac{1}{2} \left[\left. \frac{\partial^2 f}{\partial x^2}(t) \right|_{x_0, u_0} \delta x^2(t) + 2 \left. \frac{\partial^2 f}{\partial x \partial u}(t) \right|_{x_0, u_0} \delta x(t) \delta u(t) + \left. \frac{\partial^2 f}{\partial u^2}(t) \right|_{x_0, u_0} \delta u^2(t) \right] \end{aligned} \quad (\text{A.85})$$

Similarly, the output equation (A.84) can be approximated by

$$y(t) \approx g(x_0, u_0) + \left. \frac{\partial g}{\partial x}(t) \right|_{x_0, u_0} \delta x(t) + \left. \frac{\partial g}{\partial u}(t) \right|_{x_0, u_0} \delta u(t) + \frac{1}{2} \left[\left. \frac{\partial^2 g}{\partial x^2}(t) \right|_{x_0, u_0} \delta x^2(t) + 2 \left. \frac{\partial^2 g}{\partial x \partial u}(t) \right|_{x_0, u_0} \delta x(t) \delta u(t) + \left. \frac{\partial^2 g}{\partial u^2}(t) \right|_{x_0, u_0} \delta u^2(t) \right] \quad (\text{A.86})$$

If we further omit the second-order derivatives in Eqs. (A.85) and (A.86), then we are left with a pair of first-order equations, which are *linear* equations of x and u , and these equations are the same as Eqs. (A.1) and (A.2), if we make the following substitution:

$$A(t) \equiv \left. \frac{\partial f}{\partial x}(t) \right|_{x_0, u_0}, B(t) \equiv \left. \frac{\partial f}{\partial u}(t) \right|_{x_0, u_0}, C(t) \equiv \left. \frac{\partial g}{\partial x}(t) \right|_{x_0, u_0}, D(t) \equiv \left. \frac{\partial g}{\partial u}(t) \right|_{x_0, u_0} \quad (\text{A.87})$$

Strictly speaking, Eqs. (A.85) and (A.86) are valid only for a finite time interval $[t_0, t_0 + \Delta t]$. For them to be valid for an infinite time interval, that is, $[t_0, \infty]$, the nominal condition must be an exponentially stable equilibrium condition and $\dot{x}(0) = f[x_0(t_0), u_0(t_0), t_0] = 0$ [8].

A.6.1 Properties of Nonlinear Systems

Even though Eqs. (A.85) and (A.86) may be approximated as a linear system with time-varying parameters, it behaves like a nonlinear system. There are six general patterns of behavior of nonlinear systems [9], and they are the following:

1) *New frequencies*: Unlike an LTI system, whose response will contain only the frequencies of the forcing function (or input), a nonlinear system can produce a response with “new” frequencies, other than the input frequencies. These new frequencies are the harmonics or subharmonics of the input.

2) *Jump resonance*: This pattern exhibits a behavior of a discontinuity in the response that “jumps” from one part of the frequency-response curve (i.e., a magnitude plot or a phase plot in a Bode diagram) to another depending on the history of the response and the direction from which the jumps are approached. An example of jump resonance is shown in Fig. A.20.

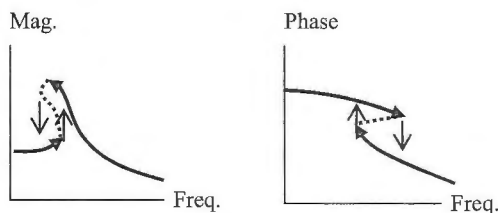


Fig. A.20 Example of jump resonance of a nonlinear system (left, magnitude plot; right, phase plot).

3) *Dependence of input shape and amplitude*: The response of a nonlinear system and its characteristics, such as the rise time, the settling time, and the stability, depend on the shape and amplitude of the input.

4) *Multiple equilibrium points*: Unlike its LTI counterpart which has a unique equilibrium point where the time derivatives of the dependent variables vanish, a nonlinear system might have more than one equilibrium point. These equilibrium points are dependent on the initial condition and the input.

5) *Nonexponential time response*: The response of an LTI system follows the exponential function, so that the response envelope formed by the peaks (and valleys) of the response is exponential; on the other hand, the response of a nonlinear system might not follow the exponential pattern of a linear system.

6) *Limit cycles*: These are oscillations of fixed frequency and amplitude. They are one of the most frequently encountered modes of behavior specific to nonlinear systems. The trajectory of a limit cycle in the time domain is shown in Fig. A.21,

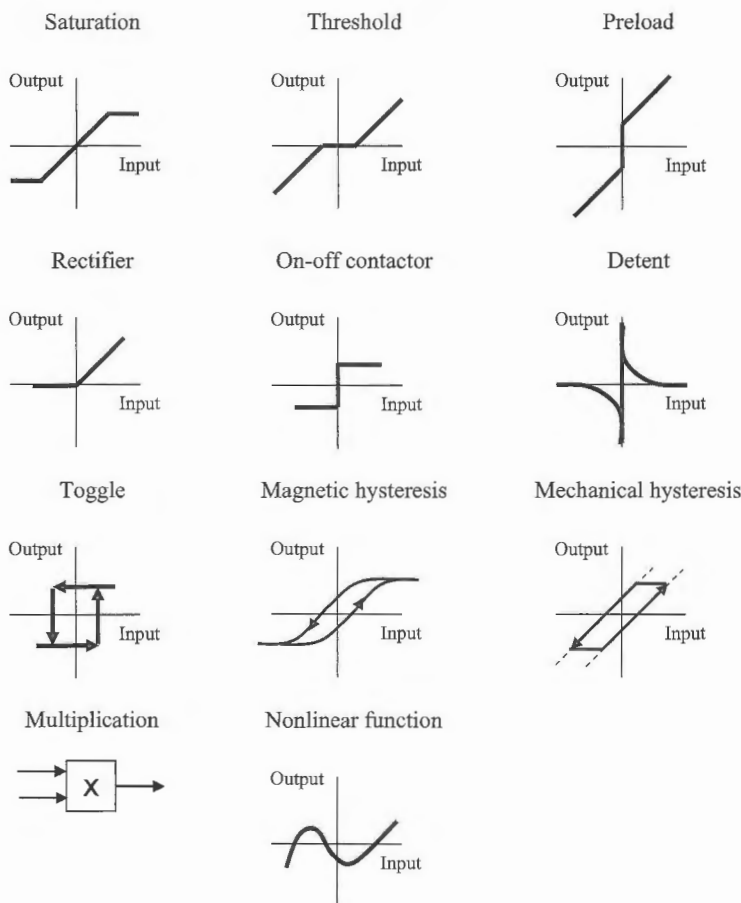


Fig. A.21 Common nonlinearities.

and the trajectory on the phase plane is a closed curve about an equilibrium point (often the origin).

A.6.2 Commonly Encountered Nonlinearities

The saying “nonlinearity is everywhere, but linearity is local” is probably a good description of the real-world problems facing control system designs.

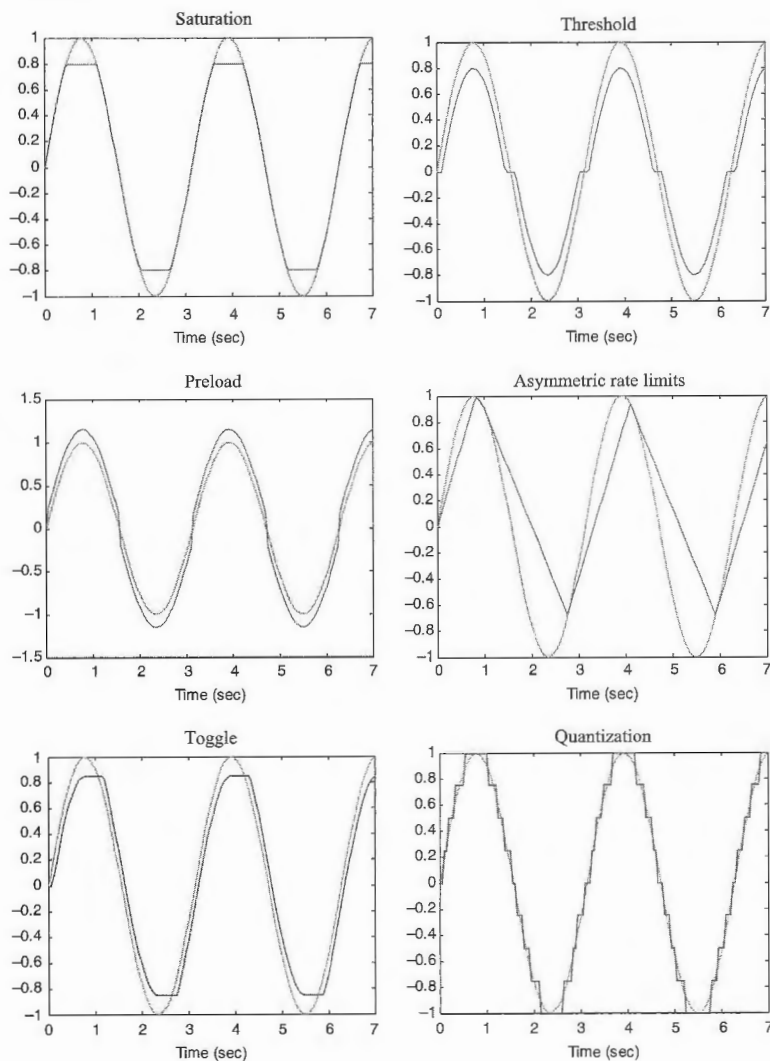


Fig. A.22 Effects of six common nonlinearities (red, input; blue, output). (See also color figure section at the back of the book.)

Experience has taught us that a control system “only” needs to work in the frequency spectrum of interest, and, whenever possible, we should use a linear system approximation to control the nonlinear system, especially in controller design. However, even in the frequency spectrum of interest, and even with a linear system approximation at hand, we are often forced to consider some nonlinear effects in the modeling and design process. The most commonly encountered nonlinear effects are [9] 1) saturation or limiting, 2) threshold or dead zone, 3) preload or friction, 4) rectifier, 5) on-off contactor or quantization, 6) detent, 7) toggle or negative deficiency, 8) hysteresis (magnetic or mechanical), 9) multiplication, and 10) nonlinear function. These nonlinear effects are illustrated in Fig. A.21.

The effects of some of these common nonlinearities are shown in Fig. A.22 by comparing the output of each of these nonlinearities with the sinusoidal input.

References

- [1] D’Azzo, J. J., and Houppis, C. H., *Linear Control System Analysis and Design—Conventional and Modern*, 2nd ed., 1981.
- [2] Dorf, R. C., and Bishop, R. H., *Modern Control Systems*, 10th ed., Prentice–Hall, Upper Saddle River, NJ, 2004.
- [3] Chen, C-T., *Introduction to Linear System Theory*, Holt, Rinehart and Winston, Philadelphia, 1970.
- [4] Cannon, R. H., Jr., *Dynamics of Physical Systems*, McGraw–Hill, New York, 1967.
- [5] Bryson, A. E., Jr., *Control of Spacecraft and Aircraft*, Princeton Univ. Press, Princeton, NJ, 1994.
- [6] Kailath, T., *Linear Systems*, Prentice–Hall, Upper Saddle River, NJ, 1980.
- [7] Ogata, K., *Modern Control Engineering*, Prentice–Hall, Upper Saddle River, NJ, 1970.
- [8] Khalil, H. K., *Nonlinear Systems*, 3rd ed., Prentice–Hall, Upper Saddle River, NJ, 2002.
- [9] Graham, D., and McRuer, D., *Analysis of Nonlinear Control Systems*, Wiley, 1961.

Appendix B*

Gas Turbine Engine Performance and Operability

B.1 Introduction

Unlikely as it might seem, turbine or jet engines merely obeying the conservation laws are perfectly capable of destroying themselves, or at least rendering themselves inoperable. Indeed, they do this more frequently than is desirable. The primary function of engine controls is therefore to allow the turbine engine to provide thrust responsively, reliably, and safely. The corresponding purpose of Appendix B is to provide a clear understanding of the underlying principles of the engine controls that make this possible and to develop and demonstrate the analytical relationships that they employ. This will be done by means of the most basic possible turbine engine cycle, namely, the single spool, fixed exhaust nozzle-area turbojet, having no compressor bleed, turbine cooling, or power TO. The steady-state flow will also be treated as a calorically perfect gas. The remarkable fact is that the behavior of this simple cycle reproduces the essential behavior of engines as complex as the afterburning, mixed-flow turbofan engine used in fighter aircraft and the high-bypass turbofan engine used in subsonic transport aircraft. The results should therefore be taken seriously.

To understand the control of gas turbine engines, one must also understand the operation of the engine. This appendix presents the basic thermodynamic, aerodynamic, and mechanical theory of engine operation and the limits they place on its performance based on [1, 2]. For ease of understanding, we will use the symbols and engine station numbering used in the industry for engine cycle analysis. Figure B.1 shows the cross section of a single-spool turbojet engine with engine stations numbered using industry standard.

The single-spool turbojet engine will be analyzed in this appendix to get a fundamental understanding of engine operation and performance with variation

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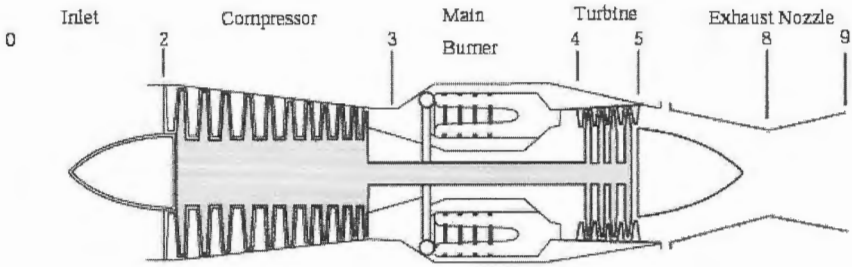


Fig. B.1 Single-spool turbojet engine.

in flight condition (altitude and Mach number) and *power level angle* (PLA). The PLA is the power request that the pilot makes of the engine and will be referred to as throttle in this appendix. The basic results of this analysis are applicable to engines as complex as the afterburning, mixed-flow turbofan engine used in fighter aircraft and the high-bypass turbofan engine used in subsonic transport aircraft.

B.2 Operational Envelopes and Standard Atmosphere

Each engine type will operate only within a certain range of altitudes and Mach numbers (velocities). Similar limitations in velocity and altitude exist for airframes. It is necessary, therefore, to match airframe and propulsion system capabilities. Figure B.2 shows the approximate velocity and altitude limits, or *corridor of flight*, within which airlift vehicles can operate. The corridor is

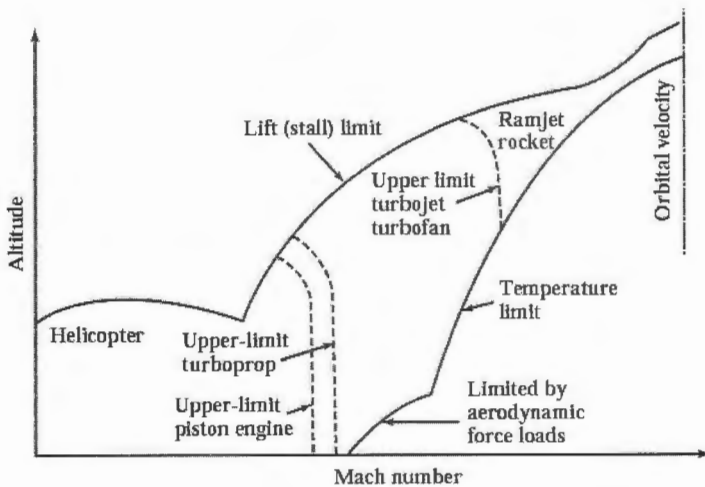


Fig. B.2 Flight limits.

bounded on the left by the *lift limit*, and on the right by a *temperature limit* and an *aerodynamic force limit*. The lift limit is determined by the maximum level-flight altitude at a given velocity. The temperature limit is set by the structural thermal limits of the material used in construction of the aircraft. At any given altitude, the maximum velocity attained is temperature limited by aerodynamic heating effects. At lower altitudes, velocity is limited by aerodynamic force loads rather than by temperatures. The location of these corridors and limits obviously depend on the assumed properties of the atmosphere, including temperature, pressure, and density. Figure B.2 is based on the U.S. Standard Atmosphere; more of which will be said in the following.

The operating regions of all aircraft lie within the flight corridor. The operating region of a particular aircraft within the corridor is determined by aircraft design, but it is a very small portion of the overall corridor. Superimposed on the flight corridor in Fig. B.2 are the operational envelopes of various powered aircraft. The operational limits of each propulsion system are determined by limitations of the components of the propulsion system and are shown in Fig. B.3 for gas turbine engines.

The analysis presented in this appendix uses the properties of the atmosphere to determine engine performance. Because these properties vary with location, season, time of day, etc., we will use the U.S. Standard Atmosphere [3] to give a known foundation for our analysis. Values of the pressure P , temperature T , and density ρ are typically given in dimensionless ratios of the properties at altitude to its value at sea level (SL), the reference value. The dimensionless ratios of pressure, temperature, and density are given the symbols δ , θ , and σ , respectively. These ratios are defined as follows:

$$\delta \equiv \frac{P}{P_{\text{std}}} \quad (\text{B.1})$$

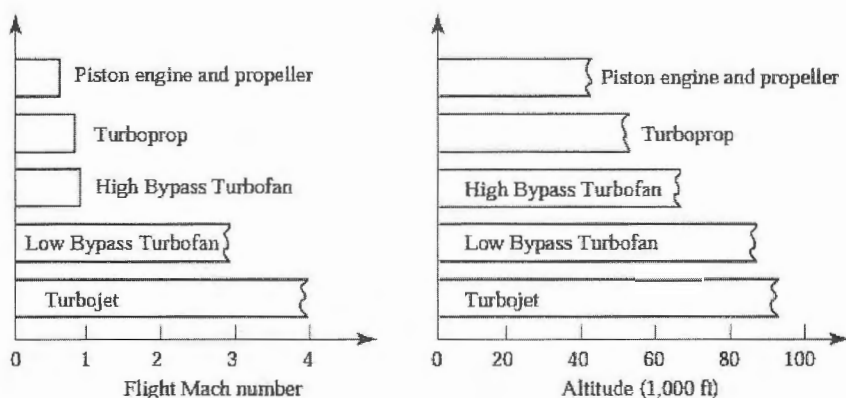


Fig. B.3 Engine operational limits.

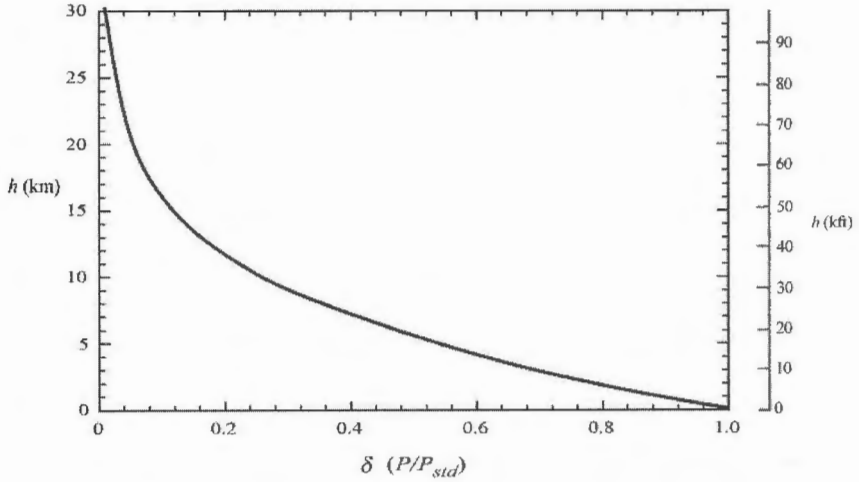


Fig. B.4 Pressure variation for U.S. Standard Atmosphere.

$$\theta \equiv \frac{T}{T_{std}} \quad (\text{B.2})$$

$$\sigma \equiv \frac{\rho}{\rho_{std}} \quad (\text{B.3})$$

where P_{std} is 14.696 psia (101.3 kPa), T_{std} is 518.7°R (288.15K), and ρ_{std} is 0.07647 lbm/ft³ (1.225 kg/m³).

Figure B.4 shown the variation of dimensionless pressure δ with altitude for this standard atmosphere. Figure B.5 shown the variation of dimensionless temperature θ with altitude for four model temperature variations: cold day, standard day, tropic day, and hot day [3–5]. The standard day is normally used in analysis. Other model days have special use such as tropic day for naval aircraft. The dimensionless density σ can be obtained from the dimensionless pressure δ of Fig. B.4 and the dimensionless temperature θ of Fig. B.5 using the dimensionless form of the perfect-gas equation

$$\sigma = \frac{\delta}{\theta} \quad (\text{B.4})$$

B.3 Basic Thermodynamics of Perfect Gas

The gas flowing through the gas turbine will be modeled as a perfect gas where the pressure P , density ρ , temperature T , and gas constant R are related by

$$P = \rho RT \quad (\text{B.5})$$

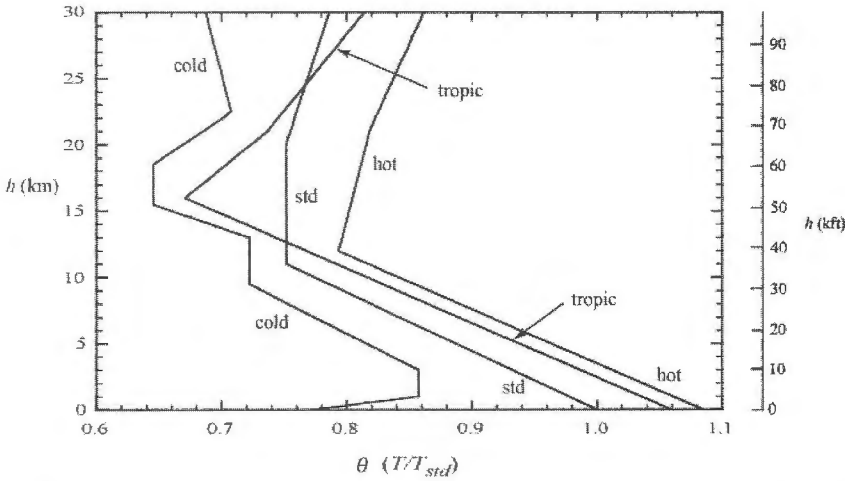


Fig. B.5 Temperature variation for U.S. Atmospheres (standard, cold, tropic, and hot).

and the internal energy u is only a function of temperature as represented by

$$u = u(T) \quad (\text{B.6})$$

For thermodynamic analysis with fluids flowing into and out of devices, the thermal energy associated with the fluid is the enthalpy h is defined as

$$h \equiv u + \frac{P}{\rho} \quad (\text{B.7})$$

From Eqs. (B.5) and (B.6), we see that the enthalpy is also only a function of temperature or

$$h = h(T) \quad (\text{B.8})$$

An infinitesimal change in internal energy du can be expressed as

$$du = c_v dT \quad (\text{B.9a})$$

where

$$c_v \equiv \frac{du}{dT} \quad (\text{B.9b})$$

and c_v is called the specific heat at constant volume. Likewise, an infinitesimal change in enthalpy dh can be expressed as

$$dh = c_p dT \quad (\text{B.10a})$$

where

$$c_p \equiv \frac{dh}{dT} \quad (\text{B.10b})$$

and c_p is called the specific heat at constant pressure. The ratio of specific heats shows up in the following analysis and is defined as

$$\gamma \equiv \frac{c_p}{c_v} \quad (\text{B.10c})$$

The analysis that follows can be simplified by using a perfect-gas model where the specific heats are constant (also called a calorically perfect gas) and using suitable average values of c_p , c_v , and γ that are representative of the temperature range being analyzed. Typical values of R , c_p , c_v , and γ for the calorically perfect gases used in this analysis are given in Table B.1. For analysis of the compression process through the engine inlet and compressor, one would use c_p of 0.240 Btu/lbm-R and γ of 1.4 as average values.

The changes of the fluid properties in a gas turbine engine are caused by energy conversion and transfer processes that occur there. These processes obey the *first and second laws of thermodynamics*, which are covered here. When analyzing the flow of gases through gas turbine components, we include changes only in internal energy and kinetic energy because all others (e.g., potential, electromagnetic, etc.) are negligible.

The *first law of thermodynamics* or *conservation of energy* can be written for a control volume in space (see Fig. B.6) with steady-state conditions (no variation in properties with respect to time) as

$$\sum_{\text{in}} \left\{ \dot{m} \left(h + \frac{V^2}{2g_c} \right) \right\} + \dot{Q}_{\text{in}} = \sum_{\text{out}} \left\{ \dot{m} \left(h + \frac{V^2}{2g_c} \right) \right\} + \dot{W}_{\text{out}} \quad (\text{B.11})$$

Table B.1 Typical gas properties for air and products of combustion

Gas	Temperature range, °R	R , ft-lbm lbm-°R	R , kJ kg-K	c_p , Btu lbm-°R	c_p , kJ kg-K	c_v , Btu lbm-°R	c_v , kJ kg-K	γ
Air	400-1000	53.34	0.286	0.240	1.004	0.171	0.718	1.4
Air	900-1700	53.34	0.286	0.258	1.079	0.189	0.794	1.36
Products of combustion ^a	1200-1800	53.40	0.287	0.276	1.157	0.207	0.870	1.33
Products of combustion ^a	1500-2500	54.09	0.290	0.295	1.234	0.226	0.945	1.3

^aFrom air and JP-8 fuel.

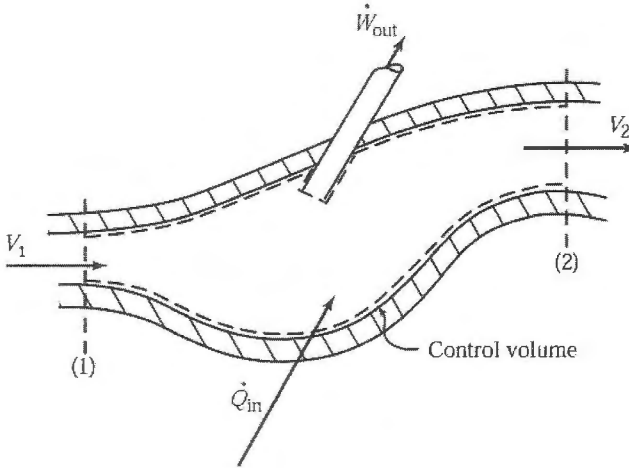


Fig. B.6 Example control volume with single inlet and exit.

where the summations refer to the flow of energy *into* and *out of* the volume with the flow of matter $\dot{m}$, $\dot{Q}_{in}$ is the rate of heat transfer (disorganized transfer of energy by a heat interaction) into the control volume, and $\dot{W}_{out}$ is the power transfer (organized transfer of energy by a work interaction) out of the control volume. Also note that g_c is a unit constant that depends on the unit system, $g_c = 1$ for Systeme International (SI) units, and $g_c = 32.174 \text{ ft-lbf/lbm-s}^2$ for British engineering units.

For a control volume with a single inlet 1 and exit 2 (see Fig. B.6), the steady flow energy equation of Eq. (B.11) becomes

$$\left(h + \frac{V^2}{2g_c} \right)_1 + q_{in} = \left(h + \frac{V^2}{2g_c} \right)_2 + w_{out} \quad (\text{B.12})$$

where $q_{in} = \dot{Q}_{in}/\dot{m}$ is the energy transfer per unit mass *into* the control volume by a heat interaction and $w_{out} = \dot{W}_{out}/\dot{m}$ is the energy transfer per unit mass *out of* the control volume by a work interaction. We will use this form of the first law in most of the analysis that follows.

The *second law of thermodynamics* places limits on what processes are possible. The ideal or model process used for most engine components (except ones where combustion occurs) is one where there is no heat interaction (also called an adiabatic process) and no losses caused by friction, shock waves, etc. The relation between inlet 1 and exit 2 properties for this ideal process with constant specific heats c_p and c_v (or suitable average values for the process) can be written as

$$\frac{P_2}{P_1} = \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}} \quad (\text{B.13})$$

and

$$\frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1} \right)^{\frac{1}{\gamma-1}} \quad (\text{B.14})$$

B.4 Total Properties and Mass Flow Parameter

The total temperature T_t , total pressure P_t , and mass flow parameter three very useful properties of the gas in a gas turbine engine that are design and analysis.

B.4.1 Stagnation or Total Temperature

The steady flow energy equation of Eq. (B.12) is

$$q_{\text{in}} - w_{\text{out}} = \left(h + \frac{V^2}{2g_c} \right)_2 - \left(h + \frac{V^2}{2g_c} \right)_1$$

For a perfect gas with constant specific heat, this becomes

$$q_{\text{in}} - w_{\text{out}} = c_p \left(T + \frac{V^2}{2g_c c_p} \right)_2 - c_p \left(T + \frac{V^2}{2g_c c_p} \right)_1$$

The quantities $h + V^2/(2g_c)$ and $T + V^2/(2g_c c_p)$ in these equations are called the *stagnation* or *total enthalpy* h_t and the *stagnation* or *total temperature* T_t , respectively. Thus

Total enthalpy:

$$h_t \equiv h + \frac{V^2}{2g_c} \quad (\text{B.15})$$

Total temperature:

$$T_t \equiv T + \frac{V^2}{2g_c c_p} \quad (\text{B.16})$$

The temperature T is sometimes called the *static temperature* (thermodynamic temperature) to distinguish it from the total temperature T_t . When $V = 0$, the static

and total temperatures are identical. From the preceding definitions, it follows that, for a perfect gas with constant specific heat,

$$\Delta h_t = c_p \Delta T_t$$

Using these new definitions, we see that the steady flow energy equation in the absence of gravity effects becomes

$$q_{\text{in}} - w_{\text{out}} = h_{t2} - h_{t1} \quad (\text{B.17})$$

or, for a perfect gas with constant specific heat,

$$q_{\text{in}} - w_{\text{out}} = c_p (T_{t2} - T_{t1}) \quad (\text{B.18})$$

If $q_{\text{in}} - w_{\text{out}} = 0$, we see from Eqs. (B.17) and (B.18) that $h_{t2} = h_{t1}$ and that, for a perfect gas with constant specific heat, $T_{t2} = T_{t1}$.

B.4.2 Stagnation or Total Pressure

If a flowing perfect gas is slowed to zero speed when $q_{\text{in}} = w_{\text{out}} = 0$, the gas attains the same final stagnation or total temperature whether it is brought to rest with or without the presence of frictional effects. The highest pressure is obtained when this slowing occurs without irreversible effects (such as friction, heat transfer, shock waves, etc.) and is referred to as the *stagnation* or *total pressure* P_t . We can use Eq. (B.13) for this process with a gas having constant specific heats and can be rewritten as

$$\frac{P_t}{P} = \left(\frac{T_t}{T} \right)^{\frac{\gamma}{\gamma-1}} \quad (\text{B.19})$$

B.4.3 T/T_t and P/P_t as Functions of Mach Number

The speed of sound a in a perfect gas is given by

$$a = \sqrt{\gamma g_c R T} \quad (\text{B.20})$$

and the Mach number is defined as the velocity divided by the local speed of sound or

$$M \equiv \frac{V}{a} \quad (\text{B.21})$$

Using the preceding relationship, we can write the Mach number as

$$M^2 = \frac{V^2}{\gamma g_c R T}$$

With the help of this expression for the Mach number, we can obtain many useful relations that give gas flow property ratios in terms of the flow Mach number alone. Two such relations for T/T_t and P/P_t are

$$\frac{T}{T_t} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1} \quad (\text{B.22})$$

$$\frac{P}{P_t} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-\gamma/(\gamma-1)} \quad (\text{B.23})$$

Equations (B.22) and (B.23) appear graphically in Fig. B.7 and are given in the compressible flow table and computer programs for user input γ value. These equations show that for each freestream Mach number (hence, for each flight Mach number of an airplane) the ratios P/P_t and T/T_t have unique values.

Both Fig. B.7 and the corresponding equations provide Mach numbers and ambient temperatures for known values of P , P_t , and T_t . For example, we are given the following in-flight measurements:

$$P = 35.4 \text{ kPa}, \quad T_t = 300 \text{ K}, \quad \text{and} \quad P_t = 60.0 \text{ kPa}$$

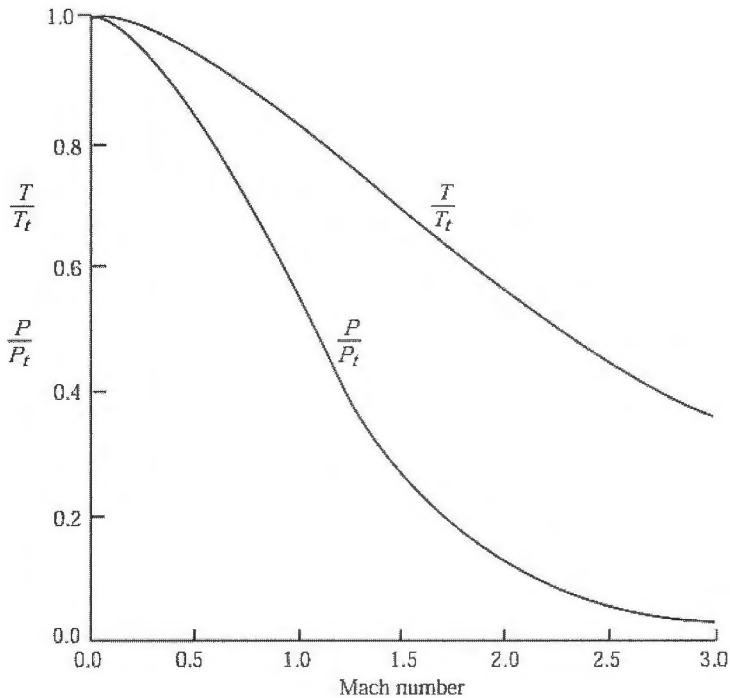


Fig. B.7 P/P_t and T/T_t vs Mach number ($\gamma = 1.4$).

From these data, $P/P_t = 0.59$. If we enter Fig. B.6 with this value of P/P_t , we find $M = 0.9$ and $T/T_t = 0.86$. Then we obtain the ambient temperature by using $T = (T/T_t) T_t = 0.86(300) = 258 \text{ K}$.

Figure B.7 shows that in a sonic ($M = 1.0$) stream of gas with $\gamma = 1.4$,

$$\frac{P}{P_t} = 0.528 \quad \text{and} \quad \frac{T}{T_t} = 0.833$$

and for supersonic flow,

$$\frac{P}{P_t} < 0.528 \quad \text{and} \quad \frac{T}{T_t} < 0.833$$

Consider the one-dimensional steady flow of a gas in a duct with T_t and P_t constant at all stations along the duct and the flow accelerated from subsonic to supersonic speeds. This means a total temperature probe will measure the same value of T_t at each duct station, and a total pressure probe will measure the same value of P_t at each station. The compressible flow can be categorized as follows:

Subsonic:

$$T > 0.883 T_t \quad P > 0.528 P_t$$

Sonic:

$$T = 0.883 T_t \quad P = 0.528 P_t$$

Supersonic:

$$T < 0.883 T_t \quad P < 0.528 P_t$$

The thermodynamic properties at the point where $M = 1$ are denoted by P^* , T^* , V^* , etc. (read as P star, etc.). In addition, the cross-sectional flow area at the $M = 1$ point is indicated by A^* . The magnitude of A^* is determined by the relation

$$A^* = \frac{\dot{m}}{\rho^* V^*} \quad (\text{B.24})$$

B.4.4 Mass Flow Parameter

The stream properties at any general station of a perfect gas with constant specific heats are related by Eqs. (B.5), (B.16), (B.19), (B.20), (B.22), (B.23), and the one-dimensional mass flow equation

$$\dot{m} = \rho A V$$

The mass flow per unit area at duct area A in the flow of a calorically perfect gas is given by the expression

$$\frac{\dot{m}}{A} = \rho V = \frac{P V}{R T} = \frac{V}{\sqrt{\gamma g_c R T}} \frac{P \sqrt{\gamma g_c}}{\sqrt{R T}} = M \sqrt{\frac{\gamma g_c}{R}} \frac{P}{P_t} \sqrt{\frac{T_t}{T}} \frac{P_t}{\sqrt{T_t}}$$

where the perfect-gas equation of state [Eq. (B.5)], the perfect-gas speed of sound equation [Eq. (B.20)], and the definition of Mach number have been used. Replacing the static/total property ratios with Eqs. (B.22) and (B.23) and rearranging, we obtain the grouping defined as the MFP:

$$\text{MFP} \equiv \frac{\dot{m}}{A} \frac{\sqrt{T_t}}{P_t} \quad (\text{B.25})$$

$$\text{MFP}(M) = \frac{\dot{m}}{A} \frac{\sqrt{T_t}}{P_t} = \sqrt{\frac{\gamma g_c}{R}} M \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma + 1}{2(\gamma - 1)}} \quad (\text{B.26})$$

Thus the MFP is a unique function of the Mach number in a calorically perfect gas. Values of the mass flow parameter are plotted in Fig. B.8 for $\gamma = 1.4$ and 1.3. From Fig. B.8, one sees that the maximum value of the mass flow parameter occurs when the Mach number is unity. Thus, for a given total temperature and pressure T_t and P_t , the maximum mass flow rate per area corresponds to a flow Mach number of 1. Because the flow rate through a given area is maximum when $M = 1$, the flow is said to be “choked” at this condition.

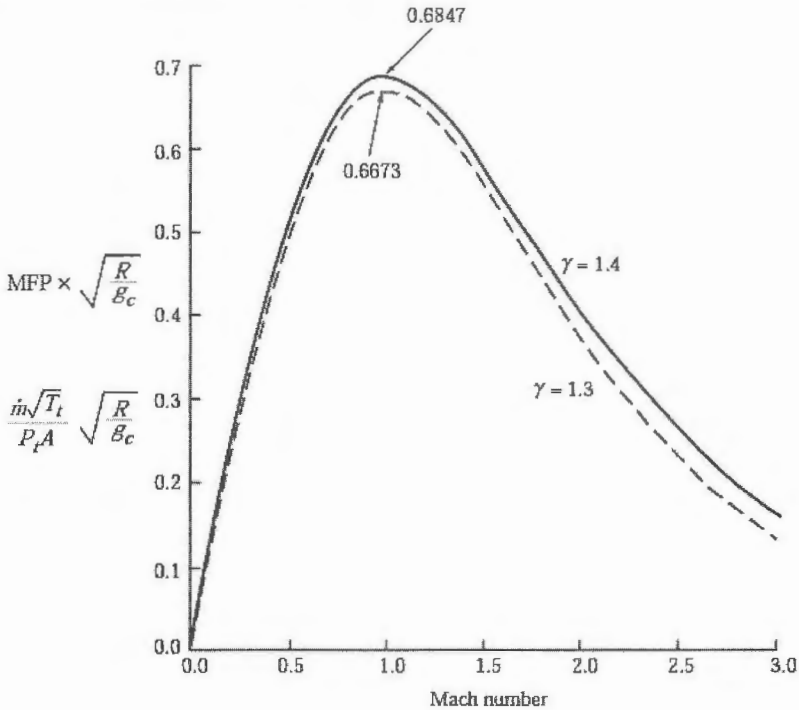


Fig. B.8 Mass flow parameter vs Mach number ($\gamma = 1.4$ and 1.3).

B.5 Dimensionless/Corrected Parameters and Component Maps

Performance (off-design) cycle analysis requires a model for the behavior of each engine component over its actual range of operation. The more accurately and completely this is done, the more reliable will be the computed results. Even though the approach (constant efficiency of rotating components and constant total pressure ratio of the other components) used in this textbook gives answers that are perfectly adequate for this type of preliminary design, it is important to know that the usual industrial practice is to use data or correlations having more accuracy and definition in the form of steady-state component “maps.” The material in this section will explain the role and usefulness of these maps and will demonstrate that the approach of this textbook is essentially correct. Indeed, the principal values of the maps are to improve the understanding of component behavior and to slightly increase the accuracy of the results.

The first step is to use dimensional analysis to identify correlating parameters that allow data taken under one set of conditions to be extended to other conditions. This is necessary because it is always impractical to accumulate experimental data for the bewildering number of possible operating conditions and because it is often impossible to reach many of them in a single, affordable facility.

The quantities of pressure and temperature are normally made dimensionless by dividing each by their respective standard sea-level static value. The dimensionless pressure and temperature are represented by δ and θ [Eqs. (B.1) and (B.2)], respectively. When total (stagnation) properties are nondimensionalized, a subscript is used to indicate the station number of that property. The only static properties made dimensionless are freestream, the symbols for which carry no subscripts. Thus,

$$\delta_i \doteq \frac{P_{ti}}{P_{std}} \quad (\text{B.27})$$

and

$$\theta_i \doteq \frac{T_{ti}}{T_{std}} \quad (\text{B.28})$$

where $P_{std} = 2116.2 \text{ lb/ft}^2$ (101.3 kPa) and $T_{std} = 518.69^\circ\text{R}$ (288.15 K).

Dimensional analysis of engine components yields many useful dimensionless and/or modified component performance parameters. Some examples of these are the compressor pressure ratio, adiabatic efficiency, Mach number at the engine face, ratio of the blade (tip) speed to the speed of sound, and the Reynolds number [2].

The “corrected mass flow rate” at engine station i used in this analysis is defined as

$$\dot{m}_{ci} \doteq \dot{m}_i \frac{\sqrt{\theta_i}}{\delta_i} \quad (\text{B.29})$$

and is related to the engine face Mach number. The “corrected engine speed” is defined as

$$N_{ci} \doteq \frac{N}{\sqrt{\theta_i}} \quad (\text{B.30})$$

and is proportional to the blade Mach number for a compressor of fixed geometry.

There is another interpretation of the corrected mass flow rate that you can find intuitively appealing. Starting from the definition of the mass flow parameter (B.25), substituting Eqs. (B.27) and (B.28) and rearranging, you will find that

$$\dot{m}_{ci} = \frac{P_{\text{std}} A_i}{\sqrt{T_{\text{std}}}} \text{MFP}_i \quad (\text{B.31})$$

Thus, $\dot{m}_{ci}$ is the amount of mass that would flow under standard conditions if A_i and MFP_i (i.e., γ and M_i) were fixed. In other words, it is the mass flow that would be measured if a given machine were tested at standard conditions with the critical similarity parameter M_i kept equal to the desired operating value.

Another corrected parameter used for engine performance is the corrected fuel flow rate for the main burner that is defined as

$$\dot{m}_{fc} = \frac{\dot{m}_f}{\delta_2 \sqrt{\theta_2}} \quad (\text{B.32})$$

where $\dot{m}_f$ is the fuel flow rate into the main burner. This is especially useful in understanding engine controls where the required corrected fuel flow rate $\dot{m}_{fc}$ is a known function of the corrected engine speed N_c . See Sec. B.6.7 for example.

It must be understood that this selection of parameters represents a first approximation to the complete set necessary to reproduce nature. This collection would not reflect, for example, the effects of viscosity (Reynolds number), humidity, or gas composition. Nevertheless, it is extremely useful and has become the propulsion community standard. When required, the effects of the mission parameters are supplied by “adjustment factors” that are based on experiment and analysis. Please note also the appearance, for the first time, of rotational speed N , a quantity of obvious significance to the structural designer.

B.5.1 Fan and Compressor Performance Maps

The performance map of a compressor or fan is normally presented using the following performance parameters: total pressure ratio, corrected mass flow rate, corrected engine speed, and adiabatic efficiency. Variations in the axial-flow velocity in response to changes in pressure cause the multistage compressor to have quite different mass flow vs pressure ratio characteristics than one of its stages. The performance map of a typical axial-flow compressor is shown in Fig. B.9.

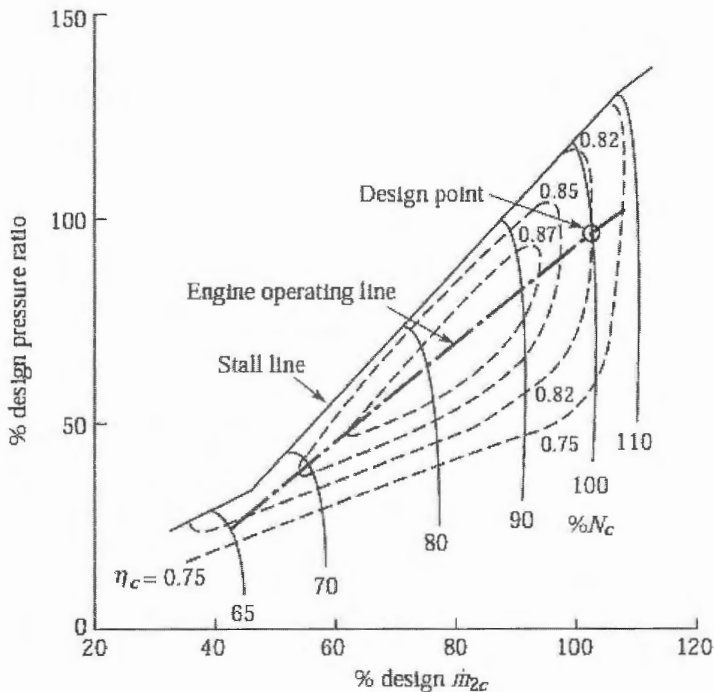


Fig. B.9 Typical compressor map.

A limitation on fan and compressor performance of special concern is the stall or surge line. Steady operation above the line is impossible, and entering the region even momentarily is dangerous to the engine and aircraft. As will be shown in the next section, the steady-state *engine operating line* is determined by the mass and power balances with the turbine.

B.5.2 Turbine Maps

In a turbine, the entry stationary airfoils, often called inlet guide vanes or nozzles, expand the entering flow and discharge it into rotating airfoils, known as rotors or blades. The flow of gas through the turbine nozzles has much in common with the flow of a compressible fluid through an exhaust nozzle, including choking at the minimum area when the backpressure is below the critical value. Therefore, the flow through the turbine nozzles controls the flow when they are choked, and the mass flow rate depends only on the inlet total pressure P_{t4} and the inlet total temperature T_{t4} . However, the flow through the turbine nozzles is a function of the turbine pressure ratio P_{t5}/P_{t4} and the turbine inlet total temperature T_{t4} when the nozzles are operating at a subcritical pressure ratio and are not choked. Conversely, the flow through the turbine nozzles depends only upon the inlet total pressure P_{t4} and the inlet total temperature T_{t4} , once choking occurs.

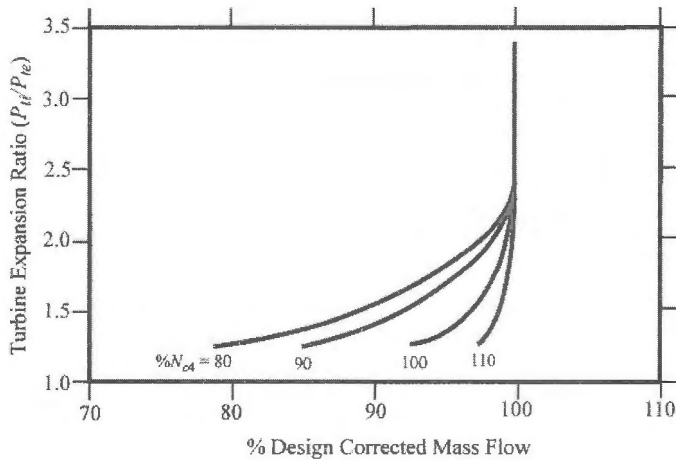


Fig. B.10 Turbine map.

The work extraction of modern gas turbines is usually so large, and P_{t5}/P_{t4} so small, that the nozzles are usually choked for the design point and the surrounding region. If not, the throat Mach number is sufficiently close to one that the mass flow parameter approximates that of choking [1]. The analytical convenience this makes possible will be capitalized upon in the next section.

The parameters that are normally used to express the performance of a turbine are total pressure ratio, corrected mass flow rate, corrected engine speed, and adiabatic efficiency.

Figure B.10 shows the turbine total-pressure expansion ratio P_{t4}/P_{t5} plotted as a function of the corrected mass flow rate ($\dot{m}_{c4} = \dot{m}_4 \sqrt{\theta_4}/\delta_4$) and corrected mechanical speed ($N_{c4} = N/\sqrt{\theta_4}$). The maximum flow of gas that can be accommodated by the nozzles when choked is clearly evident.

For reasons of size, it is desirable that the turbine mass flow rate per annulus area be as large as possible, which also points to choking and a high total pressure expansion ratio P_{t4}/P_{t5} . Because the mass flow rate is nearly independent of speed for the pressure ratios of interest, all speed characteristics collapse onto a single line when plotted vs the turbine pressure ratio P_{t4}/P_{t5} , and the turbine has the mass flow characteristics of a choked nozzle. So that only one map be required to show the performance of a turbine, the abscissa can be taken as the corrected mass flow rate times the corrected engine speed; thus, a separate plot for the turbine adiabatic efficiency is not required. A map for a typical 50% reaction (at mean radius) single-stage turbine is shown in this format in Fig. B.11.

Because the turbine adiabatic efficiency does not vary as rapidly with off-design variations as in a compressor, the turbine characteristic can be approximated for preliminary design calculations by a constant adiabatic turbine efficiency η_t and a choked mass flow characteristic. These are the characteristics that were used in the performance or off-design analysis of [1, 2].

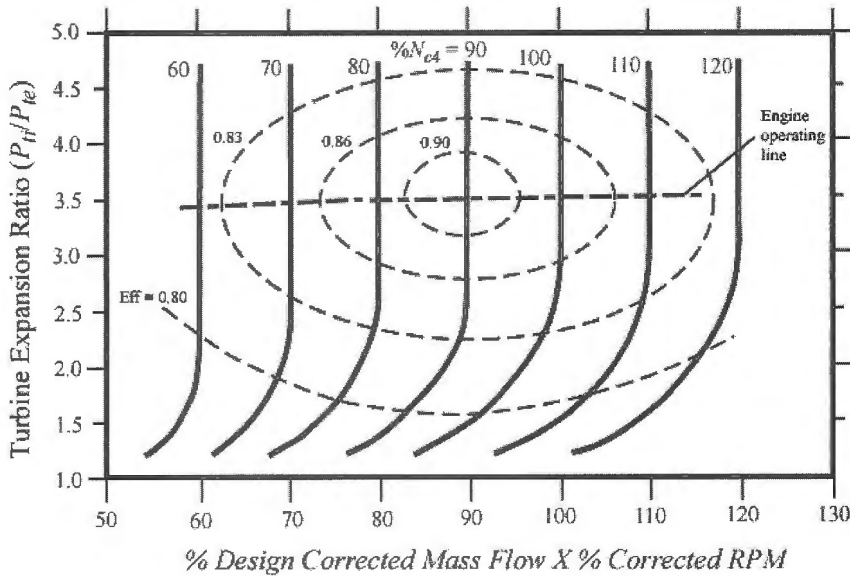


Fig. B.11 Typical turbine map.

B.6 Basic Engine Performance Model

The performance of a gas turbine engine depends on the operation of its gas generator (combination of compressor, main burner, and turbine). In this section, algebraic expressions for the steady-state performance characteristics of a single-spool turbojet engine (see Fig. B.1) are developed. For ease of analysis, we assume that the flow is steady, consisting of a gas with constant specific heat upstream of the main burner (use symbols c_{pc} and γ_c) and another gas with constant specific heat downstream of the main burner (use symbols c_{pt} and γ_t). In addition, we assume the total pressure ratio across the main burner P_{t4}/P_{t3} and the exhaust nozzle P_{t9}/P_{t5} are constant. The latter pressure ratios are caused by losses in their respective components and are therefore quite close to one.

Figures B.9 and B.11 show that the efficiencies of a typical compressor and turbine, respectively, do not change appreciably (only about 7%) over the main operating range of engine speed (70–100% corrected speed). Constant turbomachinery efficiencies were assumed as a substitute for component maps in the performance models of [1, 2]. The case of constant turbomachinery efficiency is analyzed in this section.

B.6.1 Conservation of Mass

We consider the flow through the simple single-spool turbojet engine described in Sec. B.1 having a fixed turbine nozzle throat area ($A_4 = \text{constant}$). The mass flow rate into the turbine is equal to the sum of the mass flow rate through

the compressor and the fuel flow rate into the main burner. Using the MFP, we can write

$$\dot{m}_2 + \dot{m}_f = (1 + f)\dot{m}_2 = \dot{m}_4 = \frac{P_{t4}A_4}{\sqrt{T_{t4}}} \text{MFP}(M_4)$$

where f is the fuel-to-air mass ratio in the main burner that has a maximum value of 0.067 for hydrocarbon fuels. With the help of Eqs. (B.29) and (B.31), the preceding equation yields the following expression for the compressor corrected mass flow rate:

$$\dot{m}_{c2} = \sqrt{\frac{T_{t2}}{T_{std}}} \frac{P_{std}}{P_{t2}} \frac{P_{t4}}{\sqrt{T_{t4}}} \frac{A_4}{1+f} \text{MFP}(M_4)$$

Noting that $P_{t4} = P_{t2} (P_{t3}/P_{t2})(P_{t4}/P_{t3})$ and $\theta_0 = T_{t2}/T_{std}$, we see that

$$\dot{m}_{c2} = \sqrt{\frac{\theta_0}{T_{t4}}} \frac{P_{t3}}{P_{t2}} \frac{P_{t4}}{P_{t3}} \frac{P_{std}A_4}{1+f} \text{MFP}(M_4) \quad (\text{B.33})$$

Equation (B.33) is a straight line on a compressor map for constant values of T_{t4}/θ_0 , A_4 , $(1+f)$, and M_4 . Lines of constant T_{t4}/θ_0 are plotted on a typical compressor map in Fig. B.12 for constant values of A_4 and $(1+f)$. Note that these lines start at a compressor pressure ratio of one and corrected mass flow rate of zero and are curved for low compressor pressure ratios because station 4 is unchoked. Station 4 chokes at a pressure ratio of about 2. At pressure ratios above 2, these lines are straight and appear to start at the origin (pressure ratio of zero and mass flow rate of zero). The lines of T_{t4}/θ_0 show the general pressure ratio and temperature ratio characteristics required to satisfy conservation of mass and are independent of the turbine. For a given T_{t4}/θ_0 , any point on that line will satisfy the steady-state mass conservation for engine stations 2 and 4. The actual operating point of the compressor depends on the turbine and exhaust nozzle.

Equation (B.33) can be written simply for the case when station 4 is choked ($M_4 = 1$, the normal situation in gas turbine engines) as

$$\dot{m}_{c2} = C_1 \frac{P_{t3}/P_{t2}}{\sqrt{T_{t4}/\theta_0}} \quad (\text{B.34a})$$

where

$$C_1 = \frac{P_{t4}}{P_{t3}} \frac{P_{std}A_4}{1+f} \text{MFP}(M_4 = 1) \quad (\text{B.34b})$$

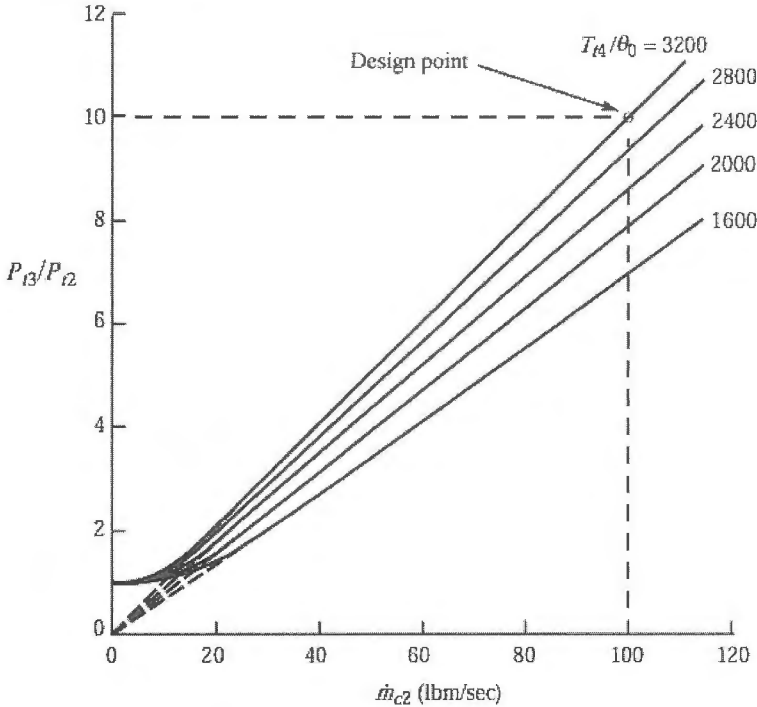


Fig. B.12 Compressor map with lines of constant T_{14}/θ_0 .

For an engine or gas generator, the specific relationship between the compressor pressure ratio and the corrected mass flow rate is called the *engine operating line* and depends on the characteristics of the turbine. The equation for this operating line is developed later in this section.

B.6.2 Turbine Characteristics

For a turbojet engine, the flow is choked ($M = 1$) in the turbine inlet guide vanes (station 4) and nearly always choked at the throat of the exhaust nozzle (station 8). Thus the corrected mass flow rate per unit area is constant at station 4 and

$$\dot{m}_4 = \frac{P_{t4} A_4}{\sqrt{T_{t4}}} \text{MFP}(M_4 = 1) \quad \dot{m}_8 = \frac{P_{t8} A_8}{\sqrt{T_{t8}}} \text{MFP}(M_8) \quad (i)$$

For a simple turbojet engine, the mass flow rate through the turbine is equal to that through the exhaust nozzle, or $\dot{m}_4 = \dot{m}_8$. Using Eq. (i) then, we have

$$\frac{\sqrt{T_{t8}/T_{t4}}}{P_{t8}/P_{t4}} = \frac{A_8}{A_4} \frac{\text{MFP}(M_8)}{\text{MFP}(M_4 = 1)}$$

Because the total pressure remain essentially constant between stations 5 and 8 and the total temperature is constant between these stations, we can rewrite the preceding equation as

$$\frac{\sqrt{T_{t5}/T_{t4}}}{P_{t5}/P_{t4}} = \frac{A_8}{A_4} \frac{\text{MFP}(M_8)}{\text{MFP}(M_4 = 1)} \quad (\text{B.35a})$$

From the definition of turbine efficiency η_t for a perfect gas with constant specific heats, we have

$$\eta_t = \frac{1 - T_{t5}/T_{t4}}{1 - (P_{t5}/P_{t4})^{(\gamma_t - 1)/\gamma_t}} \quad (\text{B.35b})$$

For choked flow at station 8 ($M_8 = 1$), constant flow area at stations 4 and 8, and constant turbine efficiency η_t , Equations (B.35a) and (B.35b) can be satisfied only by constant values of the turbine temperature ratio T_{t5}/T_{t4} and the turbine pressure ratio P_{t5}/P_{t4} . Thus, we have

$$T_{t5}/T_{t4} \text{ and } P_{t5}/P_{t4} \text{ constant for } M_8 = 1 \text{ and constant } A_4, A_8, \text{ and } \eta_t \quad (\text{B.36})$$

If the exhaust nozzle unchokes and/or the exhaust nozzle throat area is changed, then both T_{t5}/T_{t4} and P_{t5}/P_{t4} change. Decreasing the exhaust nozzle throat Mach number M_8 from choked condition increases both T_{t5}/T_{t4} and P_{t5}/P_{t4} . Increasing the exhaust nozzle throat area A_8 decreases both T_{t5}/T_{t4} and P_{t5}/P_{t4} . If the turbine efficiency η_t decreases, both T_{t5}/T_{t4} and P_{t5}/P_{t4} increase. The turbine power $\dot{W}_t$ is

$$\dot{W}_t = \dot{m}_4 c_{pt} (T_{t4} - T_{t5}) = \dot{m}_4 c_{pt} T_{t4} \left(\frac{1 - T_{t5}}{T_{t4}} \right) \quad (\text{B.37})$$

Note that the turbine power per unit mass flow $\dot{W}_t/\dot{m}_4$ is directly proportional to the main burner exit temperature T_{t4} . Also an increase in T_{t5}/T_{t4} will decrease the turbine power per unit mass flow and change the pumping characteristics of the gas generator.

B.6.3 Engine Operating Line

From the steady-state power balance between the compressor and turbine, we write

$$\dot{W}_c = \eta_m \dot{W}_t$$

where $\dot{W}_c$ is the compressor input power, $\dot{W}_t$ is the turbine output power, and η_m is the mechanical efficiency of the power shaft. This equation can be rewritten as

$$\dot{m}_2 c_{pc} (T_{t3} - T_{t2}) = \eta_m \dot{m}_2 (1 + f) c_{pt} (T_{t4} - T_{t5}) \quad (\text{B.38a})$$

or

$$\frac{T_{t3}}{T_{t2}} = 1 + \frac{T_{t4}}{T_{t2}} \frac{c_{pt}}{c_{pc}} \eta_m (1 + f) \left(\frac{1 - T_{t5}}{T_{t4}} \right) \quad (\text{B.38b})$$

where, using the compressor efficiency η_c , we have for the compressor pressure ratio

$$\frac{P_{t3}}{P_{t2}} = \left[1 + \eta_c \left(\frac{T_{t3}}{T_{t2}} - 1 \right) \right]^{\gamma_c / (\gamma_c - 1)} \quad (\text{B.38c})$$

Combining Eqs. (B.38b) and (B.38c) gives

$$\frac{P_{t3}}{P_{t2}} = \left\{ 1 + \frac{T_{t4}}{T_{t2}} \left[\eta_m \eta_c (1 + f) \left(\frac{1 - T_{t5}}{T_{t4}} \right) \frac{c_{pt}}{c_{pc}} \right] \right\}^{\gamma_c / (\gamma_c - 1)} \quad (\text{B.39})$$

where the term in the square brackets can be considered essentially a constant when T_{t5}/T_{t4} and η_c are constant. If we define a constant C_2 to be the terms within the square brackets or

$$C_2 = \eta_m \eta_c (1 + f) \left(\frac{1 - T_{t5}}{T_{t4}} \right) \frac{c_{pt}}{c_{pc}} \quad (\text{B.40})$$

then, using Eqs. (B.34a), (B.34b), (B.39), and (B.40) gives a simple algebraic equation for the compressor operating line that can be written as

$$\dot{m}_{c2} = C_1 \sqrt{C_2} \frac{P_{t3}/P_{t2}}{\sqrt{(P_{t3}/P_{t2})^{\gamma_c - 1} - 1}} \quad (\text{B.41})$$

We can plot the *engine operating line*, using Eq. (B.41), on the compressor map of Fig. B.12, giving the compressor map with operating line shown in Fig. B.13. This engine operating line shows that for each value of the temperature ratio T_{t4}/θ_0 there is one value of the compressor pressure ratio and corrected mass flow rate. One can also see that for a constant value of θ_0 , both the compressor pressure ratio and the corrected mass flow rate will increase with increases in throttle setting (increases in T_{t4}). In addition, when at constant T_{t4} , the compressor pressure ratio and corrected mass flow rate will decrease with increases in θ_0 because of higher speed and/or lower altitude {note: $\theta_0 = T_{t4}/T_{std} = (T_0/T_{std})[1 + (\gamma_c - 1)M_0^2/2] = \theta[1 + (\gamma_c - 1)M_0^2/2]$ }. The curving of the engine operating line in Fig. B.13 at pressure ratios below three is because of the exhaust nozzle being unchoked ($M_8 < 1$) at static or low-speed flight conditions ($M_0 \approx 0$), which increases the

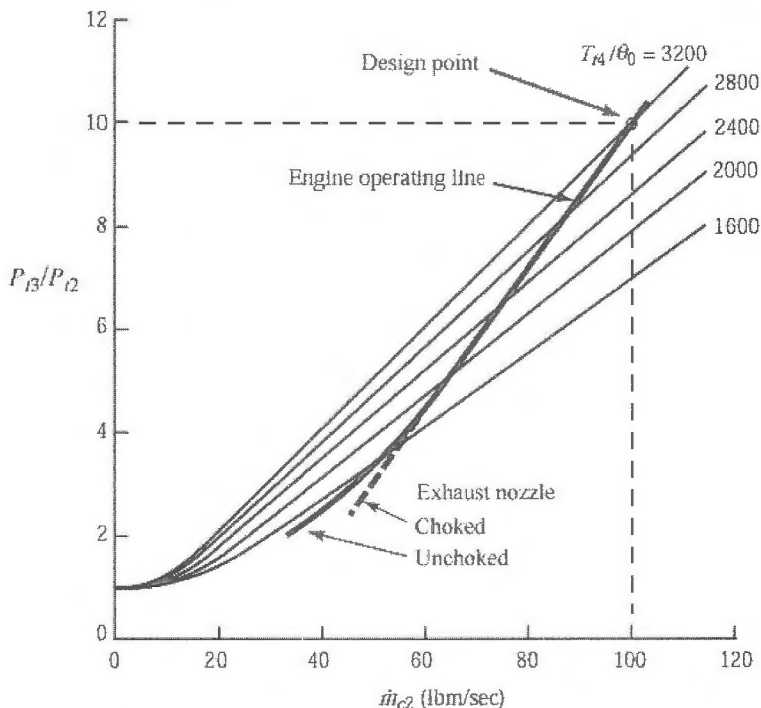


Fig. B.13 Compressor map with engine operating line.

value of T_5/T_{t4} (decreases C_2). At higher flight speeds, the inlet total pressure P_{t2} is higher, which keeps the exhaust nozzle choked at lower throttle settings, and the engine operating line is not curved (dashed line).

B.6.4 Engine Controls and Theta Break

The engine control system will control the gas-generator operation to keep the main-burner exit total temperature T_{t4} (to protect turbine life), the compressor pressure ratio P_{t3}/P_{t2} (to prevent compressor surge), rotational speed N (to protect rotating part life), compressor exit total temperature T_{t3} (to protect compressor life), and compressor exit total pressure P_{t3} (to prevent combustor rupture) from exceeding specific maximum values. Limiting T_{t4} and P_{t3}/P_{t2} has the most dominating effect and is included in the following analysis. An understanding of the influence of the engine control system on the compressor performance during changing flight conditions and throttle settings can be gained by recasting Eqs. (B.34) and (B.37) in terms of the dimensionless total temperature at station 0 (θ_0). We note that

$$T_{t0} = T_{std} \frac{T_{t0}}{T_{std}} = T_{std} \theta_0 \quad \text{and} \quad \theta_0 = \theta \left(1 + \frac{\gamma_c - 1}{2} M_0^2 \right) \quad (\text{B.42})$$

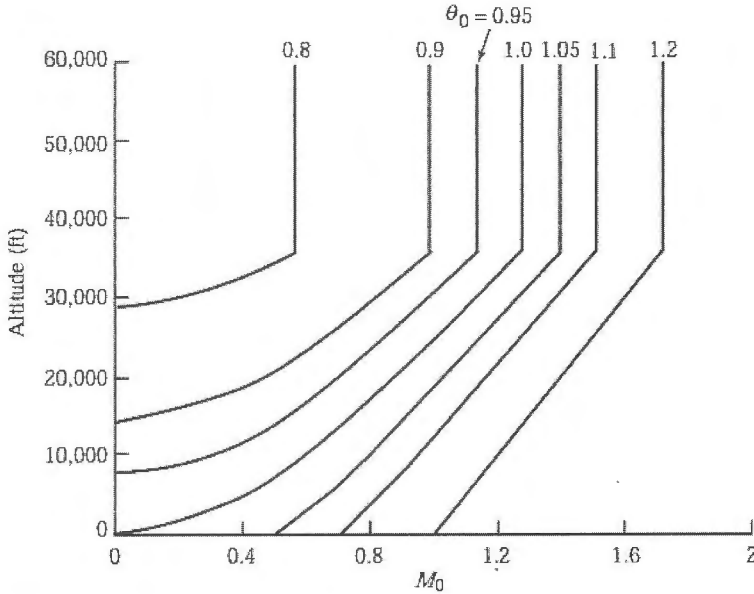


Fig. B.14 Graph shows θ_0 vs Mach number and altitude (standard day).

For ease of analysis, we use Eq. (B.42) to plot lines of θ_0 on coordinates of altitude vs Mach number. Figure B.14 gives contours of constant θ_0 for standard day atmosphere. Note that numerous combinations of Mach number and altitude give the same value of θ_0 . For example, sea-level static and 1.28 Mach at 40-kft altitude have $\theta_0 = 1.0$. Similarly, 1.0 Mach at sea level and 1.726 Mach at 40 kft altitude have $\theta_0 = 1.2$.

Using Eq. (B.42) and the fact that $T_{t2} = T_{t0}$, we can rewrite Eq. (B.39) as

$$\frac{P_{t3}}{P_{t2}} = \left(1 + \frac{T_{t4}}{\theta_0} K_1 \right)^{\gamma_c / (\gamma_c - 1)} \quad (\text{B.43})$$

where $K_1 = C_2/T_{std}$, a constant for most throttle settings. (The exhaust nozzle can unchoke at low throttle settings and low flight Mach number. Then C_2 is reduced because of increases in T_{t5}/T_{t4} .) Equation (B.43) is plotted in Fig. B.15 for a simple turbojet with $P_{t3}/P_{t2} = 15$ at $T_{t4} = 3200^\circ\text{R}$ and $\theta_0 = 1$, the maximum operating condition. The engine control system can keep the compressor pressure ratio from exceeding the maximum value of 15 ($[P_{t3}/P_{t2}]_{\max}$) and the main burner exit temperature from exceeding the maximum of 3200°R ($T_{t4 \max}$) by controlling the fuel flow rate into the main burner. The compressor pressure ratio P_{t3}/P_{t2} can be measured directly or inferred through the corrected compressor speed N_{c2} and the engine operating line (see Fig. B.9). Because T_{t4} is not measured directly, T_{t5} (the *exhaust gas temperature* or EGT) is measured, and T_{t4} inferred

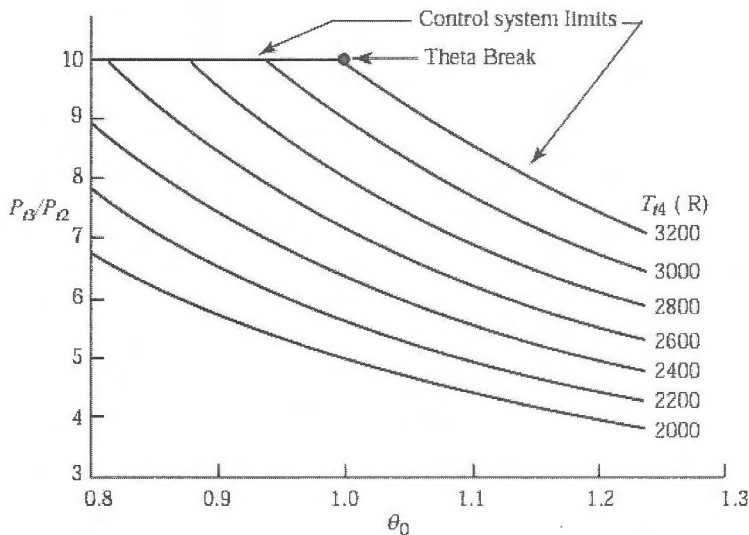


Fig. B.15 Compressor pressure ratio vs T_{t4} and θ_0 .

by noting that T_{t3}/T_{t4} is a constant at moderate and high throttle settings [see Eq. (B.36)]. From Eq. (B.43), we can see that T_{t4}/θ_0 is constant for a constant compressor pressure ratio.

$$T_{t4}/\theta_0 = \text{constant} \quad \text{for} \quad P_{t3}/P_{t2} = \text{constant} \quad (\text{B.44})$$

By using Eqs. (B.34a), (B.42), and (B.43), we can obtain the corrected mass flow rate as a function of T_{t4}/θ_0 , which can be expressed as

$$\dot{m}_{c2} = \left(\frac{\theta_0}{T_{t4}} \right)^{1/2} \left(1 + \frac{T_{t4}}{\theta_0} K_1 \right)^{\gamma_c/(\gamma_c - 1)} K_2 \quad (\text{B.45})$$

where $K_2 = C_1 \sqrt{T_{std}}$. A plot of Eq. (B.45) will be similar to Fig. B.15 for the compressor pressure ratio.

The unique point, so visually striking in Fig. B.15, at which the control logic must switch from limiting P_{t3}/P_{t2} to limiting T_{t4} , is known as the *theta break*, or $\theta_{0 \text{ break}}$. Returning briefly to Fig. B.15, you will find it very convenient to visualize that at full throttle (max PLA) and any point in the flight envelope to the left of the theta break $P_{t3}/P_{t2} = [P_{t3}/P_{t2}]_{\text{max}}$ and $T_{t4} < T_{t4 \text{ max}}$, whereas at any point to the right of the theta break $P_{t3}/P_{t2} < [P_{t3}/P_{t2}]_{\text{max}}$ and $T_{t4} = T_{t4 \text{ max}}$. The relationship of the instantaneous value of θ_0 to $\theta_{0 \text{ break}}$ has important consequences to engine cycle performance. On the one hand, when $\theta_0 < \theta_{0 \text{ break}}$ and $T_{t4} < T_{t4 \text{ max}}$, the specific thrust of the engine is less than its inherent material capabilities would make possible. On

the other hand, when $\theta_0 > \theta_{0 \text{ break}}$ and $P_{t3}/P_{t2} < [P_{t3}/P_{t2}]_{\text{max}}$, the specific fuel consumption is more than its inherent thermal efficiency would make possible.

The designer would therefore strongly prefer to have the engine always operate at or very near $\theta_0 = \theta_{0 \text{ break}}$, but this is impossible because every aircraft has a flight envelope with a range of θ_0 (see Fig. B.14). The best available compromise is to choose a $\theta_{0 \text{ break}}$ that provides the best balance of engine performance over the expected range of flight conditions.

Because early commercial and military aircraft primarily flew at or near $\theta_0 = 1$, they were successfully designed with $\theta_{0 \text{ break}} = 1$. Consequently, several generations of propulsion engineers took it for granted that aircraft engines always operated at $[P_{t3}/P_{t2}]_{\text{max}}$ and $T_{t4 \text{ max}}$ at maximum throttle (PLA) under standard sea-level static conditions. However, the special requirements of more recent aircraft such as the F-22 Raptor ($\theta_0 = 1.2$ at supercruise) have forced designers to select theta breaks different from 1.0. At maximum throttle (PLA), these new engines can operate either at $[P_{t3}/P_{t2}]_{\text{max}}$ or $T_{t4 \text{ max}}$ at standard sea-level static conditions, but never both.

B.6.5 Engine Fuel Flow

An expression for the engine fuel flow is obtained by applying energy conservation (first law of thermodynamics) to the main burner (see Fig. B.1). We have

$$\dot{m}_3 h_{t3} + \dot{m}_f \eta_b h_{\text{PR}} = \dot{m}_4 h_{t4} \quad (\text{i})$$

where $\dot{m}_3$ and $\dot{m}_4$ are the mass flow rates of gas entering and leaving the main burner, h_{t3} and h_{t4} are the total enthalpies of gas entering and leaving the main burner, $\dot{m}_f$ is the fuel flow rate entering the main burner, η_b is the combustion efficiency in the main burner, and h_{PR} is the heating value of the fuel (maximum energy released when fuel is burned at 100% efficiency). We note that the mass flow rate at engine station 4 is the sum of the gas mass flow rate at engine station 3 plus the fuel flow rate. For constant specific heats upstream and downstream of the main burner, Eq. (i) can be approximated and rewritten in terms of specific heats and total temperatures as

$$\dot{m}_3 c_{\text{pc}} T_{t3} + \dot{m}_f \eta_b h_{\text{PR}} = (\dot{m}_3 + \dot{m}_f) c_{\text{pt}} T_{t4}$$

Then the fuel-air ratio f for the main burner is

$$f \equiv \frac{\dot{m}_f}{\dot{m}_3} = \frac{c_{\text{pt}} T_{t4} - c_{\text{pc}} T_{t3}}{\eta_b h_{\text{PR}} - c_{\text{pt}} T_{t4}} \quad (\text{B.46a})$$

And the main-burner corrected fuel flow rate for $\dot{m}_2 = \dot{m}_3$ is

$$\dot{m}_{\text{fc}} \equiv \frac{\dot{m}_f}{\delta_2 \sqrt{\theta_2}} = \dot{m}_{c2} \frac{c_{\text{pt}} T_{t4} / \theta_0 - c_{\text{pc}} T_{t3} / \theta_0}{\eta_b h_{\text{PR}} - c_{\text{pt}} T_{t4}} \quad (\text{B.46b})$$

B.6.6 Variation in Engine Speed

The change in total enthalpy across a fan or compressor of fixed geometry is proportional to the rotational speed N squared [1]. For a gas with constant specific heats, we can write for the compressor

$$T_{t3} - T_{t2} = K_1 N^2 \quad (i)$$

From the work balance between compressor and turbine of Eq. (B.38a) and Eq. (i) in sec. B.6.5, we have

$$T_{t3} - T_{t2} = \eta_m (1 + f) \frac{c_{pt}}{c_{pc}} T_{t4} \left(\frac{1 - T_{t5}}{T_{t4}} \right) = K_1 N^2 \quad (ii)$$

For most of the engine operation, $\eta_m(1+f)c_{pt}/c_{pc}$, T_{t5}/T_{t4} , and K_1 are essentially constants. Thus Eq. (ii) can be rewritten as

$$N \propto \sqrt{T_{t4}} \quad (B.47)$$

For steady-state operation, the engine speed N is a measure of the total temperature at station 4 T_{t4} . Figure B.16 shows the variation in engine speed N , T_{t4} , and

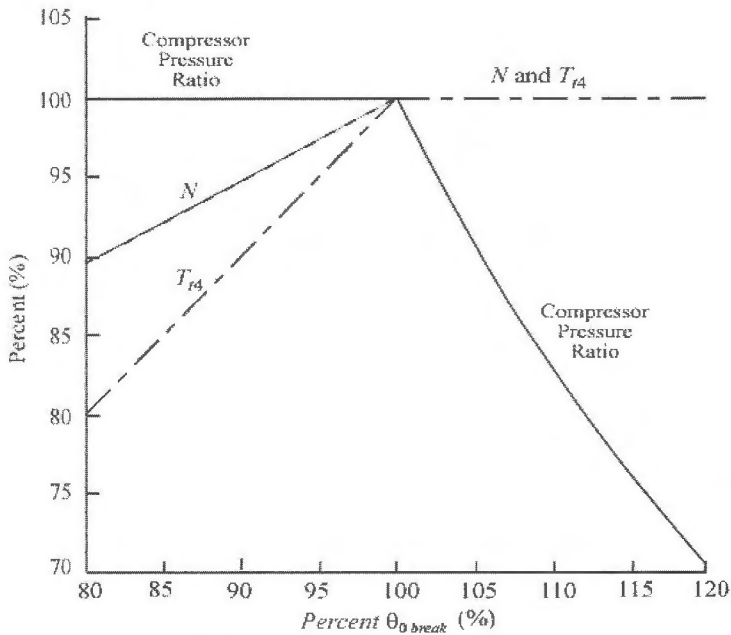


Fig. B.16 Turbojet engine at full throttle.

compressor pressure ratio with theta zero θ_0 at full throttle. Students find this representation very useful in understanding the variation of these parameters with changes in flight condition θ_0 .

Dividing Eq. (B.47) by the square root of θ_4 (T_{t4}/T_{std}) gives

$$N_{c4} = \frac{N}{\sqrt{\theta_4}} = \text{constant} \quad (\text{B.48})$$

Thus the corrected speed of the turbine N_{c4} is constant for a simple turbojet engine at any flight condition θ_0 where T_{t5}/T_{t4} is constant.

Dividing Eq. (i) by T_{t2} gives

$$\frac{T_{t3}}{T_{t2} - 1} = \frac{K_1}{T_{std}} N_{c2}^2 \quad (\text{iii})$$

Rewriting Eq. (iii) for a known reference condition (subscript R), we have

$$\left[\frac{T_{t3}}{T_{t2}} \right]_R - 1 = \frac{K_1}{T_{std}} N_{c2R}^2$$

Using this and Eq. (iii), we can write the corrected speed ratio N_{c2}/N_{c2R} as

$$\frac{N_{c2}}{N_{c2R}} = \sqrt{\frac{T_{t3}/T_{t2} - 1}{[T_{t3}/T_{t2}]_R - 1}} \quad (\text{B.49})$$

For the single-spool turbojet,

$$N_{c2} = N_{c4} \sqrt{\frac{\theta_4}{\theta_0}}$$

or

$$\frac{N_{c2}}{N_{c2R}} = \sqrt{\frac{T_{t4}/\theta_0}{[T_{t4}/\theta_0]_R}} \quad (\text{B.50})$$

Using Eqs. (B.43) and (B.50), we obtain that N_{c2} is constant for $P_{t3}/P_{t2} = \text{constant}$ with constant η 's.

B.6.7 Gas-Generator Pumping Characteristics

The pumping characteristics of a simple gas generator (compressor, main burner, and turbine) can be represented by the variation of the gas-generator's

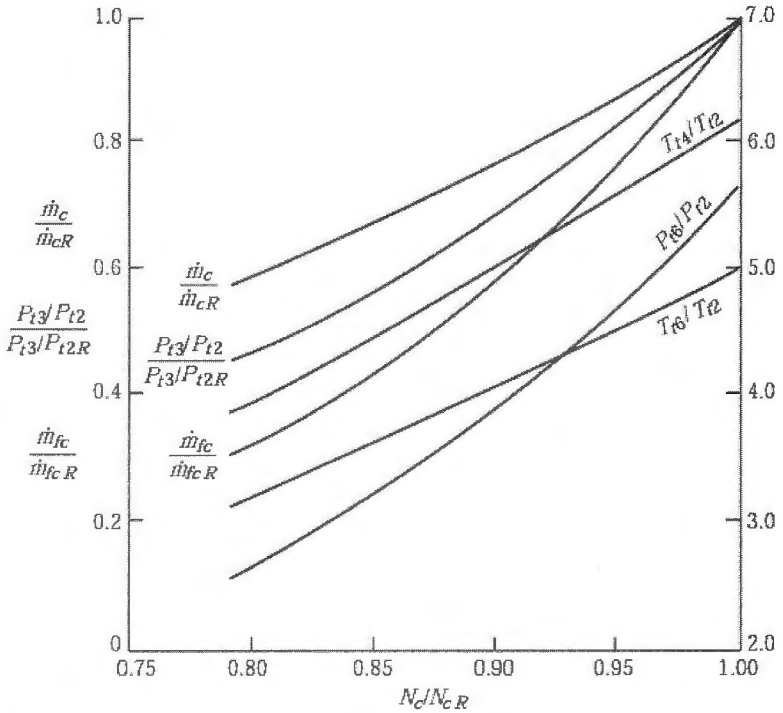


Fig. B.17 Gas-generator pumping characteristics for any throttle position.

parameter ratios with corrected compressor speed. Figure B.17 for a typical engine (from [1]) was obtained using the algebraic relationships developed in Secs. B.6.1–6.6 assuming choked exhaust nozzle. Note the one-to-one correspondence between the corrected engine speed and engine's corrected air mass flow rate, compressor pressure ratio, corrected fuel flow rate, engine total pressure ratio P_{t6}/P_{t2} , engine total temperature ratio T_{t6}/T_{t2} , and T_{t4}/T_{t2} .

B.6.8 Engine Uninstalled Thrust and Fuel Consumption

As shown in [1], the uninstalled thrust of a simple turbojet engine can be written as

$$F = \frac{1}{g_c}(\dot{m}_9 V_9 - \dot{m}_0 V_0) + A_9(P_9 - P_0) = \frac{\dot{m}_0}{g_c}[(1 + f)V_9 - V_0] + A_9(P_9 - P_0) \quad (\text{B.51})$$

where $\dot{m}_9 V_9$ and $\dot{m}_0 V_0$ are the one-dimensional rate of momentum transfer leaving and entering the uninstalled engine (see Fig. B.1). Uninstalled thrust specific fuel consumption can be written as

$$S = \frac{\dot{m}_f}{F} = \frac{f}{F / \dot{m}_0} \quad (\text{B.52})$$

where $\dot{m}_f$ is the mass flow rate of the fuel burned in the main combustor and the fuel-to-air ratio f is given by Eq. (B.46a).

For simplicity of analysis, we choose to predict the performance of a turbojet engine that has the pressure leaving the exhaust nozzle equal that of the ambient air ($P_9 = P_0$). This corresponds to maximum uninstalled thrust F and minimum thrust fuel consumption S . Thus the uninstalled thrust can be simply written as

$$F = \frac{1}{g_c} (\dot{m}_9 V_9 - \dot{m}_0 V_0) = \frac{\dot{m}_0}{g_c} [(1 + f)V_9 - V_0] \quad (\text{B.53})$$

The exit velocity V_9 can be calculated for a perfect gas with constant specific heats by

$$V_9 = \sqrt{2g_c c_{pt} T_{t9} \left[1 - \left(\frac{P_9}{P_{t9}} \right)^{(\gamma_t - 1)/\gamma_t} \right]} \quad (\text{B.54a})$$

where

$$T_{t9} = T_{t5} = T_{t4} \left(\frac{T_{t5}}{T_{t4}} \right) \quad (\text{B.54b})$$

and

$$\frac{P_{t9}}{P_9} = \frac{P_0}{P_9} \frac{P_{t0}}{P_0} \frac{P_{t2}}{P_{t0}} \frac{P_{t3}}{P_{t2}} \frac{P_{t4}}{P_{t3}} \frac{P_{t5}}{P_{t4}} \frac{P_{t9}}{P_{t5}} = \frac{P_{t0}}{P_0} \frac{P_{t2}}{P_{t0}} \frac{P_{t3}}{P_{t2}} \frac{P_{t4}}{P_{t3}} \frac{P_{t5}}{P_{t4}} \frac{P_{t9}}{P_{t5}} \quad (\text{B.54c})$$

Note that T_{t5}/T_{t4} in Eq. (B.54b) is the temperature ratio of the turbine, and the total pressure ratios in Eq. (B.54c) correspond to the following engine components: inlet or diffuser, compressor, main burner, turbine, and exhaust nozzle.

The full throttle (maximum PLA) performance of a typical simple turbojet engine obtained by this analysis (see Sec. B.6.9 for summary) is shown vs flight Mach number and altitude (standard day) in Figs. B.18a and B.18b for

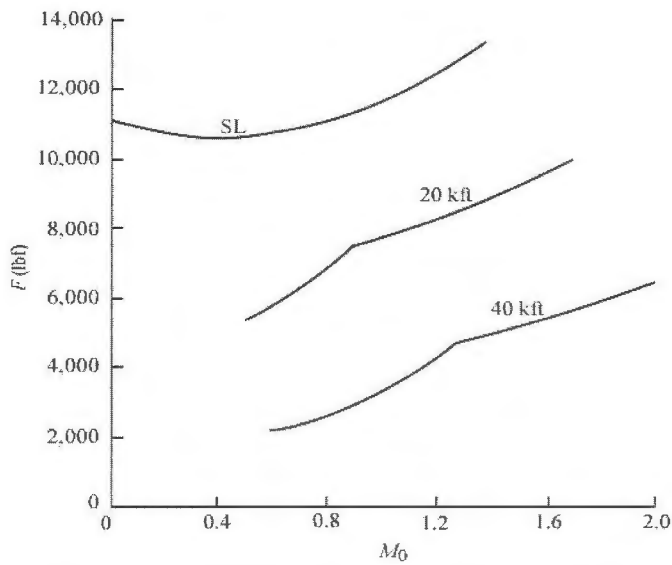


Fig. B.18a Maximum uninstalled thrust F of a turbojet vs M_0 for $\theta_0 break = 1.0$.

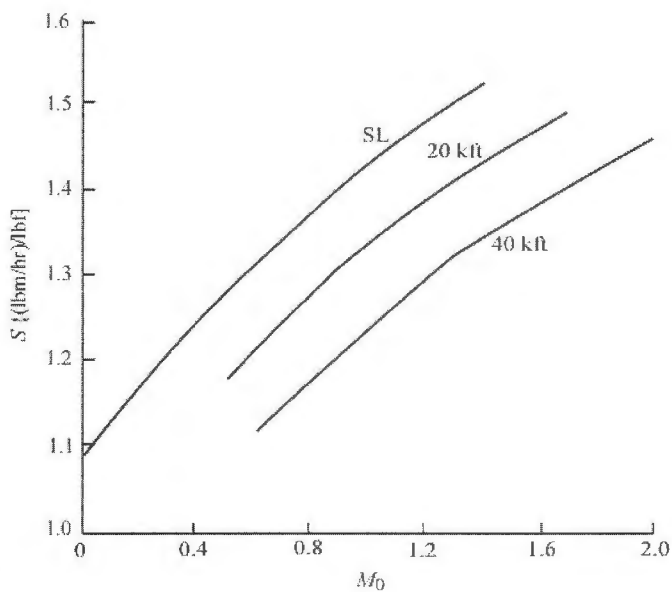


Fig. B.18b Maximum uninstalled thrust specific fuel consumption S of a turbojet vs M_0 for $\theta_0 break = 1.0$.

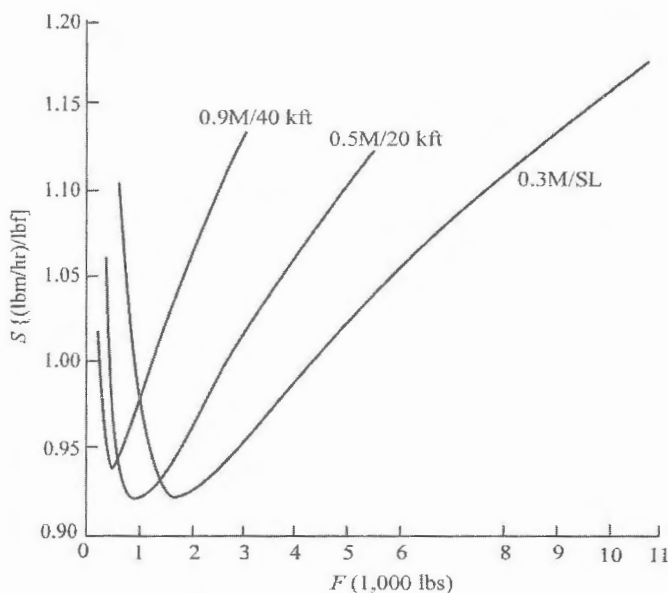


Fig. B.19 Partial throttle performance of simple turbojet at different altitudes and Mach numbers.

a theta break of 1.0. Note that the general trend is that thrust increases with Mach number and decreases with altitude. Note the thrust curves change shape (kink) where $\theta_0 = 1$, $0.9 M/20$ kft and $1.28 M/40$ kft. The trend for the fuel consumption is similar without a noticeable kink. These are the same trends obtained by more complex and accurate engine models.

The partial throttle performance of the example turbojet is shown in Fig. B.19 at different Mach numbers and altitudes. This engine model shows the reduction in fuel consumption due to improvement in propulsive efficiency until the reduction in thermal efficiency drives the fuel consumption back up.

B.6.9 Basic Nonlinear Steady-State Engine Performance **Equations for a Single-Spool Turbojet Engine**

This system of equations needs reference values (subscript R) for one operating point (can be engine design point) to be able to calculate the performance at all other operating conditions (M_0, T_0, T_{i4}). In addition, gas properties are needed.

B.6.9.1 Reference conditions

$$(P_{t2}/P_{t1})_R, (P_{t3}/P_{t2})_R, (T_{t3}/T_{t2})_R, (P_{t4}/P_{t3})_R, (P_{t5}/P_{t4})_R, (T_{t5}/T_{t4})_R, \\ (P_{t8}/P_{t5})_R, T_{t4R}, A_4, M_{8R}, \eta_c, \eta_b, \eta_t, \eta_m$$

B.6.9.2 Gas properties

$$c_{pc}, \gamma_c, c_{pt}, \gamma_t, h_{PR}$$

B.6.9.3 Operating conditions

$$M_0, T_0, T_{t4}$$

The following equations are listed in the order of calculation.

B.6.9.4 Freestream total properties

$$T_{t0} = T_0 \left(1 + \frac{\gamma_c - 1}{2} M_0^2 \right)$$

$$P_{t0} = P_0 \left(1 + \frac{\gamma_c - 1}{2} M_0^2 \right)^{\frac{\gamma_c}{\gamma_c - 1}}$$

B.6.9.5 Inlet. If $M_0 < 1$, then $P_{t1} = P_{t0}$, or else $P_{t1} = P_{t0} \{1 - 0.075(M_0 - 1)^{1.35}\}$ to account for total pressure loss caused by inlet shocks.

$$P_{t2} = \left(\frac{P_{t2}}{P_{t1}} \right)_R P_{t1}$$

where $(P_{t2}/P_{t1})_R$ is the loss in total pressure caused by friction in inlet.

$$T_{t2} = T_{t0}$$

B.6.9.6 Initial conditions

$$\frac{T_{t5}}{T_{t4}} = \left[\frac{T_{t5}}{T_{t4}} \right]_R$$

B.6.9.7 Compressor

$$\frac{T_{t3}}{T_{t2}} = 1 + \eta_m (1 + f) \frac{c_{pt}}{c_{pc}} \frac{T_{t4}}{T_{t2}} \left(\frac{1 - T_{t5}}{T_{t4}} \right)$$

$$N_{c2} = \sqrt{\frac{T_{t3}/T_{t2} - 1}{[T_{t3}/T_{t2} - 1]_R}}$$

$$\%N_R = 100 \sqrt{\frac{T_{t4}}{T_{t4R}}}$$

$$\frac{P_{t3}}{P_{t2}} = \left[1 + \eta_c \left(\frac{T_{t3}}{T_{t2}} - 1 \right) \right]^{\gamma_c / (\gamma_c - 1)}$$

$$T_{t3} = T_{t2} \left(\frac{T_{t3}}{T_{t2}} \right)$$

$$P_{t3} = P_{t2} \left(\frac{P_{t3}}{P_{t2}} \right)$$

B.6.9.8 Main burner

$$f = \frac{\dot{m}_f}{\dot{m}_0} = \frac{c_{pt} T_{t4} - c_{pc} T_{t3}}{\eta_b h_{PR} - c_{pt} T_{t4}}$$

$$P_{t4} = P_{t3} \left(\frac{P_{t4}}{P_{t3}} \right)_R$$

B.6.9.9 Turbine

$$N_{c4} = N_{c2} \sqrt{\frac{T_{t4}}{T_{t4R}} \frac{T_{t2R}}{T_{t2}}}$$

$$\left[\frac{T_{t5}}{T_{t4}} \right]_i = 1 - \eta_t \left[1 - \left(\frac{P_{t5}}{P_{t4}} \right)^{(\gamma_t - 1)/\gamma_t} \right]$$

$$\frac{P_{t8}}{P_0} = \frac{P_{t2}}{P_0} \frac{P_{t3}}{P_{t2}} \frac{P_{t4}}{P_{t3}} \frac{P_{t5}}{P_{t4}} \frac{P_{t8}}{P_{t5}}$$

If $P_{t8}/P_0 > [((\gamma_t + 1)/2)^{\gamma_t/(\gamma_t - 1)}]$, then $M_8 = 1$, or else $M_8 = \sqrt{2/(\gamma_t - 1)[(P_{t8}/P_0)^{(\gamma_t - 1)/\gamma_t} - 1]}$

$$\frac{P_{t5}}{P_{t4}} = \left(\frac{P_{t5}}{P_{t4}} \right)_R \sqrt{\frac{T_{t5}/T_{t4}}{(T_{t5}/T_{t4})_R}} \frac{\text{MFP}(\gamma_t, M_{8R})}{\text{MFP}(\gamma_t, M_8)}$$

$$\frac{T_{t5}}{T_{t4}} = 1 - \eta_t \left[1 - \left(\frac{P_{t5}}{P_{t4}} \right)^{(\gamma_t - 1)/\gamma_t} \right]$$

Go to Sec B.6.9.9 if $\text{Abs} [T_{t5}/T_{t4} - (T_{t5}/T_{t4})] \leq 0.0001$.

B.6.9.10 Exhaust nozzle (convergent nozzle). If $M_8 < 1$, then $P_8 = P_0$, or else $P_8 = P_0 (P_{t8}/P_0)^{1/(\gamma_t - 1)}$.

B.6.9.11 Mass flow rate

$$\dot{m}_0 = P_{t4} A_4 \text{ MFP } \frac{(\gamma_t, M_4 = 1)}{[(1+f)\sqrt{T_{t4}}]}$$

B.6.9.12 Overall performance

$$V_0 = M_0 \sqrt{(\gamma_c - 1) c_{pc} g_c T_0}$$

$$T_{t9} = T_{t5} = T_{t4} \left(\frac{T_{t5}}{T_{t4}} \right)$$

$$\frac{P_{t9}}{P_9} = \frac{P_{t0}}{P_0} \frac{P_{t2}}{P_{t0}} \frac{P_{t3}}{P_{t2}} \frac{P_{t4}}{P_{t3}} \frac{P_{t5}}{P_{t4}} \frac{P_{t8}}{P_{t5}}$$

$$V_9 = \sqrt{2 g_c c_{pt} T_{t9} \left[1 - \left(\frac{P_9}{P_{t9}} \right)^{(\gamma_t - 1)/\gamma_t} \right]}$$

$$F = \frac{1}{g_c} (\dot{m}_9 V_9 - \dot{m}_0 V_0) = \frac{\dot{m}_0}{g_c} [(1+f)V_9 - V_0]$$

$$S = \frac{f \dot{m}_0}{F}$$

B.7 Operability Analysis

The gas turbine engine must also be able to operate dynamically over a wide range of operating conditions such as accelerating and decelerating between different

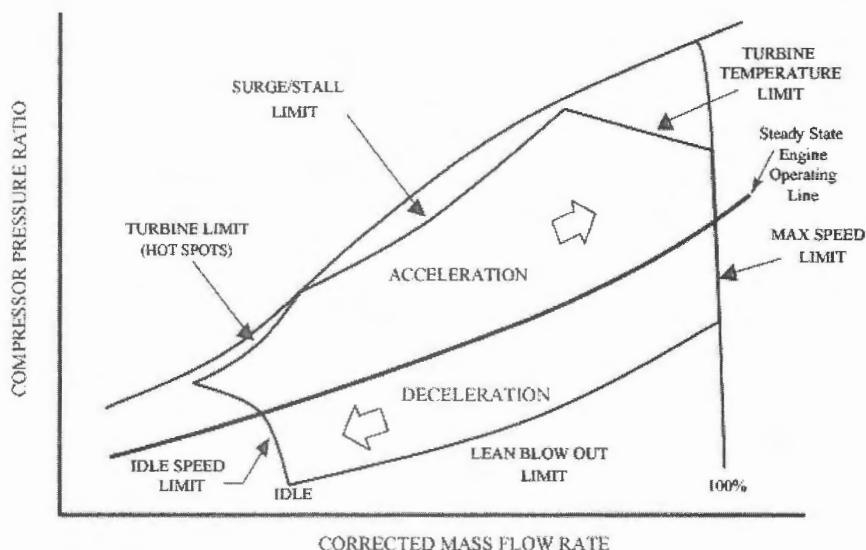


Fig. B.20 Engine operating region on compressor map.

steady-state operating points on the engine operating line (see Fig. B.20). Its operating range is bounded by the idle and maximum speeds and component operating limits. The engine control system must provide these capabilities while keeping the engine from operating in regions of instability (e.g., compressor surge and engine flame out) and/or operating at specific points within the operating region that would destroy components (e.g., blade resonance frequencies and flutter).

B.7.1 Compressor Operation

The steady efficient operation of a compressor is limited by its stall/surge limit. Steady operation at pressure ratios above this limit is not possible. Like a wing, individual compressor blades can stall, creating a rotating-stall region or cells (see Fig. B.21). This rotating-stall cell can extend over the full span of the blade (from hub to tip) or be limited to the outer radius. The resulting flow is called *surge* when the average flow rate varies widely even to the point of reverse flow through the compressor.

B.7.1.1 Stall/surge line. Figure B.9 is the steady-state component map of a typical compressor and represents the well-behaved flow through the machine. The stall/surge line limits the operation of the compressor to points below and right of it. Not shown on this map are operation of the compressor when it is in stall or surge. Heiser [6] and Greitzer [7] present excellent summary articles on compressor stall/surge, and the following material comes from Heiser.

To best understand the compressor stall/surge in an engine, one must first understand compressor stall when operating by itself (compressor not connected

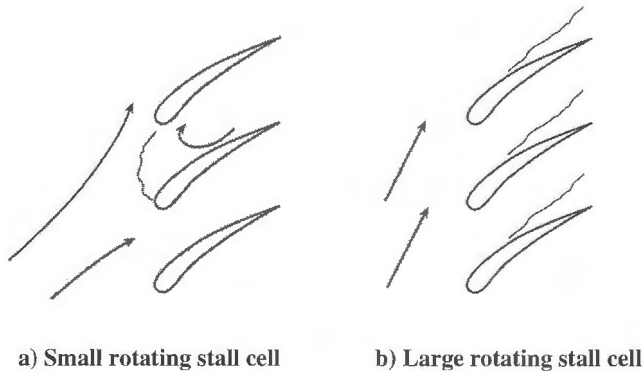


Fig. B.21 Compressor blade stall.

to the combustor). Figure B.22a shows the typical test rig for a compressor. The rotational speed is controlled by the electric motor and the mass flow rate by the exit valve. Figure B.22b shows several steady-state compressor operating points along a constant speed line. The dashed lines are the flow characteristics of the control valve. As the control valve is closed, the pressure ratio increases, and the mass flow rate decreases until the stall point is reached.

The stalled flow characteristics of the compressor can be mapped in a similar manner. Figure B.23 shows the stalled and unstalled flow characteristics of a compressor. Once the stall point a is reached, the compressor operation will flip to the stalled operation point b. Opening the control valve on the stalled compressor can move the operating point to c, and the flow will flip to the unstalled operating point d. Opening the valve further will move the operating point to e. Once on the unstalled operating line, closing the valve will move the operating point through d to a.

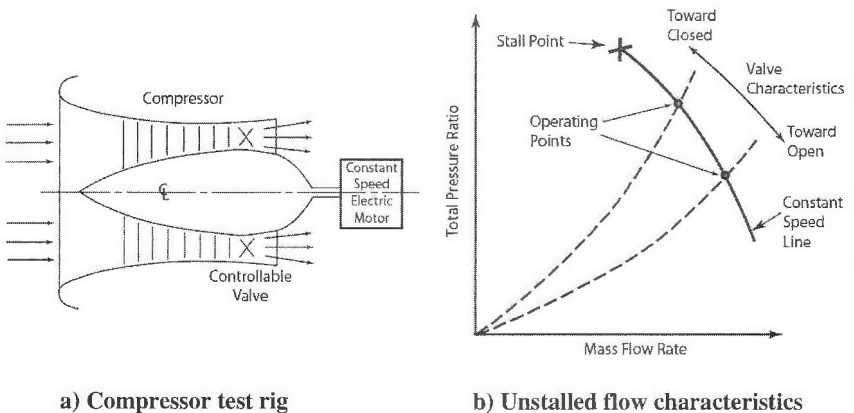


Fig. B.22 Compressor test.

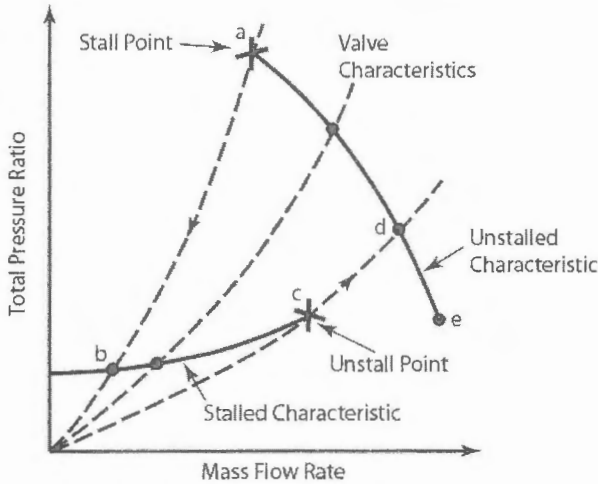


Fig. B.23 Stalled and unstalled compressor operation.

Flow when the compressor is stalled is made up of nonuniform flow. Circumferentially, there will be regions where the flow is almost completely blocked (see Fig. B.21a) and regions where the flow behaves almost normally. The blockage rotates at about one-half the speed of the rotating blades. These equilibrium states for the stalled compressor are often referred to as “rotating stall.”

When the compressor is connected to a combustor (main burner) as in a jet engine, the flow characteristics change due mainly to volumes of the combustor and compressor. These volumes act like storage tanks for the air and impact the dynamic operation of these components. As a result, here are basically three types of stall/surge: nonrecoverable (stagnation) stall, surge, and surge with reverse flow. The type of stall/surge that occurs depends upon the springiness of the compressor-combustor (main burner) system. A parameter that indicates this springiness is defined as

$$B = M_{\text{compressor tip}} \sqrt{\frac{\text{Combustor Volume}}{\text{Compressor Volume}}} \quad (\text{B.55})$$

There is a critical value of B that separates the type of behavior between nonrecoverable stall and surge, and we indicate this as B_{crit} .

Low values of the B parameter lead to the nonrecoverable stall type of behavior as shown in Fig. B.24. As shown in path of Fig. B.24, the flow starts with the stall of a normally operating compressor and ends up with a compressor operating with stalled flow throughout the compressor. This first showed up in the modern F-100 fighter engine during the 1970s. “Non-recoverable stall is characterized by loss of thrust, rising turbine temperatures, rotating stall in the high-pressure compressor,

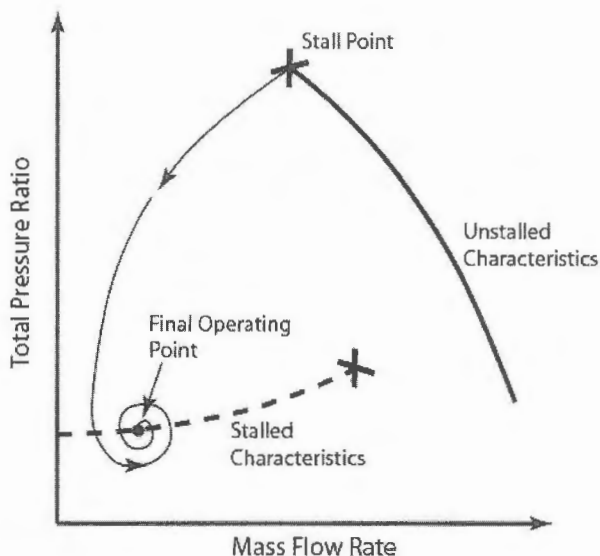


Fig. B.24 Compressor performance at a constant speed with nonrecoverable/stagnation stall path [6].

and loss of pilot control of the engine. In this mode of operation, the engine must be shut down and restarted or severe damage to the engine could result" [8].

Moderate values of the B parameter ($B > B_{crit}$) lead to the surge type of behavior shown in Fig. B.25a. Large values of this parameter ($B \gg B_{crit}$) lead to surge behavior with reverse flow as shown in Fig. B.25b.

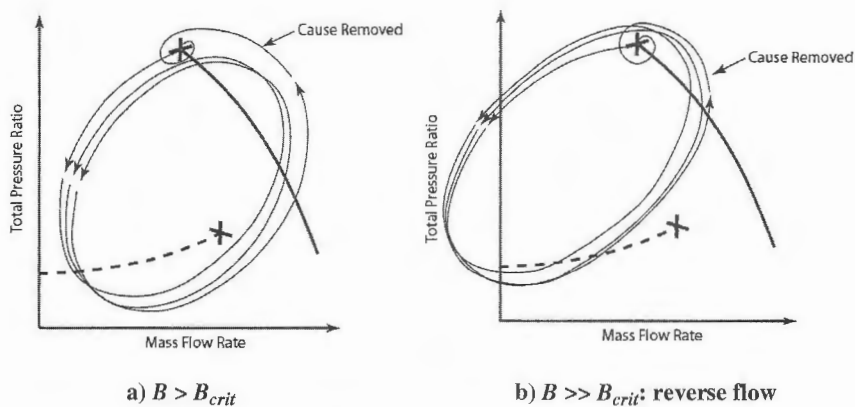


Fig. B.25 Surge cycle [6].

The large fluctuations which occur before equilibrium is restored are called surge cycles, and they appear quite violent to the observer because the often include periods of complete reverse flow which carry burning gases back out of the inlet of the engine. The length of the surge cycles is, however, so short (about 1/10 second) that the appearance is deceiving, and structural damage seldom results from them unless they are deliberately allowed to continue. In fact, surge is preferable to non-recoverable stall for at least two reasons. First, the banging noise that accompanies surge gives the pilot a clear signal of what is happening. Second, the engine will quickly return to normal operation once the cause of surge is removed, and no altitude re-start will be required [6].

The spatial and temporal distortion of the inlet air directly impacts the location of the stall/surge line on a compressor map. As the inlet flow distortion increases or the compressor airfoil surfaces become roughened by the slightest amount of grit or accumulation of dirt, the stall/surge line moves down and to the right as shown in Fig. B.26 and labeled "dirty surge line." For a given corrected mass flow rate, the difference between the pressure ratio corresponding to the surge line PR_s and that corresponding to the steady-state engine operating line PR_e divided by PR_e is called the surge margin (SM). Surge margin is used to accelerate the engine as shown in Fig. B.20—the larger the surge margin, the quicker the acceleration. As can be seen in Fig. B.26, SM is much lower for the

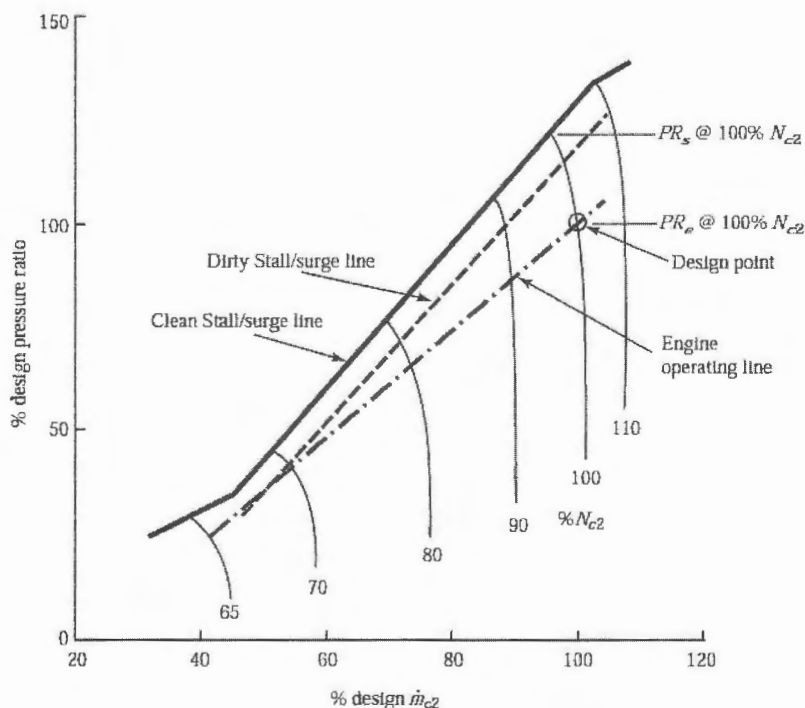


Fig. B.26 Compressor map with dirty surge line.

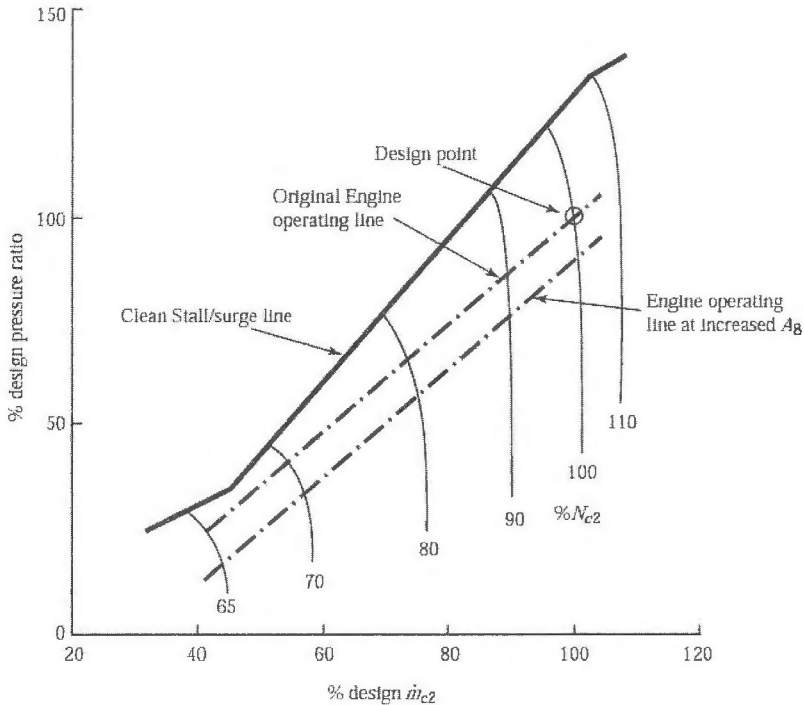


Fig. B.27 Compressor map with increased A_8 .

dirty surge line than for the “clean surge line” and is very small to nonexistent at low flow rates.

To get the needed surge margin during startup and idle, some engines have the ability to increase the exhaust nozzle throat area A_8 (see Fig. B.27). Opening the compressor bleed-air during startup and idle is another way to increase the surge margin.

B.7.1.2 Forced blade vibration and flutter. Two types of blade vibration are of major concern in compressors: forced vibration and flutter. Forced vibration of rotor blades results from passing through flow disturbances (e.g., wakes of upstream stator blades). A problem exists when the forced vibration occurs at the rotor blade’s natural frequency.

Forced vibration occurs at harmonics of the rotor’s rotational frequency, and analysis is facilitated by use of the Campbell diagram, which plots frequency vs rotor speed (see Fig. B.28). Straight lines from the origin represent the number of evenly spaced excitations per revolution. For example, three excitations (3E) at 7000 rpm have a frequency of 350 cps. The dark solid lines marked 1B, 1T, and 2B represent the natural frequency of the first bending mode, first torsional mode, and second bending mode, respectively.

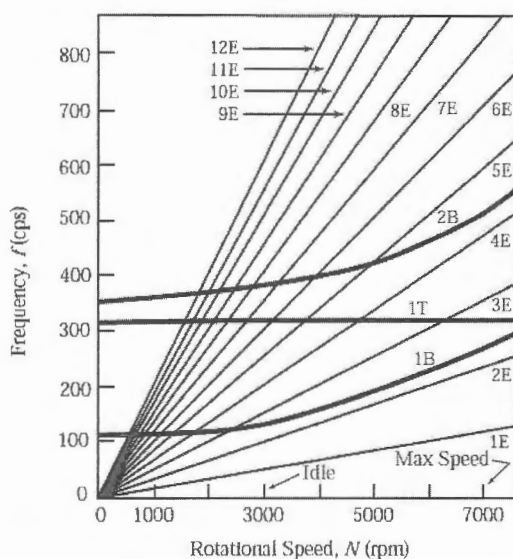


Fig. B.28 Campbell diagram [9].

and second bending mode, respectively, of blade vibration. If an engine operates too long where the natural frequency crosses the excitation frequency, the blade can be destroyed by high cycle fatigue.

Flutter (self-induced) vibration is an unsteady aerodynamic phenomenon that airfoils and/or disks can spontaneously experience, in which they vibrate at a system natural frequency and for which the driving energy is extracted from the flowing gas. This is most commonly found in fan and compressor airfoil rows and comes in many varieties [2]: supersonic stall flutter, subsonic stall flutter, supersonic unstalled flutter, and choked flutter (see Fig. B.29). Once flutter begins, the life of the parts is measured in minutes because of the large stresses and high frequency vibrations that result. Flutter must be avoided at all cost [2].

B.7.2 Turbine Operation

Turbine operation has two limits that show on the compressor map of Fig. B.20. The turbine limit for acceleration near idle speed corresponds to preventing hot spots from the combustor. This limit is managed by scheduling the maximum amount of fuel that can go to the fuel nozzles in the combustor. The turbine limit near maximum speed corresponds to limiting the temperature of the gas leaving the combustor. This limit is controlled by measuring the temperature leaving the turbine (high-pressure turbine for multispool engines) and estimating the temperature at station 4. Additional limits can be used to limit operation at blade resonance frequencies and blade flutter.

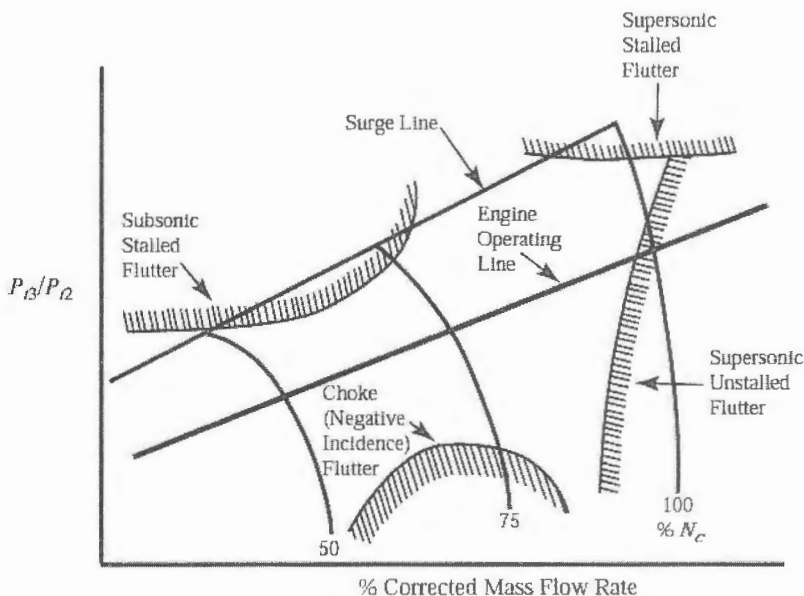


Fig. B.29 Typical flutter boundaries [9].

B.7.3 Main Burner Operation

The fuel flow rate to the main burner during steady-state operation is shown in Fig. B.30 with the arrow pointing in the direction of decreasing engine speed. If the fuel flow rate is reduced too much during deceleration, the combustor fuel-to-air ratio will decrease to a point of a nonflammable mixture, which will result in flame out. To prevent engine flame out, the fuel flow rate is scheduled for engine deceleration. Note that the lean blowout limit is also shown on Fig. B.20.

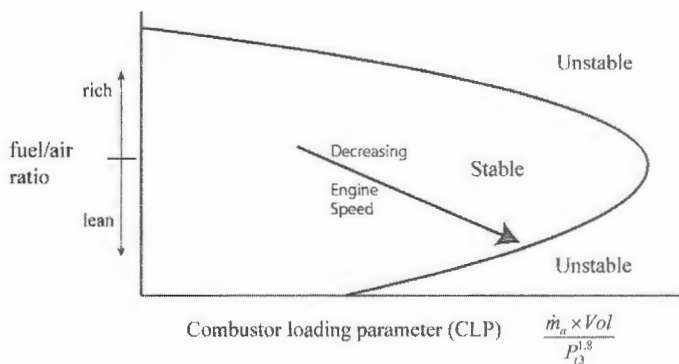


Fig. B.30 Combustor loading.

References

- [1] Mattingly, J. D., *Elements of Propulsion, Gas Turbines and Rockets*, AIAA, Reston, VA, 2006.
- [2] Mattingly, J. D., Heiser, W. H., and Pratt, D. T., *Aircraft Engine Design*, AIAA, Reston, VA, 2002.
- [3] U.S. Standard Atmosphere, U.S. Government Printing Office, Washington, D.C., Oct. 1976.
- [4] "Climatic Information to Determine Design and Test Requirements for Military Equipment," U.S. Dept. of Defense, MIL-STD-210C, Rev. C, Washington, D.C., Jan. 1997.
- [5] "Climatic Information to Determine Design and Test Requirements for Military Equipment," U.S. Dept. of Defense, MIL-STD-210A, Washington, D.C., Nov. 1958.
- [6] Heiser, W. H., "Non-Recoverable Stall," Dept. of Aeronautics, USAF Academy, USAFA TR-84-7, Colorado Springs, CO, Sept. 1984, pp. 37–48.
- [7] Greitzer, E. M., "Review—Axial Compressor Stall Phenomena," *Journal of Fluids Engineering*, Vol. 102, June 1980, pp. 134–152.
- [8] Hopf, W. R., and Steenken, W. G., "Stall Recovery Control Strategy Methodology and Results," AIAA Paper 1985-1433, July 1985.
- [9] Oates, G. C., *Aircraft Propulsion Systems Technology and Design*, AIAA, Washington, D.C., 1989, pp. 394, 410, and 411.

Index

- Abnormal conditions
 - anomaly, 208–210
 - causes, 209
 - failure, 208–210
 - fault, 208–210
- ACC. *See* Active clearance control
- Accel limits for single-spool engine, 150
- Acceleration-based (or N-dot)
 - control, 120
- Acceleration-based transient controller
 - design, 128–136
- N-dot control concept, 129–131
 - cascading (nesting) the set-point control loop, 131
 - compensator followed by an integral control, 130
 - pure integral control, 130
 - switching between the set-point controller, 131
- scheduled and N-dot controls,
 - comparison, 133–136
- for two-spool turbofan engine, 131–133
- Acceleration control, 4
 - in schedule-based transient controller design, 123–124
- Acceleration schedule, in control system
 - integration, 150–152
- Acceleration schedule reduction
 - logic, 248
- Accessories, engine, 154–158
 - categories of, 154–155
 - actuators, 155
 - connecting components, 155
 - control components, 155
 - fuel and oil flow delivery components, 155
 - power extraction or management components, 155
 - sensors, 154
 - thermal management components, 155
 - electronic control unit, 155–158
- Accuracy, 283–285
- Active clearance control (ACC),
 - 181–187
 - ACC modeling, 182–185
 - for centrifugal compressor, 183
 - closed-loop simulation model of, 186
 - linear control law design, 185–187
 - pattern factor control, 181
 - simulation model for, 184
 - tip-clearance control, 181
- Active combustion control, 196–203
 - active combustion enhancement, 196
 - active instability control, 196–197
 - actuator design, 197–198
 - actuator modeling, 198–200
 - experimental results, 200–203
 - dry-test setup, 201
 - wet-test setup, 202
 - fraction of total fuel flow, controlling, 197
 - total fuel flow directly, controlling, 197
- Active controls, 22, 171, 181, 197
- Active stall, 187–196

- Actuators, 155
 - actuator feedback sensors, 51
- Actuators, modeling of, 50–57
 - air- or gas-flow control actuators, 50
 - by a first-order lag, 51
 - fuel control actuators, 50
 - position control actuators, 50
 - See also* Hydraulic actuators;
 - Pneumatic actuators
- Adaptive acceleration control, 140–141
- Advanced controls, 171–203
 - active stall, 187–196
 - actuation challenges, 188–189
 - choked airflow model, 190
 - design of actuator, 189
 - flow supply, 188
 - model of actuator, 190–192
 - packaging, 188
 - unchoked airflow, 190
 - for optimal fuel consumption, 113–114
 - surge control, 187–196
 - See also* Active clearance control (ACC); Active combustion control; High-response actuator; Multivariable control (MVC), 171–180
- Aerodynamic force limit, 305
- Affordable capability (AC), 241–242
- AFRL. *See* Air Force Research Laboratory
- Aggressive integrator, 98, 100, 102
- Air Force Research Laboratory (AFRL)
 - Laboratory
- Aggressive integrator, 98, 100, 102
- Air Force Research Laboratory (AFRL)
 - model, 58–59
 - in the area of fluid mechanics, 59
 - in the area of heat transfer, 59
 - in the area of thermodynamics, 59
 - in Matlab®/Simulink®, 60
- Air- or gas-flow control actuators, 50
- Airframe design, compatibility
 - issues, 86–88
- Alarm rules, 234–235
 - instability pattern, 234
 - mixture pattern, 234
 - stratification pattern, 234
 - trend pattern, 234
- Algorithmic approaches, 219–220
 - complexity of, 221
 - data-driven algorithms, 220
 - fidelity of, 221
 - monitoring algorithm design, 220
 - physics-based algorithms, 220–222
- Ambiguity group, 227
- Anomalies, 208–210
- Artificial neural network (ANN), 226
- Asymptotic stability, 288
 - critically damped, 288
 - overdamped, 288
 - underdamped, 288
- Attenuation or roll-off, 295
- Augmented state variable, 211
- Automatic control systems, 5
 - basic functions, 6
 - closed-loop, 5–6
 - fundamentals of, 265–301
 - linear dynamic systems
 - representation, 267–276
 - See also* Closed-loop control; Open-loop control
 - limit protection control, 6
 - open-loop, 5–6
 - set-point control, 6
 - transient control, 6
- Autotrim, 149–150
- Availability, 208
- Averaging, 297
- Balanced realization, model reduction
 - by, 76–79
 - weighted balanced realization, 78
- Bayesian method, 224–225
- Bleed flow control concept, 255–257
- Bleed valves, 50
 - opening, 257
- Bode plot method, 175
- Bounded-input stability, 288
- Bounded-output stability, 288
- Built-in-test (BIT) logics, 149
- Cambell diagram, 343
- Capabilities, monitoring, 216–217
- Capability, 181
- Capacitance position sensors, 184
- Cascaded control loops for turboshaft engine, 112

- Cascading the set-point control
 - loop, 131
- CFMI CFM56, 30
- Characteristic locus (CL) methods, 16
- Choked airflow model, 190
- Choking, 56
- Closed-loop control system, 5–6,
 - 265–267
 - functions, 266
 - limit protection control, 266
 - set-point control, 266
 - transient control, 266
 - gas turbine engine control systems, 7
 - vs open-loop controls, 266
 - order of, 271
 - poles of, 271
 - in TF form, 270
 - zeros of, 271
- Closed-loop performance simulation
 - during large transients, 163–165
- Closed-loop sensitivity function, 285
- Closed-loop simulation model
 - of ACC, 186
- Closed-loop transfer function (CLTF), 270
- C-MAPSS (Commercial Modular Aero Propulsion System Simulation), 59
 - main elements, 61
- Combustion instabilities, 196–197
- Compatibility issues, engine and
 - airframe, 86–88
- Complexity, control system, 22–25
 - in number of control variables, 22–24
 - in number of measured variables, 25
- Component damages, 245–248
- Component maps, 315–319
- Compressible flow, 313
 - sonic, 313
 - subsonic, 313
 - supersonic, 313
- Compressor operation, 337–343
 - compressor blade stall, 338
 - compressor test, 338
 - flutter, 342–343
 - forced blade vibration, 342–343
 - stall/surge line, 337–342
 - stalled compressor operation, 339
 - unstalled compressor operation, 339
- Compressor performance maps, 316–317
- Condition-based maintenance (CBM)
 - concept, 212, 214
- Condition-based plus maintenance
 - concept, 214
- Conservation of mass, 319–321
- Conservative integrator, 98–99
- Control charts, 229
- Control components, 155
- Control design for turboshaft engines, 109–114
- Control envelope for aircraft
 - turbine engine, 10
- Control laws, 3, 5, 276–279
 - derivative control law, 278
 - integral control law, 277
 - proportional control law, 277
 - proportional-integral-derivative (PID) control law, 278–279
- Control limits, 231
- Control modes, 152–154
 - controlled variables, 154
 - durability, 154
 - operability, 154
 - performance, 154
 - deceleration schedule, 153
 - durability, 152–153
 - functional control modes, 152
 - operability, 152–153
 - performance, 152
 - system control modes, 152
- Control schedules concept, 121–122
- Control system integration, 143–169
 - autotrim, 149–150
 - control modes, 152–154
 - controller synthesis, 158–159
 - engine accessories, 154–158
 - See also* Accessories
 - idle power schedule, 146–147
 - part power, 147–149
 - power setting logic, 143, 144–150
 - power setting schedules, 145
 - takeoff power schedule, 145–146

- Control system integration (*Continued*)
 - transient schedules, 150–152
 - trimming function, 149–150
 - See also* One-spool engine controllers
- Control systems, 1–32
 - challenges facing, 1
 - complexity, 22–25
 - components, 3
 - control envelope for aircraft turbine engine, 10
 - electronics period (from 1970 to 1989), 11, 15–18
 - flight research and advanced controls, 20–22
 - active controls, 22
 - in extending inlet distortion tolerance, 21
 - life-extending controls (LEC), 22
 - growth period (from 1950 to 1969), 11, 12–14
 - historical development, 9–32
 - See also* Engines, historical
 - infancy or start-up period (from 1940 to 1949), 11–12
 - integration period (from 1990 to mid-2000s), 11, 18–20
 - monitoring, 2
 - terminology for, 5–7
 - See also* Automatic control systems; Gas turbine engine control systems
- Controllability, 282–283
- Controller components, 3, 5
 - See also* Acceleration control; Schedule-based control
- Convolution integral, 281
- Cooperating neural network, 226–227
- Corner frequency, 69
- Corridor of flight, 304
- Cost per flight hour (CPFH), 214
- Cost-capability index (CCI), 241
- Covariance matrices, 222
- Criticality, 207
- Critically damped asymptotic stability, 288
- Customer deck, 38
- Cycle decks, 37–38
- Damage (rate) algorithm, 247
- Damage sensors, 51
- Damage state variable, 211
- Damage-adaptive controls, 252
- Data entry plug, 150
- Data-driven algorithms, 220
- Decel limits for single-spool engine, 150
- Deceleration control, in schedule-based transient controller design, 124
- Deceleration schedule, in control system integration, 152–153
- Deceleration surge in transient controls, 128
- Delta fuel flow DW_f , 159
- De-rate, 150
- Derivative control law, 278
- Detent, 299
- Deviation, 209
 - residual, 209
- Diagnostics, monitoring concept, 210–212
- Different speed error (DN) thresholds, 248
- Digital electronic engine control (DEEC), 156
- Dimensionless/corrected parameters, 315–319
 - fan and compressor performance maps, 316–317
- Dirty surge line, 341
- Distortion-tolerant control system, 21
- Dominant dynamics, 69–70
 - in primary control bandwidth, 72–79
- Droop lines, 122
- Durability, in control modes, 152–153
- Dynamic analysis, 69–92
 - compatibility issues, 86–88
 - helicopter rotor and turboshaft engine, 88–90
 - integrated engine and vehicle dynamics, 79–90
 - turboshaft engine, 82–86
 - GE T700 turboshaft engine, 83
 - variations in engine dynamics, 90–92
- See also* Dominant dynamics

- Dynamic engine models, 37, 39–47
 - See also* Pressure dynamics; Shaft dynamics; Temperature dynamics
- Dynamic engine performance, 13–14
- Dynamic simulation model of high-response fuel actuator, 199
- Dynamics, of engine, modeling, 47–50
 - linear-time-invariant (LTI) system, 50
 - transient cycle deck, 48
 - two-spool afterburning turbofan, 49
 - See also* Actuators; Sensor dynamics
- Electrohydraulic servovalve (EHSV), 51
- Electronics period (from 1970 to 1989),
 - control system development, 11, 15–18
 - electronic control units (ECU), 15
 - F100 DEEC (digital electronic engine control), 16
 - line replaceable unit (LRU) level, 15
 - multivariable control design methods, 16
 - propulsion controls in, 30–32
 - single-input and single-output (SISO) approach, 16
 - supervisory control units, 15
- Engine operating line, 322–324
- Engine specific deck, 38
- Engines, historical, 25–32
 - CFM1 CFM56, 30
 - evolution of engine controls, 26
 - Garrett TFE731, 30
 - GE 90 engine, 25
 - GE CF6, 29
 - GE CJ610, 29
 - GE CJ805, 29
 - GE F101, 30
 - GE F110-100, 30
 - GE F110-129, 31
 - GE I-16, 27
 - GE I-A, 27
 - GE J-33, 27
 - GE J35, 28
 - GE J47, 28
 - GE J79, 28
 - GE J85, 28
 - GE J85/Lift fan, 29
 - GE J93, 29
 - GE TF39, 29
 - GE TG-100/T31, 27
 - GE YF120, 31
 - GE90, 31
 - IAE V2500, 31
 - PW 6000, 32
 - PW F100, 30
 - PW F100-220, 31
 - PW F100-229, 31
 - PW F119, 31
 - PW F135, 31
 - PW J52, 28
 - PW J57, 28
 - PW J58, 29
 - PW JT3, 29
 - PW JT8D, 29
 - PW JT9D, 29
 - PW JT9D-7R4, 30
 - PW TF30, 29
 - PW2037, 30
 - Trent 1000, 32
 - Westinghouse J30, 28
- Equation of motion (EOM), 72
- Estimation, 222
- Event driven or failure driven
 - maintenance concept, 214
- Exhaust gas temperature (EGT), 40, 325
- F100 DEEC sensors, 156–157
- F100 turbofan engine, 71, 172
 - comparison of MVC methods, 179–180
 - frequency-domain methods, 175–178
 - linear-time-invariant (LTI) models of, 172
 - input variables of, 173
 - output variables of, 173
 - state variables of, 173
 - open-loop poles of, 91
 - power change response requirements, 172
 - reduced-order model, 174
 - A matrix elements in, 174
 - B matrix elements in, 174–175
 - C matrix elements in, 175
 - D matrix elements in, 175

- F100 turbofan engine (*Continued*)
 - thrust responses of, 91
 - time-domain methods, 178–179
- F135 engine, 19
- F414 engine, 157–158
- Failures, 208–210
- False alarms, 215, 242
- Fan and compressor performance
 - maps, 316–317
- Fast-response control for helicopter engines, 253–263
 - integrated engine-helicopter model, 257–260
 - Kalman filter techniques, 262
 - MVC design techniques, 262
 - proportional control law design, 260–263
 - See also* Turbine bleed control mode
- Faults, 208–210
- Feedback control for power turbine speed, 110–111
- Feed-forward control for rotor load anticipation, 111–113
- Filter bank, 224
- Final-value theorem of Laplace transform, 285
- First law of thermodynamics, 308
- First-order model of hydraulic actuator, 53–54
- First-order system, 268
- Flat-rate power schedule, 145–146
- Fliegner's approximation, 190
- Flight idle schedule, 146–147
- Flight limits, 304
 - aerodynamic force limit, 305
 - lift limit, 305
 - temperature limit, 305
- Flight research and advanced controls, 20–22
- Flutter, 342–343
- Forced blade vibration, 342–343
- Forward-loop transfer function (FLTF), 271
- Frequency spectrums of interest, 71–72
- Frequency-domain methods, 16
 - Frequency-domain MVC methods, 175–178
 - root-locus method, 175
 - transfer functions (TFs) in, 175–177
 - Frequency-domain representation, 286–287
 - of control system, 269–273
 - in shaft dynamics, 41
 - Frequency-response design method, 13, 294–296
 - for one-spool engine, 96, 103–104
 - closed-loop speed response, 103
 - Fuel and oil flow delivery
 - components, 155
 - Fuel consumption, 330–333
 - Fuel control actuators, 50
 - Fuel flow, engine, 327
 - Fuel metering valves, 50
 - Functional control modes, in control system integration, 152
 - Functions, monitoring, 217–219
 - Fusion, 217
- Gain scheduling, in transient controls, 125–127
- Garrett TFE731, 30
- Gas turbine engine control systems, 7–9
 - simplified engine control system, 8
- Gas turbine engine performance and operability, 303–344
 - basic engine performance model, 319–336
 - engine controls, 324–327
 - engine fuel flow, 327
 - engine operating line, 322–324
 - engine uninstalled thrust and fuel consumption, 330–333
 - gas-generator pumping characteristics, 329–330
 - theta break, 324–327
 - turbine characteristics, 321–322
 - variation in engine speed, 328–329
 - component maps, 315–319
 - dimensionless/corrected parameters, 315–319
 - operational envelopes, 304–306

- standard atmosphere, 304–306
- turbine maps, 317–319
- Gas-generator pumping characteristics, 329–330
- Gas-path sensors, 51
- GE 90 engine, 25
- GE CF6, 29
- GE CJ610, 29
- GE CJ805, 29
- GE F101, 30
- GE F110 engine, 48
- GE F110-100, 30
- GE F110-129, 31
- GE I-16, 27
- GE I-A, 10, 27
- GE J-33, 27
- GE J35, 28
- GE J47, 11, 28
- GE J79, 28
- GE J85, 28
- GE J85/Lift fan, 29
- GE J93, 29
- GE T700 turboshaft engine, 83
 - fifth-order model of, 85
 - sixth-order model of, 86
- GE TF39, 29
- GE TG-100/T31, 27
- GE90, 31
- GoldCare®, 214
- Ground idle schedule, 146
- Ground-based monitoring, 216
- Growth period (from 1950 to 1969),
 - control system development, 11, 12–14
 - aerodynamically connected two-spool engine—PW J57, 12
 - classical control design methods, 13
 - dynamic performance, 13
 - frequency-response design methods, 13
 - hydromechanical controls (HMCs), 12
 - propulsion controls in, 28
 - single-loop design methods, 13
 - steady-state engine performance model, 13
 - time-domain design methods, 13
- Health and usage monitoring system (HUMS), 215
- Health management, 207–235
 - management, 207–208
 - reliability, 207–208
 - safety, 207–208
 - See also* Monitoring, engine
- Health monitoring, 239–263
- Helicopter rotor, 88–90
- Heuristic physical model, 226
- High pressure compressor (HPC), 38
- High Stability Engine Control (HISTEC) program, 21
- High-fidelity engine simulations, 57–61
 - NASA generic turbofan engine model, 59–61
 - U.S. Air Force Generic Turbofan Engine Model, 58–59
- High-gain control, 253
- Highly Integrated Digital Electronic Control (HIDEC)
 - program, 20
- High-pressure (HP) turbine gas temperature, 39
- High-response actuator, 191–192
 - compressor rig test of, 193
 - dynamic test of, 194
 - wind tunnel test of, 193
- High-response fuel actuator, dynamic simulation model of, 199
- H-infinity design method, 186
- Hybrid model estimation, 225–227
- Hydraulic actuators, 51–56
 - first-order model of, 53–54
 - loadbearing actuator, 51
 - second-order model of, 54–55
 - servovalve, 51
 - third-order model of, 54–55
 - See also* Multistage hydraulic actuator
- Hydromechanical controls (HMCs), 12
- Idle power schedule, 146–147
 - flight idle schedule, 146
 - ground idle schedule, 146
- Impulse function, 284–285
- Impulse response matrix, 281

- Infancy or start-up period (from 1940 to 1949), control system development, 11–12
 - control laws, 11
 - J47 turbojet engine, 11
 - propulsion controls in, 27–28
 - time-domain design methods, 11
- Inherent reliability, 208
- Inlet distortion tolerance, engine control
 - research in extending, 21
- Inlet guide vane position (IGV), 75
- Instability pattern, 234
- Integral control law, 277
- Integrated control, 239–263
 - affordable capability (AC), 241–242
 - capabilities, 240–245
 - fast-response control for helicopter engines, 253–263
 - life-extending control, 245–249
 - requirements, 240–245
 - See also* Integration architecture; Safety assurance control
- Integrated engine, 79–90
- Integrated engine-helicopter model, 257–260
- Integrated flight and propulsion control (IFPC), 17
- Integrated propulsion control system (IPCS), 20
- Integration architecture, 240–241
- Integration period (from 1990 to mid-2000s), control system development, 11, 18–20
- Integrator wind-up (IWU) protection, 125, 127–128
- Inverse Laplace transform, 281
- Inverse nyquist array (INA) method, 16, 175
- J47 turbojet engine, 11
- Jump resonance of nonlinear system, 298
- Kalman filter (KF), 222, 223–224
- Kalman-filtering approach, 20–21
- Laplace transform method, 281–282
- Least-square estimation, 222–223
- LEC. *See* Life-extending control
- Life-extending control (LEC), 22, 239, 245–249
 - component damages, 245–248
 - for a commercial turbofan engine, 246
 - control design, 248–249
- Life-forecasting (or prediction), 247
- Lifing algorithms, 247
- Lift limit, 305
- Limit cycles, 299
- Limit protections, controller design for, 6, 136–139, 266
 - for a cascaded primary control loop, 138
 - exceeding the maximum compressor exit static pressure P_3 , 137
 - exceeding the maximum EGT limit in turbines, 137
 - exceeding the maximum HP-shaft physical speed N_2 limit, 136
 - exceeding the maximum LP-shaft physical speed N_1 , 136
- Limit-reset control, 252–253
- Line replaceable unit (LRU), 15
- Linear control design methods, 185–187, 291–296
 - frequency-response design method, 294–296
 - root-locus design method, 292–293
- Linear dynamic systems analysis, 279–296
 - solution of state equation, 279–281
 - homogeneous solution, 279
 - particular solution that satisfies the entire state equation, 279
 - stability, 288–291
- Linear dynamic systems representation, 267–276
 - additivity, 267
 - frequency-domain representation, 269–273
 - homogeneity, 267
- Linear engine models, derivation of, 61–66
 - extraction from nonlinear simulations, 62–64
 - identification of models from experimental/operational data, 64–66

- nonparametric models, 64
- parametric models, 64
- Linear quadratic Gaussian (LQG)
 - method, 178
- Linear quadratic regulator (LQR)
 - methods, 16, 178, 260
- Linearization, 297
- Linear-time-invariant (LTI) models, 268
 - of F100 engine, 172
 - input variables of, 173
 - output variables of, 173
 - state variables of, 173
- Logistic response time (LRT), 244
- Loop gain, 97, 271
- Low-cycle fatigue of disks, 245
- Lower control limit (LCL), 229
- Low-pressure compressor (LPC), 38
- Low-pressure turbine (LPT), 38
- LQG synthesis, 262
- LTS101 electronic engine control
 - logic diagram, 159
- Lumped-parameter approach, 37
- Lyapunov stability, 288
- Mach number, 113
 - T/T_t and P/P_t as functions of, 311–313
- Magnetic hysteresis, 299
- Maintenance concepts, 212–214
 - condition based, 214
 - condition-based maintenance (CBM) concept, 212
 - condition-based plus, 214
 - event driven or failure driven, 214
 - reliability-centered-maintenance (RCM) concept, 212
 - time based, 214
 - usage based, 214
- Maintenance man-hour (MMH), 208
- Management, 207–208
- Manual trim, 149
- Marginally stable system, 289
- Mass flow parameter (MFP), 313–314
- Mass-spring-damper model, 87
- MATLAB®/Simulink®, 19
- MATRIXx®/SystemBuild, 19
- Mean time between maintenance (MTBM), 208
- Mean time between unscheduled removals (MTBUR), 208
- Mean-value control chart, 230
- Measured by mean time to repair (MTTR), 208
- Mechanical hysteresis, 299
- Military power, 145
- Missed-detection rate, 242
- Mixture pattern, 234
- Modal state variable, 276
- Mode switching, 6
- Model reduction, 69–92
 - by singular value decomposition (balanced realization), 76–79
 - by spectral decomposition, 73–76
 - inlet guide vane position (IGV), 75
- Model reference control method, 178
- Modeling and simulation, engine, 37–66
 - high-fidelity engine simulations, 57–61
 - See also* Dynamic engine models; Steady-state engine models
- Module matching, 213
- Monitoring, engine, 207–235
 - capabilities, 216–217
 - diagnostics, 210–212
 - functions, monitoring, 217–219
 - algorithmic approaches, 219–220
 - software architecture, 219
 - hybrid model estimation, 225–227
 - monitoring algorithm design, 220
 - monitoring system design, 215–220
 - ground-based monitoring, 216
 - off-board system architecture, 216
 - onboard system architecture, 216
 - system architecture, 216
 - system requirements, 216
 - probability estimation, 224–225
 - prognostics, 210–212
 - See also* Health management; Least-square estimation; Maintenance concepts; Support concepts; Trend monitoring
- Monitoring, system requirements for, 242–245
- Multi-input, multi-output (MIMO) system, 268

- Multiple equilibrium points, 299
- Multiplication, 299
- Multistage hydraulic actuator, 52
- Multivariable control (MVC) design
 - methods, 16, 171–180
 - MVC design theme problem, 172–175
- Multivariable frequency-domain methods, 16
- MVC. *See* Multivariable control
- NASA generic turbofan engine
 - model, 59–61
- N-dot acceleration schedule, 248
- N-dot control concept, 129–131
 - accel schedules superimposed on, 134
 - decel schedules superimposed on, 134
 - pure integral control, 130
- N-dot controller, in two-spool engine
 - controllers design, 167–169
- Newtonian mechanics, 39
- Newton–Raphson method, 13–14
- Nonexponential time response, 299
- Nonlinear considerations in transient
 - controls, 124–128
 - deceleration surge, 128
 - gain scheduling, 125–127
 - integrator wind-up (IWU) protection, 127–128
- Nonlinear design techniques, 37
- Nonlinear dynamic systems, 296–301
 - averaging, 297
 - commonly encountered nonlinearities, 300–301
 - linearization, 297
 - perturbation theory, 297
 - properties of, 298–300
 - dependence of input shape and amplitude, 299
 - jump resonance, 298
 - limit cycles, 299
 - multiple equilibrium points, 299
 - new frequencies, 298
 - nonexponential time response, 299
- Nonlinear function, 299
- Nonlinear simulations, linear models
 - extraction from, 62–64
- Nonlinear steady-state engine
 - performance equations for a single-spool turbojet engine, 333–336
- Nontraditional control systems, 70
 - See also* Advanced controls
- Numerical Propulsion System Simulation (NPSS), 58
- Observability, 282–283
- Oil and debris sensors, 51
- One-spool engine controllers, 96–104
 - closed-loop performance simulation during large transients, 163–165
 - design of, 159–165
 - frequency response method, 96, 103–104
 - input variables, 159
 - delta fuel flow DW_f , 159
 - shaft speed, 159
 - piecewise linear model of, 160
 - root-locus design, 96, 98–103
 - set-point controllers
 - design of, 160–161
 - synthesis of, 161–162
 - shaft dynamics of, 40
 - synthesis of, 159–165
 - synthesized large-transient controller for, 162
 - transient controller
 - design of, 161
 - synthesis of, 161–162
- On-off contactor, 299
- Open System Architecture for Condition Based Maintenance (OSA-CBM), 217–218
- Open-loop control system, 5–6, 265–267
 - in TF form, 269
- Open-loop step response, 259
- Open-loop transfer function (OLTF), 97, 271
- Operability analysis, 336–344
- Operability, in control modes, 152–153
- Operational envelopes, 304–306
- Operational limits, 305
- Operational reliability, 208
- Operational uncertainties, 292

- Optimal fuel consumption, adaptive control for, 113–114
- Original equipment manufacturers (OEMs), 214
- Overdamped asymptotic stability, 288
- Overshoot, 286
- Part power, in control system
 - integration, 147–149
 - segmented part-power setting schedule, 149
 - segmented schedule, 148
 - speed reference vs PLA, 148
 - thrust vs speed, 148
- Pattern factor control, 181
- Performance, 283–287
 - accuracy, 283–285
 - in control modes, 152
 - performance cycle analysis, 315
 - performance model, 319–336
 - performance-seeking control (PSC), 21
 - response, 286
 - frequency-domain response, 286–287
 - time-domain response, 286
- Perturbation analysis, 41
- Perturbation theory, 297
- Phase lag, 289
- Physical system, controller, 5
- Physics-based algorithms, 220
- Physics-based modeling, increased accuracy through, 232–234
 - corrected variables, 232
 - dependent variable, 232
- Pneumatic actuators, 56–57
- Position control actuators, 50
- Power level angle (PLA), 147
- Power plug, 150
- Power setting logic, 143, 144–150
 - gas turbine engine control logic, 144
- Power setting schedules, 145
- Power-by-the-hour (PBH), 214
- Preload, 299
- Pressure dynamics, mass storage effect, 43–46
- Primary control bandwidth, dominant dynamics in, 72–79
- Probability estimation, Bayesian method, 224–225
- Prognostics and health management (PHM), 210
- Prognostics, monitoring concept, 210–212
- Proportional control law, 260–263, 277
- Proportional-integral-derivative (PID) control law, 278–279
- Proportional-plus-integral (PI) control-law, 96
 - for two-spool engine without actuator dynamics, 107–108
- Proportional-plus-integral-plus-derivative-(PID) control law, 97
 - for two-spool engine with actuator dynamics, 108–109
- Propulsion flight research, 20
- Propulsion-controlled aircraft (PCA) research, 21, 81
- Pulse-width-modulated (PWM) servo amplifier, 184
- PW 6000, 32
- PW F100, 30
- PW F100-220, 31
- PW F100-229, 31
- PW F119 engine, 19
- PW F135 engine, 19, 31
- PW J52, 28
- PW J57, 28
- PW J58, 29
- PW JT3, 29
- PW JT8D, 29
- PW JT9D, 29
- PW JT9D-7R4, 30
- PW TF30, 29
- Ramp function, 284–285
- Rectifier, 299
- Reliability, 207–208
 - inherent reliability, 208
 - level of, measurement, 208
 - operational reliability, 208
- Reliability-centered-maintenance (RCM) concept, 212
- Remaining useful life (RUL), 212, 242–244
 - vs usage, 247

- Removals, engine, 208
 - scheduled removals, 208
 - unscheduled removals, 208
- Residual deviation, 209
- Response, 286
 - time-domain response, 286
- Restrictive state transition matrix, 280
- Riccati equation, 179
- Rise time, 286
- Rolls-Royce's TotalCare®, 214
- Root-locus design method, 175, 292–293
 - for one-spool engine, 96, 98–103
 - aggressive integrator (Case 2), 98, 100
 - conservative integrator (Case 1), 98–99
- Rotating stall, 339
- Rotor load anticipation, feed-forward control for, 111–113
- Rotorcraft engine considerations, 139–141
 - adaptive acceleration control and surge recovery, 140–141
 - power management, 139–140
- RUL. *See* Remaining useful life
- Safety, 207–208
- Safety assurance control, 249–253
 - closed-loop response of, 252
 - component degradations, 252
 - high-gain control, 253
 - limit-reset control, 252–253
- Saturation, 299
- Saturation nonlinearity, 125
 - for transient schedules, 121
- Schedule-based control, 4, 120
- Schedule-based transient controller
 - design, 120–124
 - acceleration control, 123–124
 - control schedules concept, 121–122
 - deceleration control, 124
 - saturation nonlinearities for, 121
- Scheduled removals, 208
- Second law of thermodynamics
- Second-order mass-spring-damper system, 182
- Second-order model of hydraulic actuator, 54–55
- Segmented part-power setting schedule, 149
- Sensor dynamics, modeling of, 50–57
 - actuator feedback sensors, 51
 - damage sensors, 51
 - gas-path sensors, 51
 - goals, 51
 - oil and debris sensors, 51
 - vibration sensors, 51
- Sensors, 154
- Servovalve, model of, 55–56
- Set-point automatic control, 6
- Set-point controllers, 266
 - design of, 95–118, 160–161
 - one-spool engines, 96–104
 - PI control law for, 161
 - practical considerations for, 114–118
 - engine responses under different control laws, 114–116
 - using fuel ratio unit as control variable, 116–118
 - speed control loop for, 97
 - two-spool engines, 104–109
- Settling time, 286
- Shaft dynamics
 - frequency-domain representation, 41
 - inertia effect, 39–43
 - of one-spool engines, 40
 - time-domain representation, 41
 - by a two-disk system, 39
 - of two-spool engines, 42–43
- Shaft speed, 159
- Shewhart solution, 229
- Short-takeoff-and-vertical-landing (STOVL) control technology, 18
- Similarity transformation, 276
- Simulation model of unstable system, 290
- Single-input and single-output (SISO) approach, 16
- Single-input, single-output (SISO) system, 268
- Single-loop design methods, 13
- Single-spool engine
 - accel limits for, 150
 - decel limits for, 150

- Single-spool turbojet engine, 303–304
 - main burner operation, 344
 - nonlinear steady-state engine
 - performance equations for, 333–336
 - compressor, 334–335
 - equations in order of calculation, 334
 - exhaust nozzle, 336
 - gas properties, 334
 - initial conditions, 334
 - inlet, 334
 - main burner, 335
 - mass flow rate, 336
 - operating conditions, 334
 - overall performance, 336
 - reference conditions, 333
 - turbine, 335–336
 - operability analysis, 336–344
 - See also* Compressor operation
 - turbine operation, 343–344
- Singular value decomposition.
 - See* Balanced realization
- Software architecture, 219
- Sonic compressible flow, 313
- Spectral decomposition, model
 - reduction by, 73–76
- Speed control loop for set-point controller, 97
- Speed of engine, variation, 328–329
- Spool valve/Electro-hydraulic servovalve (EHSV), 51
- Stability, 288–291
 - asymptotic stability, 288
 - bounded-input stability, 288
 - bounded-output stability, 288
- Stagnation or total pressure, 311
- Stagnation or total temperature, 310–311
- Stall/surge line, 337–342
- Stalled compressor operation, 339
- Standard atmosphere, 304–306
- State equation, solution of, 279–281
 - homogeneous solution, 279
- State transition matrix, 280
- State-space realization, 276
- State-variable feedback (SVFB), 261
- Static temperature, 310
- Steady-state cycle deck, 17
- Steady-state engine models, 13, 37–38
- Step function, 284–285
- Stratification pattern, 234
- Subsonic compressible flow, 313
- Supersonic compressible flow, 313
- Support concepts, 214–215
 - bill-and-material, 214
 - power-by-the-hour (PBH), 214
 - Rolls-Royce's TotalCare®, 214
- Surge control, 187–196
- Surge cycle, 340
 - dirty surge line, 341
- Surge margin (SM), 341
- Surge margin stacking, 151
- Surge schedule, in control system integration, 151
- Synthesis, 161
- System control modes, in control system integration, 152
- System identification (system ID), in linear engine models, 64–66
- System requirements for monitoring, 242–245
 - acceptable prognostic capability, 244
- T/T_t and P/P_t as functions of Mach number, 311–313
- Takeoff (TO) power schedule, in control system integration, 145–146
 - flat-rate schedule, 145
 - temperature schedule, 145
- Temperature dynamics
 - energy storage effect, 46–47
 - associated heat transfer, 46
 - direct change of the thermodynamic state, 46
 - slow temperature dynamics, 46
- Temperature limit, 305
- Temperature power schedule, 145–146
- Terminology for control systems, 5–7
 - physical system, 5
- TF. *See* Transfer-function
- Thermal management components, 155
- Thermal mechanical fatigue (TMF), 245
- Thermo-acoustic instabilities, 197

- Thermodynamics of perfect gas, 306–310
 - first law of thermodynamics, 308
 - mass flow parameter (MFP), 310–319
 - second law of thermodynamics, 309
 - stagnation or total pressure, 311
 - stagnation or total temperature, 310–311
 - total properties, 310–319
- Theta break, 326
- Third-order model of hydraulic
 - actuator, 54–55
- Threshold, 299
- Time-based maintenance concept, 214
- Time constant, 69
- Time history, 286
- Time-domain design methods, 13
 - during infancy period, 11
- Time-domain MVC methods, 178–179
 - linear quadratic-regulator (LQR)
 - method, 178
 - model reference control method, 178
- Time-domain representation
 - of control system, 273–276
 - in shaft dynamics, 41
- Time-domain response, 286
 - overshoot, 286
 - rise time, 286
 - settling time, 286
- Tip-clearance control, 181–182
- Toggle, 299
- Topping governors, 120, 137
 - integrating approaches, 139
- Total accumulated cycle (TAC), 153
- Total support concept, 214
- Total Support Package (TSP®), 214
- Tracking error, 284
- Traditional control systems, 70
- Transfer-function (TF) form
 - closed-loop control system in, 270
 - open-loop control system in, 269
 - partial fraction form of, 272
 - pole-zero form, 272
 - polynomial form of, 271
 - residues of, 273
- Transient automatic control, 6
- Transient control design, 119–141,
 - 161, 266
 - rotorcraft engine considerations,
 - 139–141
- See also* Acceleration-based transient
 - controller design; Limit
 - protections; Nonlinear
 - considerations; Schedule-based
 - transient controller design
- Transient cycle deck, 17
- Transient schedules, in control system
 - integration, 150–152
 - acceleration schedule, 150–152
 - deceleration schedule, 152
 - surge schedule, 151
- Trend monitoring from periodically
 - recorded data, 227–235
 - alarm rules, 234–235
 - increased accuracy through physics-
 - based modeling, 232–234
 - mean-value control chart, 230
 - thresholds, 229–232
 - trend charts, 229–232
- Trend pattern, 234
- Trent 1000, 32
- Trimming function, in control system
 - integration, 149–150
 - de-rate and, 150
 - manual trim, 149
- Turbine bleed control mode, 255–257
 - gain-magnifying control, 255
 - limit-reset control, 255
- Turbine characteristics, 321–322
- Turbine maps, 317–319
- Turbine operation, 343–344
- Turbofan, 9
- Turbojet, 9
- Turboshaft engine, 88–90
 - control design for, 109–114
 - adaptive control for optimal fuel
 - consumption, 113–114
 - cascaded control loops for, 112
 - feedback control for power turbine
 - speed, 110–111
 - feed-forward control for rotor load
 - anticipation, 111–113
 - dynamics, 82–86
- Two-spool engines, 38, 168–169
 - air volumes in, 44–45
 - controller design for, 104–109
 - EPR control loop for, 106
 - general control loop for, 106

- PI control law design, 107–108
- PID control law design, 108–109
- design of, 165–169
- EPR controller, design of, 167–168
- gas volumes in, 44–45
- N-dot control design for, 131–133
- N-dot controller, design of, 167–169
- shaft dynamics of, 42–43
- synthesis of, 165–169
- TF model of, 166
- U.S. Air Force Generic Turbofan Engine Model, 58–59
- Unbalanced torque performance analysis, 41
- Unchoked airflow, 190
- Underdamped asymptotic stability, 288
- Uninstalled thrust, 330–333
- Unscheduled removals, 208
- Unstalled compressor operation, 339
- Upper control limit (UCL), 229
- Usage based maintenance concept, 214
- Vehicle dynamics, 79–90
- Vibration sensors, 51
- Weighted balanced realization, 78
- Westinghouse J30, 28

Supporting Materials

Many of the topics introduced in this book are discussed in more detail in other AIAA publications. For a complete listing of titles in the AIAA Education Series, as well as other AIAA publications, please visit <http://www.aiaa.org>.

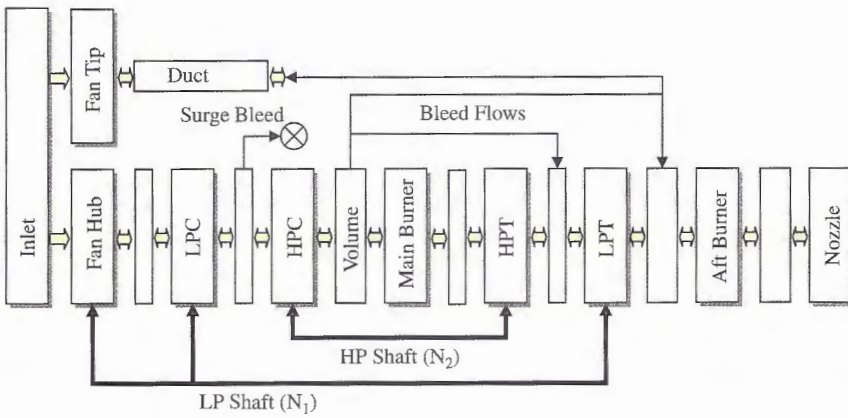


Fig. 2.9 Computational flow diagram of a two-spool afterburning turbofan.

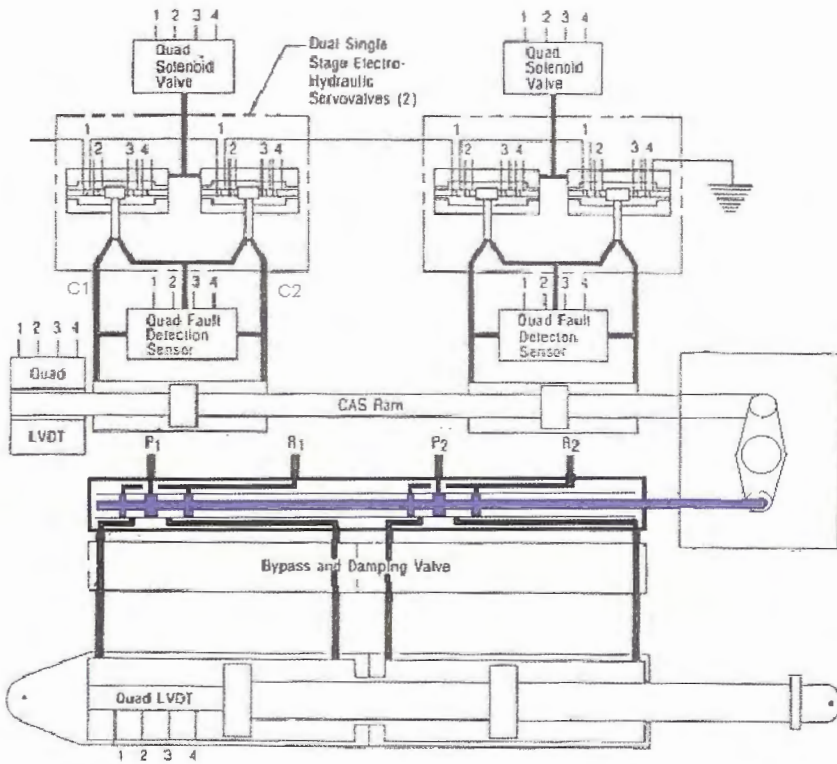


Fig. 2.11 Example hydraulic actuator (courtesy of Boeing Phantom Works).

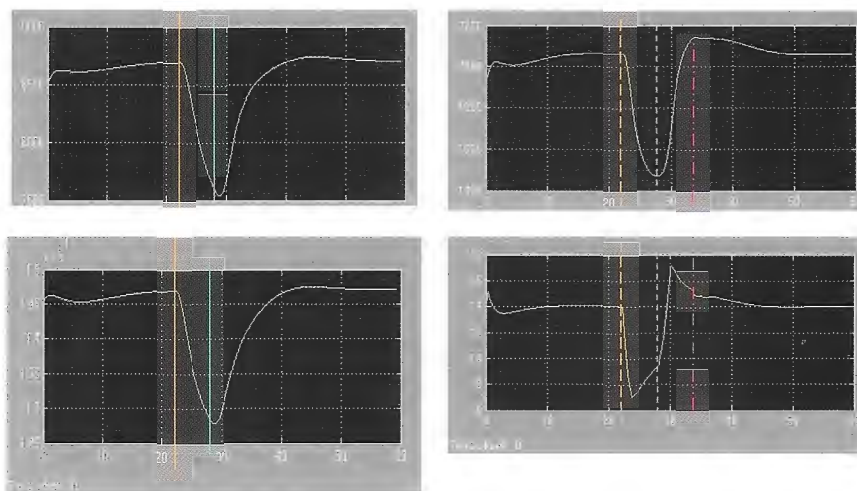


Fig. 2.18 Transient simulation of the AFRL generic engine model (N_1 in top left, N_2 in bottom left, T_{45} in top right, W_f/P_3 in bottom right).

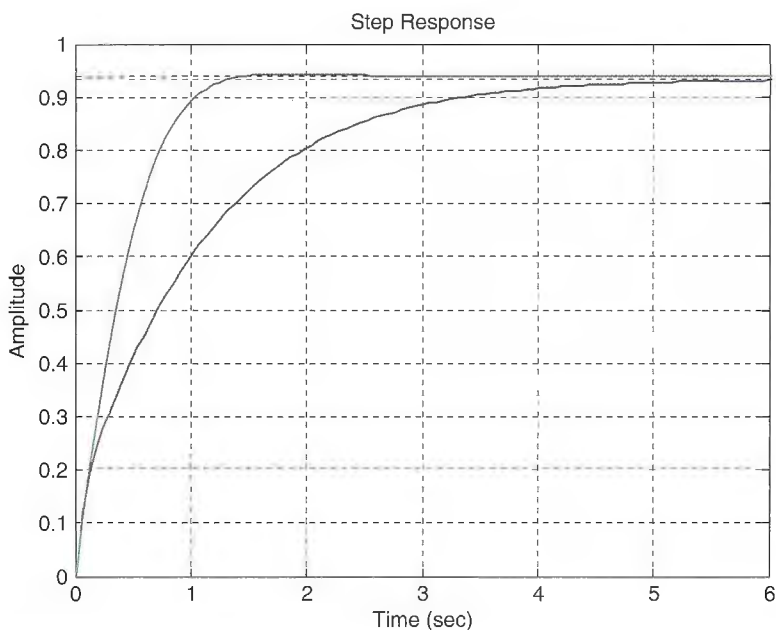


Fig. 3.13 Comparison of thrust responses of the F100 engine at idle (blue line) and intermediate power (green line).

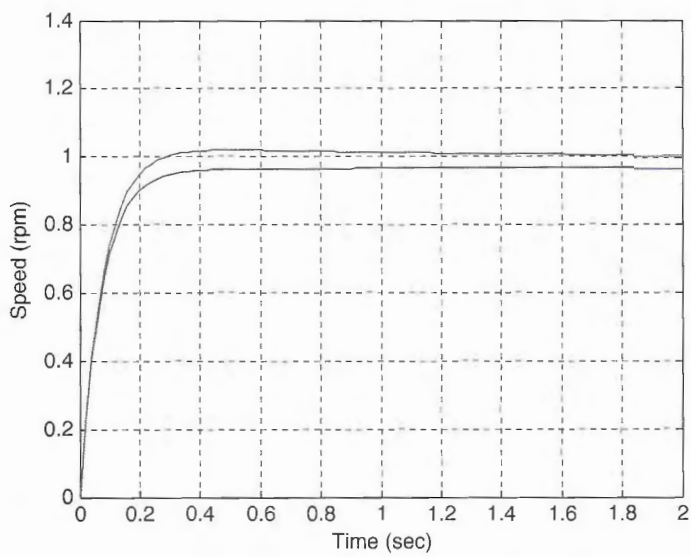


Fig. 4.6 Comparison of the closed-loop responses of the speed with a high-gain controller for both design cases (case 1 in blue, case 2 in green).

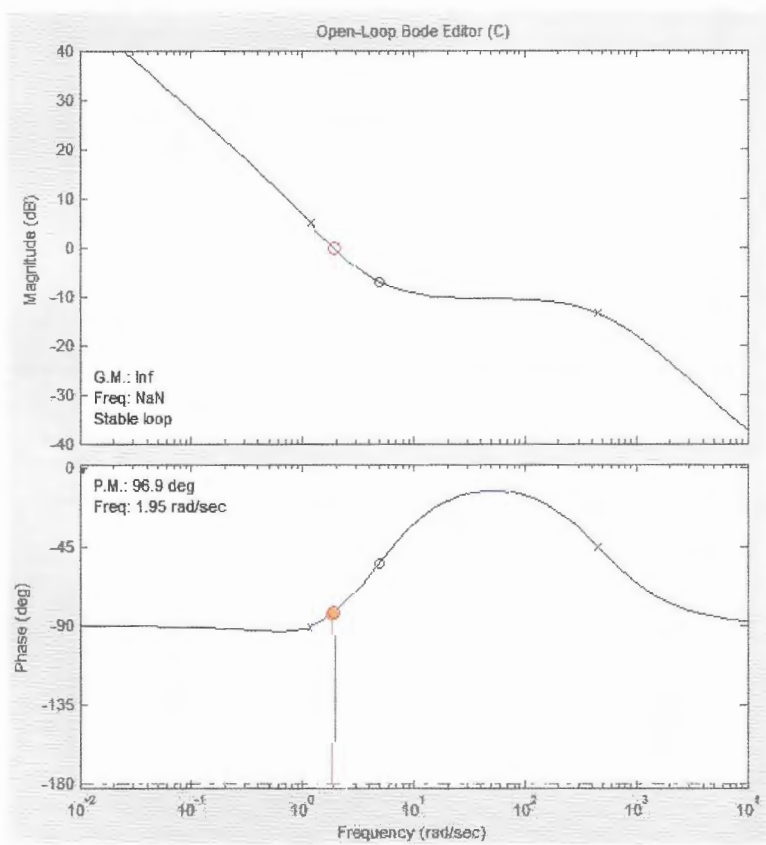


Fig. 4.14 Bode plot of a PI control loop for a two-spool engine.

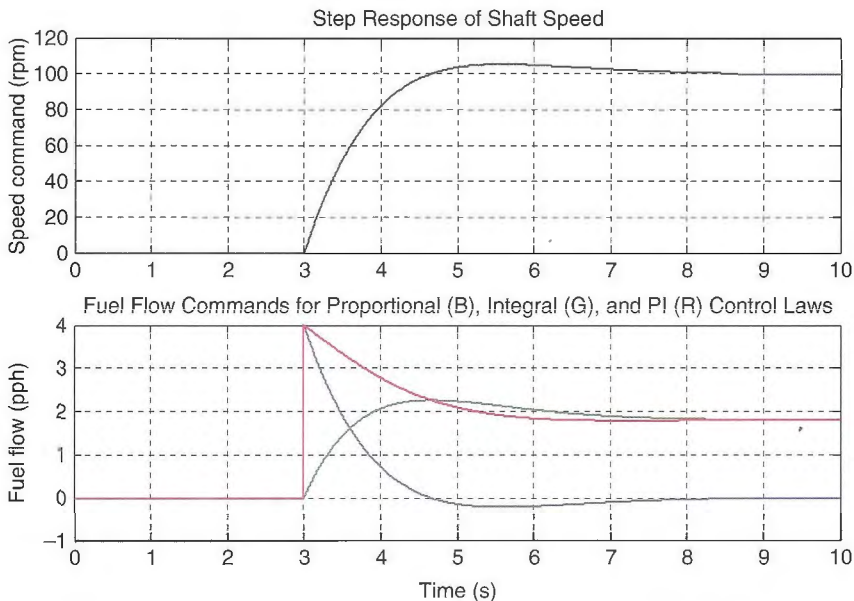


Fig. 4.22 Fuel control commands for proportional, integral, and PI control laws in design case 2 of the single-spool engine.

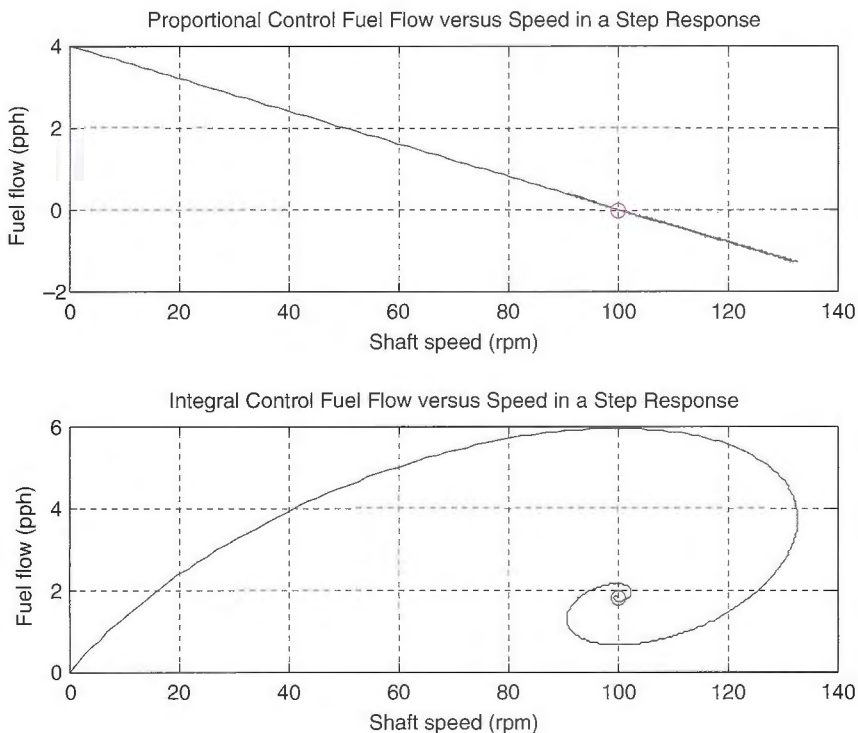


Fig. 4.24 Comparison of the parametric relationship of fuel control command vs engine speed between a proportional control and an integral control.

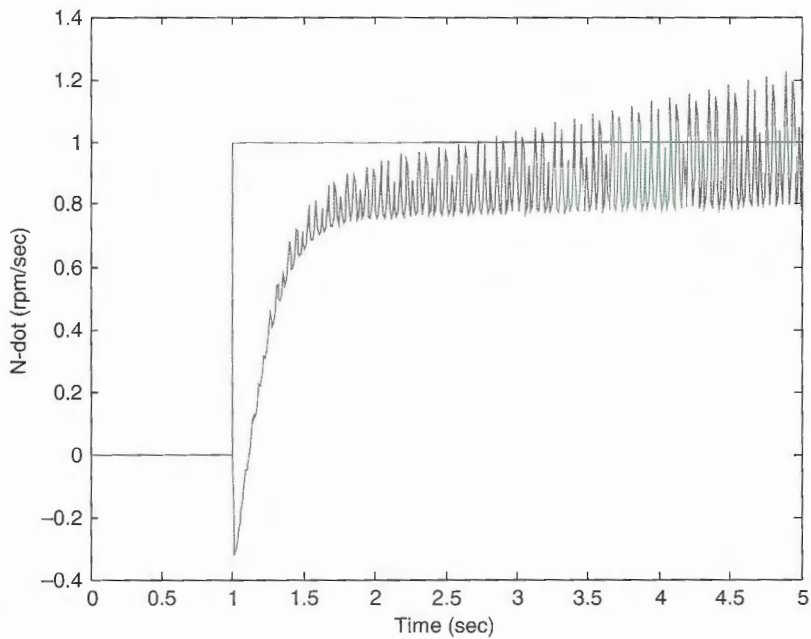


Fig. 5.15 Comparison between the calculated and low-pass-filtered $\dot{N}_1$ (in green) and the $\dot{N}_1$ reference command (in blue).

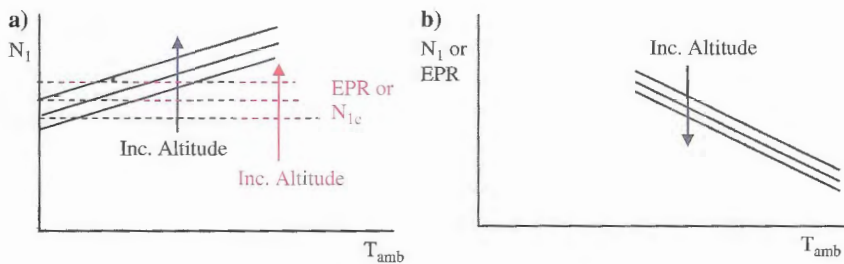


Fig. 6.2 (a) Flat-rate schedule. (b) Temperature schedule.

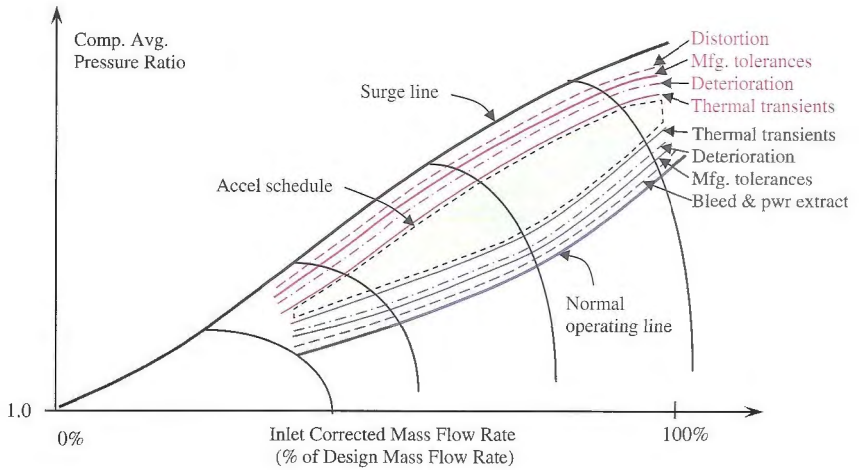


Fig. 6.10 Surge margin stacking and the reduced acceleration region (in green).

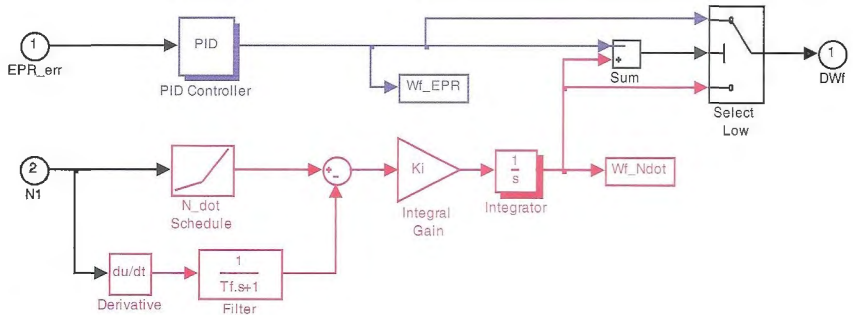


Fig. 6.28 Combined EPR control and N-dot control for a hypothetical two-spool engine.

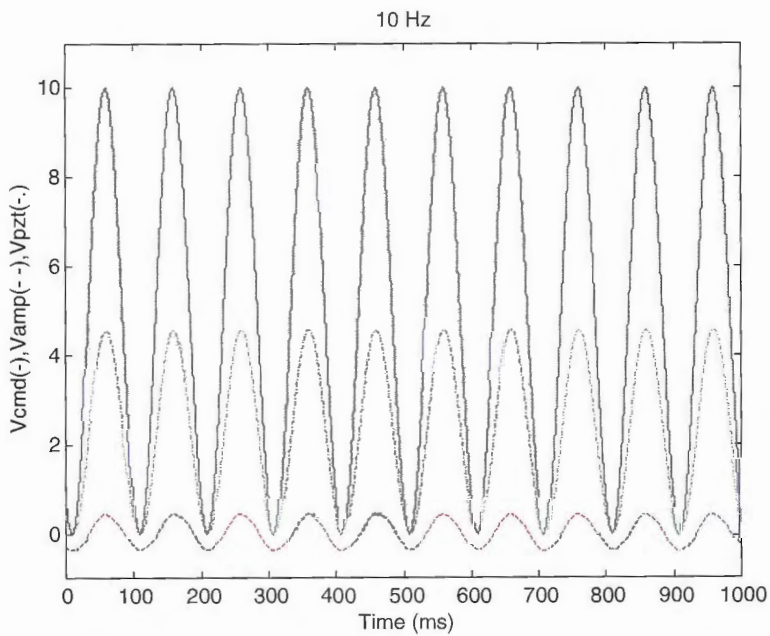


Fig. 7.16 Translator command and displacement signals at 10 Hz.

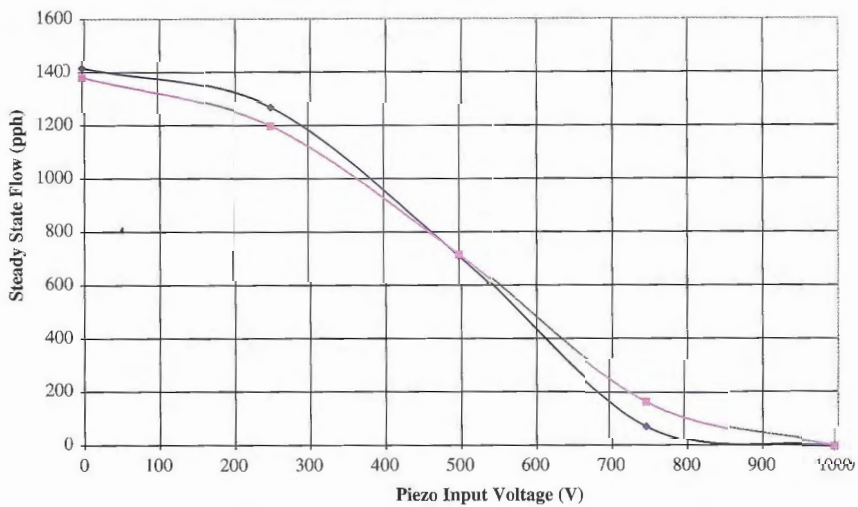


Fig. 7.24 Steady-state fuel flows for valve 1 (blue) and valve 2 (red) vs prime-mover input voltage at 300 psig.

Computation demand

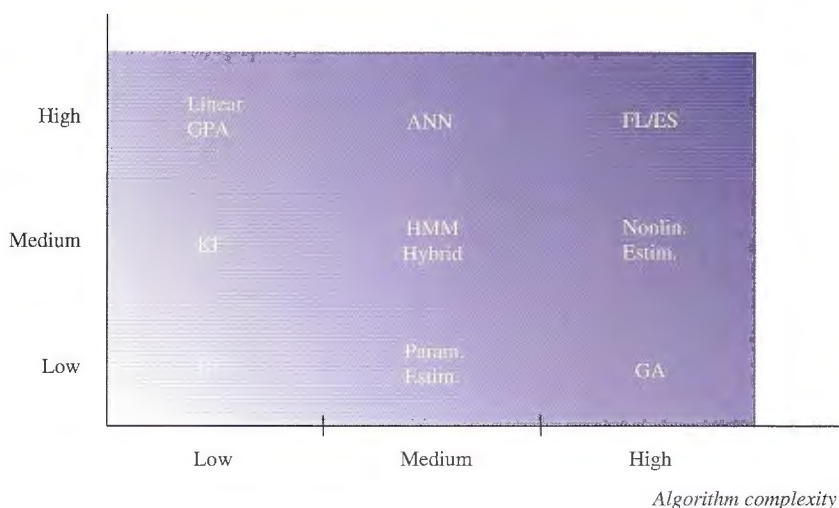


Fig. 8.9 Computational performance of GPA diagnostic algorithms.

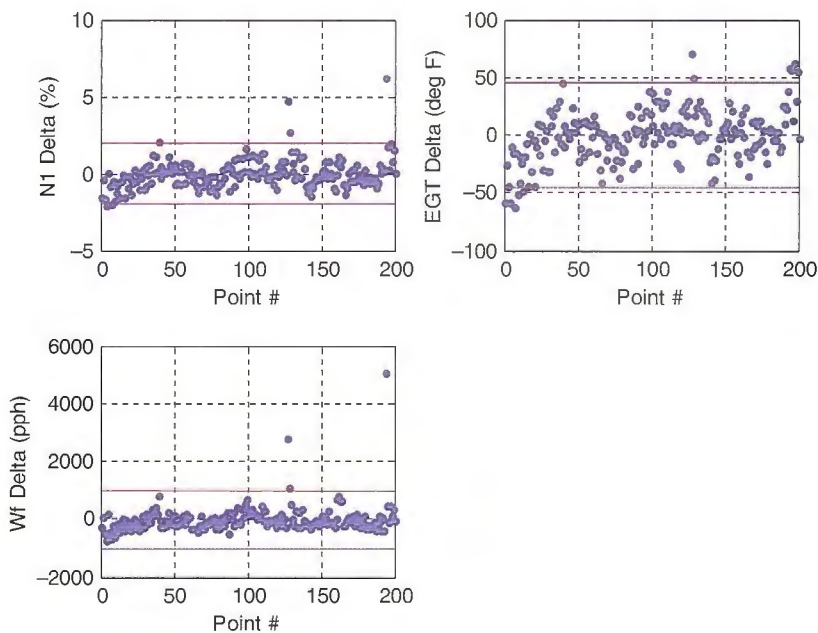


Fig. 8.13 Trend charts for the fan speed, the turbine exhaust temperature, and the fuel flow.

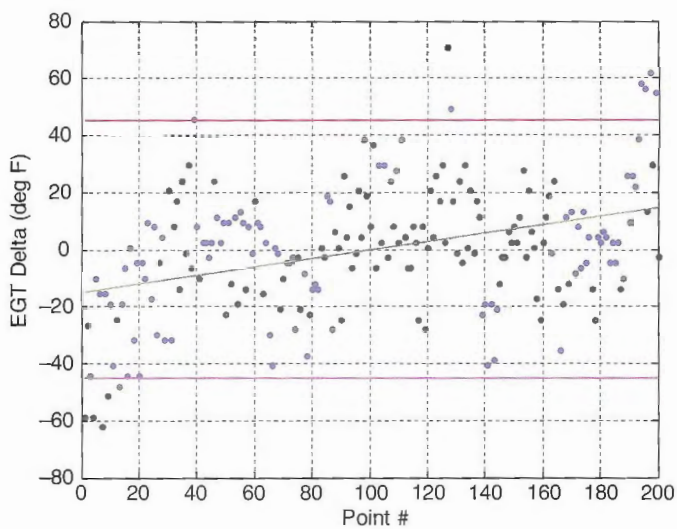


Fig. 8.14 Control limits (in red) and the trend line (in green) on EGT trend chart.

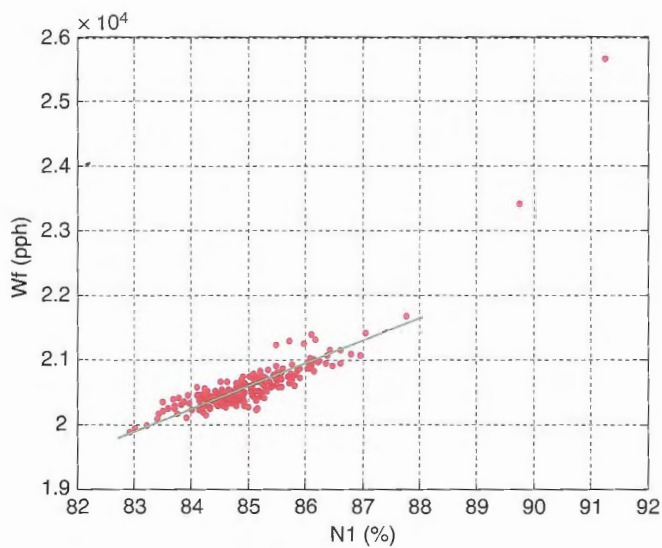


Fig. 8.15 Fuel flow rate vs N_1 .

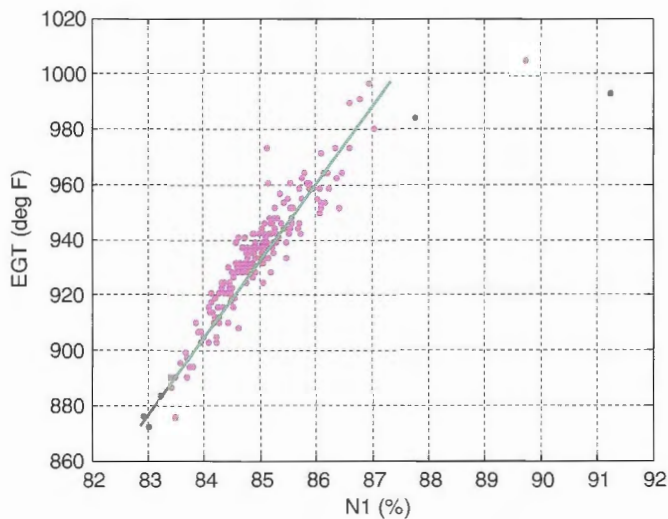


Fig. 8.16 Turbine temperature vs N_1 .

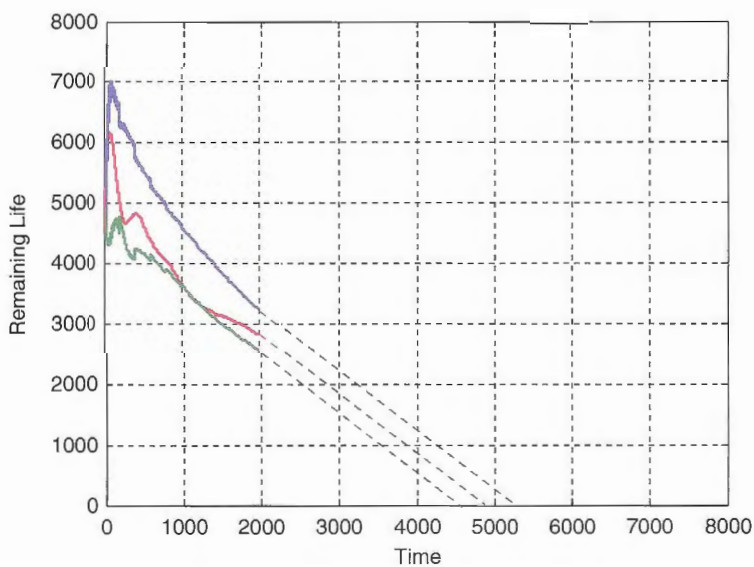


Fig. 9.4 RUL forecast with the confidence band.

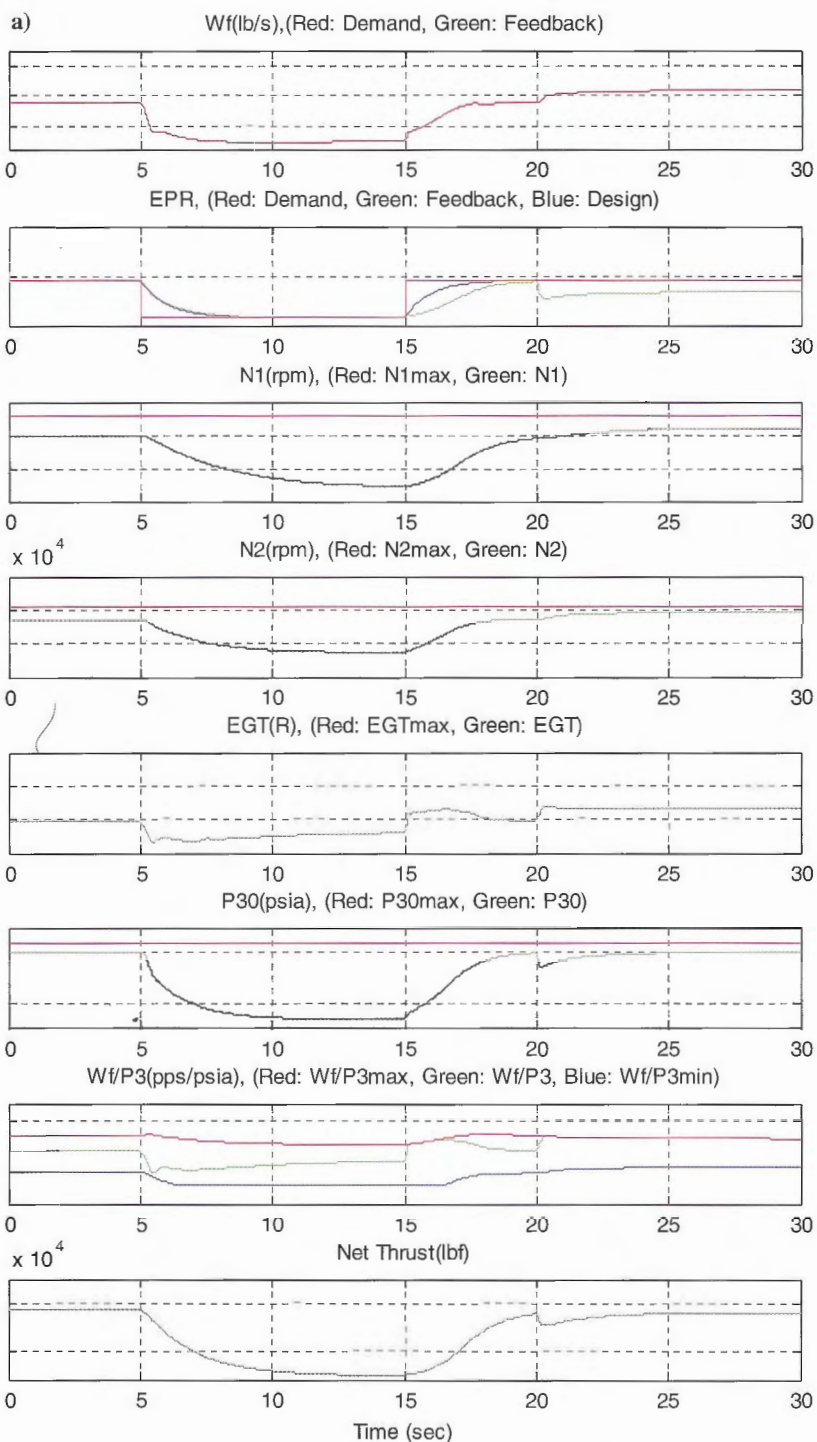


Fig. 9.11 (a) Engine response with a normal controller.

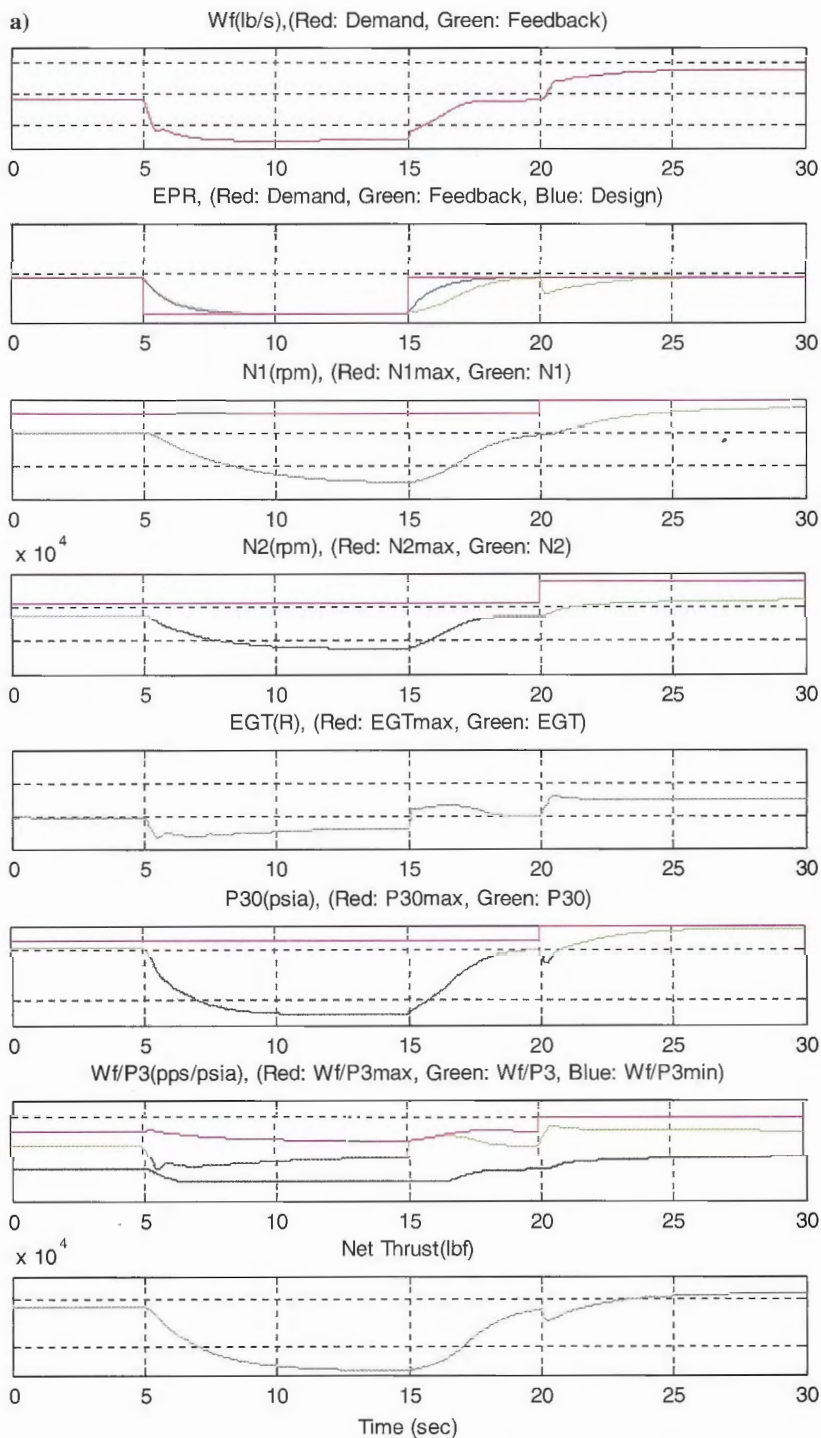


Fig. 9.12 (a) Engine response with a limit-reset controller.

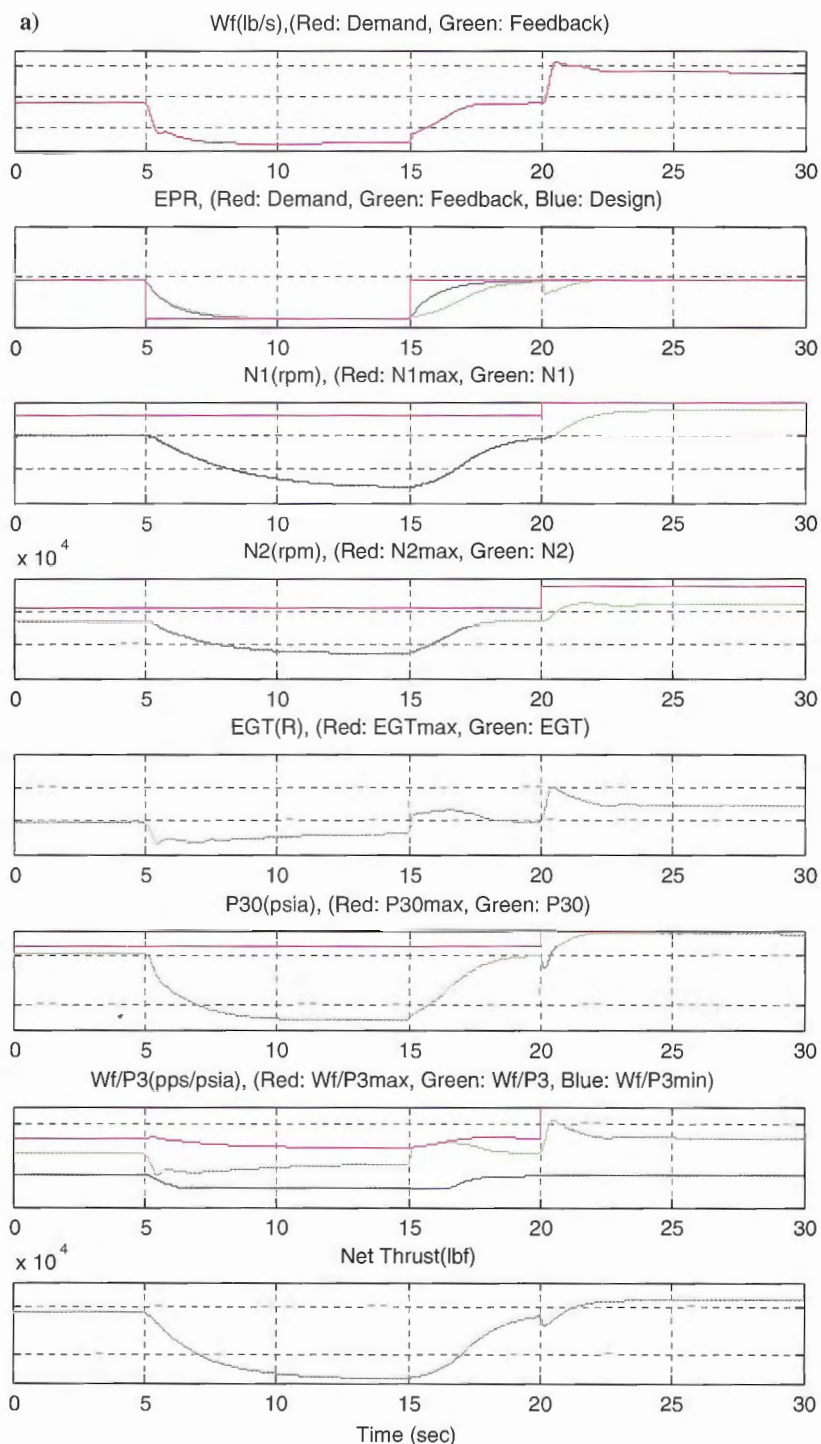


Fig. 9.13 (a) Engine response with a gain-magnified controller.

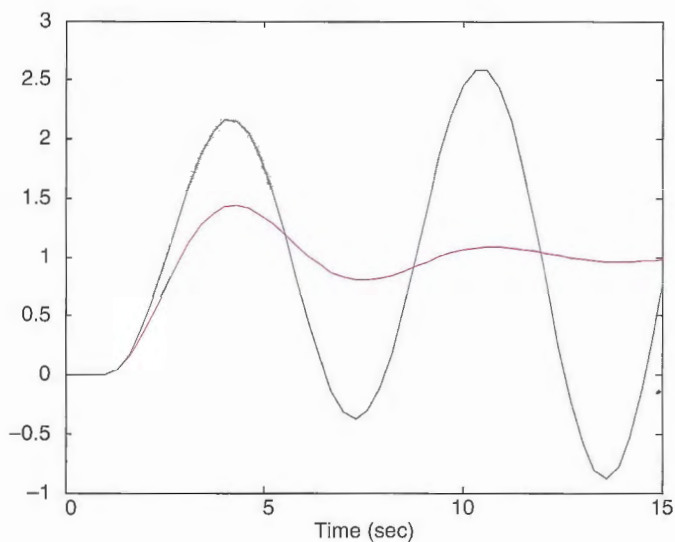


Fig. A.13 Example of an unstable system—output response (red, stable; blue, unstable).

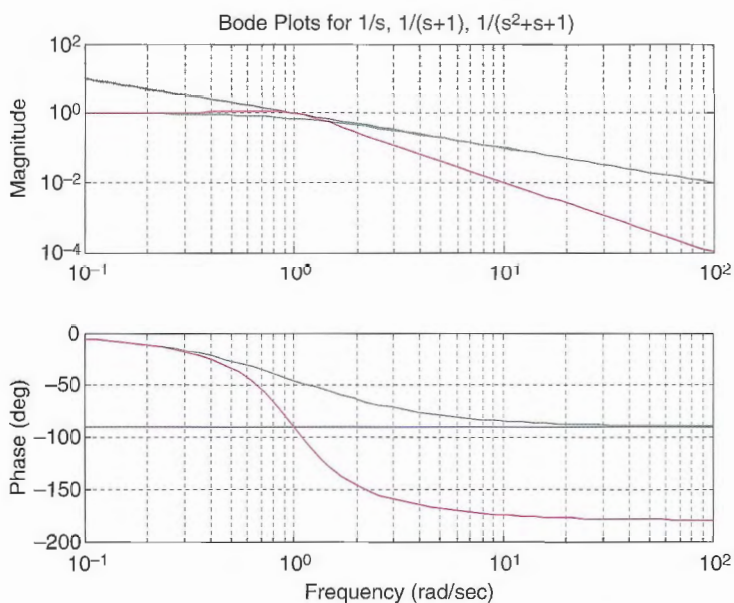


Fig. A.17 Comparison of the frequency responses of three basic TFs. (The integrator is in blue, the first-order lag is in green, and the second-order TF is in red.)

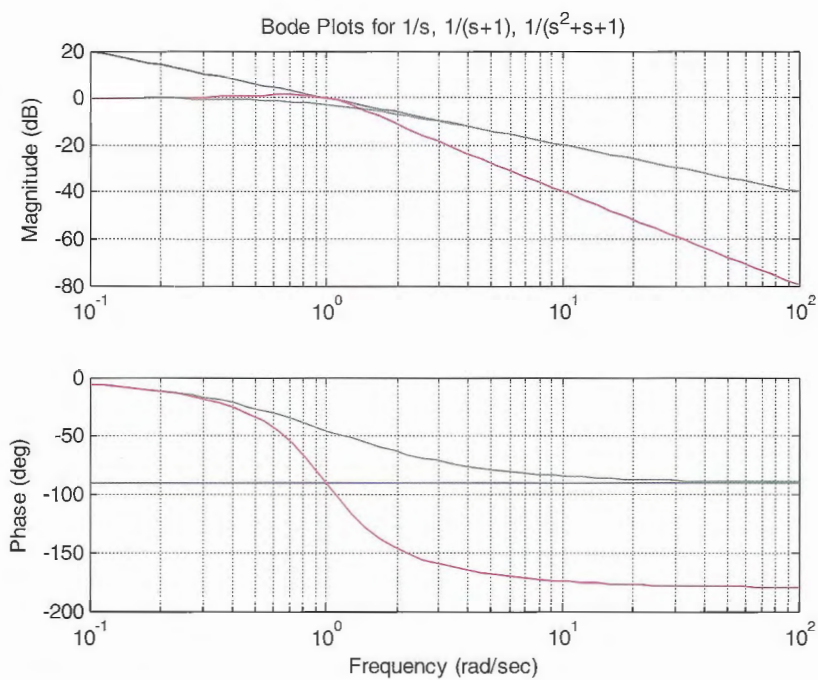


Fig. A.18 Comparison of the frequency responses of three basic TFs with magnitudes in the unit of decibels. (The integrator is in blue, the first-order lag is in green, and the second-order TF is in red.)

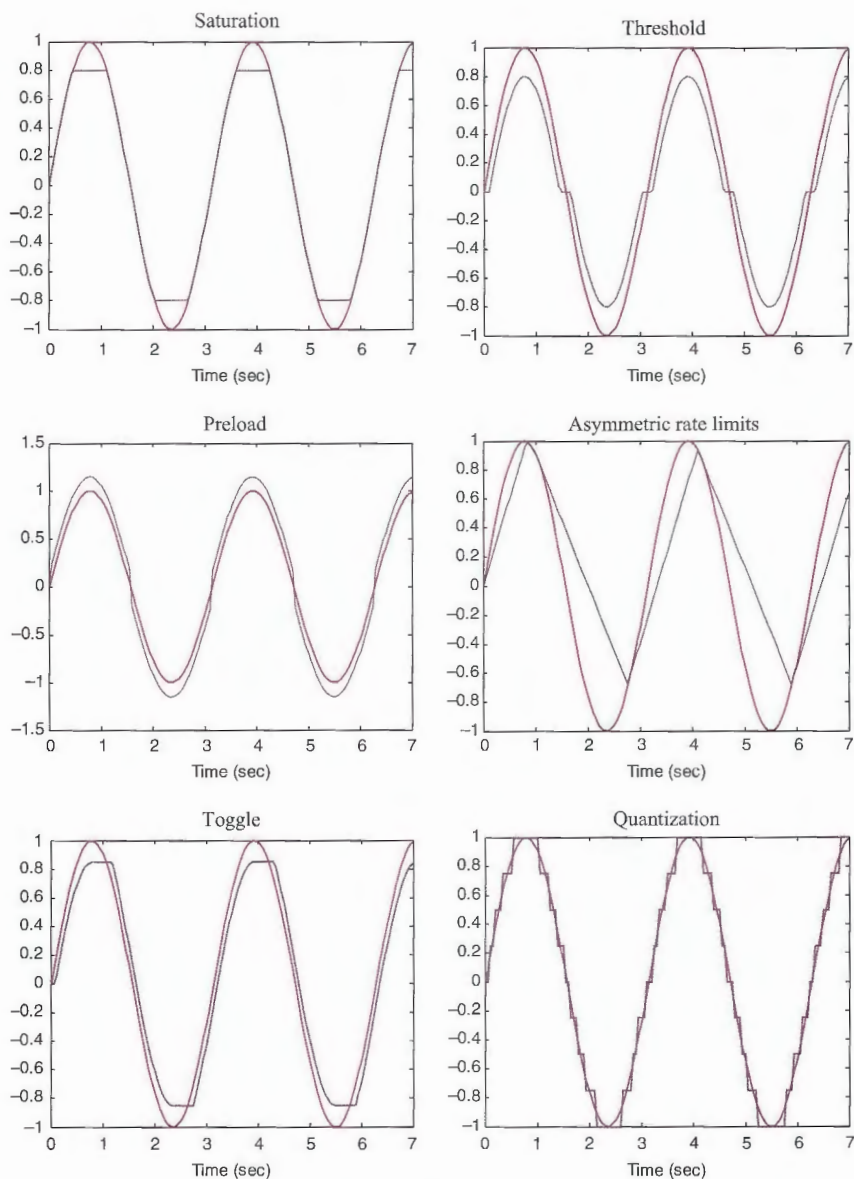


Fig. A.22 Effects of six common nonlinearities (red, input; blue, output).

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ABOUT THE BOOK

This book covers the design of engine control and monitoring systems, with a dual interest in both turbofan and turboshaft engines. It focuses on four areas of interest: 1) modeling of engine dynamics, 2) application of specific control design methods to gas turbine engines, 3) advanced control concepts, and 4) engine condition monitoring.

While the material in this book is applicable to all air, land, and sea-based gas turbines, the majority of the book is concerned with aviation gas turbines. The emphasis is on aviation gas turbine controls because it is believed that most of the issues in the control design of other gas turbine control systems will be included in the substantial challenges inherent in aviation gas turbine control designs. The book is intended to provide a one-stop resource for understanding the design of modern turbine engine control and monitoring systems. Because gas turbine engine control and monitoring systems have successfully employed applications of nonlinear control and system health management, this book can also be used as a reference for the engineers and researchers who are designing control and monitoring systems for other industrial equipment and systems.

The book is based on a course on gas turbine engine controls developed and taught by Dr. Jaw and Dr. Mattingly during 2005–2007 to Air Force, Navy, and Army engineers. In addition to six chapters on engine control analysis and design, chapters are included on integration with aircraft systems, advanced engine control concepts, and engine monitoring and health management.



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